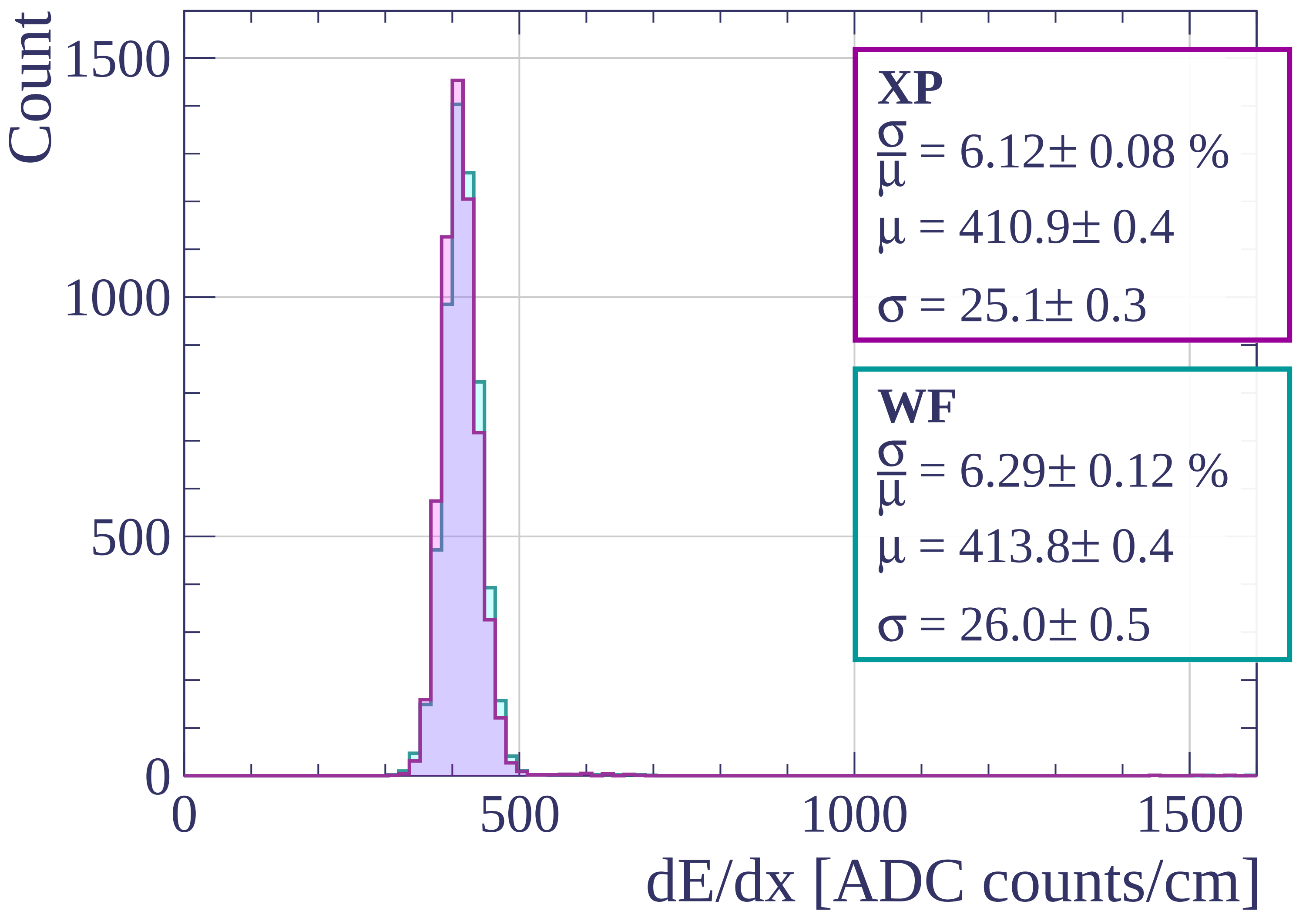
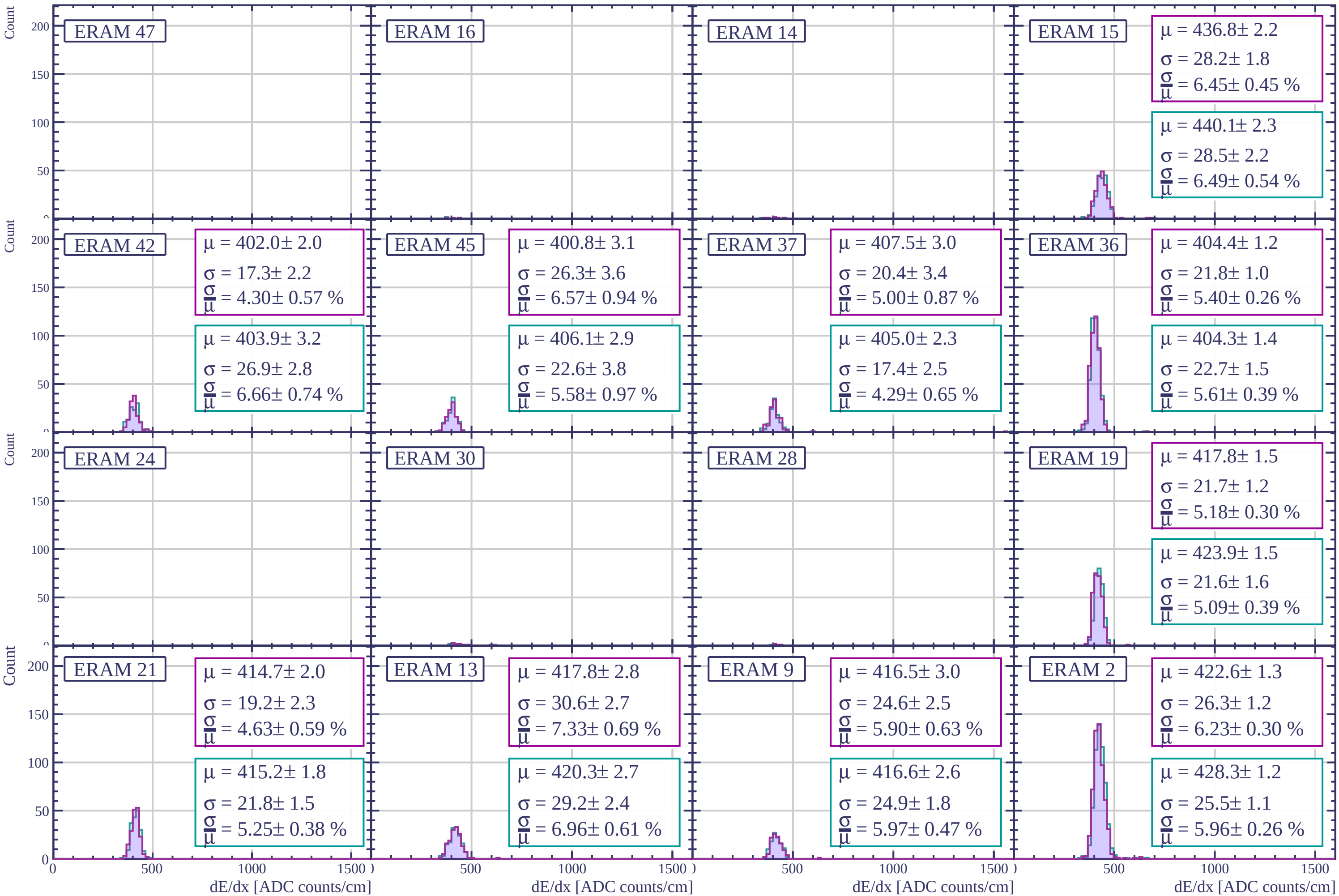
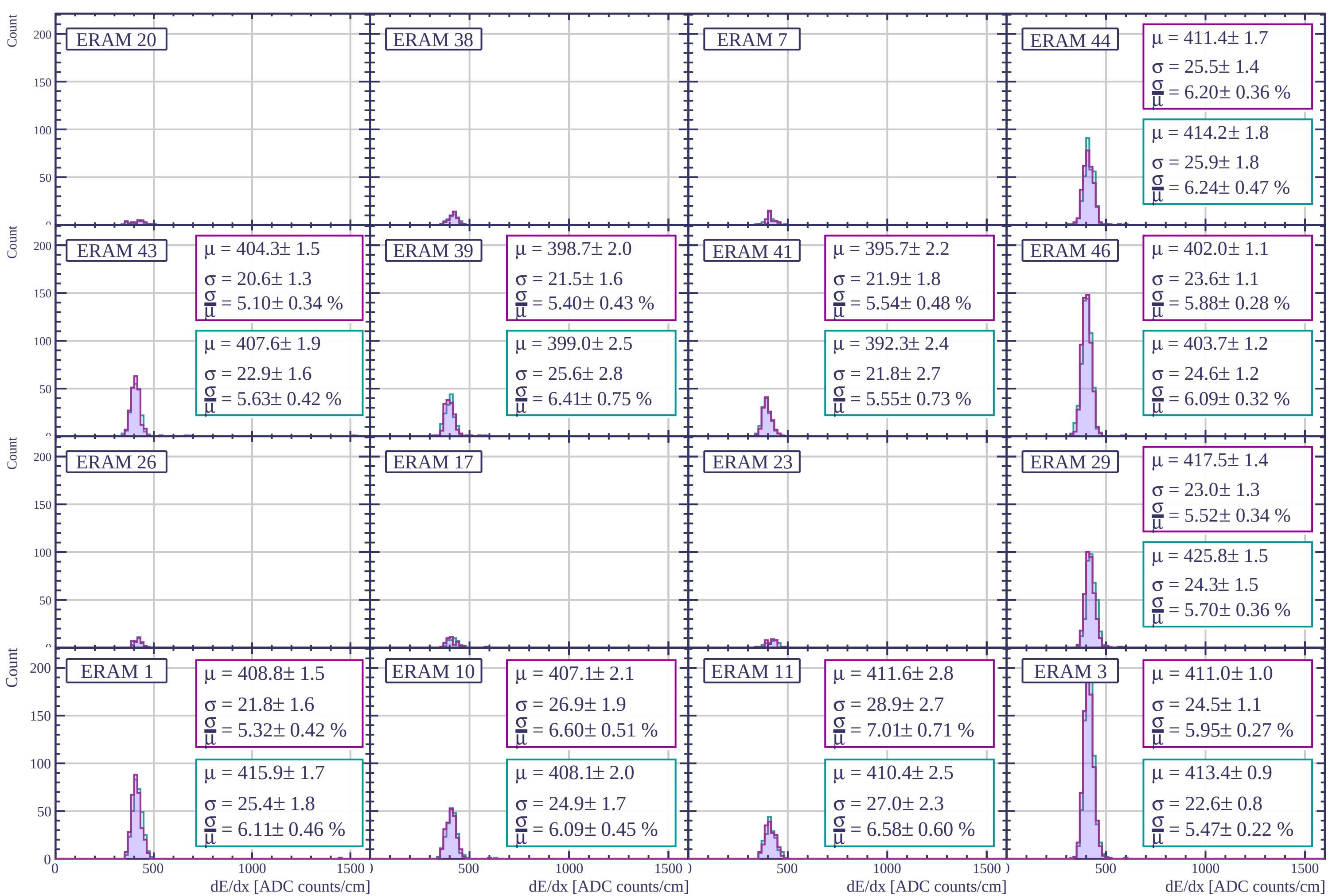


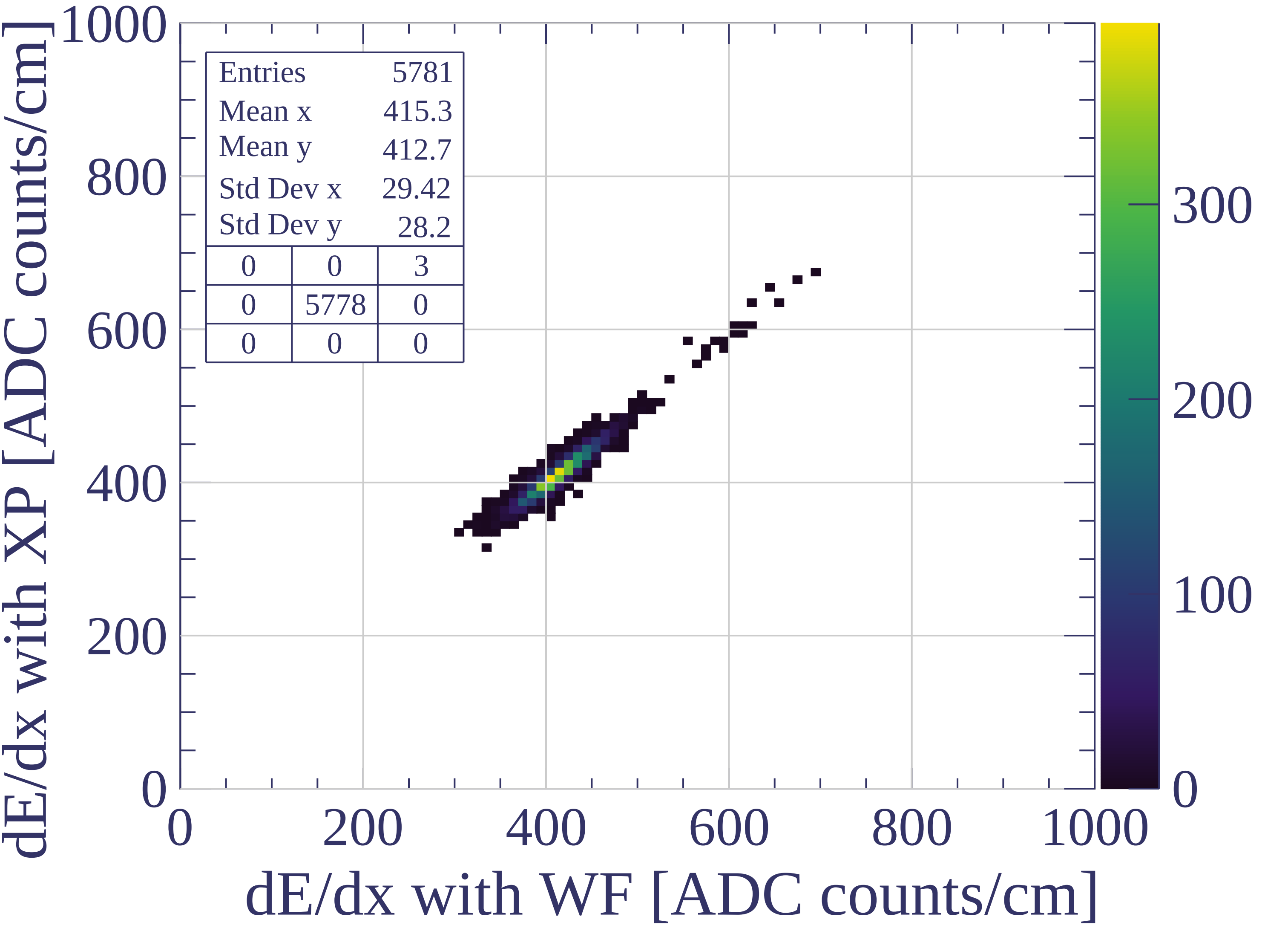
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 6.12± 0.08 %
μ = 410.9± 0.4
σ = 25.1± 0.3
WF
$\frac{\sigma}{\mu}$ = 6.29± 0.12 %
μ = 413.8± 0.4
σ = 26.0± 0.5

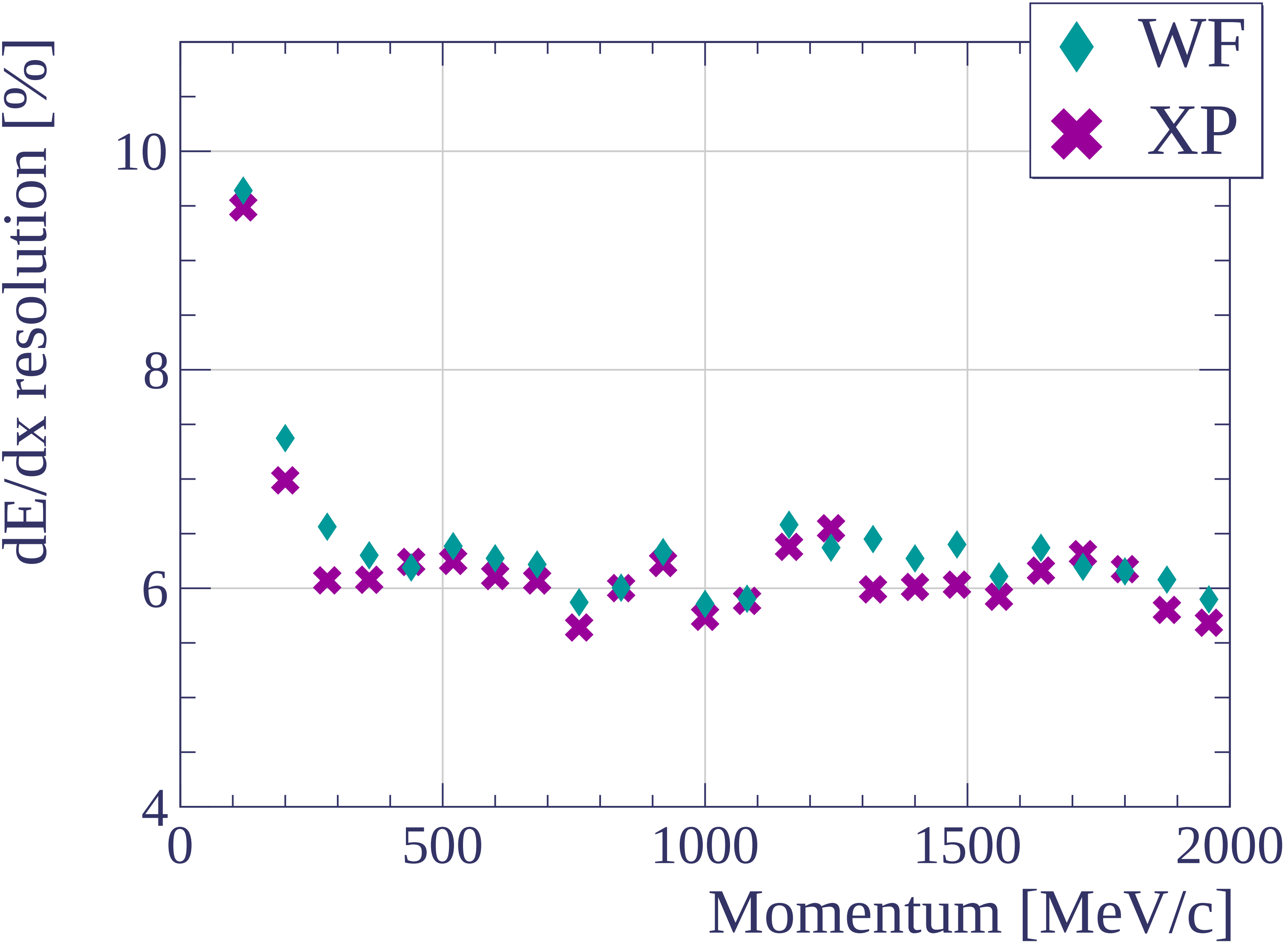


ERAM 47
ERAM 16
ERAM 14
ERAM 15
μ = 436.8± 2.2
σ = 28.2± 1.8
σ/μ = 6.45± 0.45 %
μ = 440.1± 2.3
σ = 28.5± 2.2
σ/μ = 6.49± 0.54 %
ERAM 42
μ = 402.0± 2.0
σ = 17.3± 2.2
σ/μ = 4.30± 0.57 %
μ = 403.9± 3.2
σ = 26.9± 2.8
σ/μ = 6.66± 0.74 %
ERAM 45
μ = 400.8± 3.1
σ = 26.3± 3.6
σ/μ = 6.57± 0.94 %
μ = 406.1± 2.9
σ = 22.6± 3.8
σ/μ = 5.58± 0.97 %
ERAM 37
μ = 407.5± 3.0
σ = 20.4± 3.4
σ/μ = 5.00± 0.87 %
μ = 405.0± 2.3
σ = 17.4± 2.5
σ/μ = 4.29± 0.65 %
ERAM 36
μ = 404.4± 1.2
σ = 21.8± 1.0
σ/μ = 5.40± 0.26 %
μ = 404.3± 1.4
σ = 22.7± 1.5
σ/μ = 5.61± 0.39 %
ERAM 24
ERAM 30
ERAM 28
ERAM 19
μ = 417.8± 1.5
σ = 21.7± 1.2
σ/μ = 5.18± 0.30 %
μ = 423.9± 1.5
σ = 21.6± 1.6
σ/μ = 5.09± 0.39 %
ERAM 21
μ = 414.7± 2.0
σ = 19.2± 2.3
σ/μ = 4.63± 0.59 %
μ = 415.2± 1.8
σ = 21.8± 1.5
σ/μ = 5.25± 0.38 %
ERAM 13
μ = 417.8± 2.8
σ = 30.6± 2.7
σ/μ = 7.33± 0.69 %
μ = 420.3± 2.7
σ = 29.2± 2.4
σ/μ = 6.96± 0.61 %
ERAM 9
μ = 416.5± 3.0
σ = 24.6± 2.5
σ/μ = 5.90± 0.63 %
μ = 416.6± 2.6
σ = 24.9± 1.8
σ/μ = 5.97± 0.47 %
ERAM 2
μ = 422.6± 1.3
σ = 26.3± 1.2
σ/μ = 6.23± 0.30 %
μ = 428.3± 1.2
σ = 25.5± 1.1
σ/μ = 5.96± 0.26 %
Count
dE/dx [ADC counts/cm]

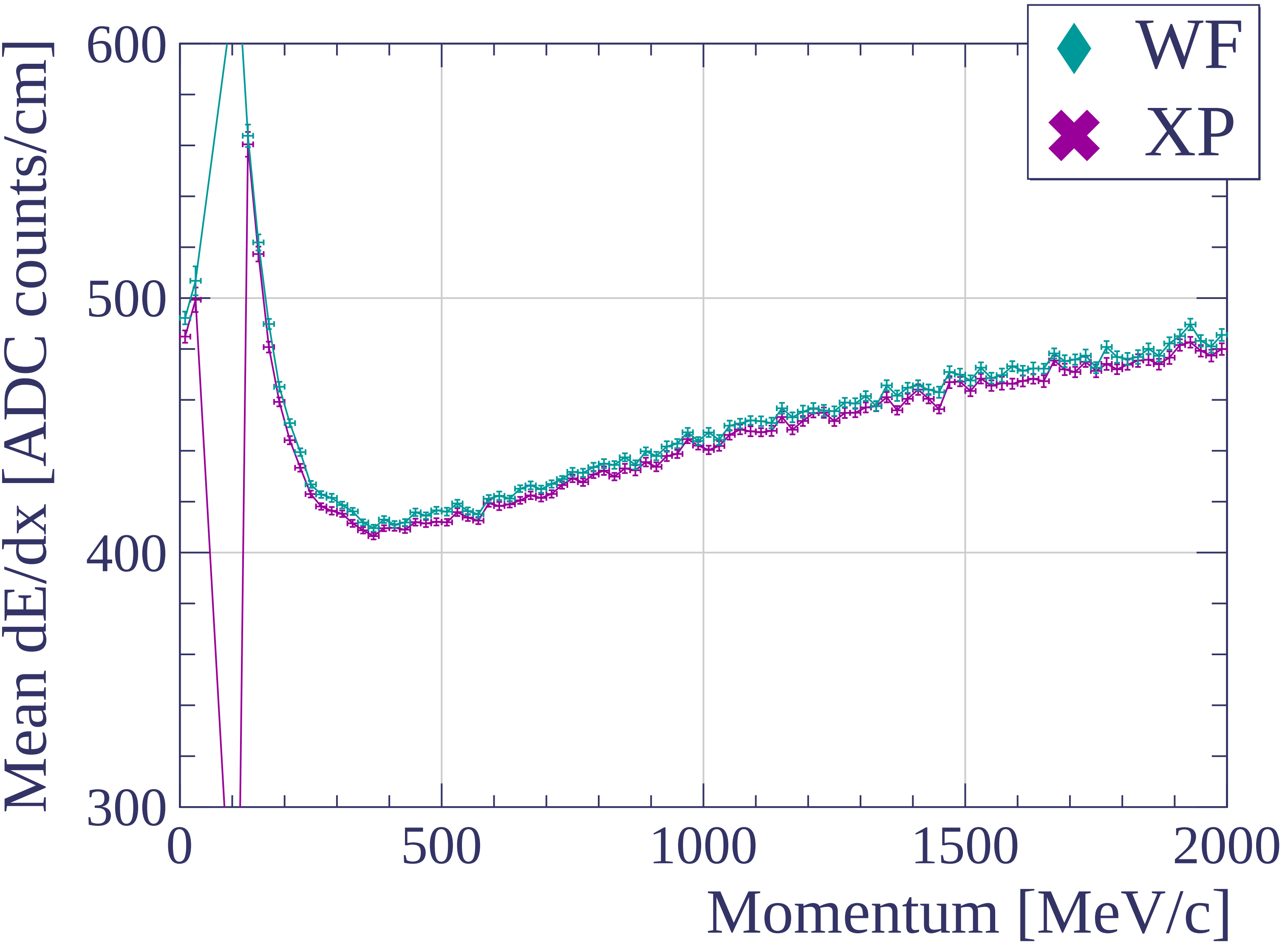


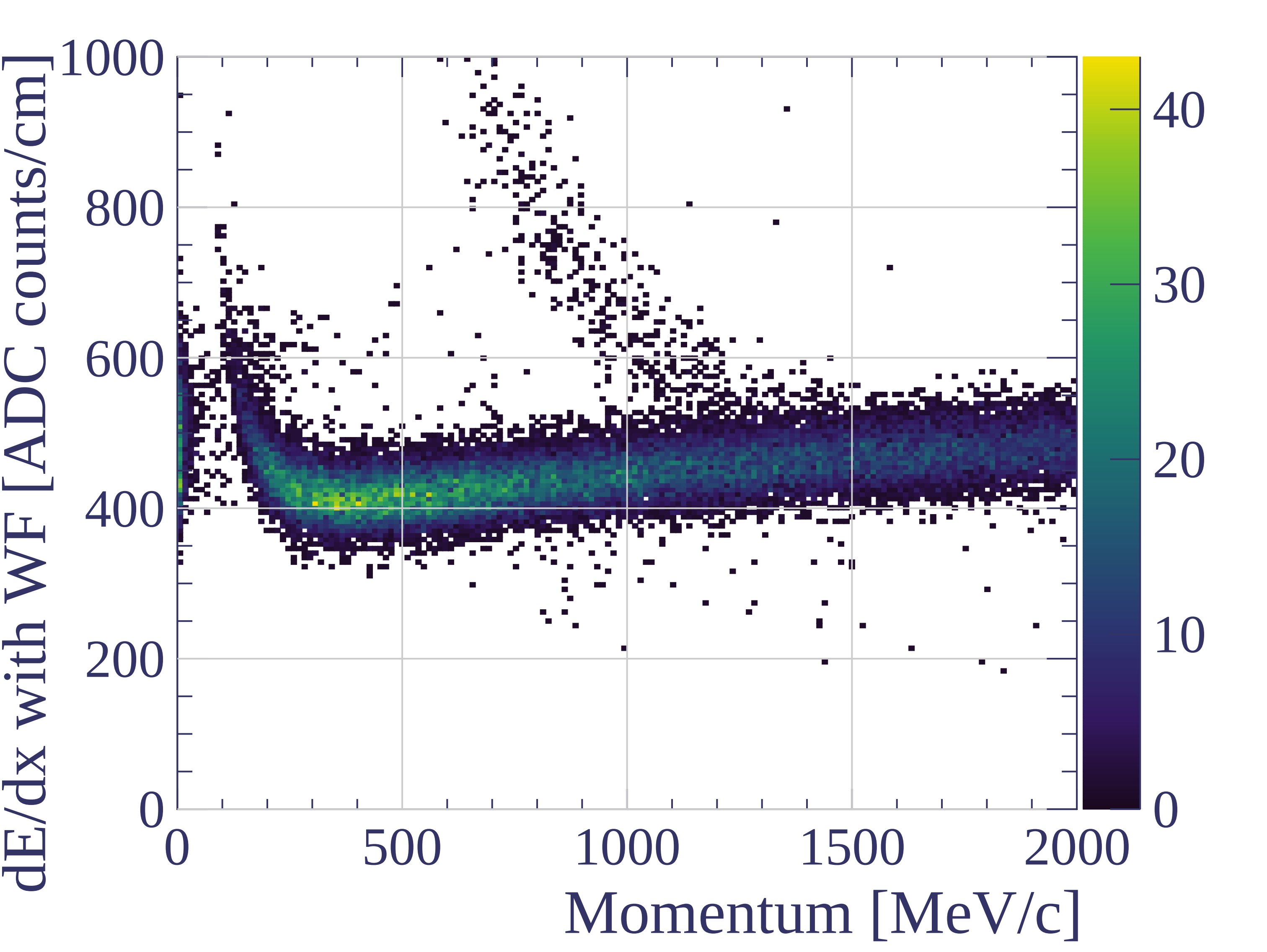
ERAM 20
ERAM 38
ERAM 7
ERAM 44
μ = 411.4± 1.7
σ = 25.5± 1.4
σ/μ = 6.20± 0.36 %
μ = 414.2± 1.8
σ = 25.9± 1.8
σ/μ = 6.24± 0.47 %
ERAM 43
μ = 404.3± 1.5
σ = 20.6± 1.3
σ/μ = 5.10± 0.34 %
μ = 407.6± 1.9
σ = 22.9± 1.6
σ/μ = 5.63± 0.42 %
ERAM 39
μ = 398.7± 2.0
σ = 21.5± 1.6
σ/μ = 5.40± 0.43 %
μ = 399.0± 2.5
σ = 25.6± 2.8
σ/μ = 6.41± 0.75 %
ERAM 41
μ = 395.7± 2.2
σ = 21.9± 1.8
σ/μ = 5.54± 0.48 %
μ = 392.3± 2.4
σ = 21.8± 2.7
σ/μ = 5.55± 0.73 %
ERAM 46
μ = 402.0± 1.1
σ = 23.6± 1.1
σ/μ = 5.88± 0.28 %
μ = 403.7± 1.2
σ = 24.6± 1.2
σ/μ = 6.09± 0.32 %
ERAM 26
ERAM 17
ERAM 23
ERAM 29
μ = 417.5± 1.4
σ = 23.0± 1.3
σ/μ = 5.52± 0.34 %
μ = 425.8± 1.5
σ = 24.3± 1.5
σ/μ = 5.70± 0.36 %
ERAM 1
μ = 408.8± 1.5
σ = 21.8± 1.6
σ/μ = 5.32± 0.42 %
μ = 415.9± 1.7
σ = 25.4± 1.8
σ/μ = 6.11± 0.46 %
ERAM 10
μ = 407.1± 2.1
σ = 26.9± 1.9
σ/μ = 6.60± 0.51 %
μ = 408.1± 2.0
σ = 24.9± 1.7
σ/μ = 6.09± 0.45 %
ERAM 11
μ = 411.6± 2.8
σ = 28.9± 2.7
σ/μ = 7.01± 0.71 %
μ = 410.4± 2.5
σ = 27.0± 2.3
σ/μ = 6.58± 0.60 %
ERAM 3
μ = 411.0± 1.0
σ = 24.5± 1.1
σ/μ = 5.95± 0.27 %
μ = 413.4± 0.9
σ = 22.6± 0.8
σ/μ = 5.47± 0.22 %
Count
0
50
100
150
200
500
1000
1500
dE/dx [ADC counts/cm]


Entries 5781
Mean x 415.3
Mean y 412.7
Std Dev x 29.42
Std Dev y 28.2
0 0 3
0 5778 0
0 0 0
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]
0
200
400
600
800
1000
100
300

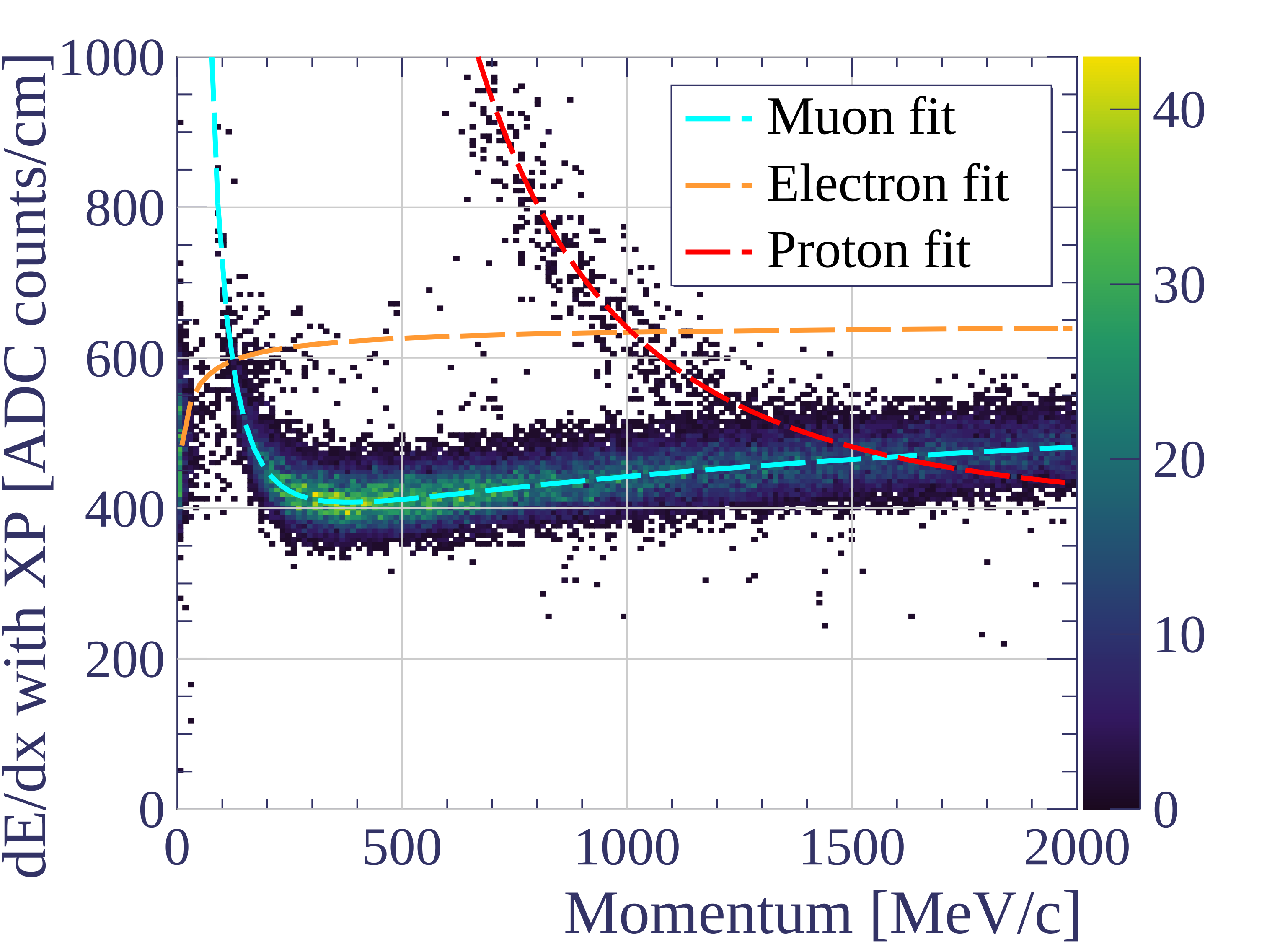


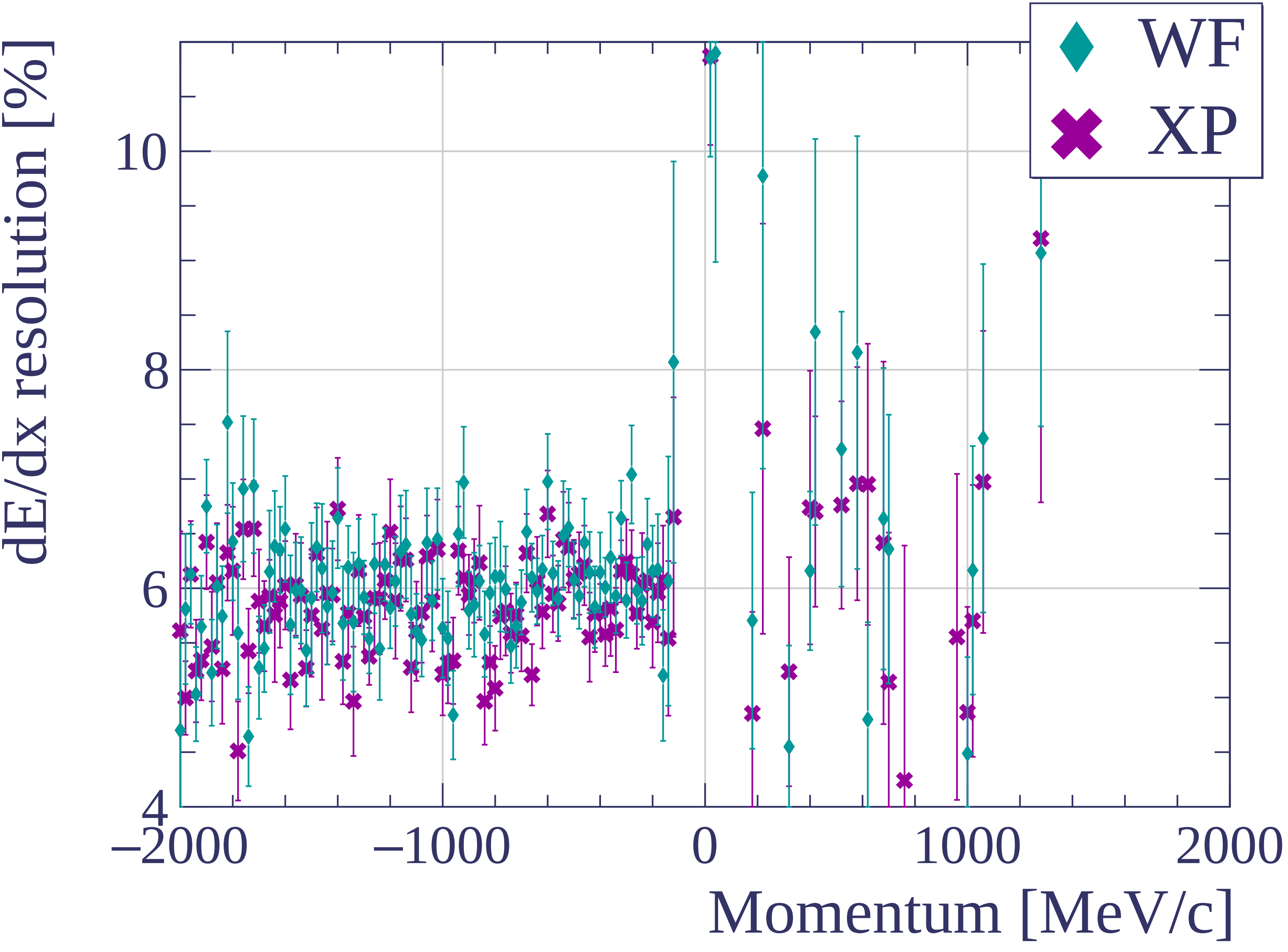
dE/dx resolution [%]
4
6
8
10
0
500
1000
1500
2000
Momentum [MeV/c]
WF
XP


WF
XP
Mean dE/dx [ADC counts/cm]
Momentum [MeV/c]
600
500
400
300
0
500
1000
1500
2000

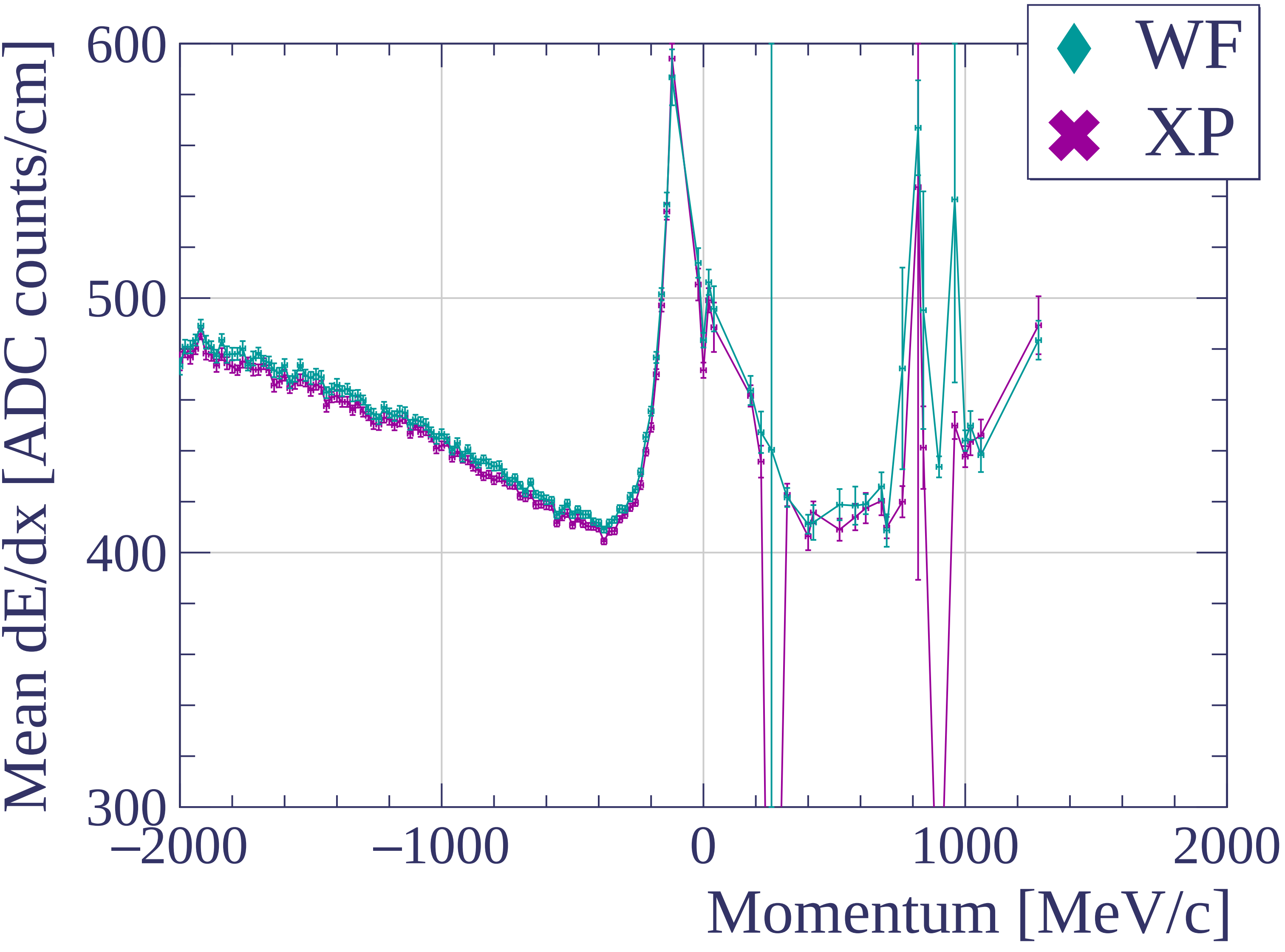


dE/dx with WF [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
500
1000
1500
2000
10
20
30
40

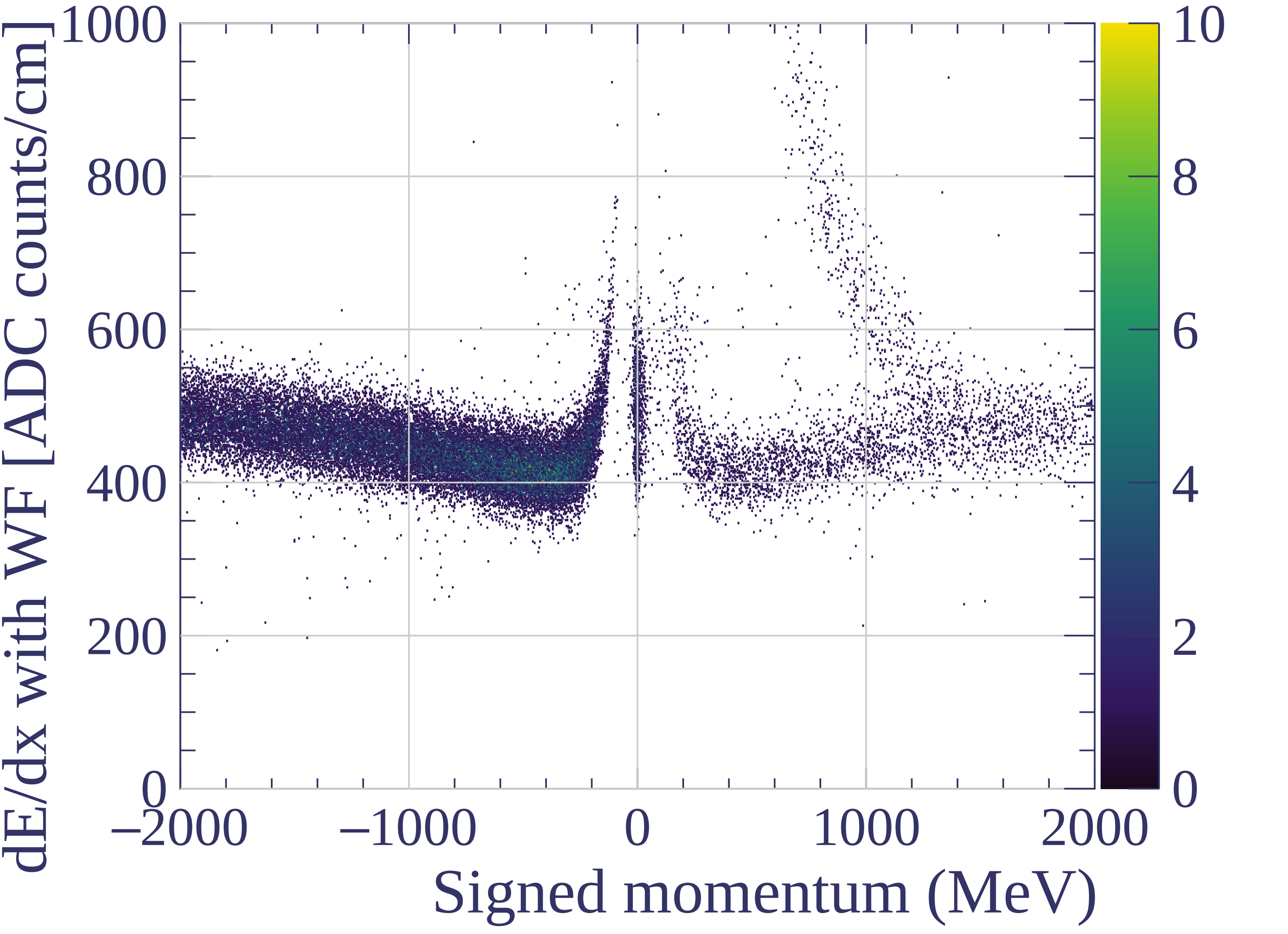

Muon fit
Electron fit
Proton fit
dE/dx with XP [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
0
500
1000
1500
2000
0
10
20
30
40



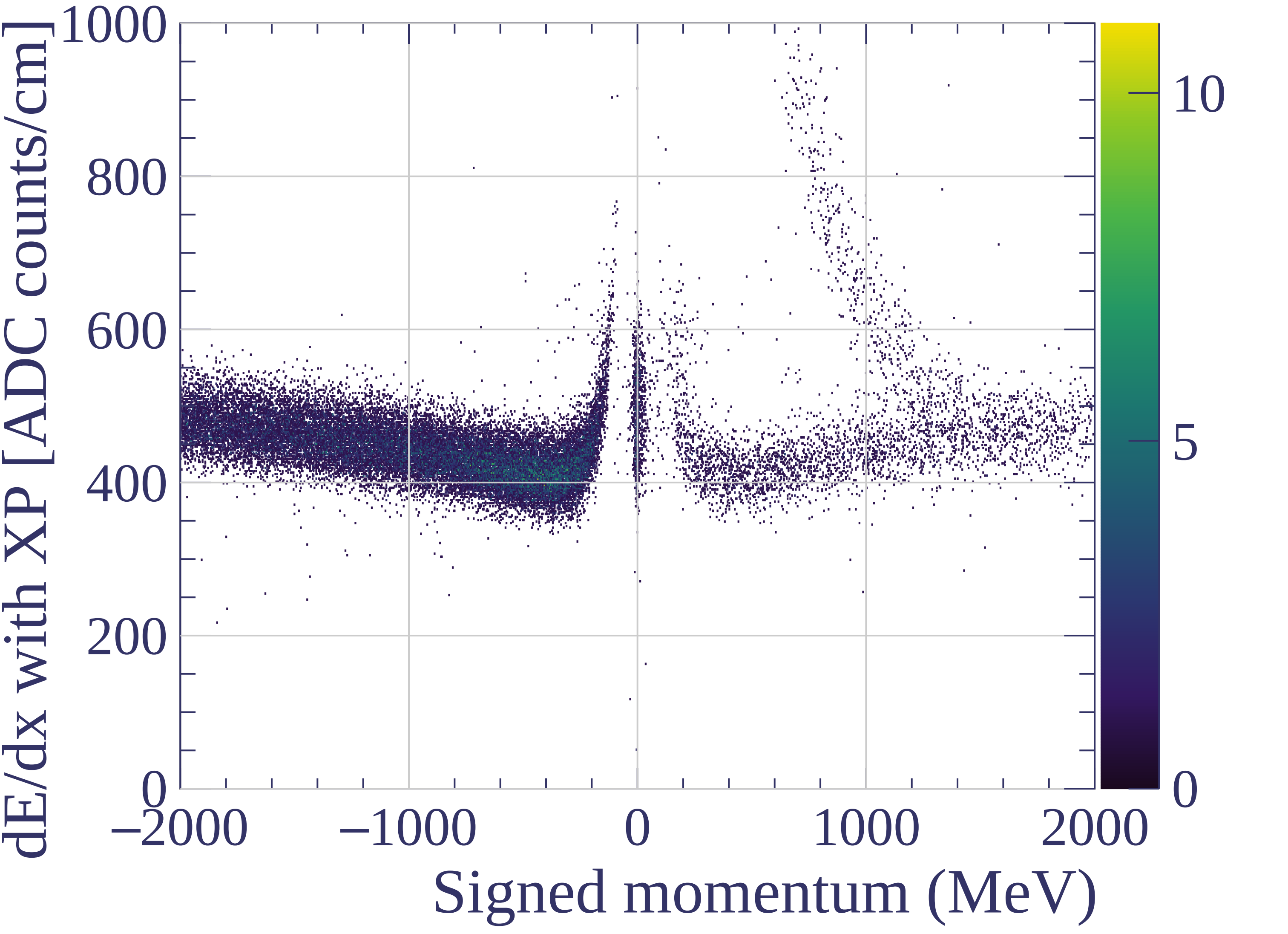
WF
XP
dE/dx resolution [%]
10
8
6
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]

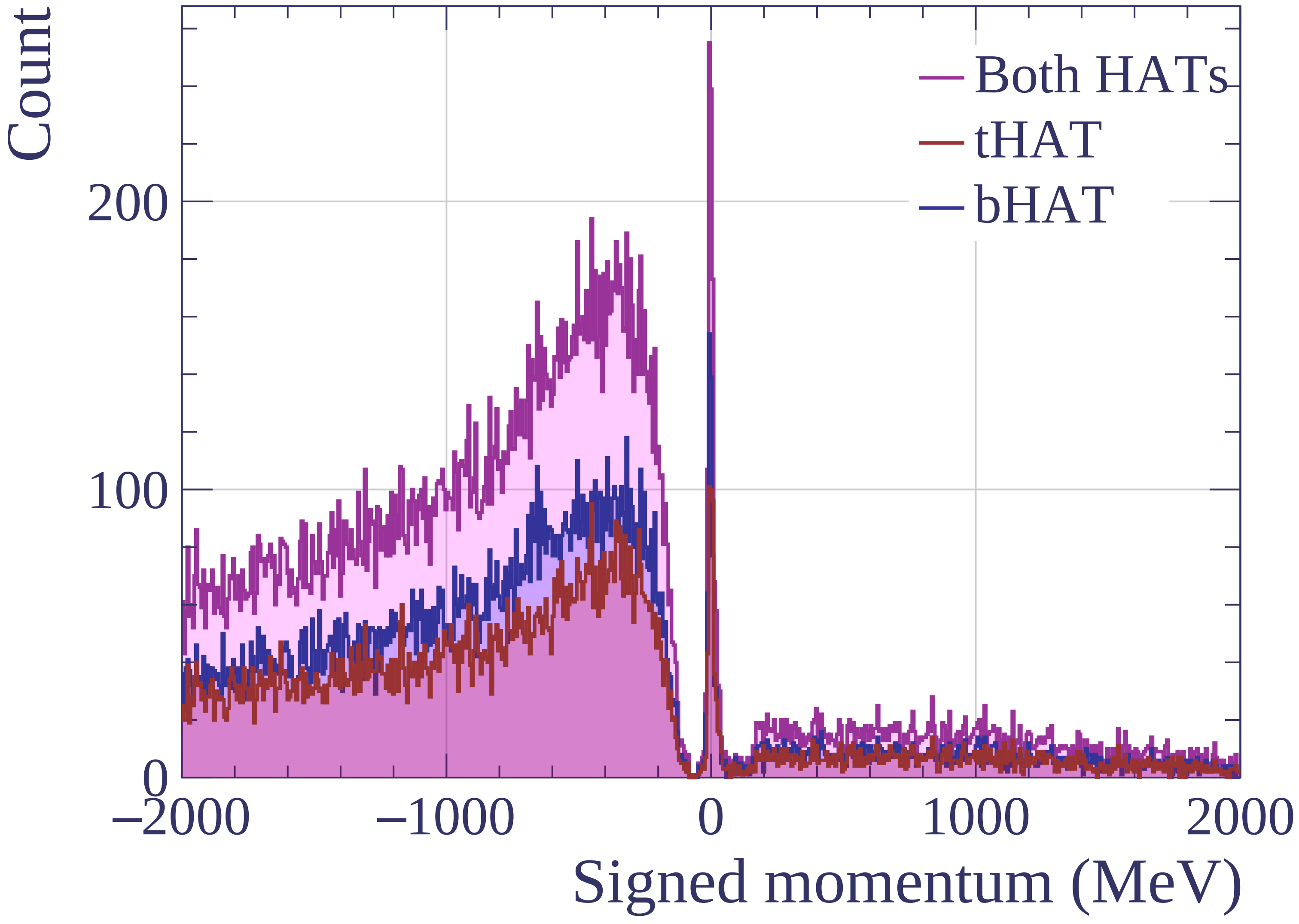


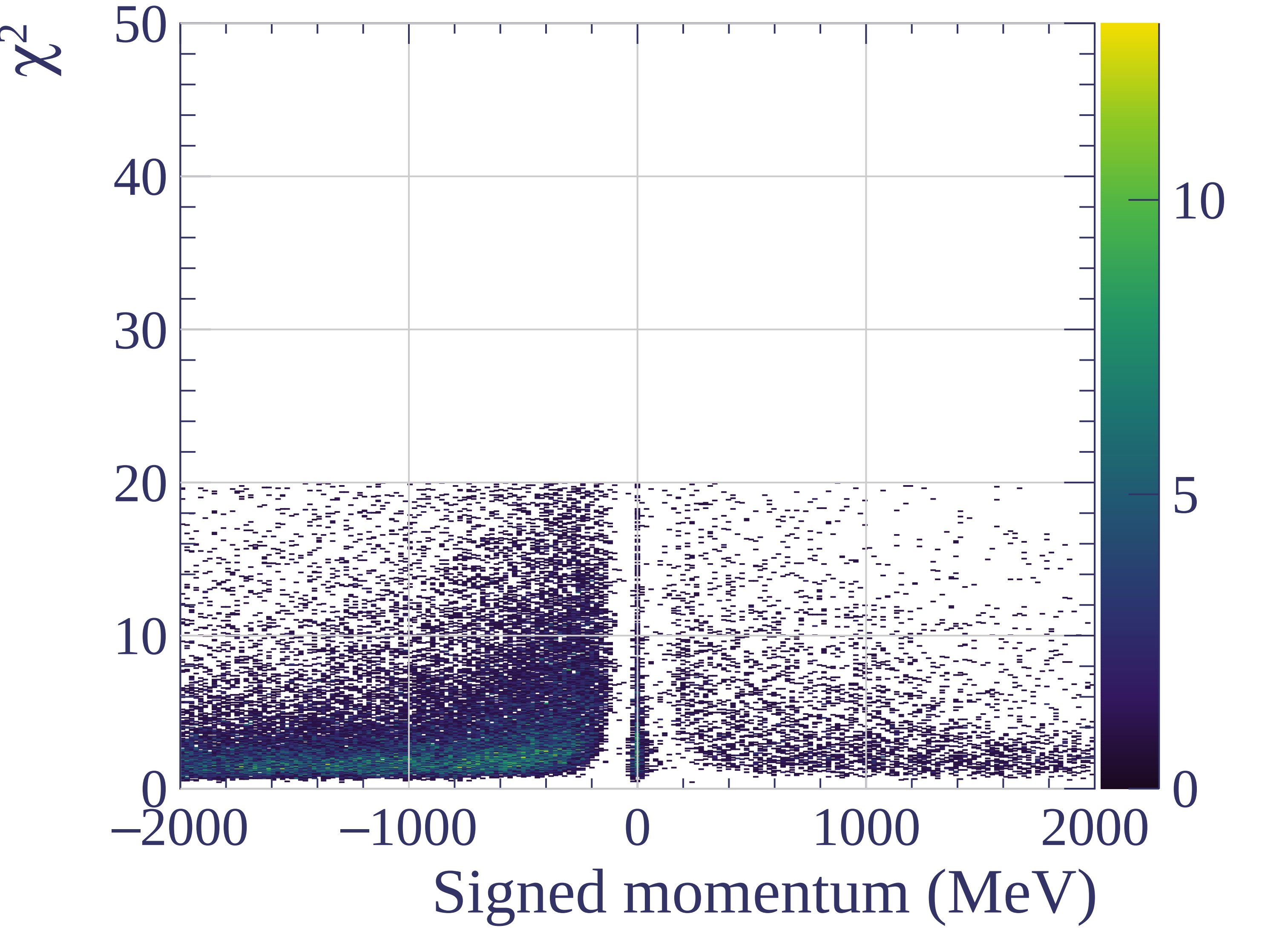
WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
−2000
−1000
0
1000
2000
Momentum [MeV/c]



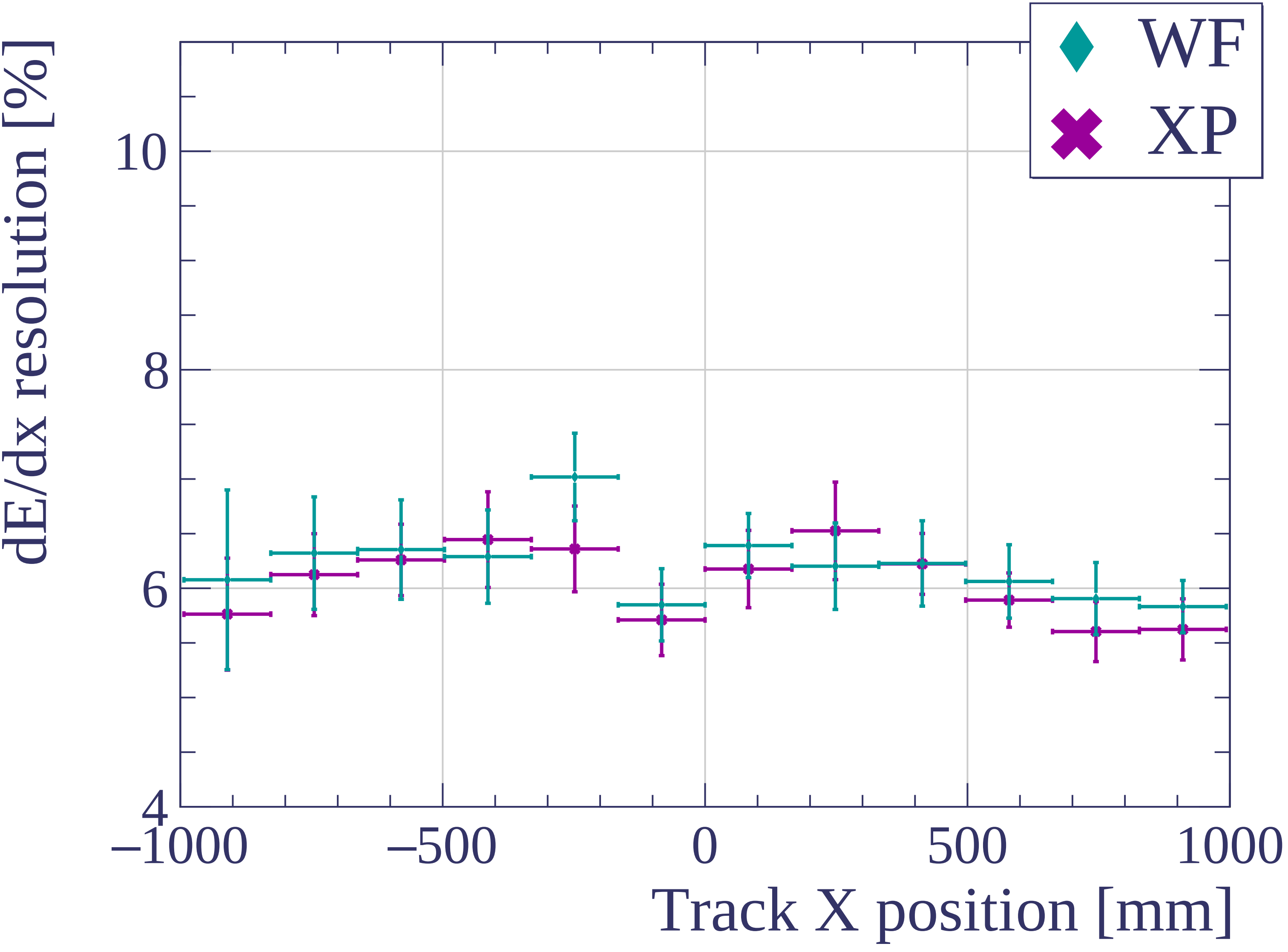
dE/dx with WF [ADC counts/cm]
Signed momentum (MeV)
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
10
8
6
4
2
0


dE/dx with XP [ADC counts/cm]
Signed momentum (MeV)
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
10
5
0

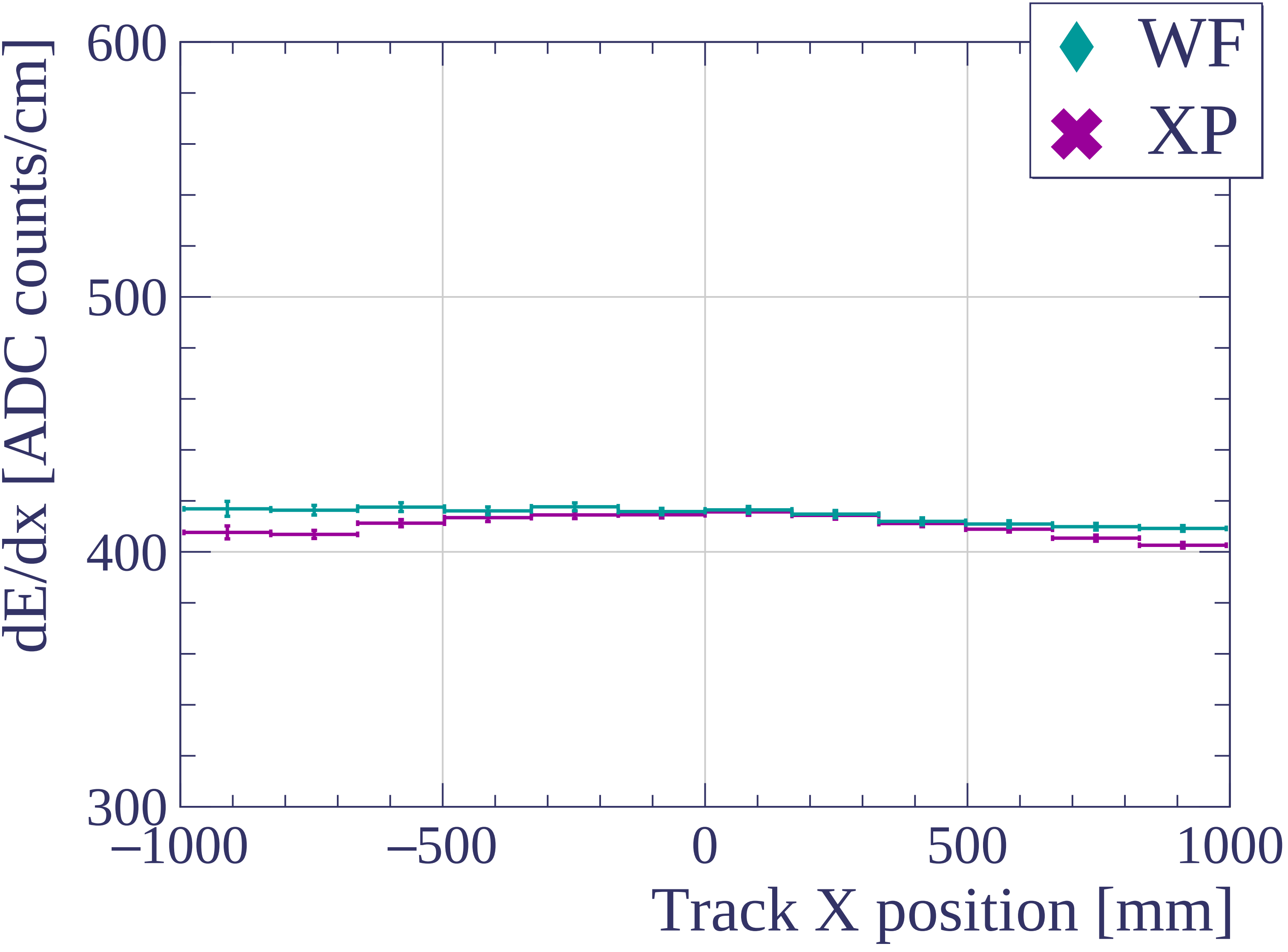

Count
200
100
0
–2000
–1000
0
1000
2000
Signed momentum (MeV)
Both HATs
tHAT
bHAT



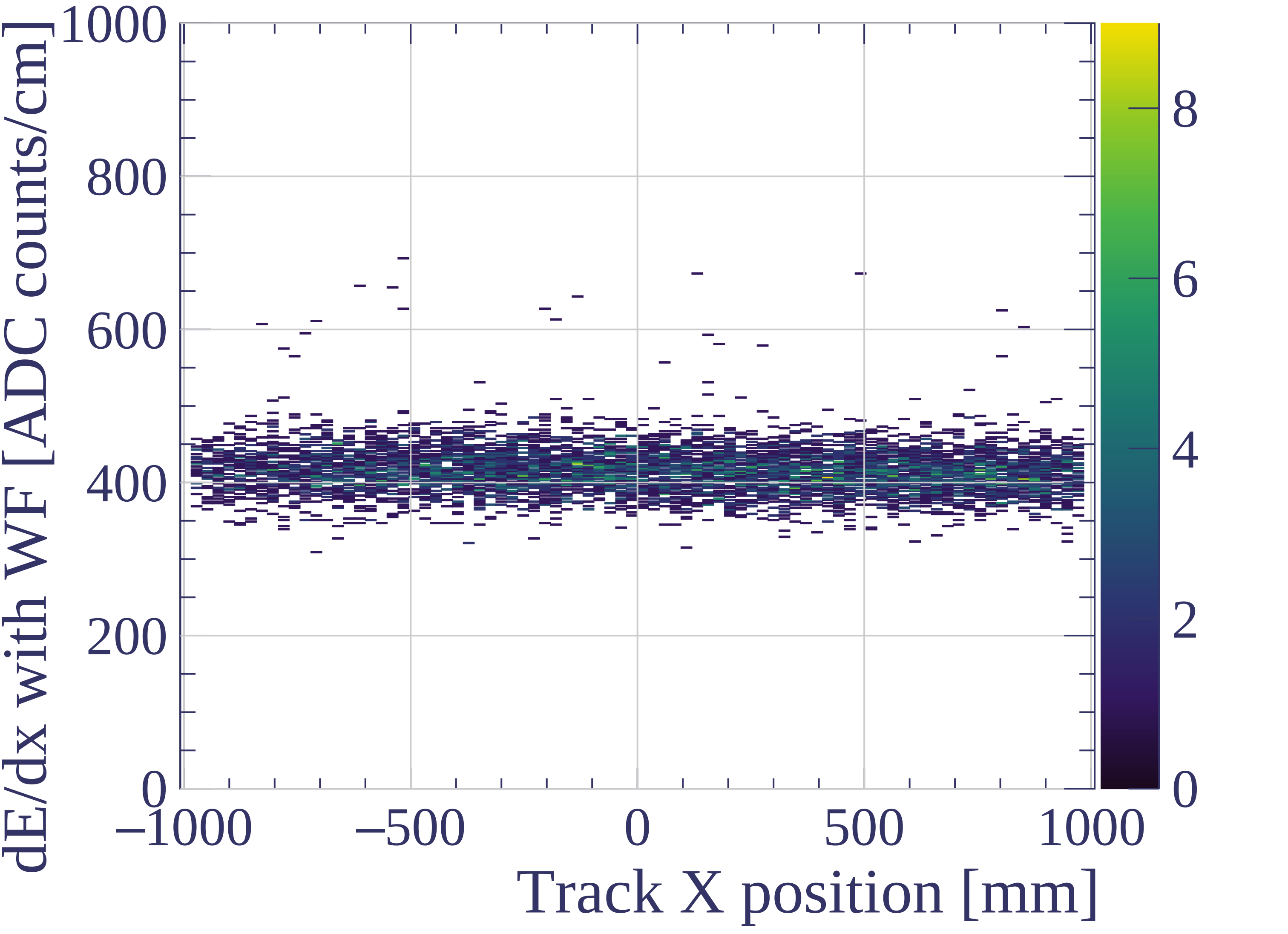
χ^2
50
40
30
20
10
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
10
5
0



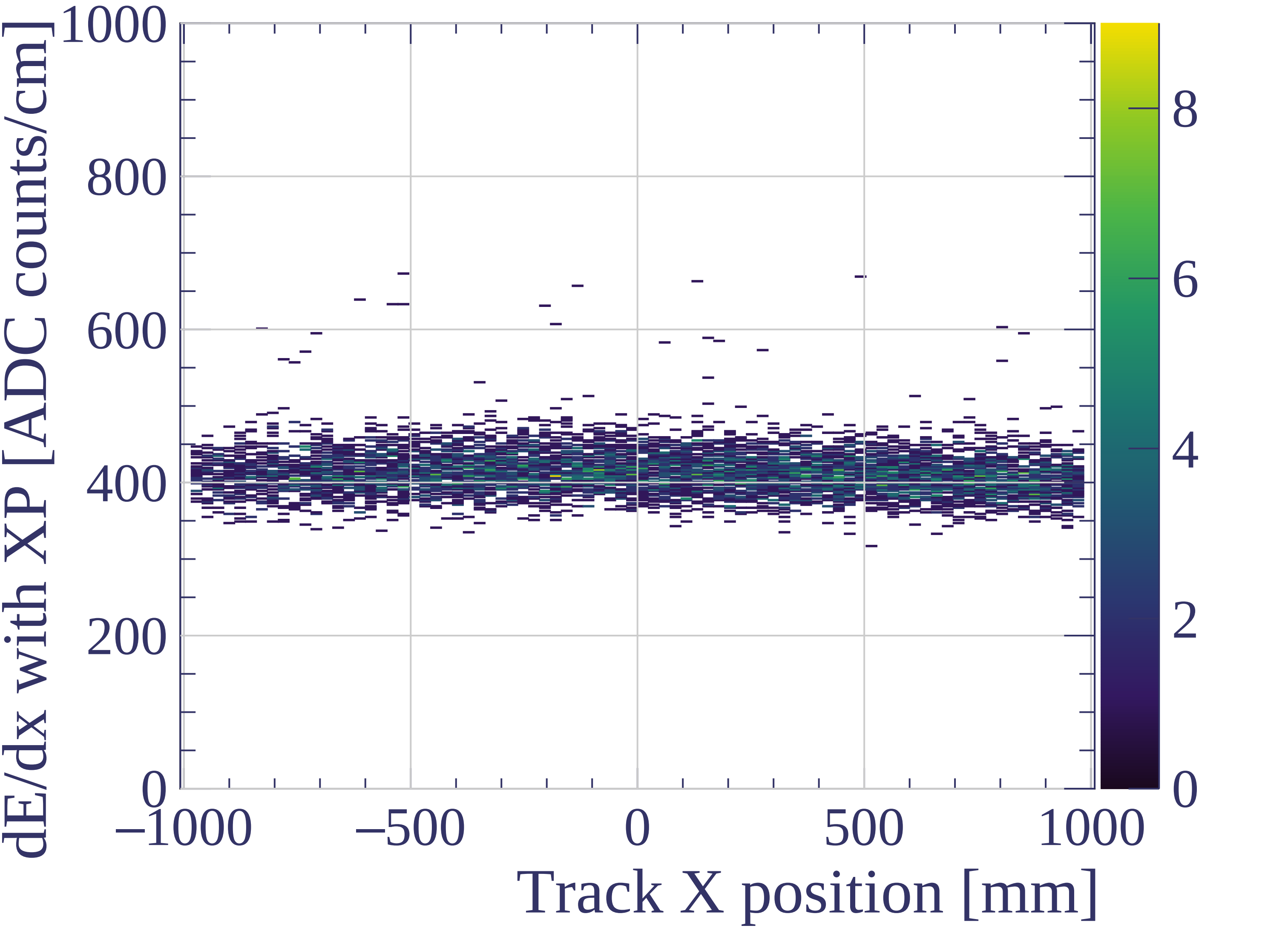
WF
XP
dE/dx resolution [%]
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]

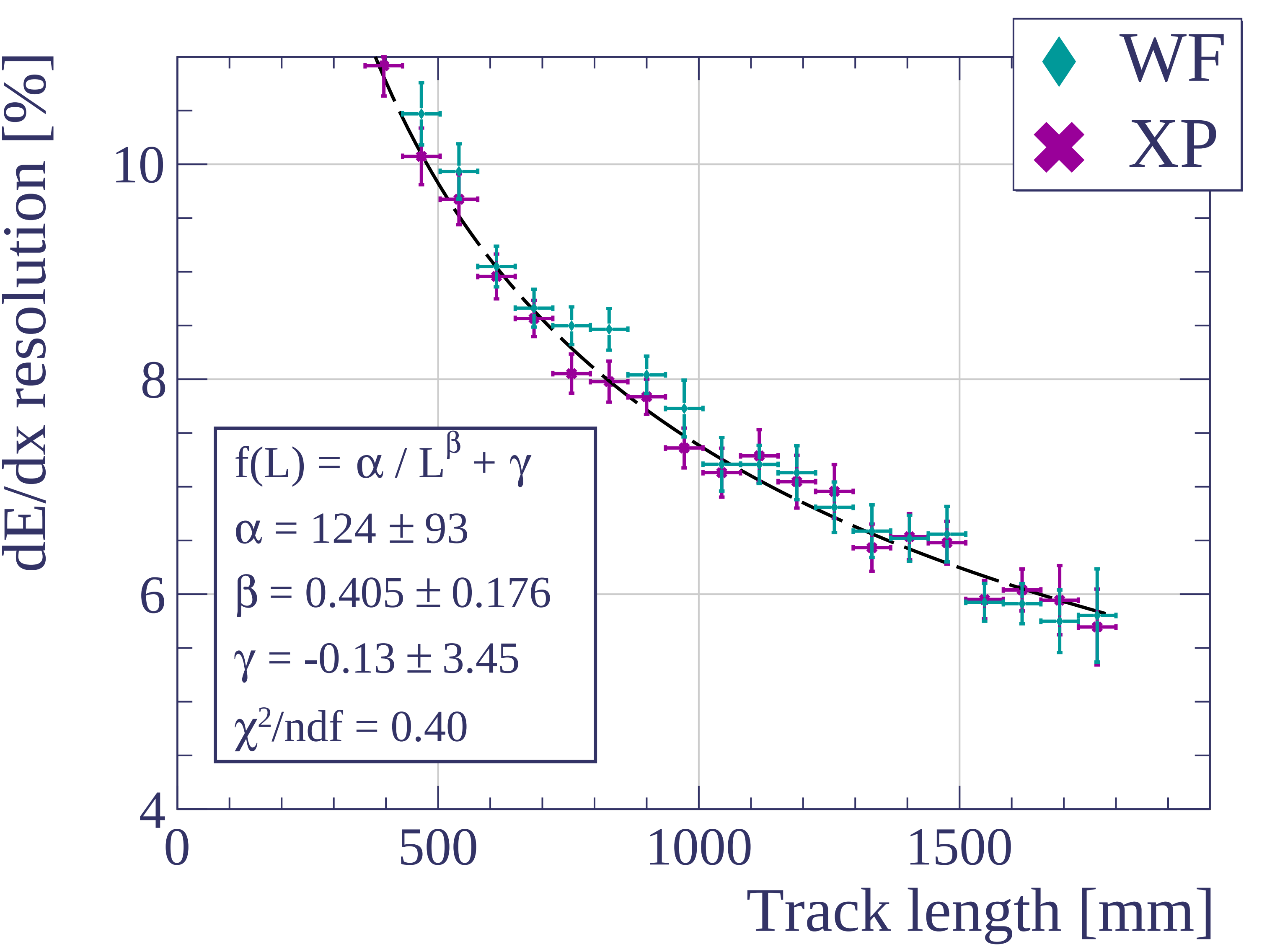


WF
XP
dE/dx [ADC counts/cm]
600
500
400
300
−1000
−500
0
500
1000
Track X position [mm]

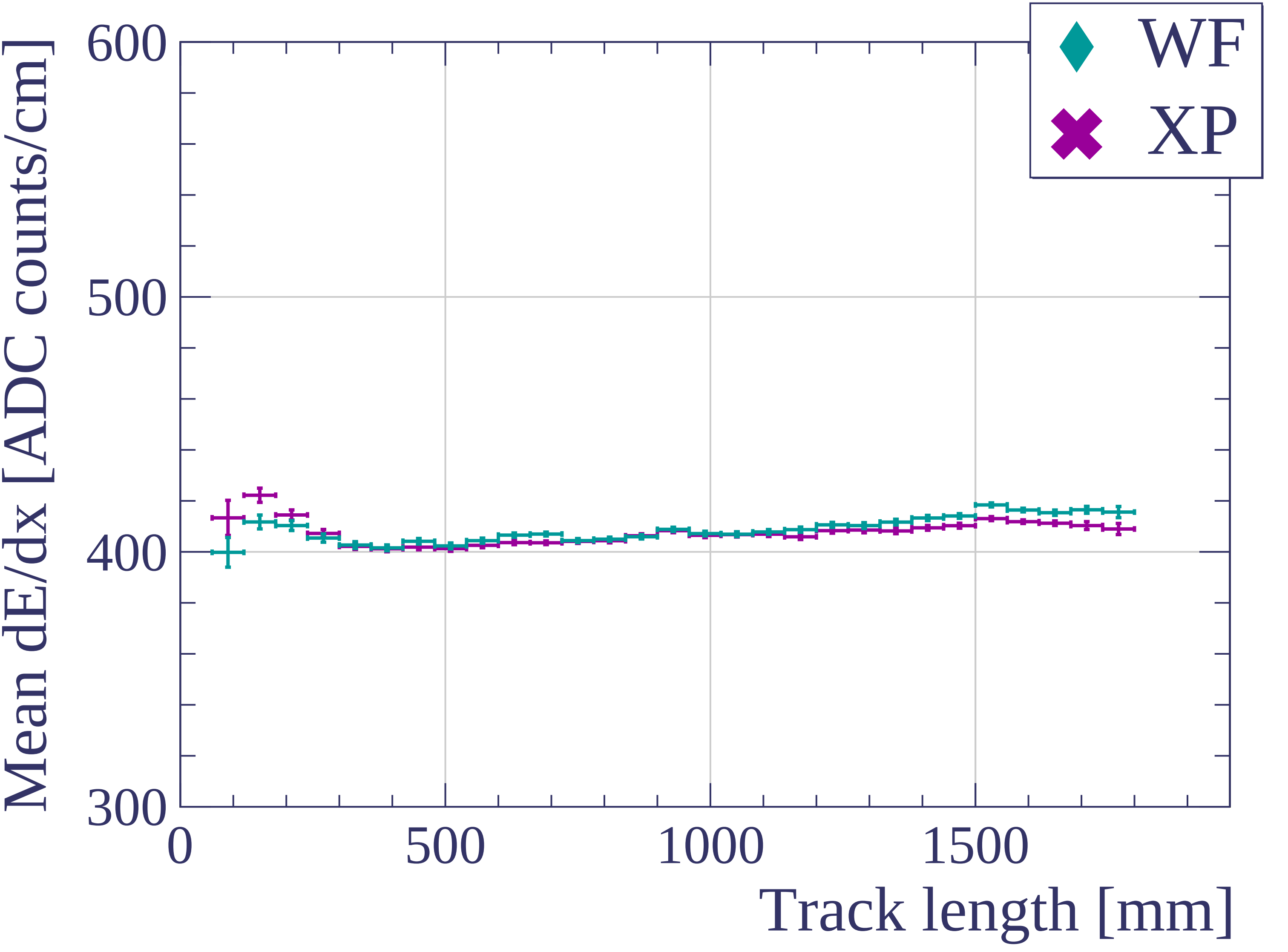


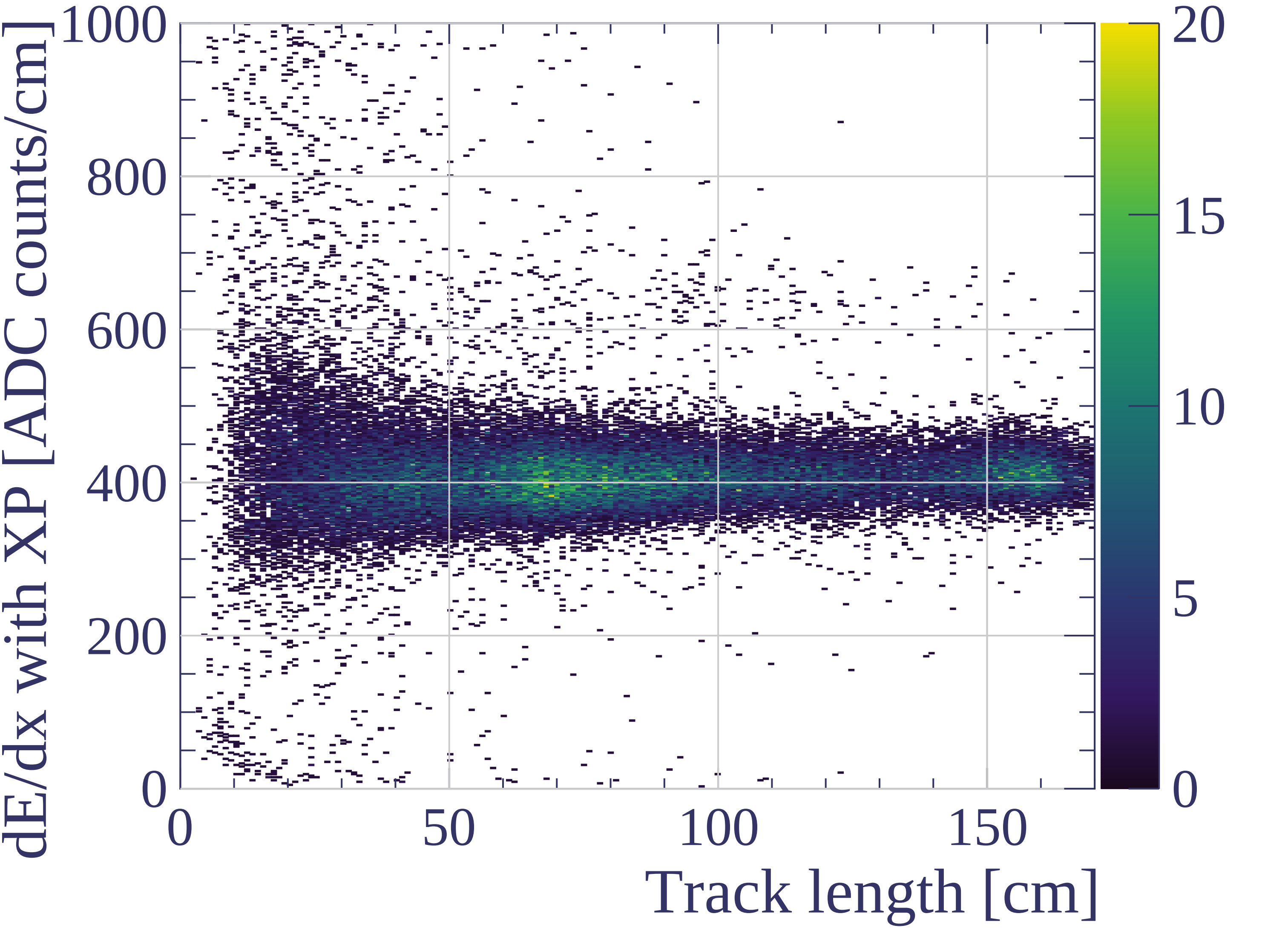
dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
8
6
4
2
0


dE/dx with XP [ADC counts/cm]
Track X position [mm]
0
200
400
600
800
1000
−1000
−500
0
500
1000
0
2
4
6
8

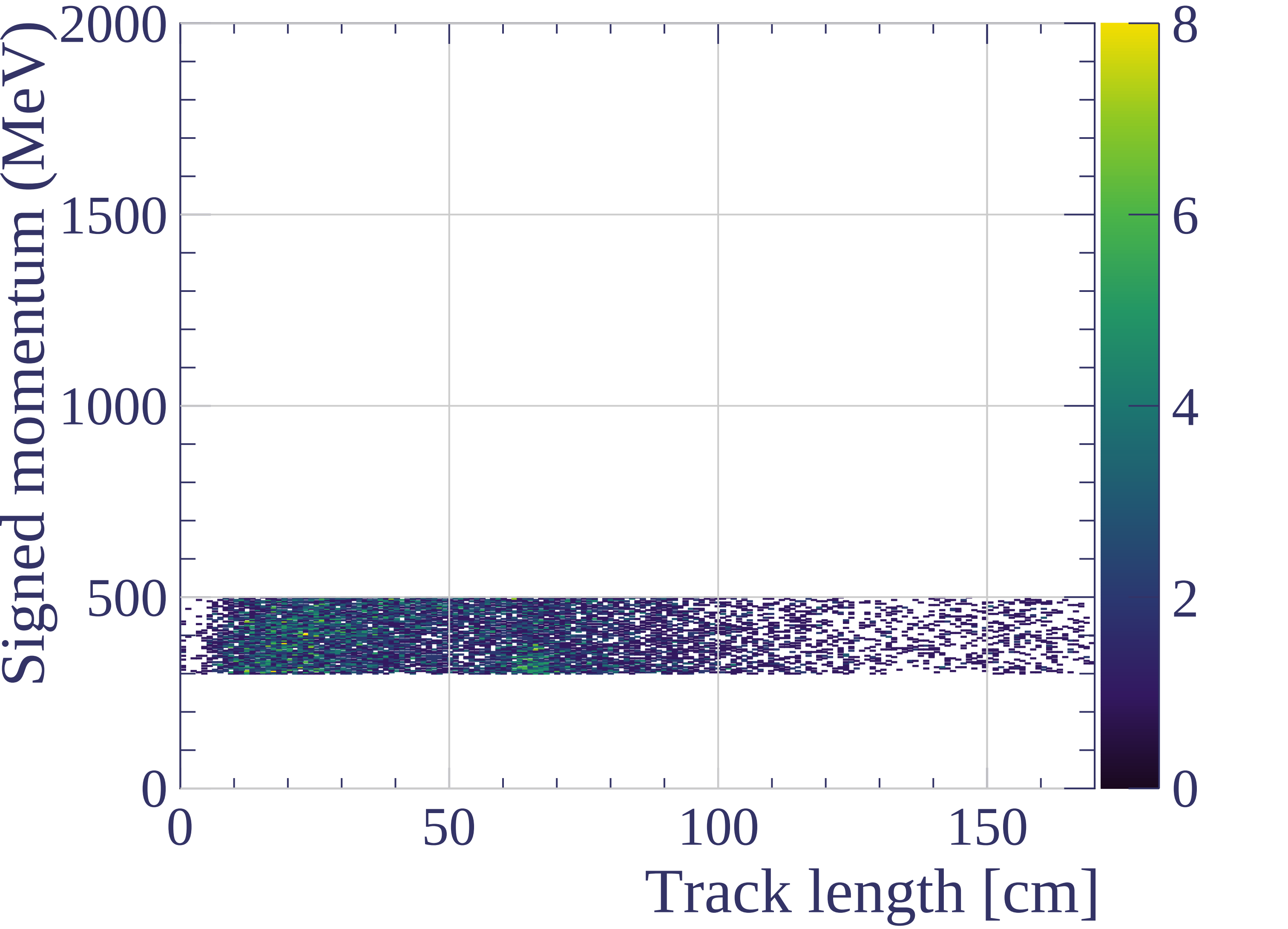


WF
XP
f(L) = α / L^β + γ
α = 124 ± 93
β = 0.405 ± 0.176
γ = -0.13 ± 3.45
χ²/ndf = 0.40
dE/dx resolution [%]
Track length [mm]
4
6
8
10
0
500
1000
1500

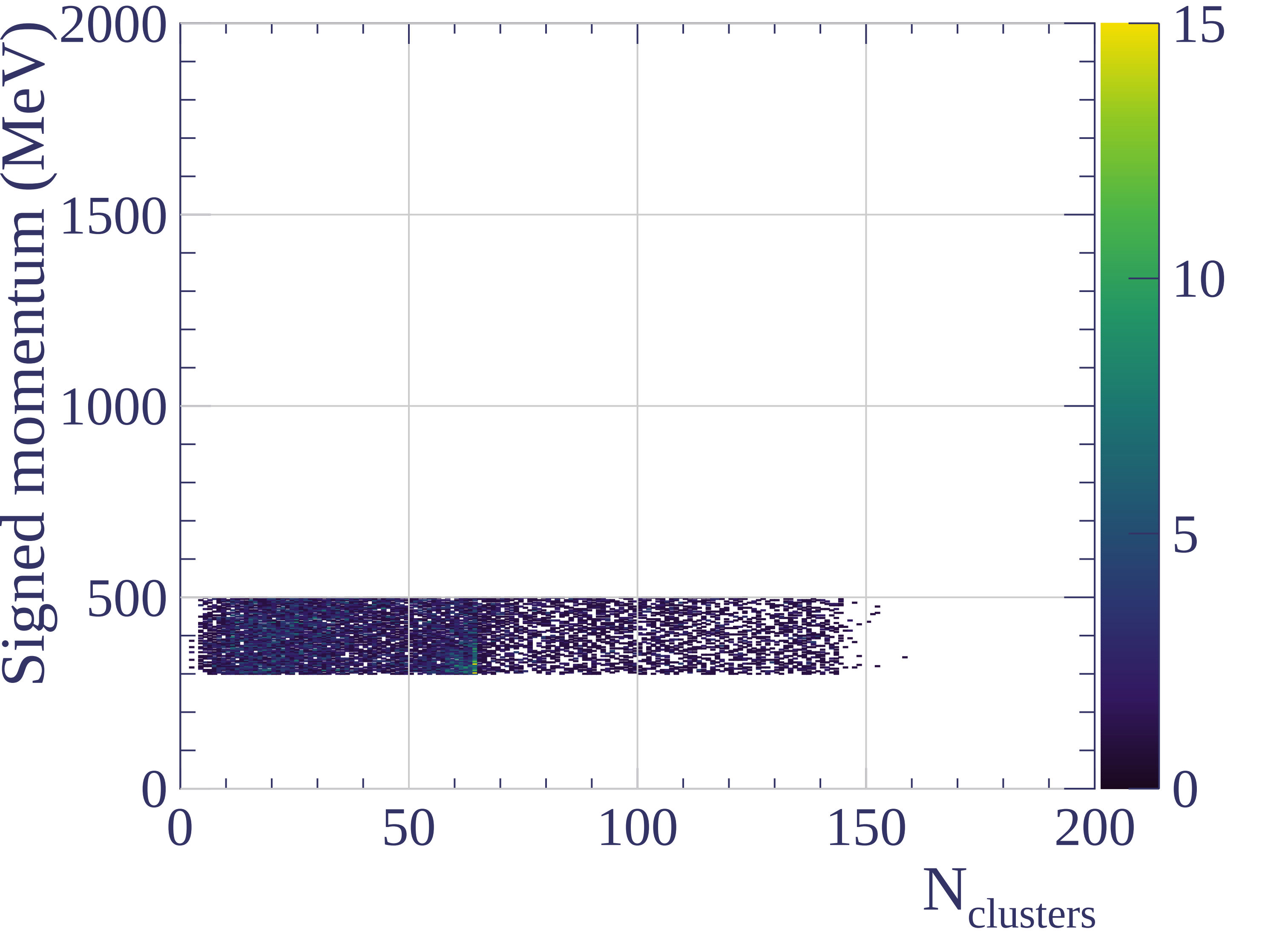

WF
XP
Mean dE/dx [ADC counts/cm]
Track length [mm]
600
500
400
300
0
500
1000
1500

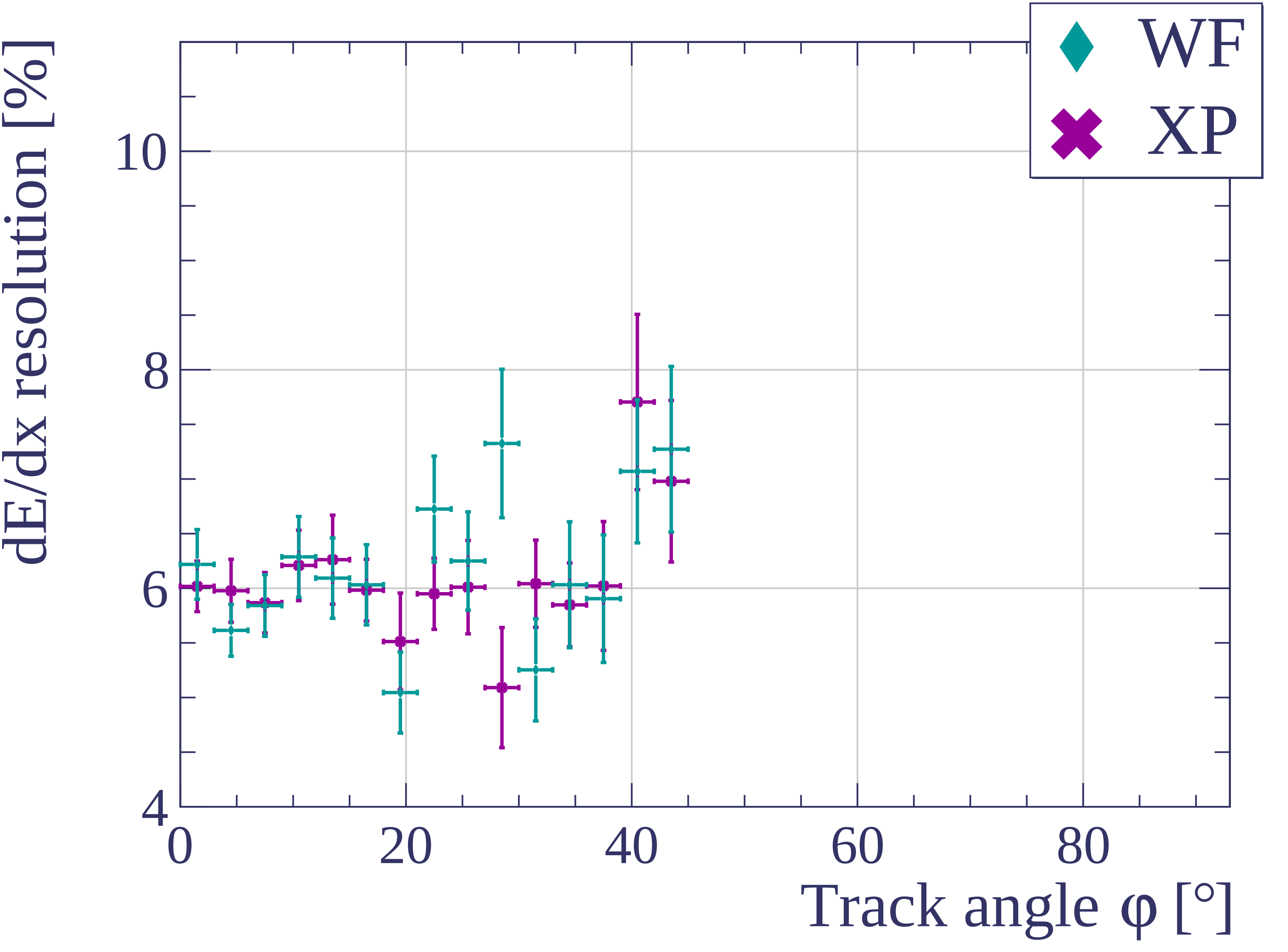


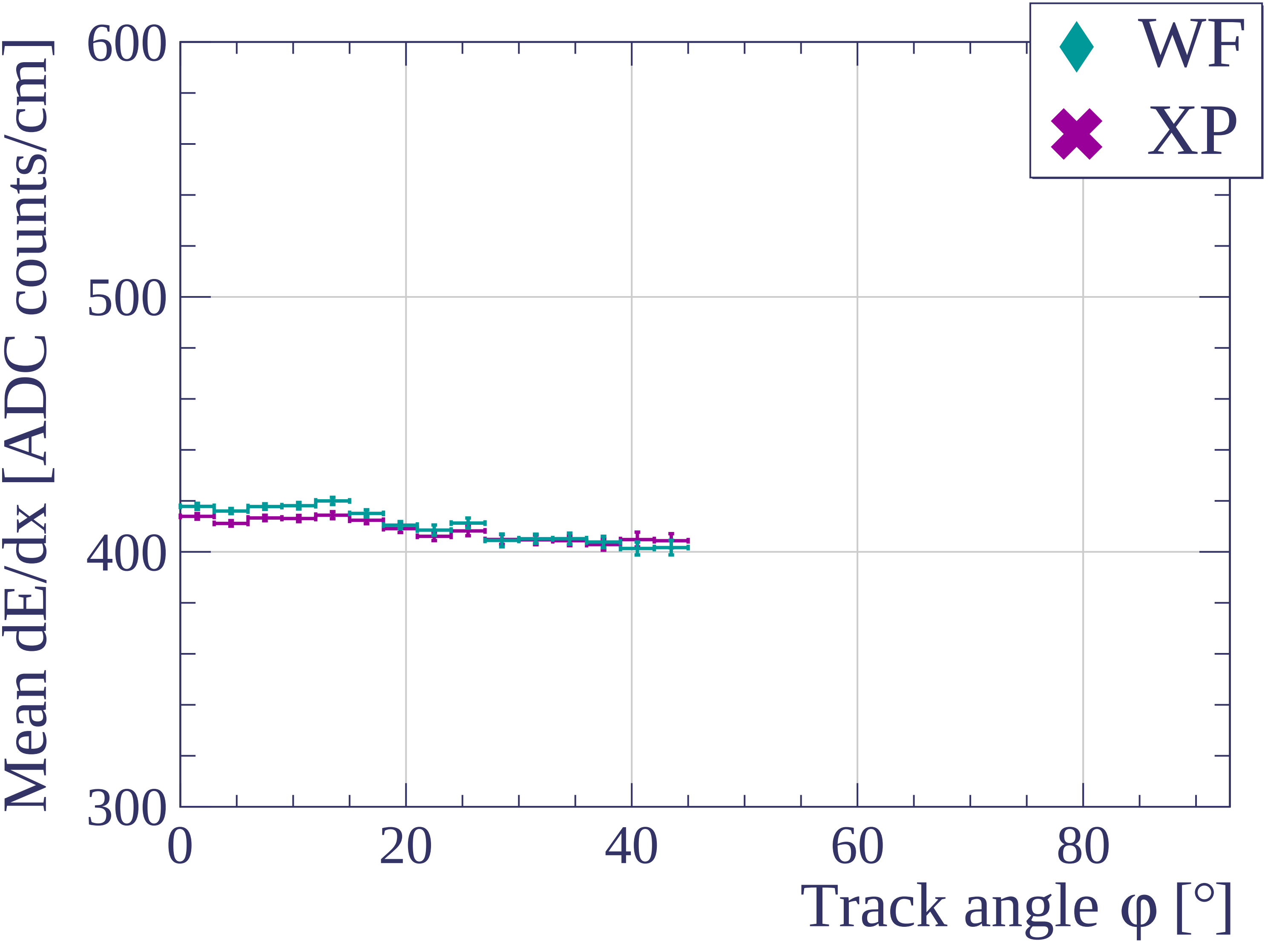
dE/dx with XP [ADC counts/cm]
Track length [cm]
0
200
400
600
800
1000
0
50
100
150
0
5
10
15
20

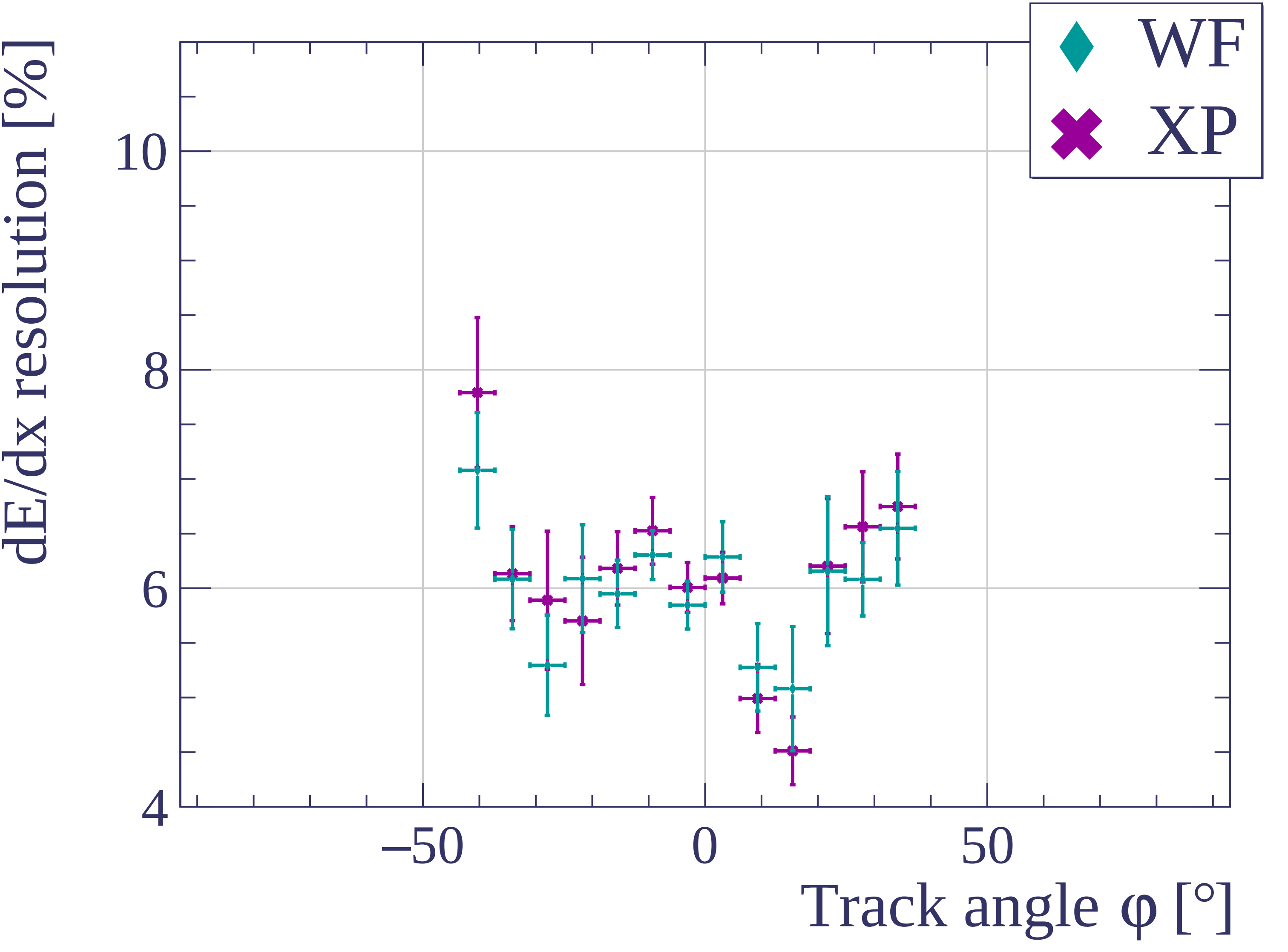


Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
Track length [cm]
8
6
4
2
0

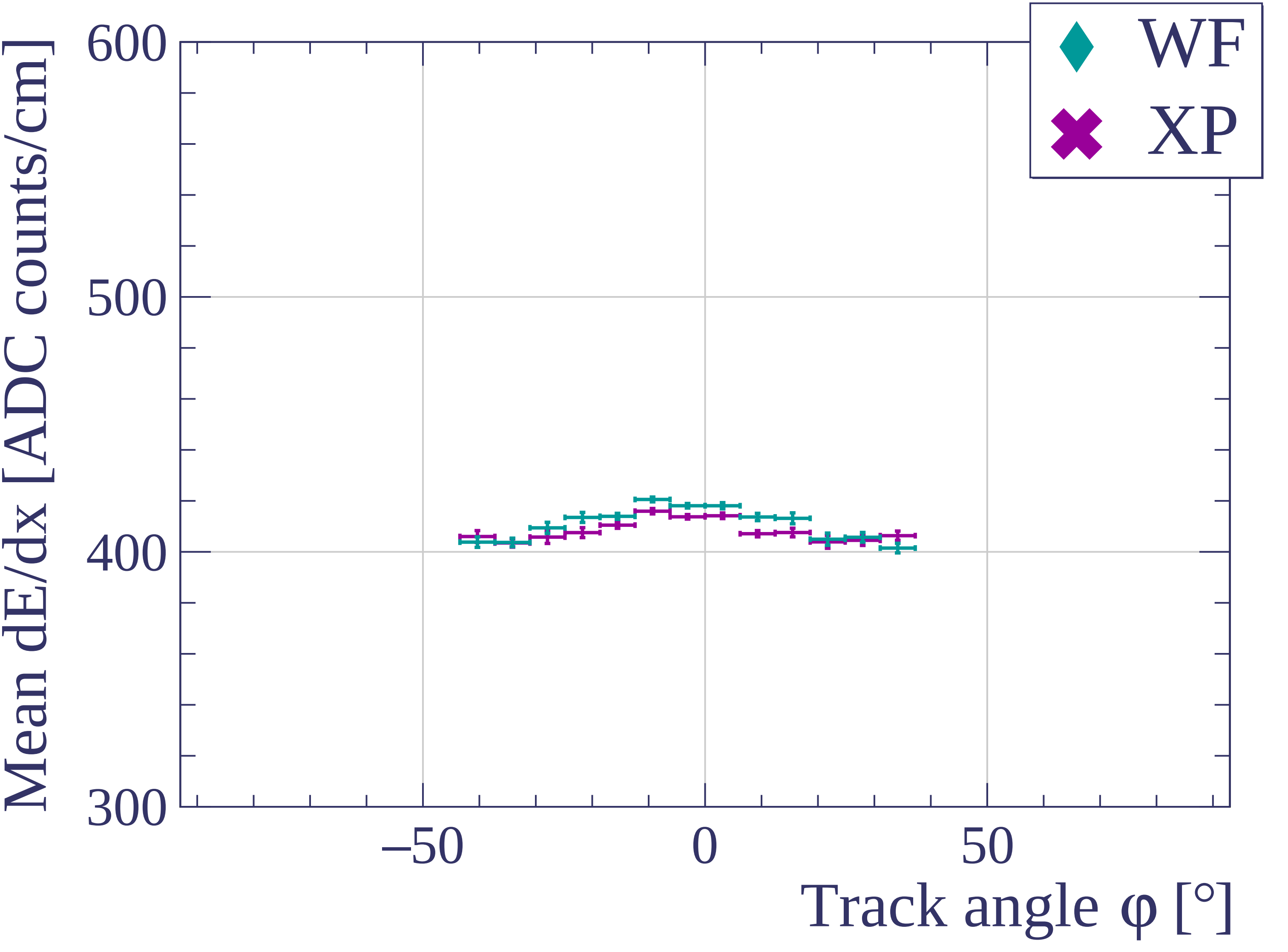

Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
200
N clusters
15
10
5
0


WF
XP
dE/dx resolution [%]
4
6
8
10
0
20
40
60
80
Track angle φ [°]

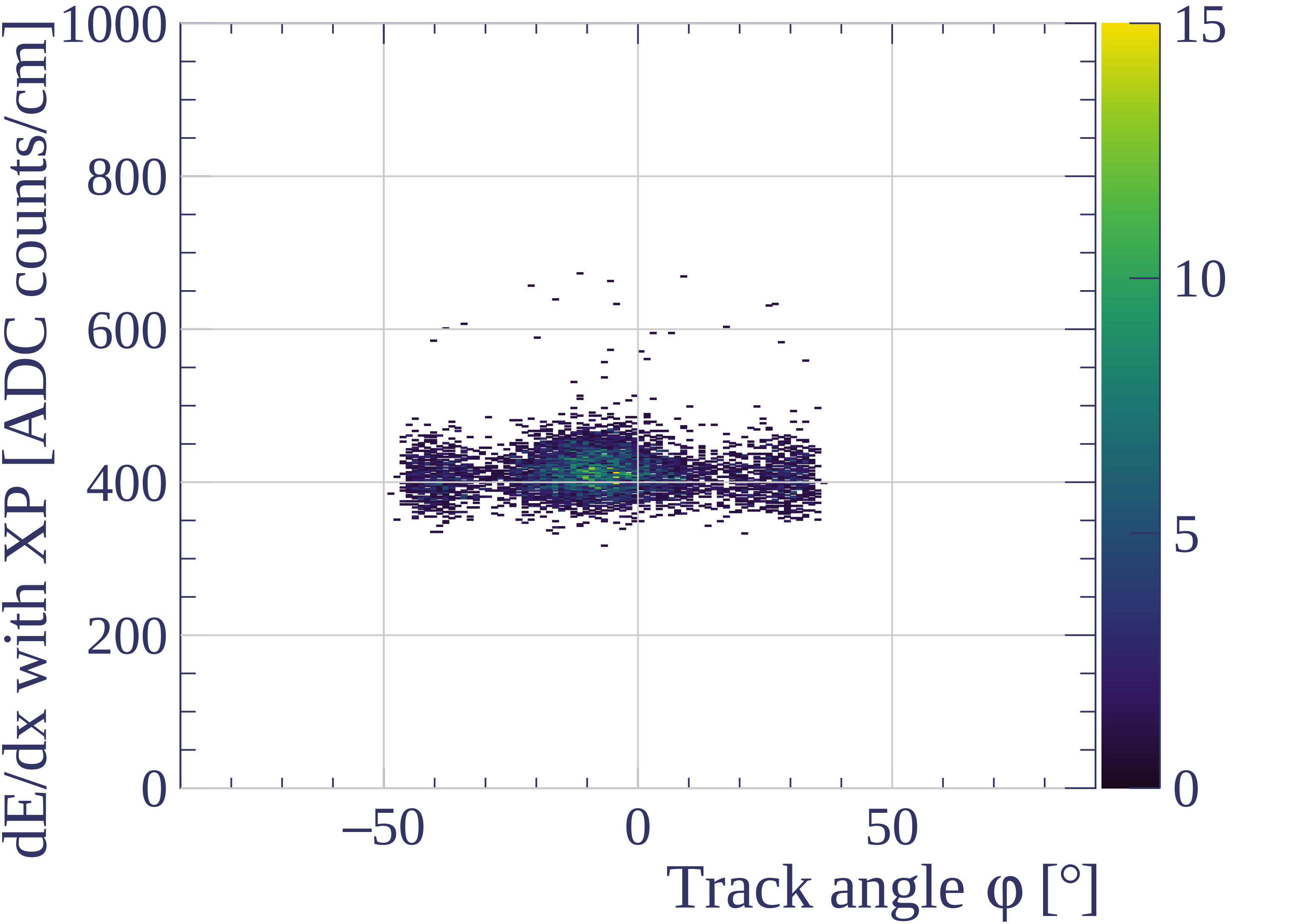

WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
20
40
60
80
Track angle φ [°]



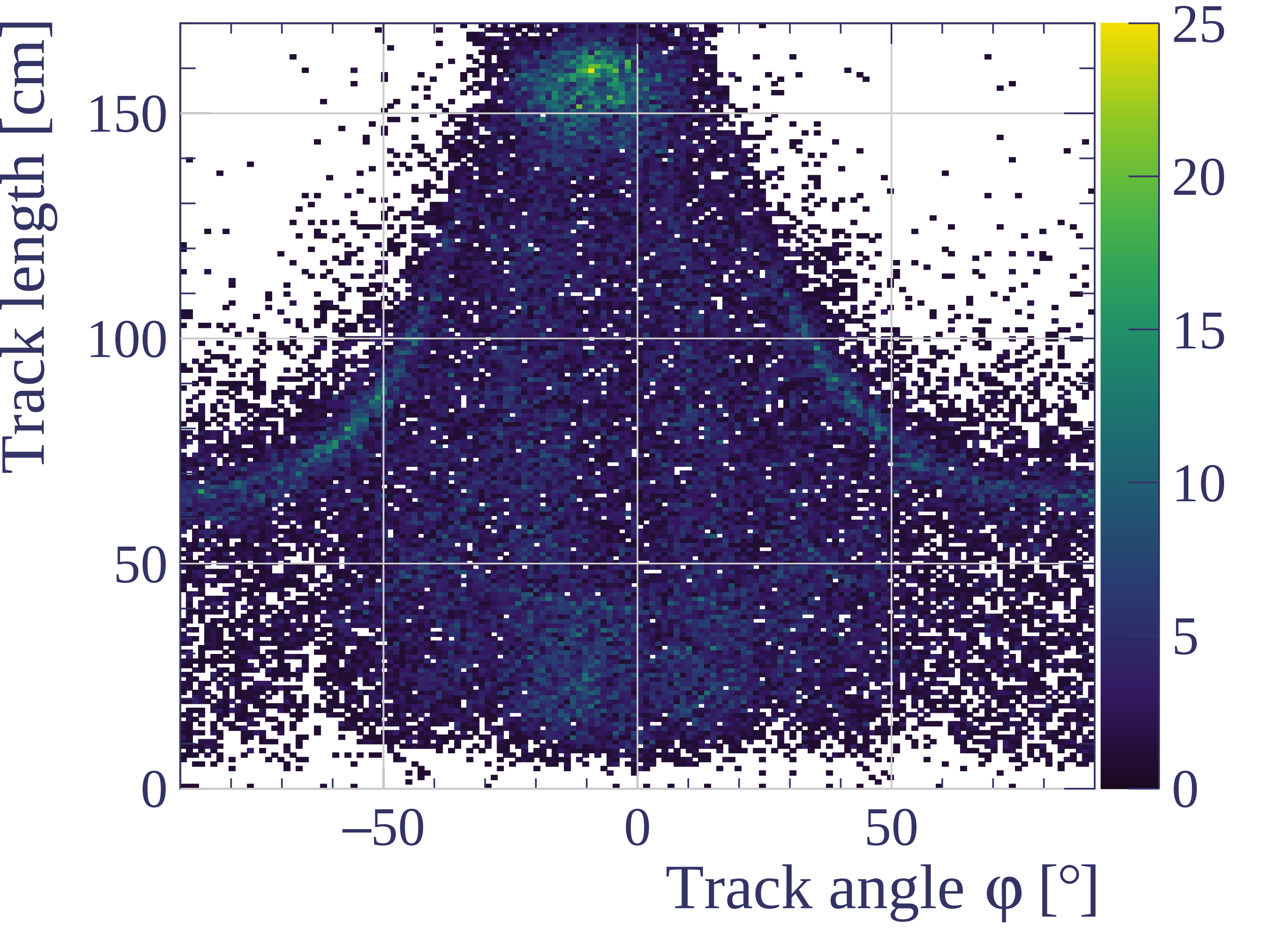
WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle φ [°]



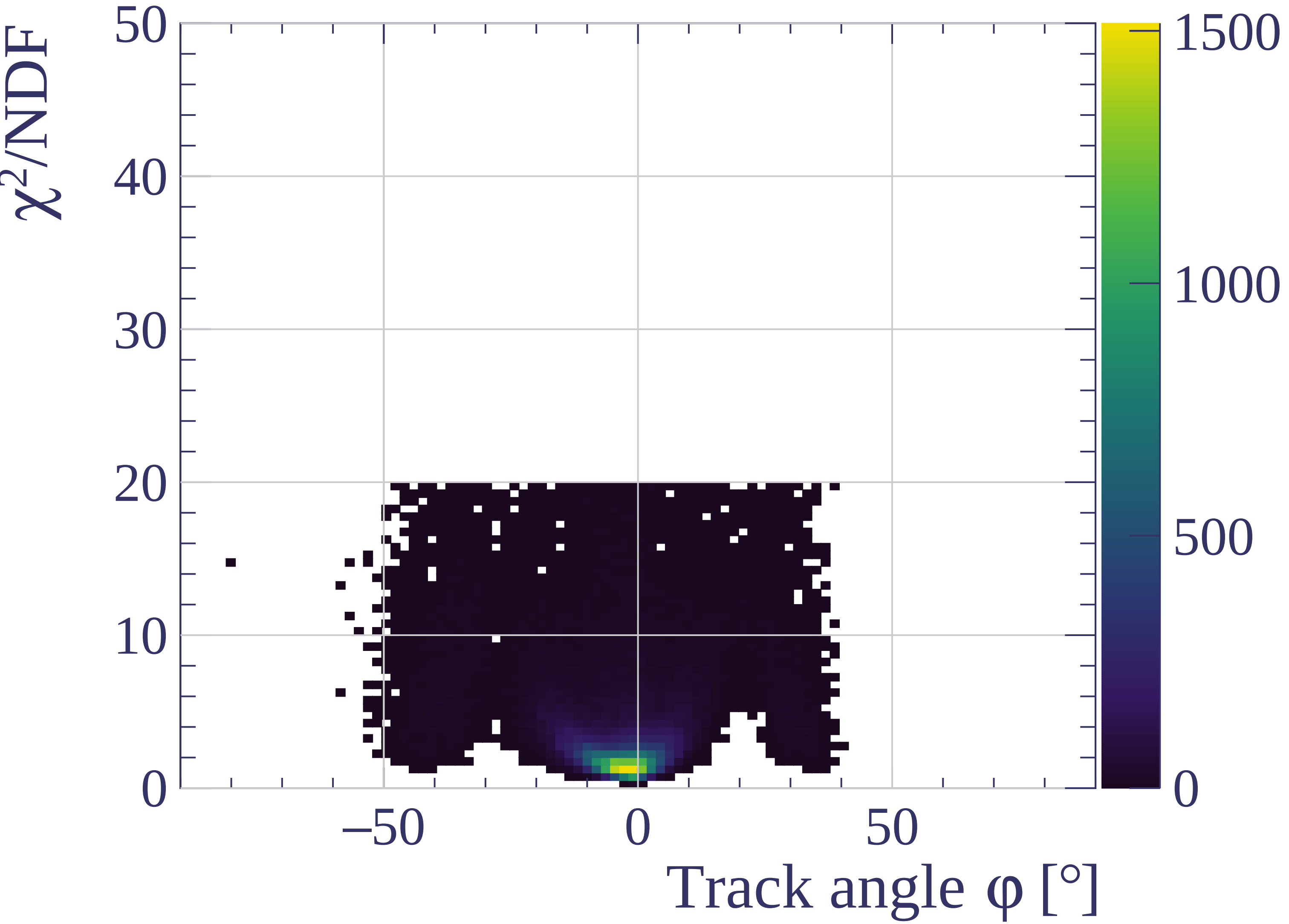
WF
XP
600
500
400
300
−50
0
50
Mean dE/dx [ADC counts/cm]
Track angle φ [°]



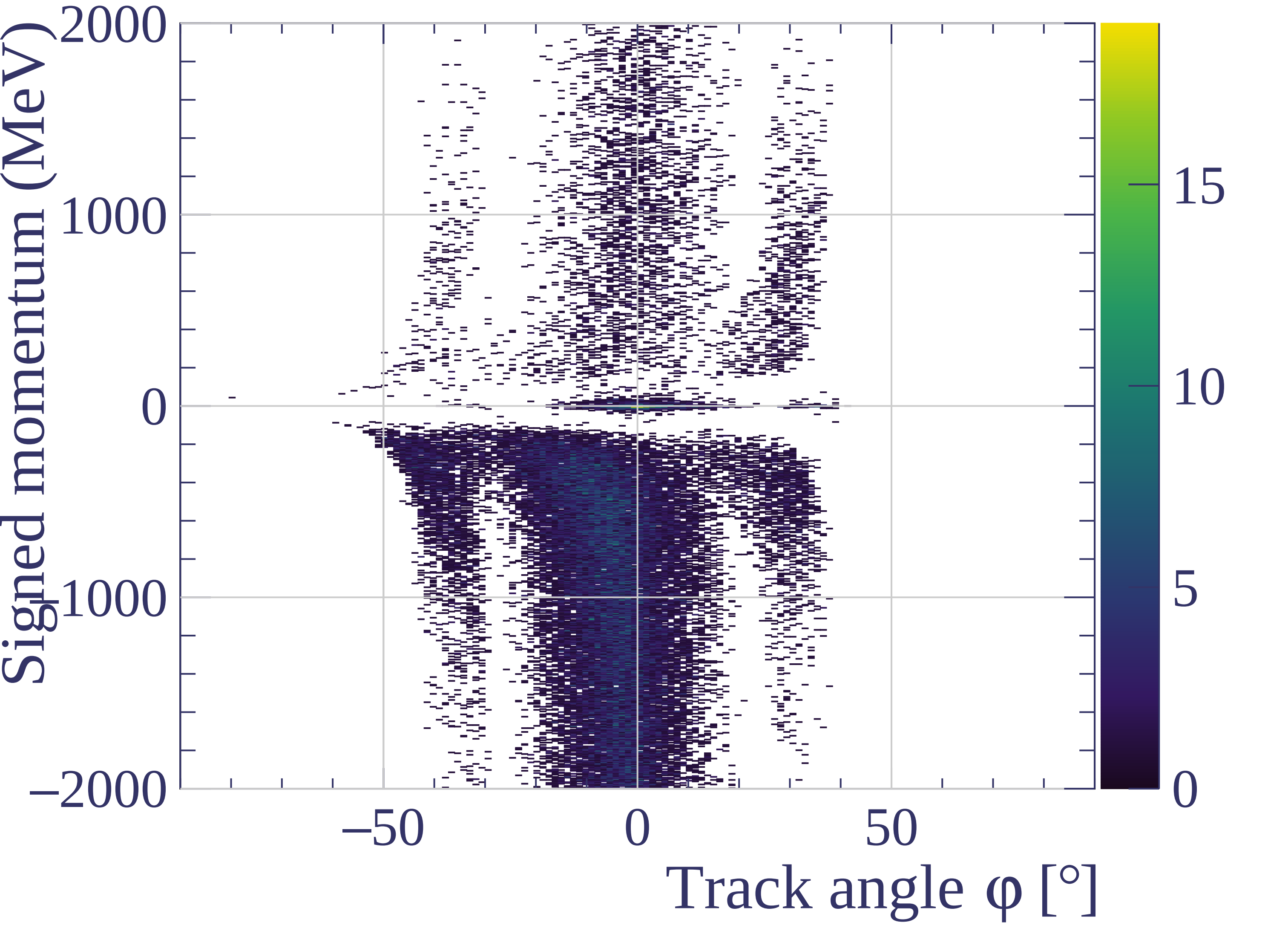
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle φ [°]
15
10
5
0



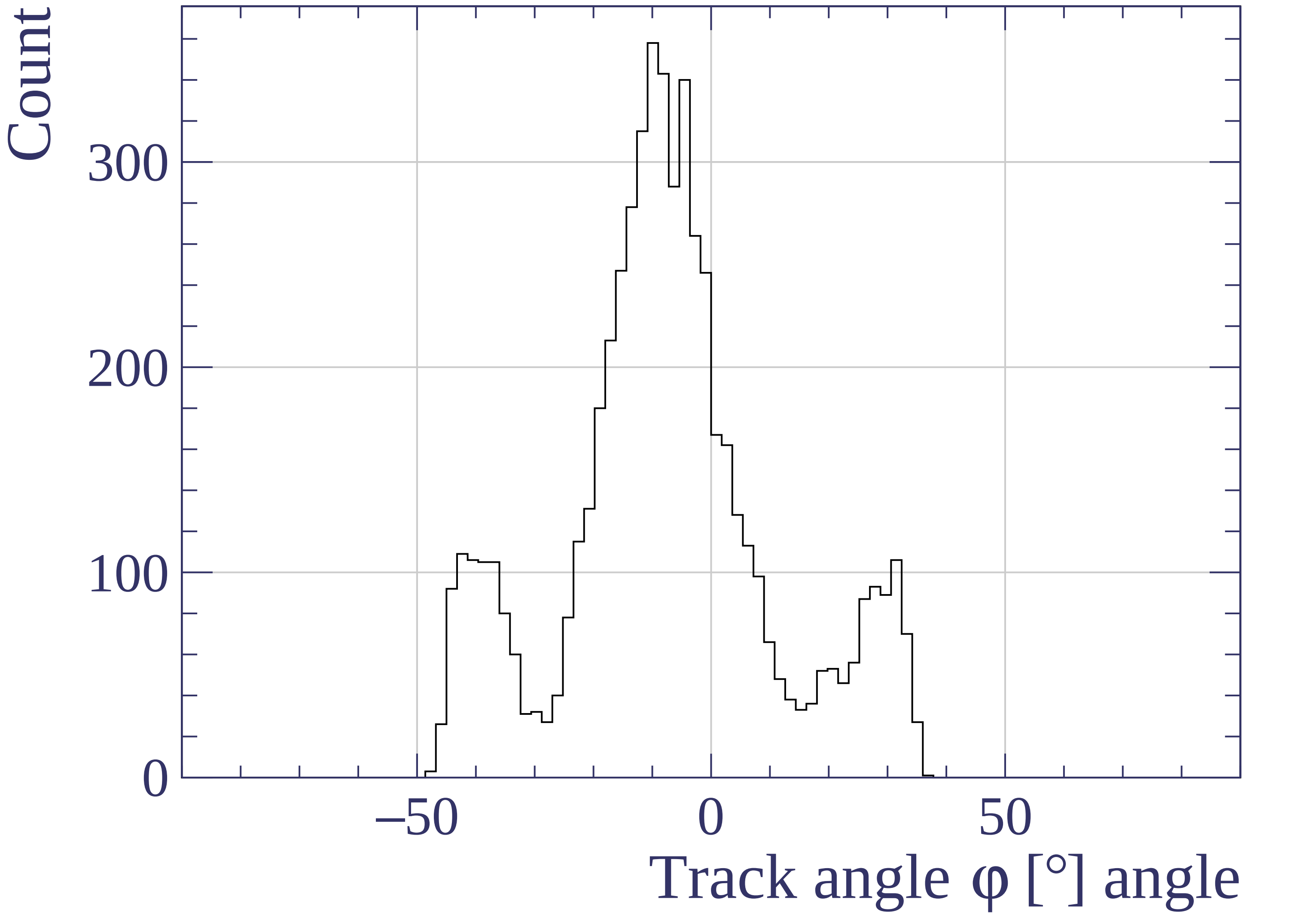
Track length [cm]
150
100
50
0
−50
0
50
Track angle φ [°]
25
20
15
10
5
0



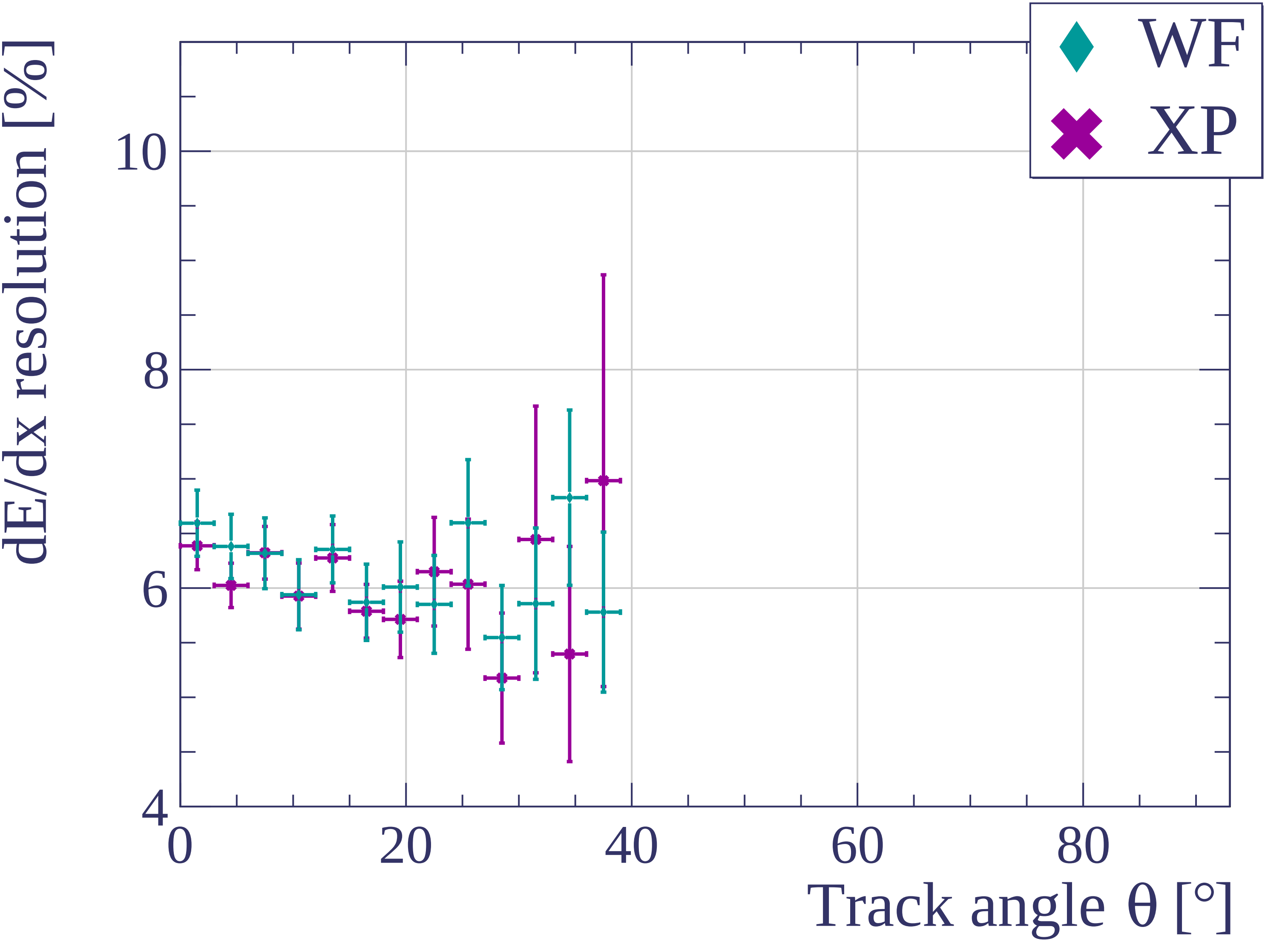
χ^2/NDF
50
40
30
20
10
0
−50
0
50
Track angle φ [°]
1500
1000
500
0



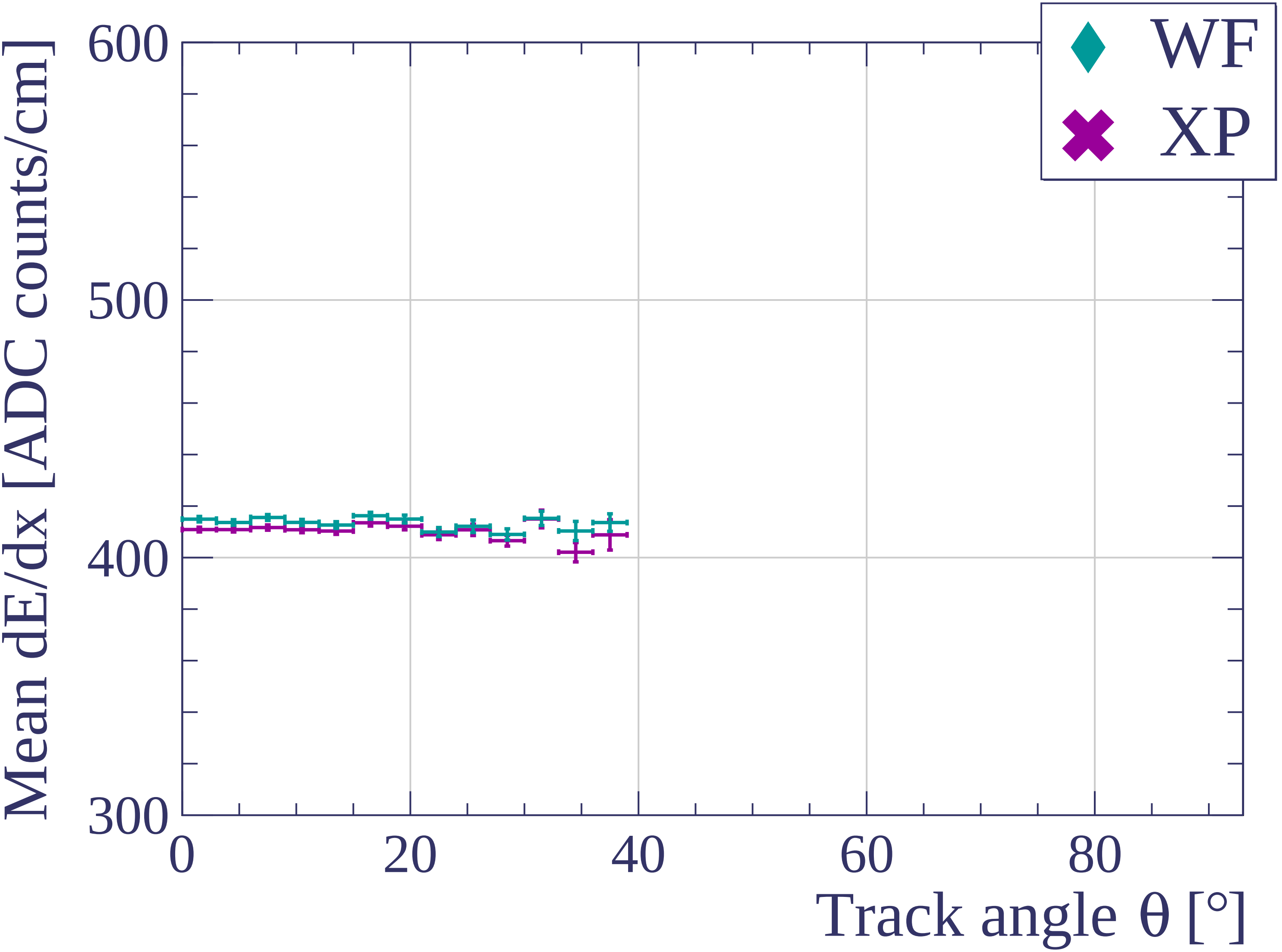
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle φ [°]
15
10
5
0



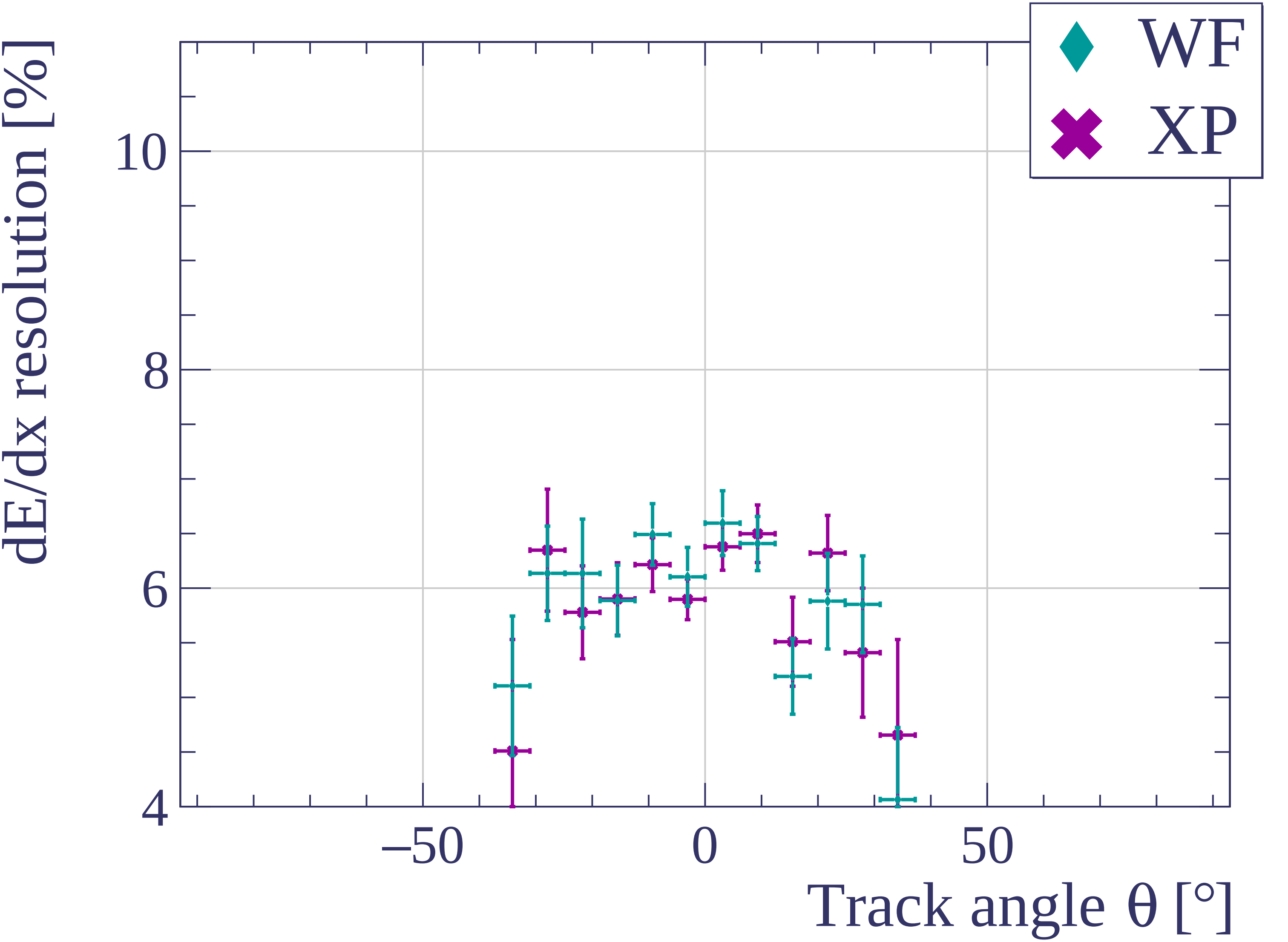
Count
300
200
100
0
−50
0
50
Track angle φ [°] angle



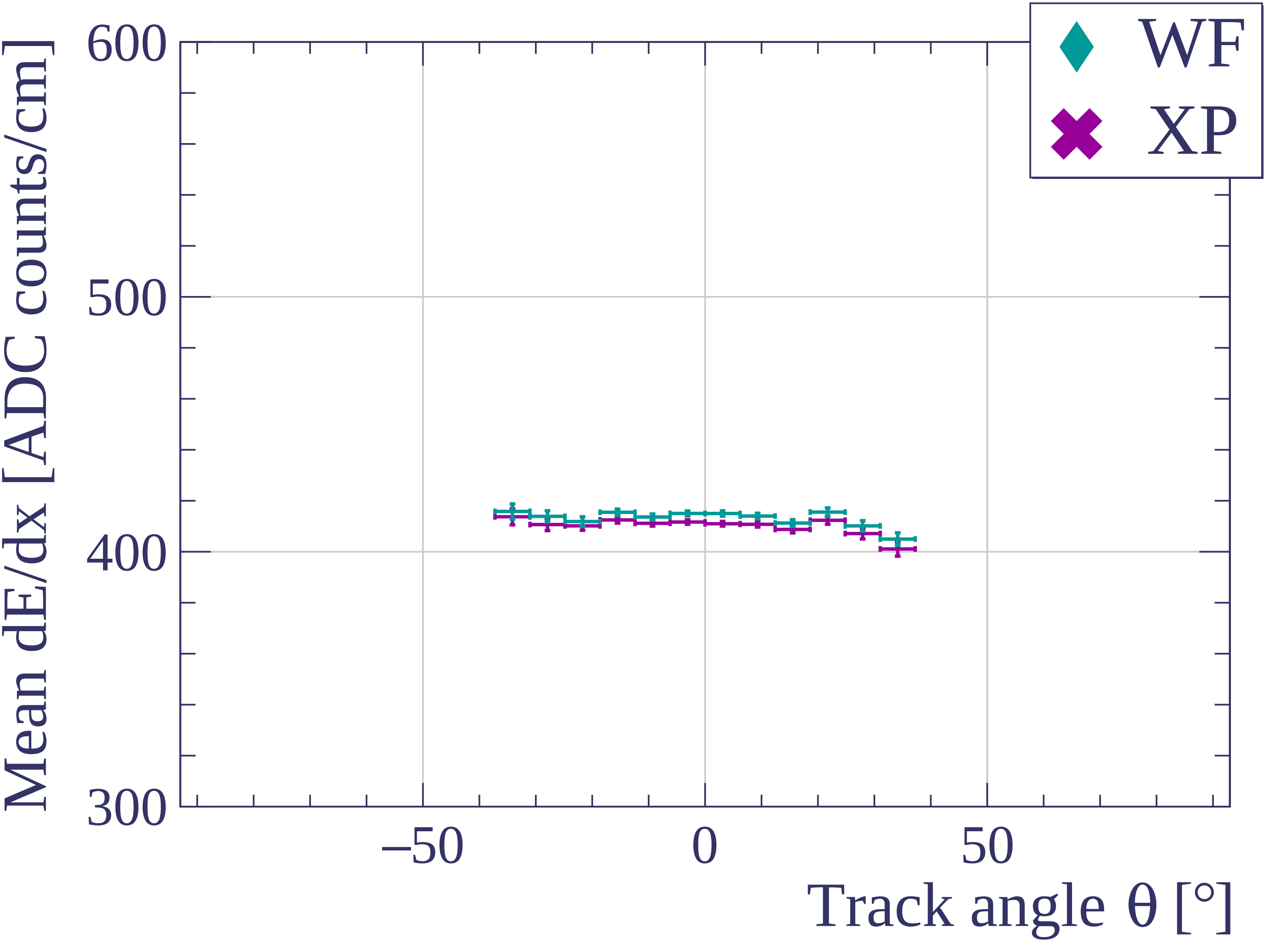
WF
XP
dE/dx resolution [%]
Track angle θ [°]
4
6
8
10
0
20
40
60
80



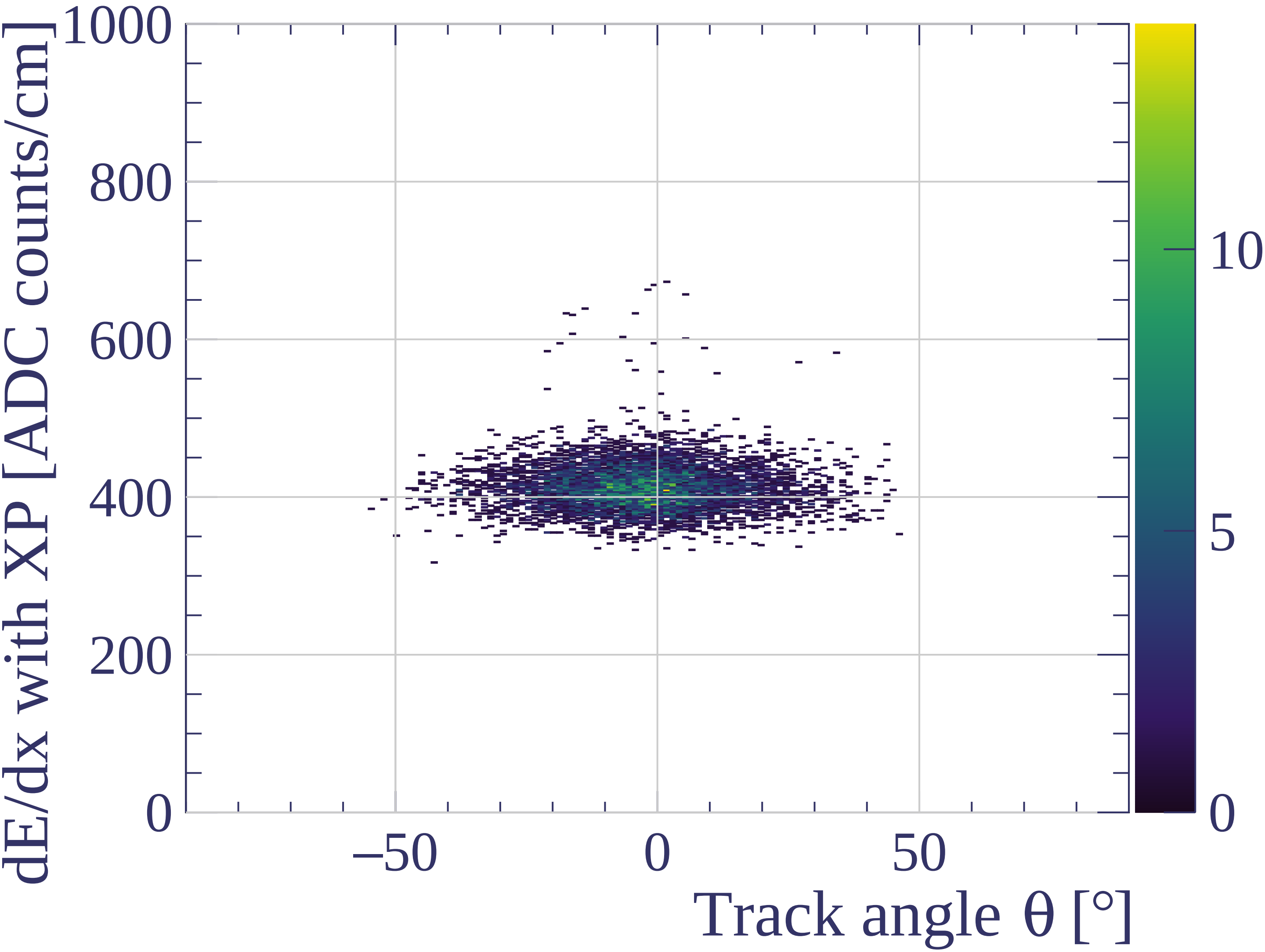
WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
20
40
60
80
Track angle θ [°]



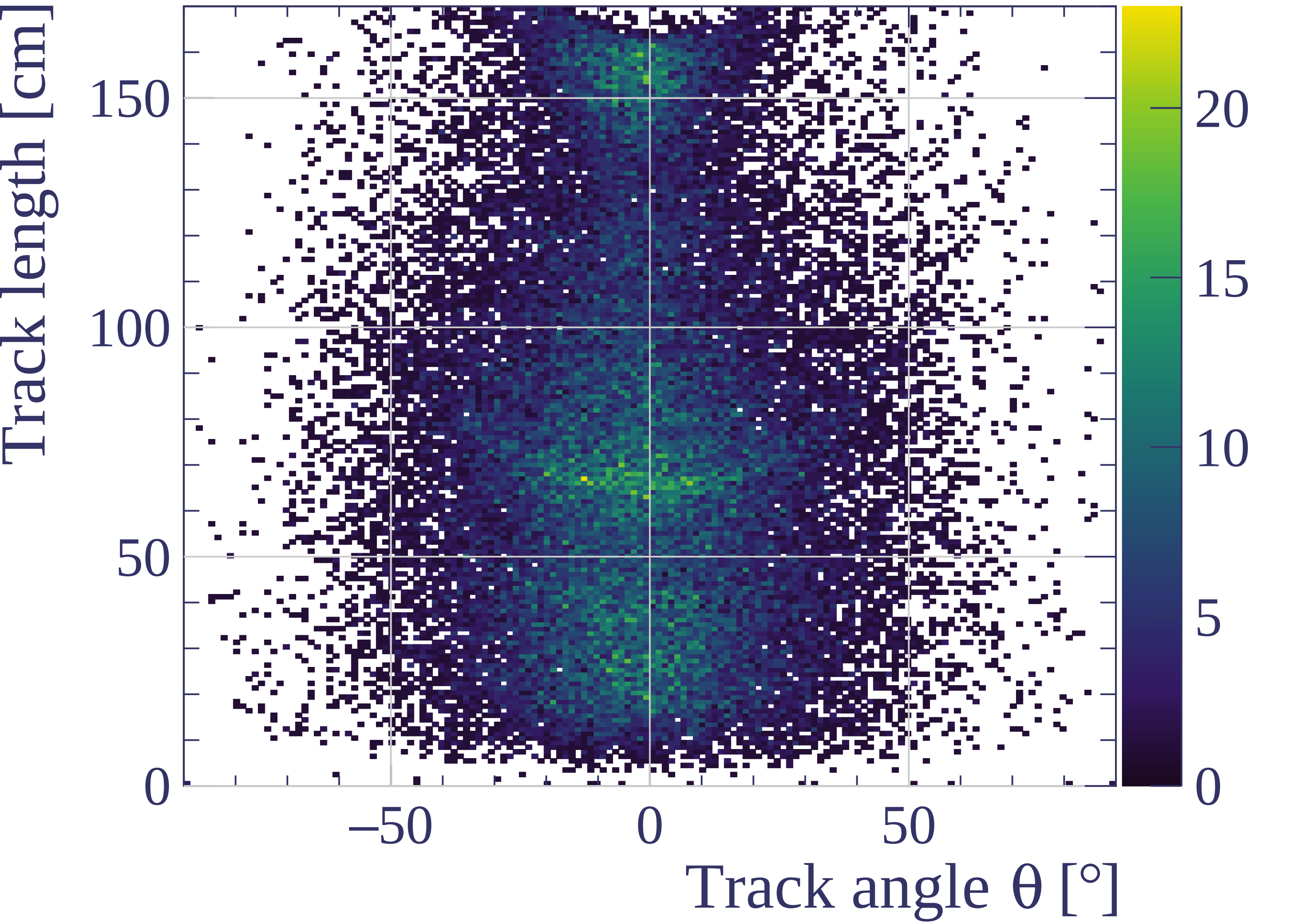
WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle θ [°]

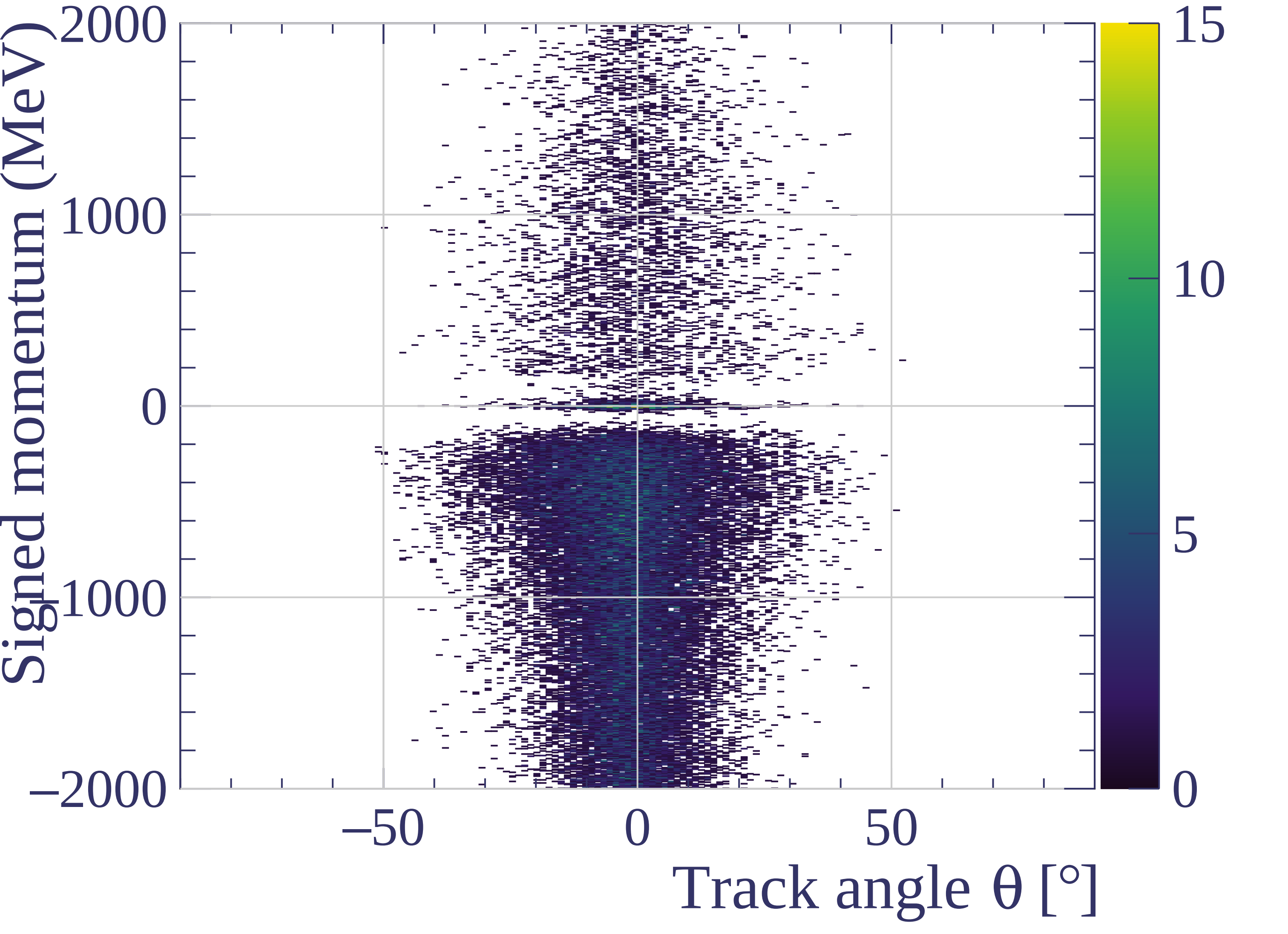


WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
−50
0
50
Track angle θ [°]

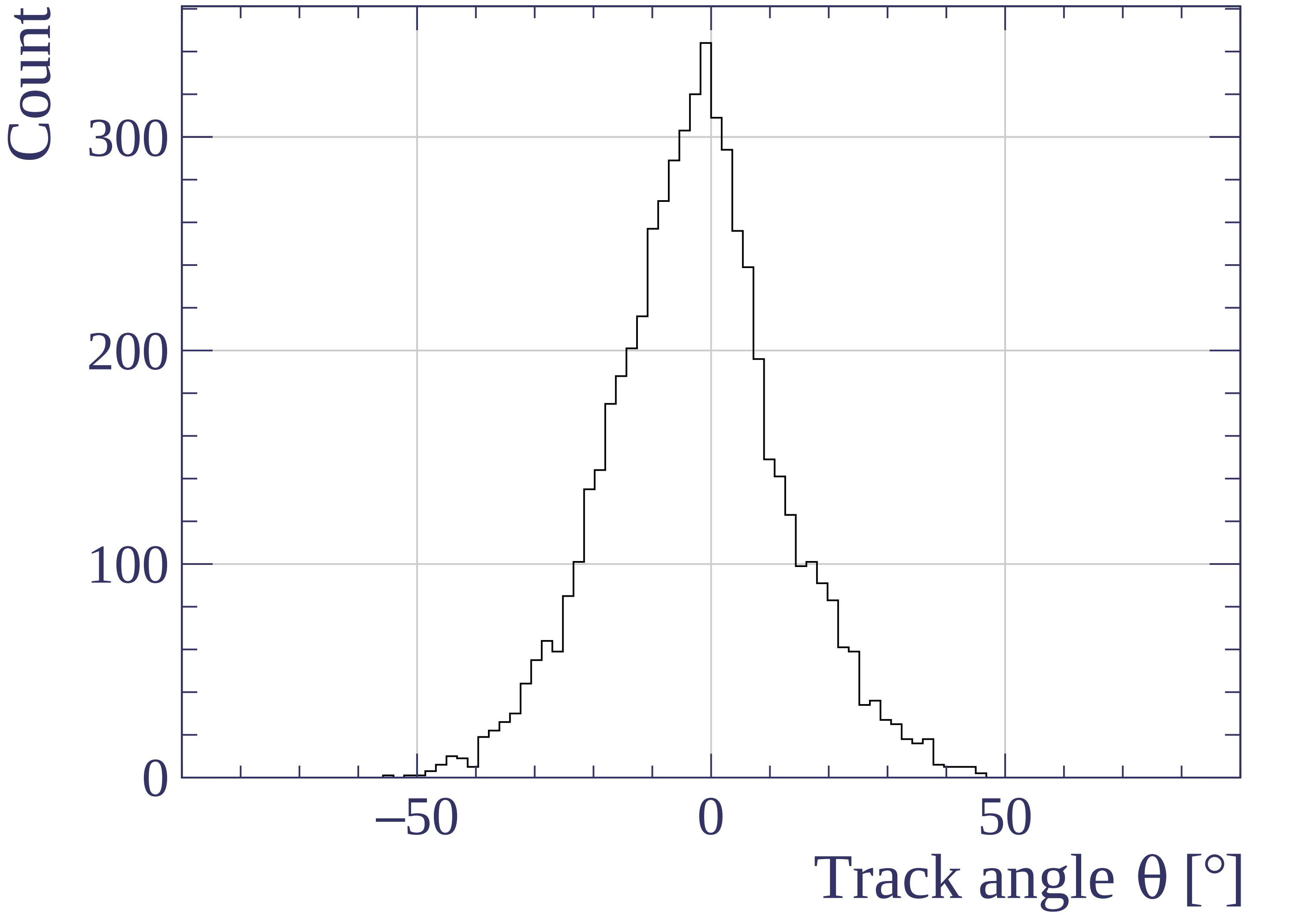


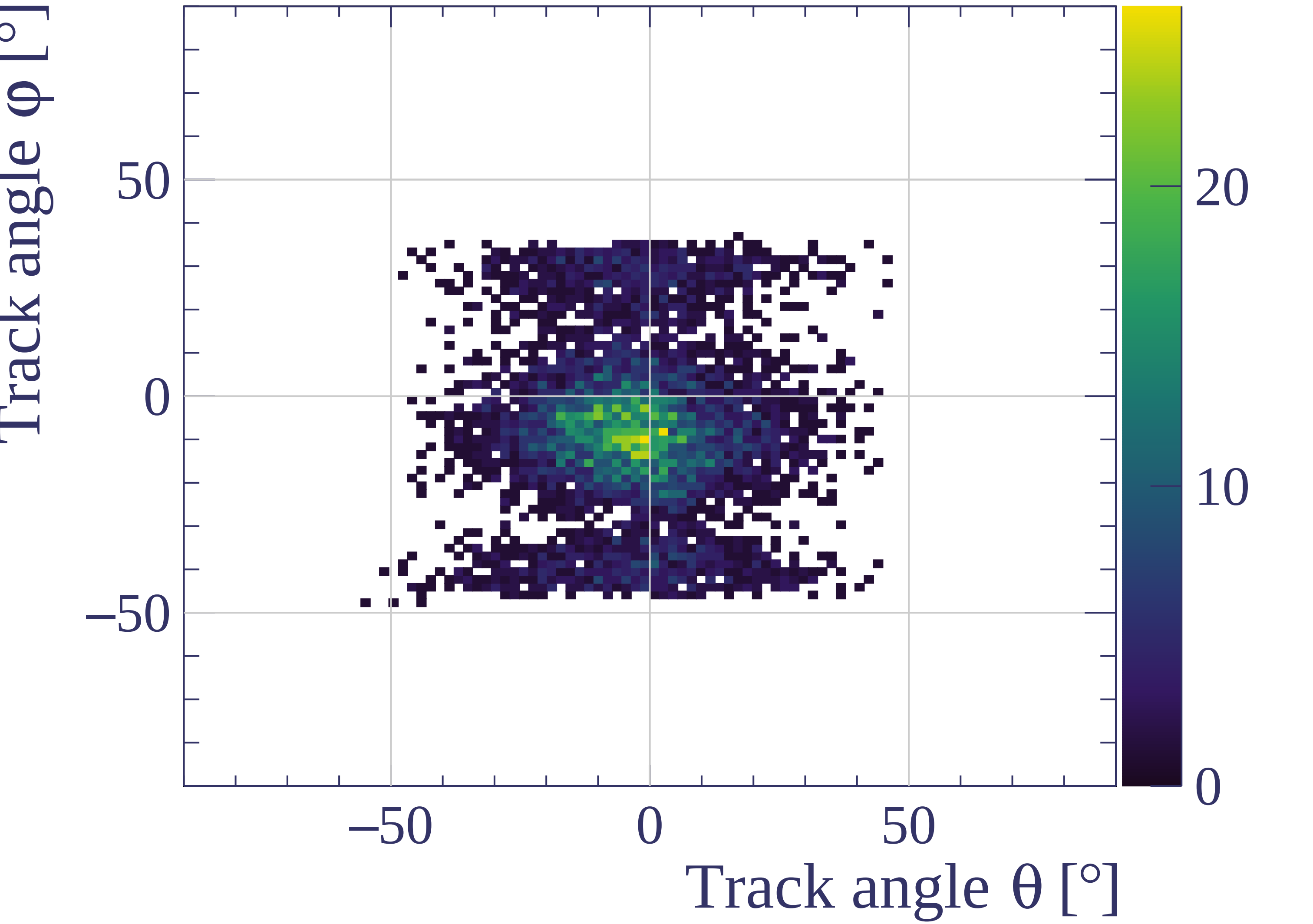
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle θ [°]
10
5
0


Track length [cm]
0
50
100
150
−50
0
50
Track angle θ [°]
0
5
10
15
20

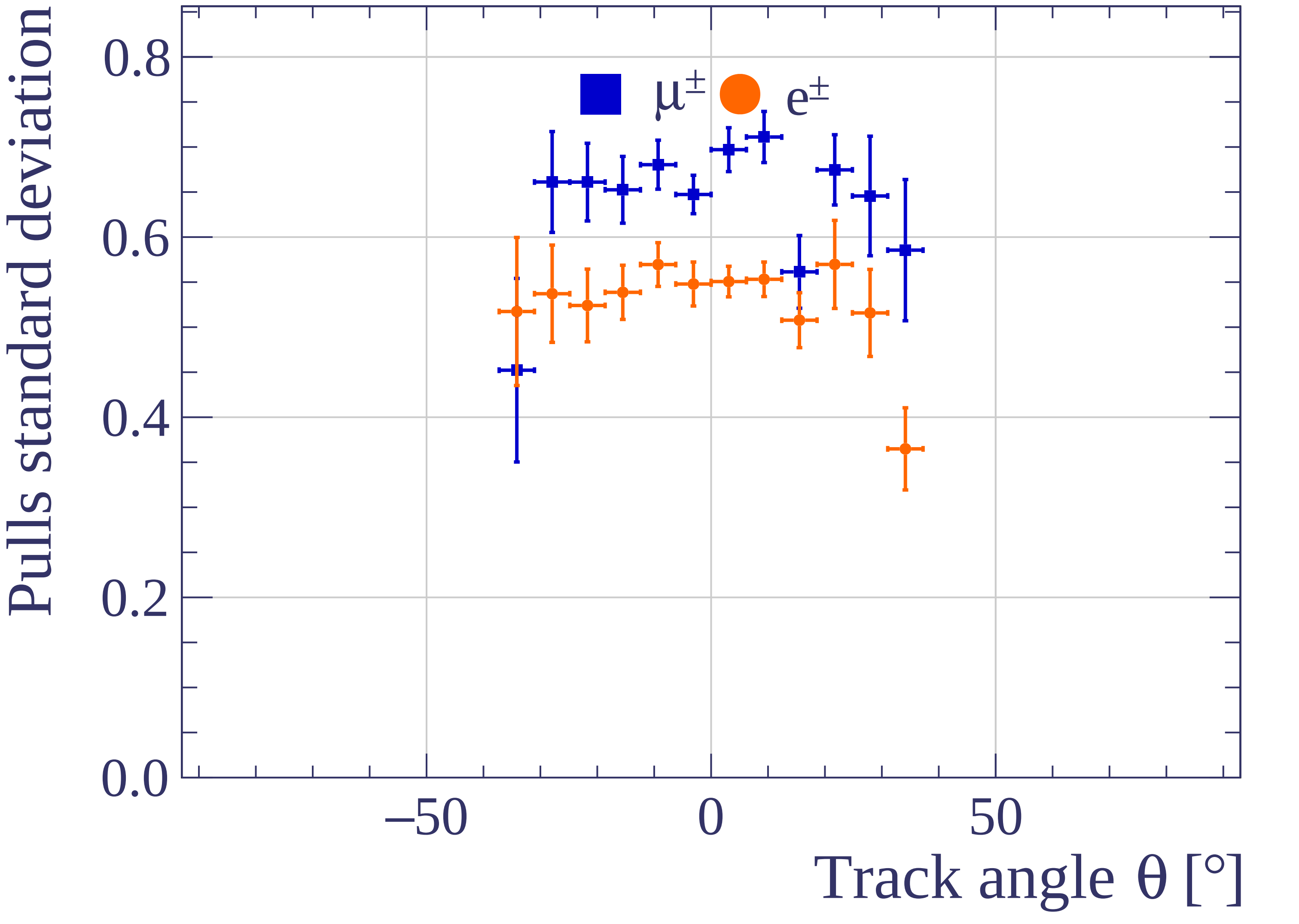


2000
1000
0
−1000
−2000
Signed momentum (MeV)
−50
0
50
Track angle θ [°]
15
10
5
0

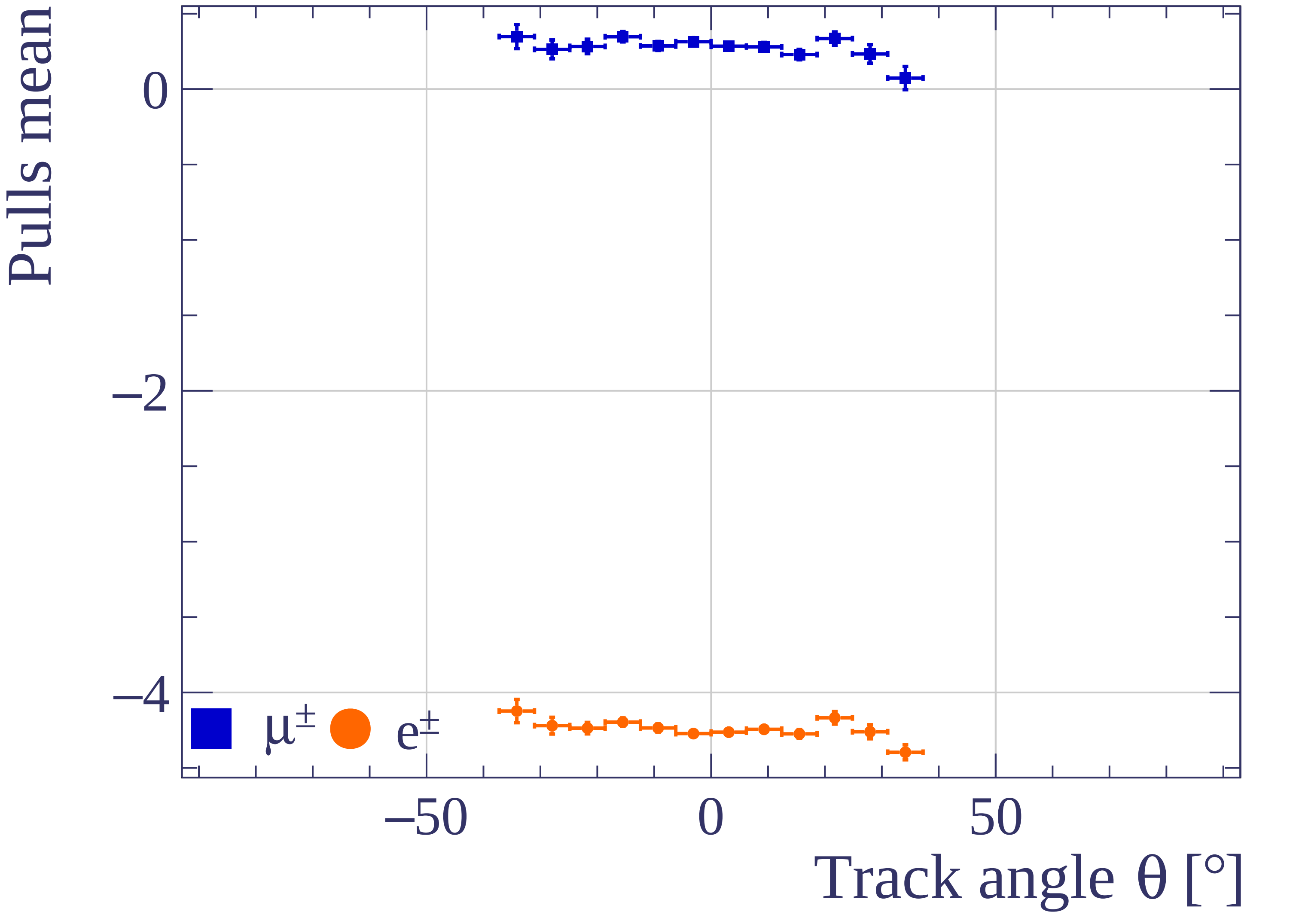

Count
300
200
100
0
−50
0
50
Track angle θ [°]



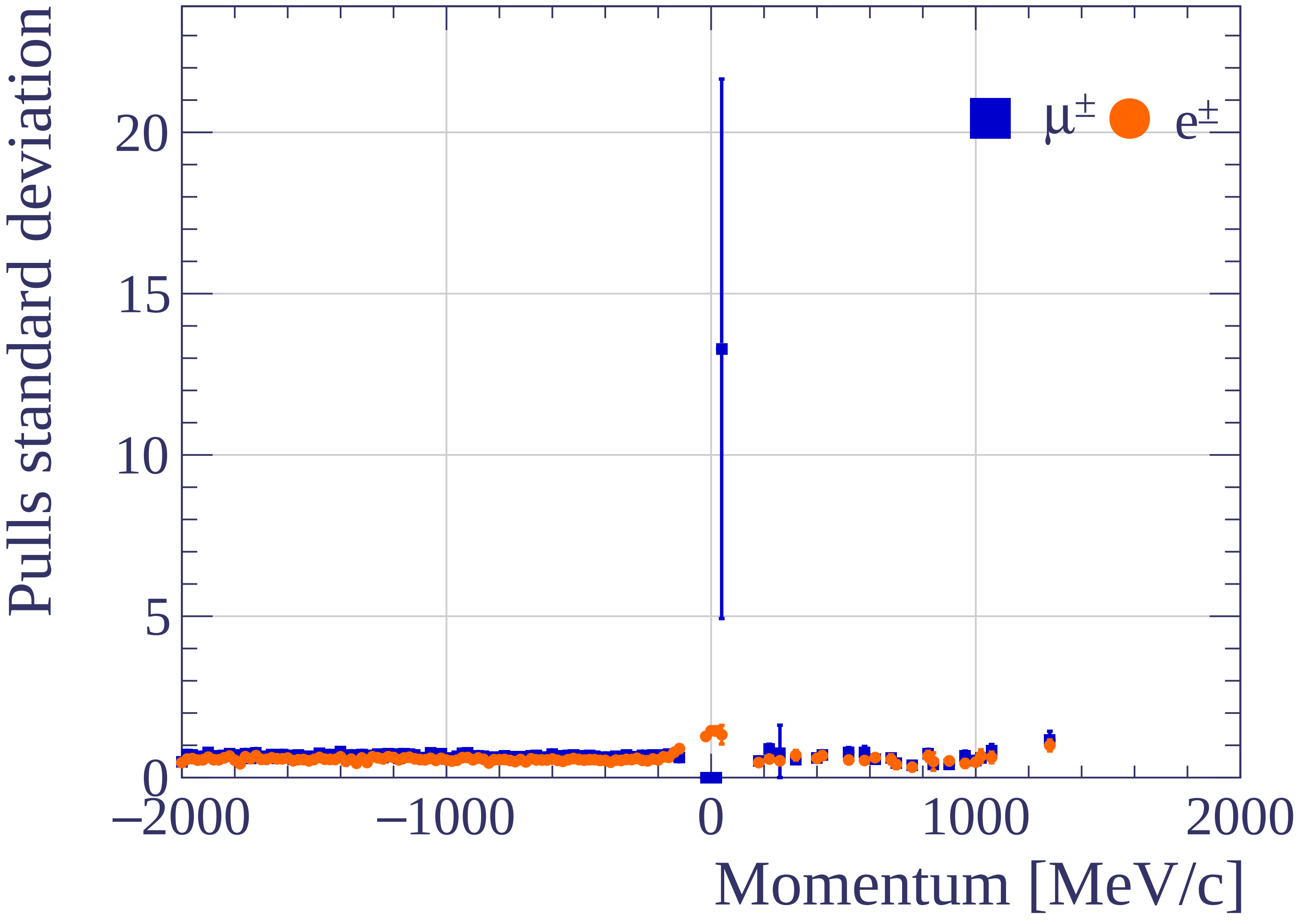
Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
20
10
0



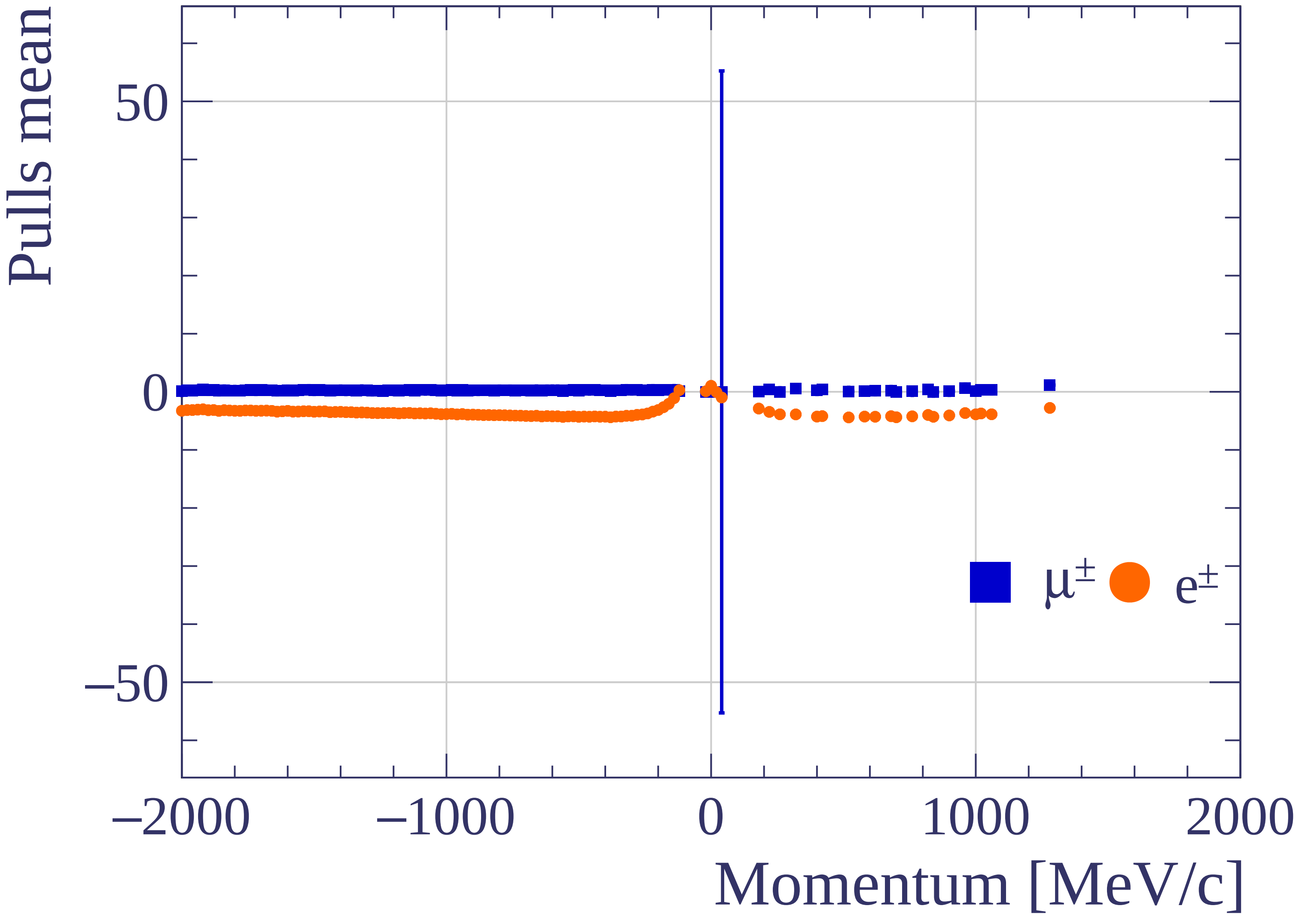
Pulls standard deviation
0.0
0.2
0.4
0.6
0.8
−50
0
50
Track angle θ [°]
μ±
e±



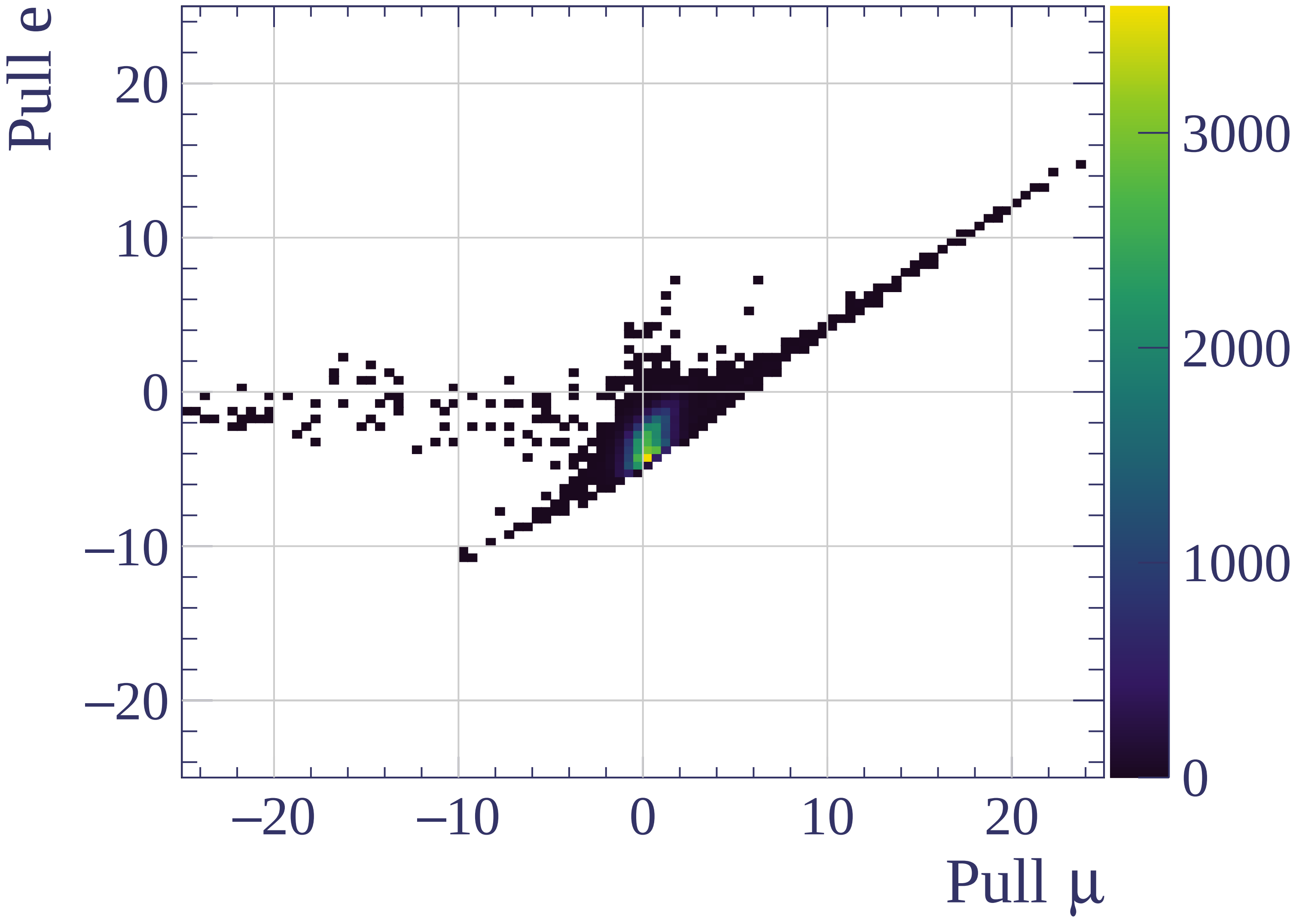
Pulls mean
0
−2
−4
μ±
e±
−50
0
50
Track angle θ [°]



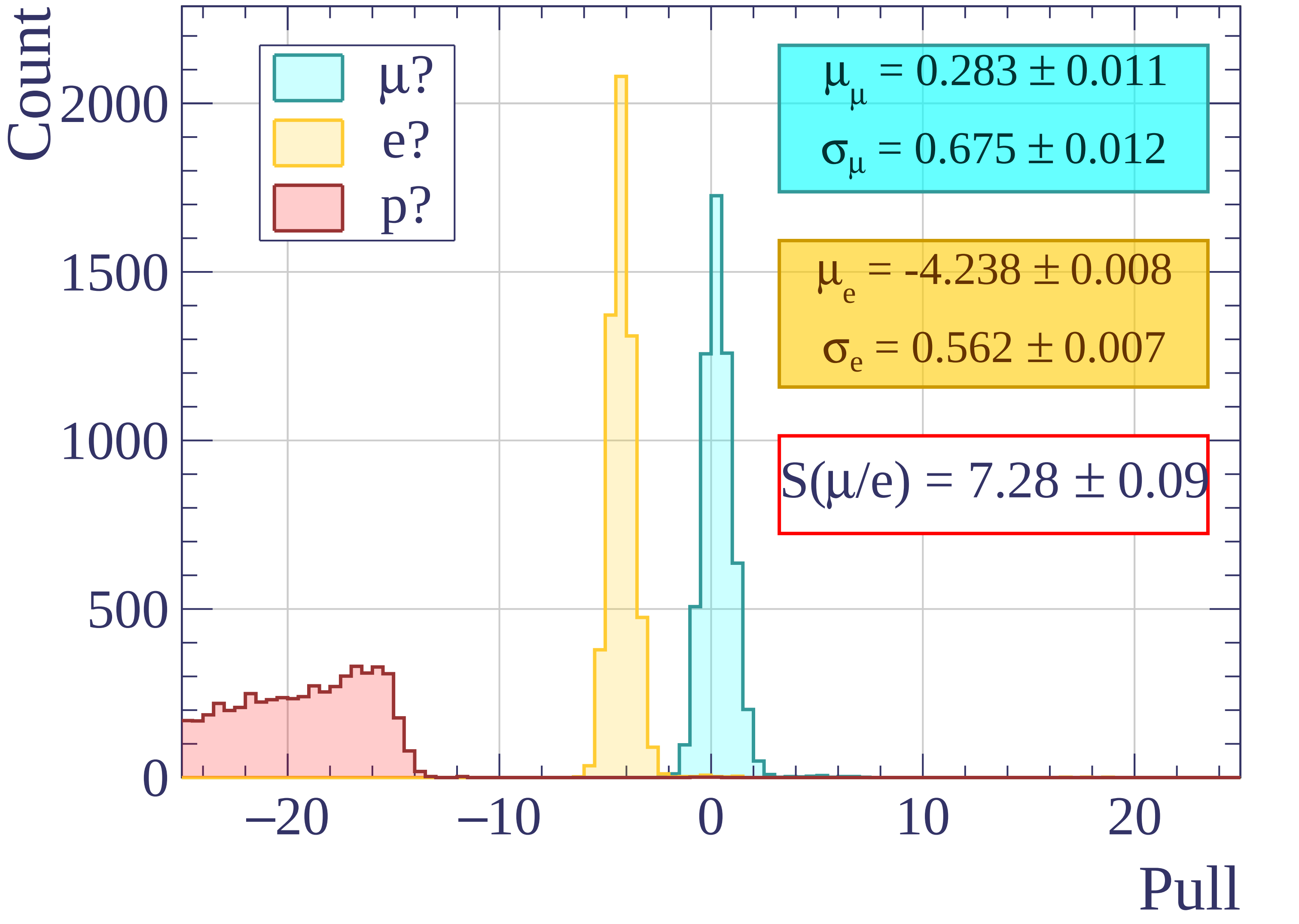
Pulls standard deviation
Momentum [MeV/c]
μ±
e±
0
5
10
15
20
−2000
−1000
0
1000
2000



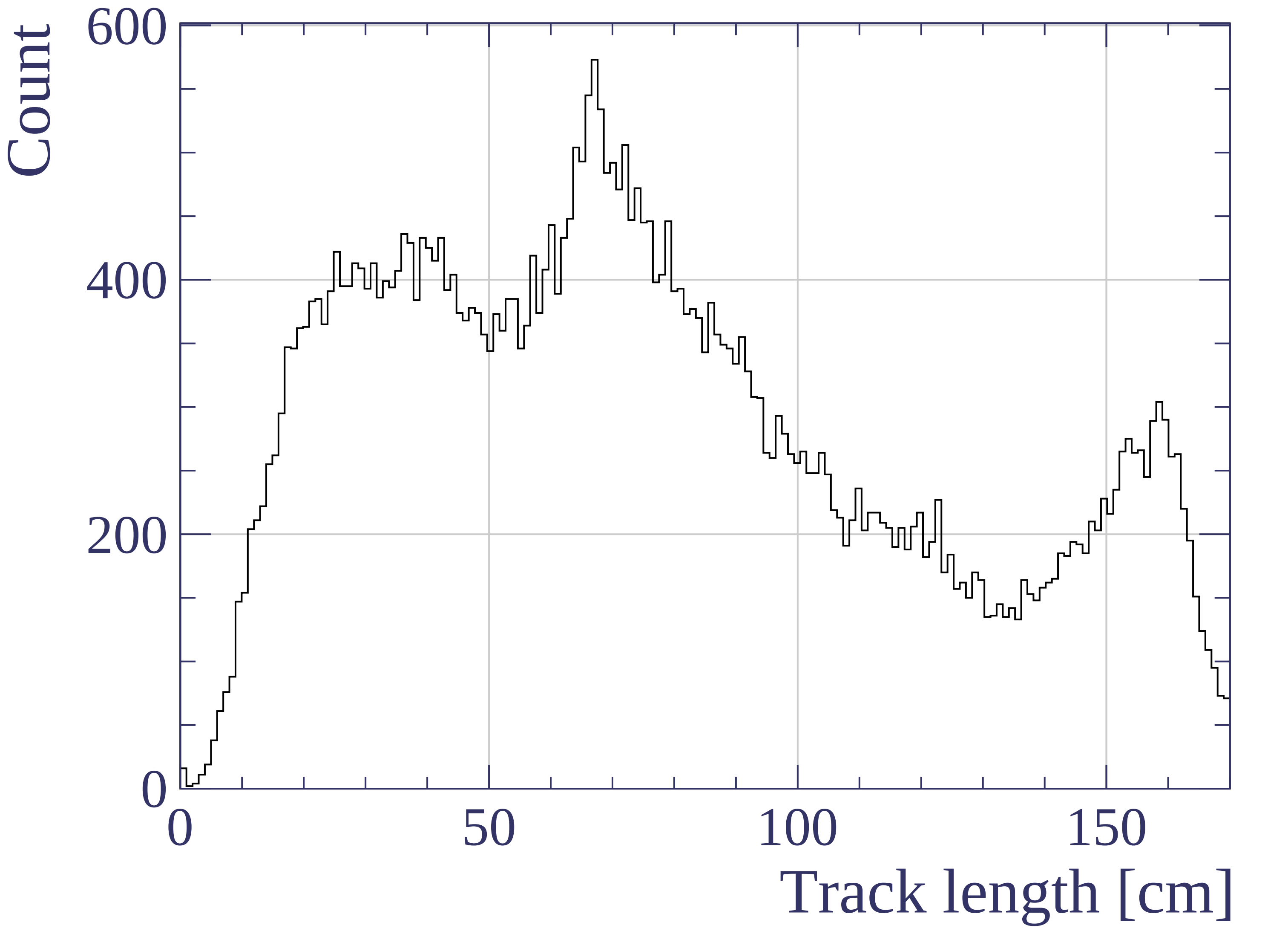
Pulls mean
50
0
−50
−2000
−1000
0
1000
2000
Momentum [MeV/c]
μ±
e±

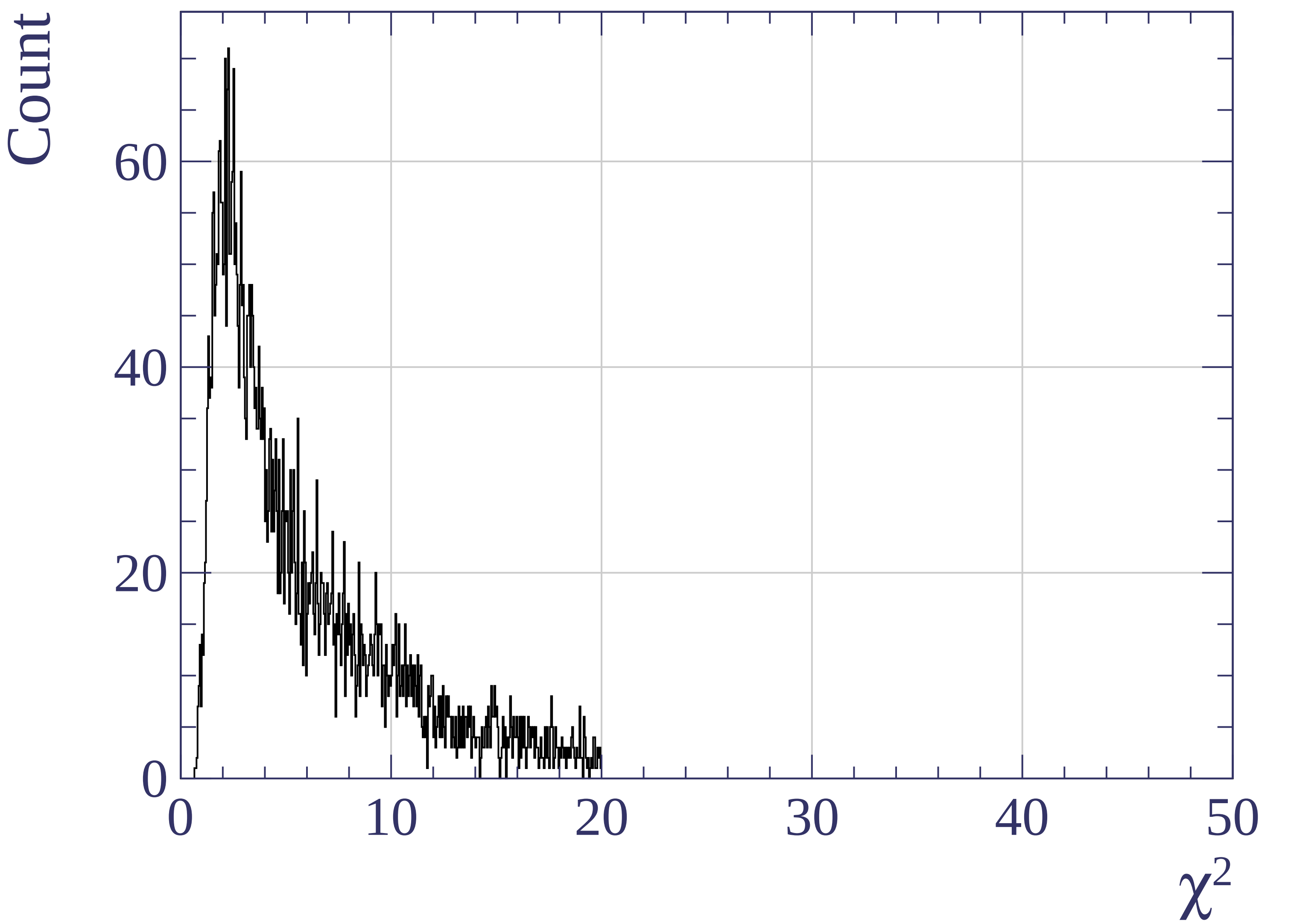


Pull e
20
10
0
−10
−20
−20
−10
0
10
20
Pull μ
3000
2000
1000
0

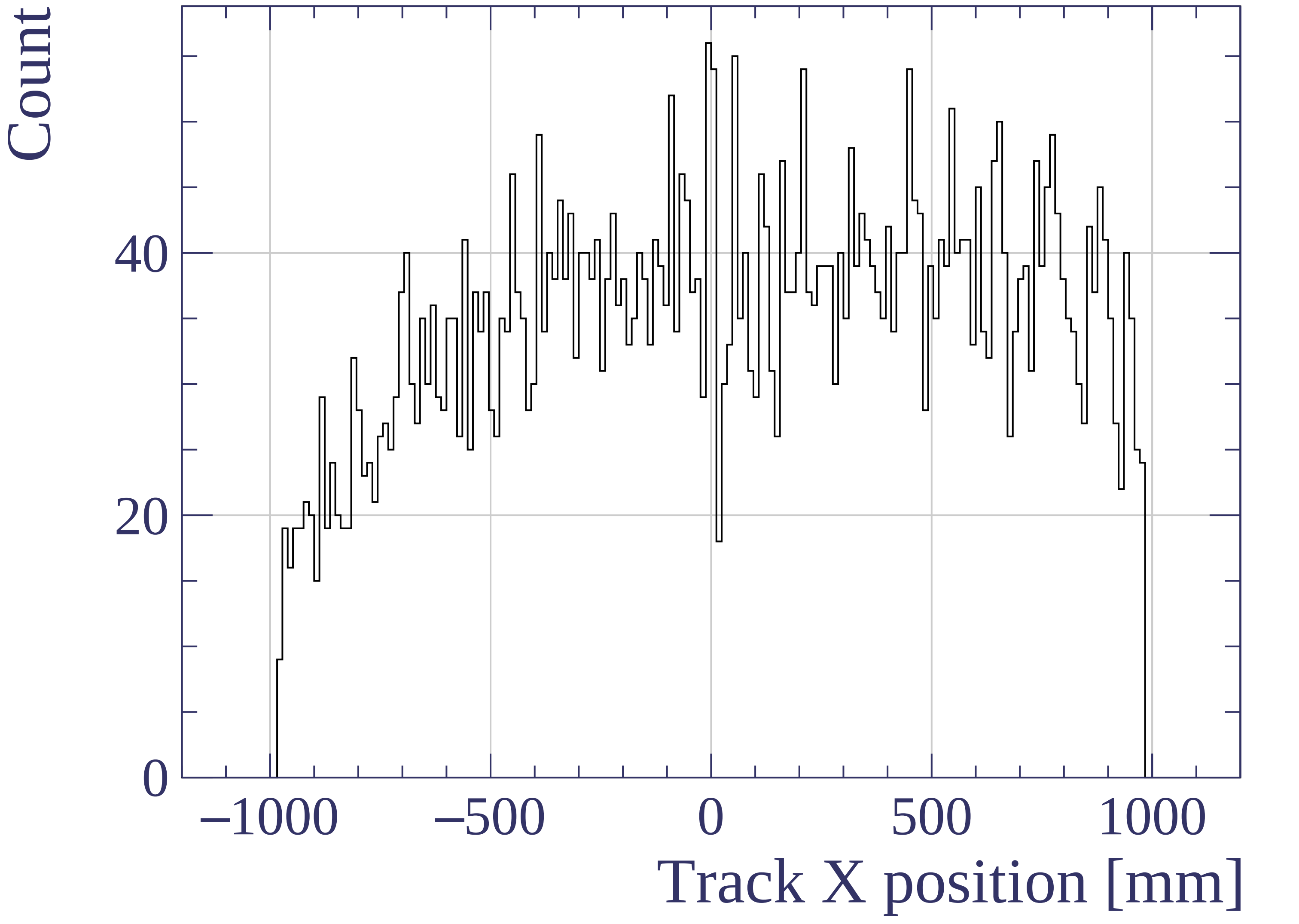


Count
Pull
μ?
e?
p?
$\mu_\mu = 0.283 \pm 0.011$
$\sigma_\mu = 0.675 \pm 0.012$
$\mu_e = -4.238 \pm 0.008$
$\sigma_e = 0.562 \pm 0.007$
$S(\mu/e) = 7.28 \pm 0.09$
2000
1500
1000
500
0
−20
−10
0
10
20

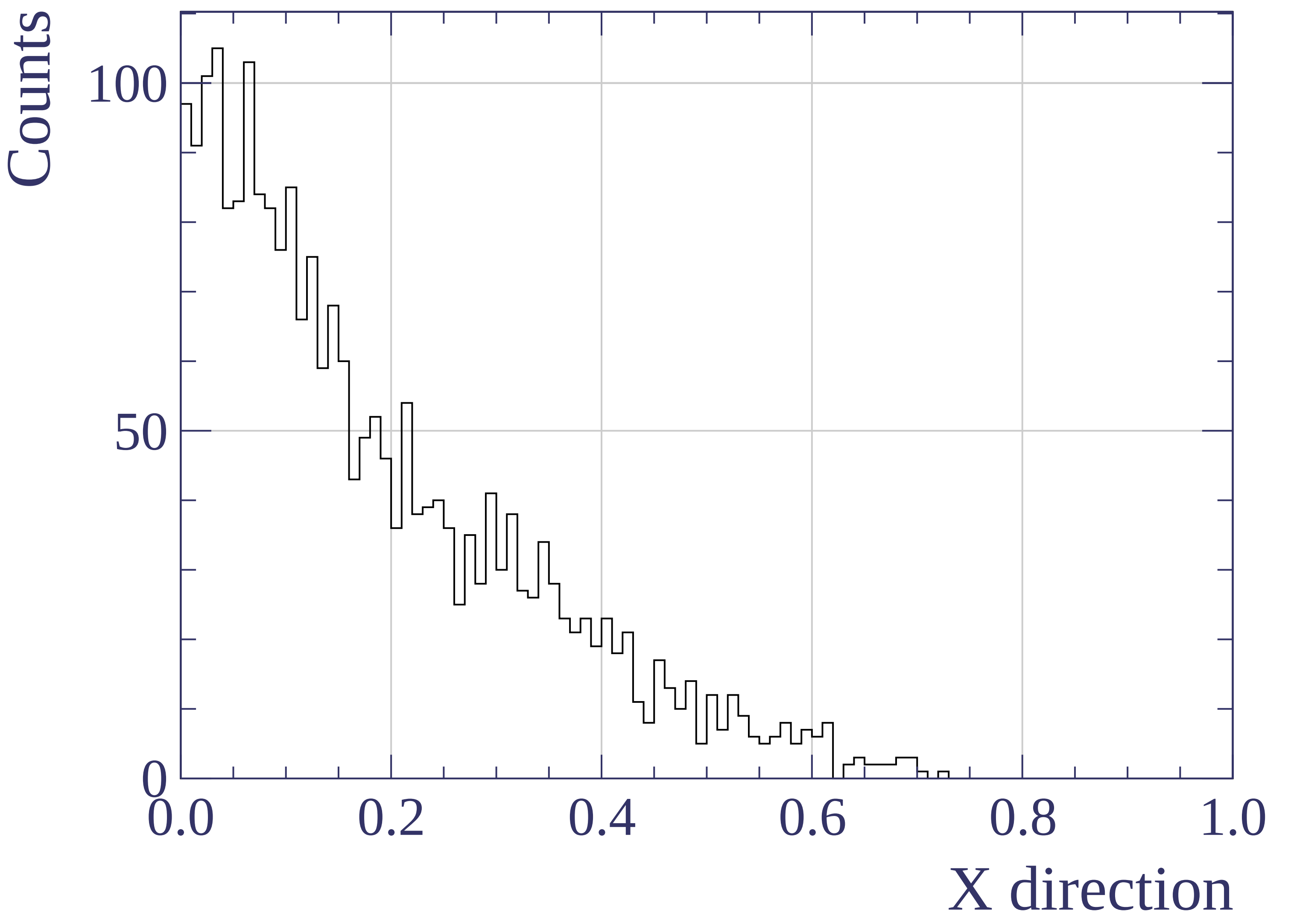

Count
600
400
200
0
0
50
100
150
Track length [cm]



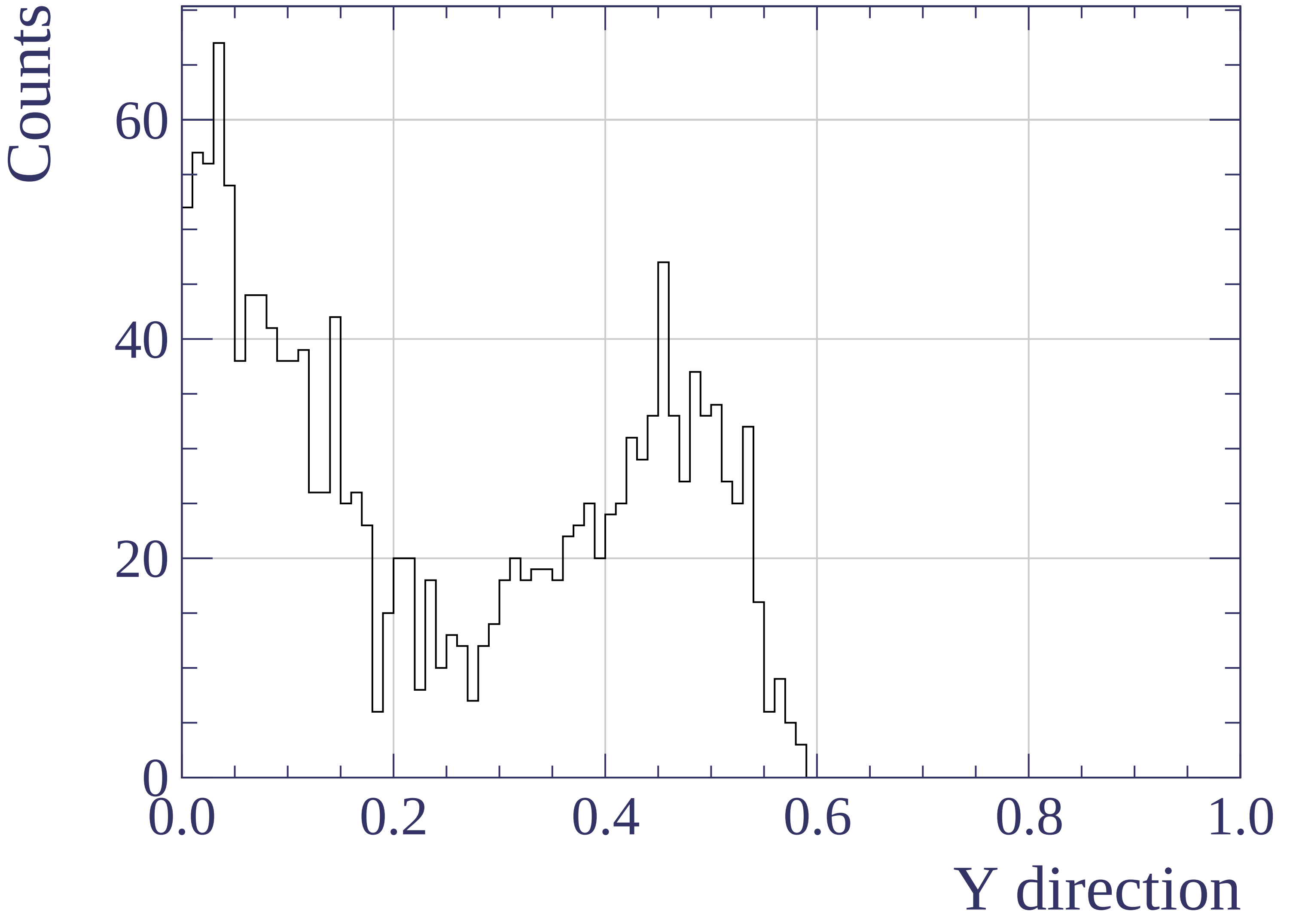
Count
0
20
40
60
0
10
20
30
40
50
χ^2

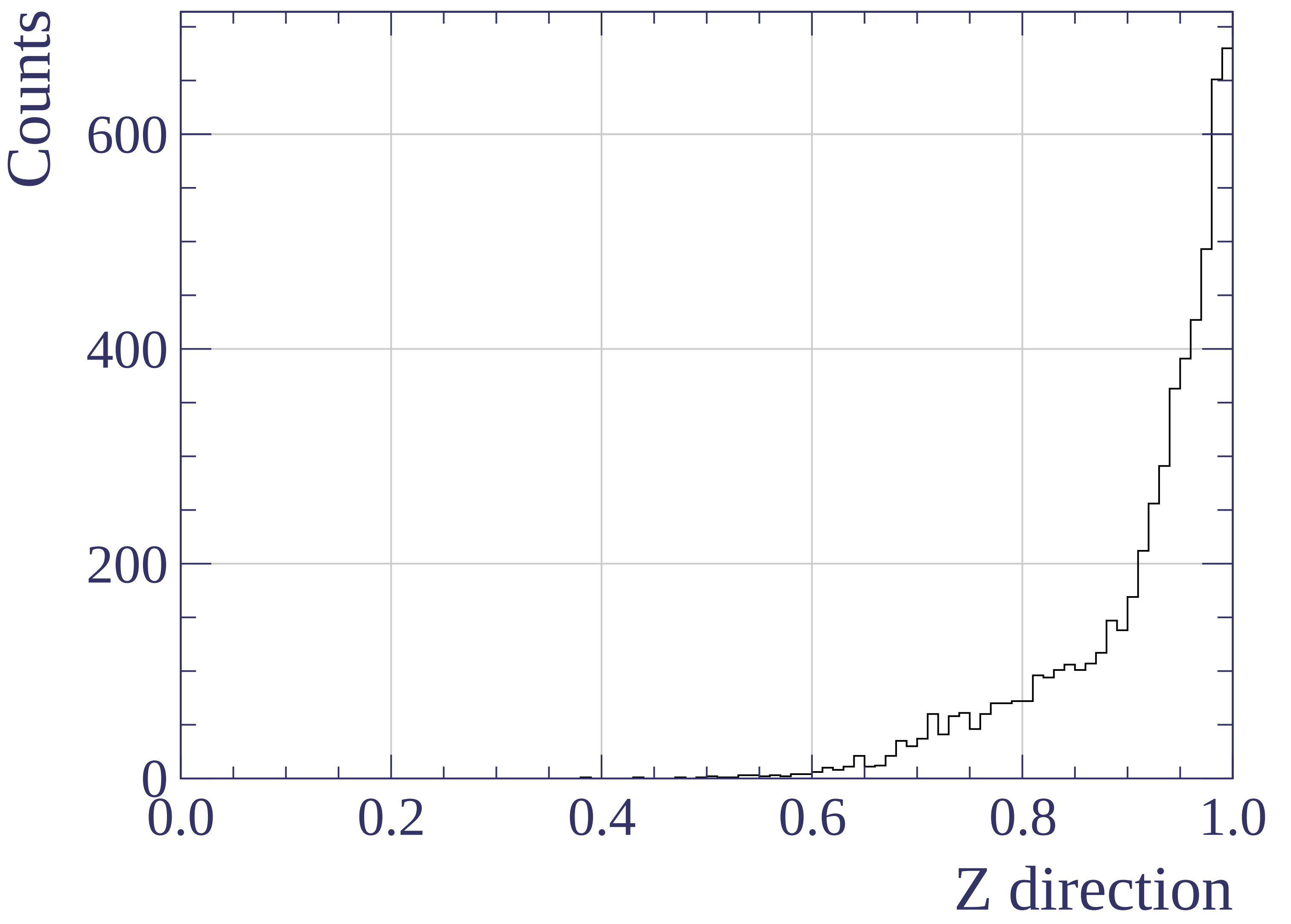


Count
0
20
40
−1000
−500
0
500
1000
Track X position [mm]

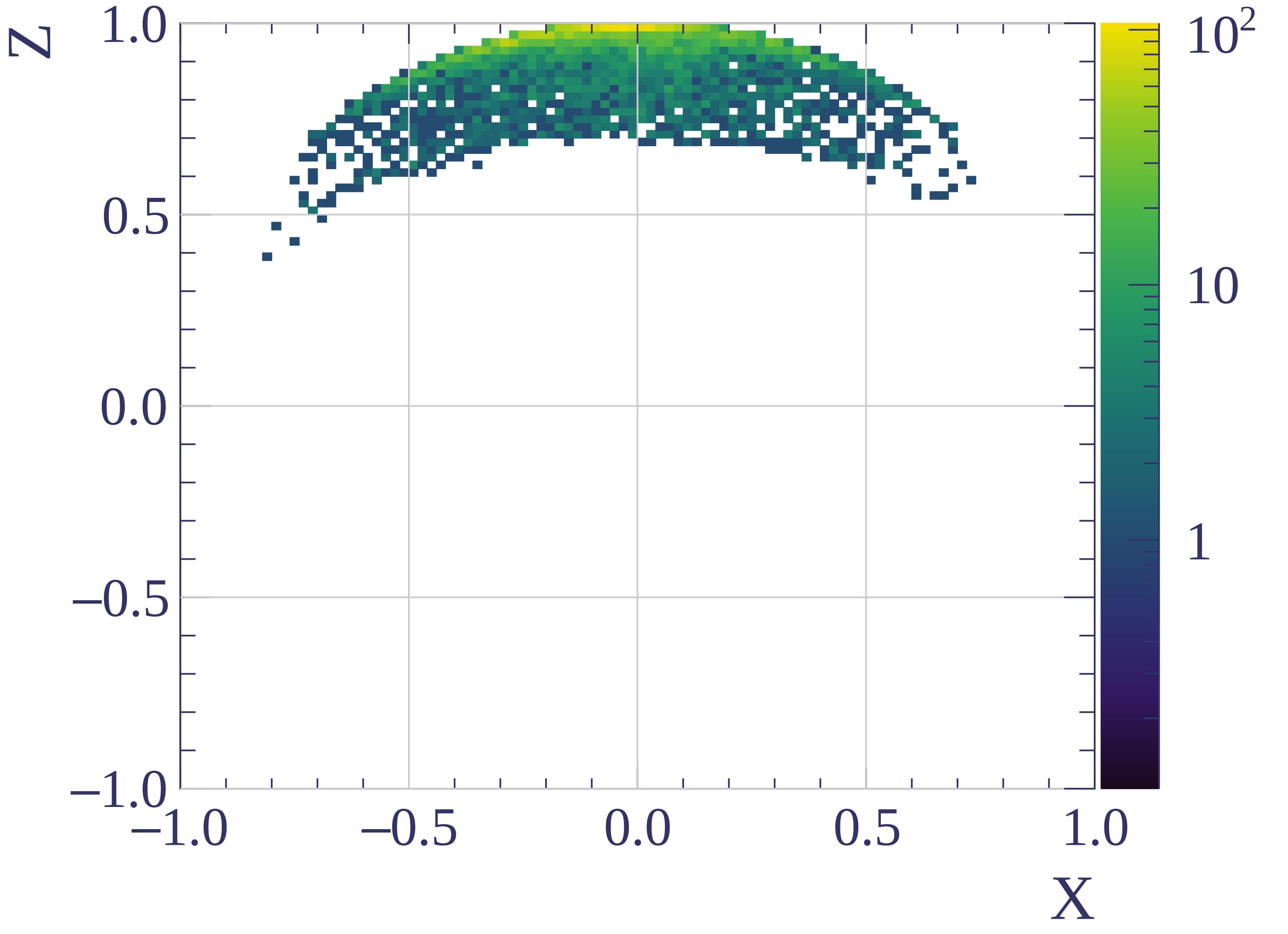


Counts
100
50
0
0.0
0.2
0.4
0.6
0.8
1.0
X direction

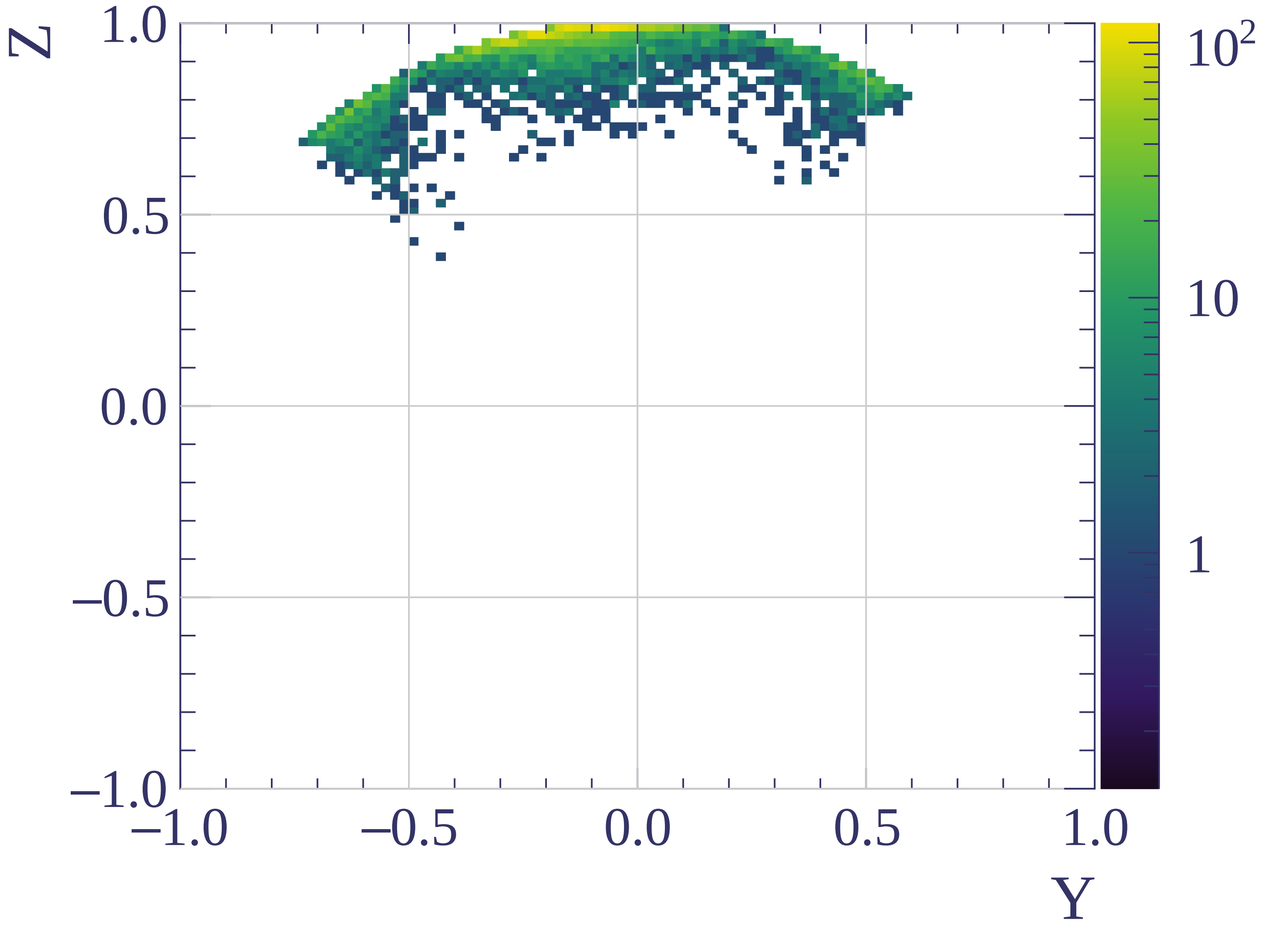

Counts
60
40
20
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



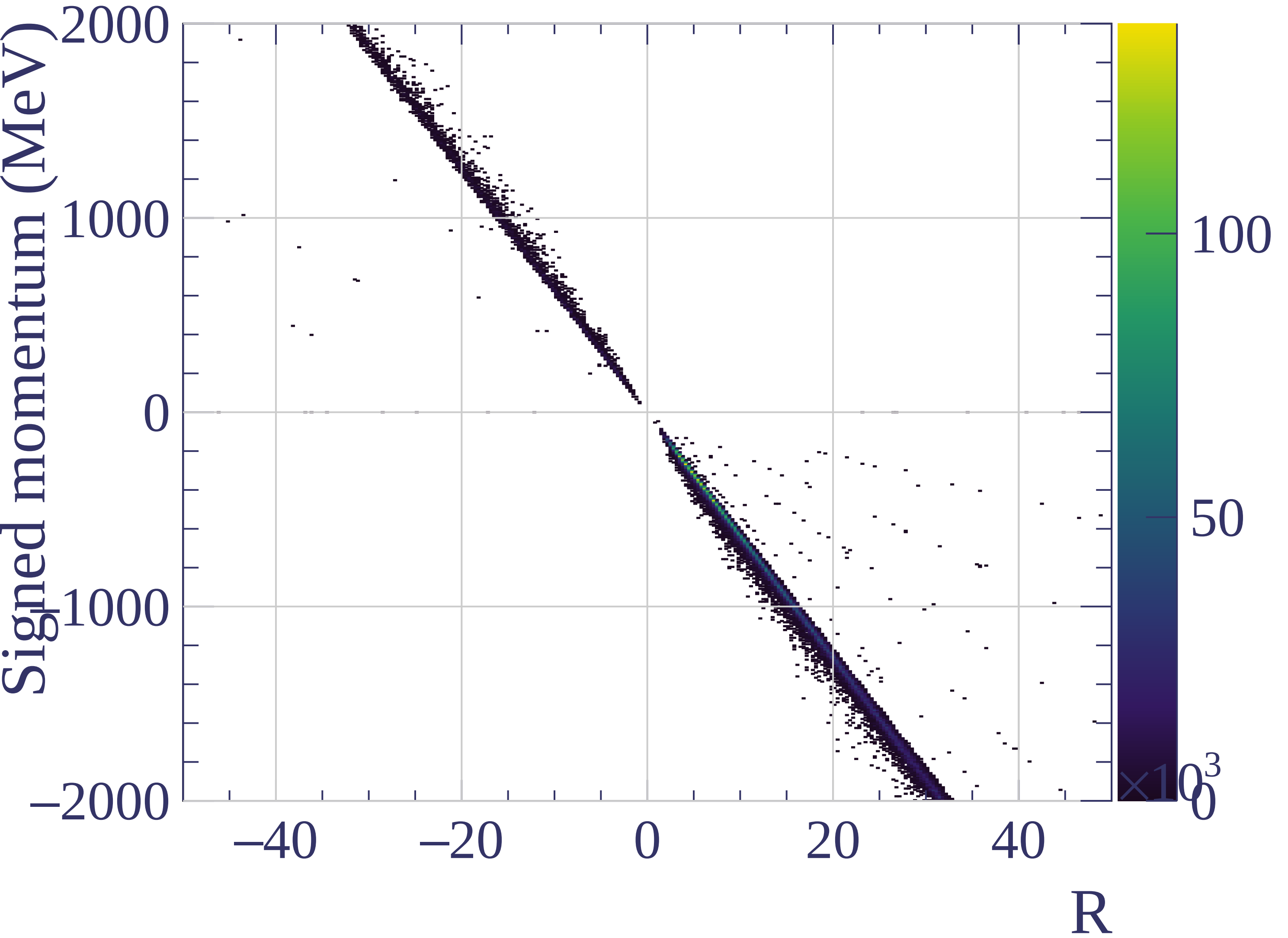
Counts
600
400
200
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction



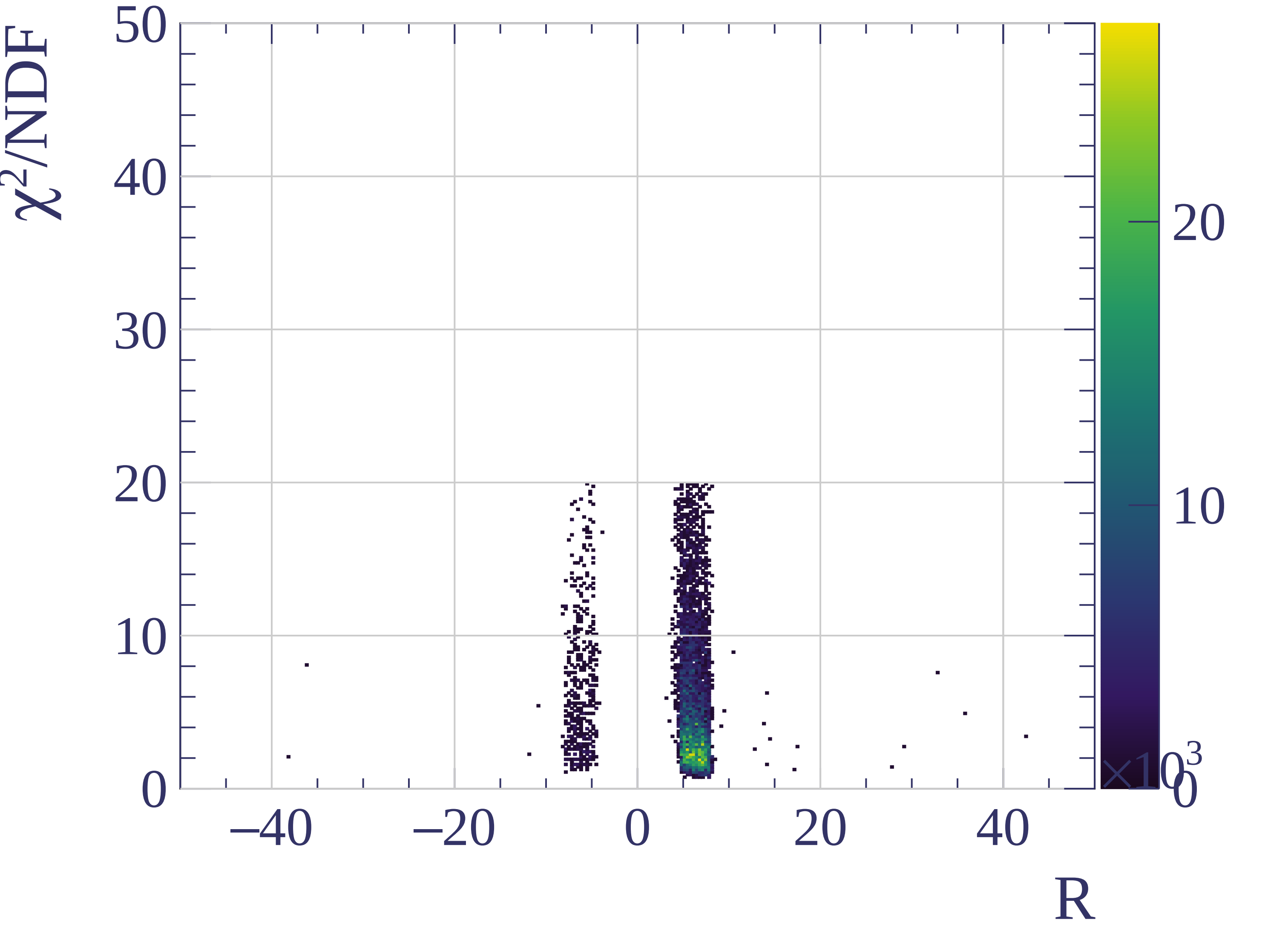
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
10^2
10
1



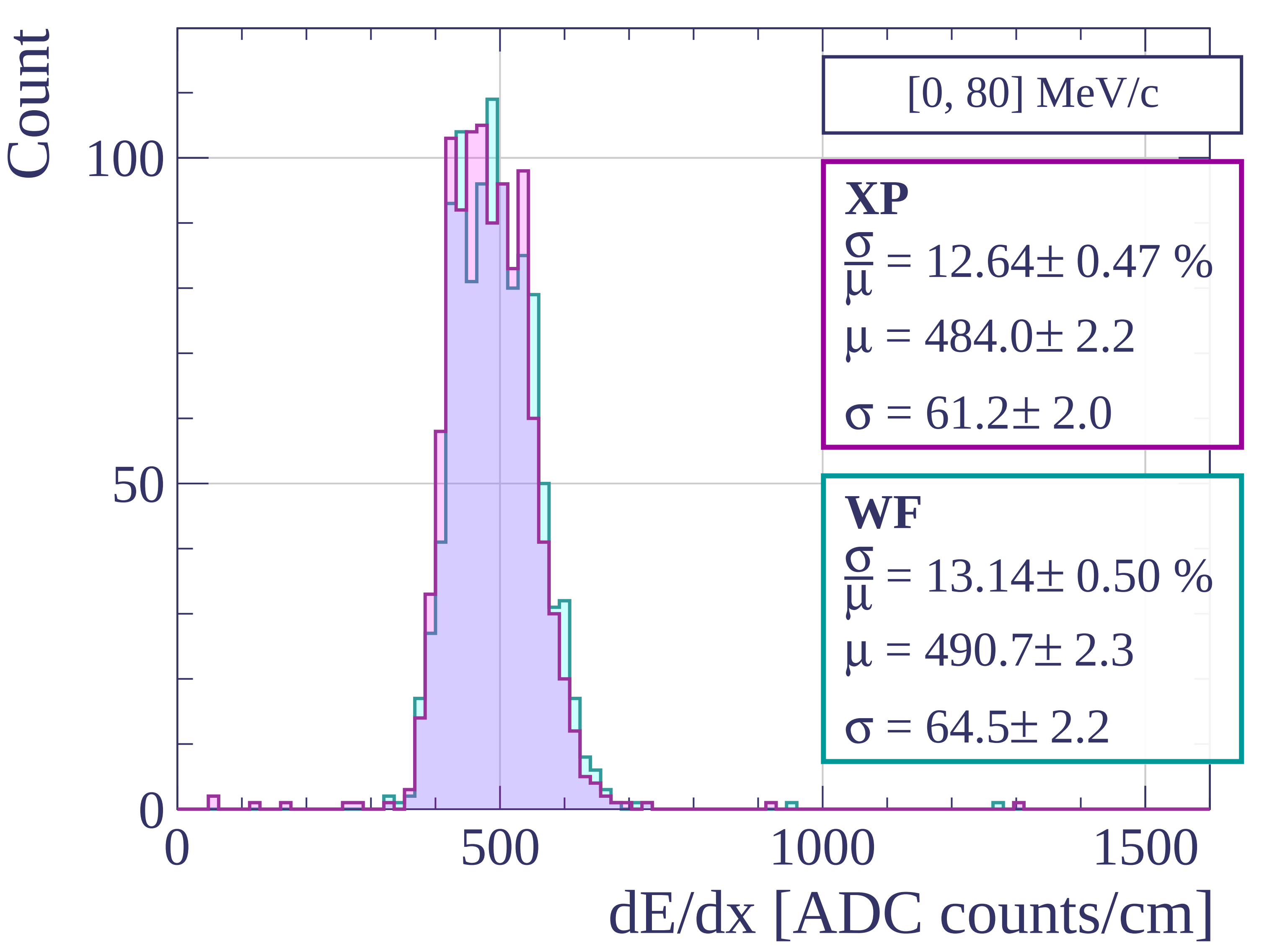
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
10^2
10
1



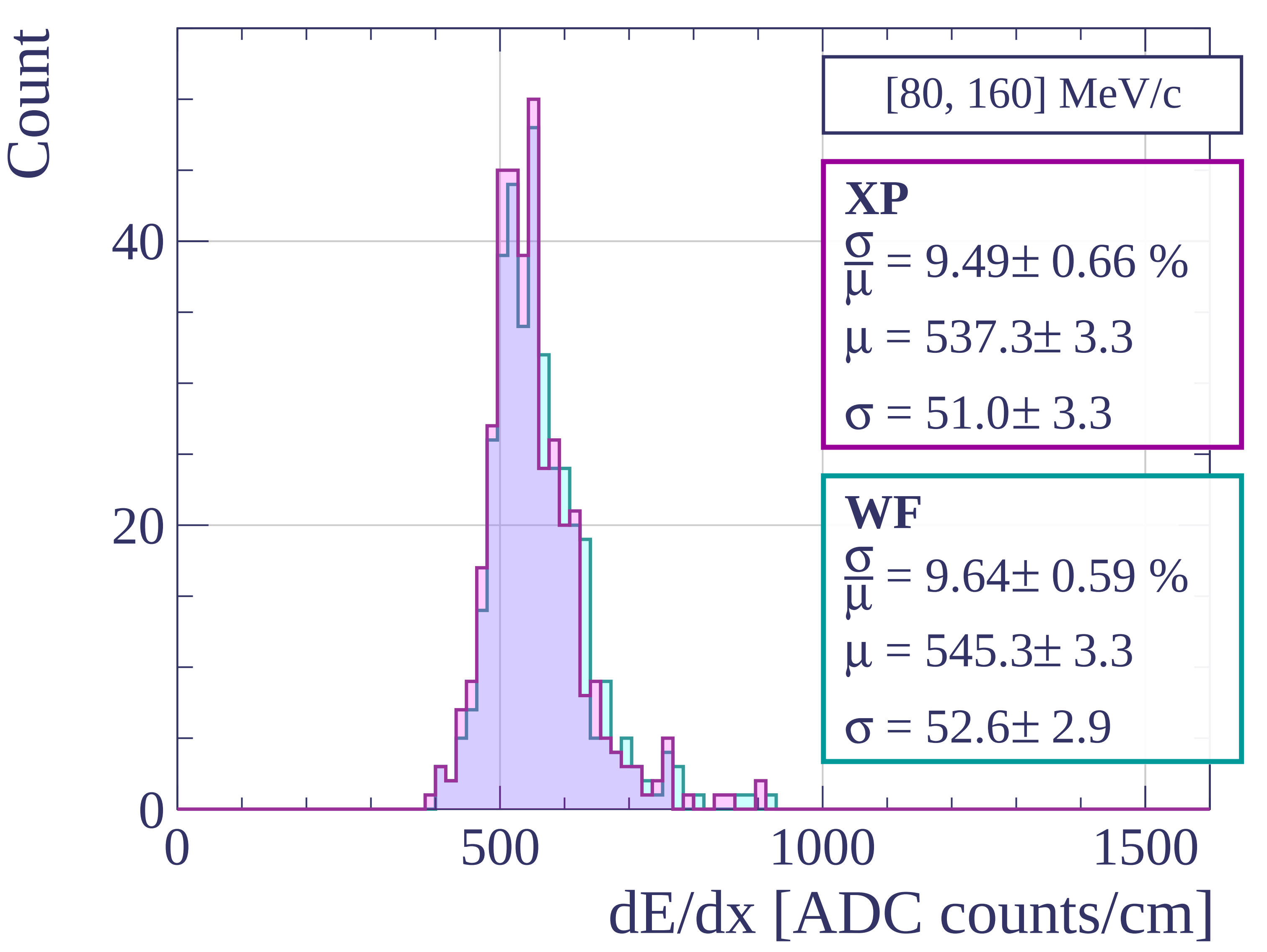
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−40
−20
0
20
40
R
100
50
0
$\times10^3$



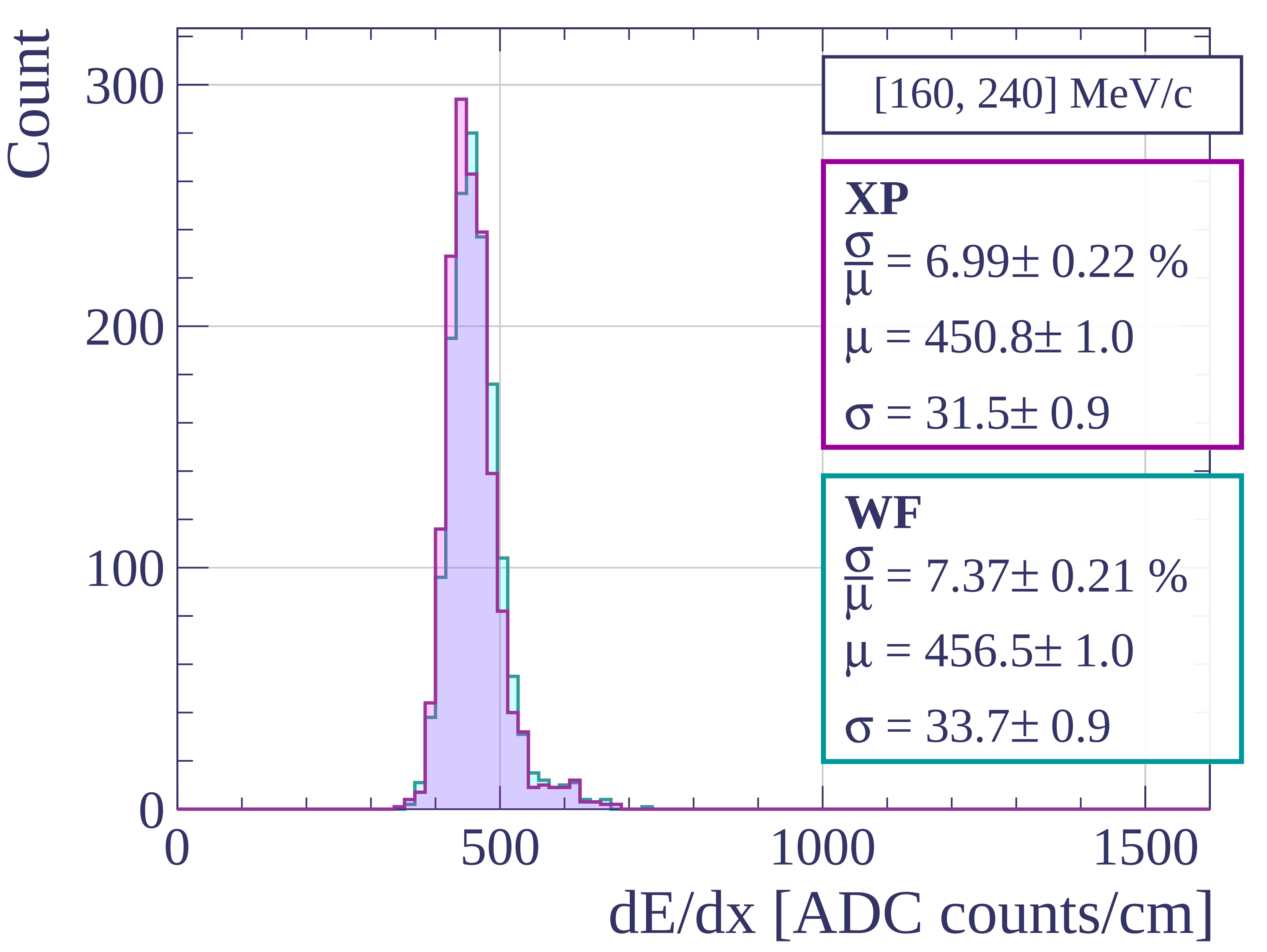
χ^2/NDF
50
40
30
20
10
0
−40
−20
0
20
40
R
20
10
0
×10³



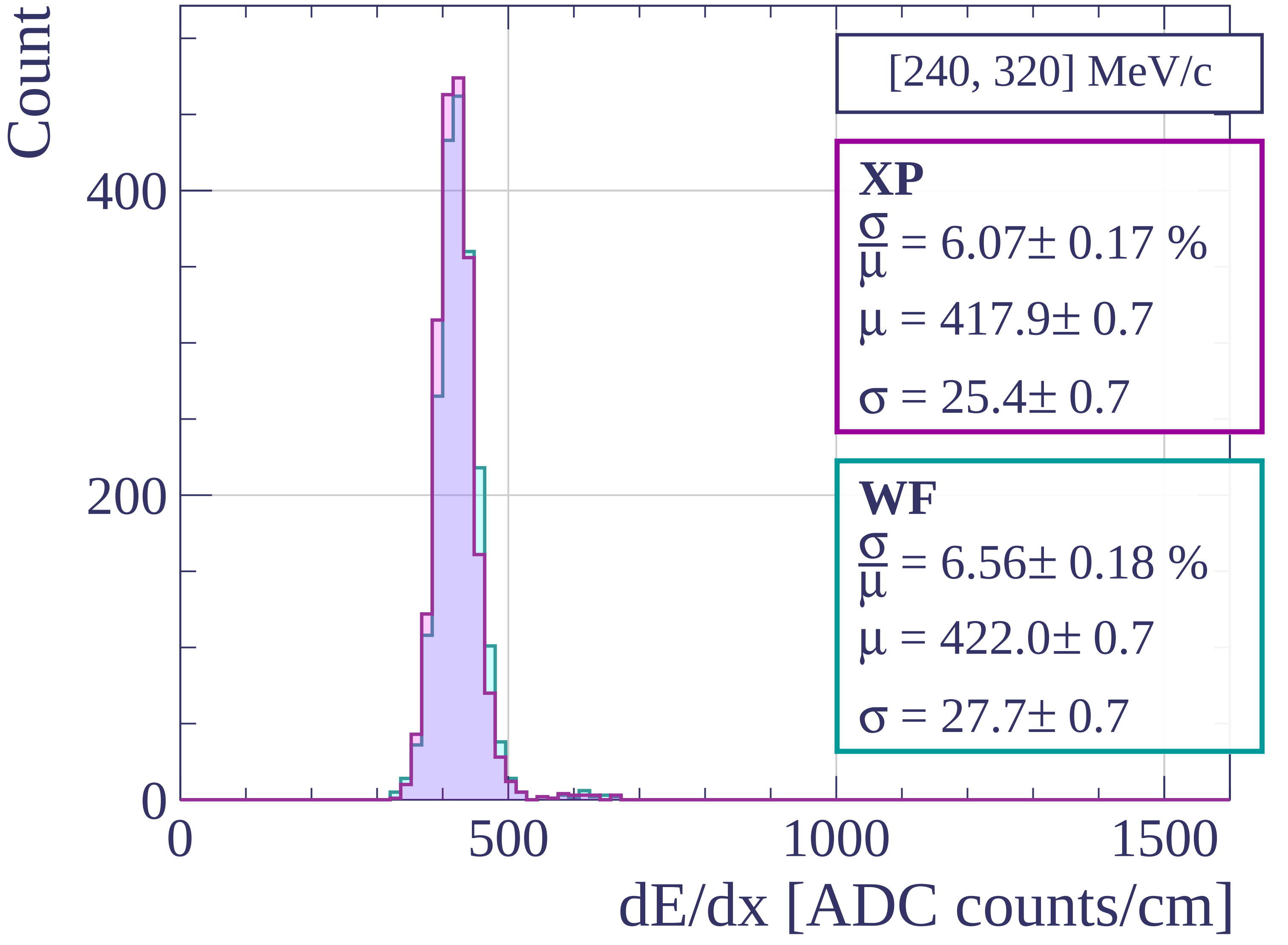
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 80] MeV/c
XP
σ/μ = 12.64± 0.47 %
μ = 484.0± 2.2
σ = 61.2± 2.0
WF
σ/μ = 13.14± 0.50 %
μ = 490.7± 2.3
σ = 64.5± 2.2

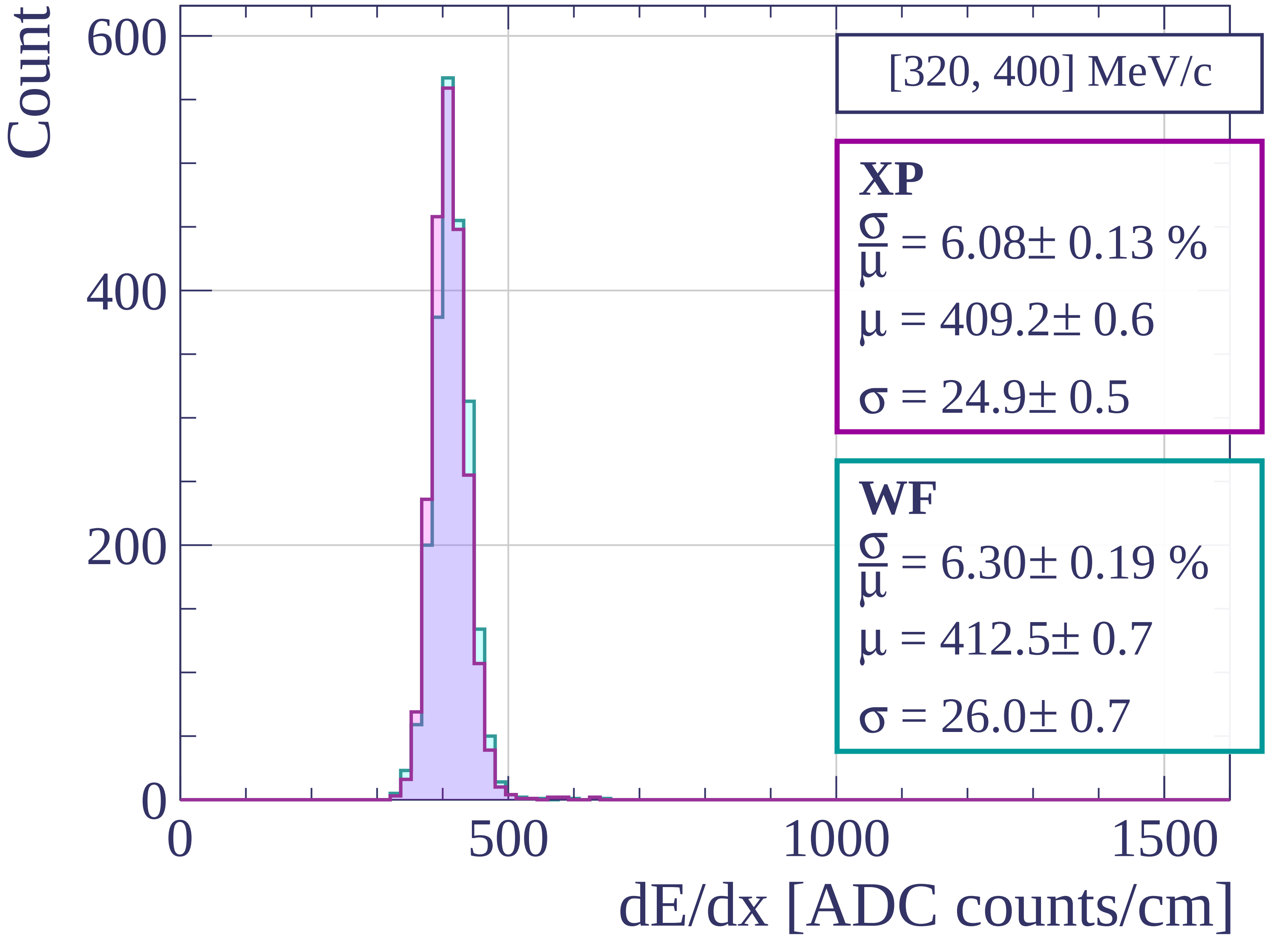


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[80, 160] MeV/c
XP
σ/μ = 9.49± 0.66 %
μ = 537.3± 3.3
σ = 51.0± 3.3
WF
σ/μ = 9.64± 0.59 %
μ = 545.3± 3.3
σ = 52.6± 2.9

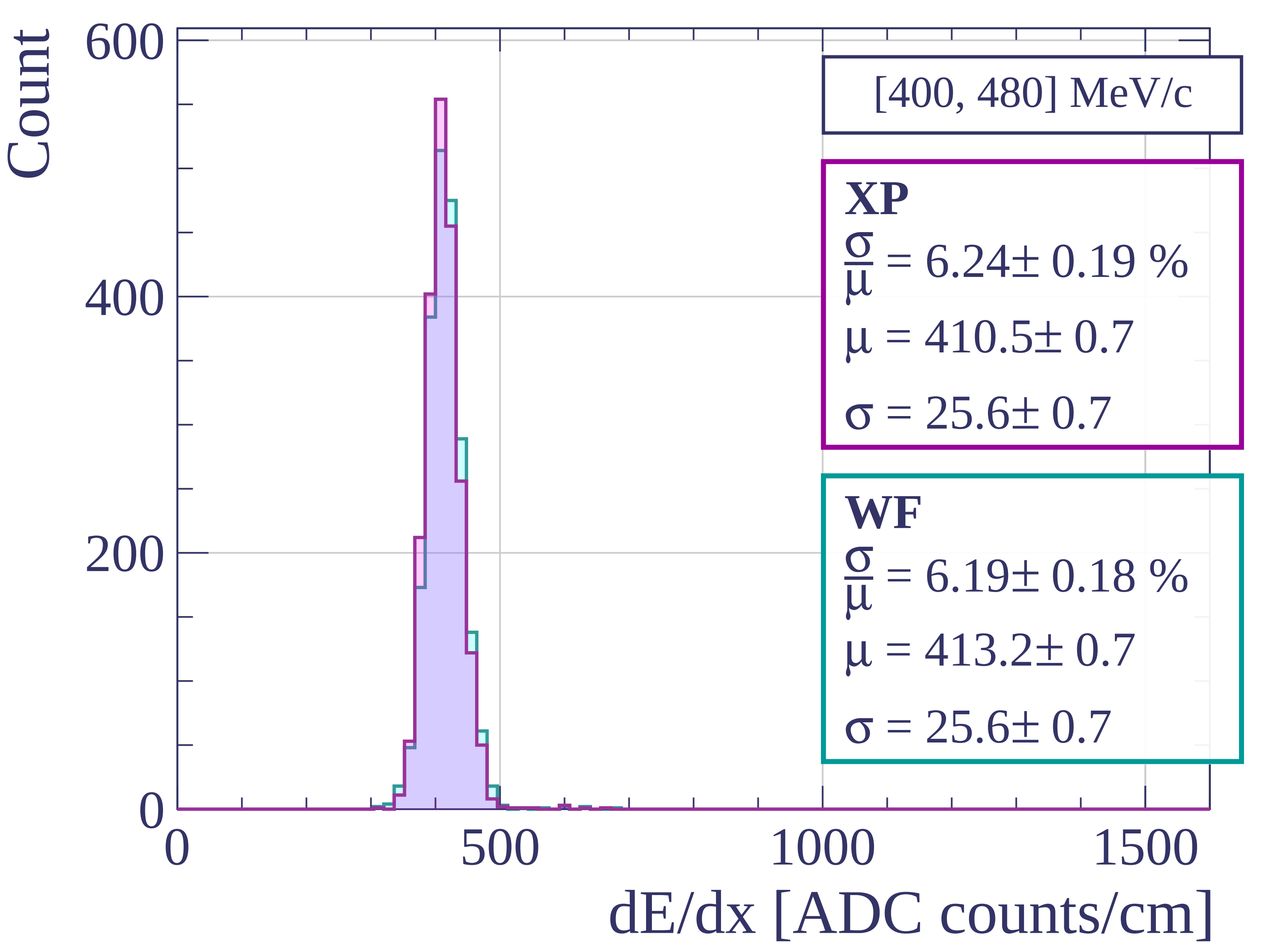


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 240] MeV/c
XP
σ/μ = 6.99± 0.22 %
μ = 450.8± 1.0
σ = 31.5± 0.9
WF
σ/μ = 7.37± 0.21 %
μ = 456.5± 1.0
σ = 33.7± 0.9

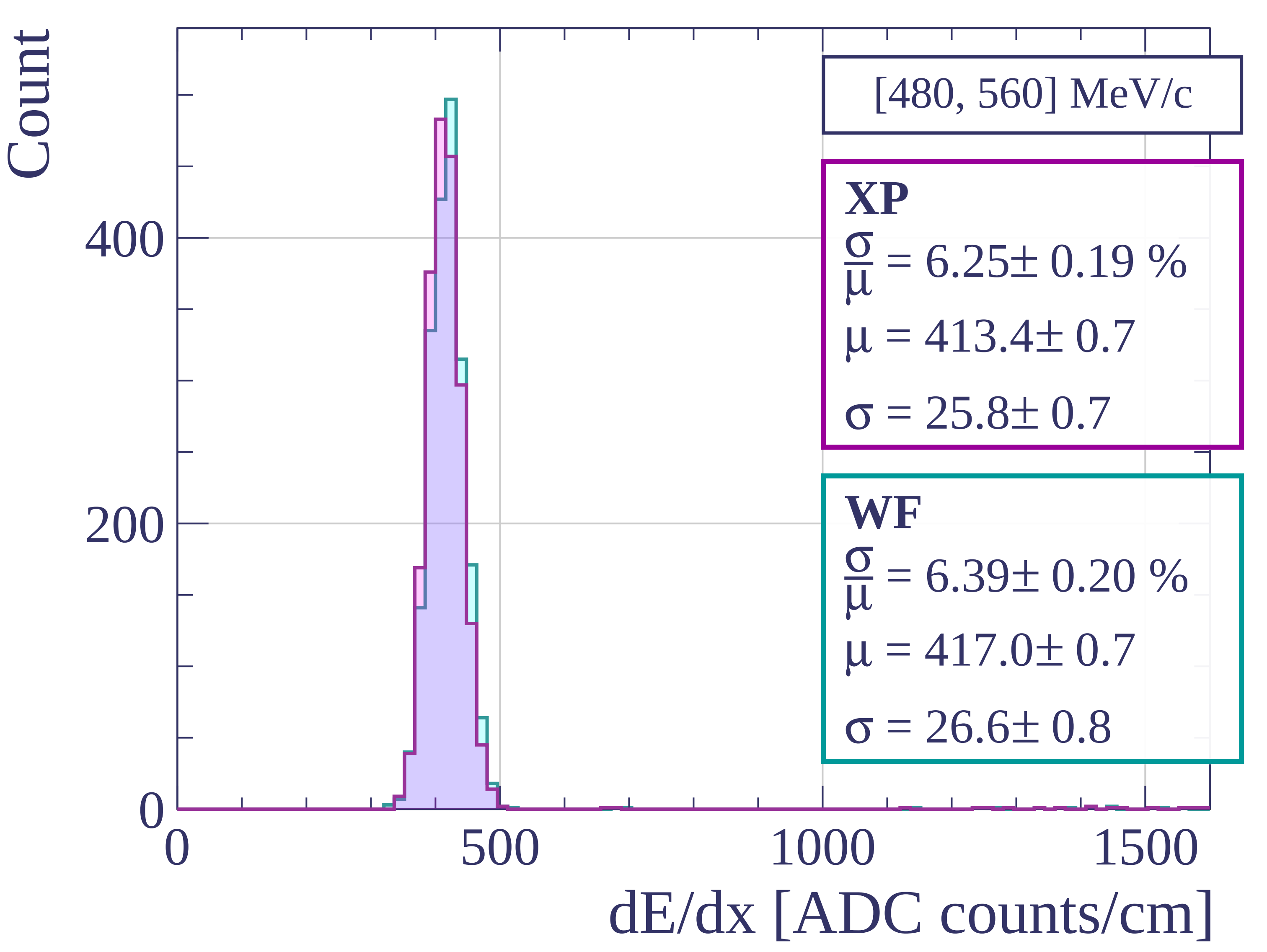

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.07± 0.17 %
μ = 417.9± 0.7
σ = 25.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 6.56± 0.18 %
μ = 422.0± 0.7
σ = 27.7± 0.7

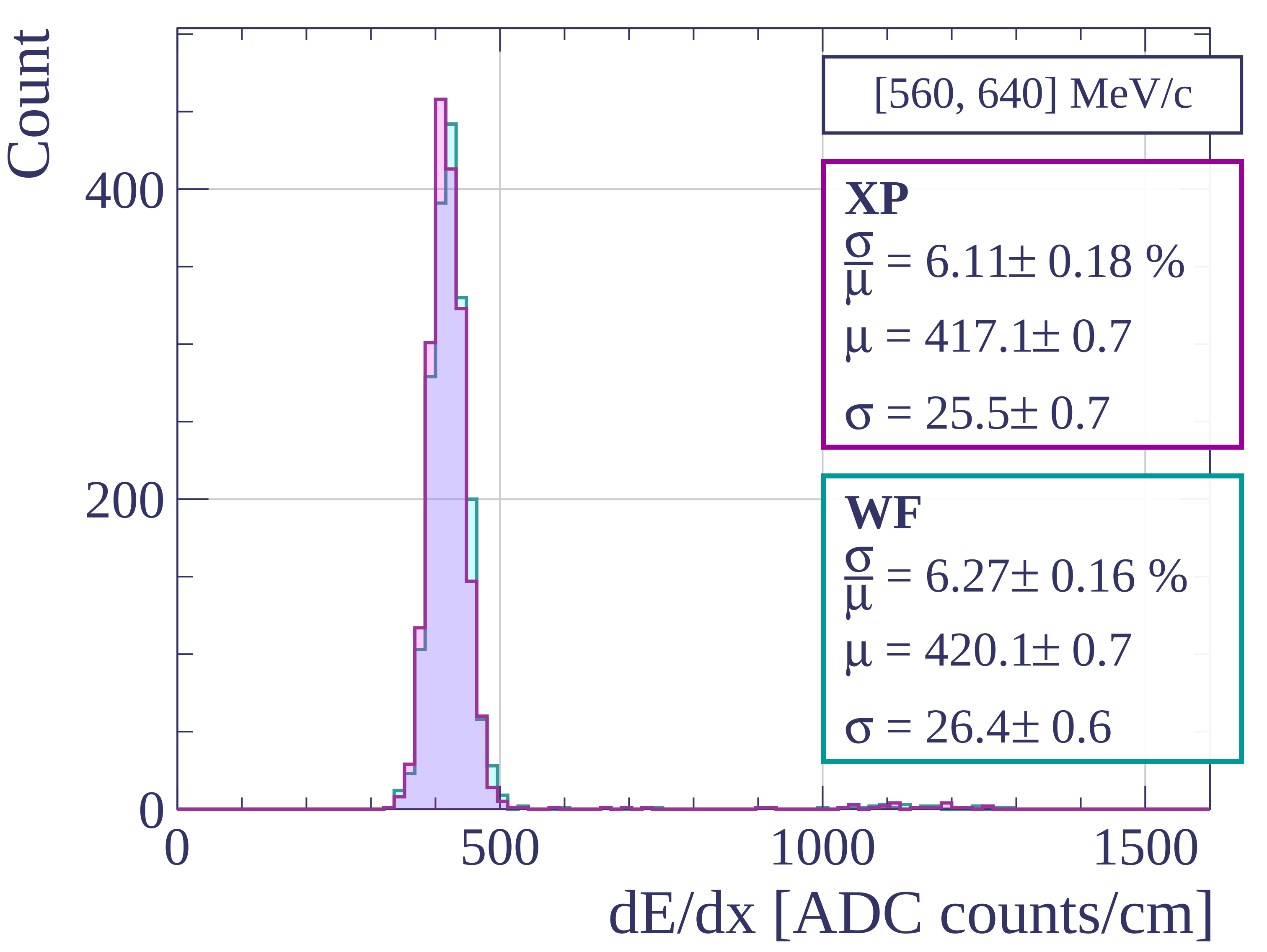


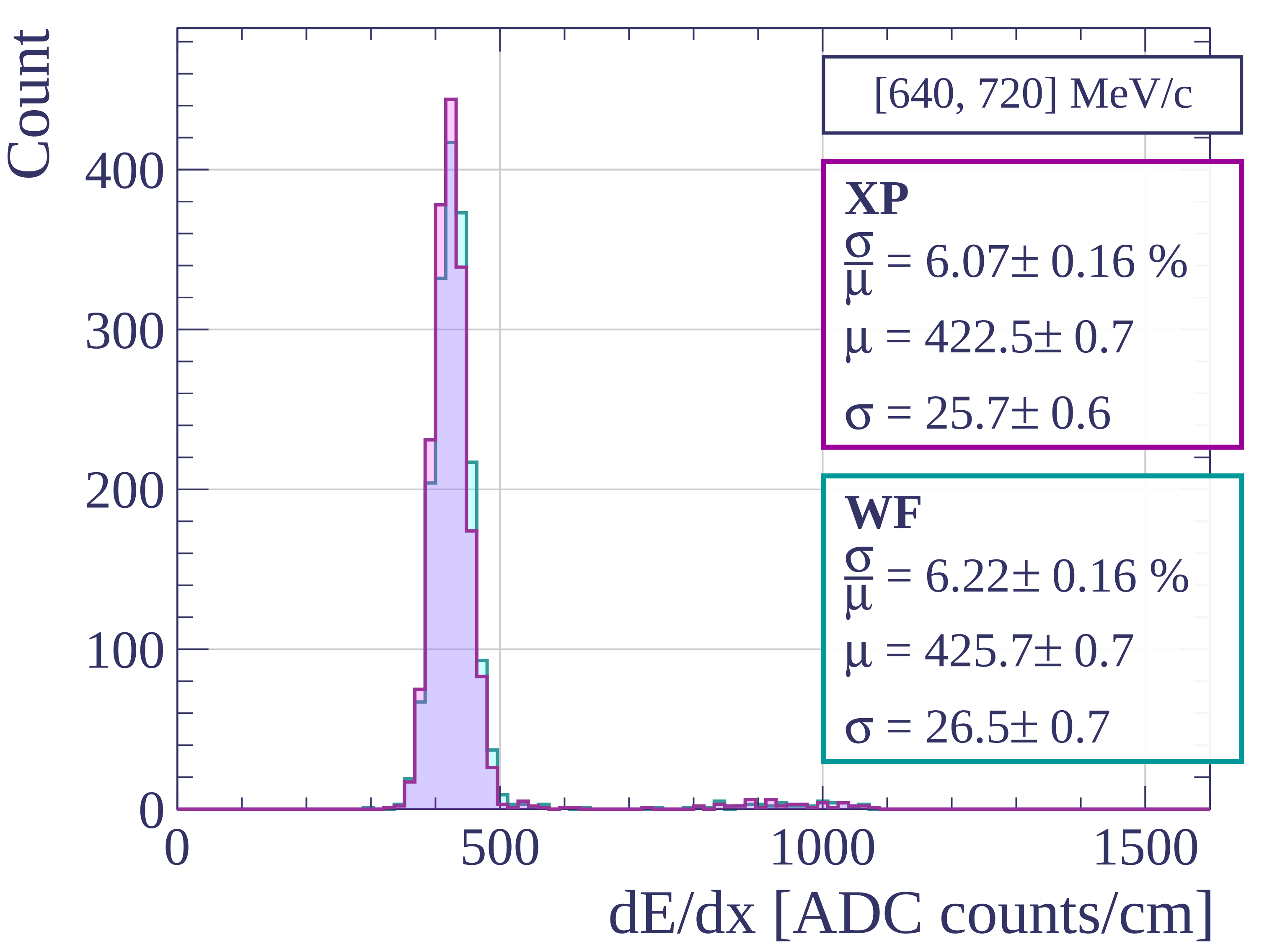
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[320, 400] MeV/c
XP
σ/μ = 6.08± 0.13 %
μ = 409.2± 0.6
σ = 24.9± 0.5
WF
σ/μ = 6.30± 0.19 %
μ = 412.5± 0.7
σ = 26.0± 0.7



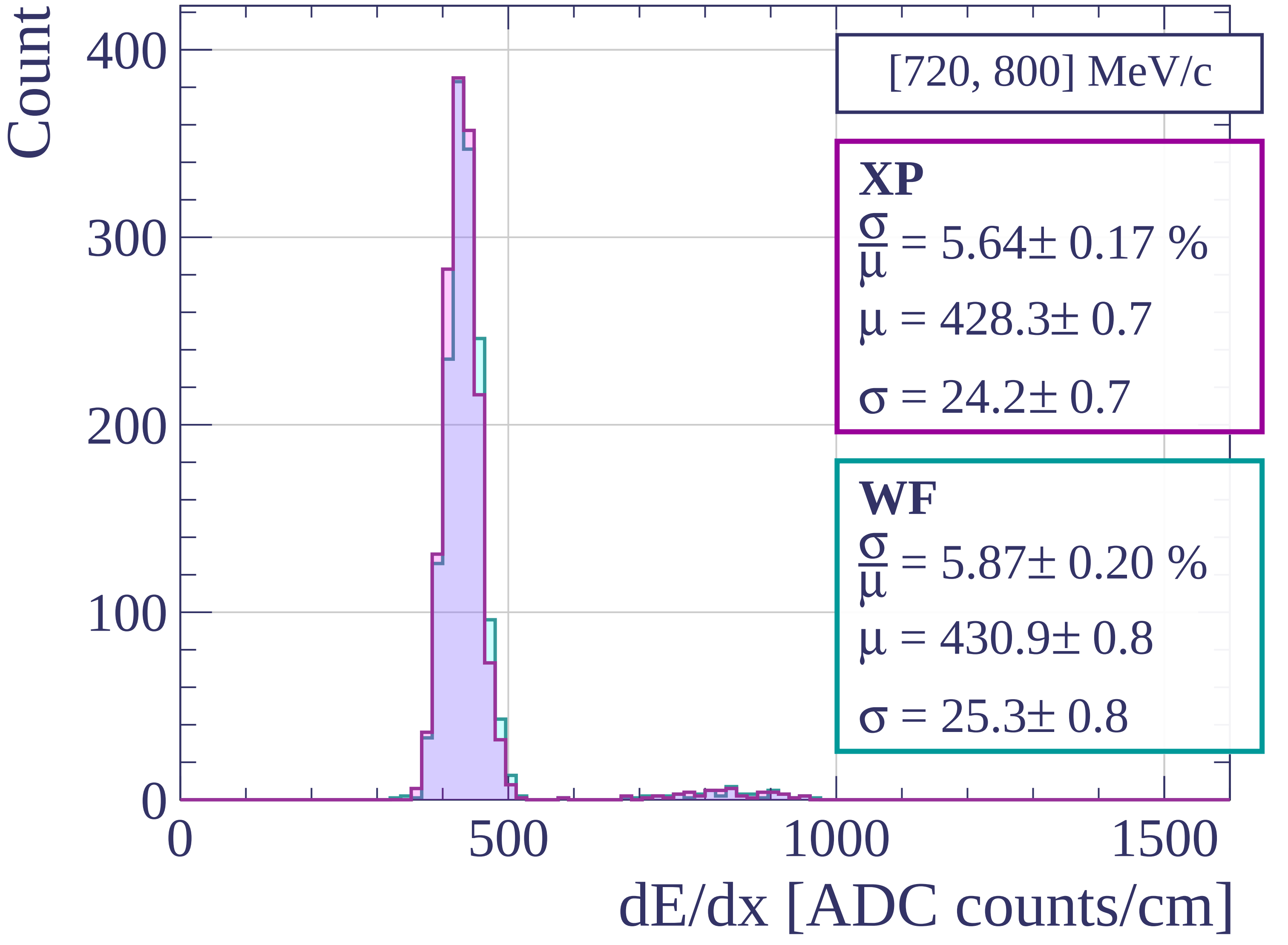
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.24± 0.19 %
μ = 410.5± 0.7
σ = 25.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 6.19± 0.18 %
μ = 413.2± 0.7
σ = 25.6± 0.7


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.25± 0.19 %
μ = 413.4± 0.7
σ = 25.8± 0.7
WF
$\frac{\sigma}{\mu}$ = 6.39± 0.20 %
μ = 417.0± 0.7
σ = 26.6± 0.8

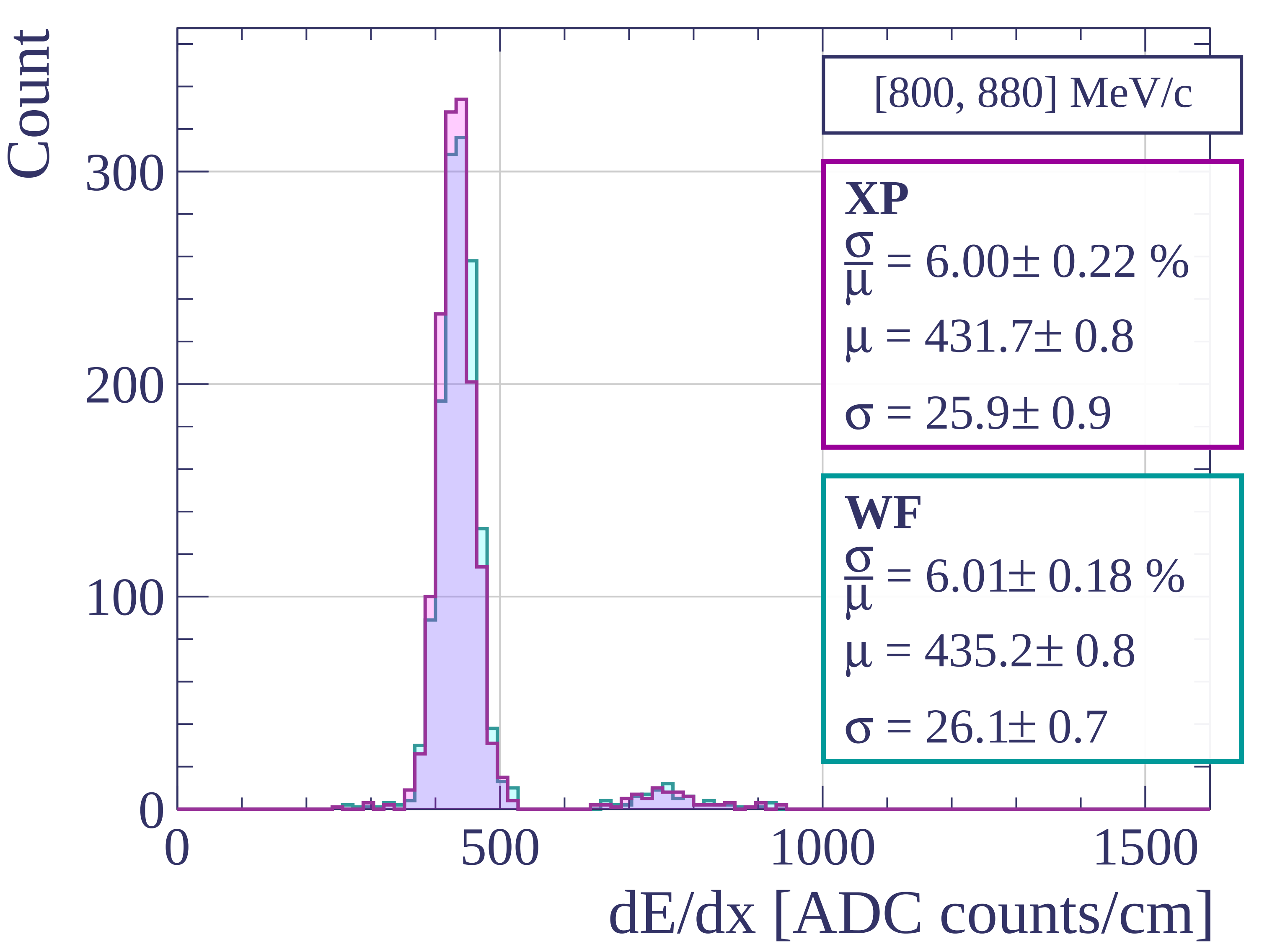

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 640] MeV/c
XP
σ/μ = 6.11± 0.18 %
μ = 417.1± 0.7
σ = 25.5± 0.7
WF
σ/μ = 6.27± 0.16 %
μ = 420.1± 0.7
σ = 26.4± 0.6



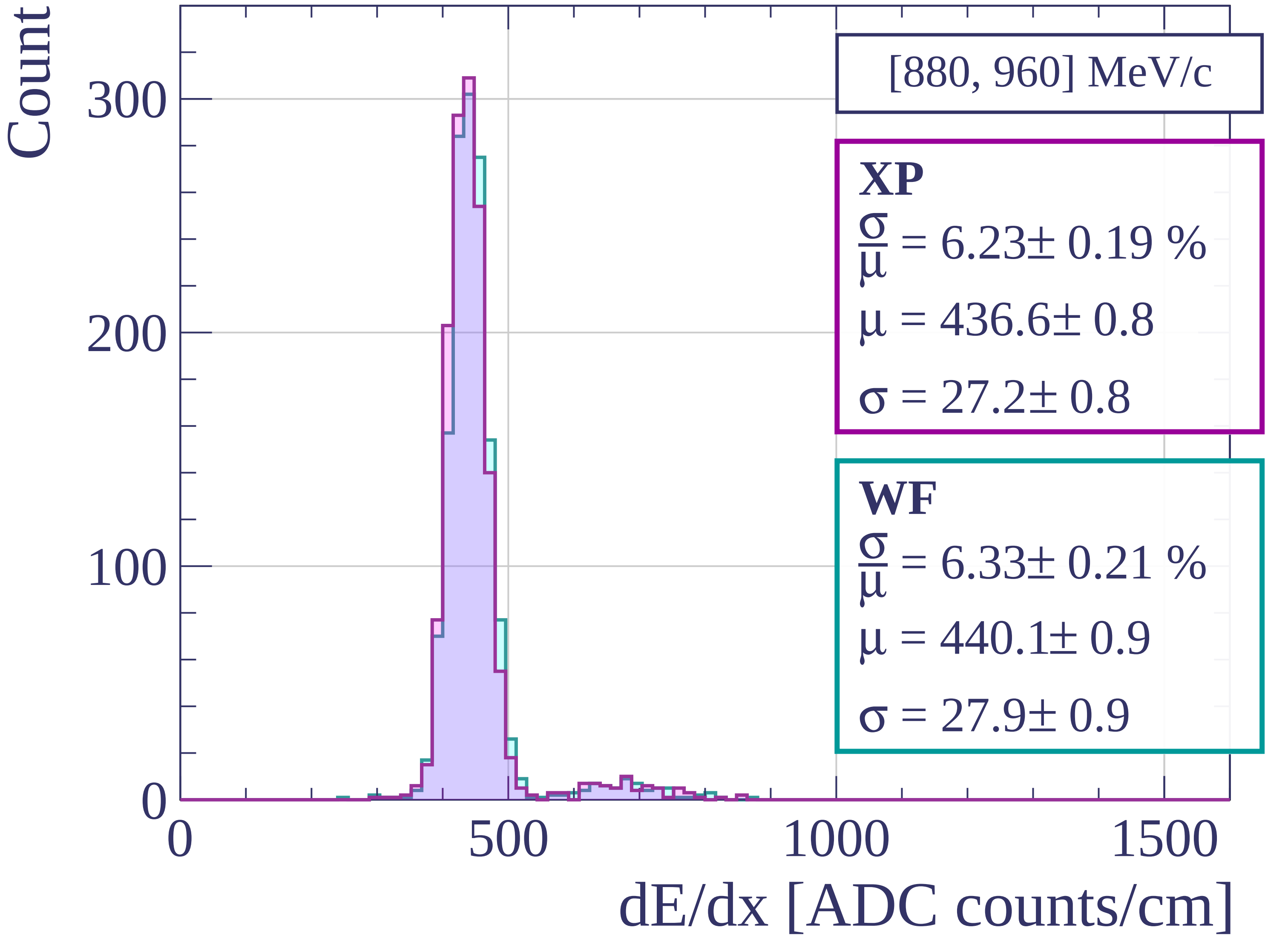
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[640, 720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.07± 0.16 %
μ = 422.5± 0.7
σ = 25.7± 0.6
WF
$\frac{\sigma}{\mu}$ = 6.22± 0.16 %
μ = 425.7± 0.7
σ = 26.5± 0.7

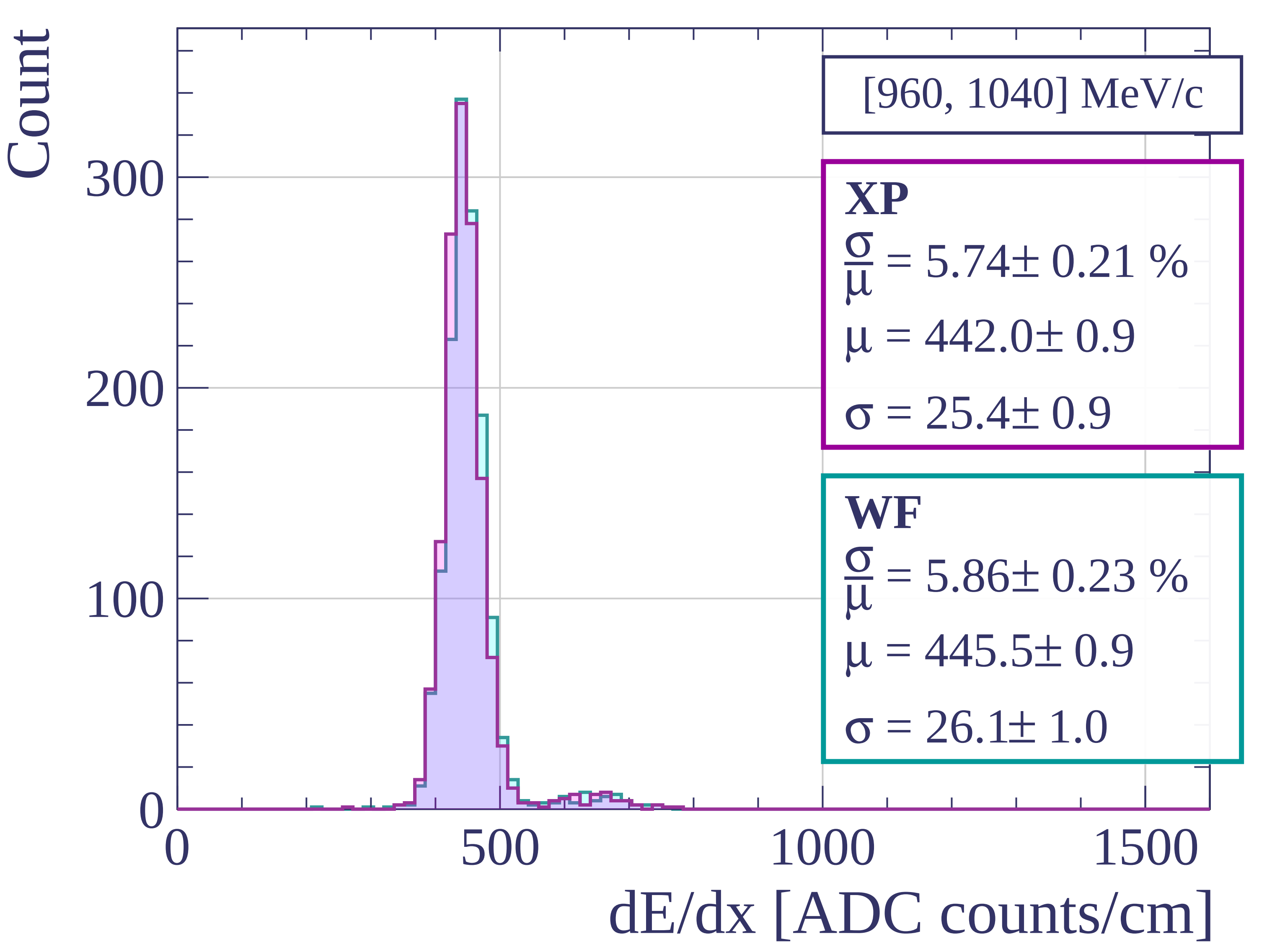


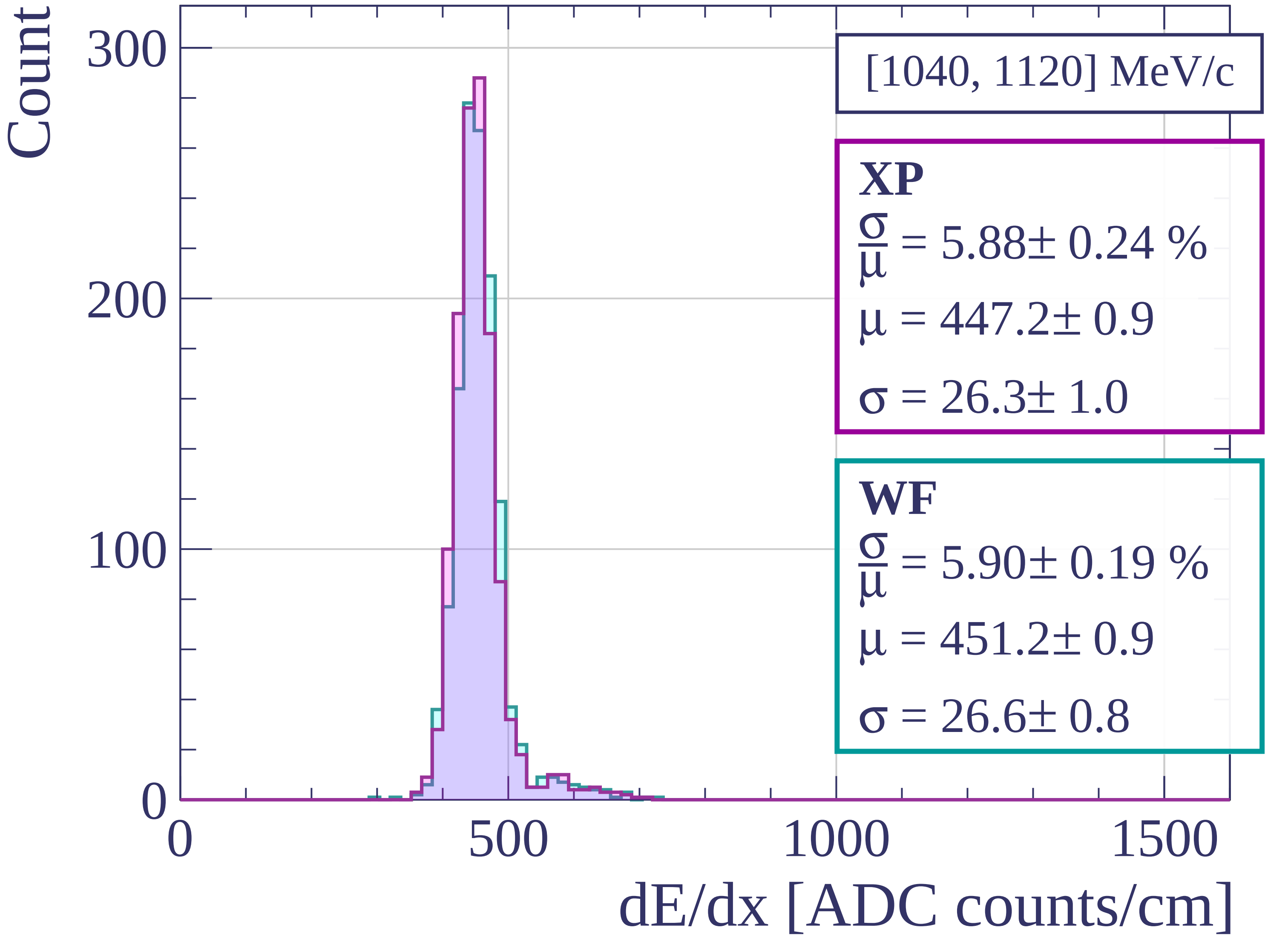
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[720, 800] MeV/c
XP
σ/μ = 5.64± 0.17 %
μ = 428.3± 0.7
σ = 24.2± 0.7
WF
σ/μ = 5.87± 0.20 %
μ = 430.9± 0.8
σ = 25.3± 0.8

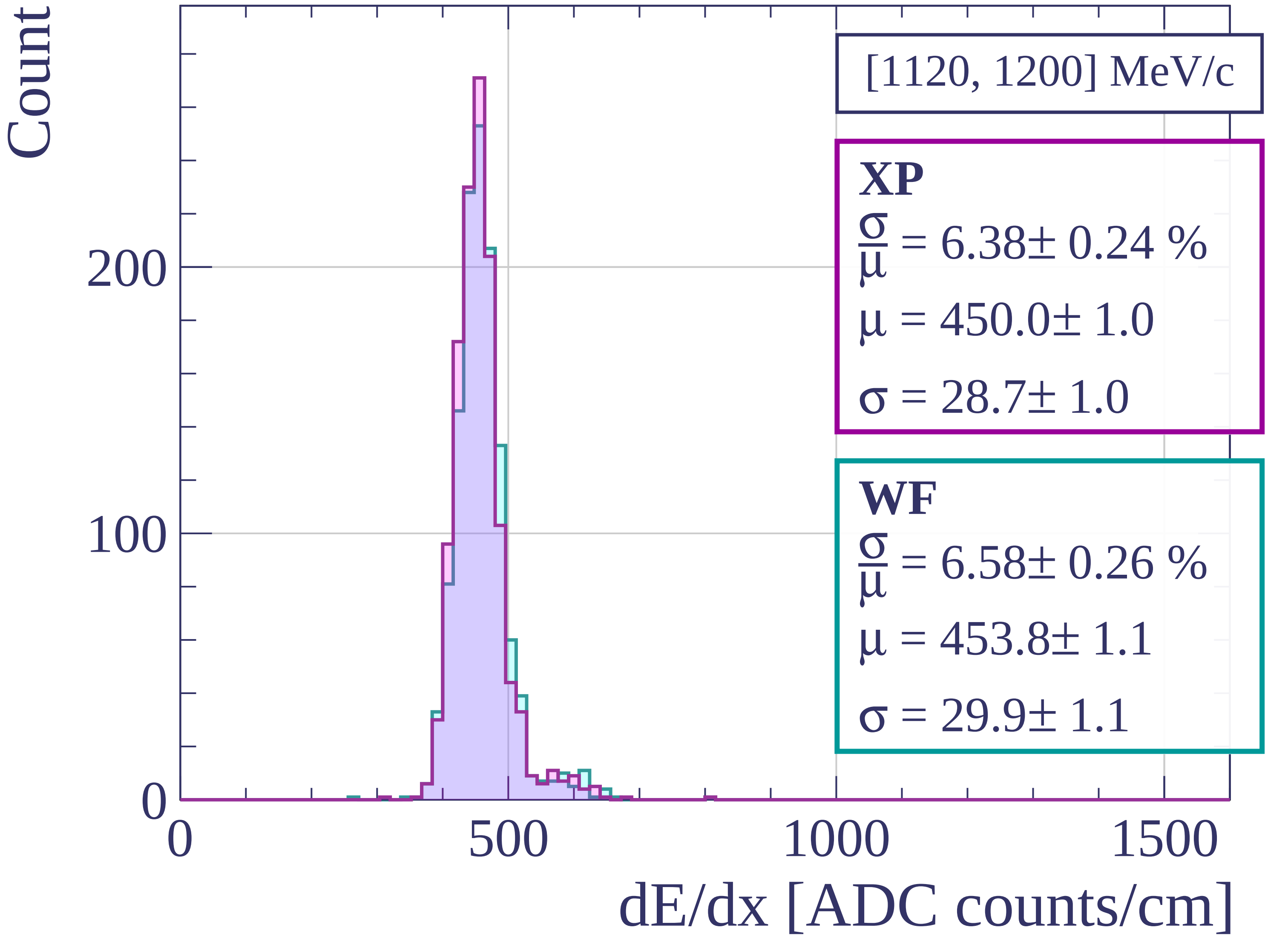


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.00± 0.22 %
μ = 431.7± 0.8
σ = 25.9± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.01± 0.18 %
μ = 435.2± 0.8
σ = 26.1± 0.7

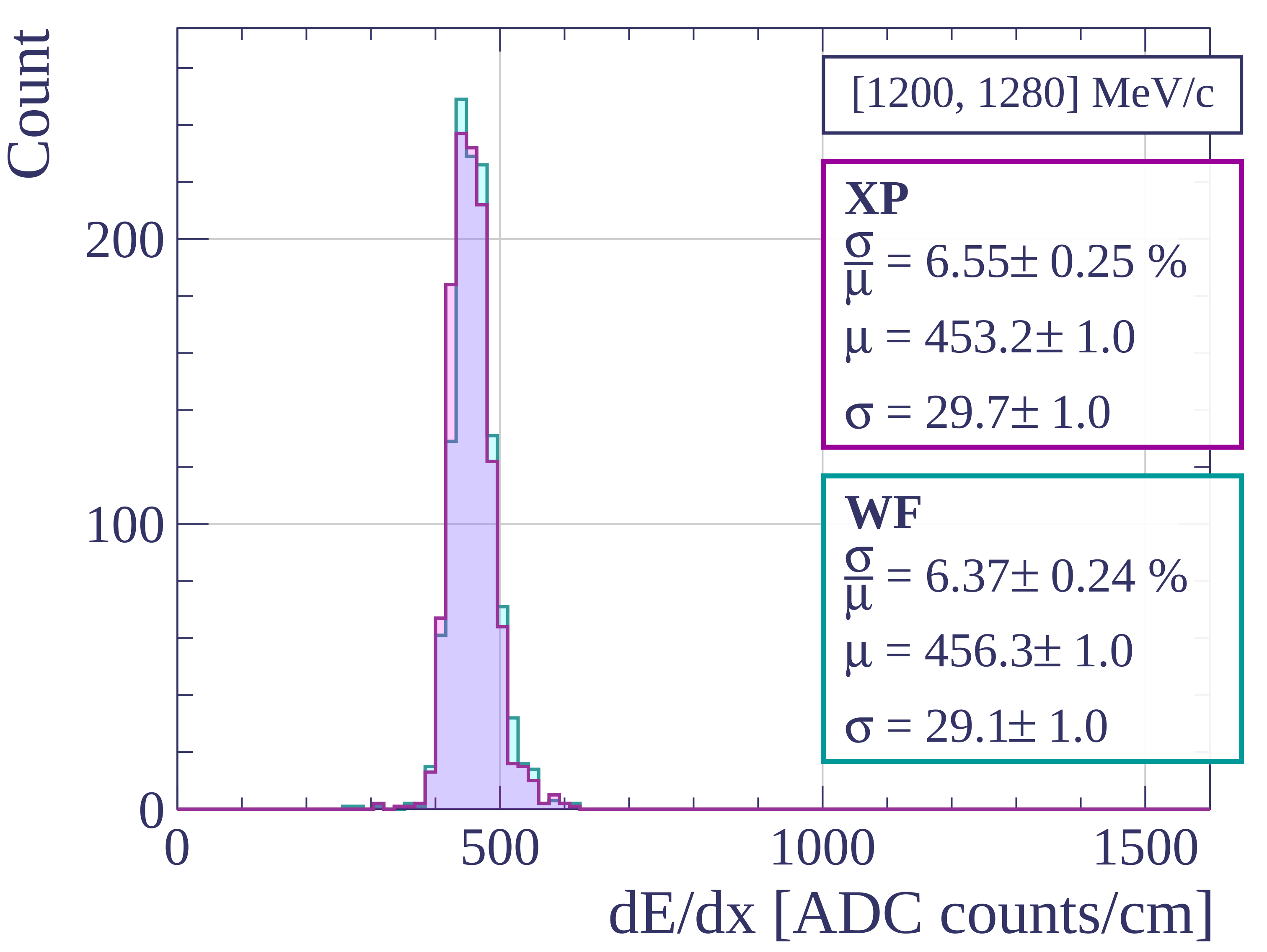

Count
dE/dx [ADC counts/cm]
[880, 960] MeV/c
XP
σ/μ = 6.23± 0.19 %
μ = 436.6± 0.8
σ = 27.2± 0.8
WF
σ/μ = 6.33± 0.21 %
μ = 440.1± 0.9
σ = 27.9± 0.9


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[960, 1040] MeV/c
XP
σ/μ = 5.74± 0.21 %
μ = 442.0± 0.9
σ = 25.4± 0.9
WF
σ/μ = 5.86± 0.23 %
μ = 445.5± 0.9
σ = 26.1± 1.0

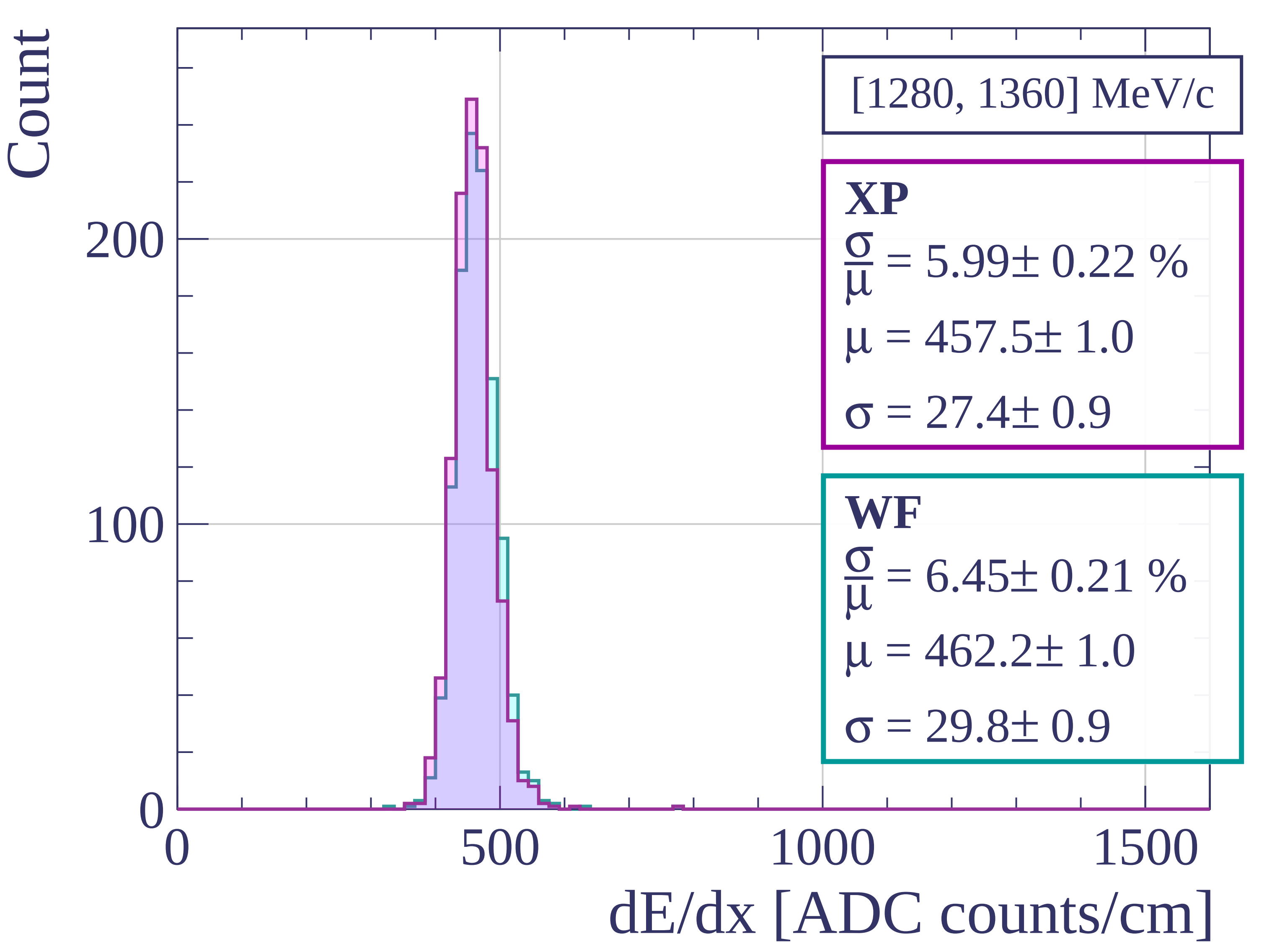

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1040, 1120] MeV/c
XP
σ/μ = 5.88± 0.24 %
μ = 447.2± 0.9
σ = 26.3± 1.0
WF
σ/μ = 5.90± 0.19 %
μ = 451.2± 0.9
σ = 26.6± 0.8

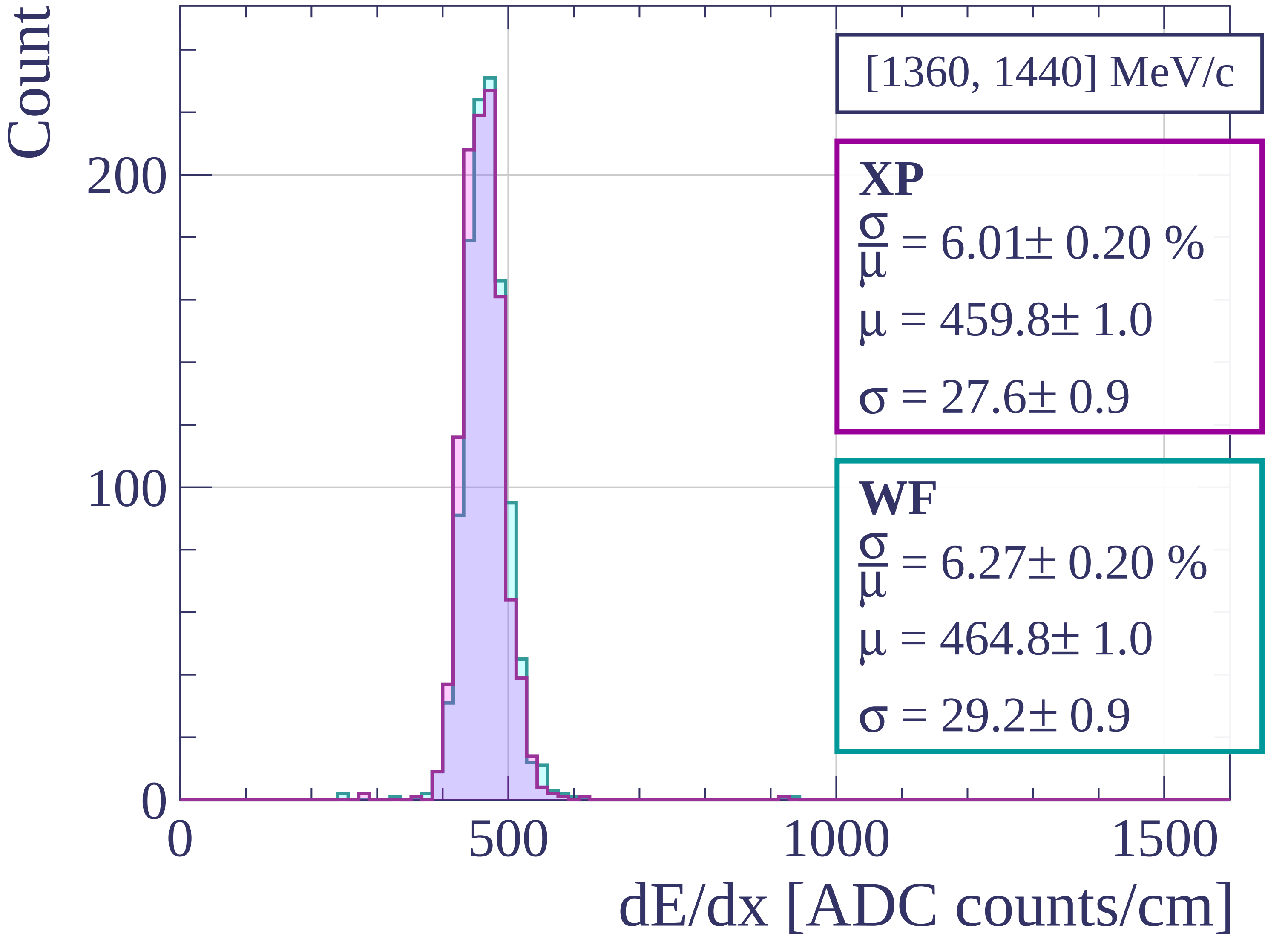


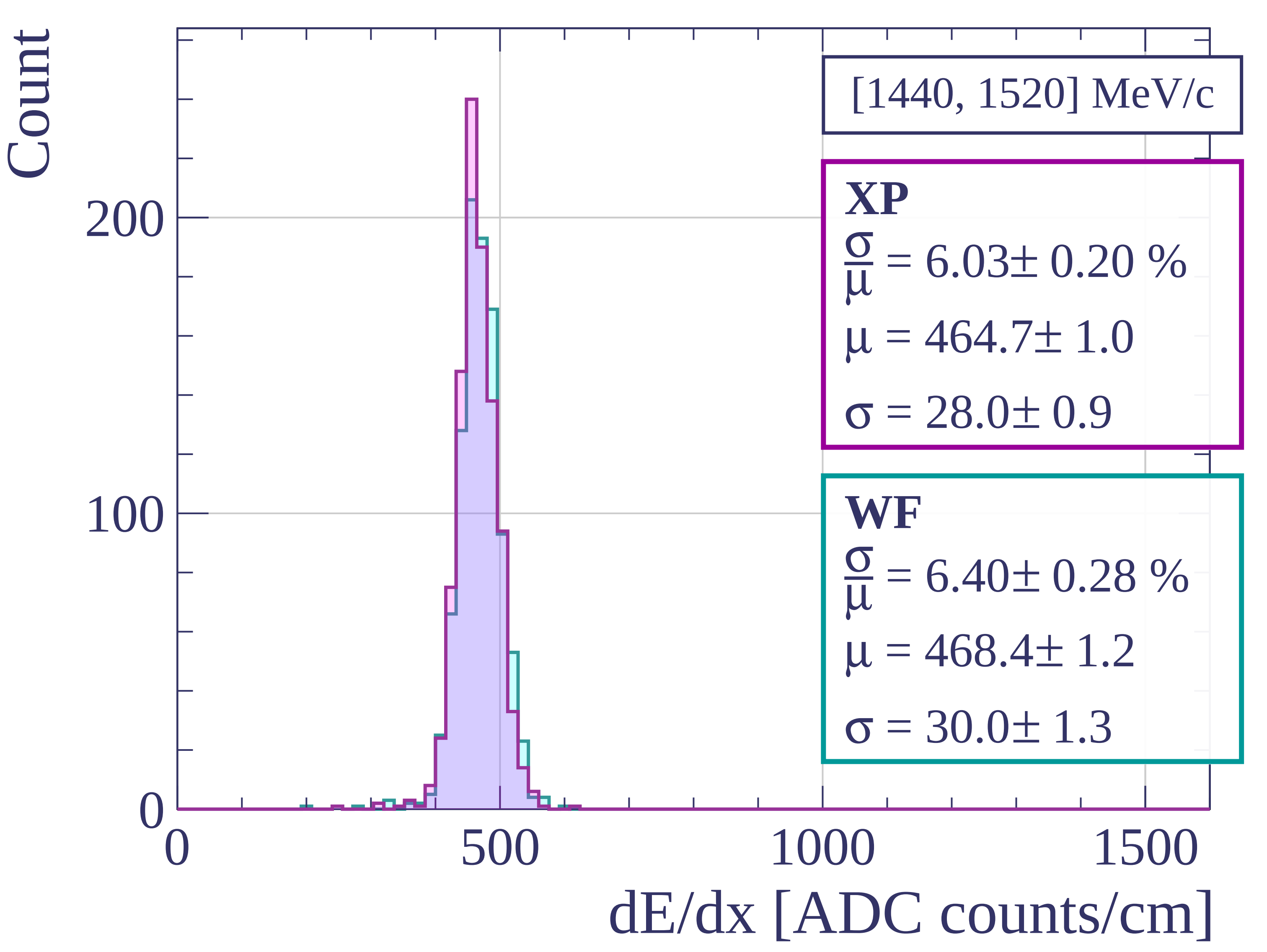
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1120, 1200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.38± 0.24 %
μ = 450.0± 1.0
σ = 28.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.58± 0.26 %
μ = 453.8± 1.1
σ = 29.9± 1.1

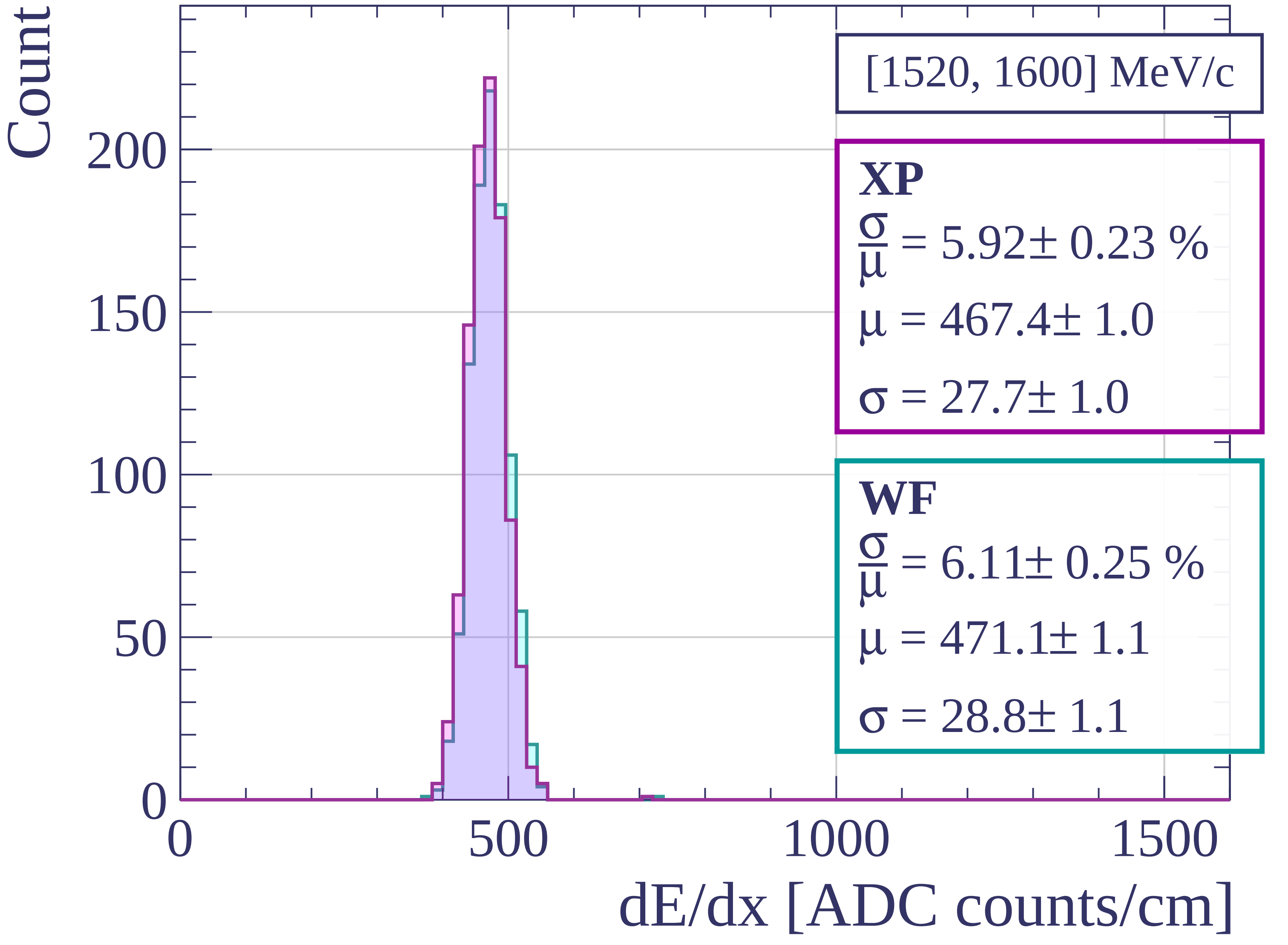


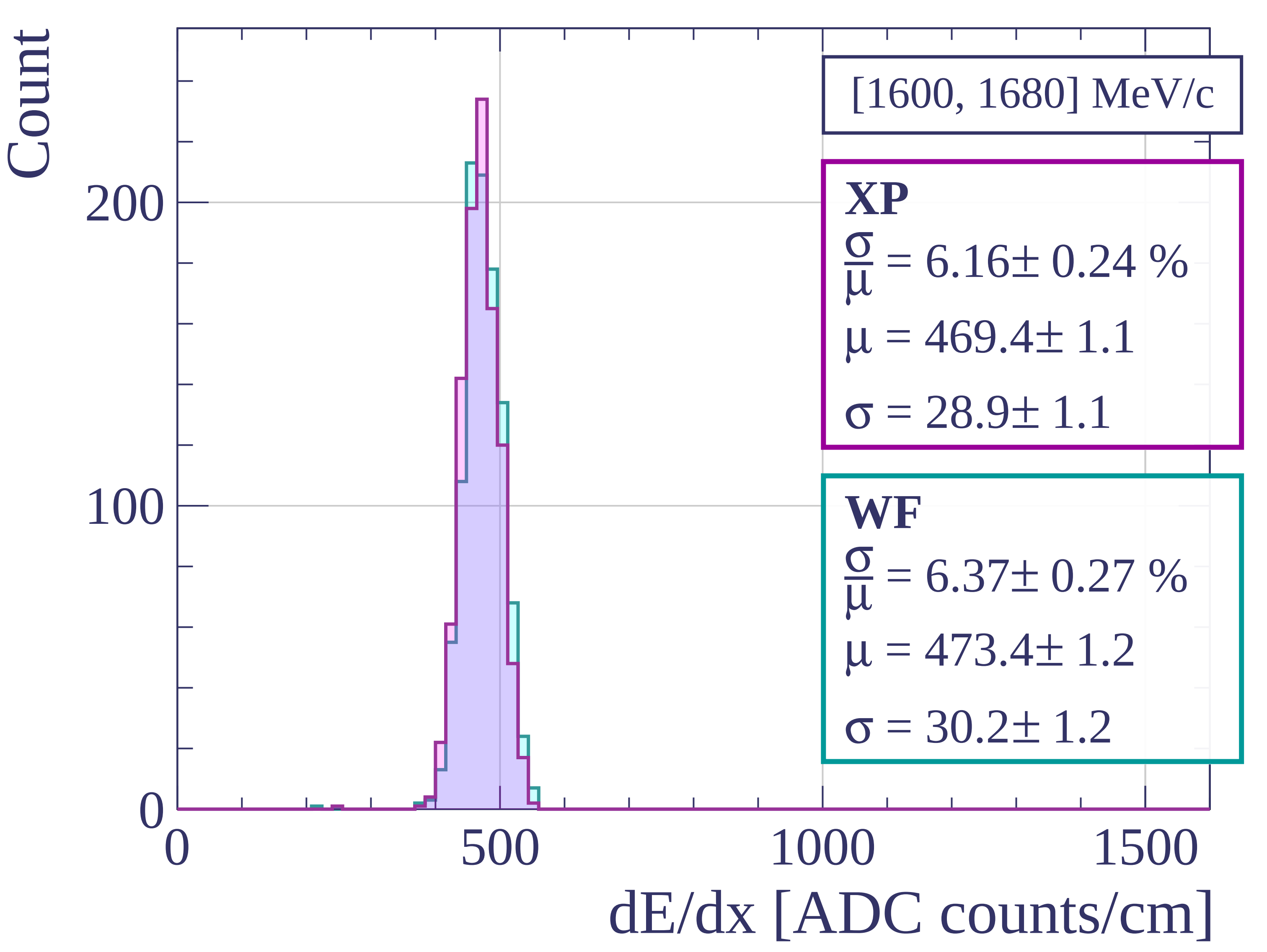
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1200, 1280] MeV/c
XP
σ/μ = 6.55± 0.25 %
μ = 453.2± 1.0
σ = 29.7± 1.0
WF
σ/μ = 6.37± 0.24 %
μ = 456.3± 1.0
σ = 29.1± 1.0


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1280, 1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.99± 0.22 %
μ = 457.5± 1.0
σ = 27.4± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.45± 0.21 %
μ = 462.2± 1.0
σ = 29.8± 0.9

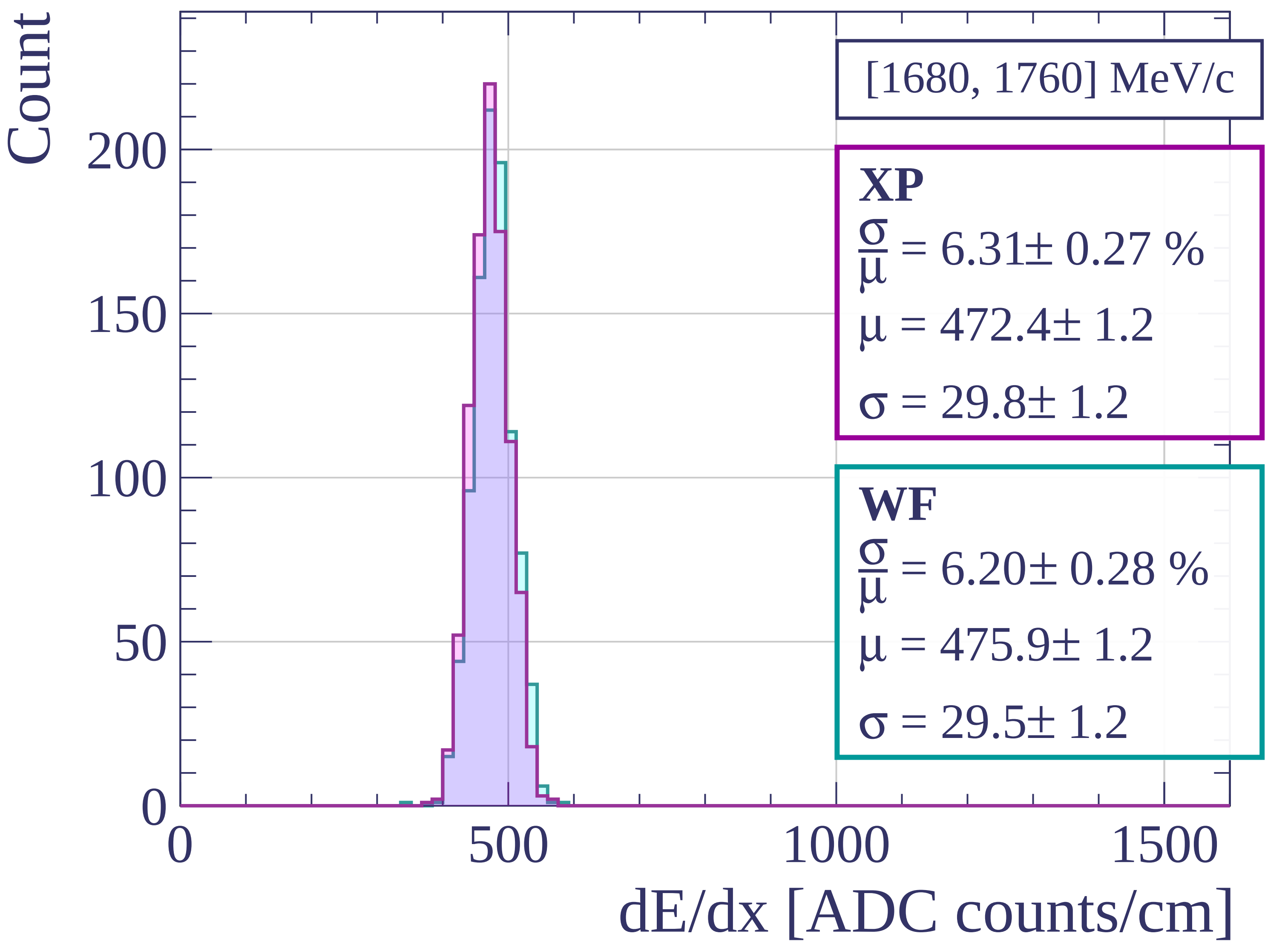

Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1360, 1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.01± 0.20 %
μ = 459.8± 1.0
σ = 27.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.27± 0.20 %
μ = 464.8± 1.0
σ = 29.2± 0.9


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1440, 1520] MeV/c
XP
σ/μ = 6.03± 0.20 %
μ = 464.7± 1.0
σ = 28.0± 0.9
WF
σ/μ = 6.40± 0.28 %
μ = 468.4± 1.2
σ = 30.0± 1.3

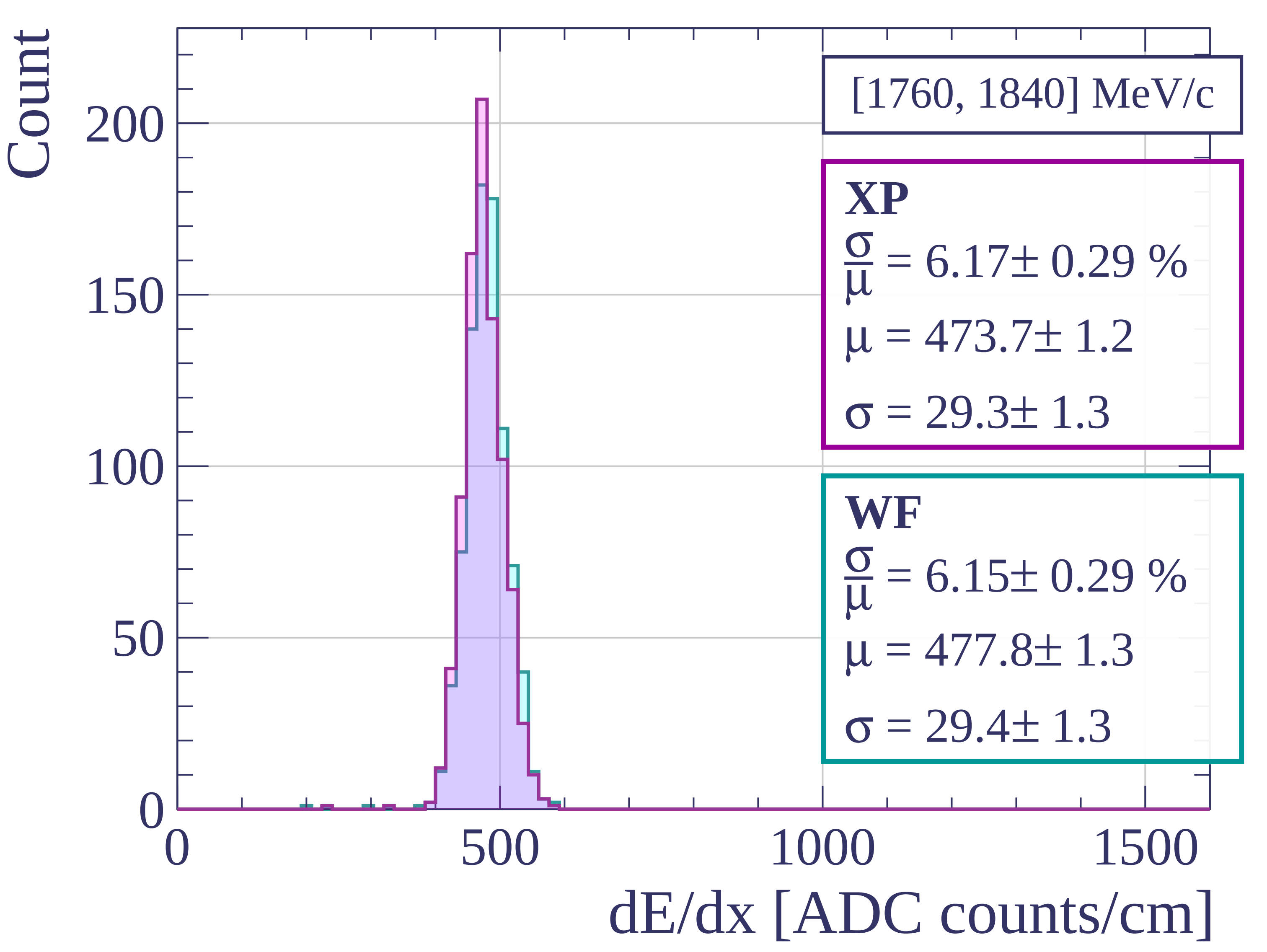

Count
0
50
100
150
200
0
500
1000
1500
dE/dx [ADC counts/cm]
[1520, 1600] MeV/c
XP
σ/μ = 5.92± 0.23 %
μ = 467.4± 1.0
σ = 27.7± 1.0
WF
σ/μ = 6.11± 0.25 %
μ = 471.1± 1.1
σ = 28.8± 1.1

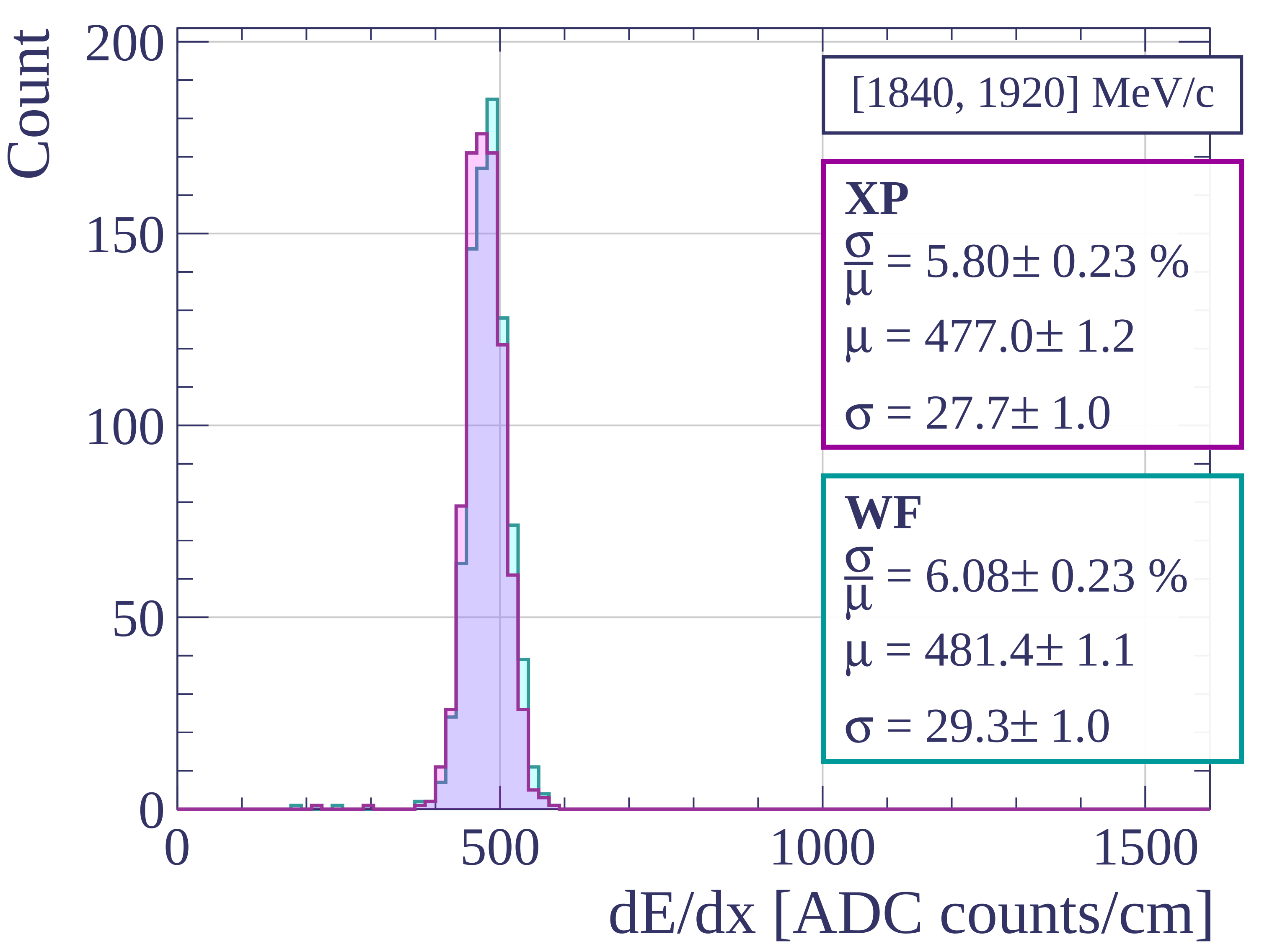


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.16± 0.24 %
μ = 469.4± 1.1
σ = 28.9± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.37± 0.27 %
μ = 473.4± 1.2
σ = 30.2± 1.2

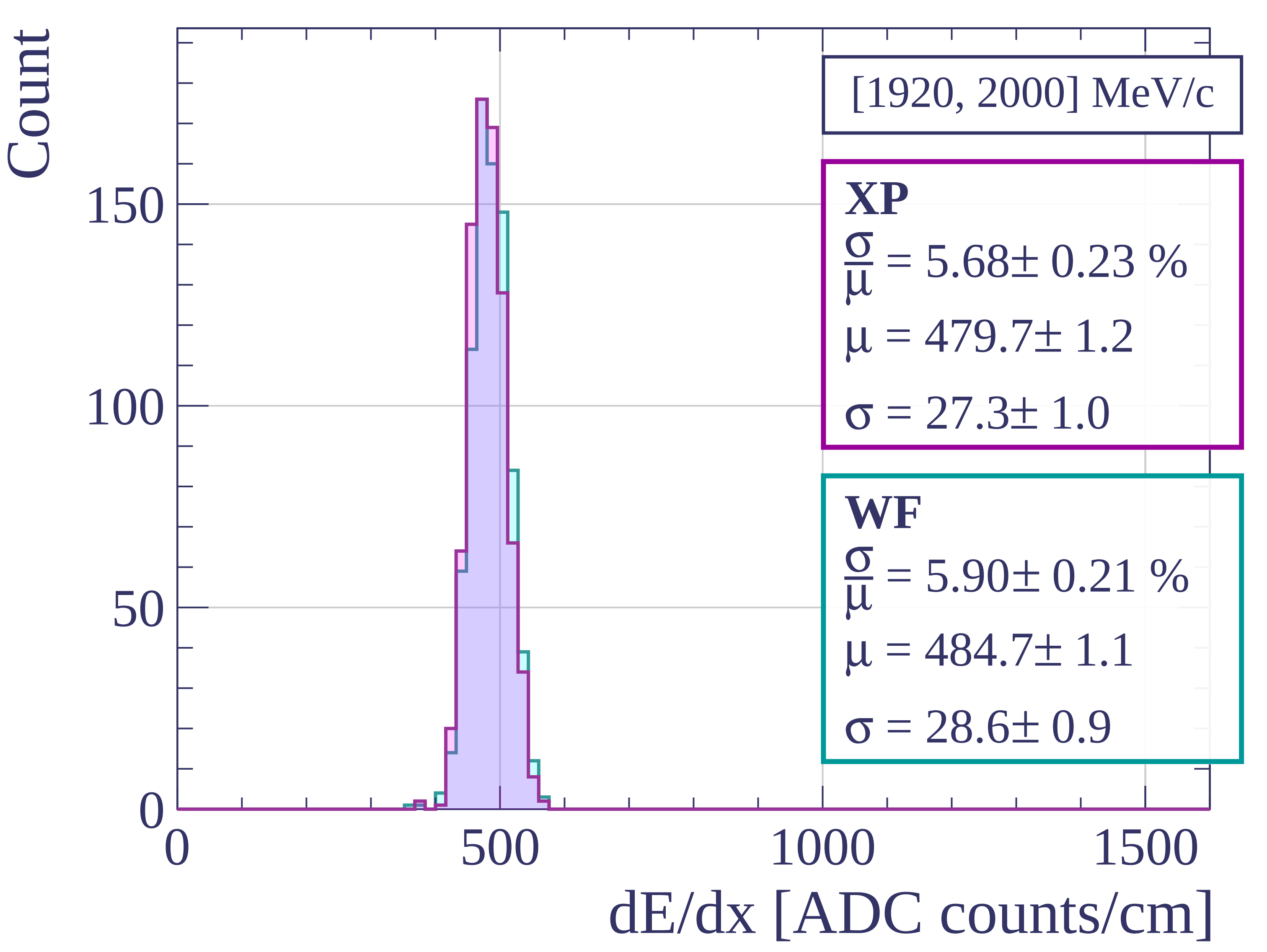


Count
dE/dx [ADC counts/cm]
0
50
100
150
200
0
500
1000
1500
[1680, 1760] MeV/c
XP
σ/μ = 6.31± 0.27 %
μ = 472.4± 1.2
σ = 29.8± 1.2
WF
σ/μ = 6.20± 0.28 %
μ = 475.9± 1.2
σ = 29.5± 1.2

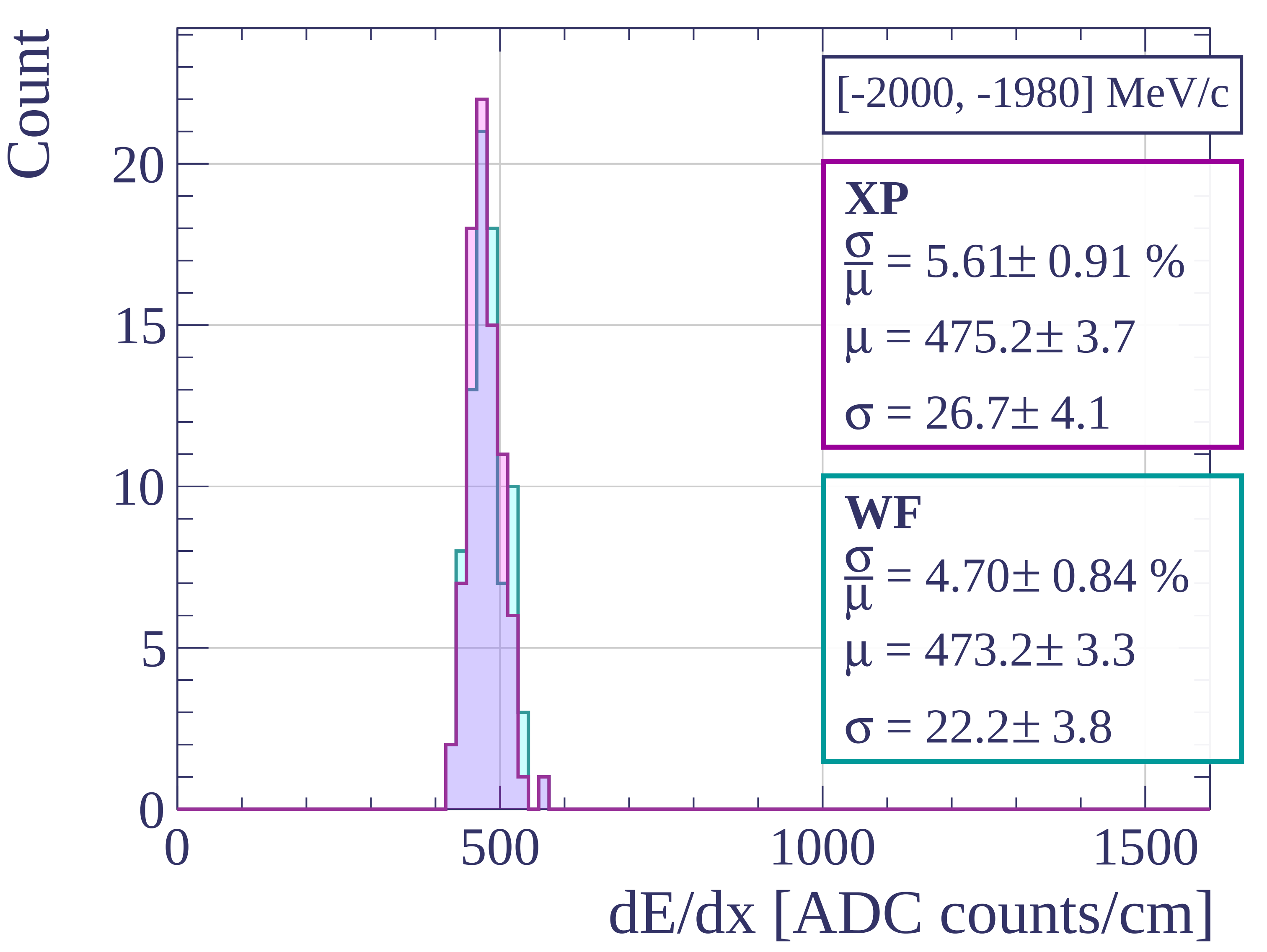

Count
dE/dx [ADC counts/cm]
[1760, 1840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.17± 0.29 %
μ = 473.7± 1.2
σ = 29.3± 1.3
WF
$\frac{\sigma}{\mu}$ = 6.15± 0.29 %
μ = 477.8± 1.3
σ = 29.4± 1.3
0
50
100
150
200
0
500
1000
1500

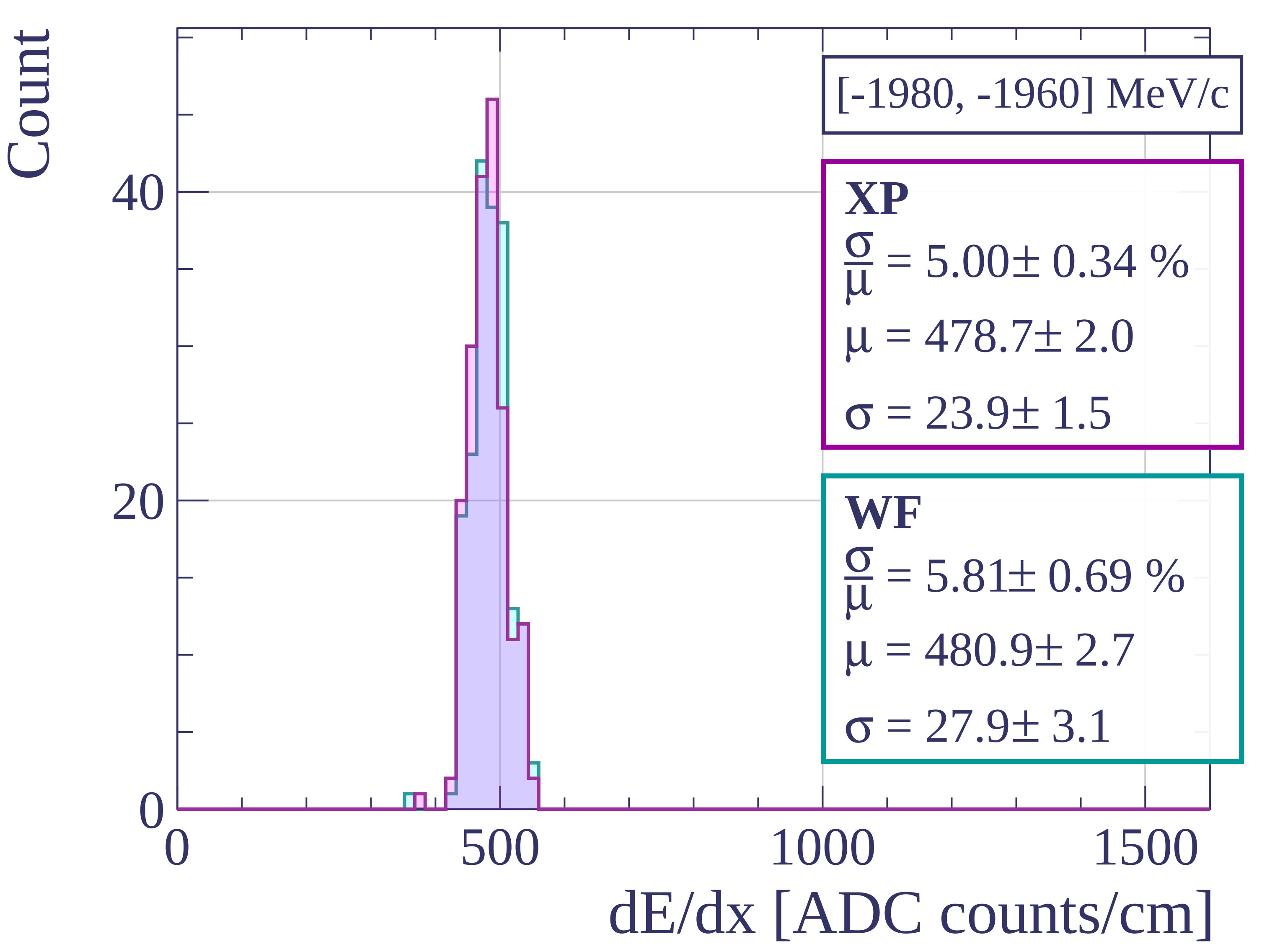


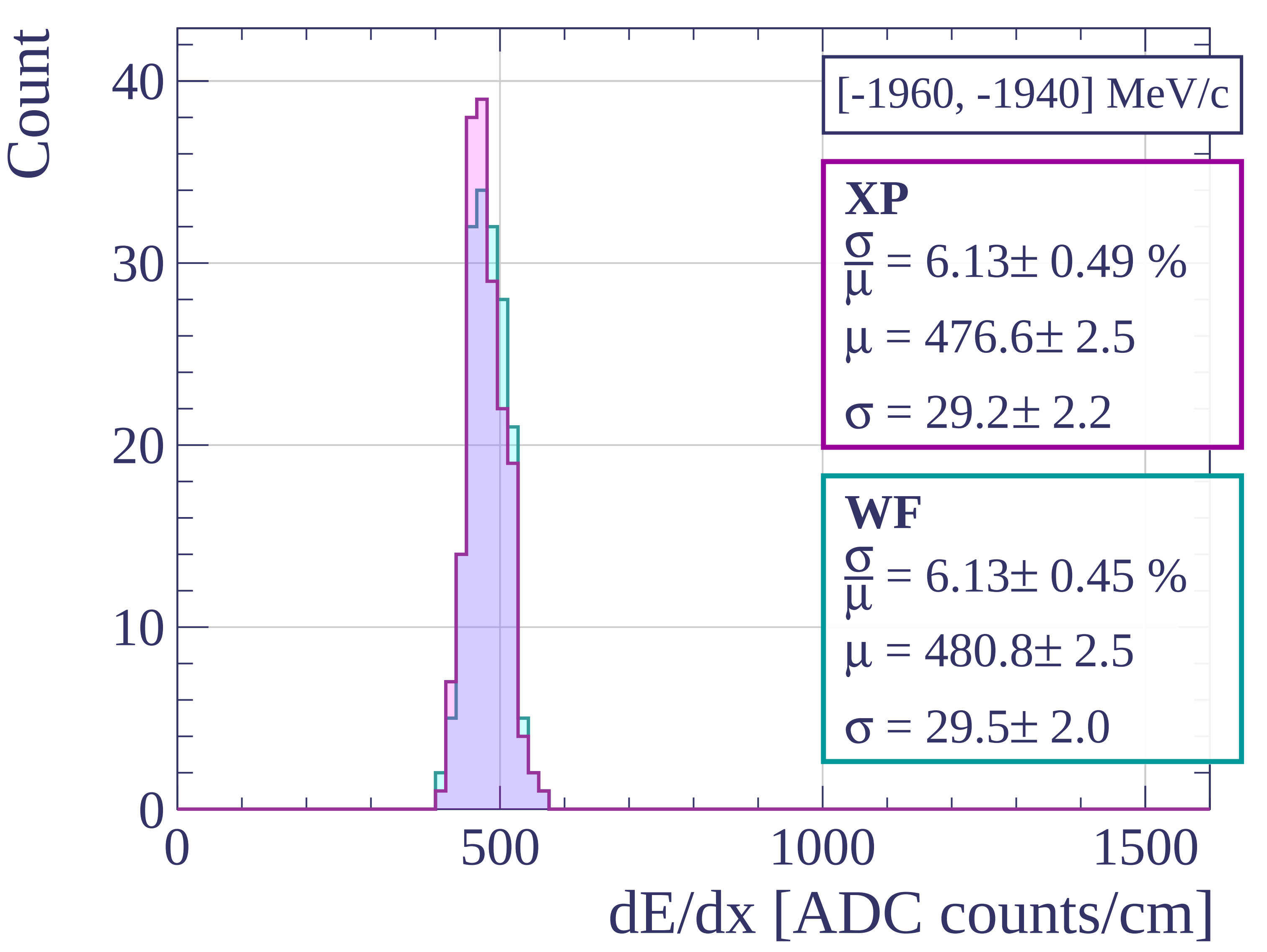
Count
dE/dx [ADC counts/cm]
[1840, 1920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.80± 0.23 %
μ = 477.0± 1.2
σ = 27.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.08± 0.23 %
μ = 481.4± 1.1
σ = 29.3± 1.0

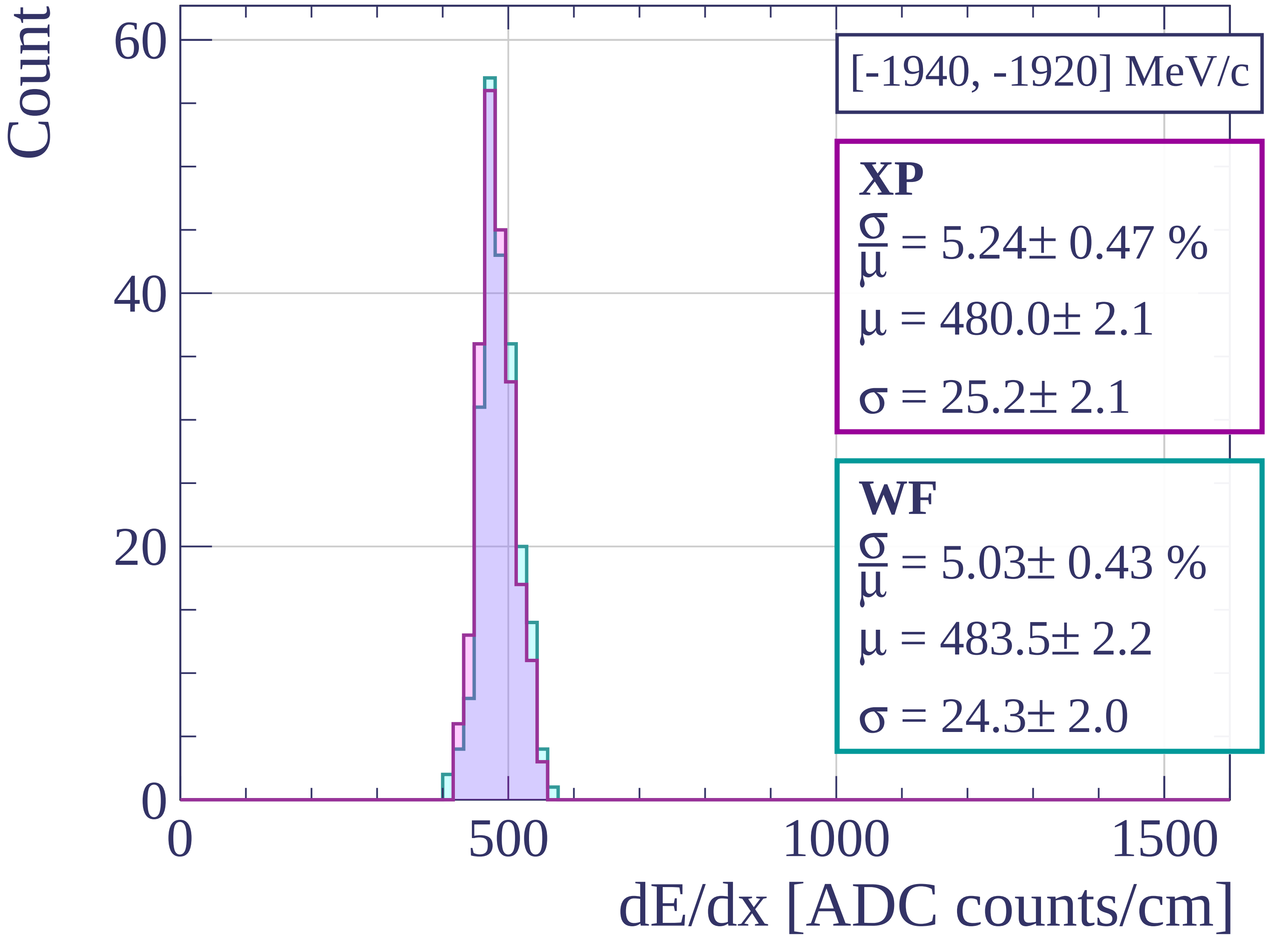


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1920, 2000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.68± 0.23 %
μ = 479.7± 1.2
σ = 27.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 5.90± 0.21 %
μ = 484.7± 1.1
σ = 28.6± 0.9

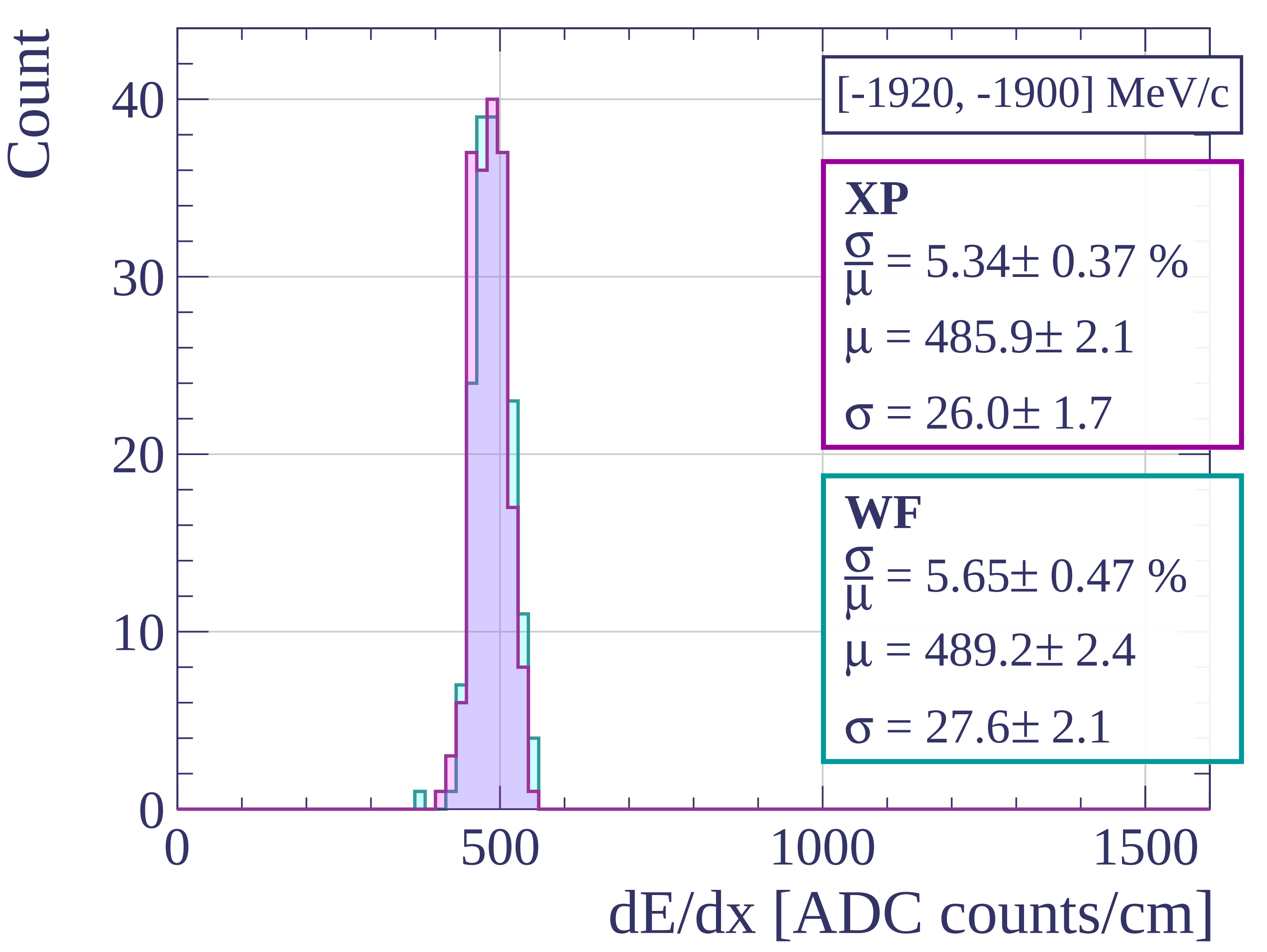

Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
5
10
15
20
[-2000, -1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.61± 0.91 %
μ = 475.2± 3.7
σ = 26.7± 4.1
WF
$\frac{\sigma}{\mu}$ = 4.70± 0.84 %
μ = 473.2± 3.3
σ = 22.2± 3.8


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1980, -1960] MeV/c
XP
σ/μ = 5.00± 0.34 %
μ = 478.7± 2.0
σ = 23.9± 1.5
WF
σ/μ = 5.81± 0.69 %
μ = 480.9± 2.7
σ = 27.9± 3.1

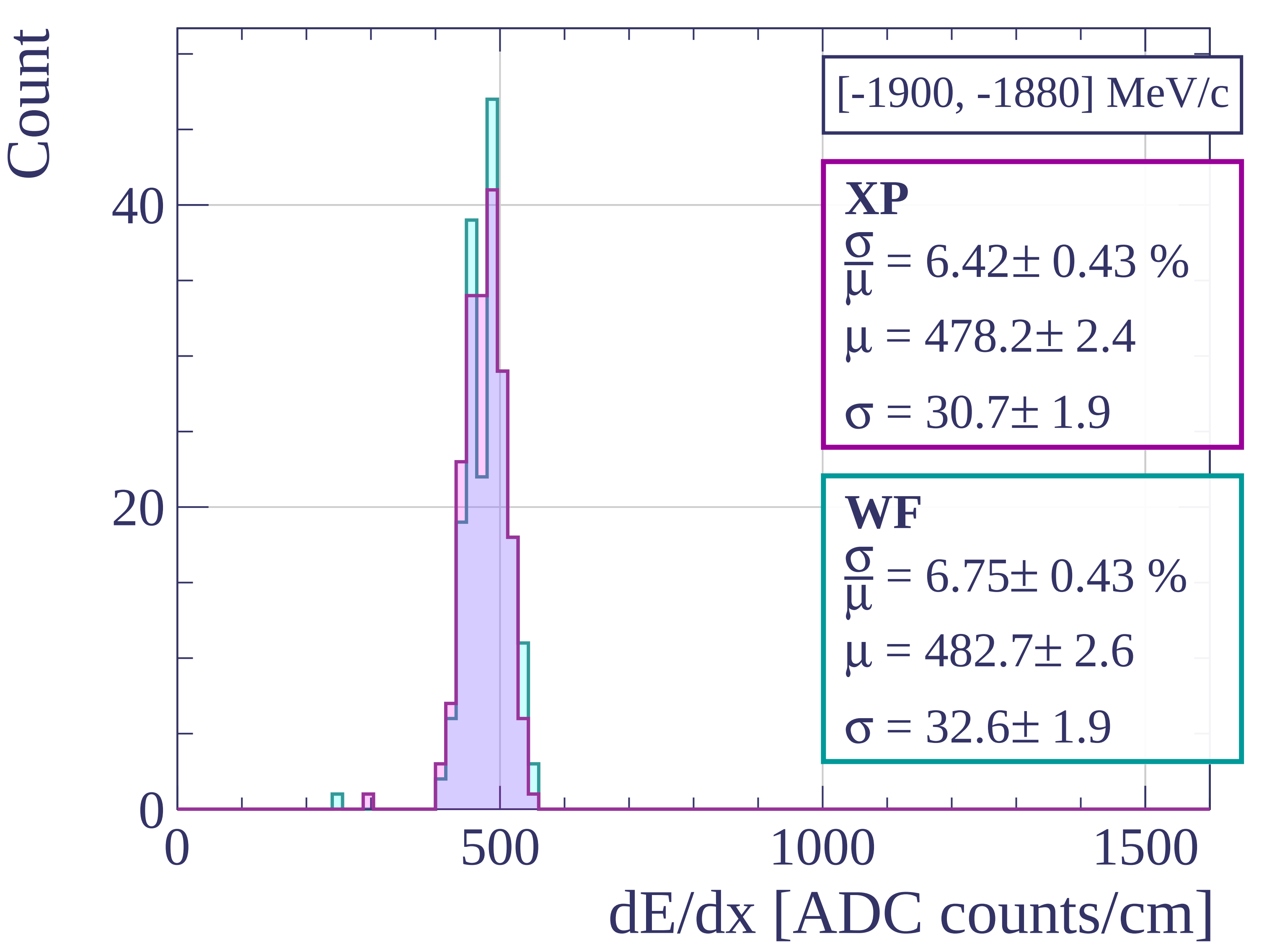

Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1960, -1940] MeV/c
XP
σ/μ = 6.13± 0.49 %
μ = 476.6± 2.5
σ = 29.2± 2.2
WF
σ/μ = 6.13± 0.45 %
μ = 480.8± 2.5
σ = 29.5± 2.0



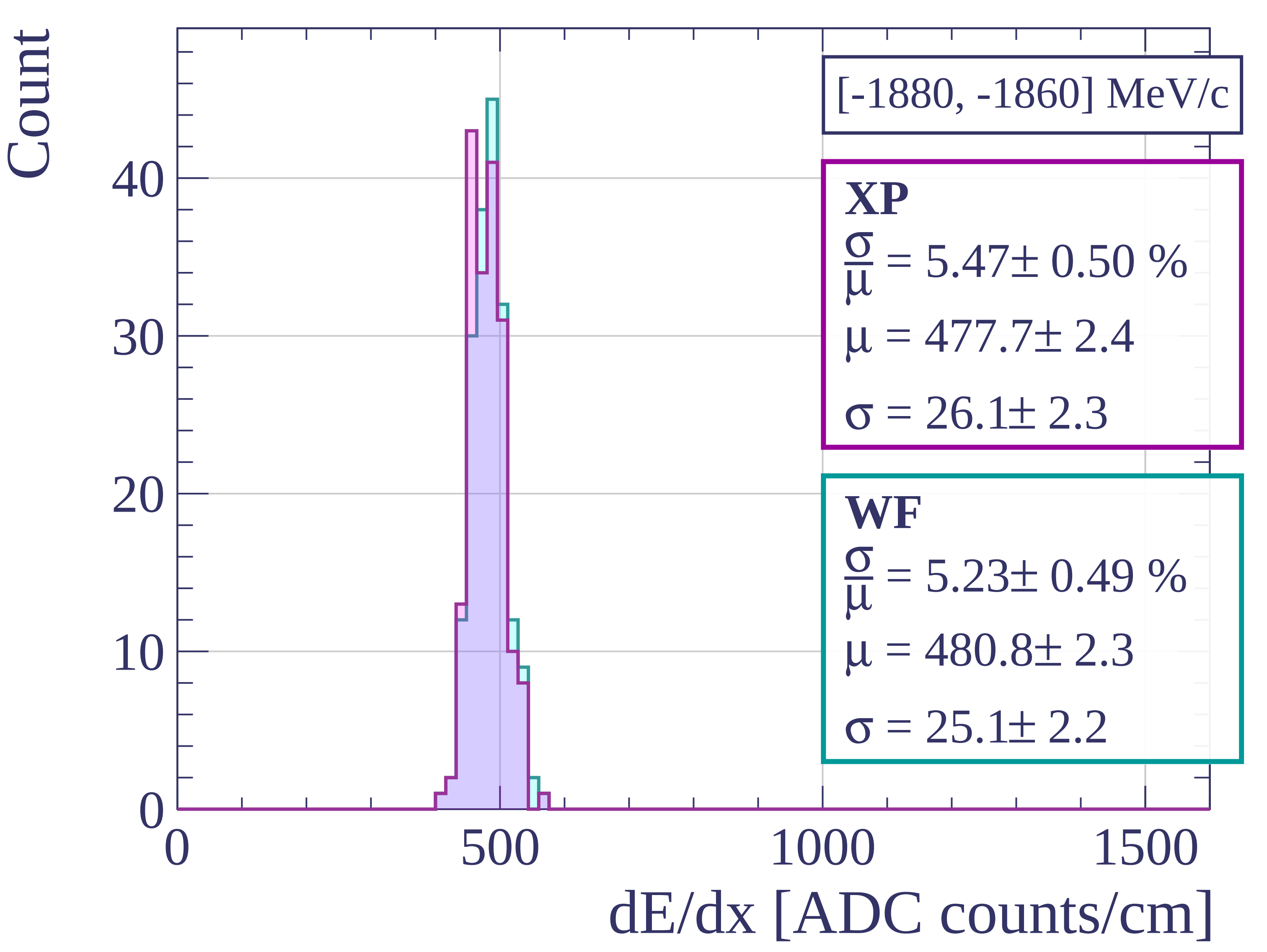
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1940, -1920] MeV/c
XP
σ/μ = 5.24± 0.47 %
μ = 480.0± 2.1
σ = 25.2± 2.1
WF
σ/μ = 5.03± 0.43 %
μ = 483.5± 2.2
σ = 24.3± 2.0



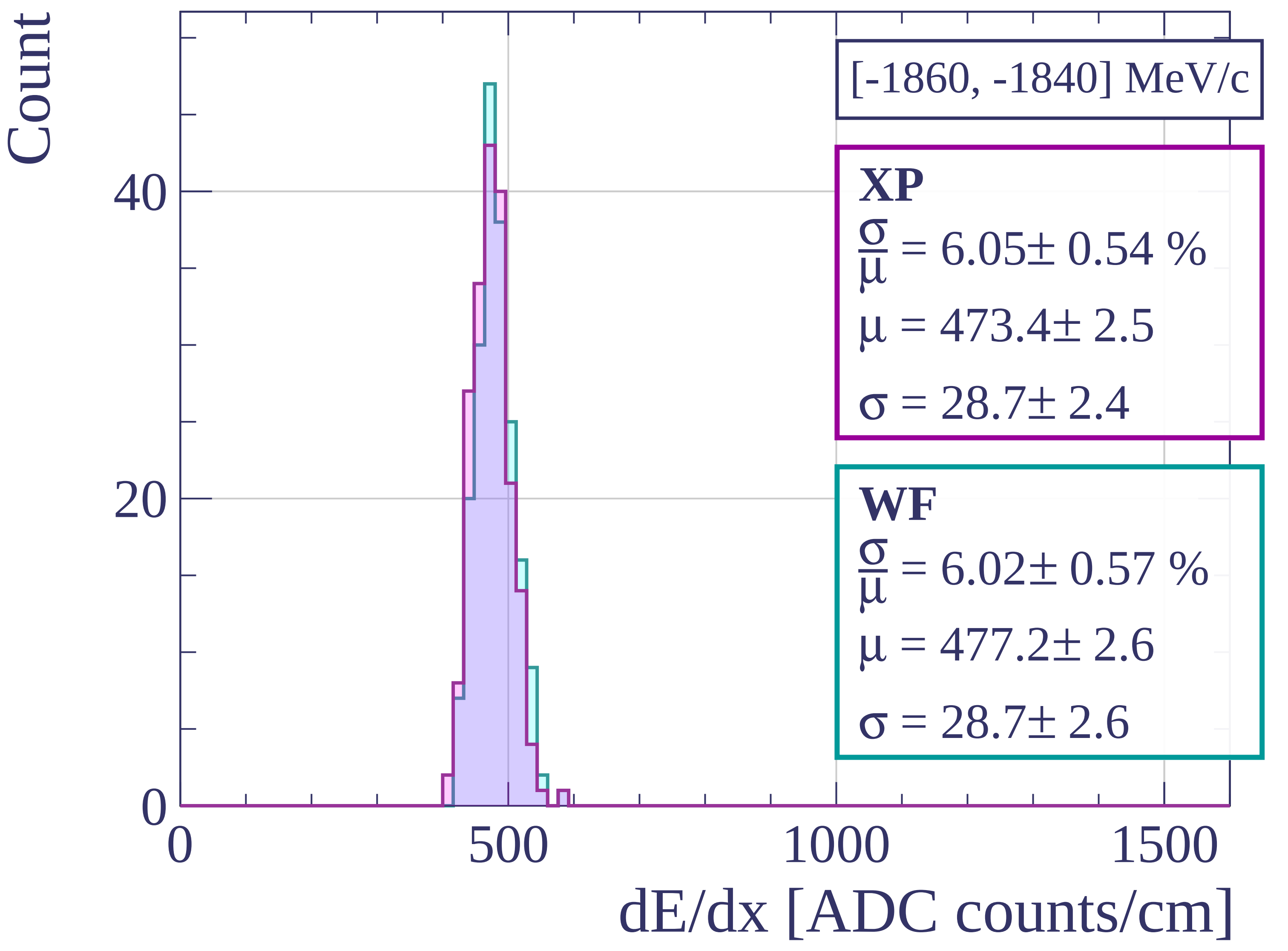
Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1920, -1900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.34± 0.37 %
μ = 485.9± 2.1
σ = 26.0± 1.7
WF
$\frac{\sigma}{\mu}$ = 5.65± 0.47 %
μ = 489.2± 2.4
σ = 27.6± 2.1



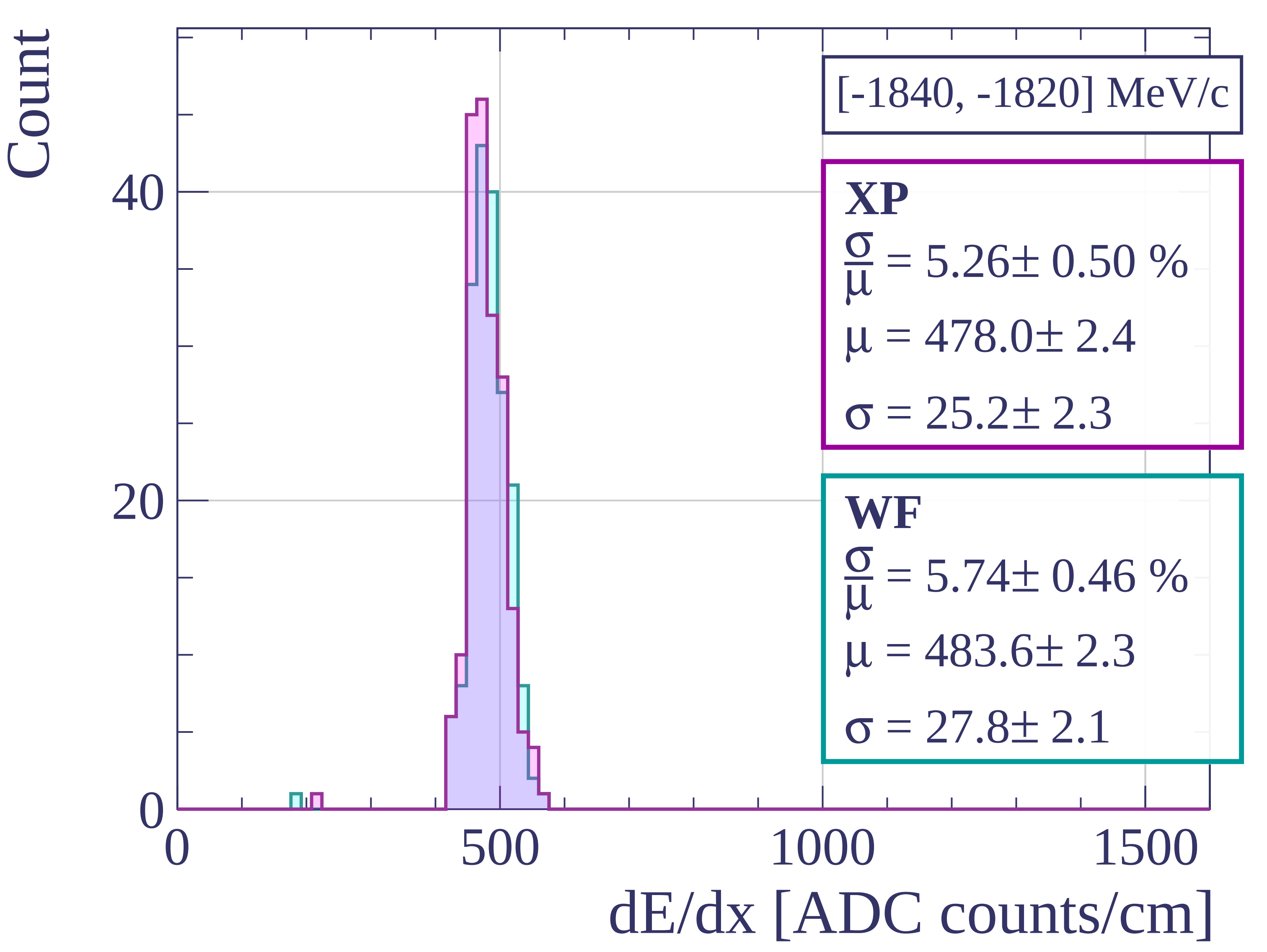
Count
dE/dx [ADC counts/cm]
0
20
40
0
500
1000
1500
[-1900, -1880] MeV/c
XP
σ/μ = 6.42± 0.43 %
μ = 478.2± 2.4
σ = 30.7± 1.9
WF
σ/μ = 6.75± 0.43 %
μ = 482.7± 2.6
σ = 32.6± 1.9



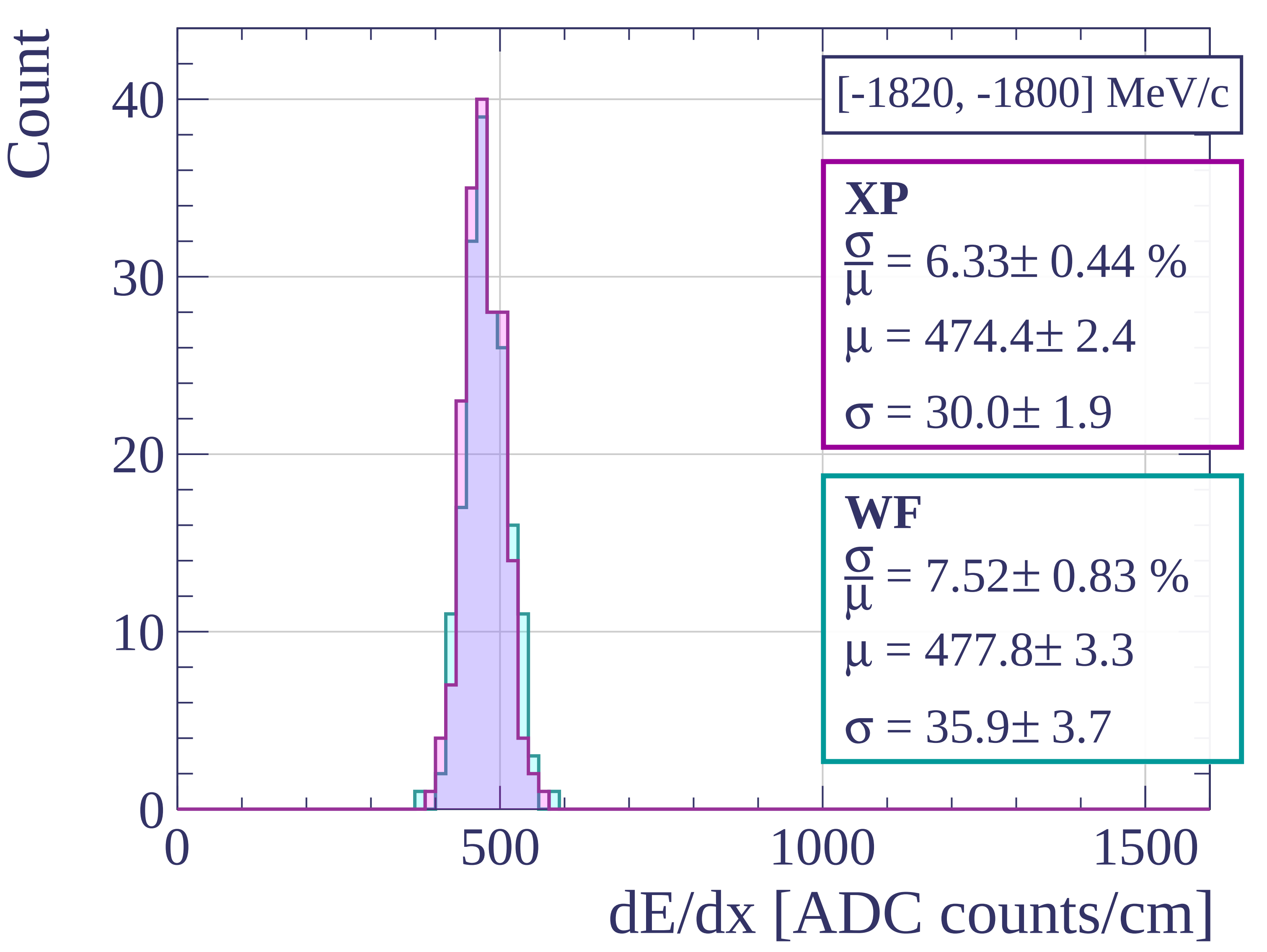
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
10
20
30
40
[-1880, -1860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.47± 0.50 %
μ = 477.7± 2.4
σ = 26.1± 2.3
WF
$\frac{\sigma}{\mu}$ = 5.23± 0.49 %
μ = 480.8± 2.3
σ = 25.1± 2.2



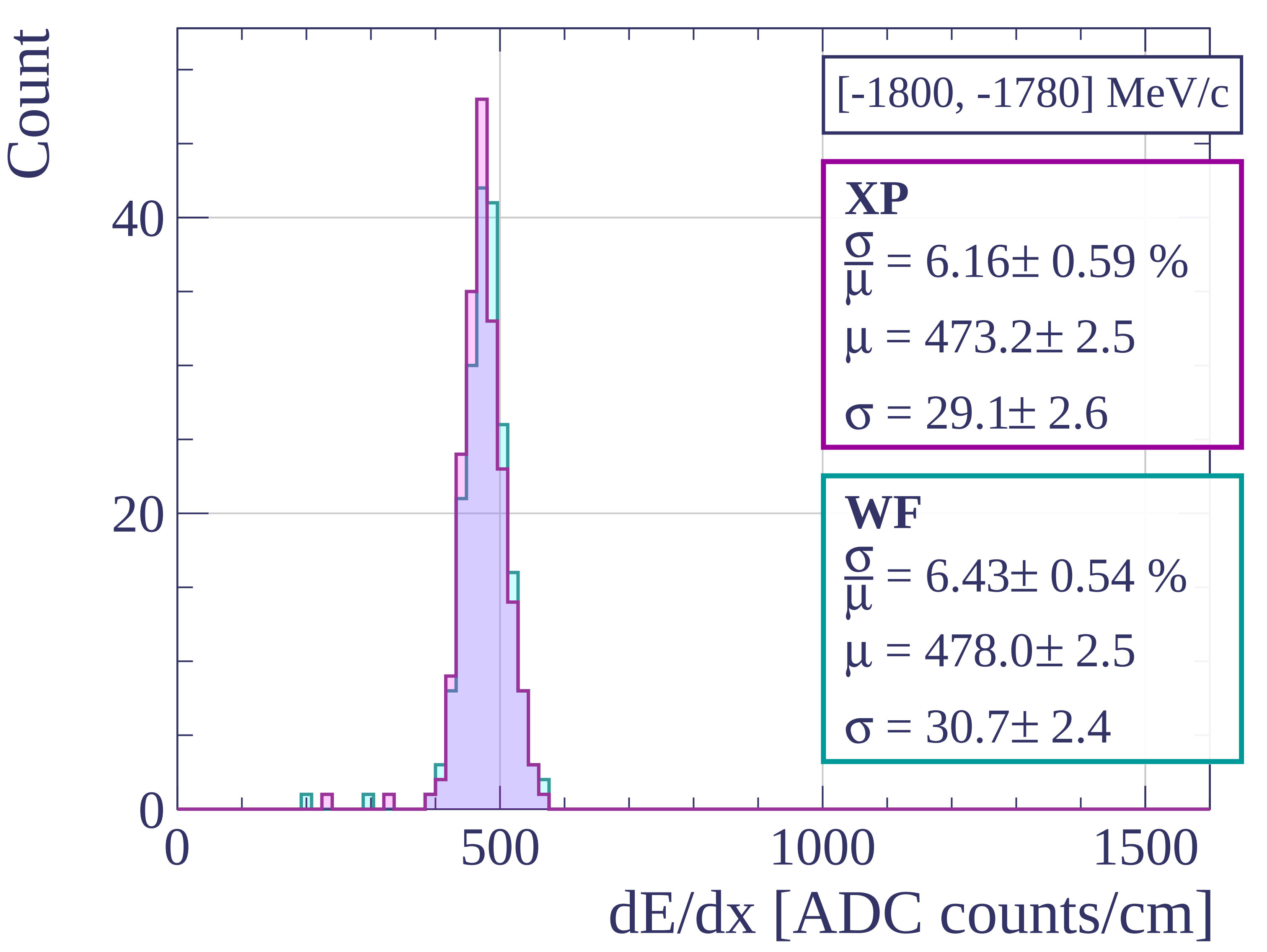
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1860, -1840] MeV/c
XP
σ/μ = 6.05± 0.54 %
μ = 473.4± 2.5
σ = 28.7± 2.4
WF
σ/μ = 6.02± 0.57 %
μ = 477.2± 2.6
σ = 28.7± 2.6



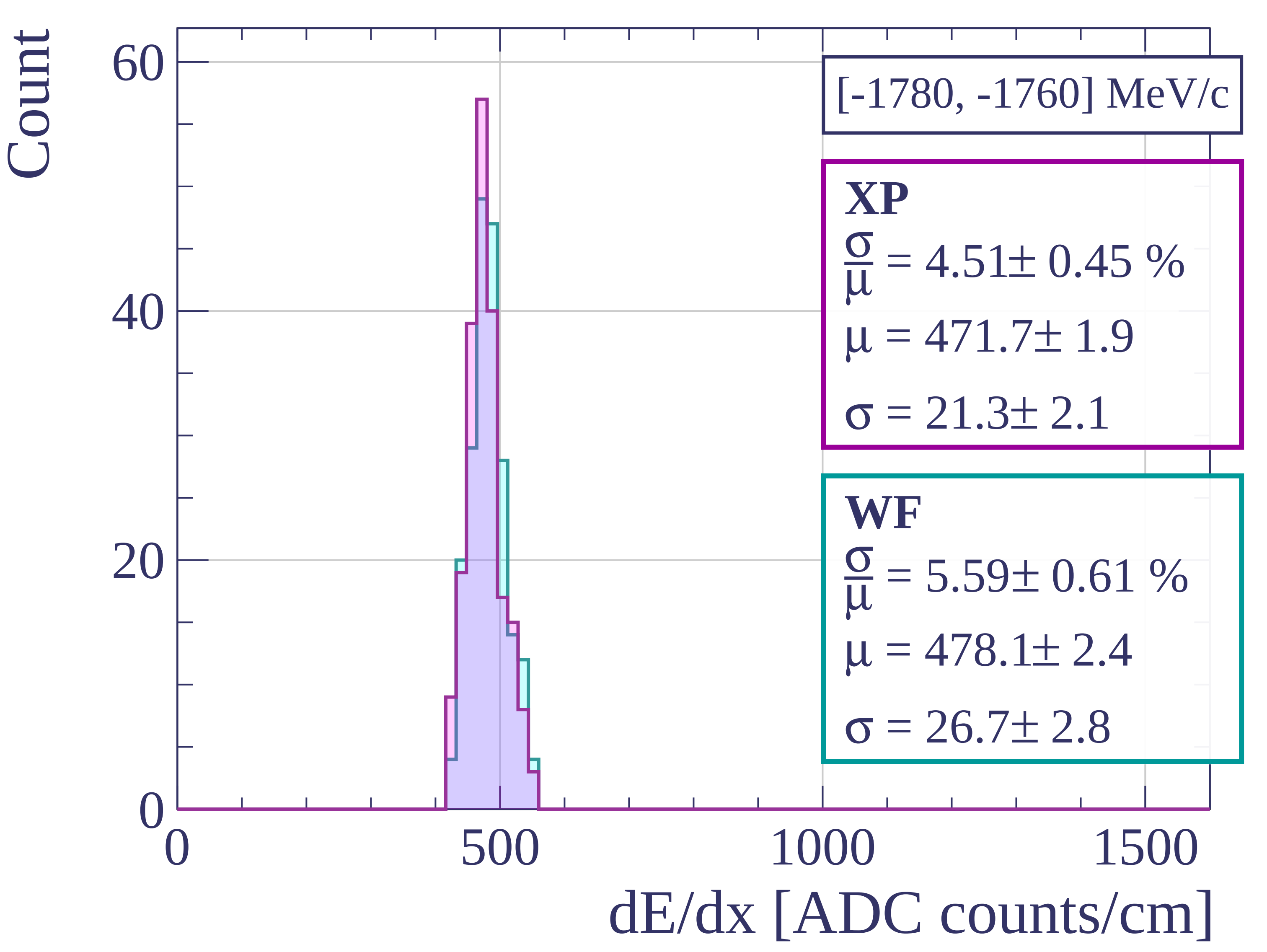
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
20
40
[-1840, -1820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.26± 0.50 %
μ = 478.0± 2.4
σ = 25.2± 2.3
WF
$\frac{\sigma}{\mu}$ = 5.74± 0.46 %
μ = 483.6± 2.3
σ = 27.8± 2.1

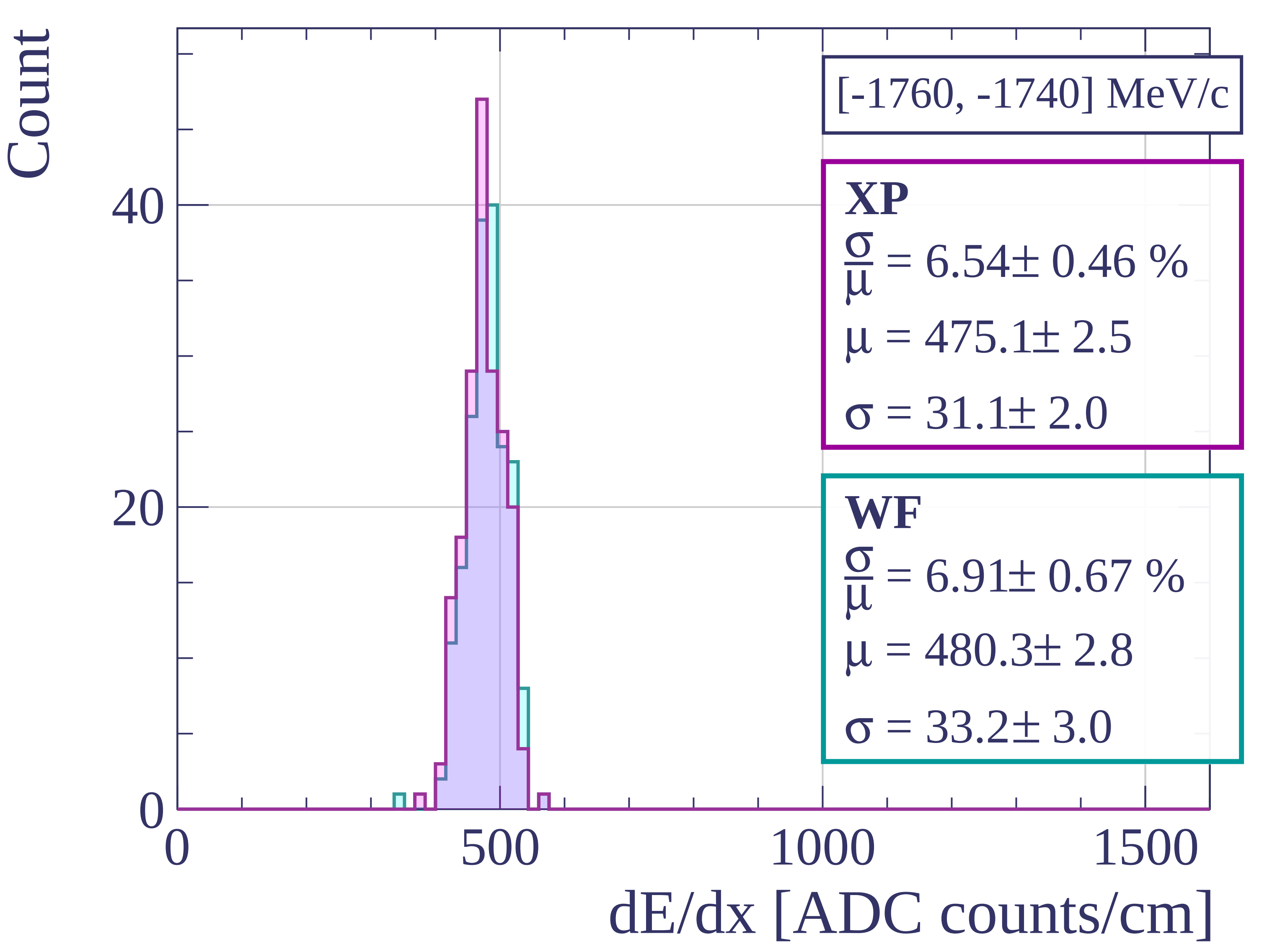


Count
dE/dx [ADC counts/cm]
0
10
20
30
40
0
500
1000
1500
[-1820, -1800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.33± 0.44 %
μ = 474.4± 2.4
σ = 30.0± 1.9
WF
$\frac{\sigma}{\mu}$ = 7.52± 0.83 %
μ = 477.8± 3.3
σ = 35.9± 3.7

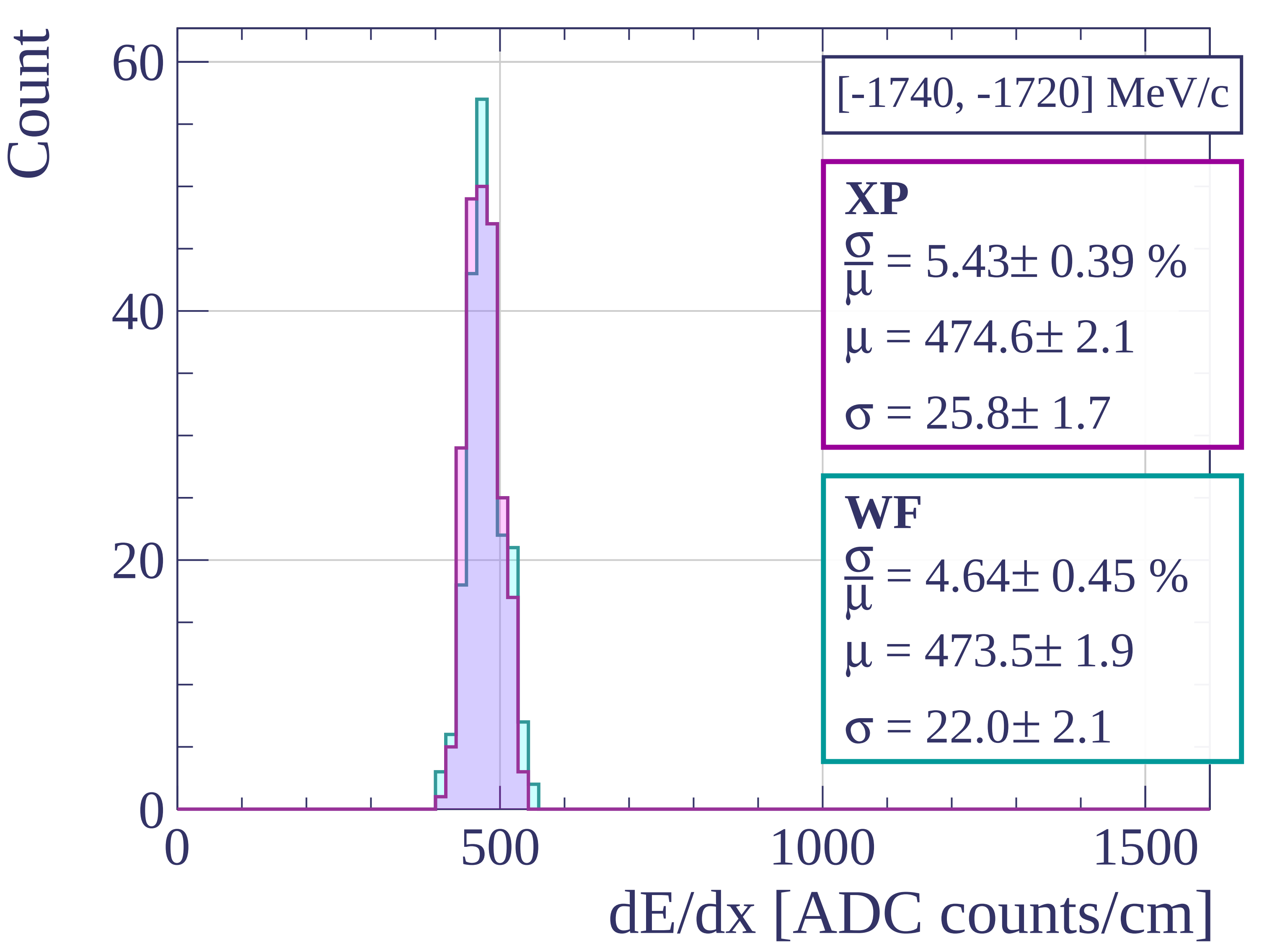


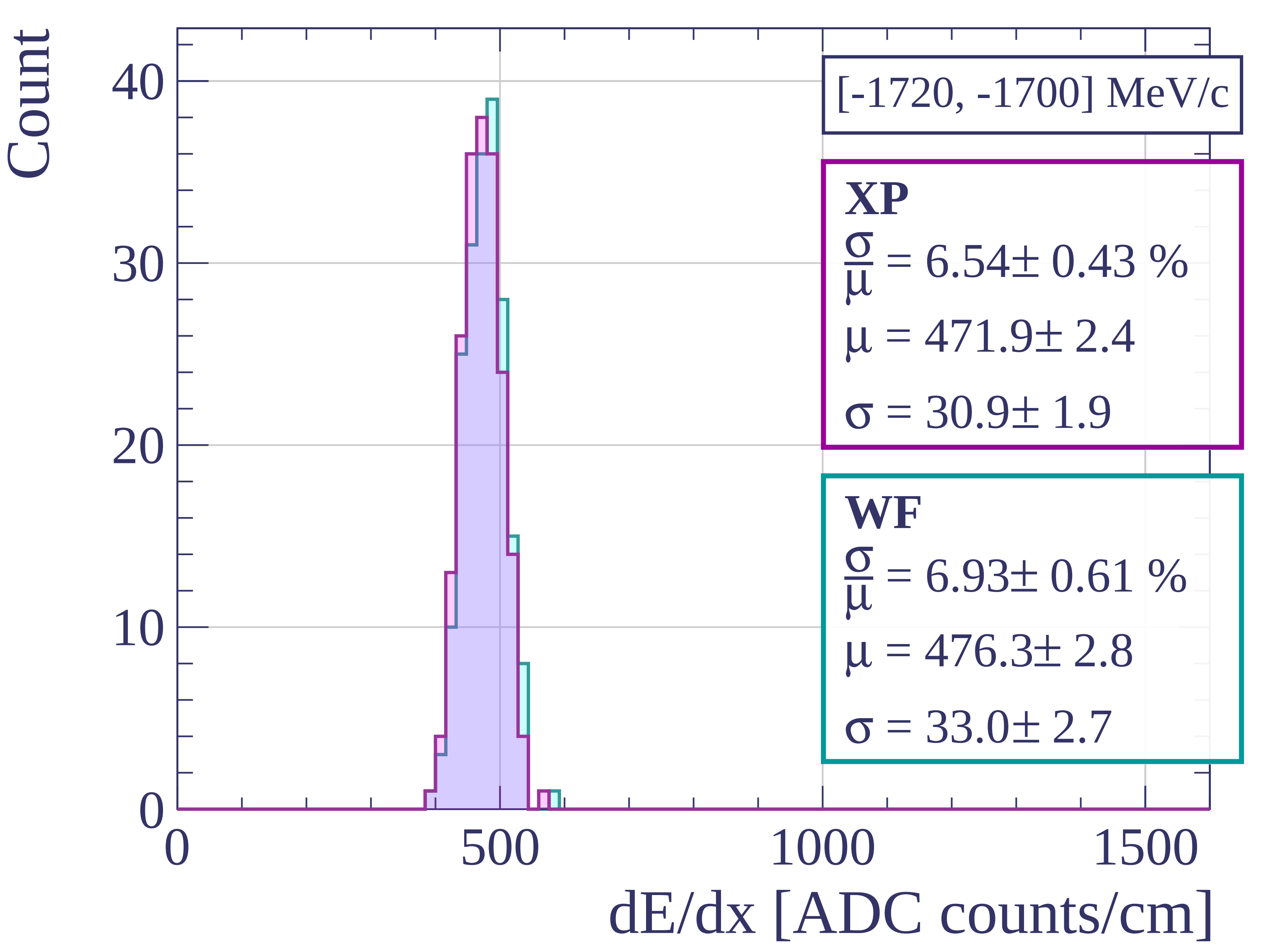
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1800, -1780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.16± 0.59 %
μ = 473.2± 2.5
σ = 29.1± 2.6
WF
$\frac{\sigma}{\mu}$ = 6.43± 0.54 %
μ = 478.0± 2.5
σ = 30.7± 2.4


Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[-1780, -1760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.51± 0.45 %
μ = 471.7± 1.9
σ = 21.3± 2.1
WF
$\frac{\sigma}{\mu}$ = 5.59± 0.61 %
μ = 478.1± 2.4
σ = 26.7± 2.8

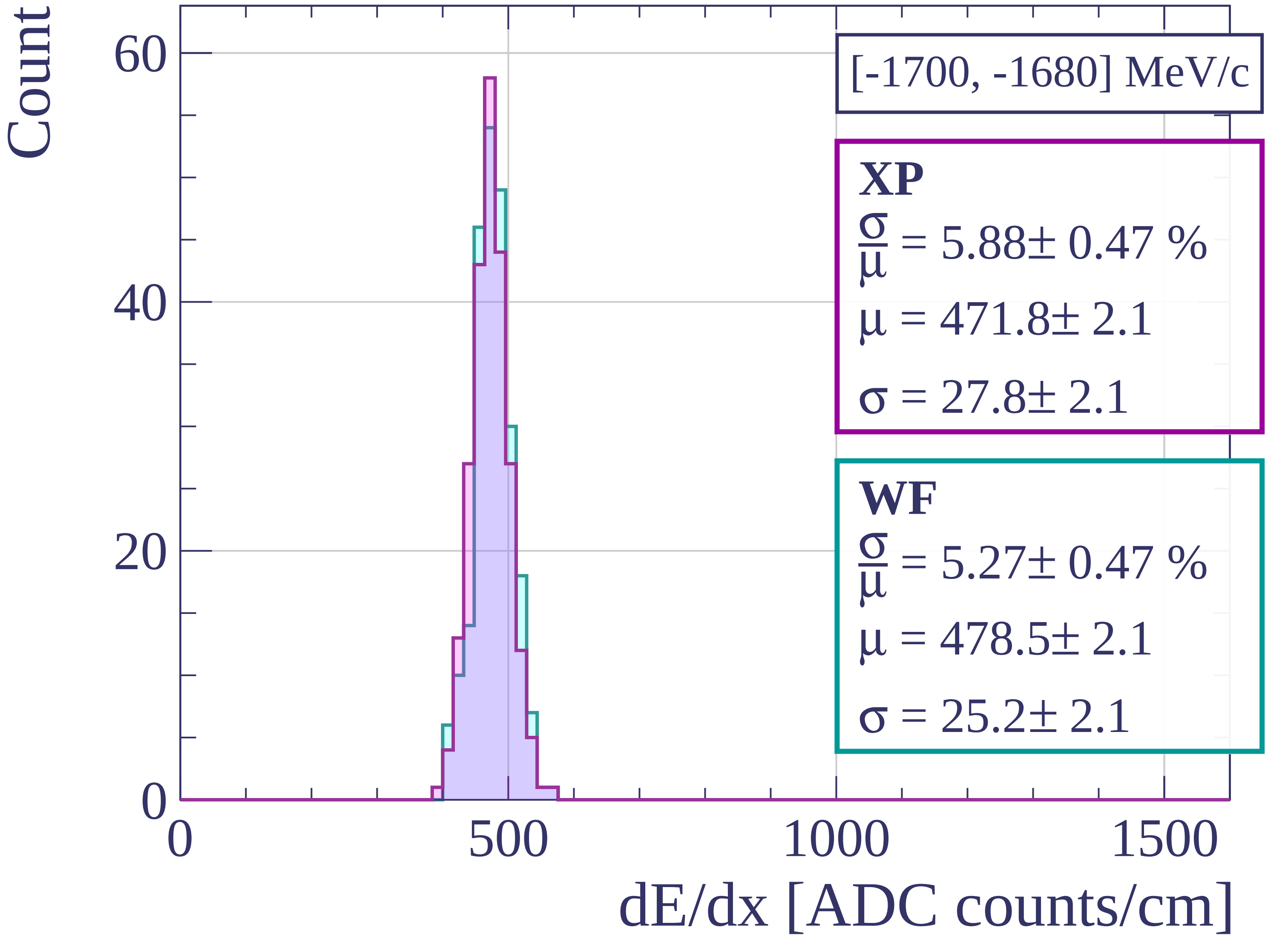


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1760, -1740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.54± 0.46 %
μ = 475.1± 2.5
σ = 31.1± 2.0
WF
$\frac{\sigma}{\mu}$ = 6.91± 0.67 %
μ = 480.3± 2.8
σ = 33.2± 3.0

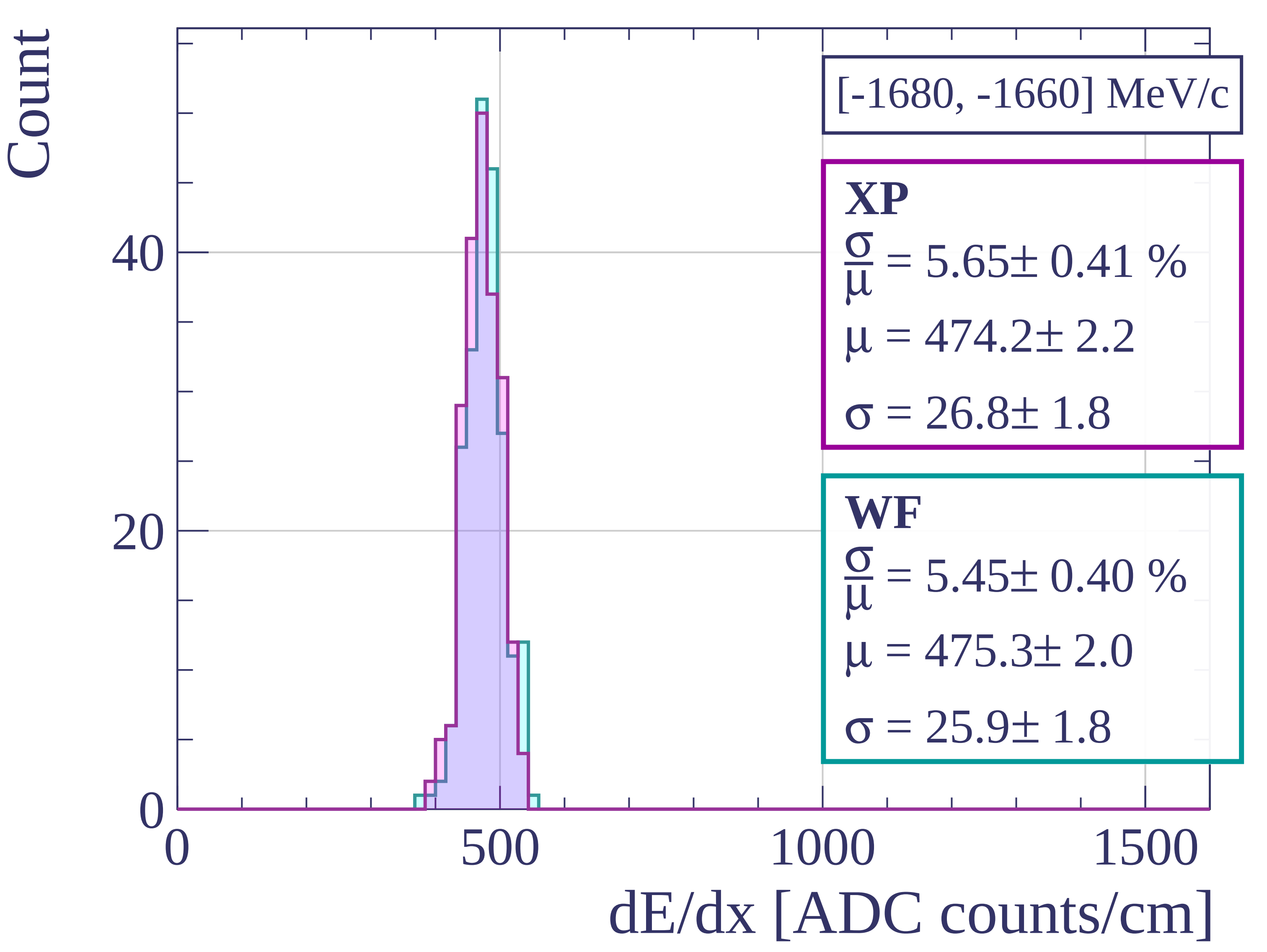

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1740, -1720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.43± 0.39 %
μ = 474.6± 2.1
σ = 25.8± 1.7
WF
$\frac{\sigma}{\mu}$ = 4.64± 0.45 %
μ = 473.5± 1.9
σ = 22.0± 2.1

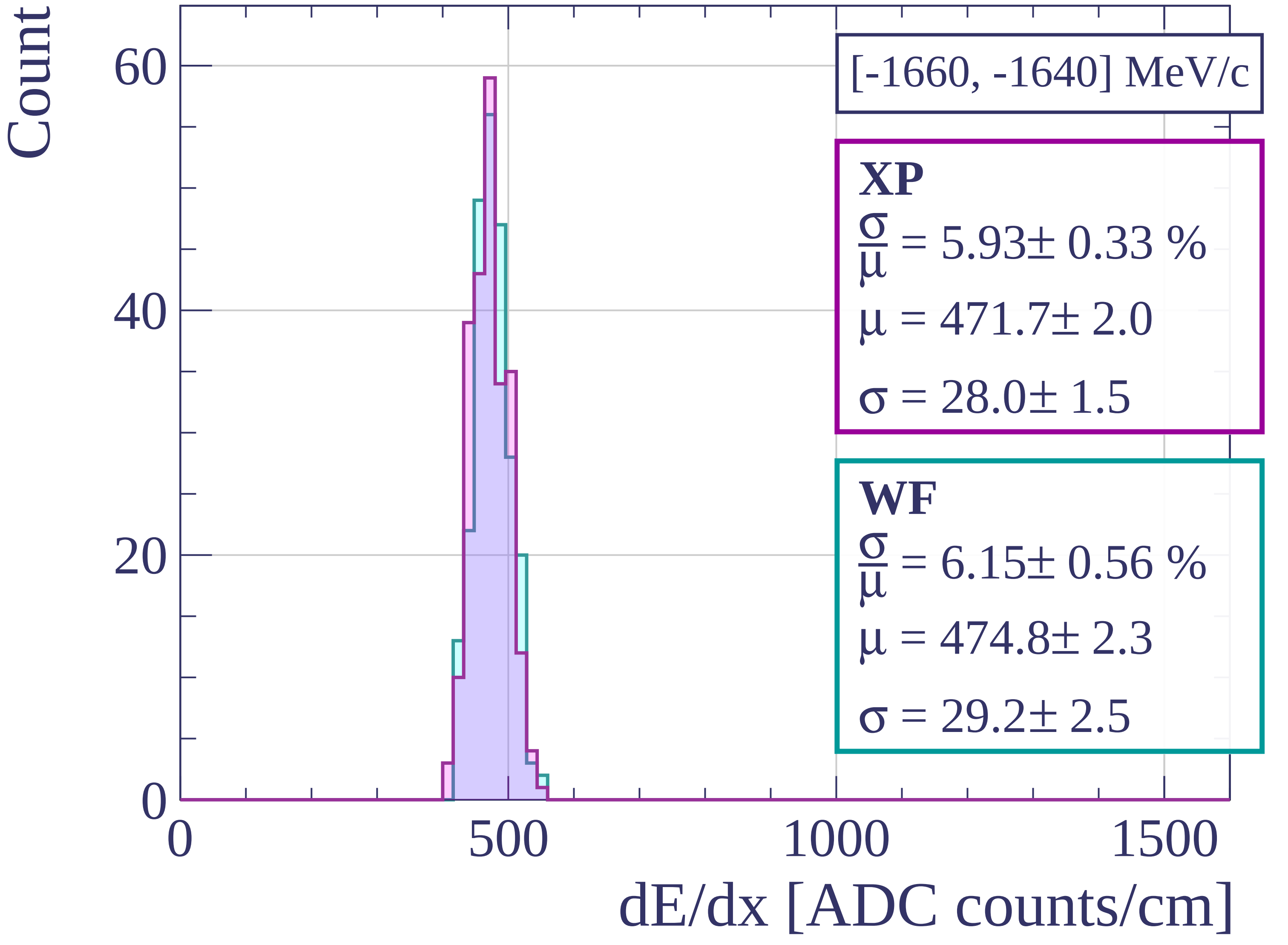


Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1720, -1700] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.54± 0.43 %
μ = 471.9± 2.4
σ = 30.9± 1.9
WF
$\frac{\sigma}{\mu}$ = 6.93± 0.61 %
μ = 476.3± 2.8
σ = 33.0± 2.7

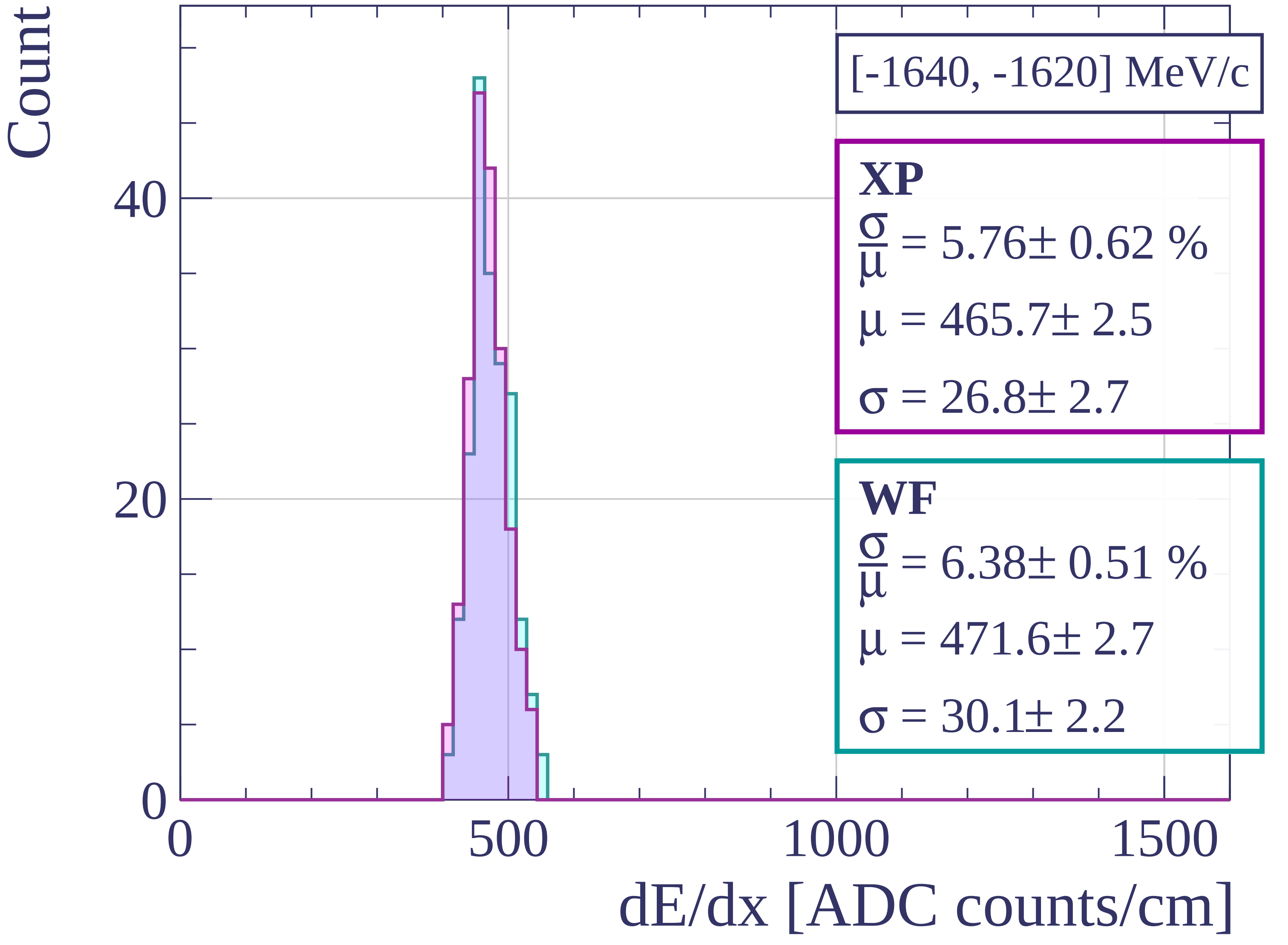


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1700, -1680] MeV/c
XP
σ/μ = 5.88± 0.47 %
μ = 471.8± 2.1
σ = 27.8± 2.1
WF
σ/μ = 5.27± 0.47 %
μ = 478.5± 2.1
σ = 25.2± 2.1

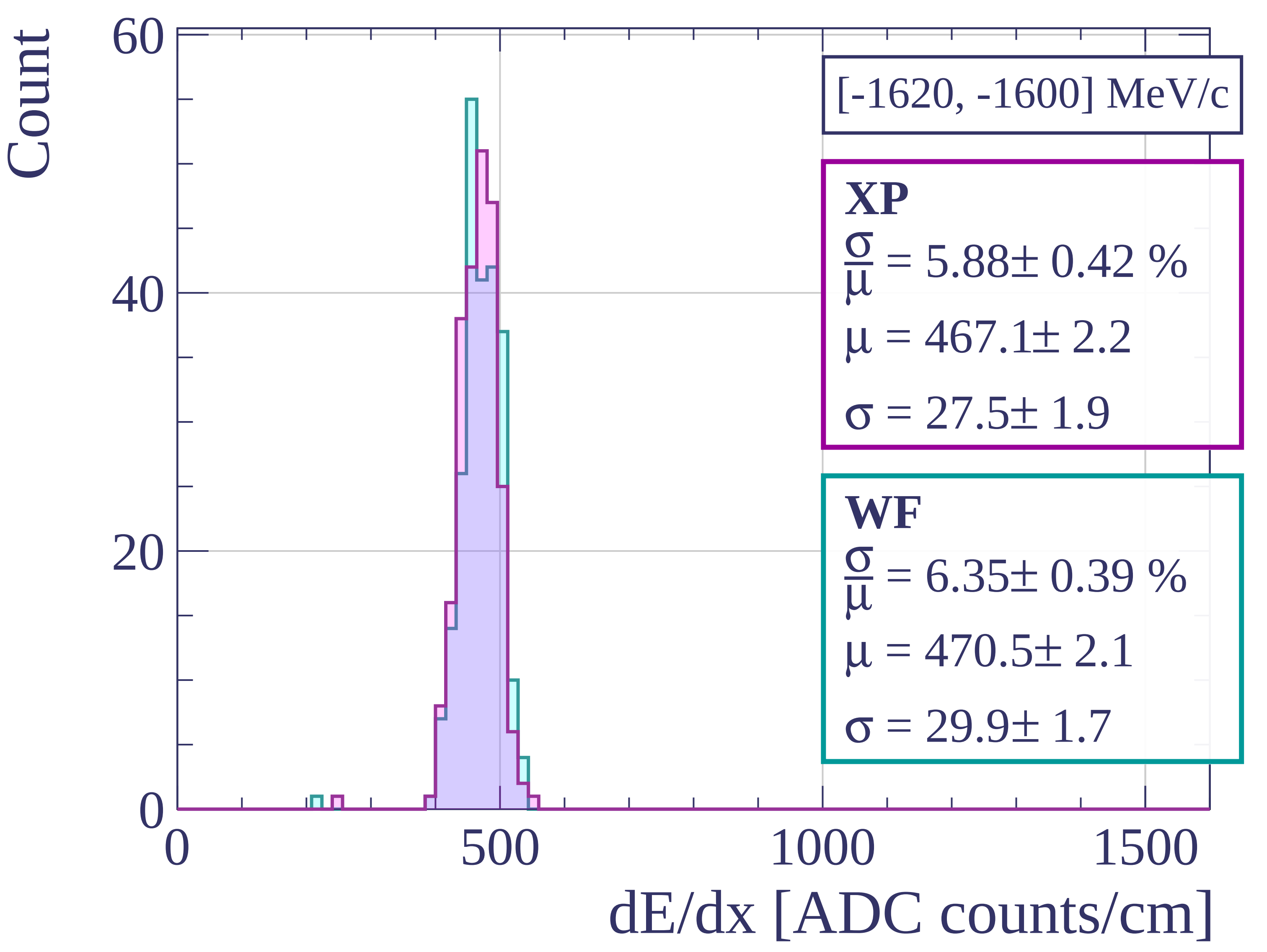

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1680, -1660] MeV/c
XP
σ/μ = 5.65± 0.41 %
μ = 474.2± 2.2
σ = 26.8± 1.8
WF
σ/μ = 5.45± 0.40 %
μ = 475.3± 2.0
σ = 25.9± 1.8



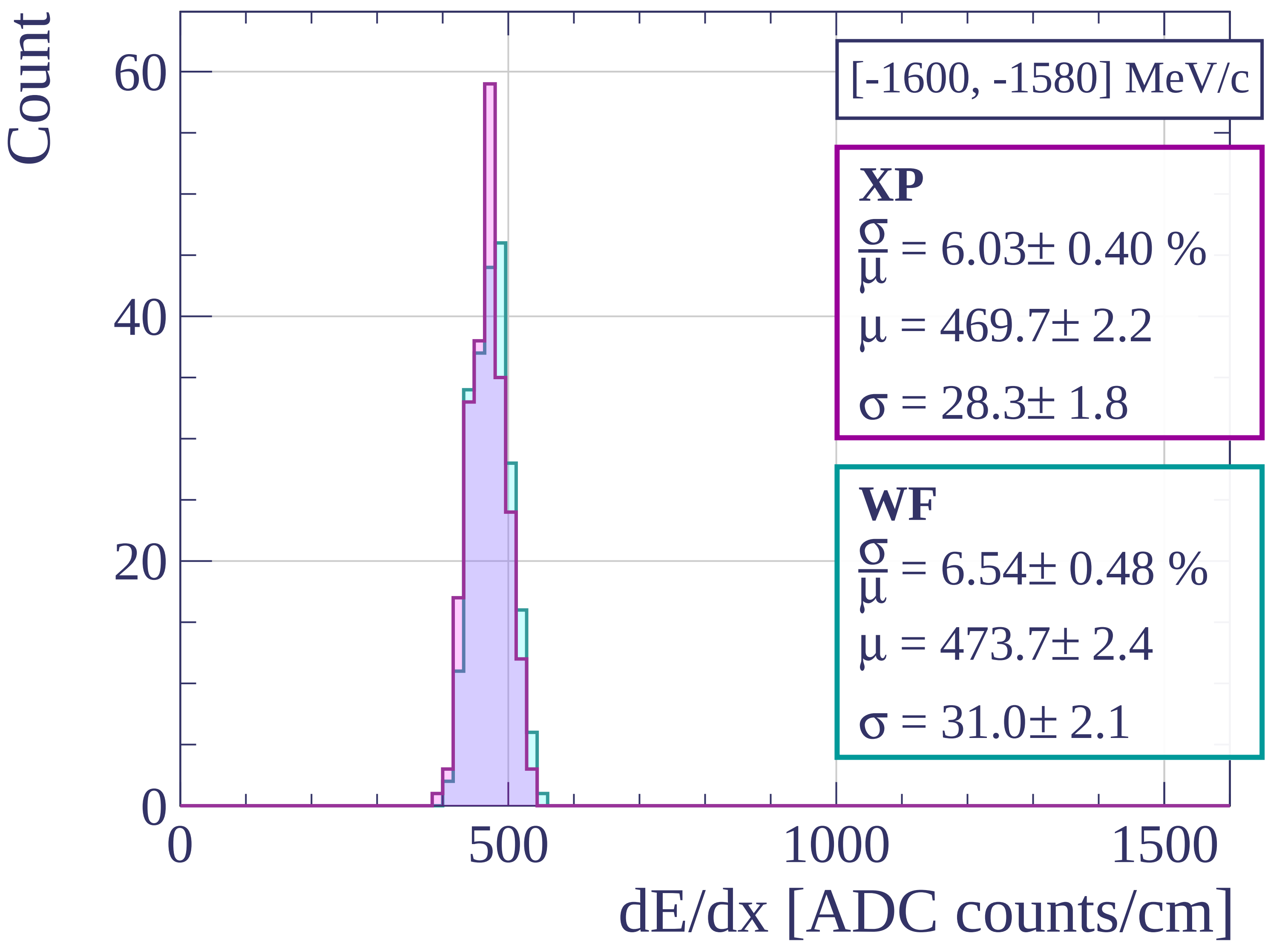
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1660, -1640] MeV/c
XP
σ/μ = 5.93± 0.33 %
μ = 471.7± 2.0
σ = 28.0± 1.5
WF
σ/μ = 6.15± 0.56 %
μ = 474.8± 2.3
σ = 29.2± 2.5



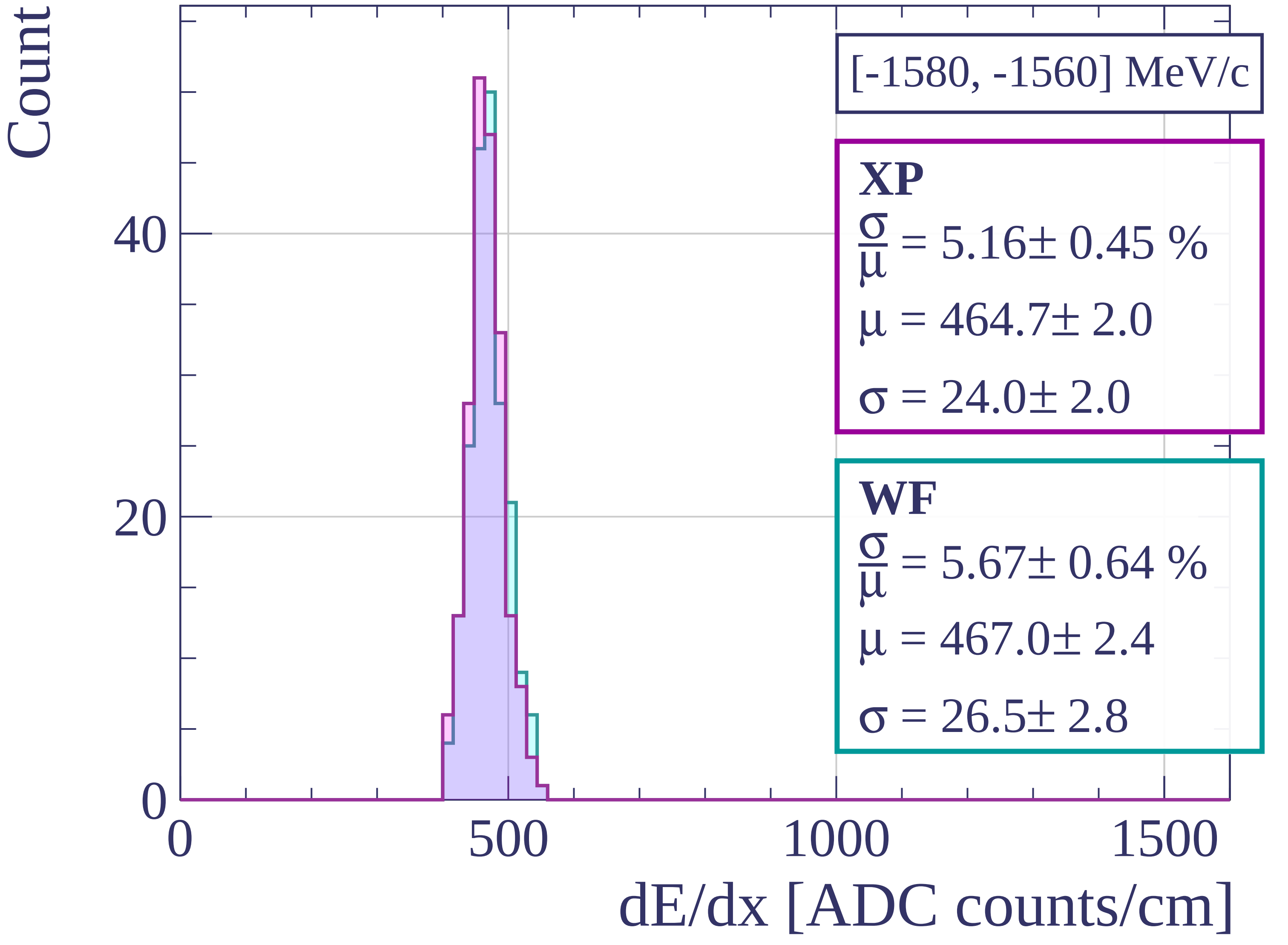
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1640, -1620] MeV/c
XP
σ/μ = 5.76± 0.62 %
μ = 465.7± 2.5
σ = 26.8± 2.7
WF
σ/μ = 6.38± 0.51 %
μ = 471.6± 2.7
σ = 30.1± 2.2



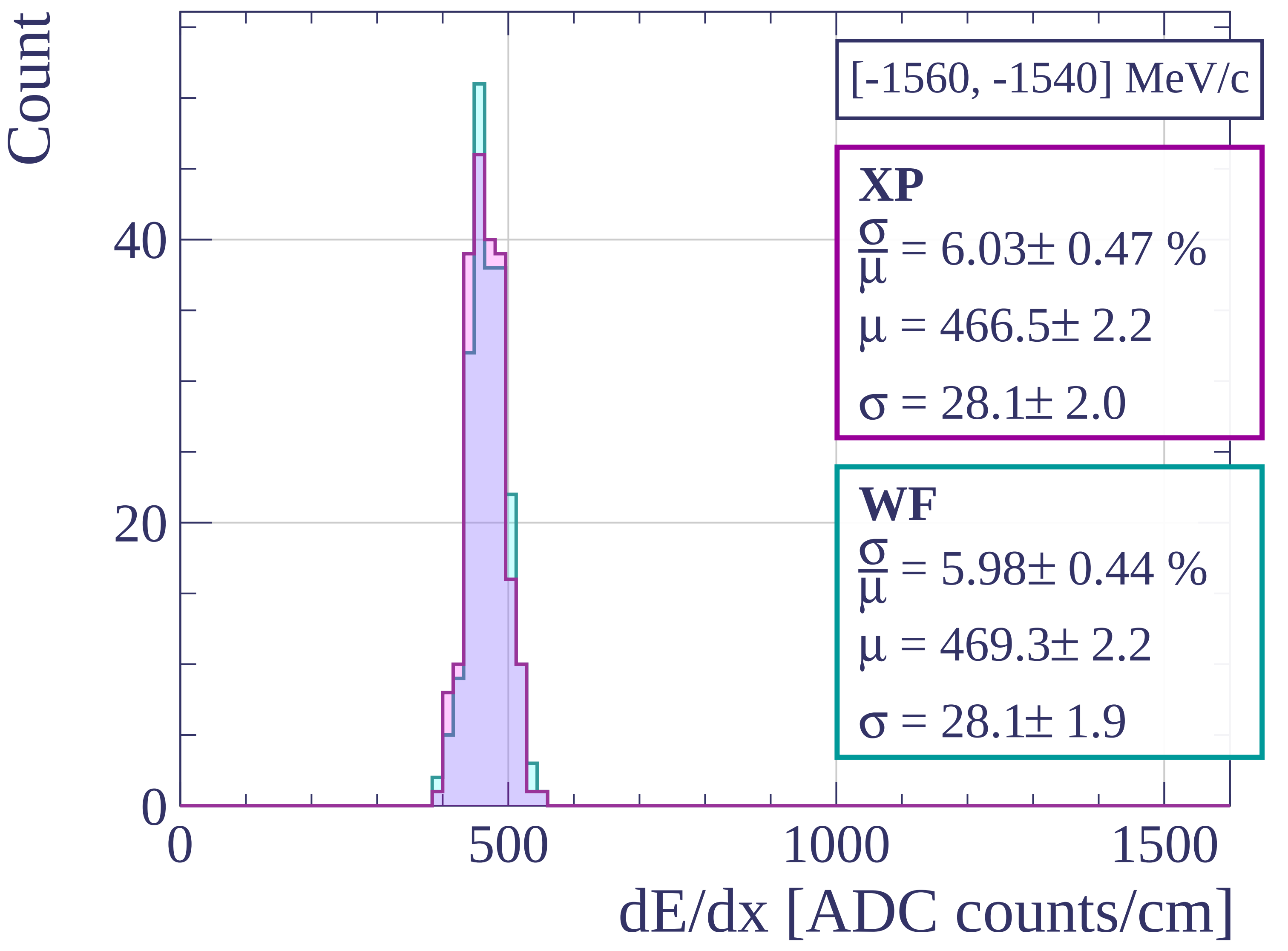
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1620, -1600] MeV/c
XP
σ/μ = 5.88± 0.42 %
μ = 467.1± 2.2
σ = 27.5± 1.9
WF
σ/μ = 6.35± 0.39 %
μ = 470.5± 2.1
σ = 29.9± 1.7



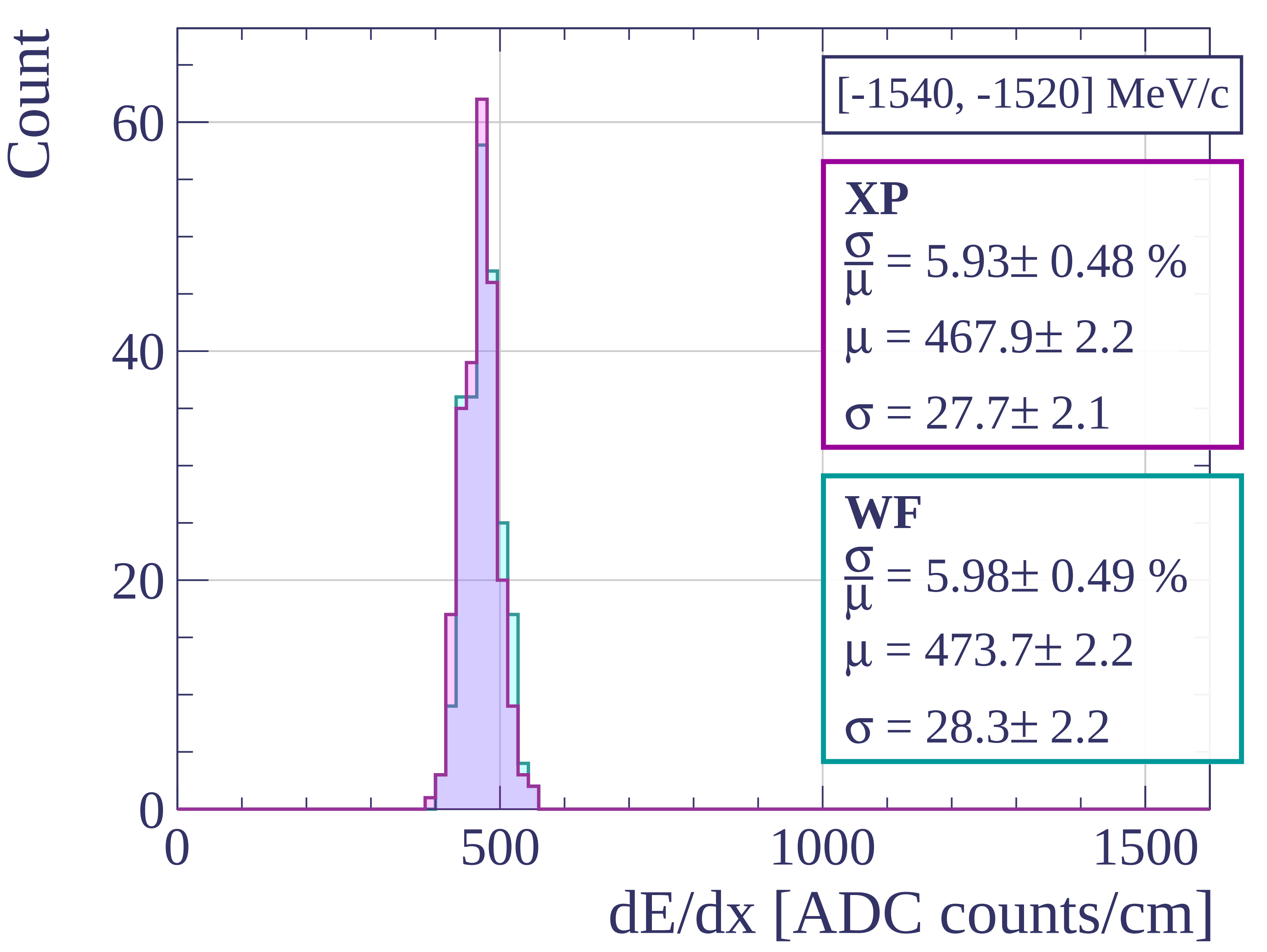
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1600, -1580] MeV/c
XP
σ/μ = 6.03± 0.40 %
μ = 469.7± 2.2
σ = 28.3± 1.8
WF
σ/μ = 6.54± 0.48 %
μ = 473.7± 2.4
σ = 31.0± 2.1

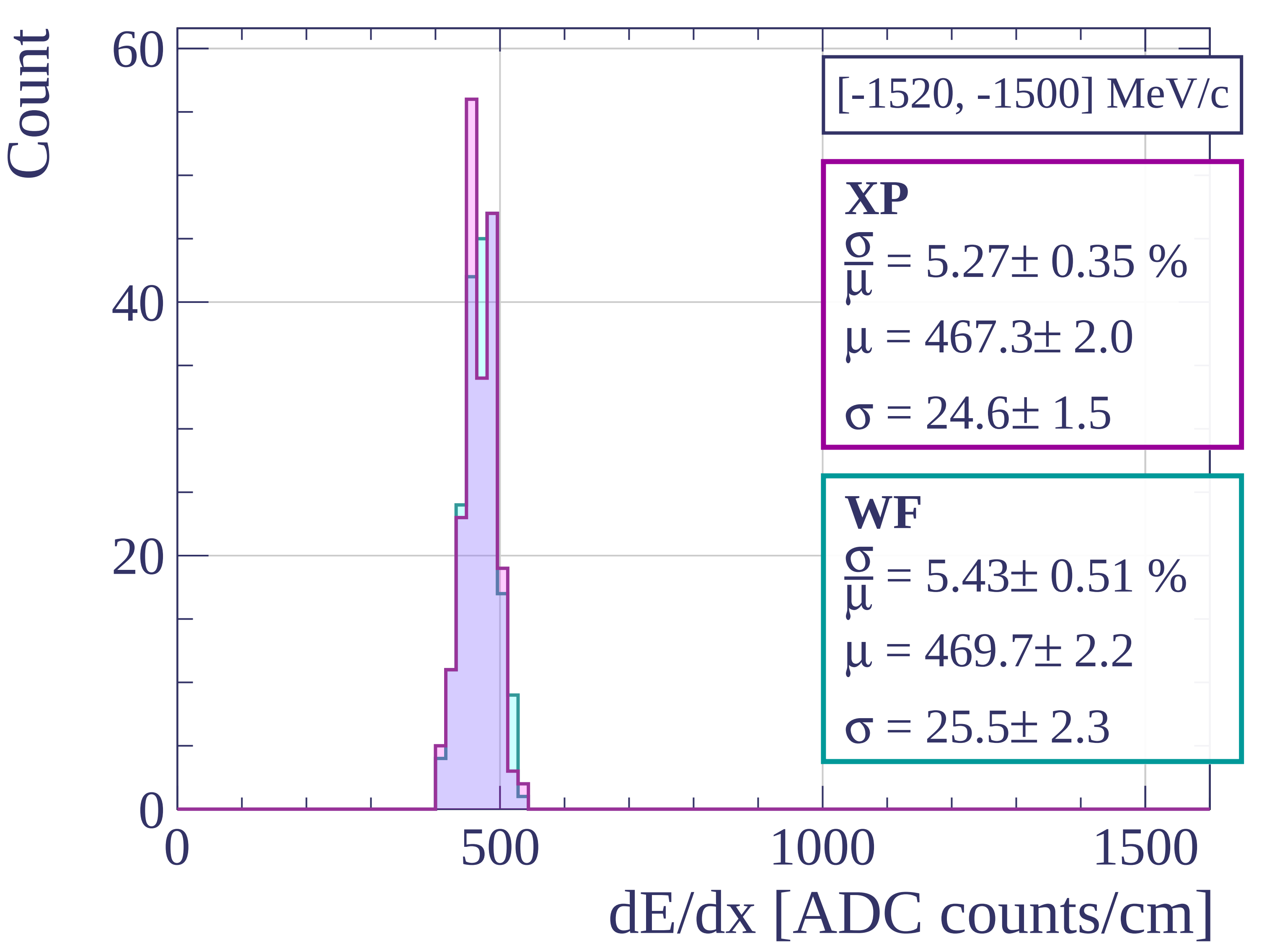


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1580, -1560] MeV/c
XP
σ/μ = 5.16± 0.45 %
μ = 464.7± 2.0
σ = 24.0± 2.0
WF
σ/μ = 5.67± 0.64 %
μ = 467.0± 2.4
σ = 26.5± 2.8

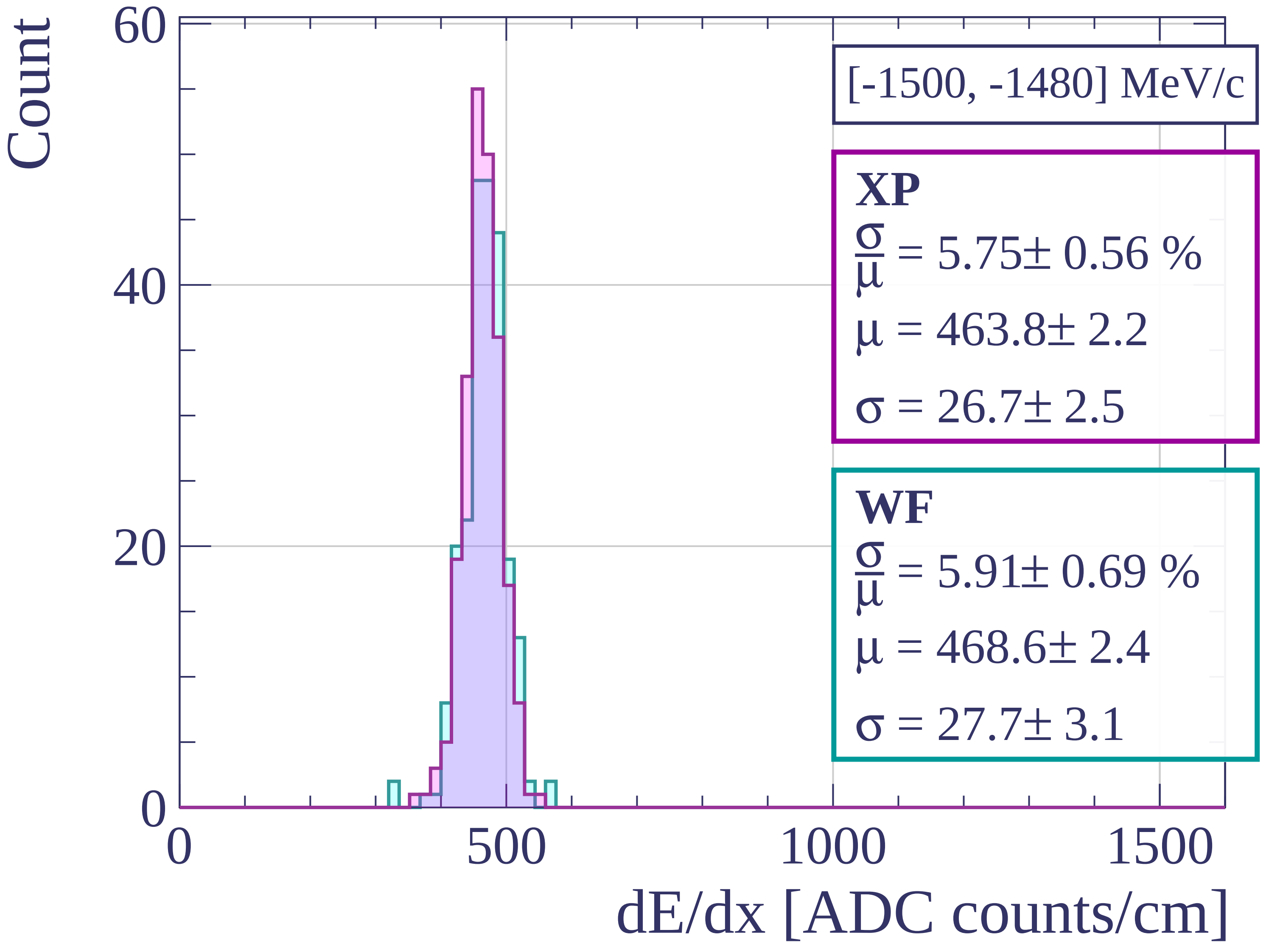


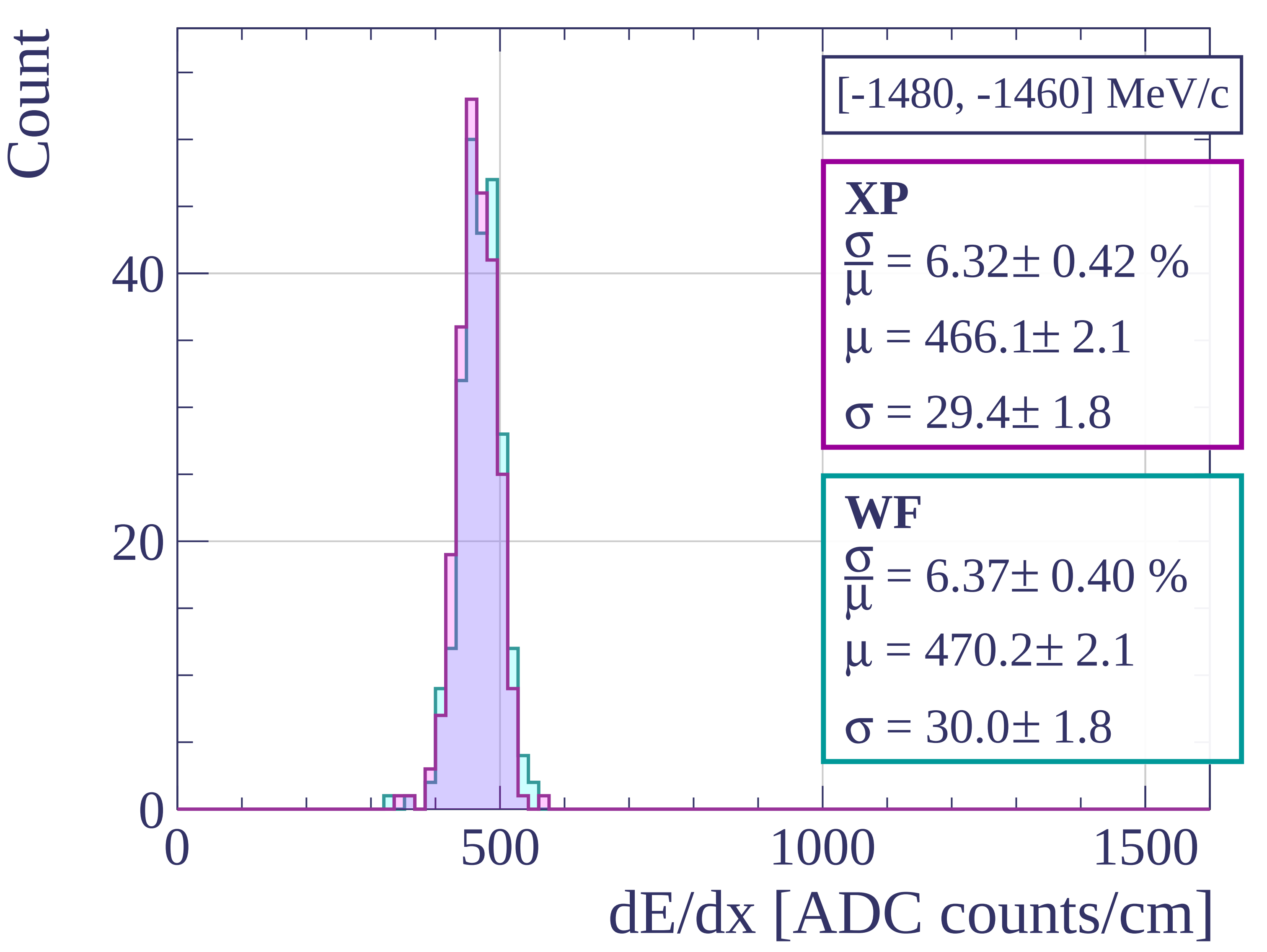
Count
[-1560, -1540] MeV/c
XP
σ/μ = 6.03± 0.47 %
μ = 466.5± 2.2
σ = 28.1± 2.0
WF
σ/μ = 5.98± 0.44 %
μ = 469.3± 2.2
σ = 28.1± 1.9
0
20
40
0
500
1000
1500
dE/dx [ADC counts/cm]

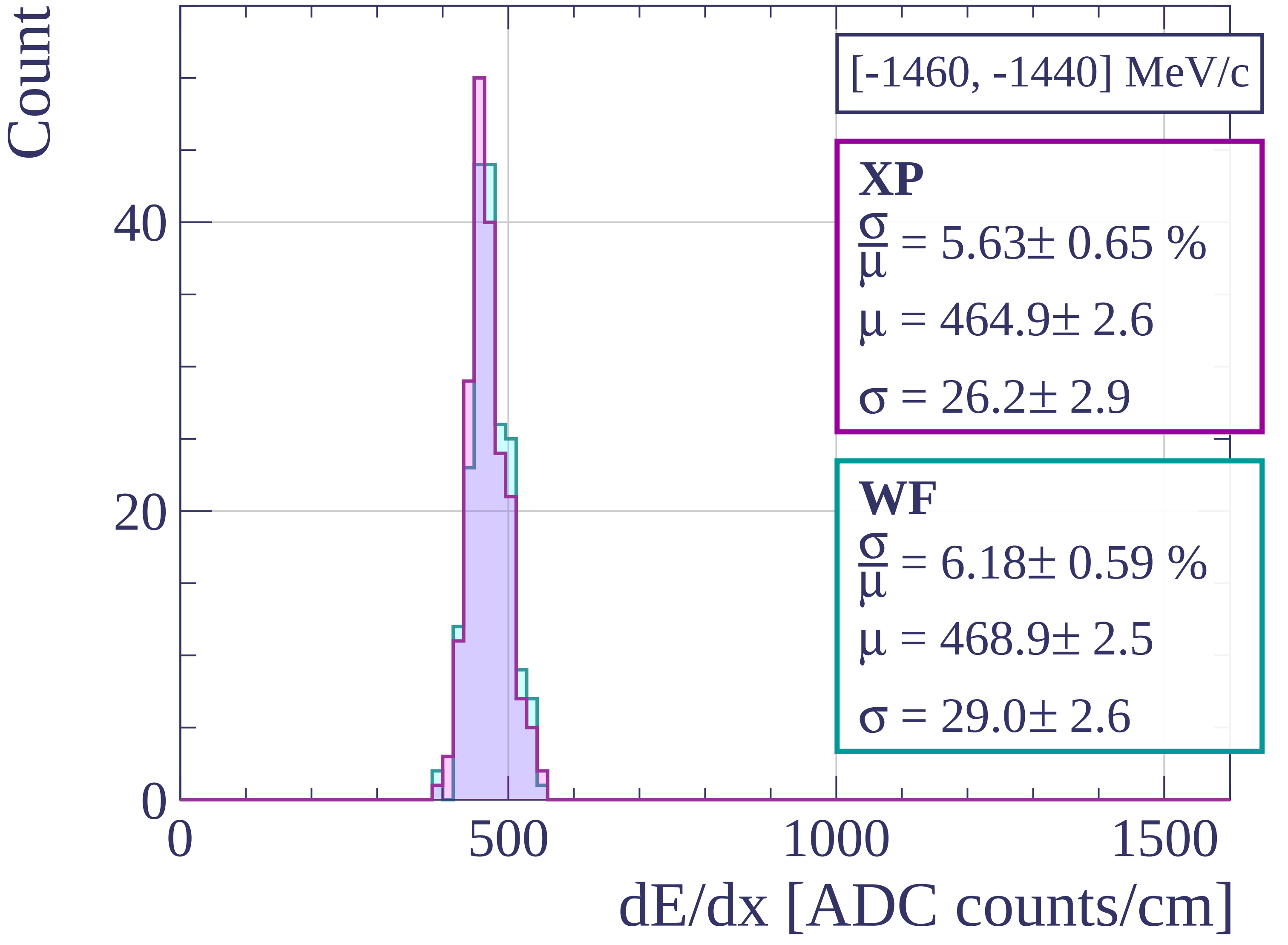

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1540, -1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.93± 0.48 %
μ = 467.9± 2.2
σ = 27.7± 2.1
WF
$\frac{\sigma}{\mu}$ = 5.98± 0.49 %
μ = 473.7± 2.2
σ = 28.3± 2.2

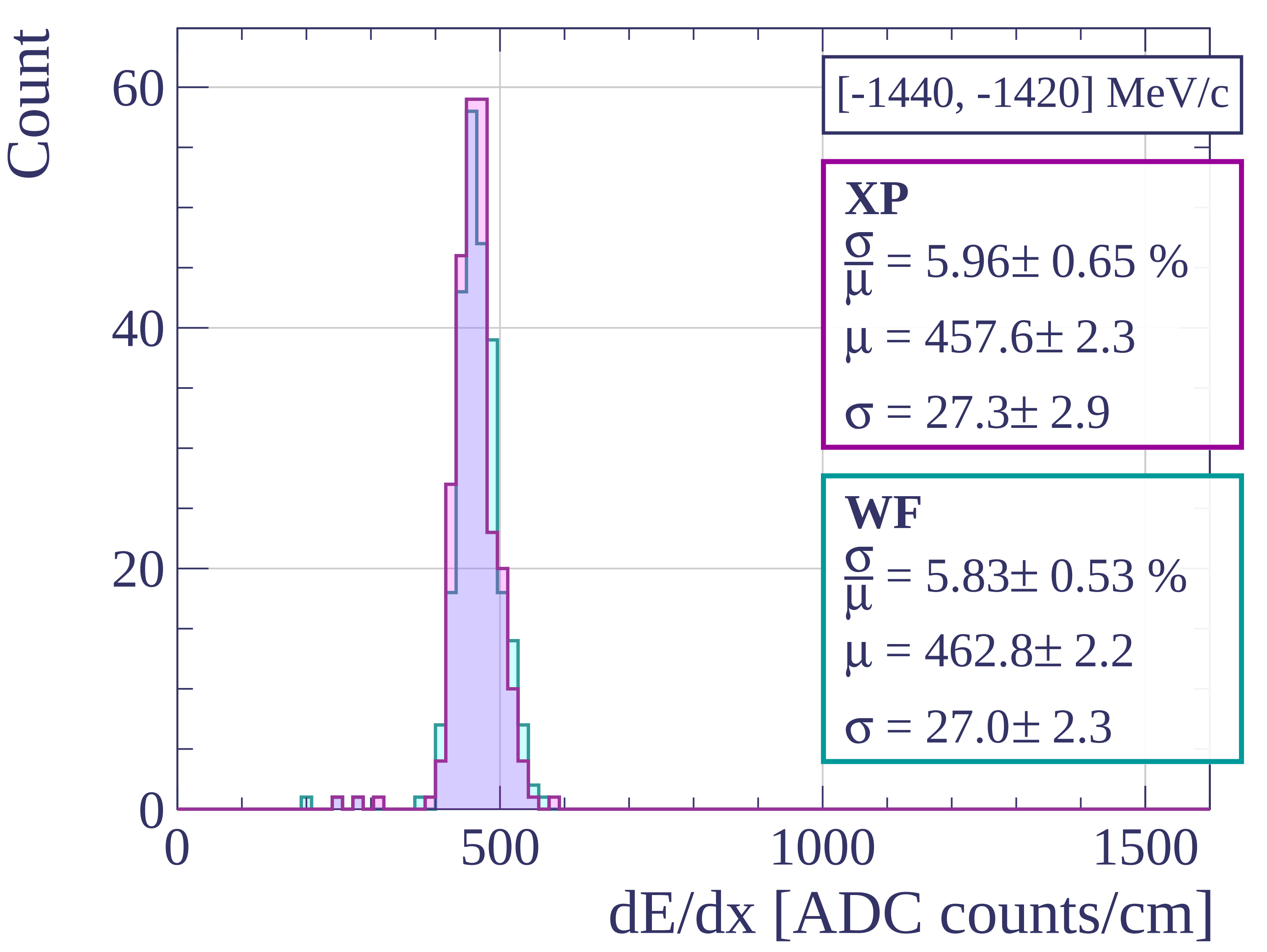


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1520, -1500] MeV/c
XP
σ/μ = 5.27± 0.35 %
μ = 467.3± 2.0
σ = 24.6± 1.5
WF
σ/μ = 5.43± 0.51 %
μ = 469.7± 2.2
σ = 25.5± 2.3

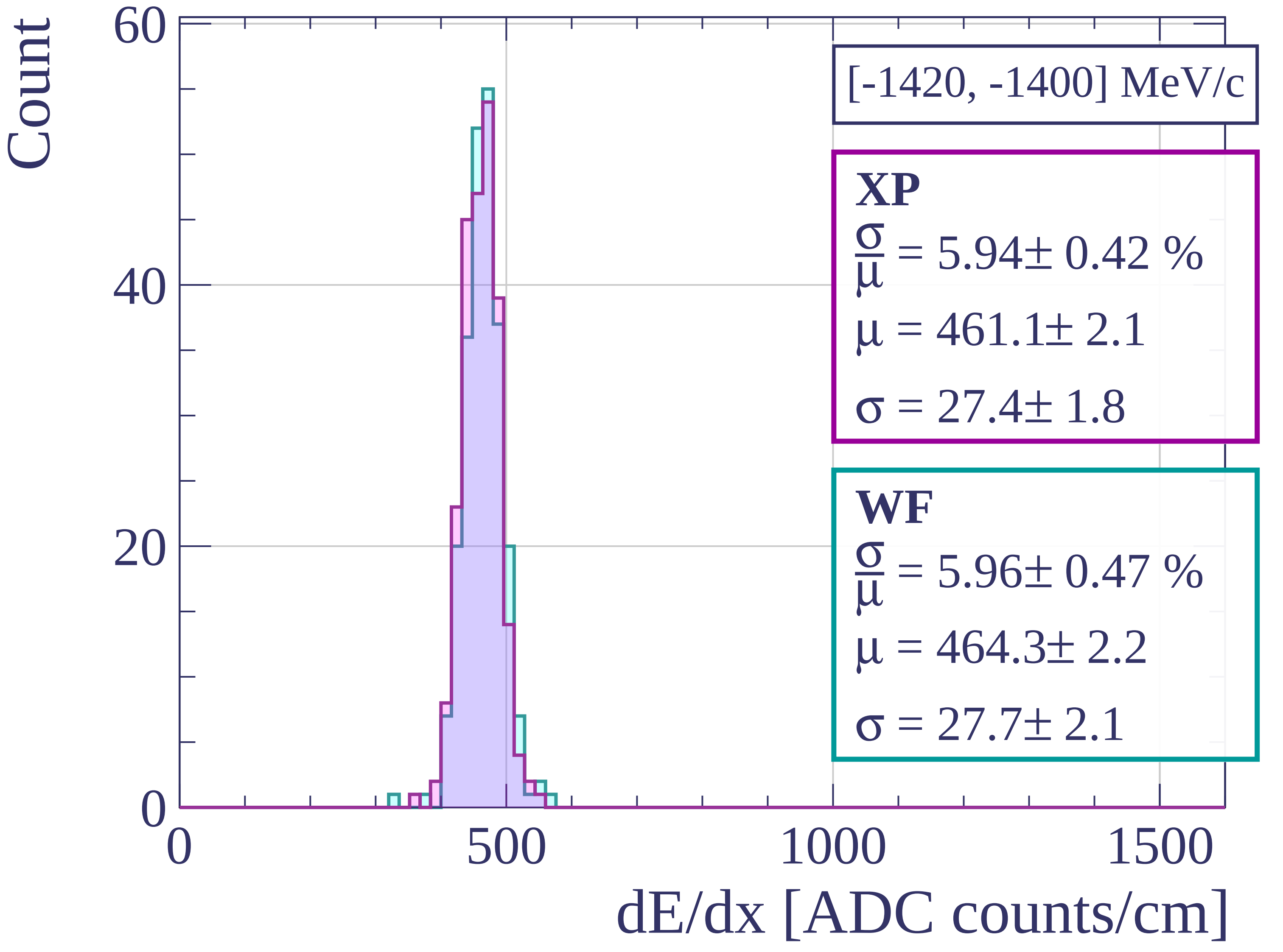

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1500, -1480] MeV/c
XP
σ/μ = 5.75± 0.56 %
μ = 463.8± 2.2
σ = 26.7± 2.5
WF
σ/μ = 5.91± 0.69 %
μ = 468.6± 2.4
σ = 27.7± 3.1

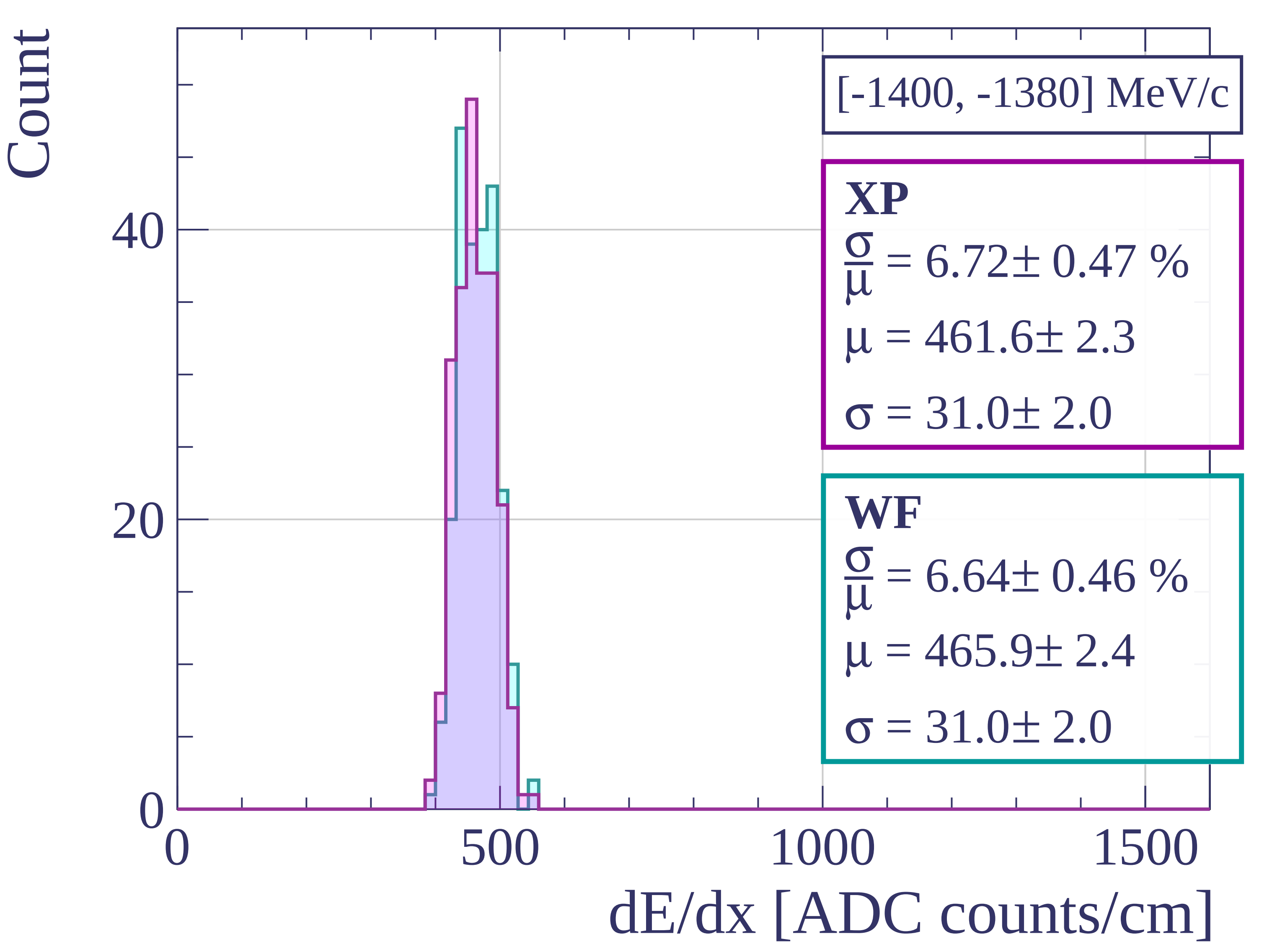

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1480, -1460] MeV/c
XP
σ/μ = 6.32± 0.42 %
μ = 466.1± 2.1
σ = 29.4± 1.8
WF
σ/μ = 6.37± 0.40 %
μ = 470.2± 2.1
σ = 30.0± 1.8

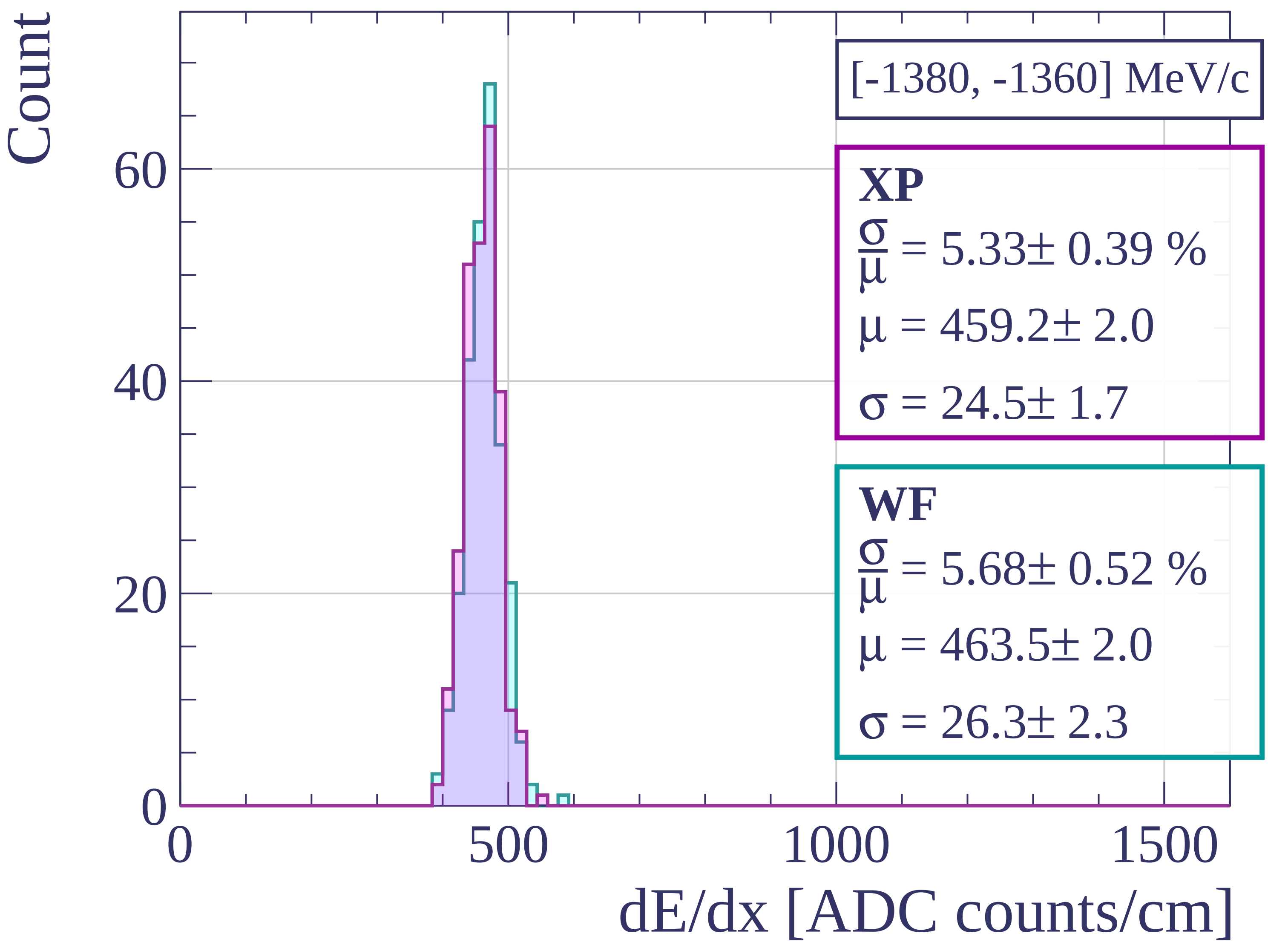

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1460, -1440] MeV/c
XP
σ/μ = 5.63± 0.65 %
μ = 464.9± 2.6
σ = 26.2± 2.9
WF
σ/μ = 6.18± 0.59 %
μ = 468.9± 2.5
σ = 29.0± 2.6

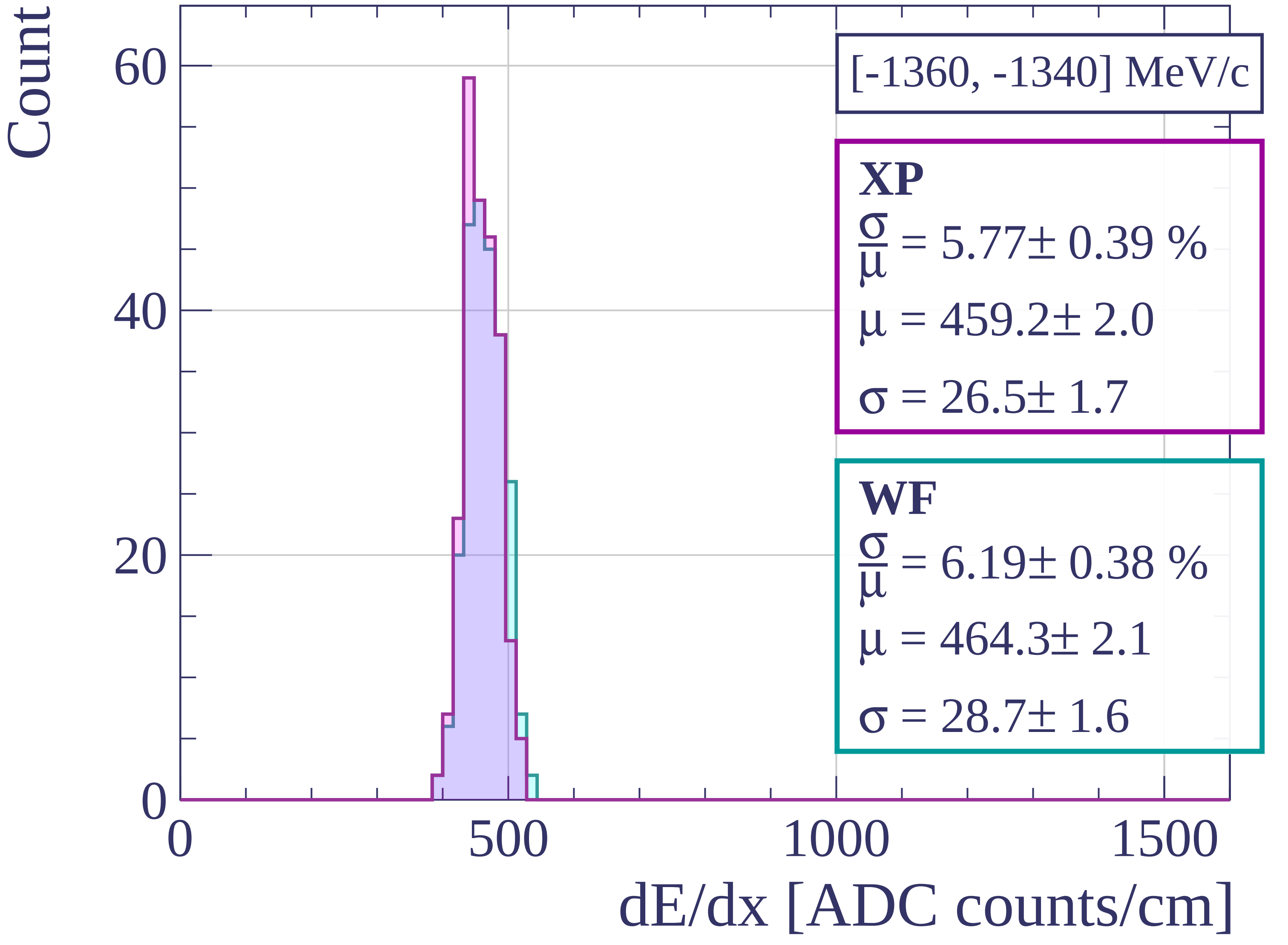


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1440, -1420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.96± 0.65 %
μ = 457.6± 2.3
σ = 27.3± 2.9
WF
$\frac{\sigma}{\mu}$ = 5.83± 0.53 %
μ = 462.8± 2.2
σ = 27.0± 2.3

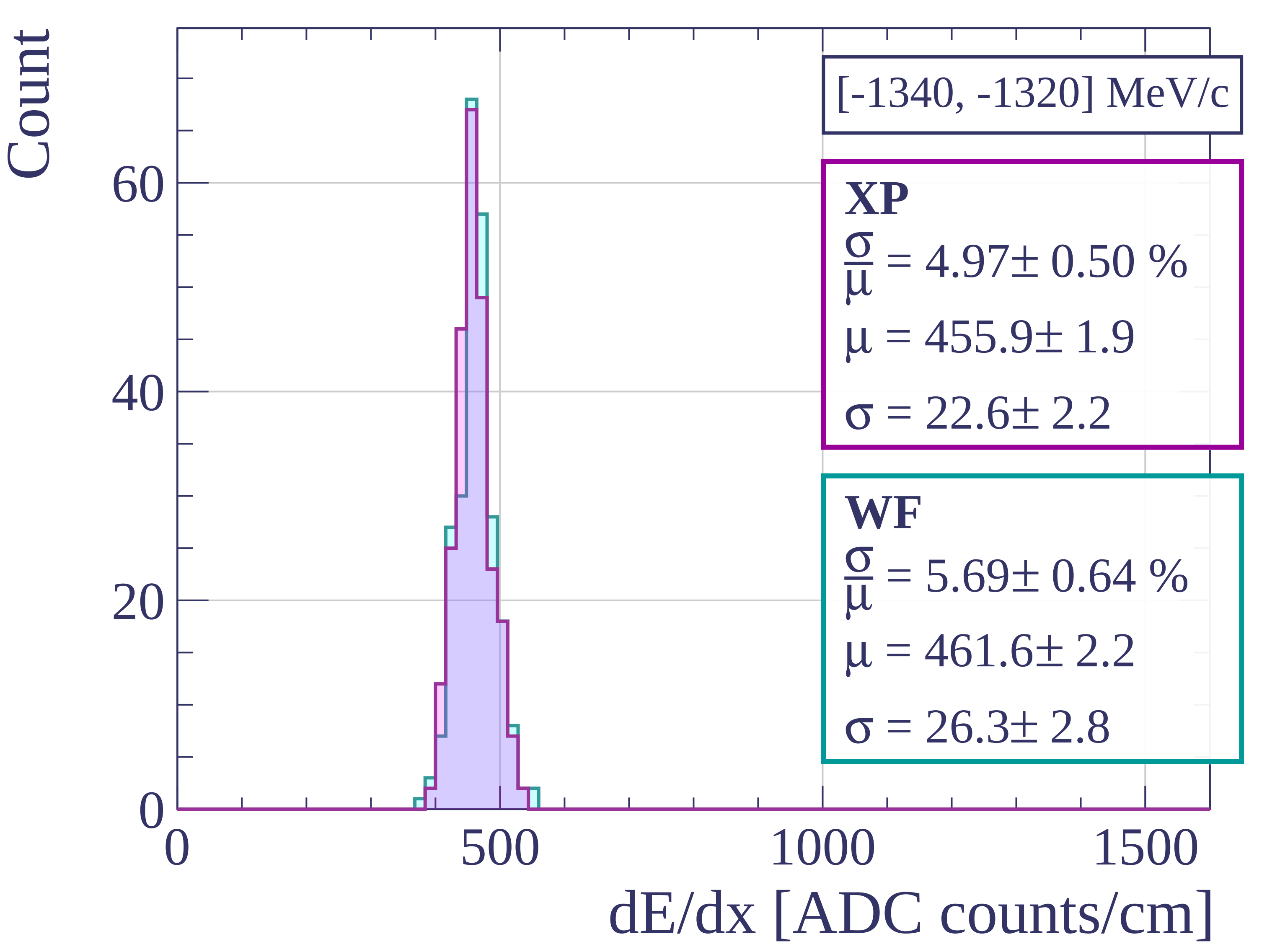

Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[-1420, -1400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.94± 0.42 %
μ = 461.1± 2.1
σ = 27.4± 1.8
WF
$\frac{\sigma}{\mu}$ = 5.96± 0.47 %
μ = 464.3± 2.2
σ = 27.7± 2.1


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1400, -1380] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.72± 0.47 %
μ = 461.6± 2.3
σ = 31.0± 2.0
WF
$\frac{\sigma}{\mu}$ = 6.64± 0.46 %
μ = 465.9± 2.4
σ = 31.0± 2.0

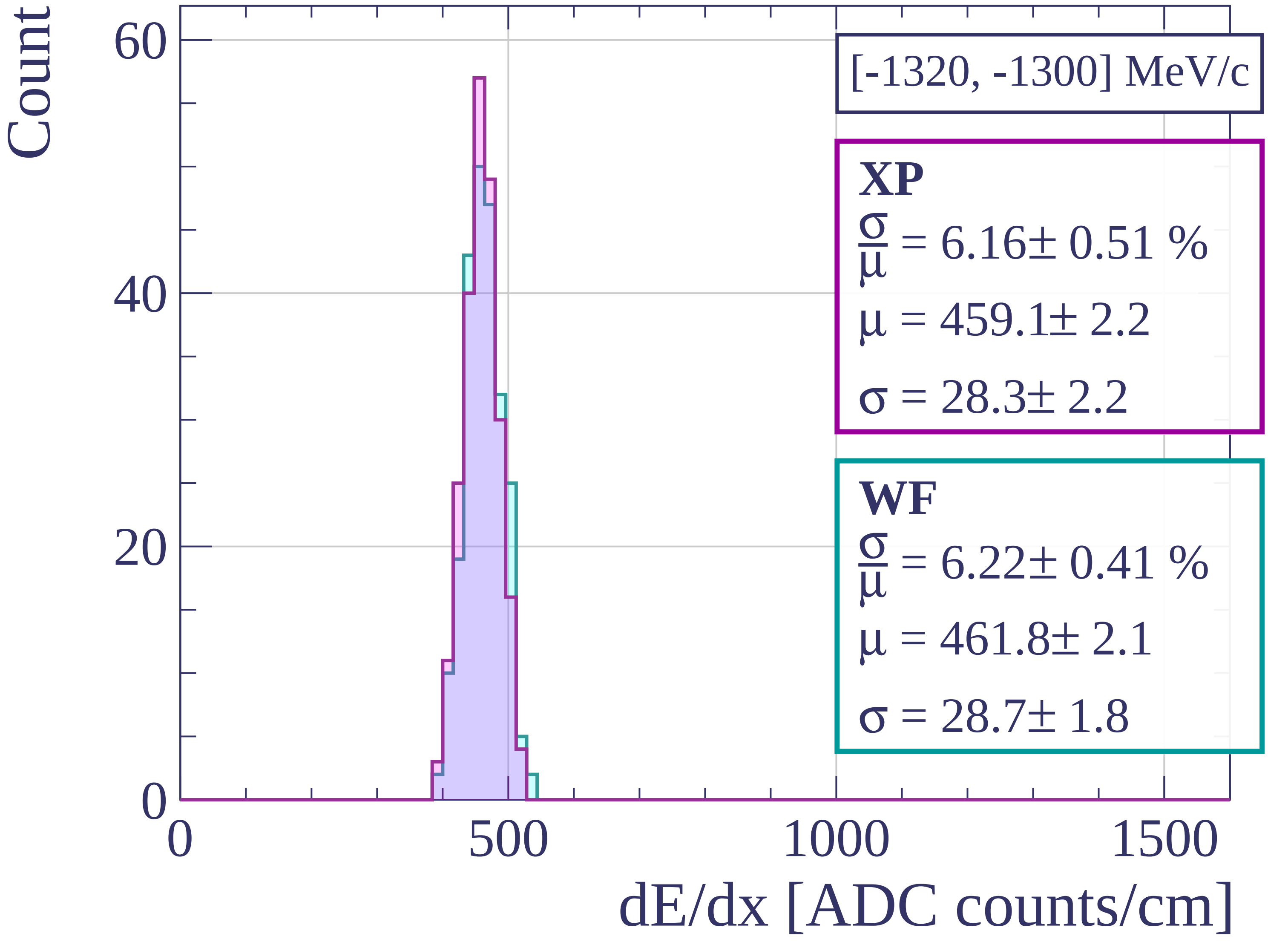

Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1380, -1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.33± 0.39 %
μ = 459.2± 2.0
σ = 24.5± 1.7
WF
$\frac{\sigma}{\mu}$ = 5.68± 0.52 %
μ = 463.5± 2.0
σ = 26.3± 2.3

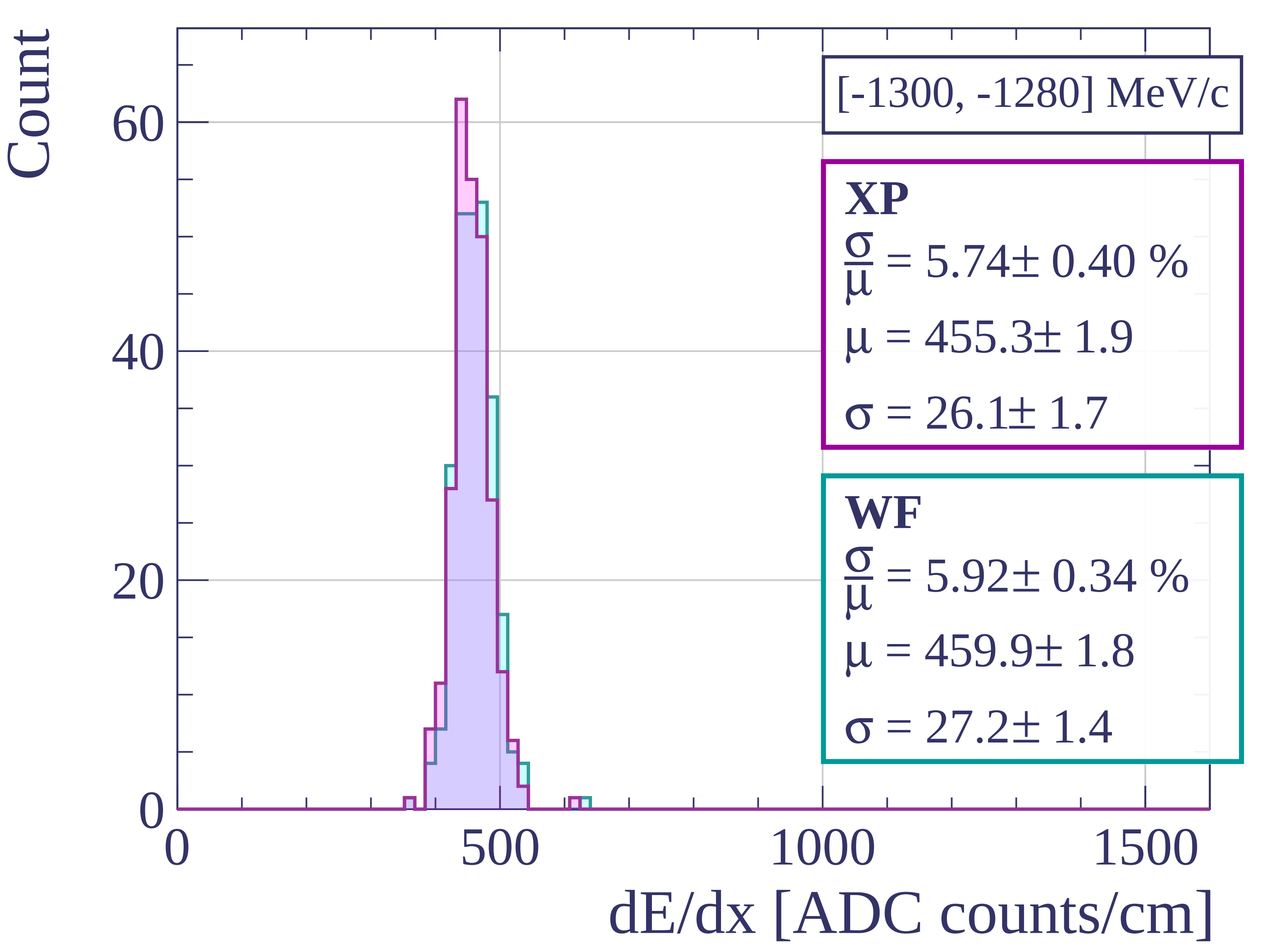


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1360, -1340] MeV/c
XP
σ/μ = 5.77± 0.39 %
μ = 459.2± 2.0
σ = 26.5± 1.7
WF
σ/μ = 6.19± 0.38 %
μ = 464.3± 2.1
σ = 28.7± 1.6

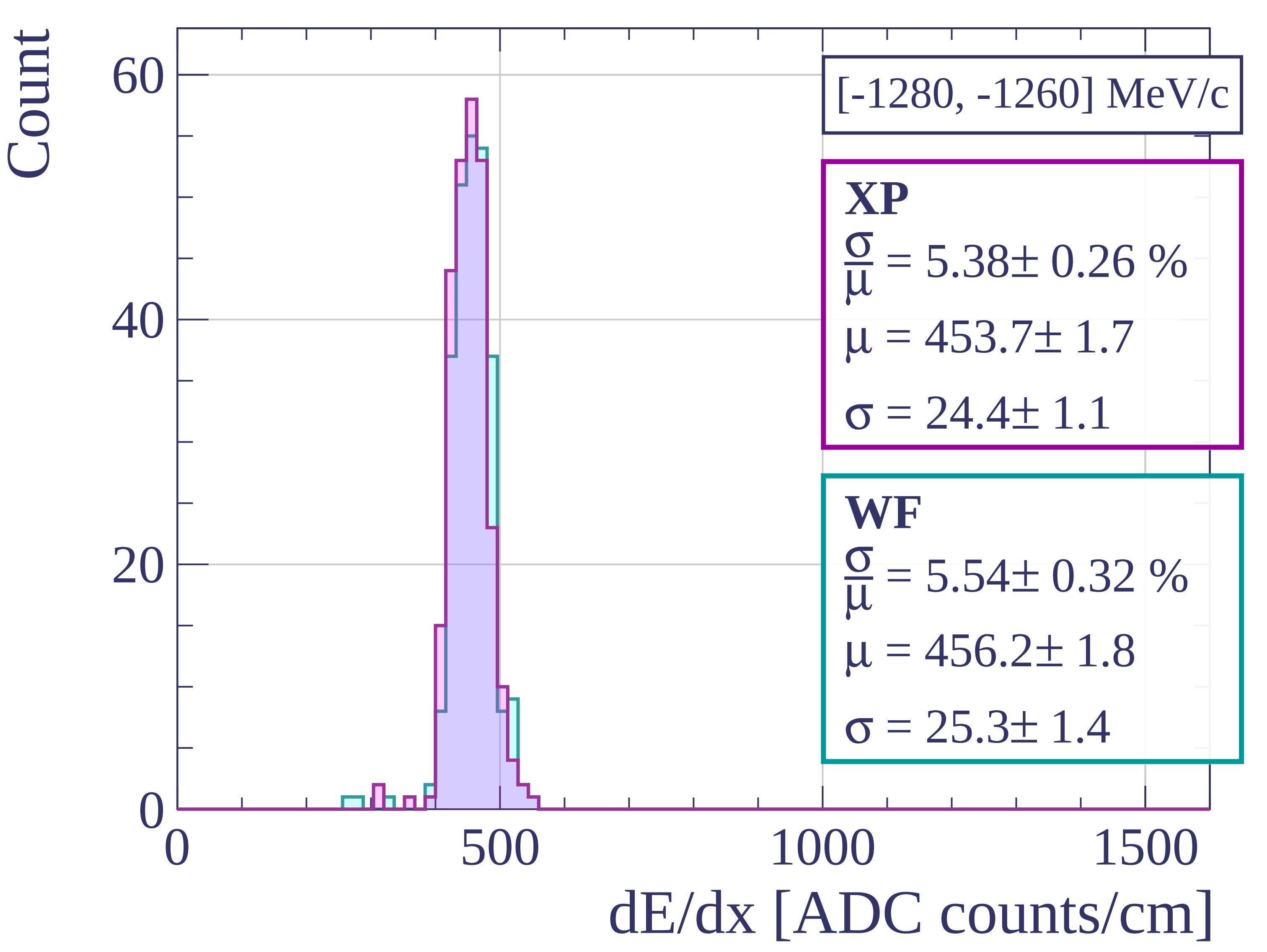


Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1340, -1320] MeV/c
XP
σ/μ = 4.97± 0.50 %
μ = 455.9± 1.9
σ = 22.6± 2.2
WF
σ/μ = 5.69± 0.64 %
μ = 461.6± 2.2
σ = 26.3± 2.8

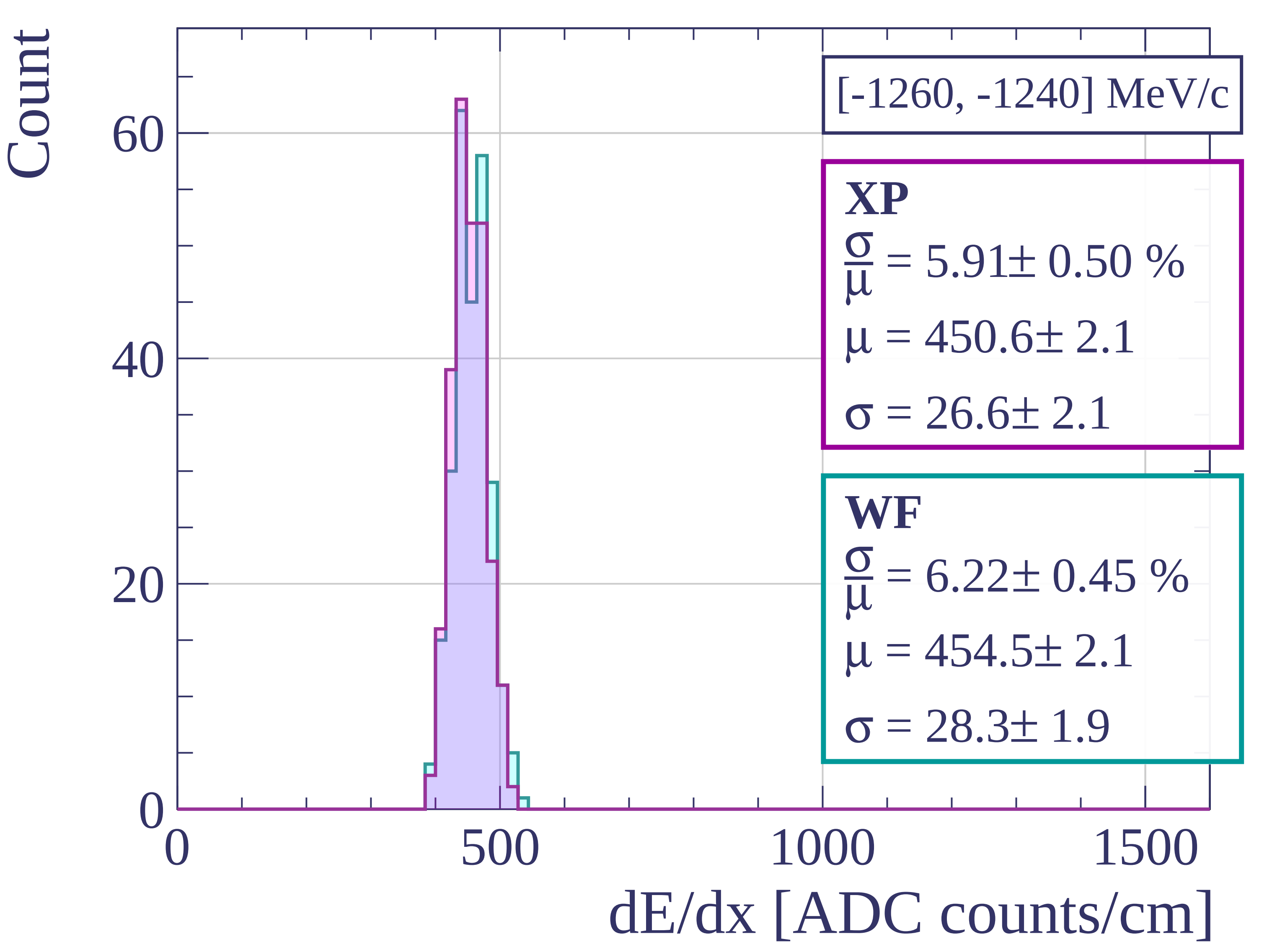

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1320, -1300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.16± 0.51 %
μ = 459.1± 2.2
σ = 28.3± 2.2
WF
$\frac{\sigma}{\mu}$ = 6.22± 0.41 %
μ = 461.8± 2.1
σ = 28.7± 1.8

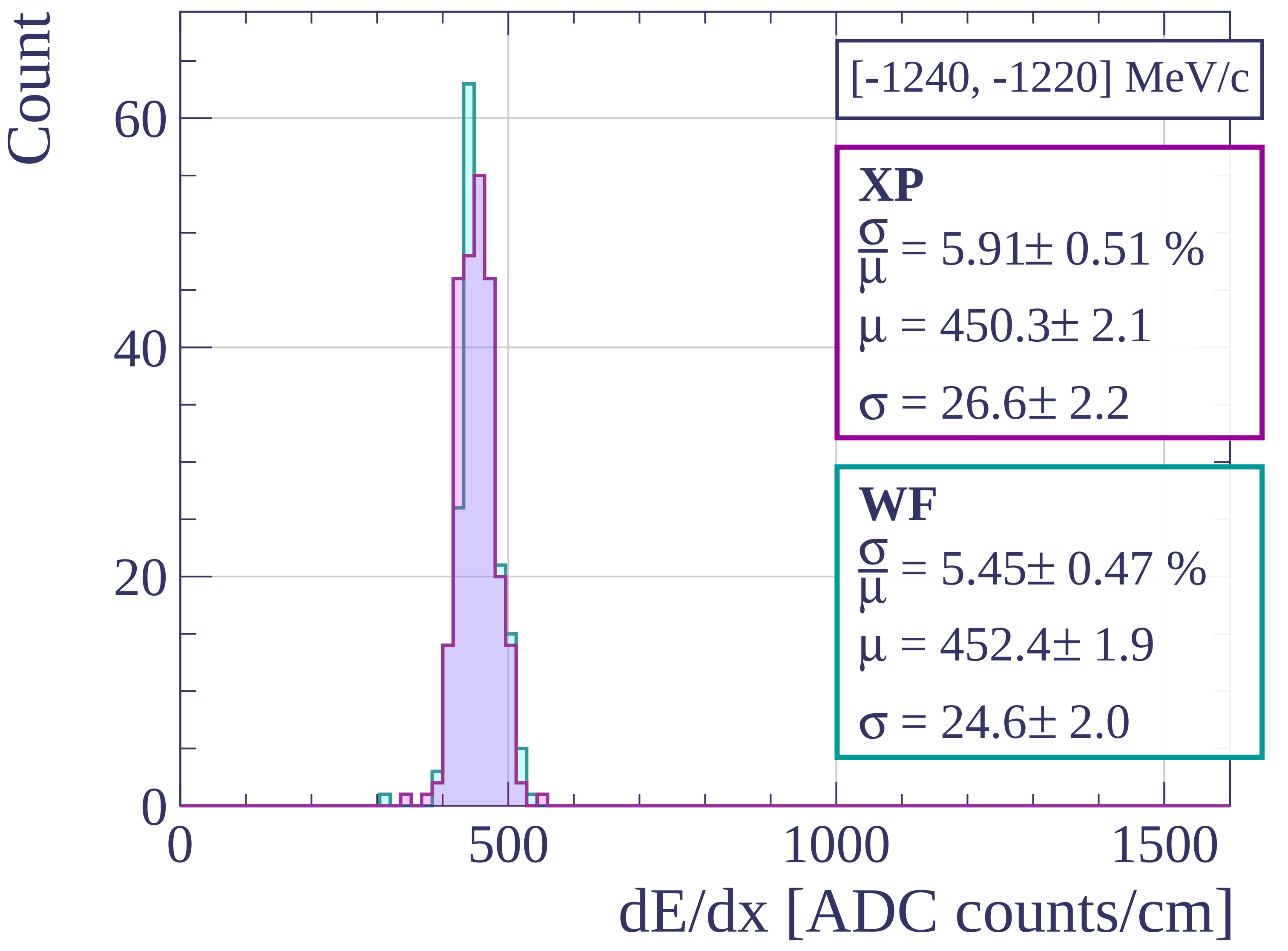


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1300, -1280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.74± 0.40 %
μ = 455.3± 1.9
σ = 26.1± 1.7
WF
$\frac{\sigma}{\mu}$ = 5.92± 0.34 %
μ = 459.9± 1.8
σ = 27.2± 1.4

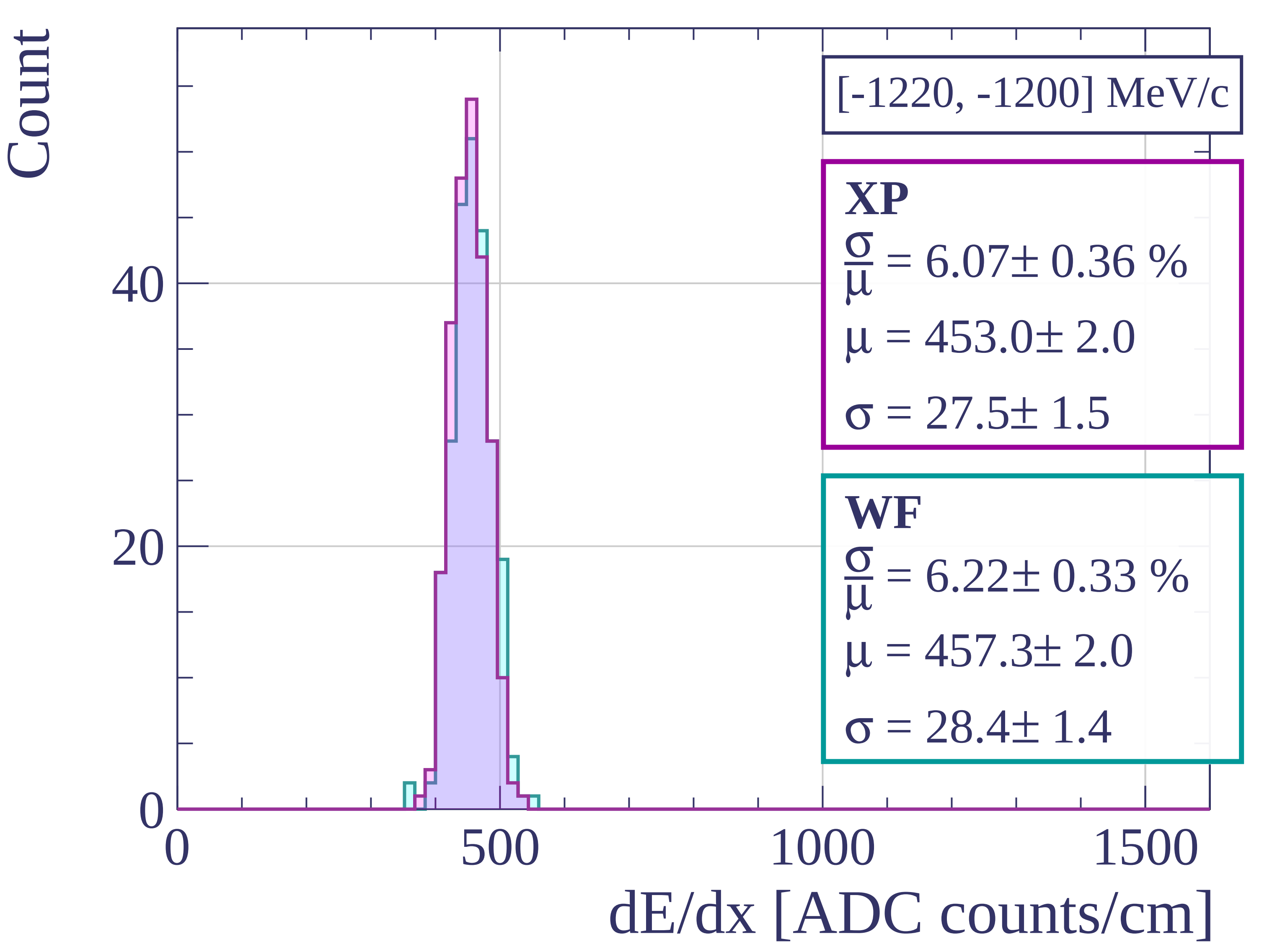


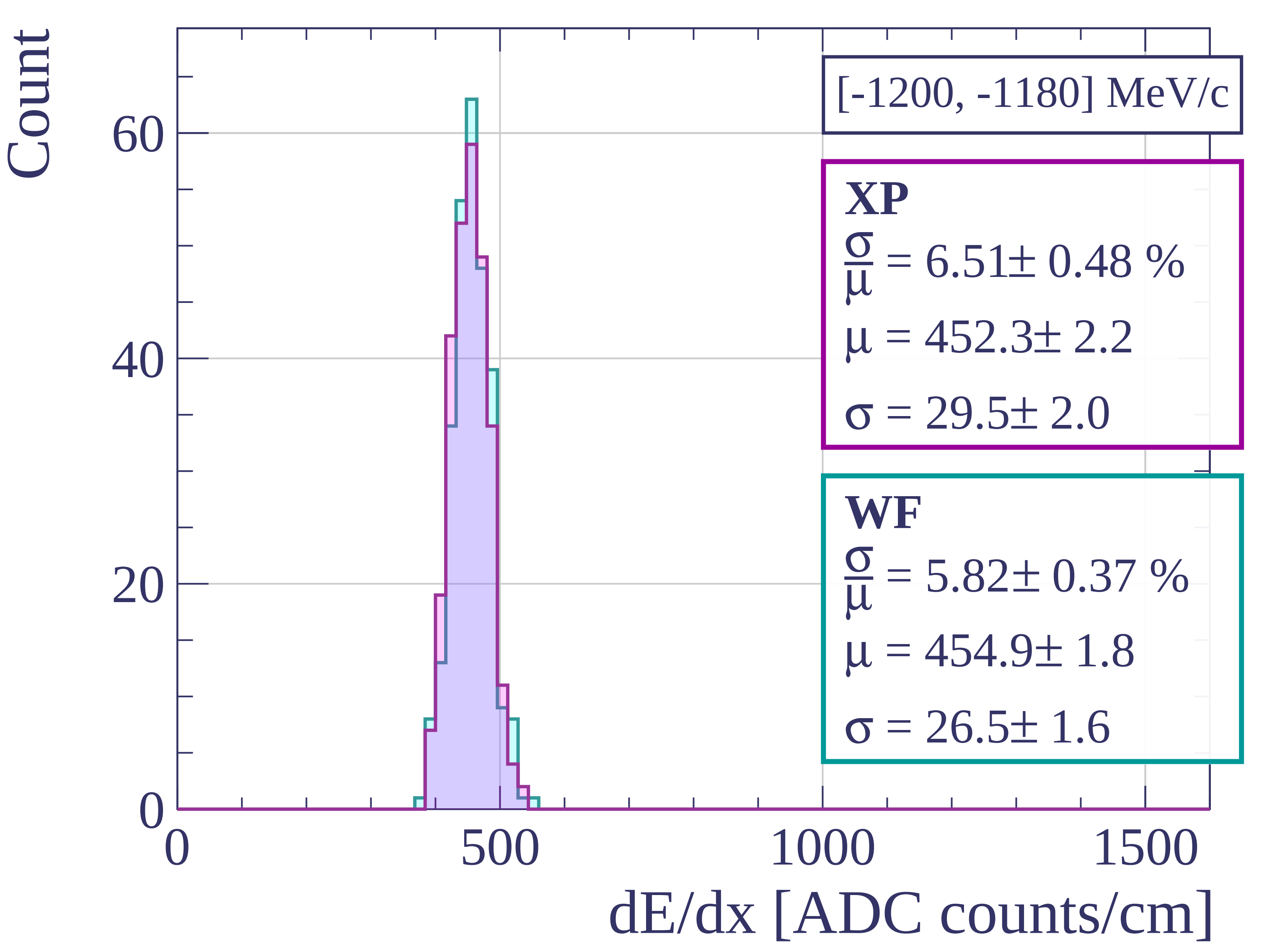
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1280, -1260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.38± 0.26 %
μ = 453.7± 1.7
σ = 24.4± 1.1
WF
$\frac{\sigma}{\mu}$ = 5.54± 0.32 %
μ = 456.2± 1.8
σ = 25.3± 1.4


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1260, -1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.91± 0.50 %
μ = 450.6± 2.1
σ = 26.6± 2.1
WF
$\frac{\sigma}{\mu}$ = 6.22± 0.45 %
μ = 454.5± 2.1
σ = 28.3± 1.9

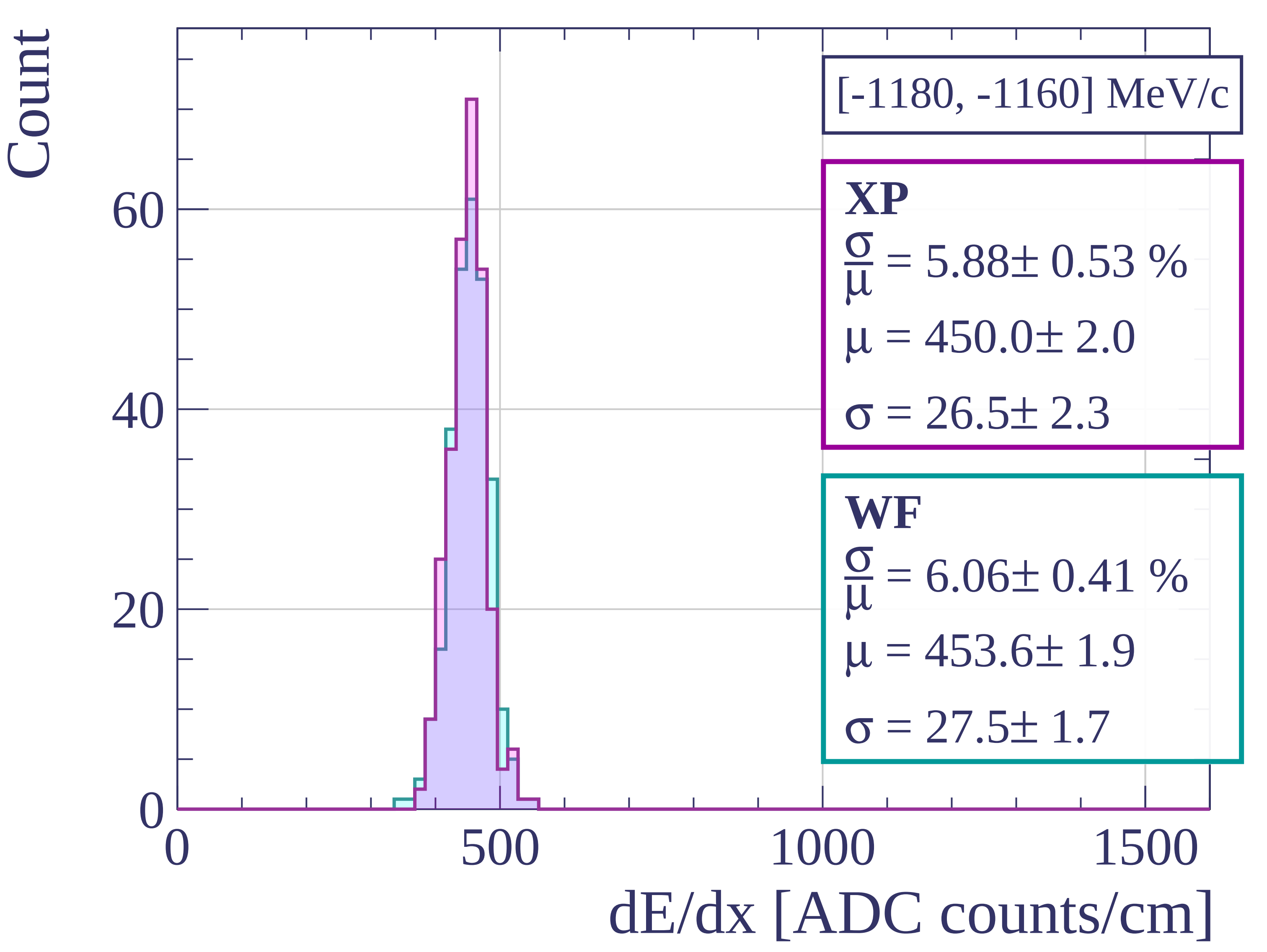


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1240, -1220] MeV/c
XP
σ/μ = 5.91± 0.51 %
μ = 450.3± 2.1
σ = 26.6± 2.2
WF
σ/μ = 5.45± 0.47 %
μ = 452.4± 1.9
σ = 24.6± 2.0

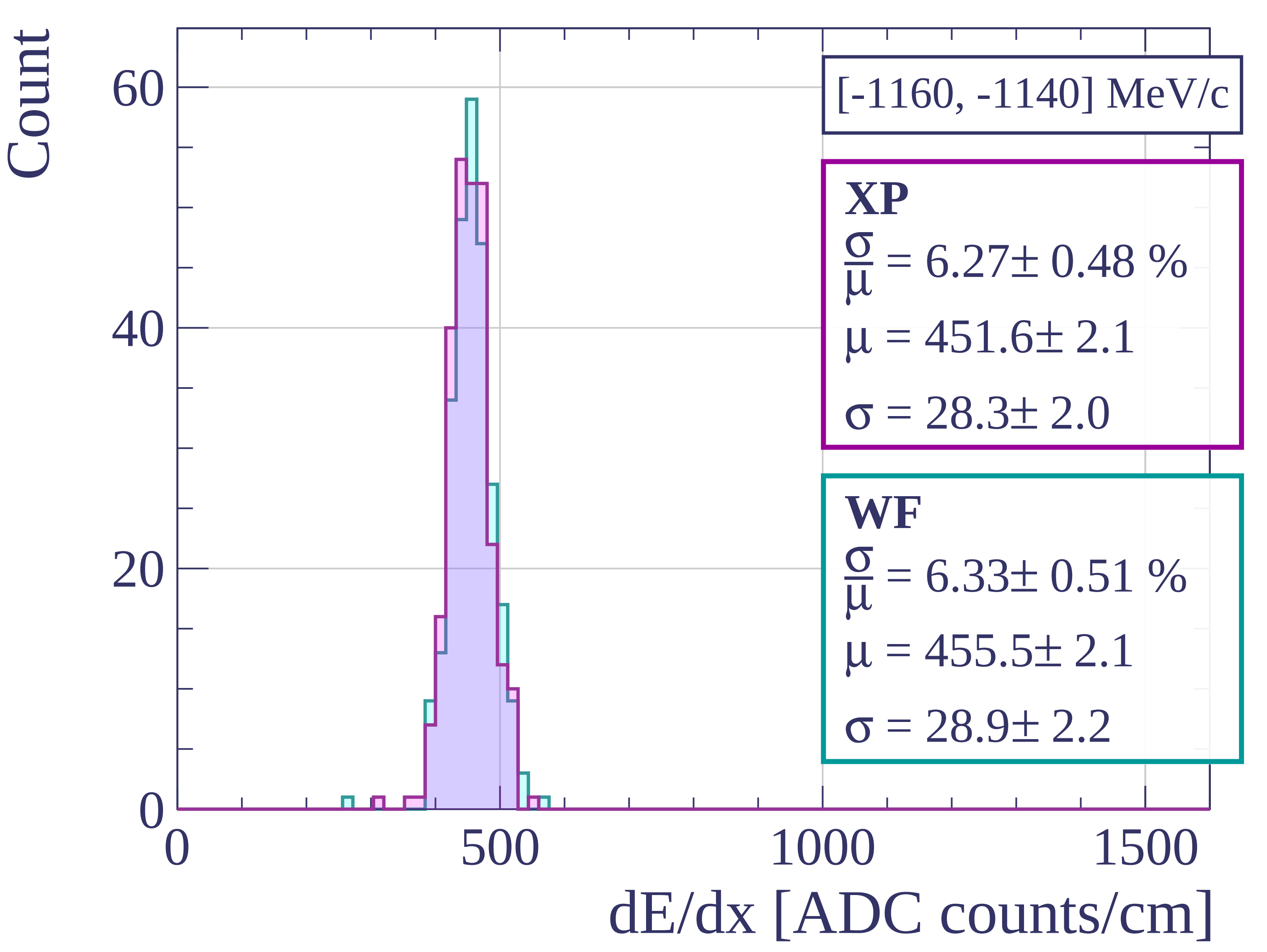

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1220, -1200] MeV/c
XP
σ/μ = 6.07± 0.36 %
μ = 453.0± 2.0
σ = 27.5± 1.5
WF
σ/μ = 6.22± 0.33 %
μ = 457.3± 2.0
σ = 28.4± 1.4



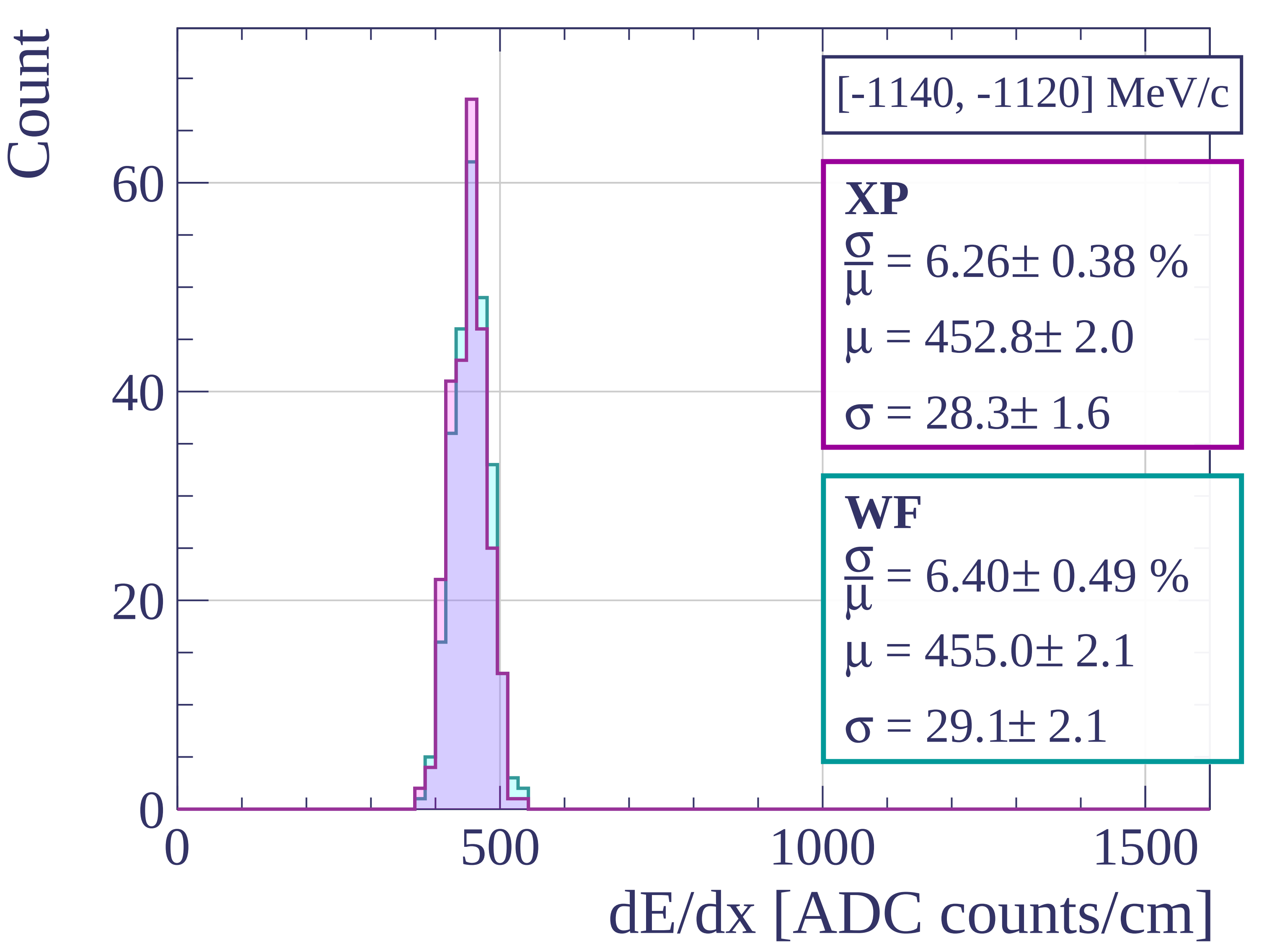
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1200, -1180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.51± 0.48 %
μ = 452.3± 2.2
σ = 29.5± 2.0
WF
$\frac{\sigma}{\mu}$ = 5.82± 0.37 %
μ = 454.9± 1.8
σ = 26.5± 1.6

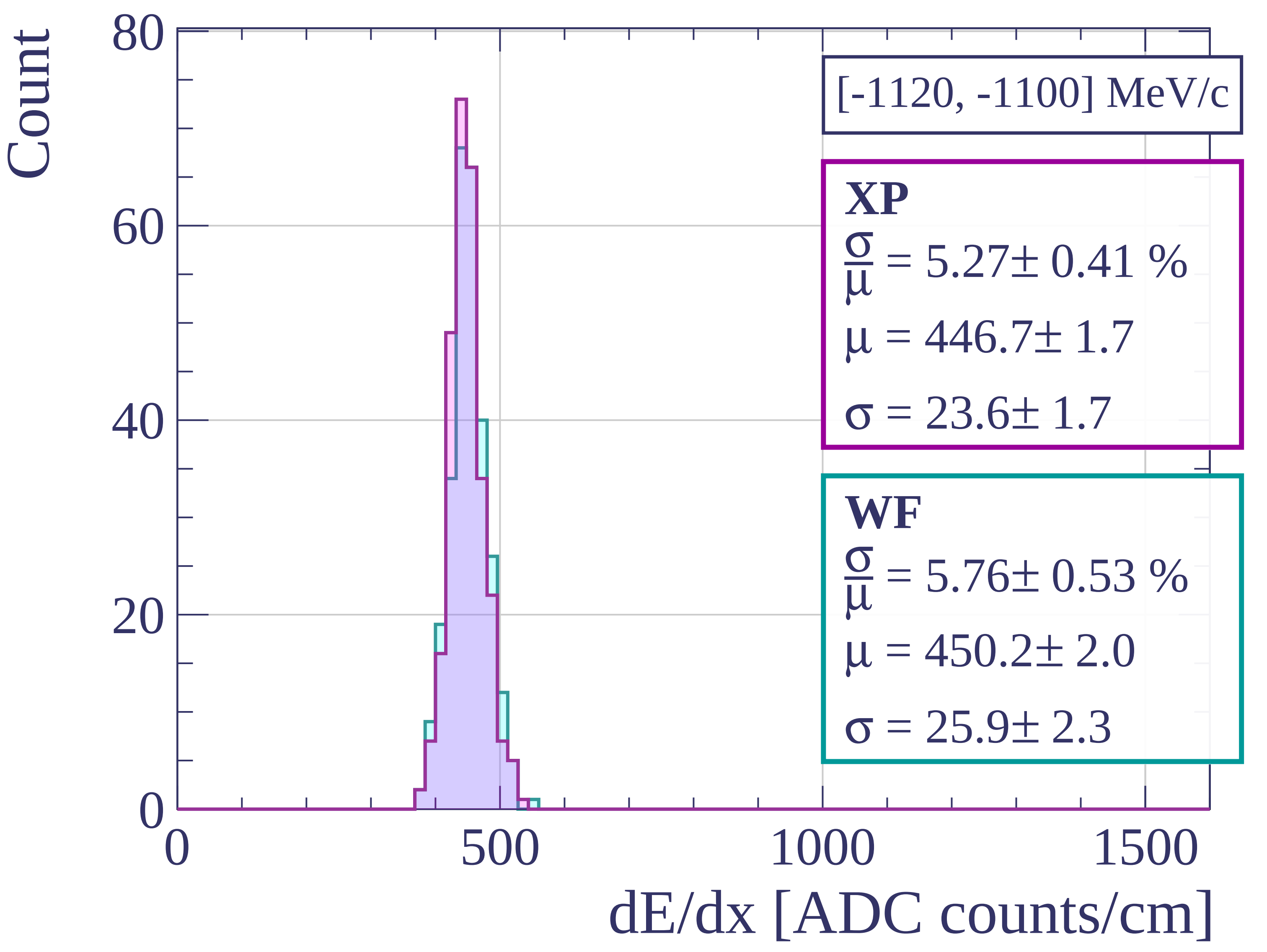


Count
dE/dx [ADC counts/cm]
[-1180, -1160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.88± 0.53 %
μ = 450.0± 2.0
σ = 26.5± 2.3
WF
$\frac{\sigma}{\mu}$ = 6.06± 0.41 %
μ = 453.6± 1.9
σ = 27.5± 1.7
0
20
40
60
0
500
1000
1500

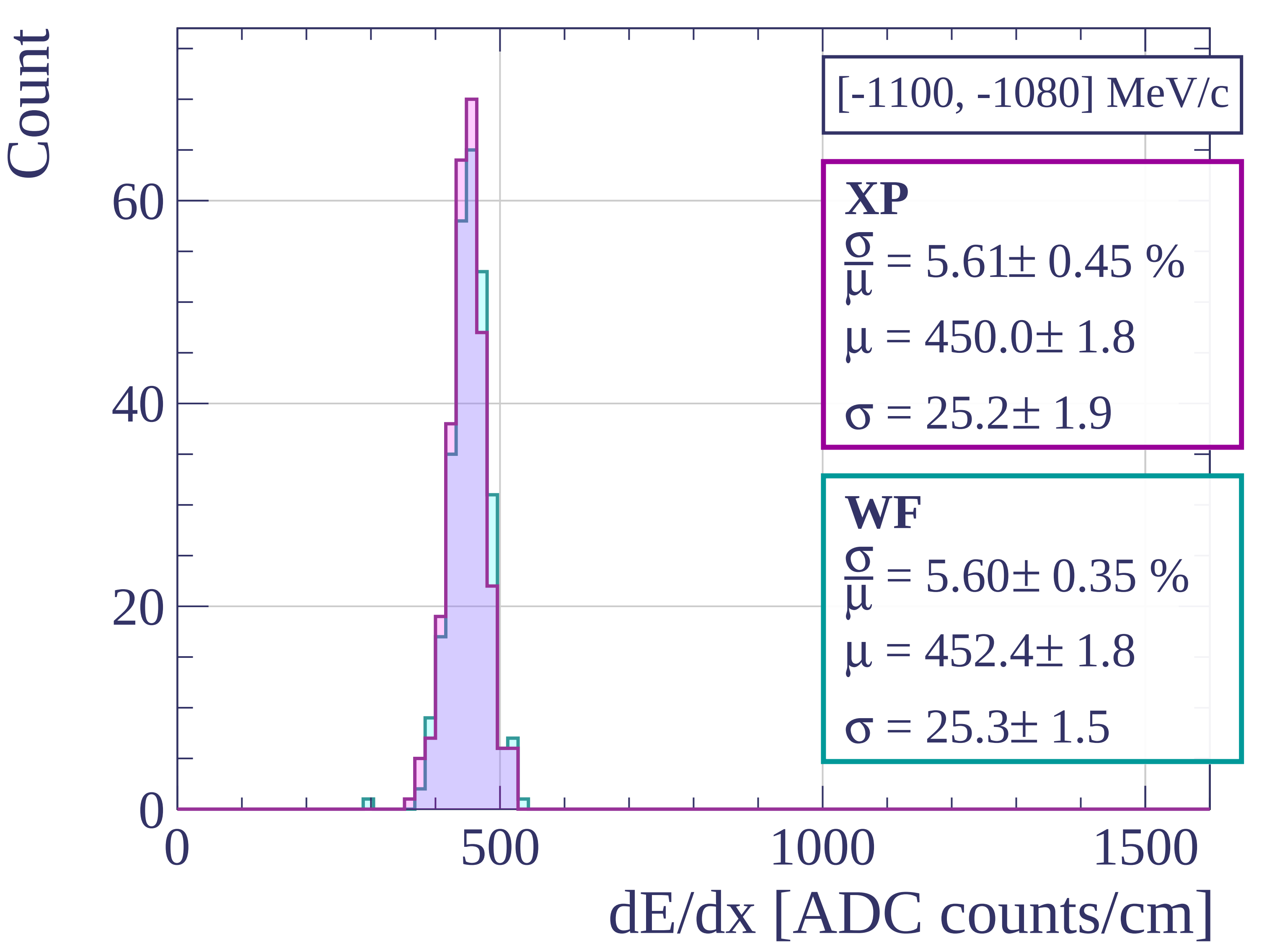


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1160, -1140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.27± 0.48 %
μ = 451.6± 2.1
σ = 28.3± 2.0
WF
$\frac{\sigma}{\mu}$ = 6.33± 0.51 %
μ = 455.5± 2.1
σ = 28.9± 2.2

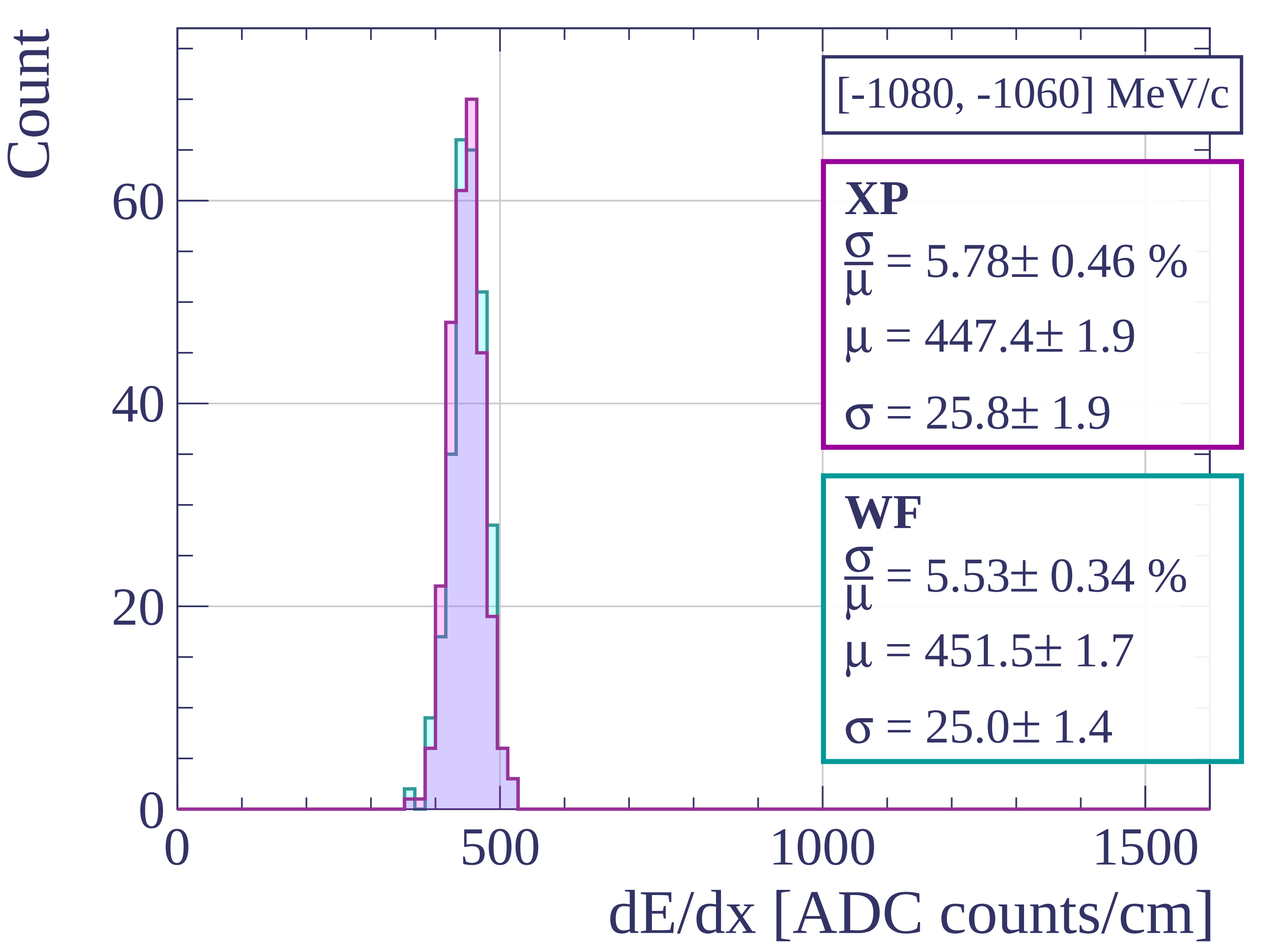

Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[-1140, -1120] MeV/c
XP
σ/μ = 6.26± 0.38 %
μ = 452.8± 2.0
σ = 28.3± 1.6
WF
σ/μ = 6.40± 0.49 %
μ = 455.0± 2.1
σ = 29.1± 2.1

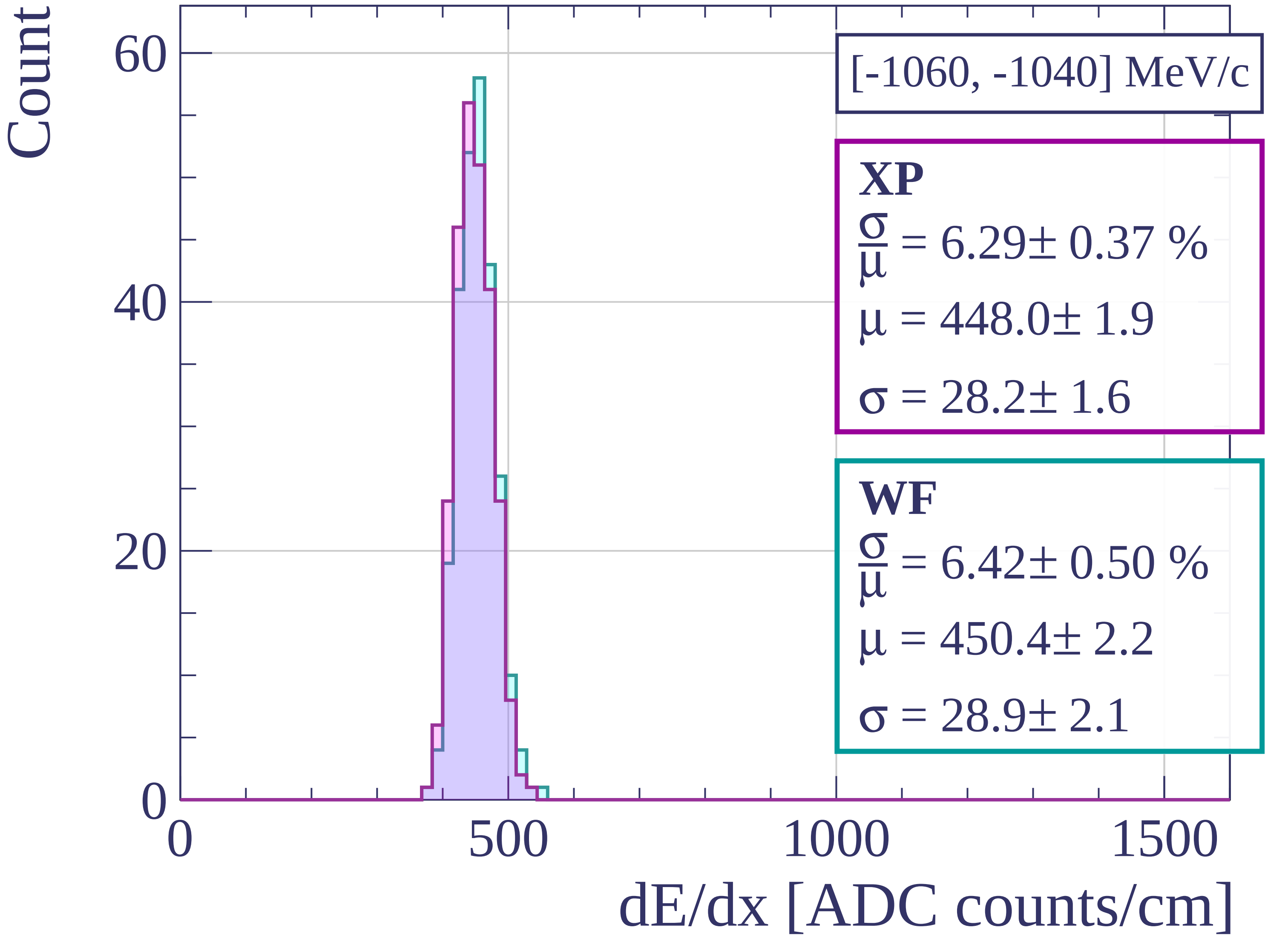


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
20
40
60
80
[-1120, -1100] MeV/c
XP
σ/μ = 5.27± 0.41 %
μ = 446.7± 1.7
σ = 23.6± 1.7
WF
σ/μ = 5.76± 0.53 %
μ = 450.2± 2.0
σ = 25.9± 2.3

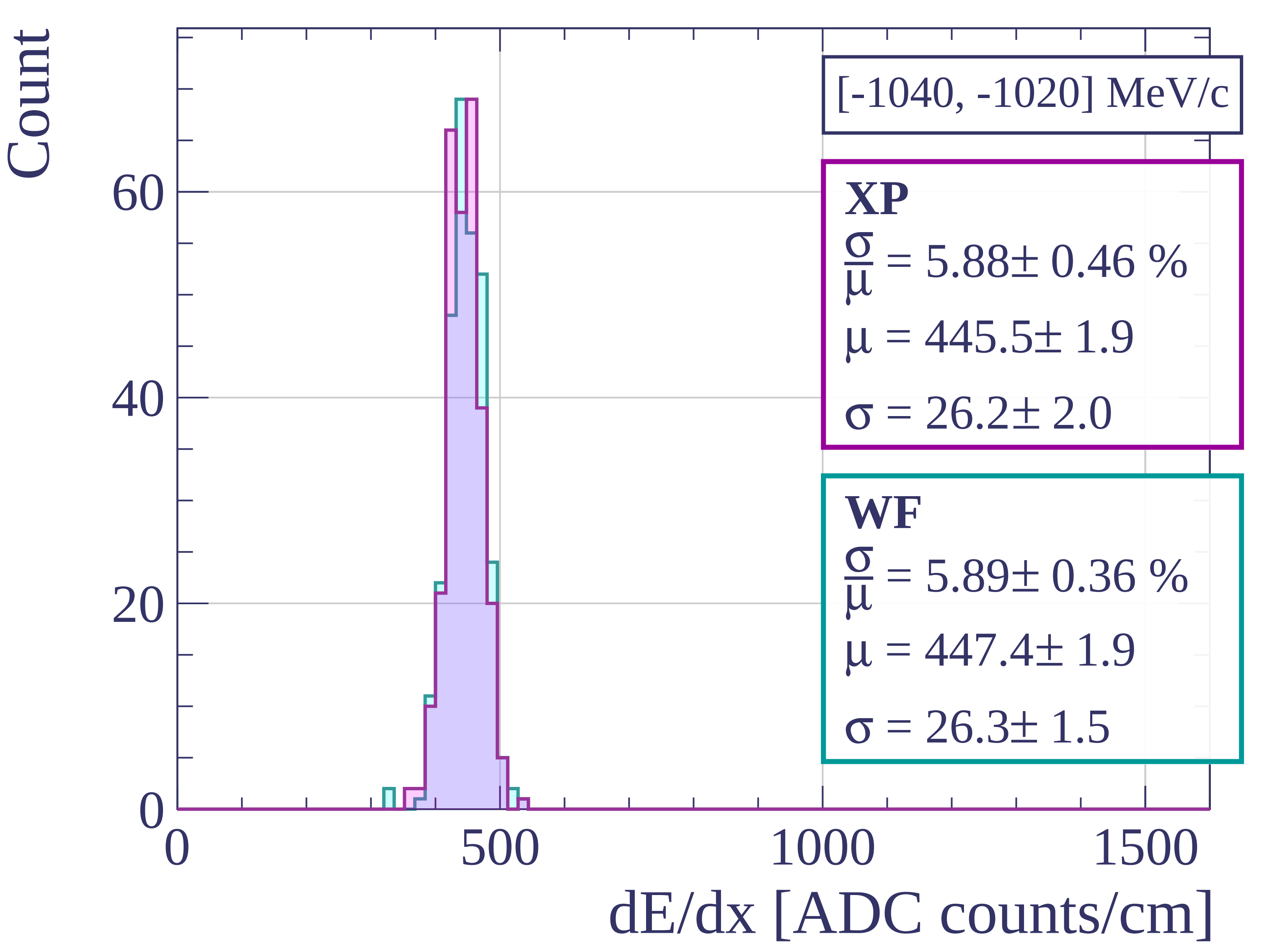


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1100, -1080] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.61± 0.45 %
μ = 450.0± 1.8
σ = 25.2± 1.9
WF
$\frac{\sigma}{\mu}$ = 5.60± 0.35 %
μ = 452.4± 1.8
σ = 25.3± 1.5

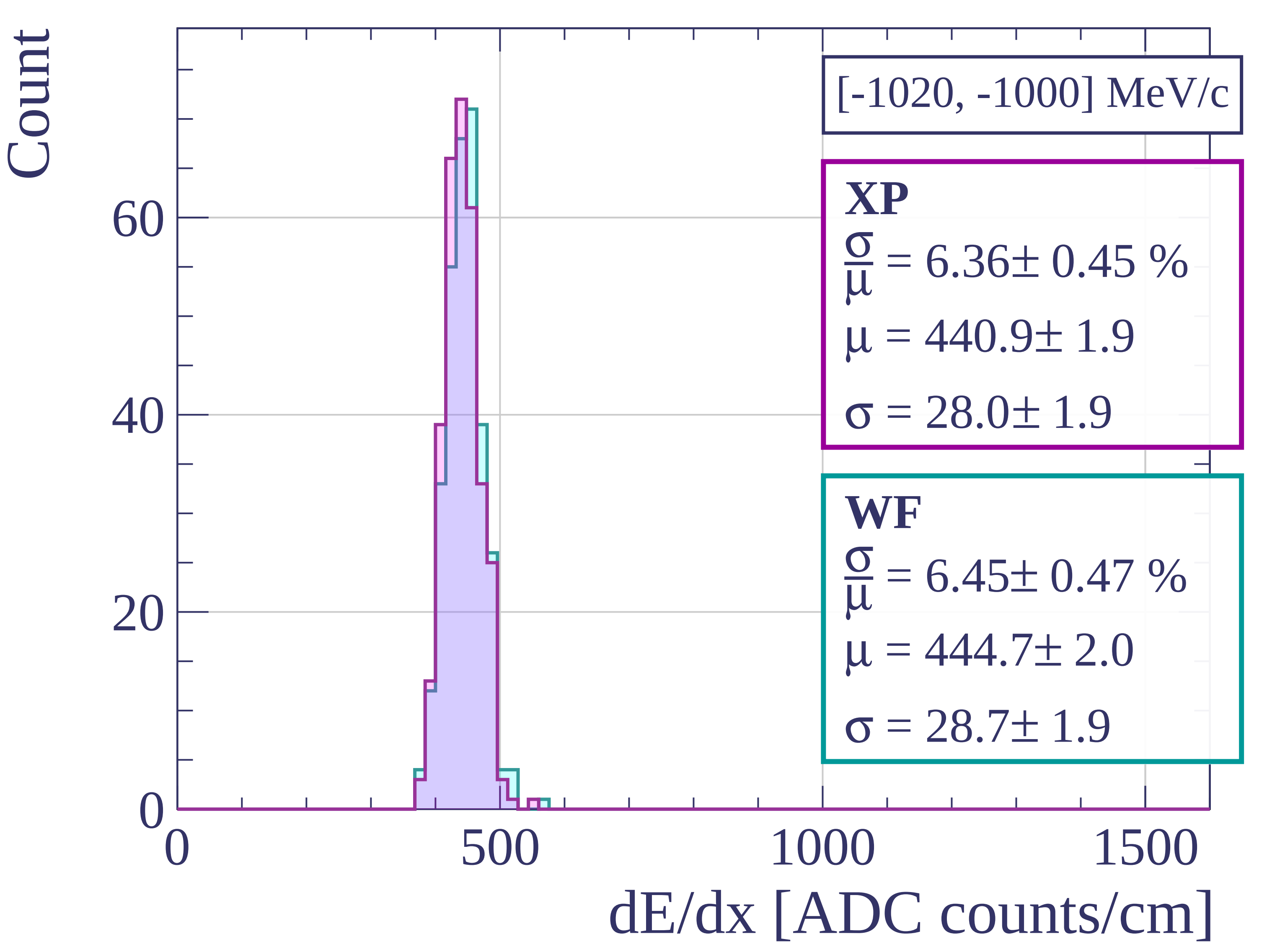

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1080, -1060] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.78± 0.46 %
μ = 447.4± 1.9
σ = 25.8± 1.9
WF
$\frac{\sigma}{\mu}$ = 5.53± 0.34 %
μ = 451.5± 1.7
σ = 25.0± 1.4



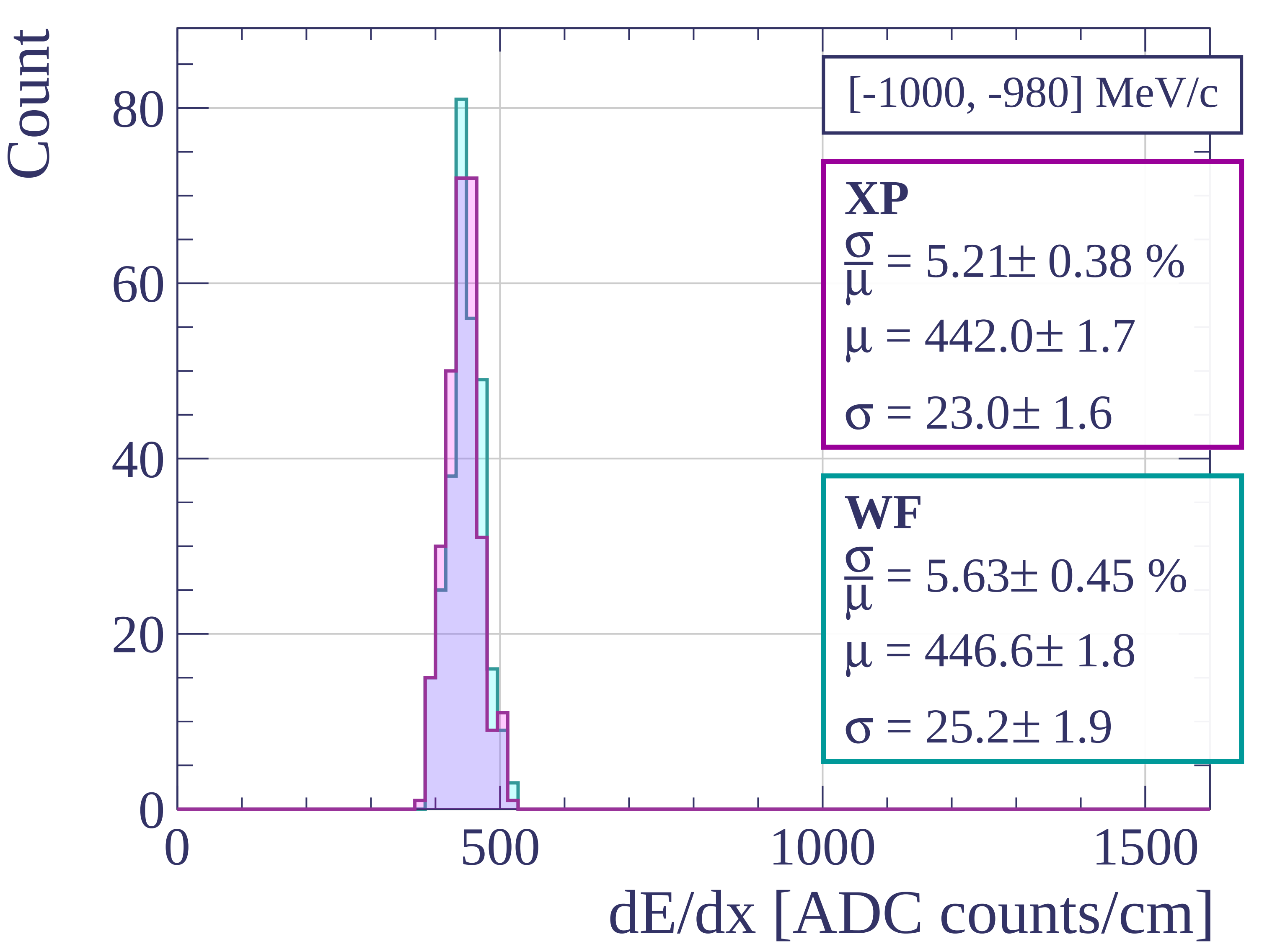
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1060, -1040] MeV/c
XP
σ/μ = 6.29± 0.37 %
μ = 448.0± 1.9
σ = 28.2± 1.6
WF
σ/μ = 6.42± 0.50 %
μ = 450.4± 2.2
σ = 28.9± 2.1



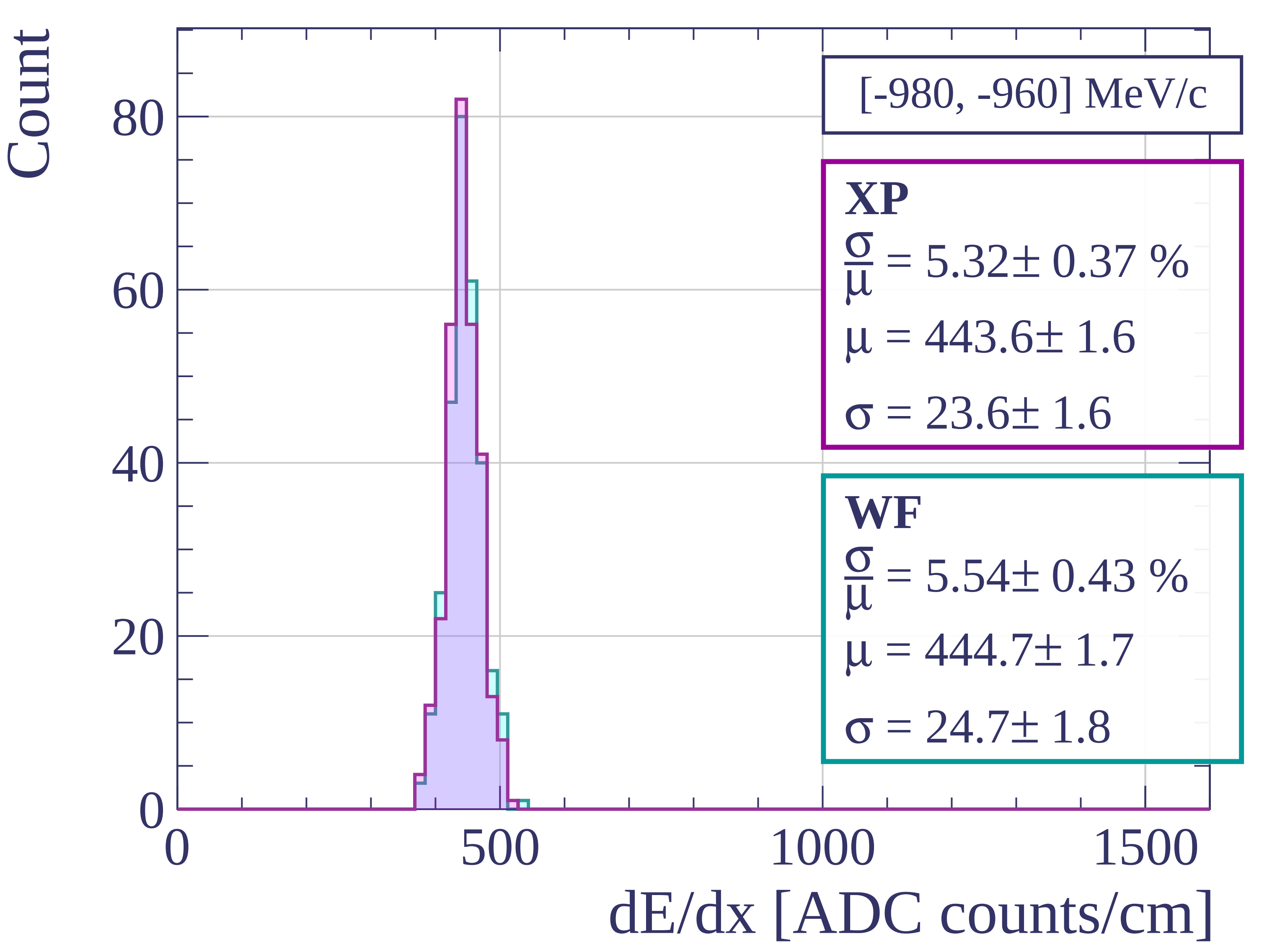
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1040, -1020] MeV/c
XP
σ/μ = 5.88± 0.46 %
μ = 445.5± 1.9
σ = 26.2± 2.0
WF
σ/μ = 5.89± 0.36 %
μ = 447.4± 1.9
σ = 26.3± 1.5



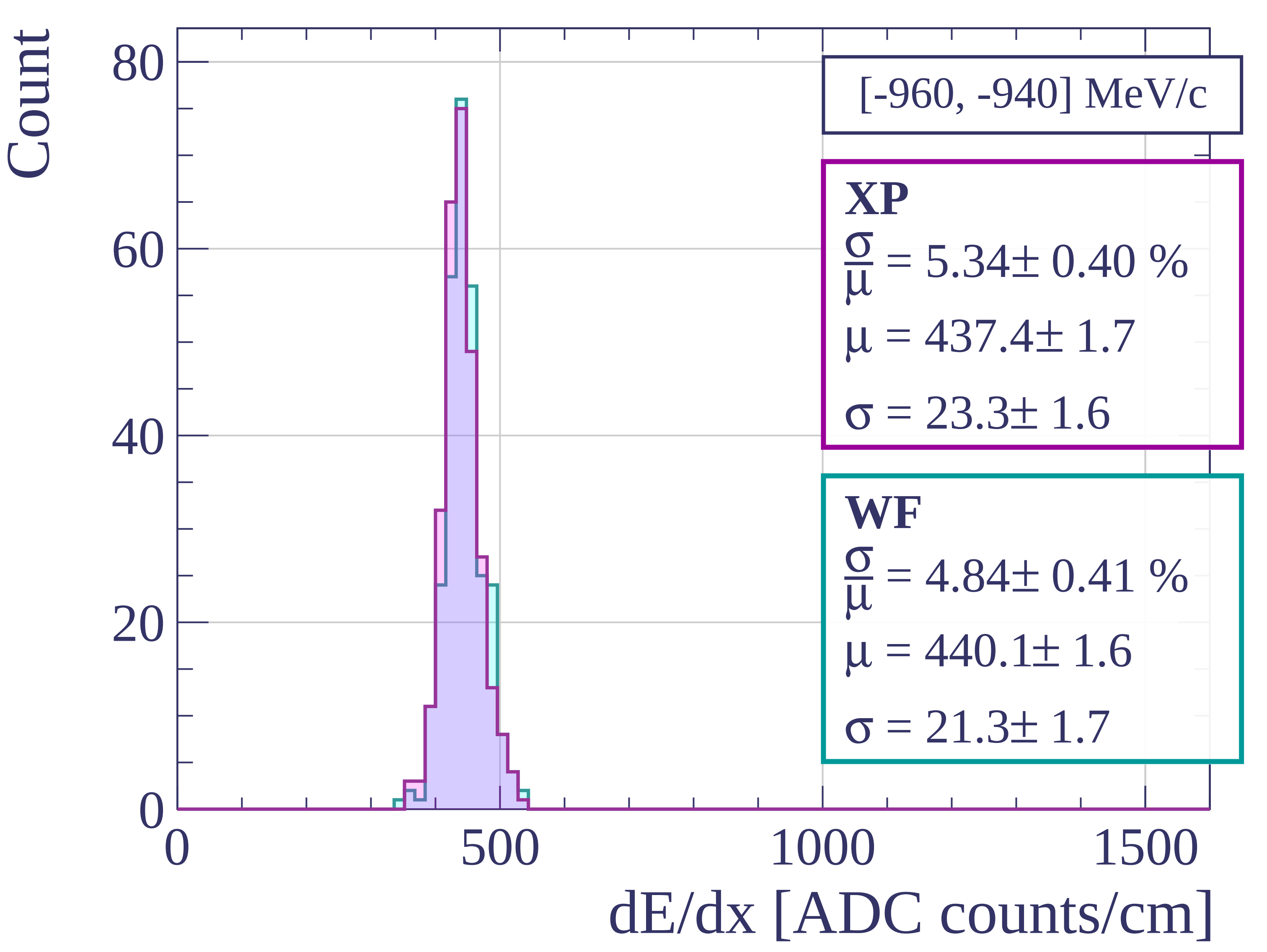
Count
dE/dx [ADC counts/cm]
[-1020, -1000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.36± 0.45 %
μ = 440.9± 1.9
σ = 28.0± 1.9
WF
$\frac{\sigma}{\mu}$ = 6.45± 0.47 %
μ = 444.7± 2.0
σ = 28.7± 1.9

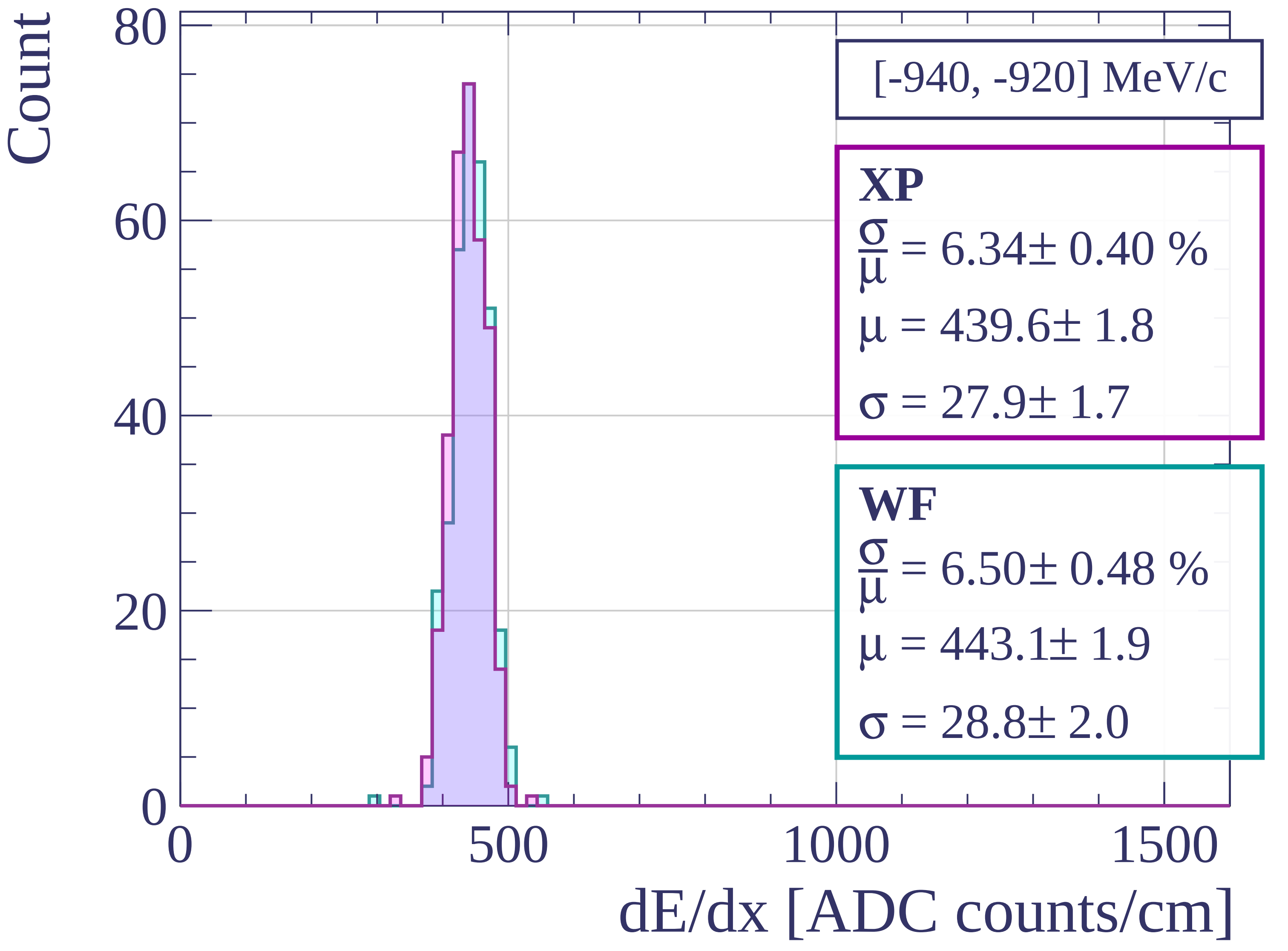


Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[-1000, -980] MeV/c
XP
σ/μ = 5.21± 0.38 %
μ = 442.0± 1.7
σ = 23.0± 1.6
WF
σ/μ = 5.63± 0.45 %
μ = 446.6± 1.8
σ = 25.2± 1.9

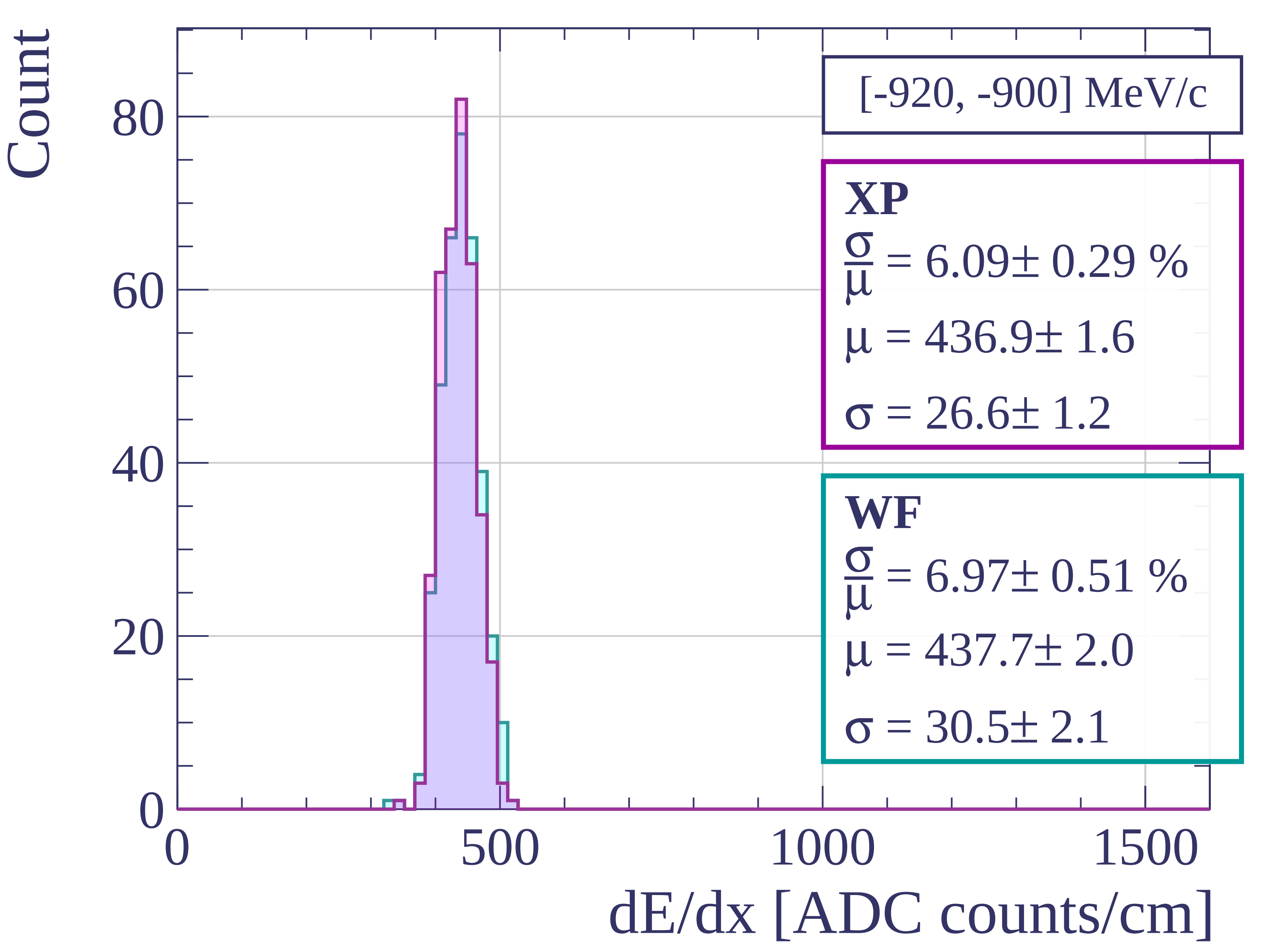


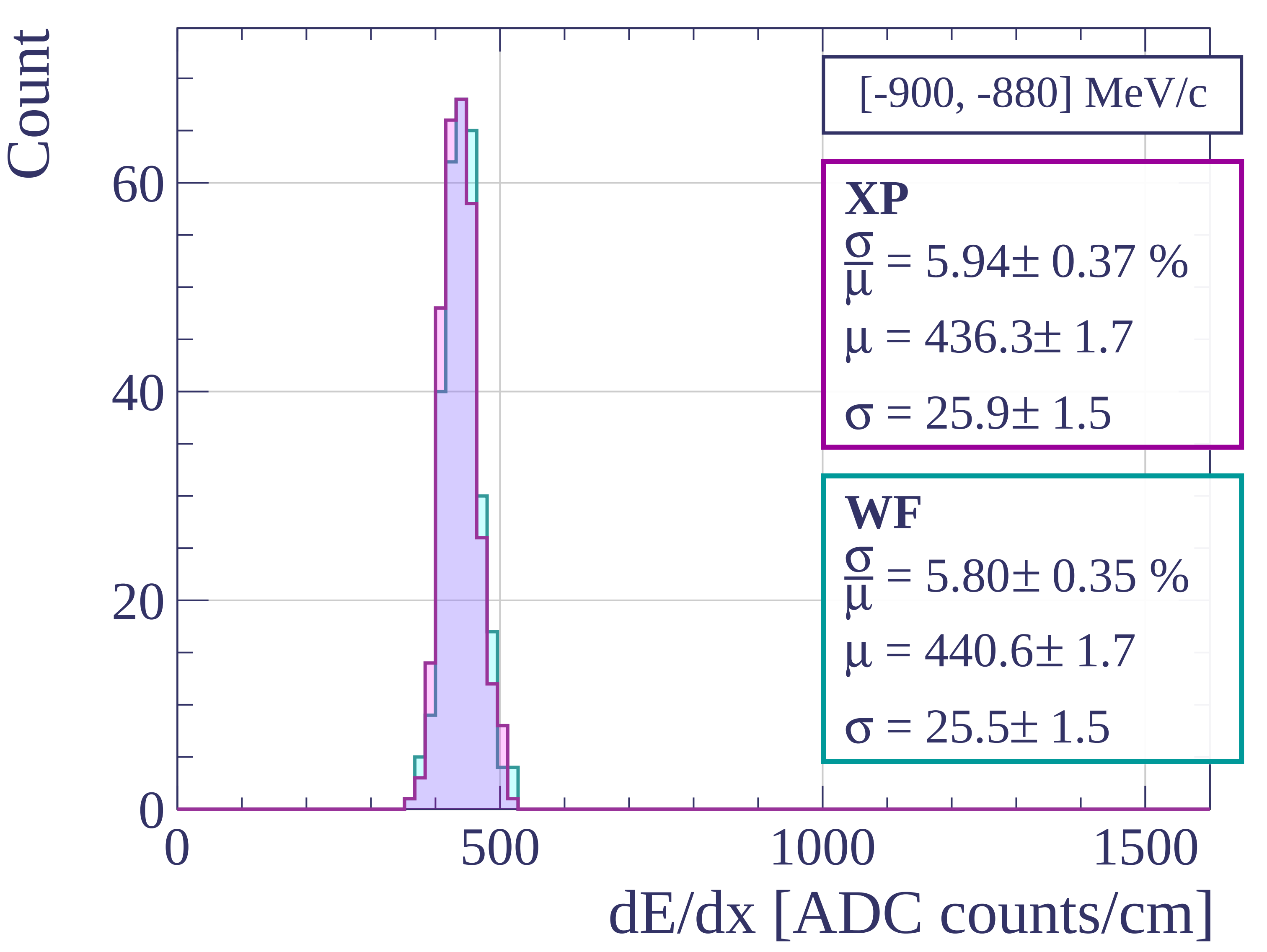
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-980, -960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.32± 0.37 %
μ = 443.6± 1.6
σ = 23.6± 1.6
WF
$\frac{\sigma}{\mu}$ = 5.54± 0.43 %
μ = 444.7± 1.7
σ = 24.7± 1.8


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-960, -940] MeV/c
XP
σ/μ = 5.34± 0.40 %
μ = 437.4± 1.7
σ = 23.3± 1.6
WF
σ/μ = 4.84± 0.41 %
μ = 440.1± 1.6
σ = 21.3± 1.7

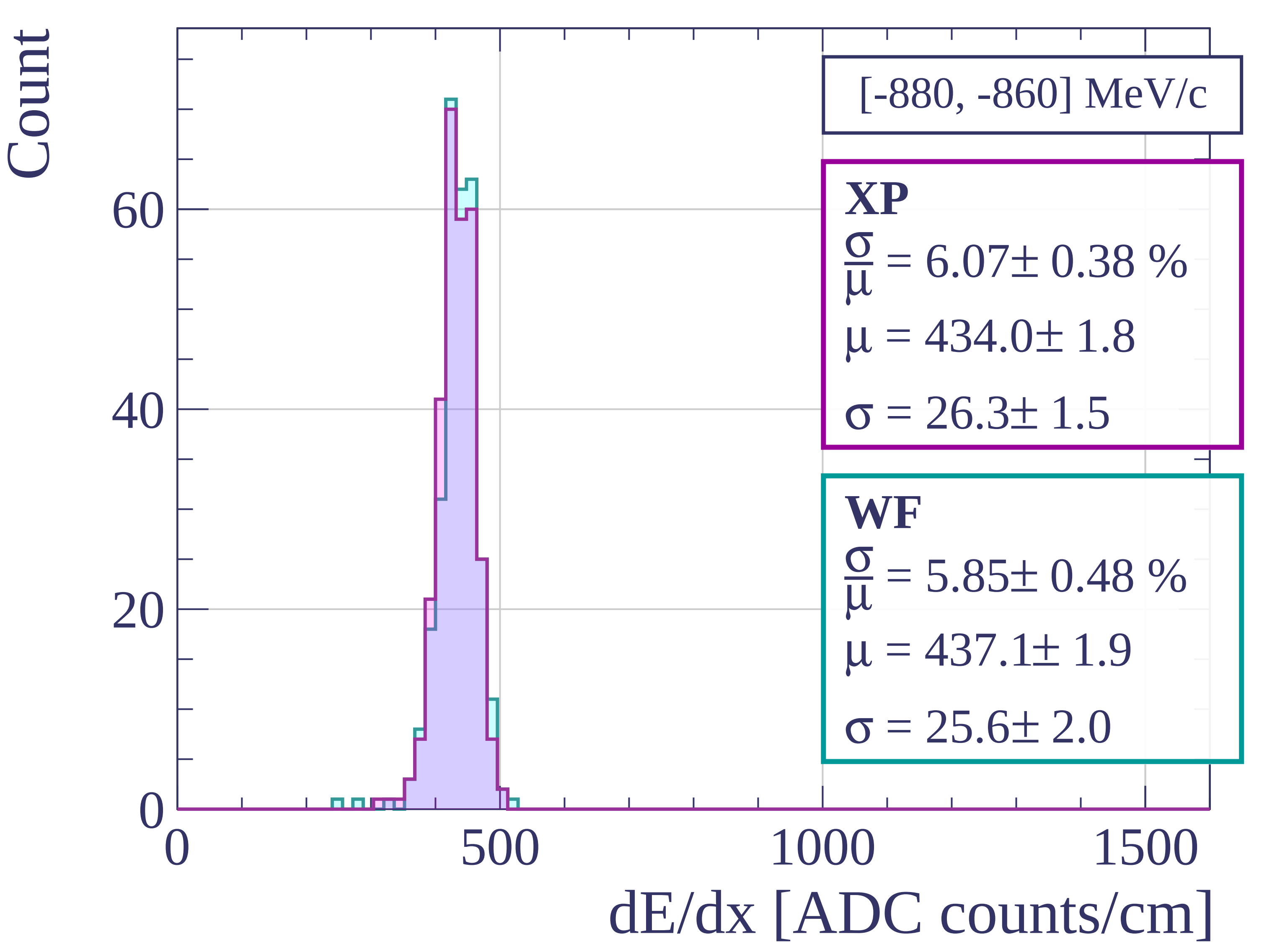


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
20
40
60
80
[-940, -920] MeV/c
XP
$\frac{\sigma}{\mu} = 6.34 \pm 0.40$ %
$\mu = 439.6 \pm 1.8$
$\sigma = 27.9 \pm 1.7$
WF
$\frac{\sigma}{\mu} = 6.50 \pm 0.48$ %
$\mu = 443.1 \pm 1.9$
$\sigma = 28.8 \pm 2.0$

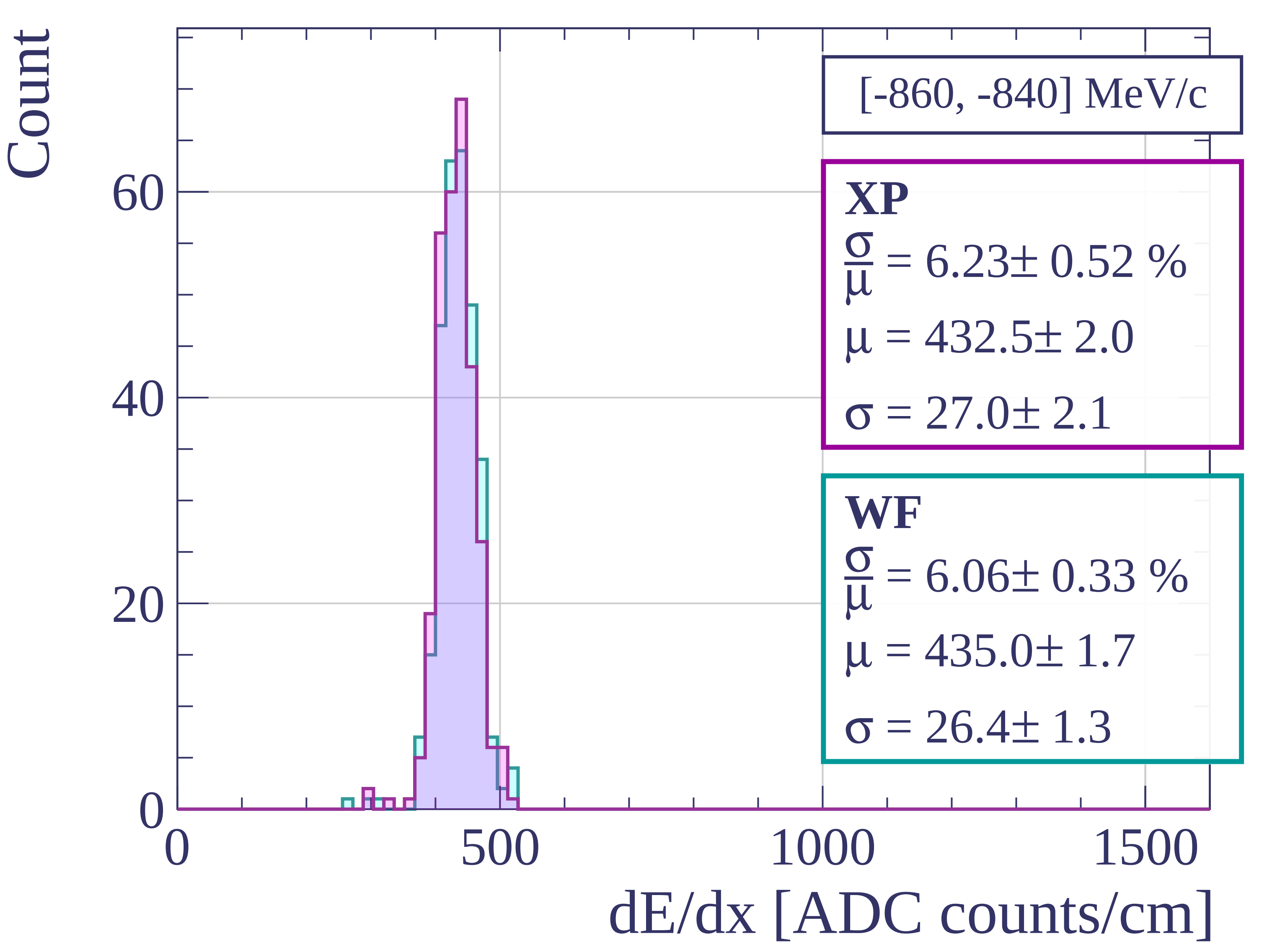

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-920, -900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.09± 0.29 %
μ = 436.9± 1.6
σ = 26.6± 1.2
WF
$\frac{\sigma}{\mu}$ = 6.97± 0.51 %
μ = 437.7± 2.0
σ = 30.5± 2.1



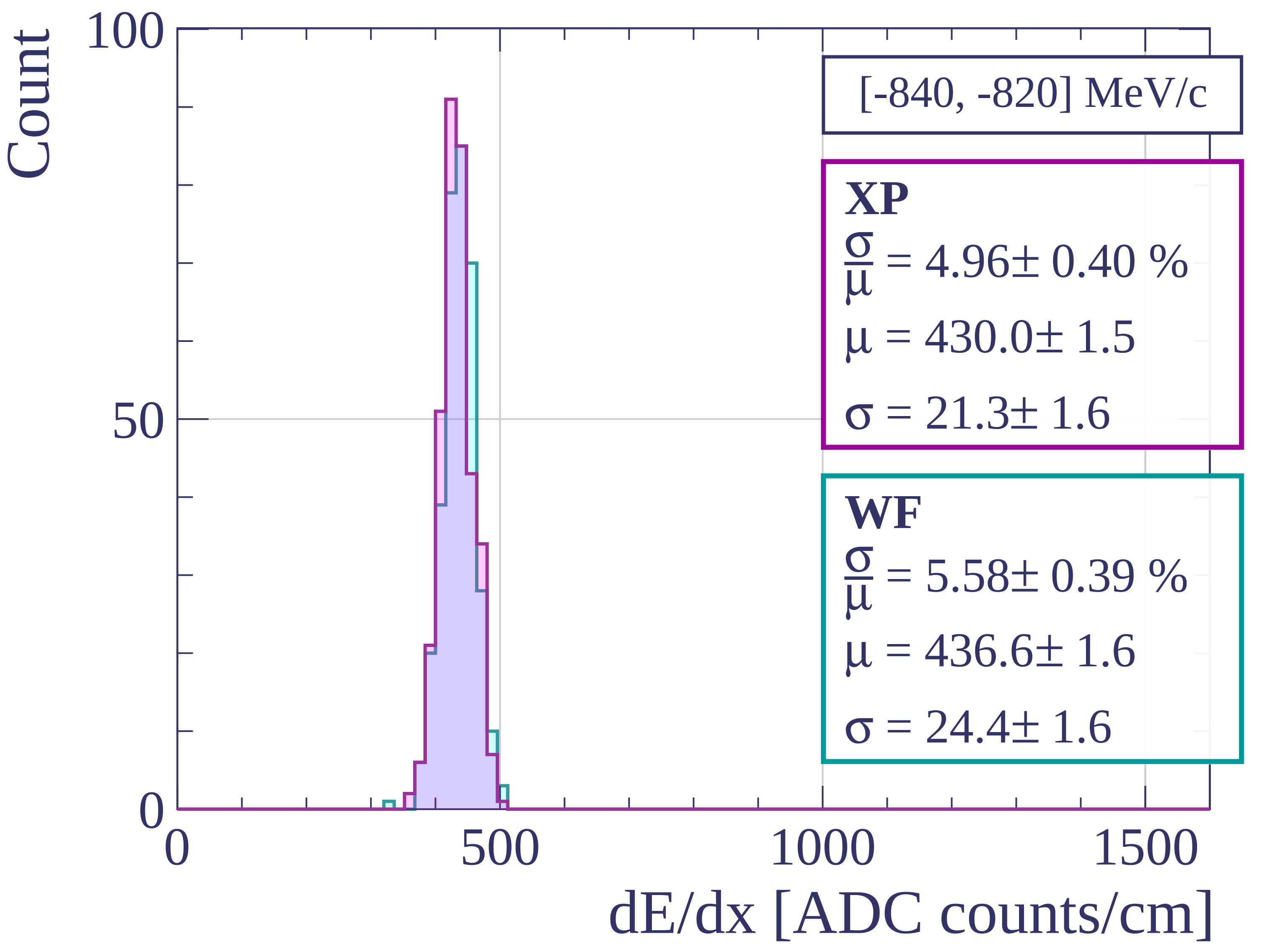
Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[-900, -880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.94± 0.37 %
μ = 436.3± 1.7
σ = 25.9± 1.5
WF
$\frac{\sigma}{\mu}$ = 5.80± 0.35 %
μ = 440.6± 1.7
σ = 25.5± 1.5



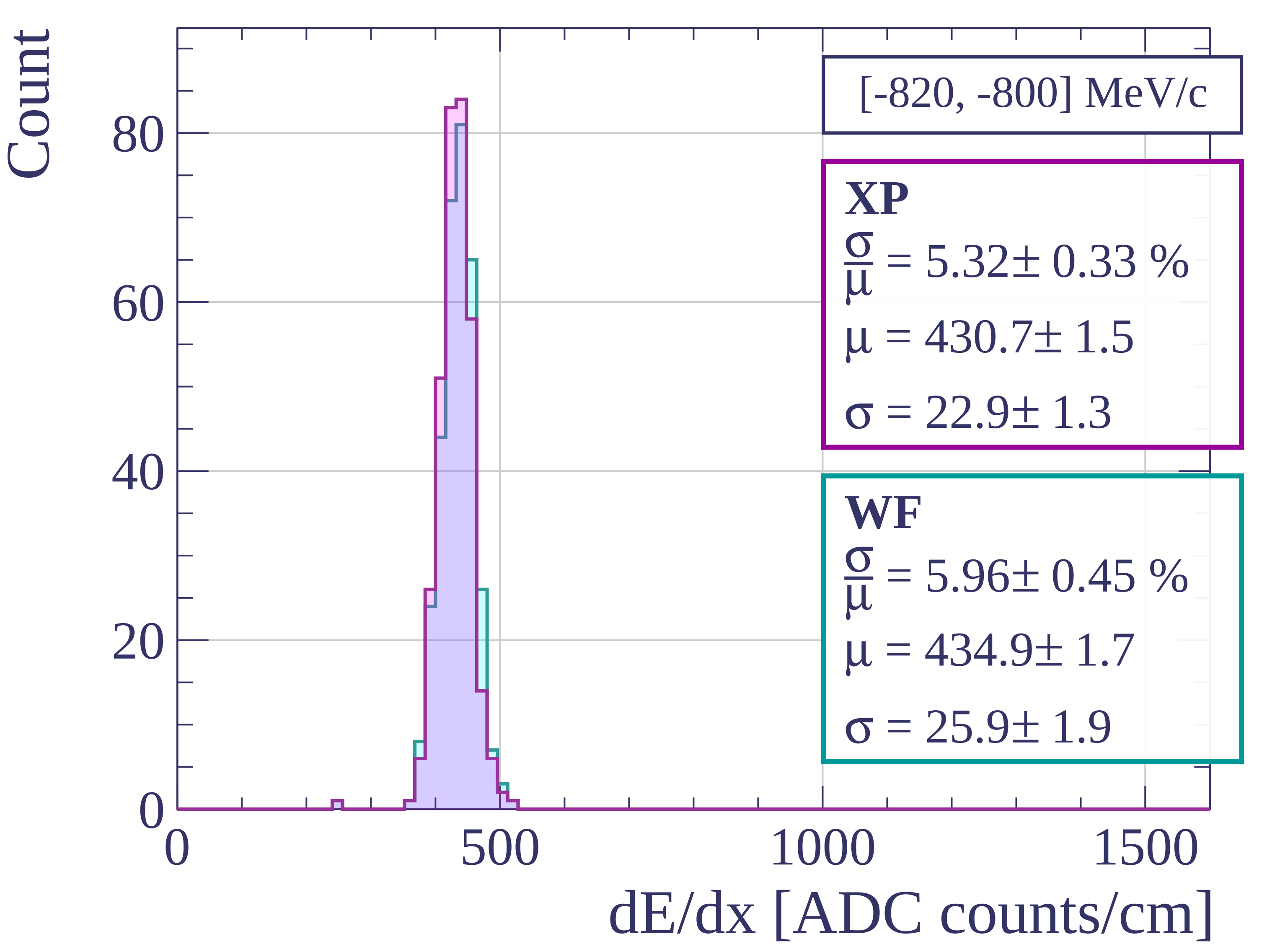
Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[-880, -860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.07± 0.38 %
μ = 434.0± 1.8
σ = 26.3± 1.5
WF
$\frac{\sigma}{\mu}$ = 5.85± 0.48 %
μ = 437.1± 1.9
σ = 25.6± 2.0



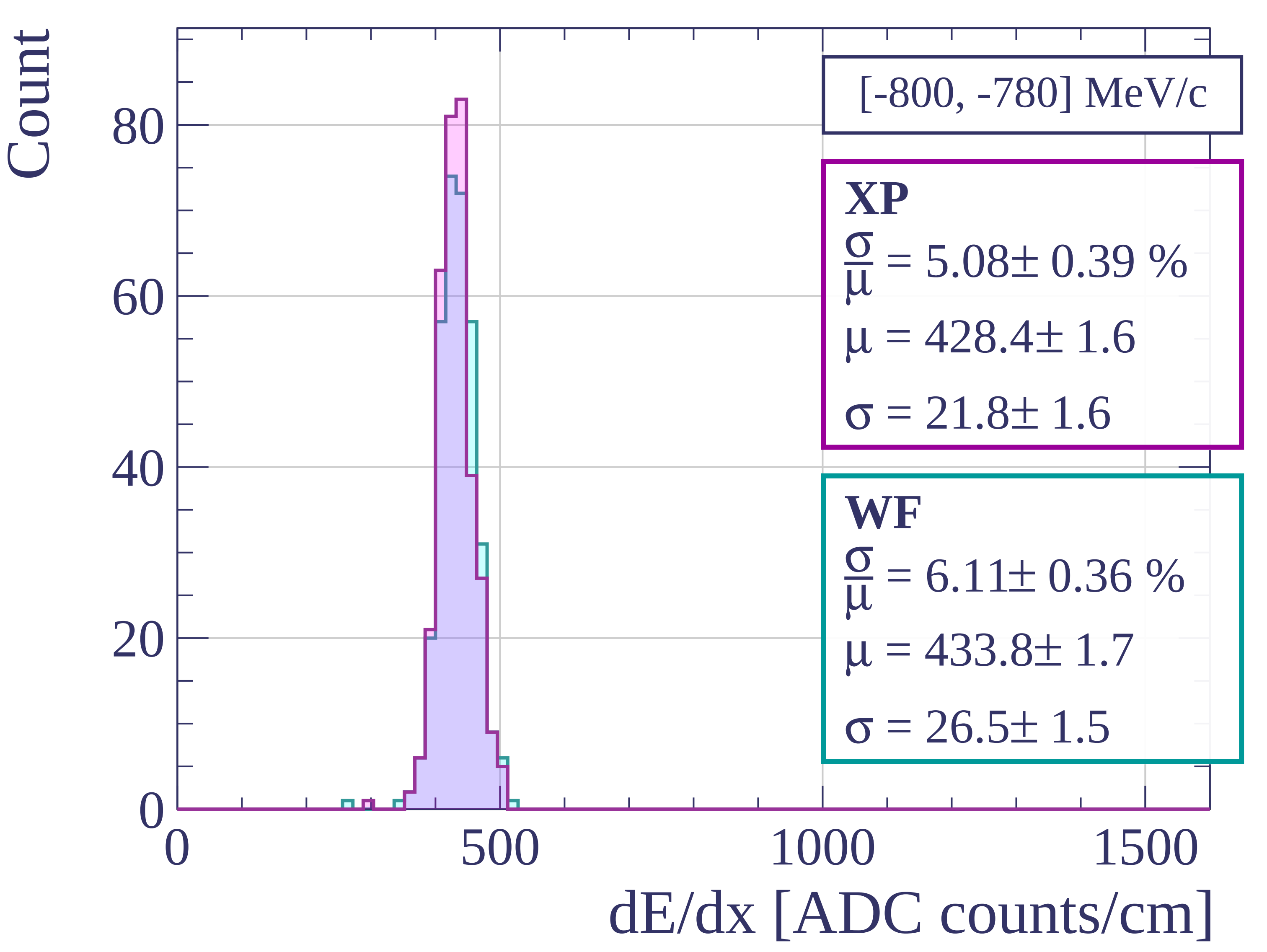
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-860, -840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.23± 0.52 %
μ = 432.5± 2.0
σ = 27.0± 2.1
WF
$\frac{\sigma}{\mu}$ = 6.06± 0.33 %
μ = 435.0± 1.7
σ = 26.4± 1.3

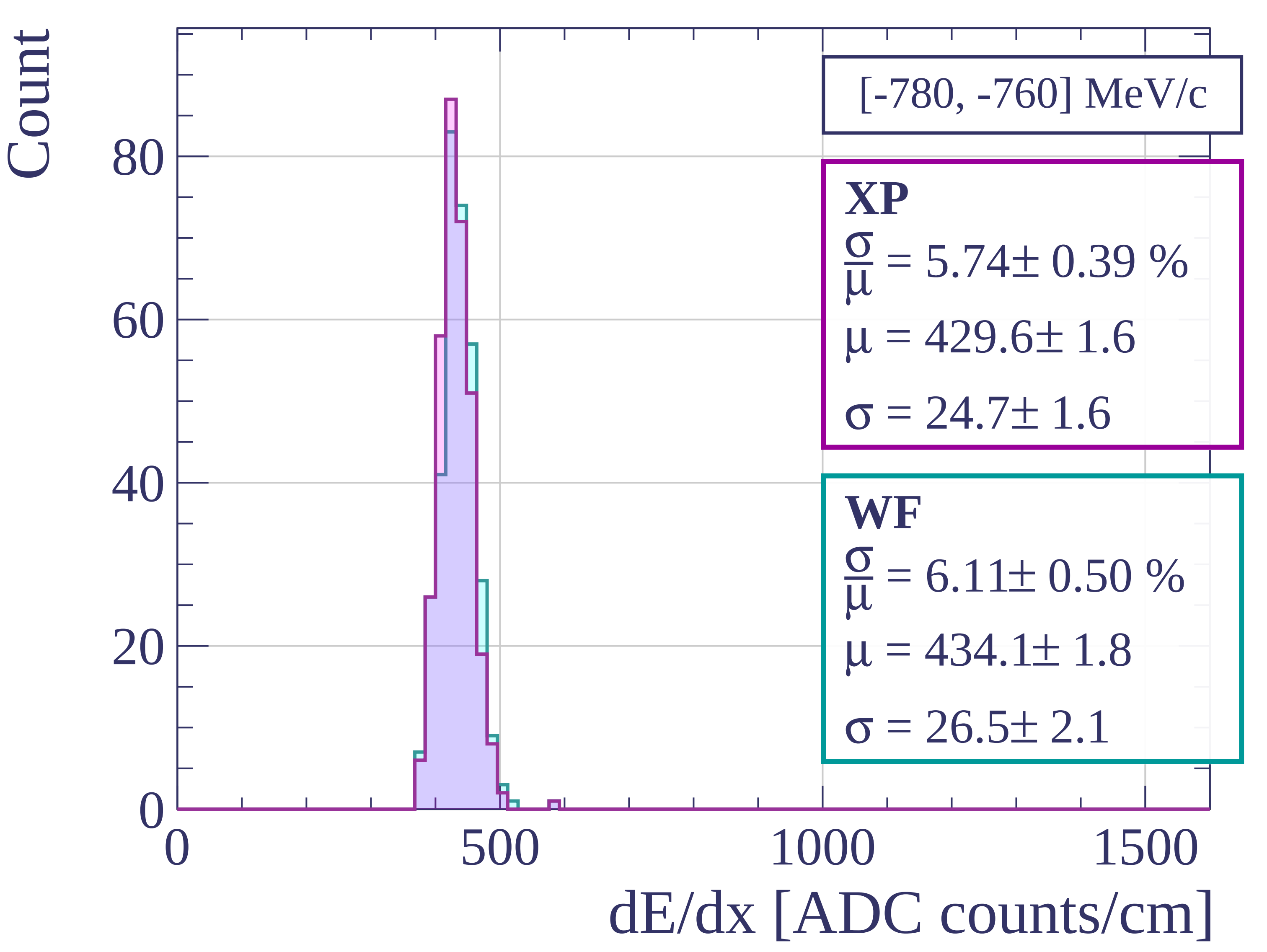


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-840, -820] MeV/c
XP
σ/μ = 4.96± 0.40 %
μ = 430.0± 1.5
σ = 21.3± 1.6
WF
σ/μ = 5.58± 0.39 %
μ = 436.6± 1.6
σ = 24.4± 1.6

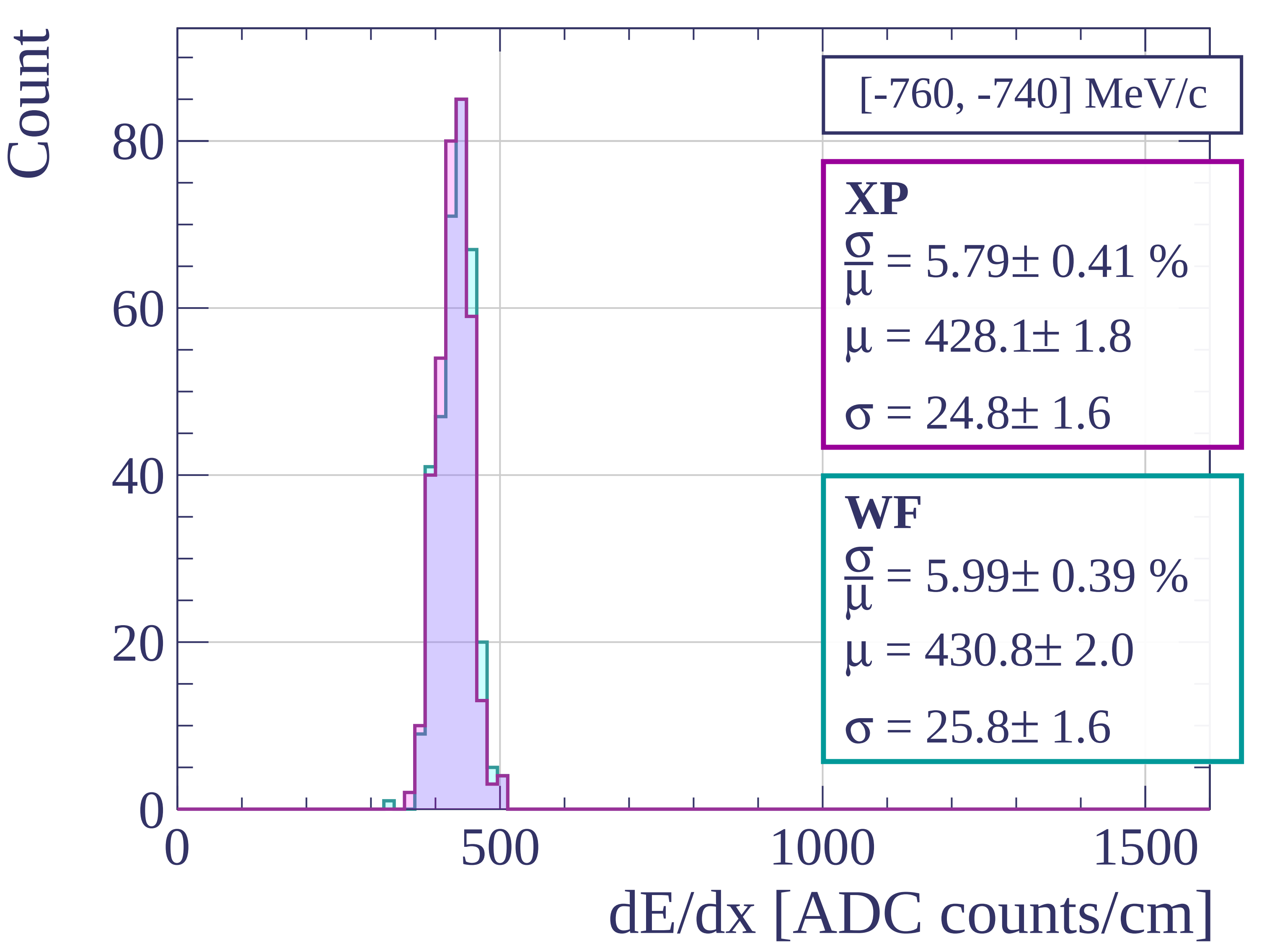


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-820, -800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.32± 0.33 %
μ = 430.7± 1.5
σ = 22.9± 1.3
WF
$\frac{\sigma}{\mu}$ = 5.96± 0.45 %
μ = 434.9± 1.7
σ = 25.9± 1.9

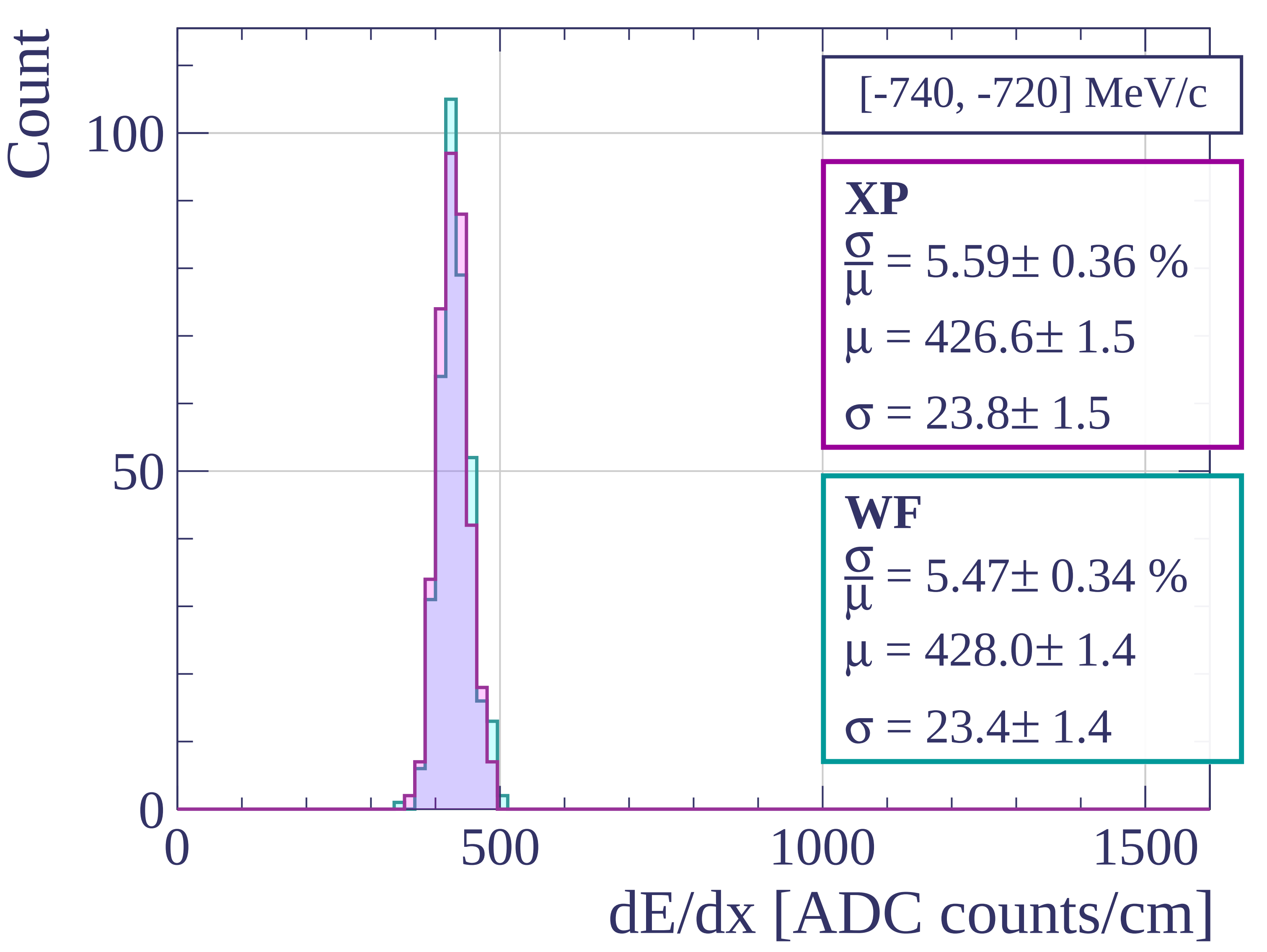

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-800, -780] MeV/c
XP
σ/μ = 5.08± 0.39 %
μ = 428.4± 1.6
σ = 21.8± 1.6
WF
σ/μ = 6.11± 0.36 %
μ = 433.8± 1.7
σ = 26.5± 1.5

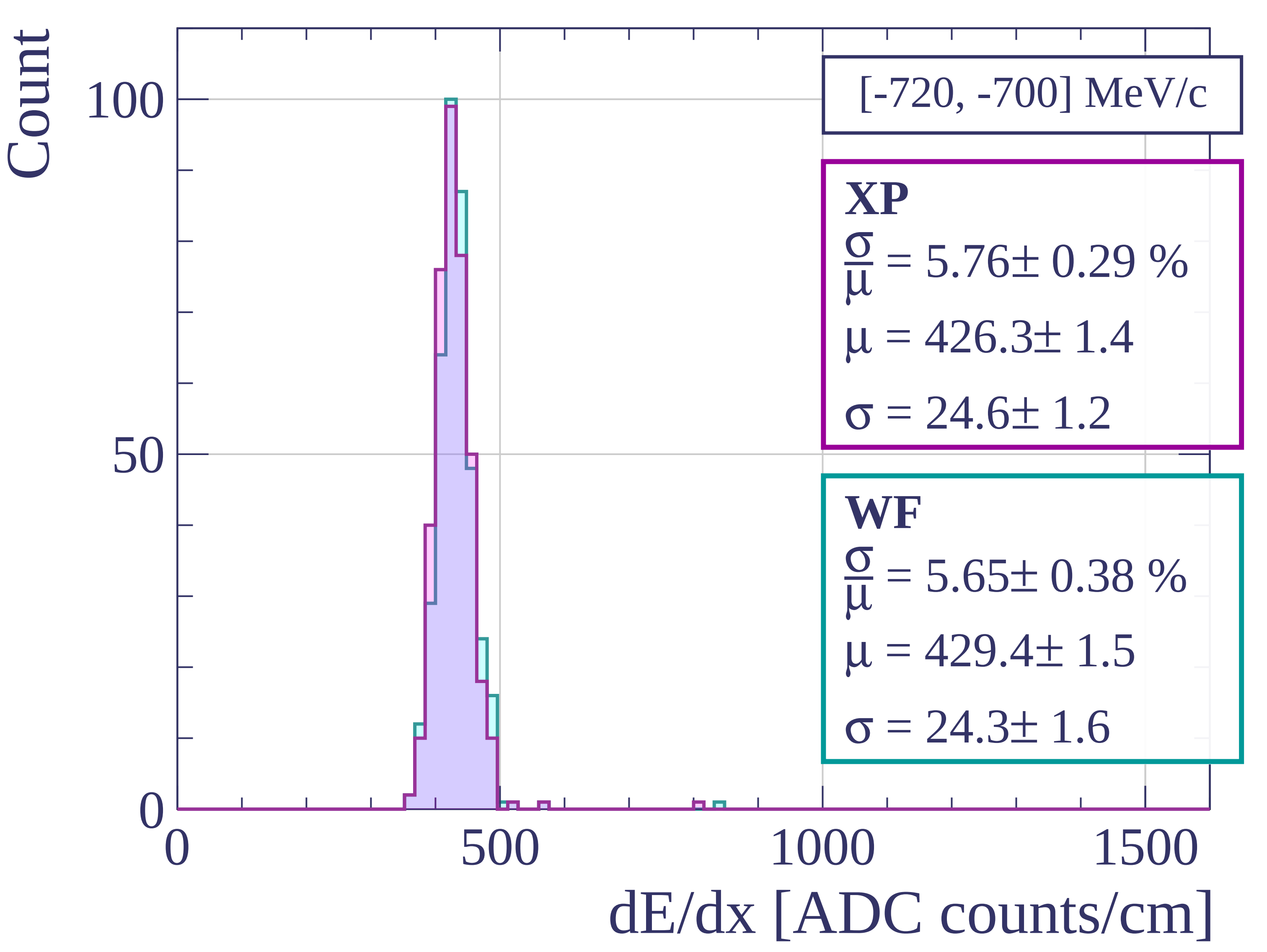


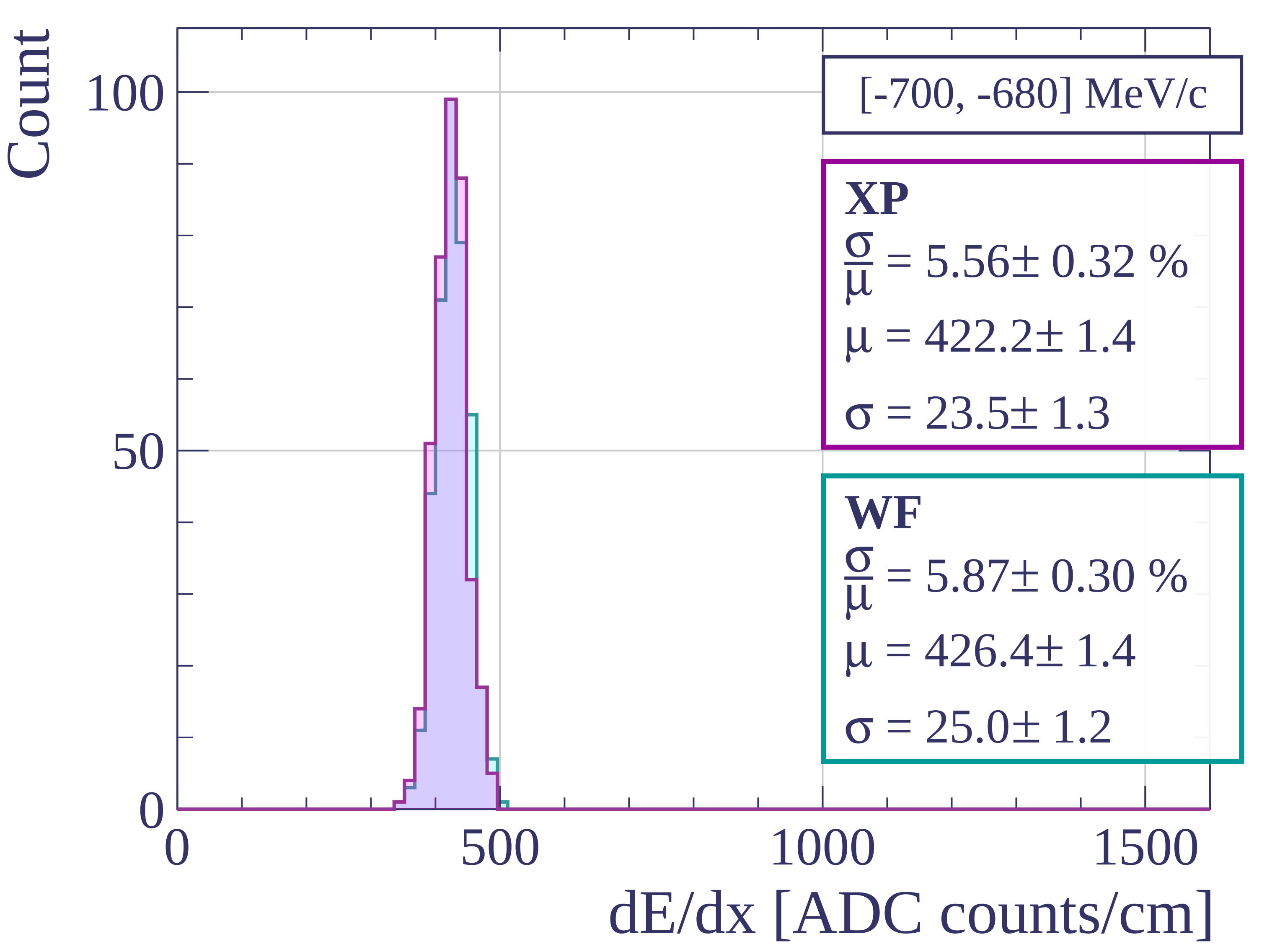
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[-780, -760] MeV/c
XP
σ/μ = 5.74± 0.39 %
μ = 429.6± 1.6
σ = 24.7± 1.6
WF
σ/μ = 6.11± 0.50 %
μ = 434.1± 1.8
σ = 26.5± 2.1



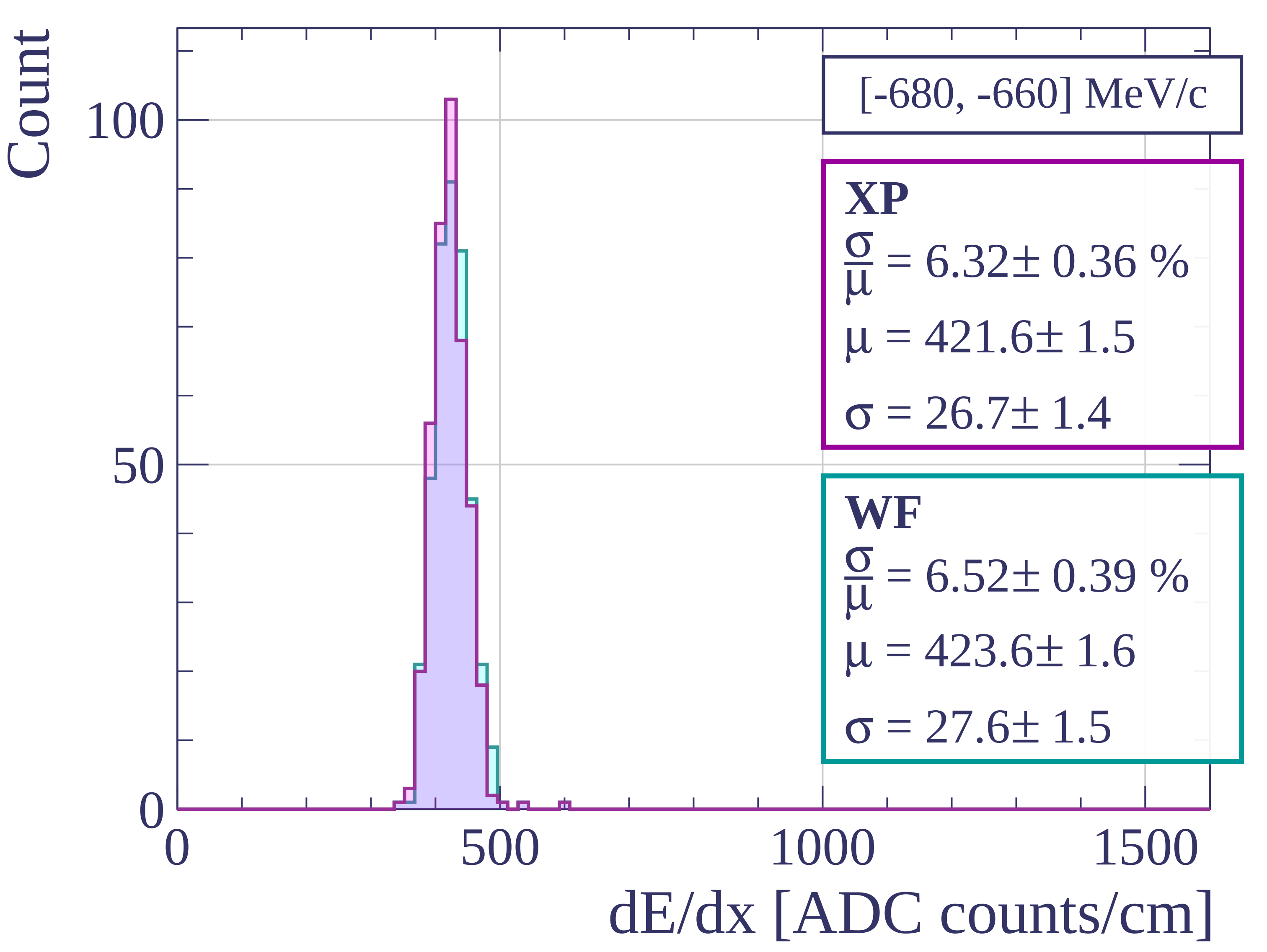
Count
[-760, -740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.79± 0.41 %
μ = 428.1± 1.8
σ = 24.8± 1.6
WF
$\frac{\sigma}{\mu}$ = 5.99± 0.39 %
μ = 430.8± 2.0
σ = 25.8± 1.6
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]

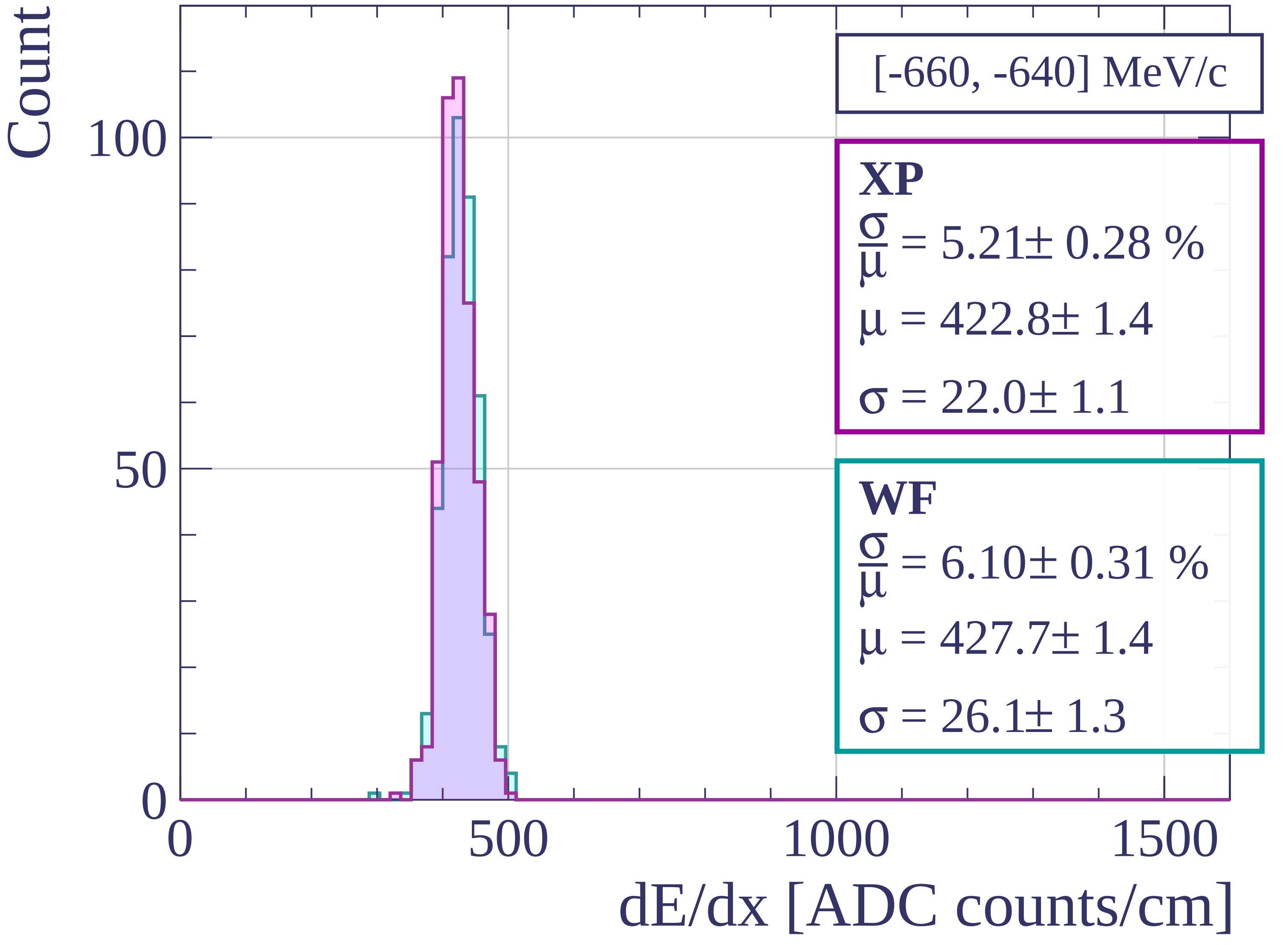

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-740, -720] MeV/c
XP
σ/μ = 5.59± 0.36 %
μ = 426.6± 1.5
σ = 23.8± 1.5
WF
σ/μ = 5.47± 0.34 %
μ = 428.0± 1.4
σ = 23.4± 1.4


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-720, -700] MeV/c
XP
σ/μ = 5.76± 0.29 %
μ = 426.3± 1.4
σ = 24.6± 1.2
WF
σ/μ = 5.65± 0.38 %
μ = 429.4± 1.5
σ = 24.3± 1.6

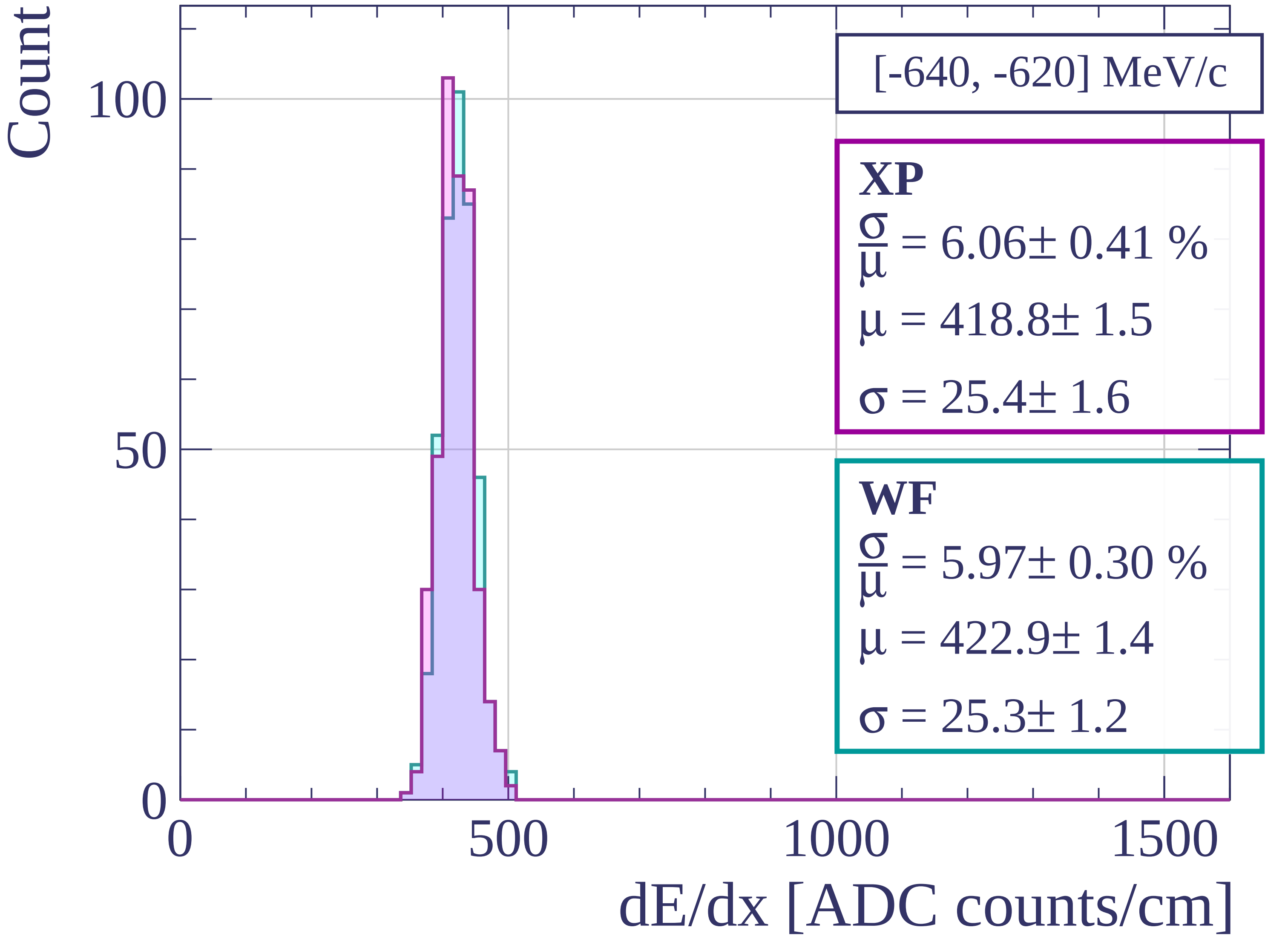


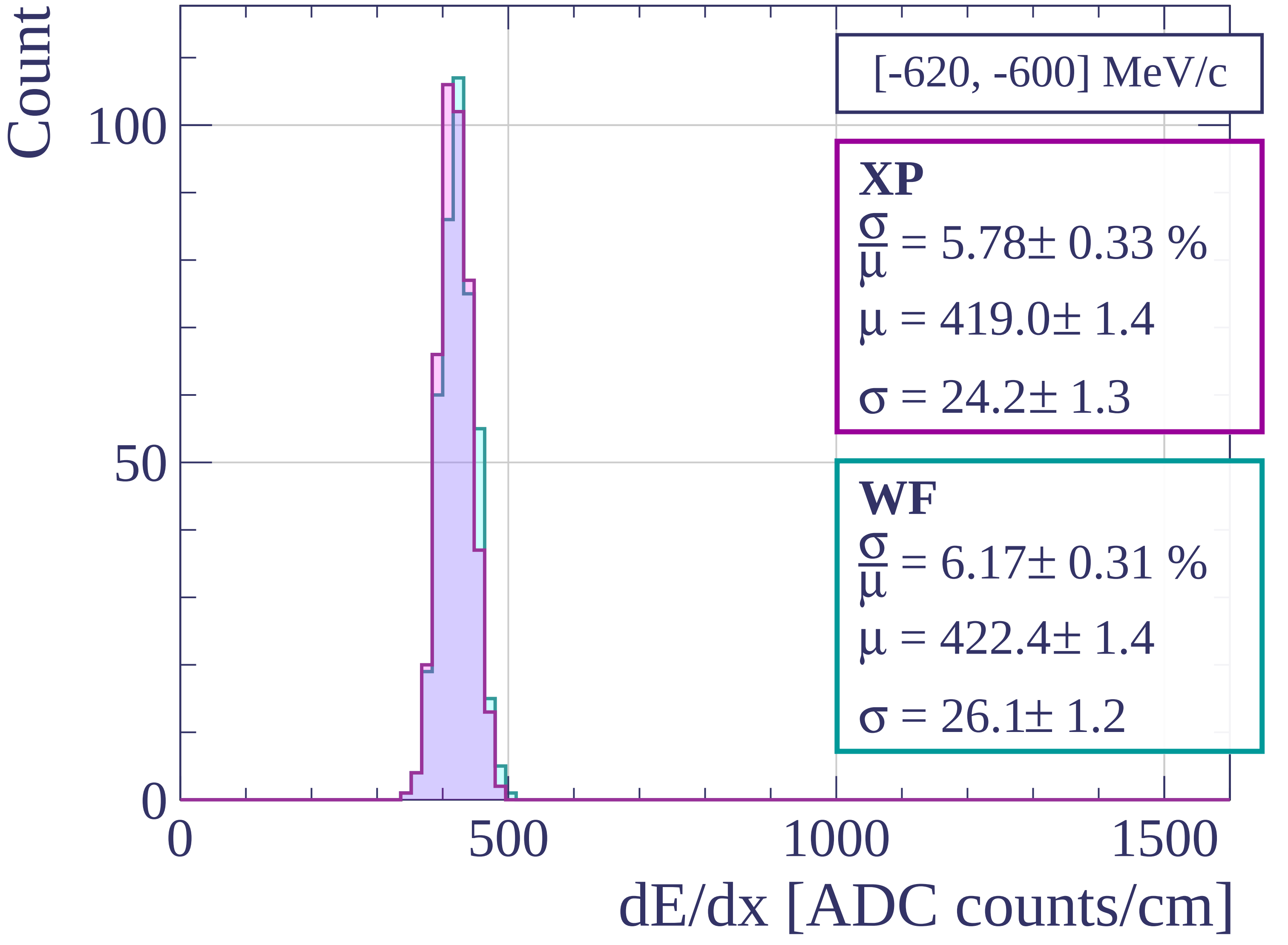
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-700, -680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.56± 0.32 %
μ = 422.2± 1.4
σ = 23.5± 1.3
WF
$\frac{\sigma}{\mu}$ = 5.87± 0.30 %
μ = 426.4± 1.4
σ = 25.0± 1.2

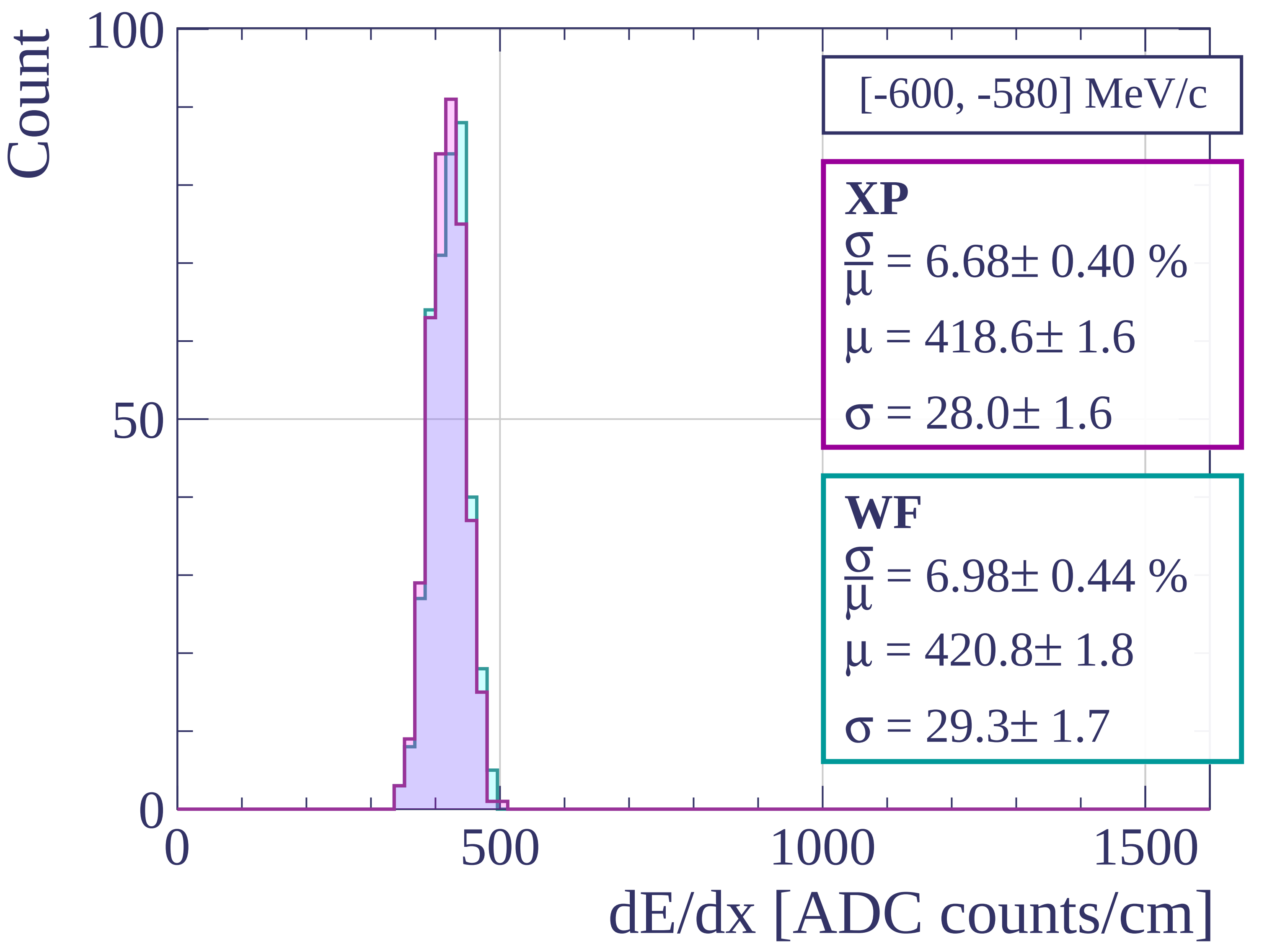

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-680, -660] MeV/c
XP
σ/μ = 6.32± 0.36 %
μ = 421.6± 1.5
σ = 26.7± 1.4
WF
σ/μ = 6.52± 0.39 %
μ = 423.6± 1.6
σ = 27.6± 1.5



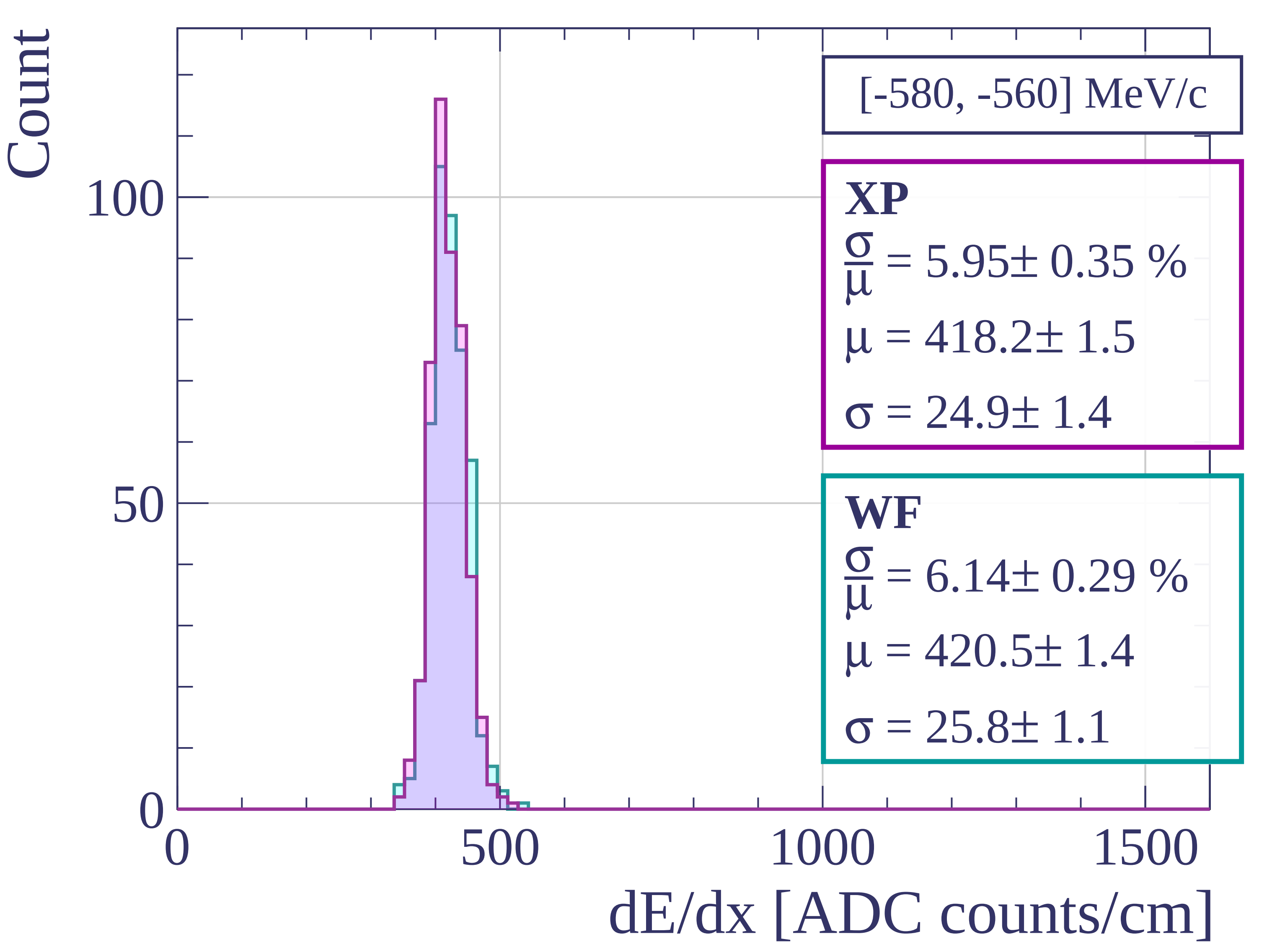
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-660, -640] MeV/c
XP
σ/μ = 5.21± 0.28 %
μ = 422.8± 1.4
σ = 22.0± 1.1
WF
σ/μ = 6.10± 0.31 %
μ = 427.7± 1.4
σ = 26.1± 1.3


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-640, -620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.06± 0.41 %
μ = 418.8± 1.5
σ = 25.4± 1.6
WF
$\frac{\sigma}{\mu}$ = 5.97± 0.30 %
μ = 422.9± 1.4
σ = 25.3± 1.2

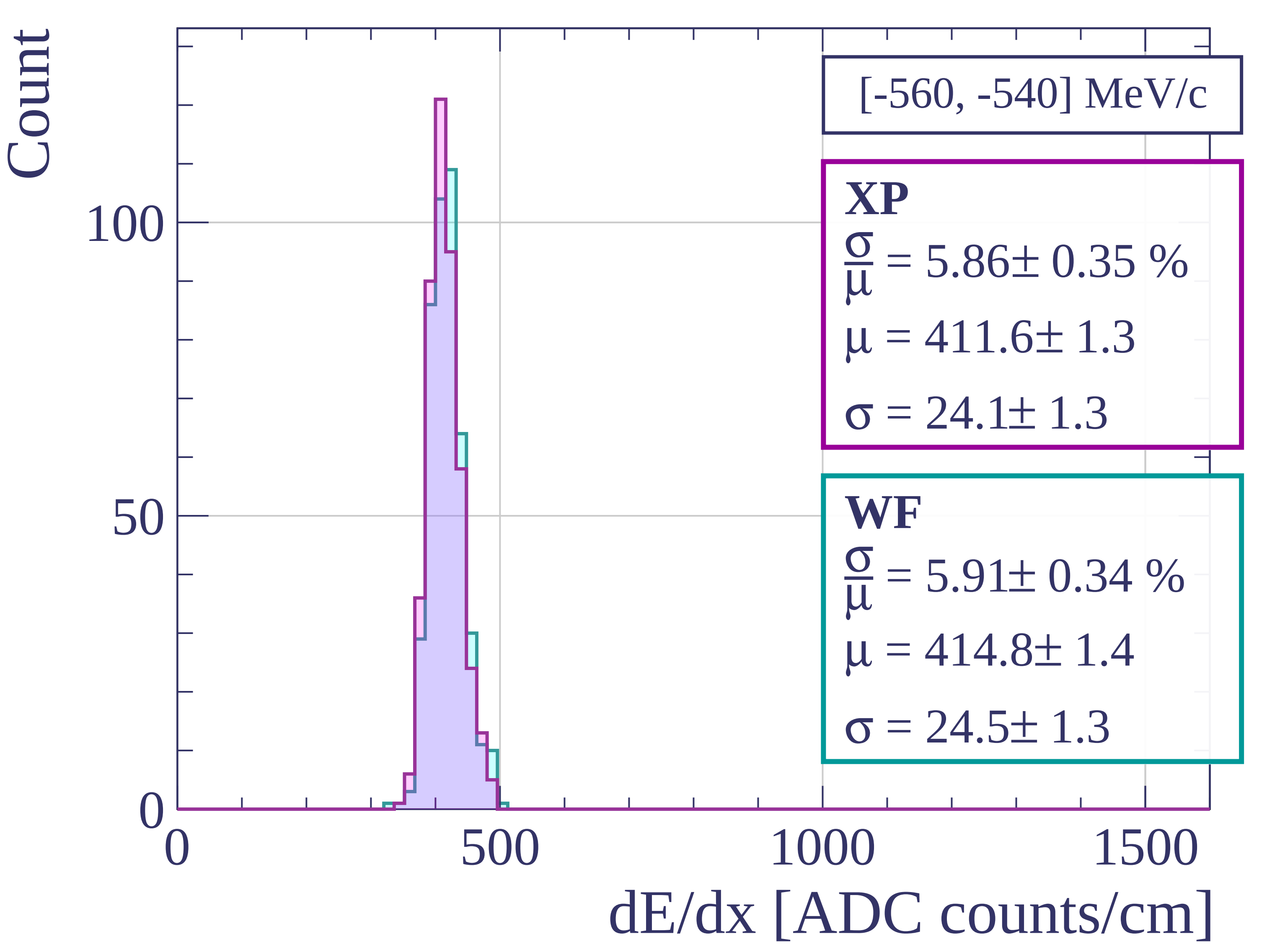

[-620, -600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.78± 0.33 %
μ = 419.0± 1.4
σ = 24.2± 1.3
WF
$\frac{\sigma}{\mu}$ = 6.17± 0.31 %
μ = 422.4± 1.4
σ = 26.1± 1.2
Count
0
50
100
0
500
1000
1500
dE/dx [ADC counts/cm]

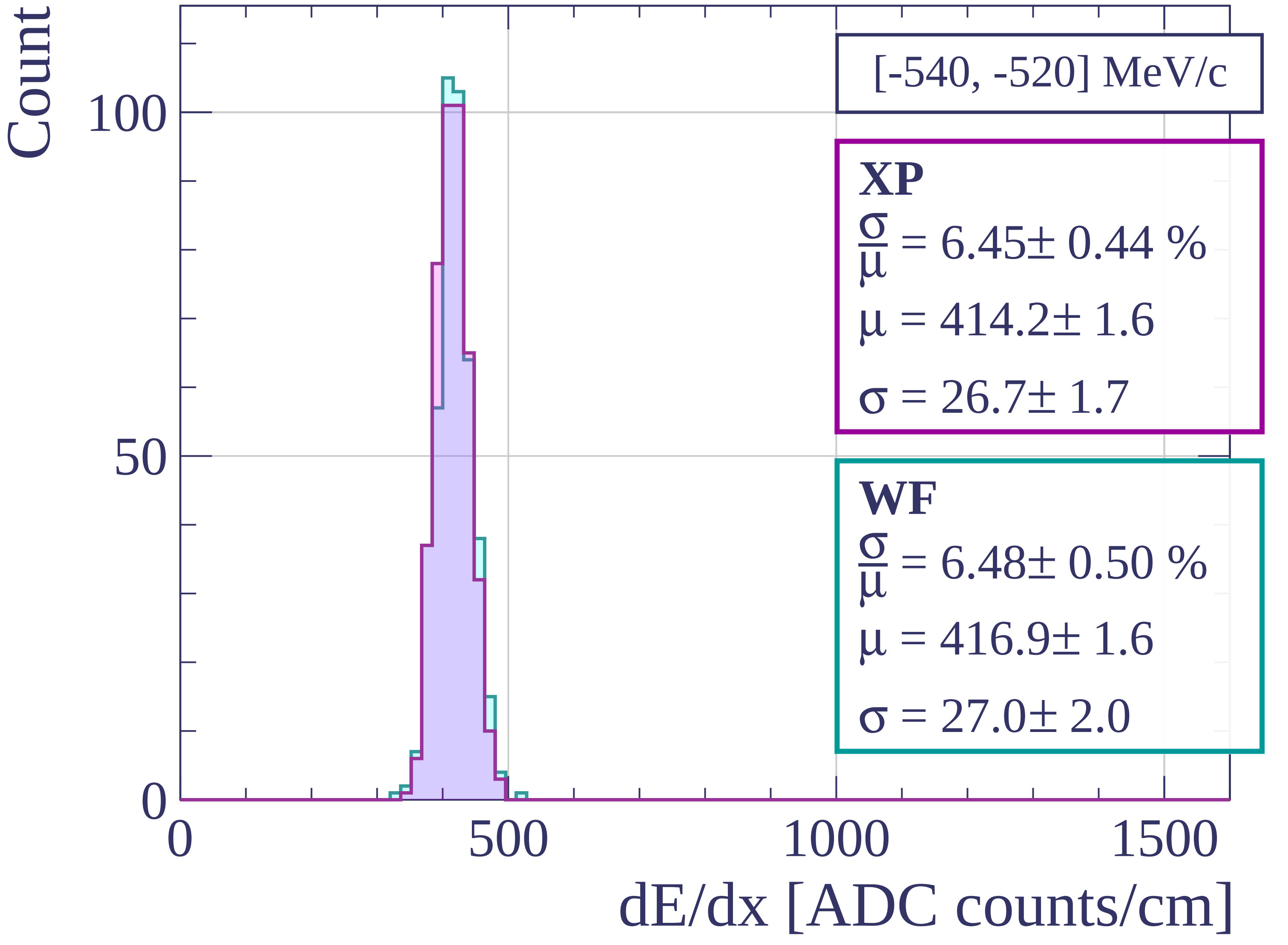


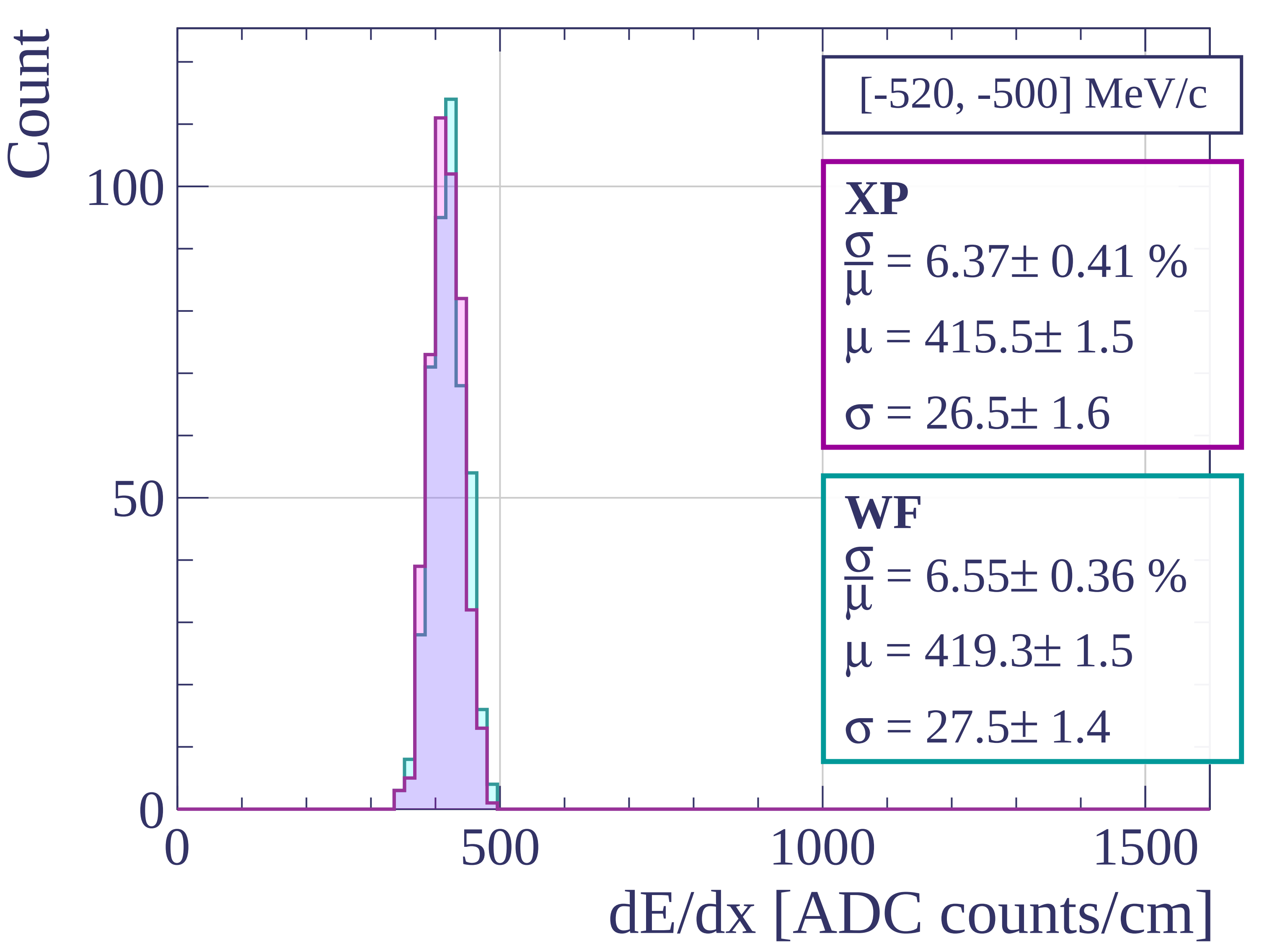
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-600, -580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.68± 0.40 %
μ = 418.6± 1.6
σ = 28.0± 1.6
WF
$\frac{\sigma}{\mu}$ = 6.98± 0.44 %
μ = 420.8± 1.8
σ = 29.3± 1.7

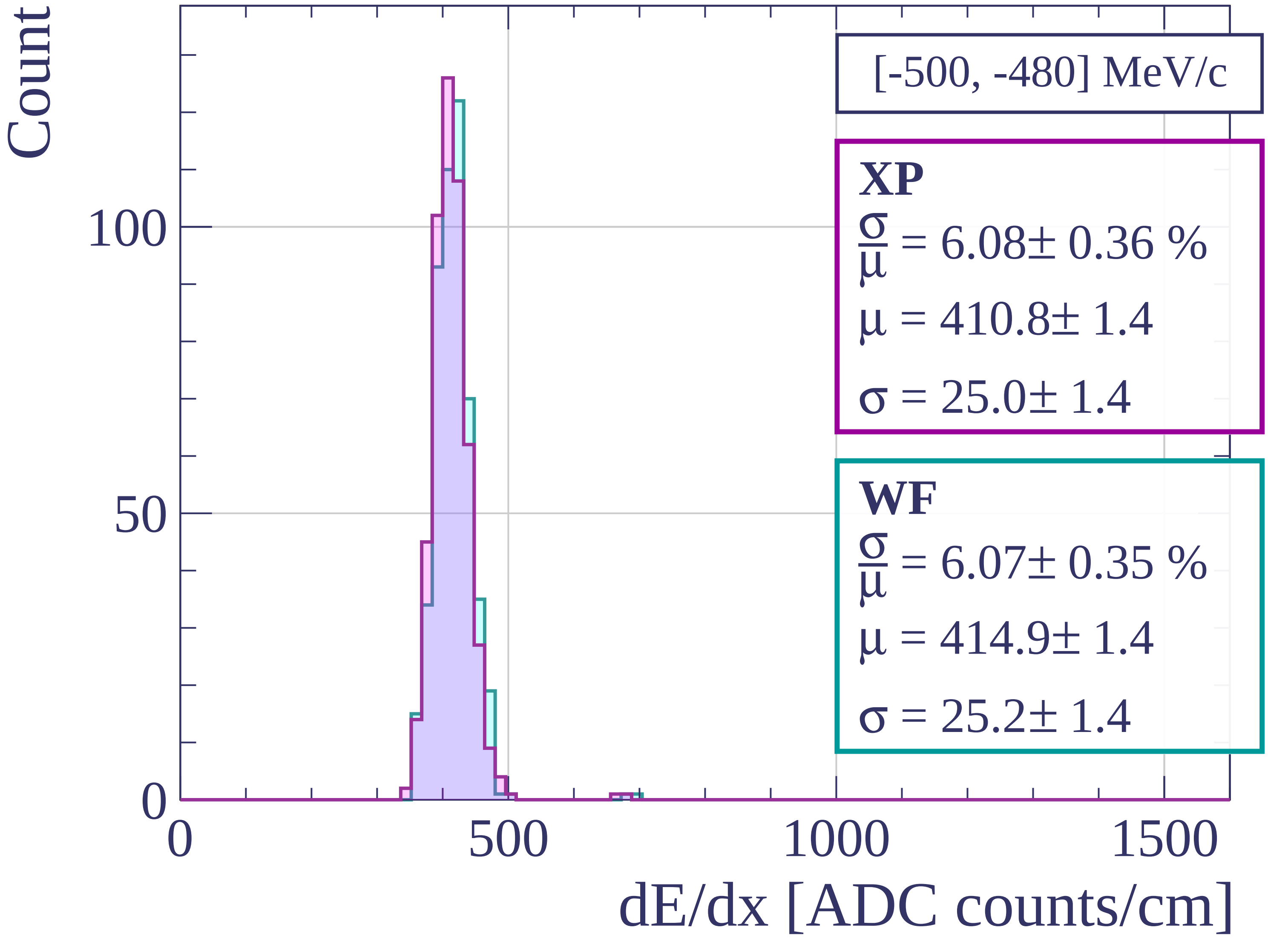


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-580, -560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.95± 0.35 %
μ = 418.2± 1.5
σ = 24.9± 1.4
WF
$\frac{\sigma}{\mu}$ = 6.14± 0.29 %
μ = 420.5± 1.4
σ = 25.8± 1.1

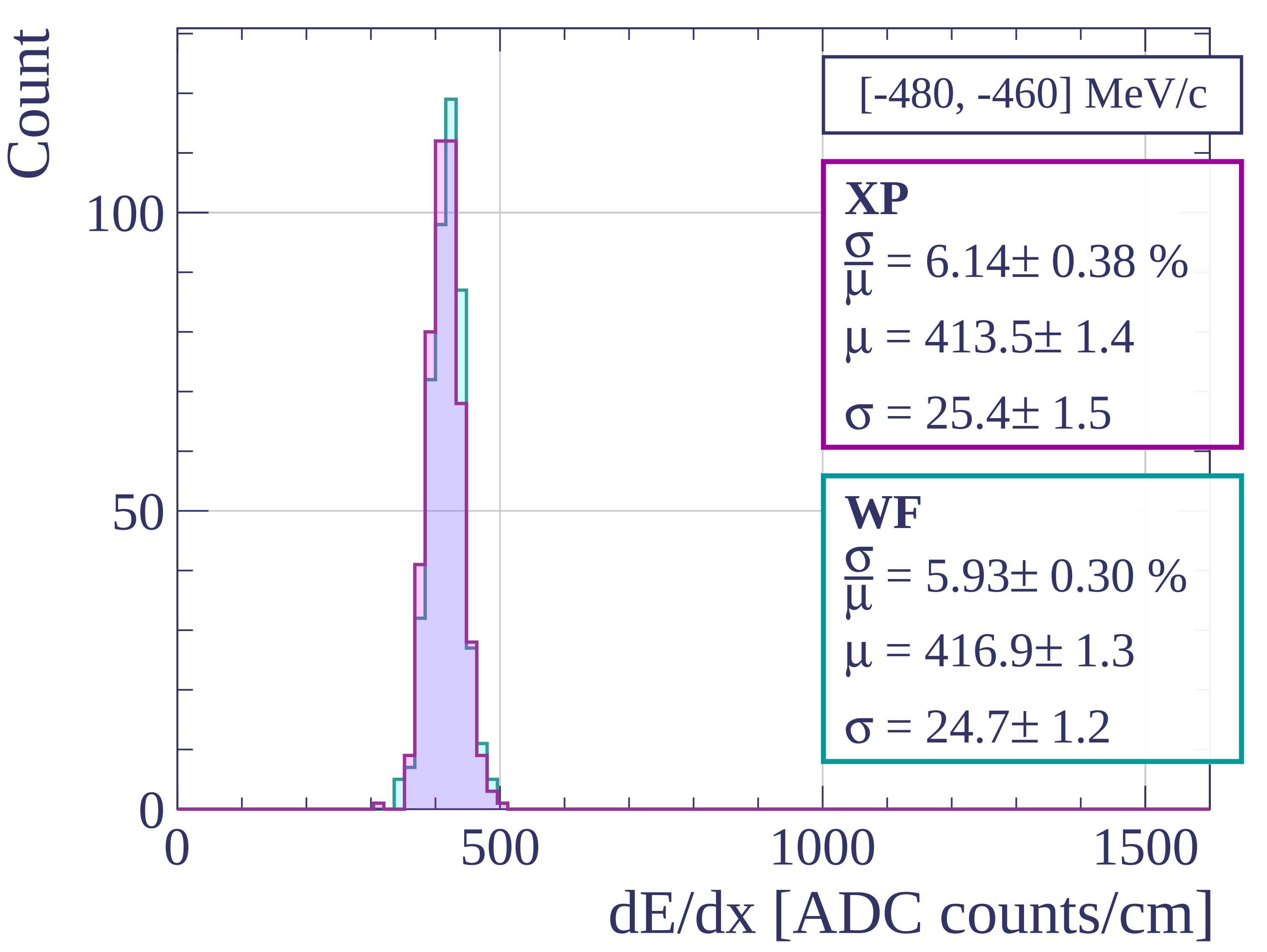

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-560, -540] MeV/c
XP
σ/μ = 5.86± 0.35 %
μ = 411.6± 1.3
σ = 24.1± 1.3
WF
σ/μ = 5.91± 0.34 %
μ = 414.8± 1.4
σ = 24.5± 1.3

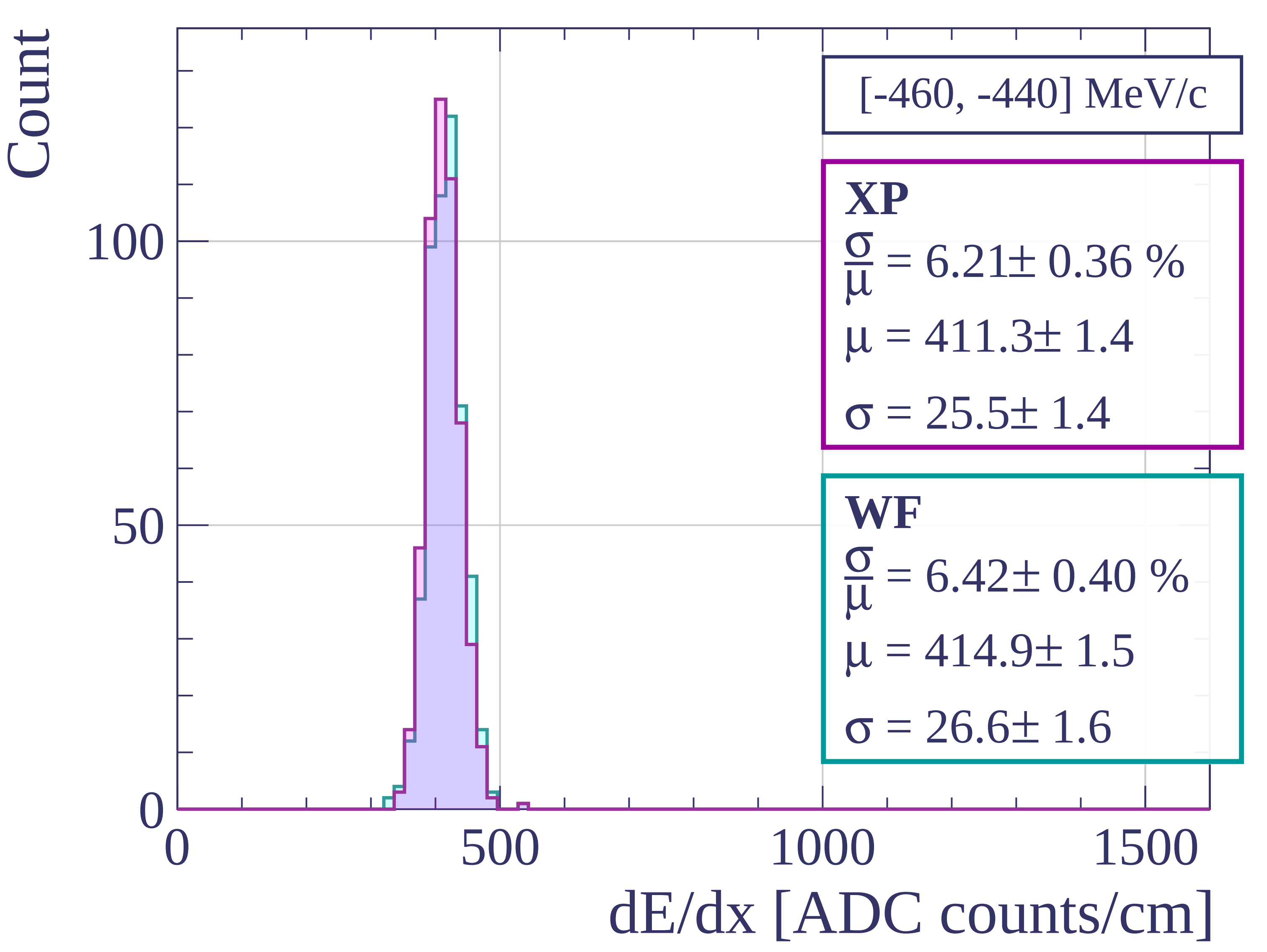

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-540, -520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.45± 0.44 %
μ = 414.2± 1.6
σ = 26.7± 1.7
WF
$\frac{\sigma}{\mu}$ = 6.48± 0.50 %
μ = 416.9± 1.6
σ = 27.0± 2.0

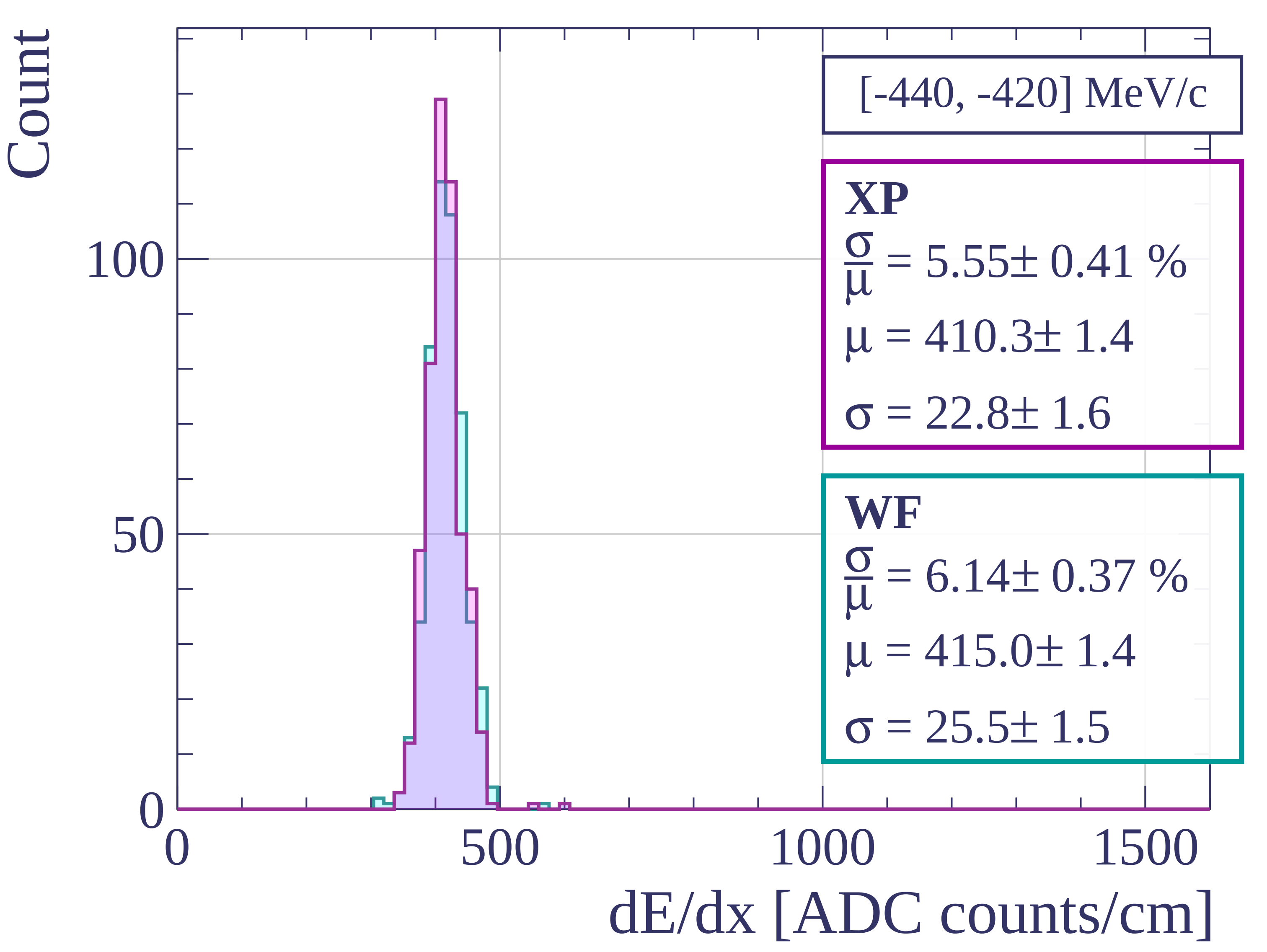

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-520, -500] MeV/c
XP
σ/μ = 6.37± 0.41 %
μ = 415.5± 1.5
σ = 26.5± 1.6
WF
σ/μ = 6.55± 0.36 %
μ = 419.3± 1.5
σ = 27.5± 1.4



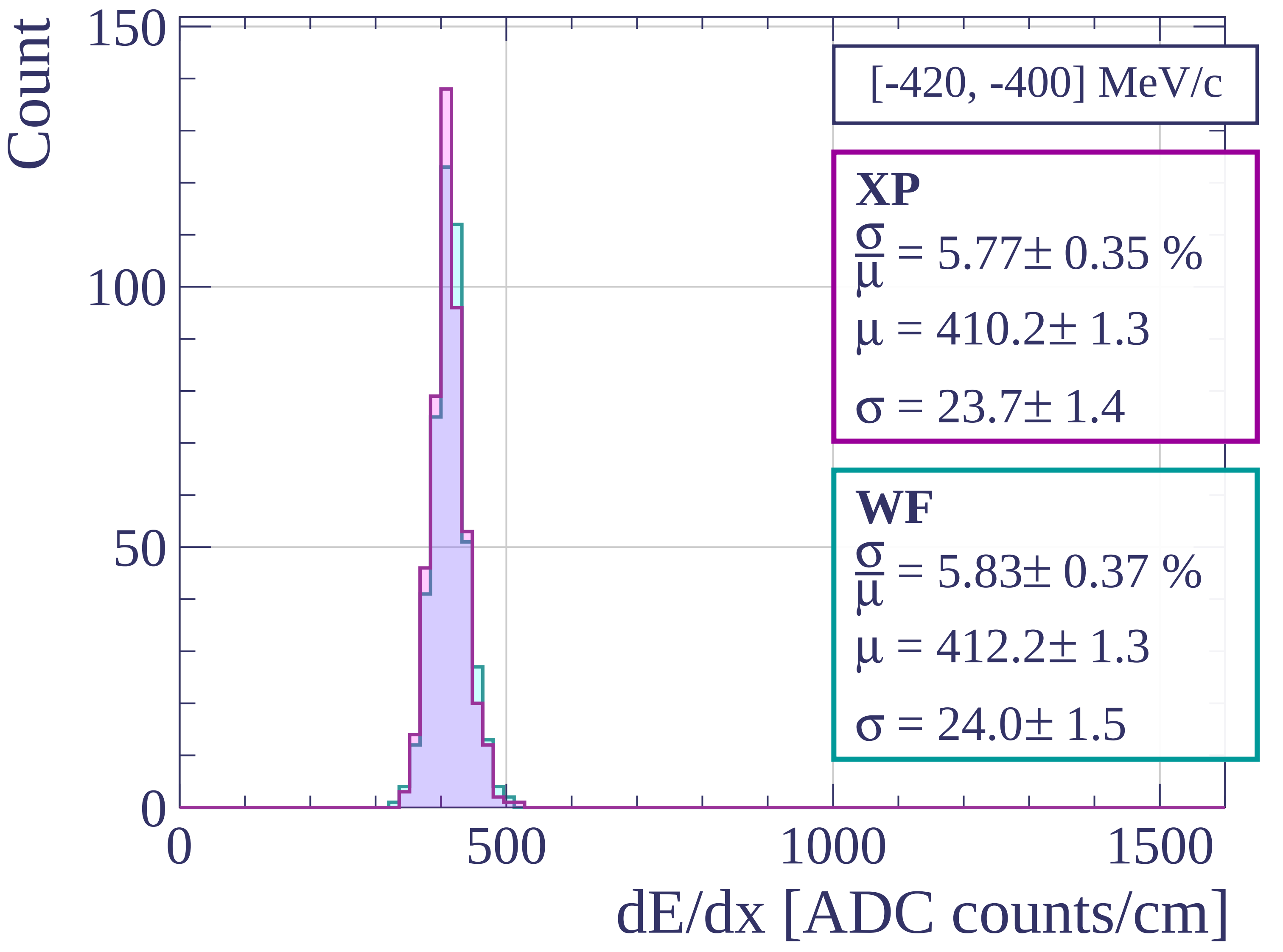
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-500, -480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.08± 0.36 %
μ = 410.8± 1.4
σ = 25.0± 1.4
WF
$\frac{\sigma}{\mu}$ = 6.07± 0.35 %
μ = 414.9± 1.4
σ = 25.2± 1.4


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-480, -460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.14± 0.38 %
μ = 413.5± 1.4
σ = 25.4± 1.5
WF
$\frac{\sigma}{\mu}$ = 5.93± 0.30 %
μ = 416.9± 1.3
σ = 24.7± 1.2

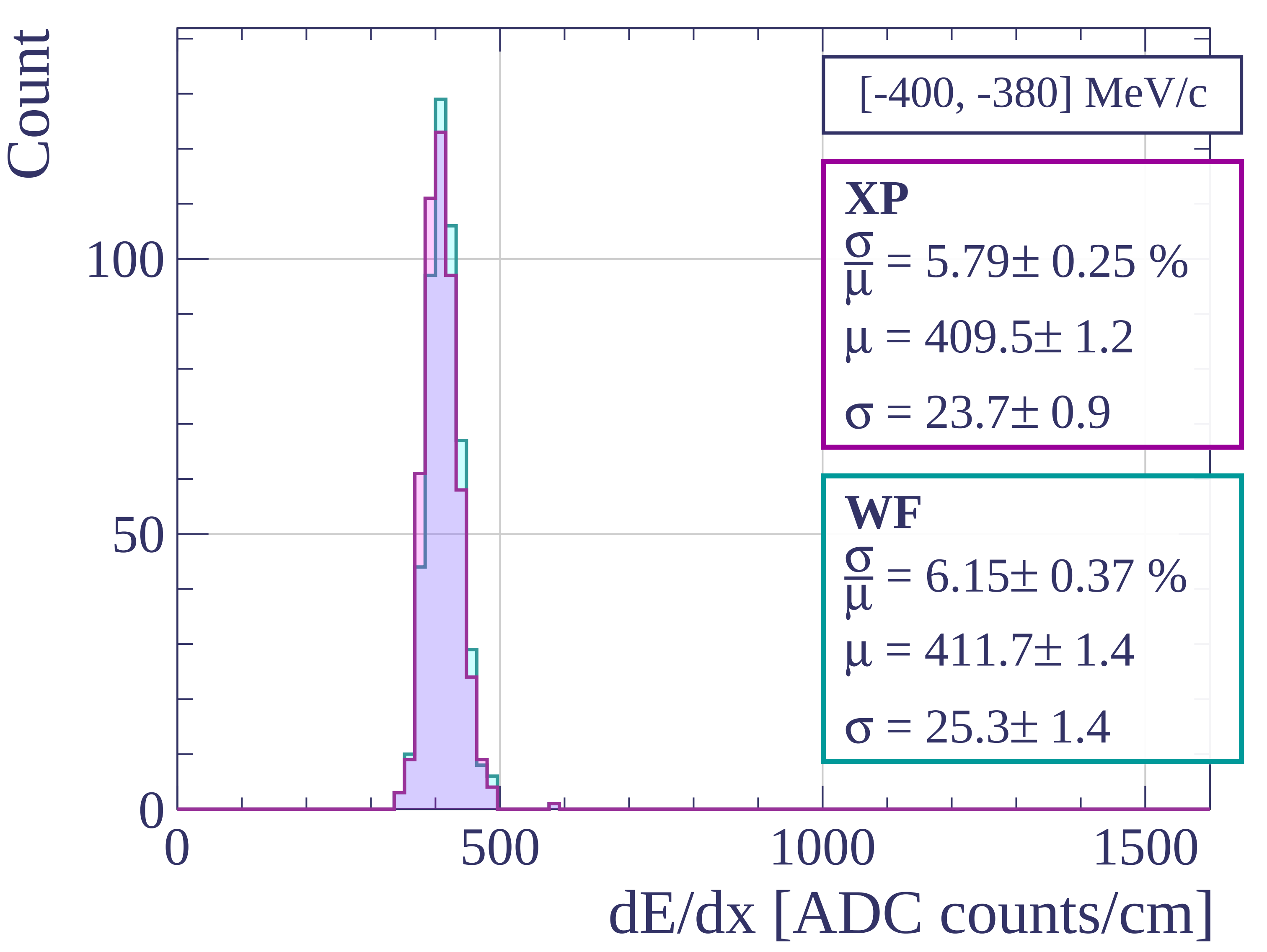

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-460, -440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.21± 0.36 %
μ = 411.3± 1.4
σ = 25.5± 1.4
WF
$\frac{\sigma}{\mu}$ = 6.42± 0.40 %
μ = 414.9± 1.5
σ = 26.6± 1.6



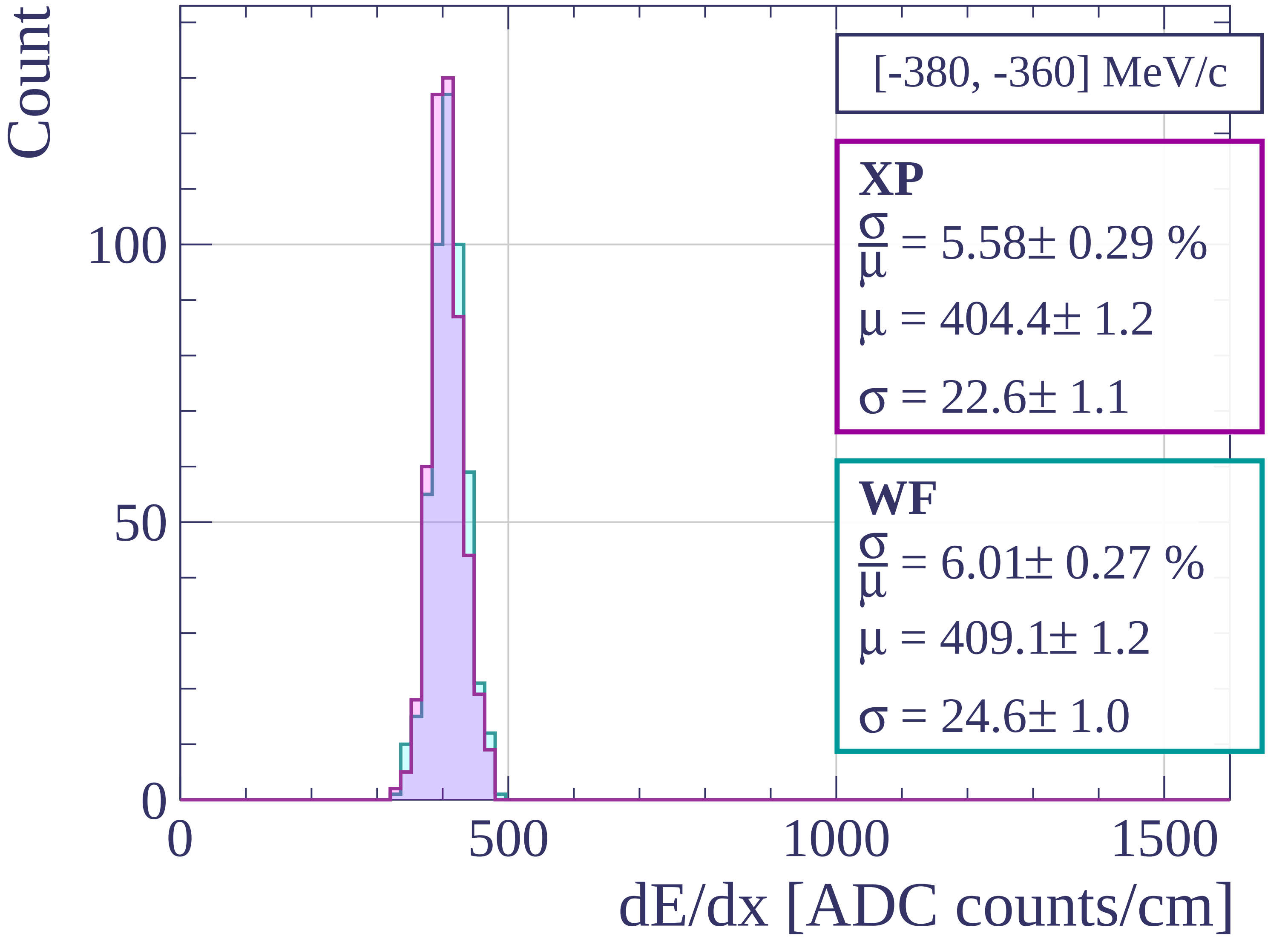
Count
dE/dx [ADC counts/cm]
[-440, -420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.55± 0.41 %
μ = 410.3± 1.4
σ = 22.8± 1.6
WF
$\frac{\sigma}{\mu}$ = 6.14± 0.37 %
μ = 415.0± 1.4
σ = 25.5± 1.5
0
50
100
0
500
1000
1500

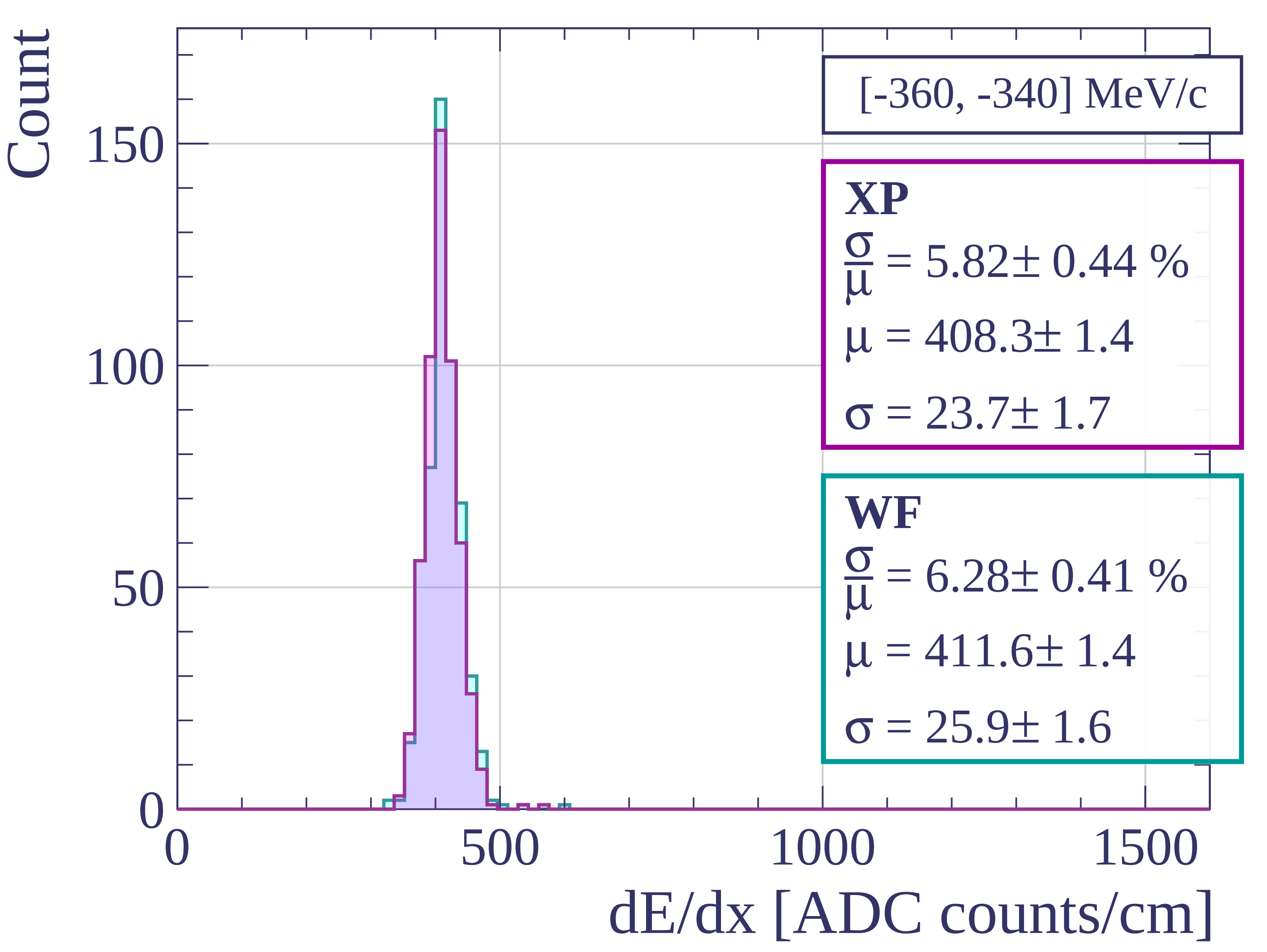


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-420, -400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.77± 0.35 %
μ = 410.2± 1.3
σ = 23.7± 1.4
WF
$\frac{\sigma}{\mu}$ = 5.83± 0.37 %
μ = 412.2± 1.3
σ = 24.0± 1.5

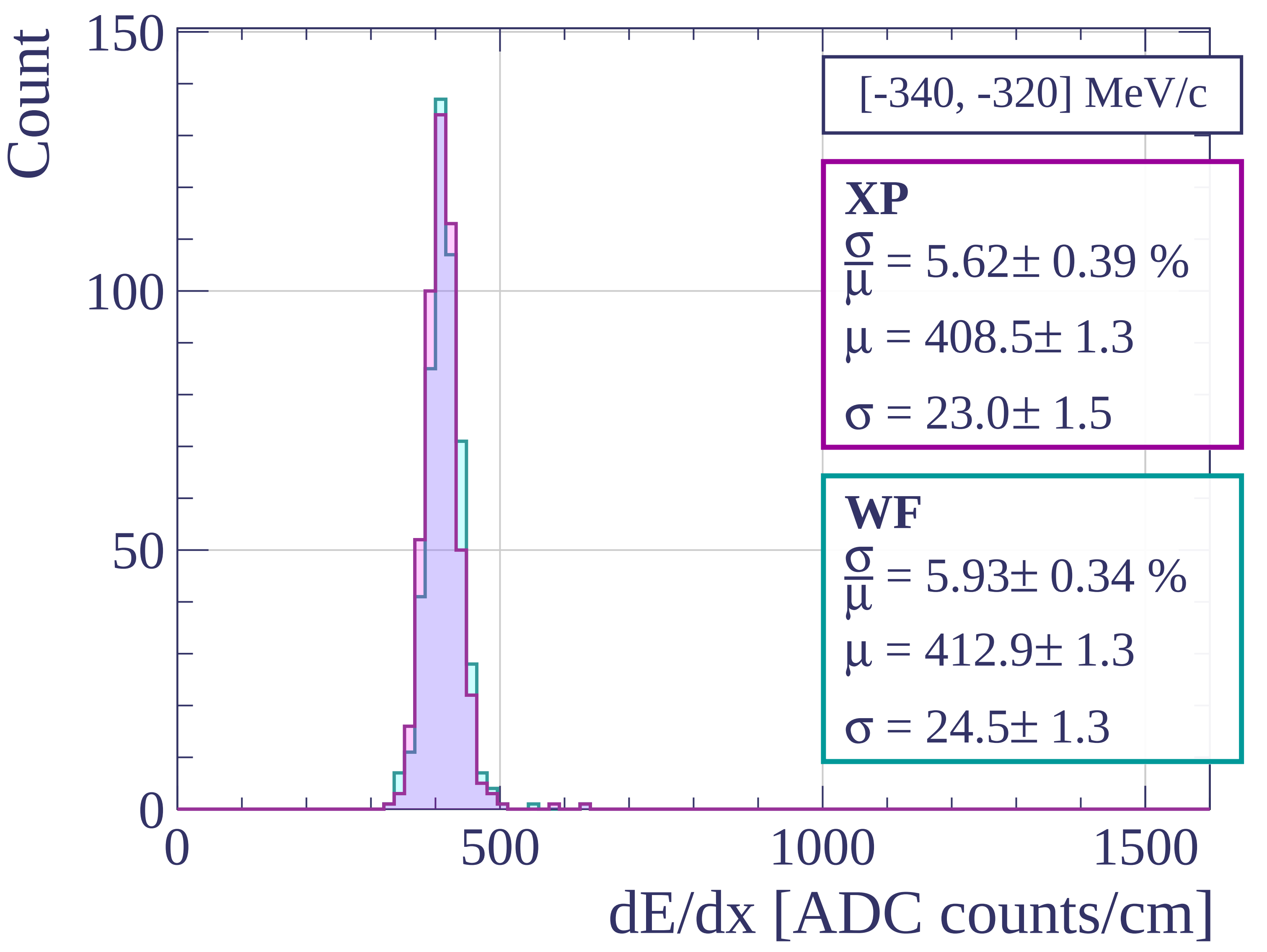


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-400, -380] MeV/c
XP
σ/μ = 5.79± 0.25 %
μ = 409.5± 1.2
σ = 23.7± 0.9
WF
σ/μ = 6.15± 0.37 %
μ = 411.7± 1.4
σ = 25.3± 1.4

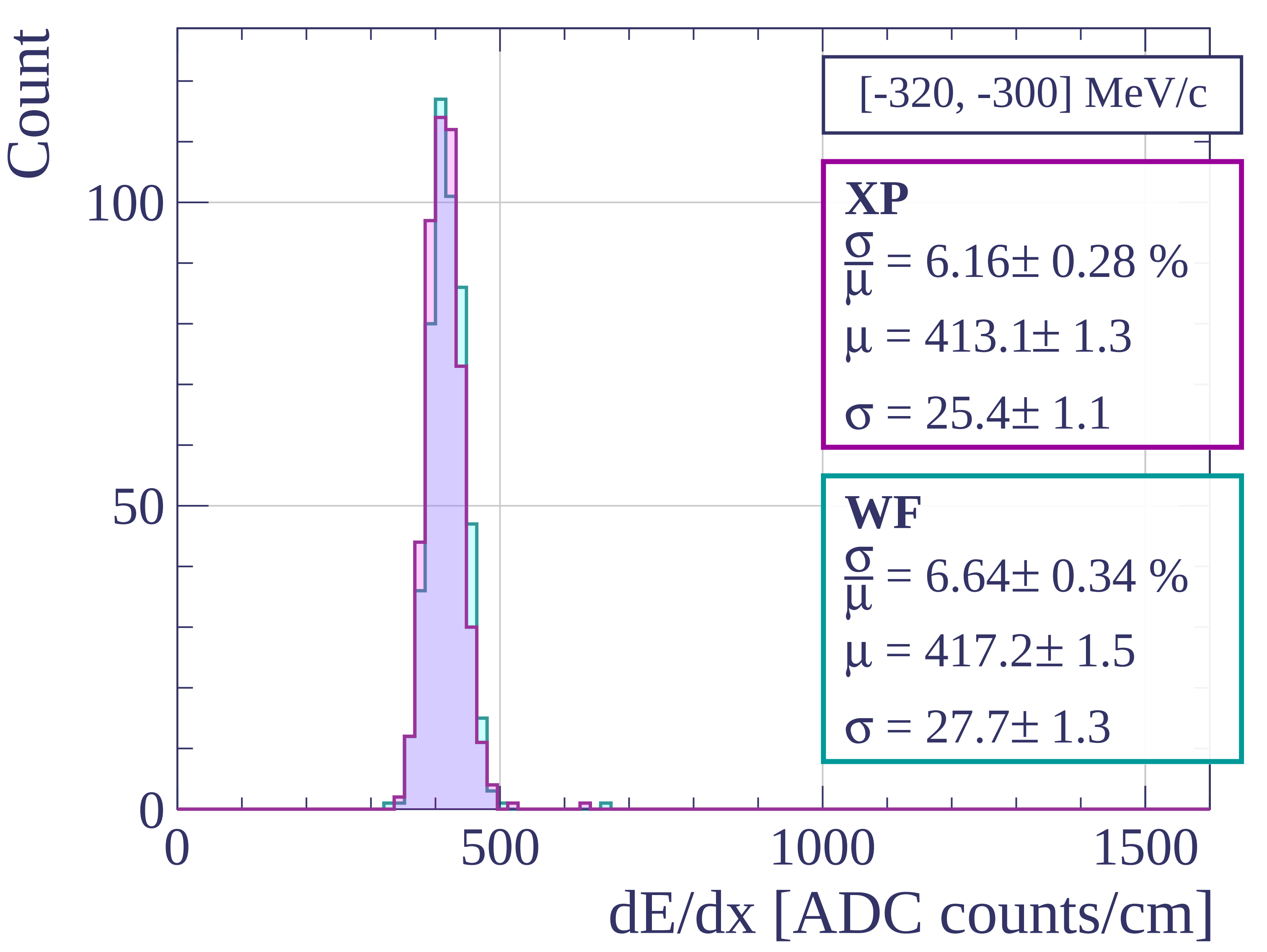

[-380, -360] MeV/c
XP
σ/μ = 5.58± 0.29 %
μ = 404.4± 1.2
σ = 22.6± 1.1
WF
σ/μ = 6.01± 0.27 %
μ = 409.1± 1.2
σ = 24.6± 1.0
Count
dE/dx [ADC counts/cm]
0
50
100
500
1000
1500



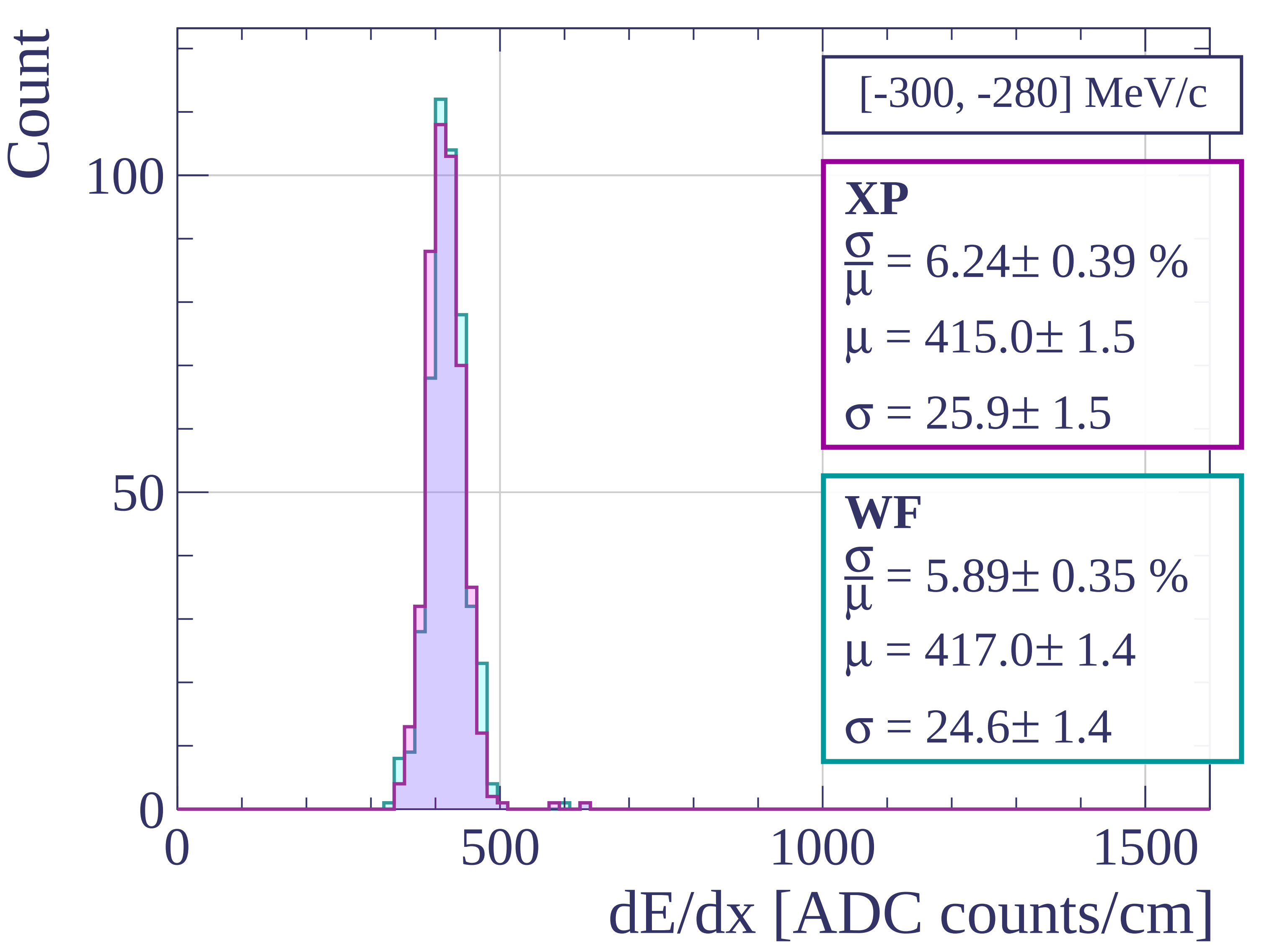
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-360, -340] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.82± 0.44 %
μ = 408.3± 1.4
σ = 23.7± 1.7
WF
$\frac{\sigma}{\mu}$ = 6.28± 0.41 %
μ = 411.6± 1.4
σ = 25.9± 1.6



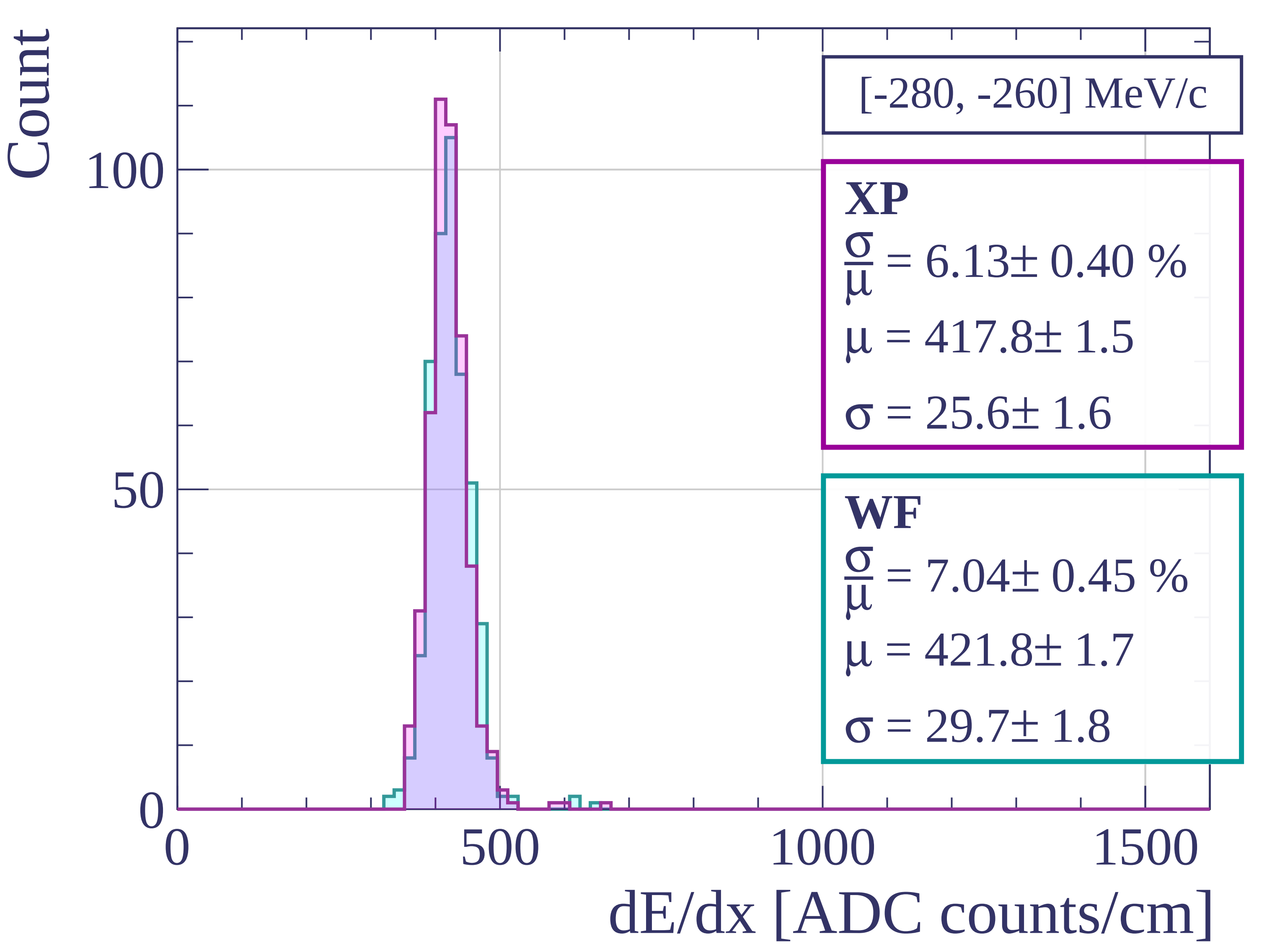
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-340, -320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.62± 0.39 %
μ = 408.5± 1.3
σ = 23.0± 1.5
WF
$\frac{\sigma}{\mu}$ = 5.93± 0.34 %
μ = 412.9± 1.3
σ = 24.5± 1.3



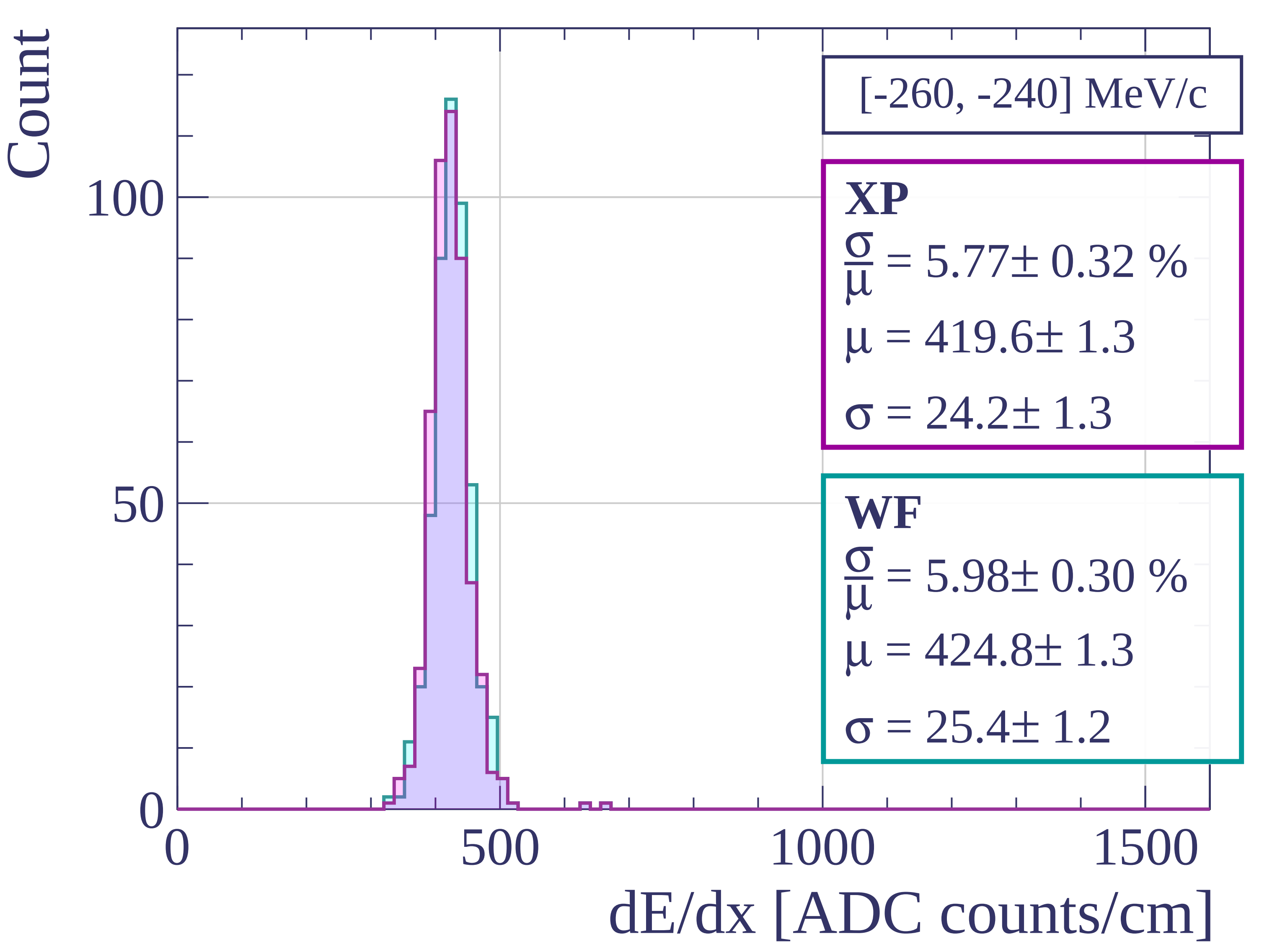
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-320, -300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.16± 0.28 %
μ = 413.1± 1.3
σ = 25.4± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.64± 0.34 %
μ = 417.2± 1.5
σ = 27.7± 1.3



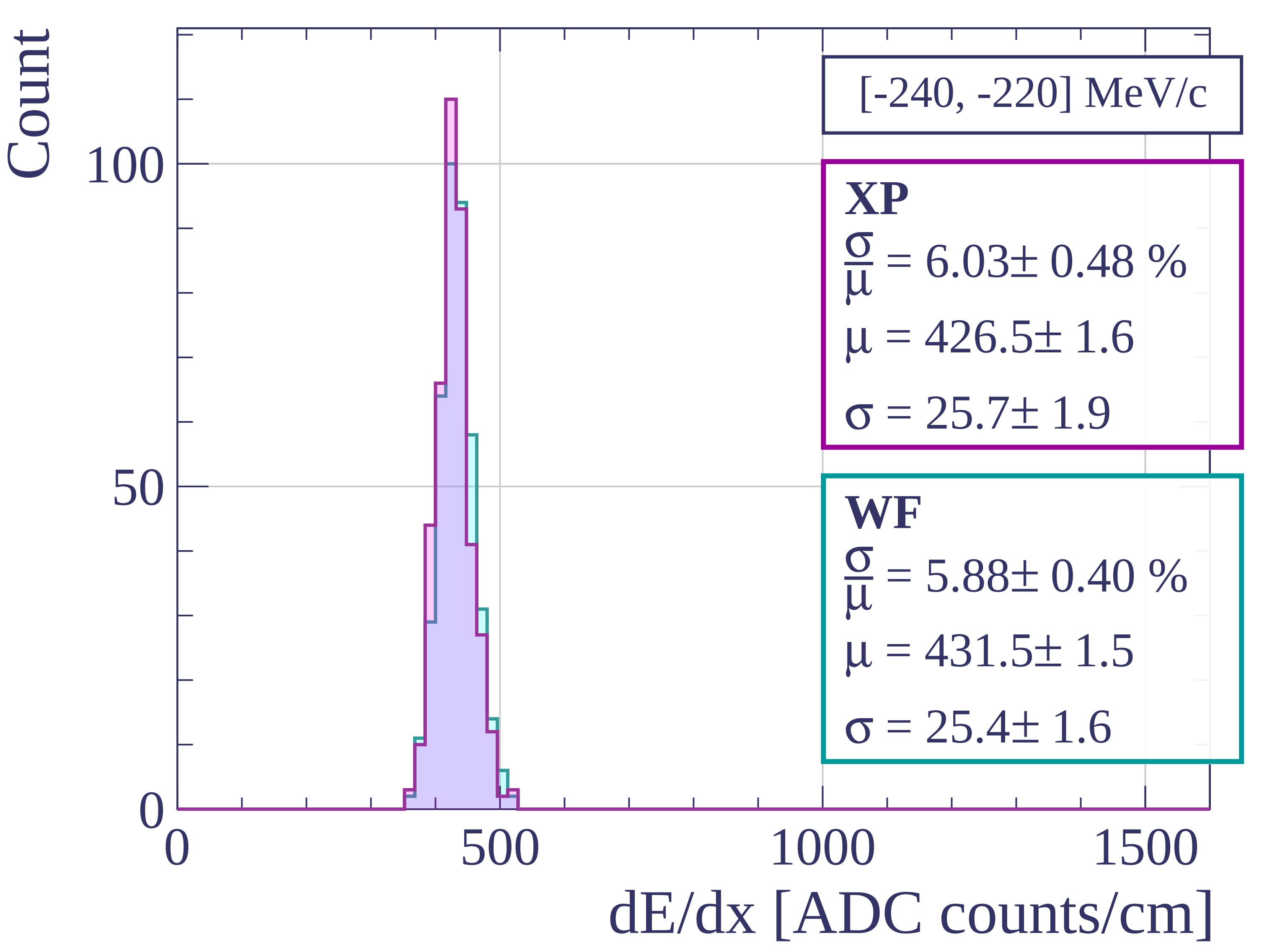
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-300, -280] MeV/c
XP
σ/μ = 6.24± 0.39 %
μ = 415.0± 1.5
σ = 25.9± 1.5
WF
σ/μ = 5.89± 0.35 %
μ = 417.0± 1.4
σ = 24.6± 1.4

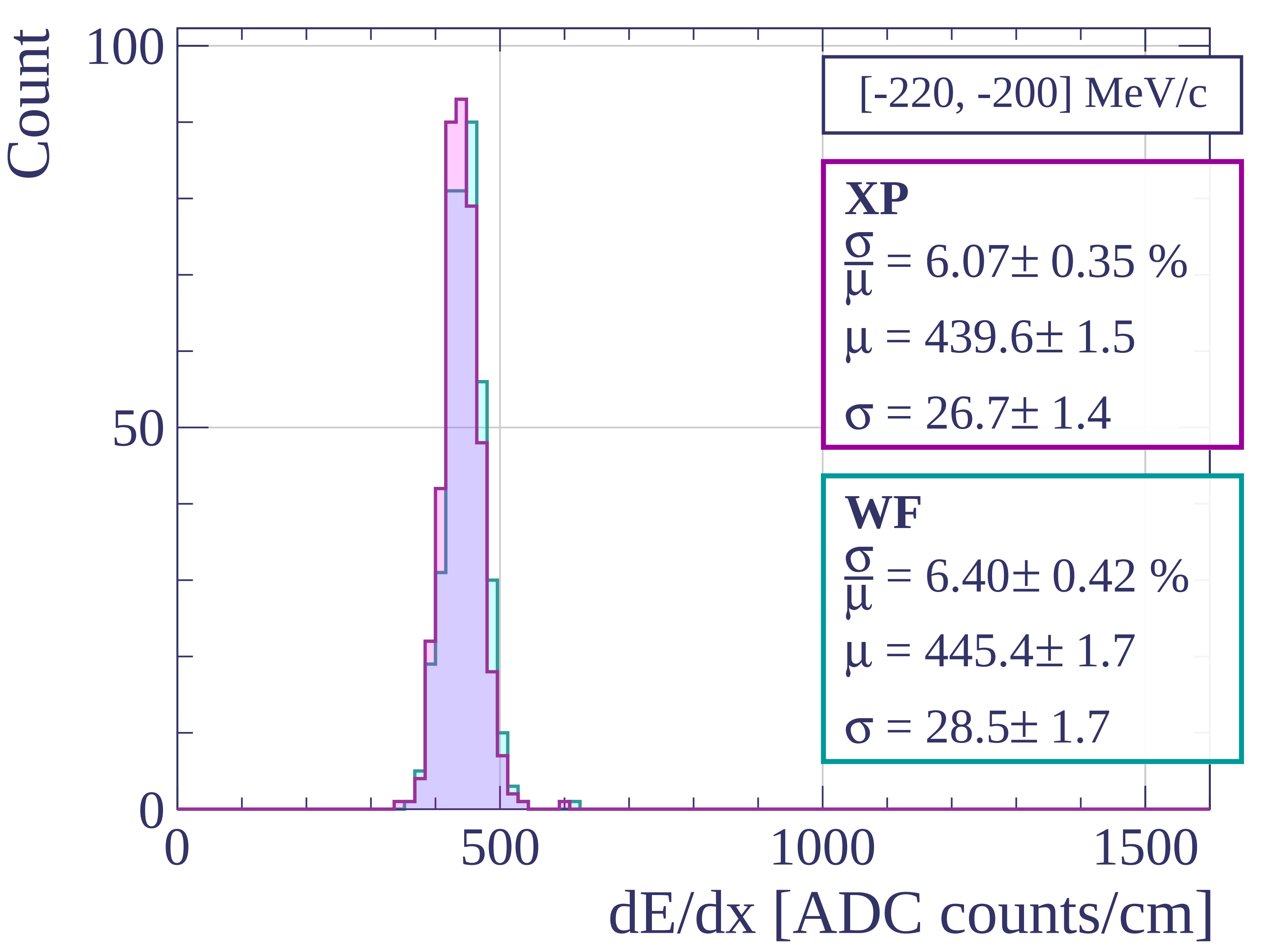


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
50
100
[-280, -260] MeV/c
XP
σ/μ = 6.13± 0.40 %
μ = 417.8± 1.5
σ = 25.6± 1.6
WF
σ/μ = 7.04± 0.45 %
μ = 421.8± 1.7
σ = 29.7± 1.8

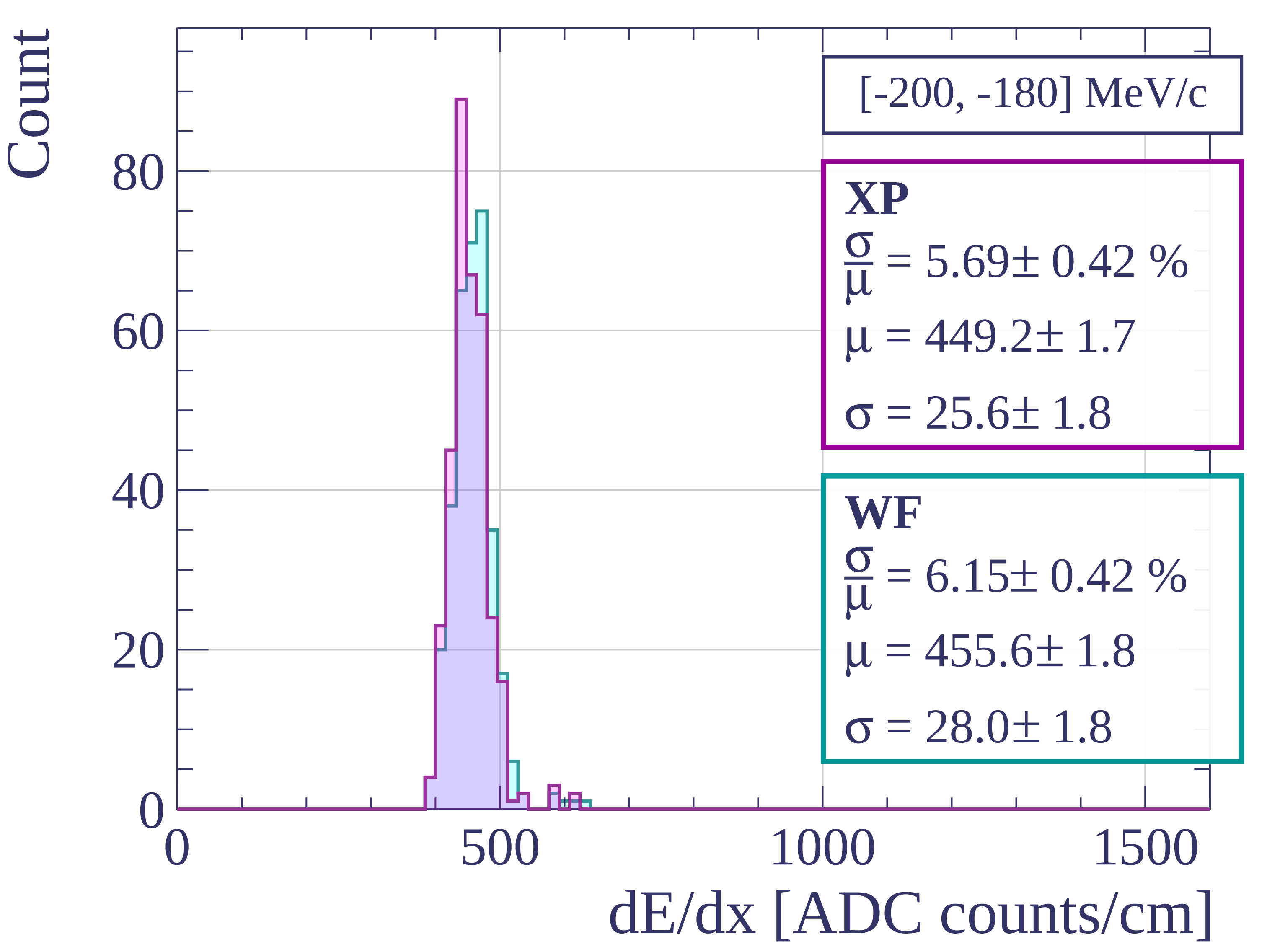


Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
[-260, -240] MeV/c
XP
σ/μ = 5.77± 0.32 %
μ = 419.6± 1.3
σ = 24.2± 1.3
WF
σ/μ = 5.98± 0.30 %
μ = 424.8± 1.3
σ = 25.4± 1.2

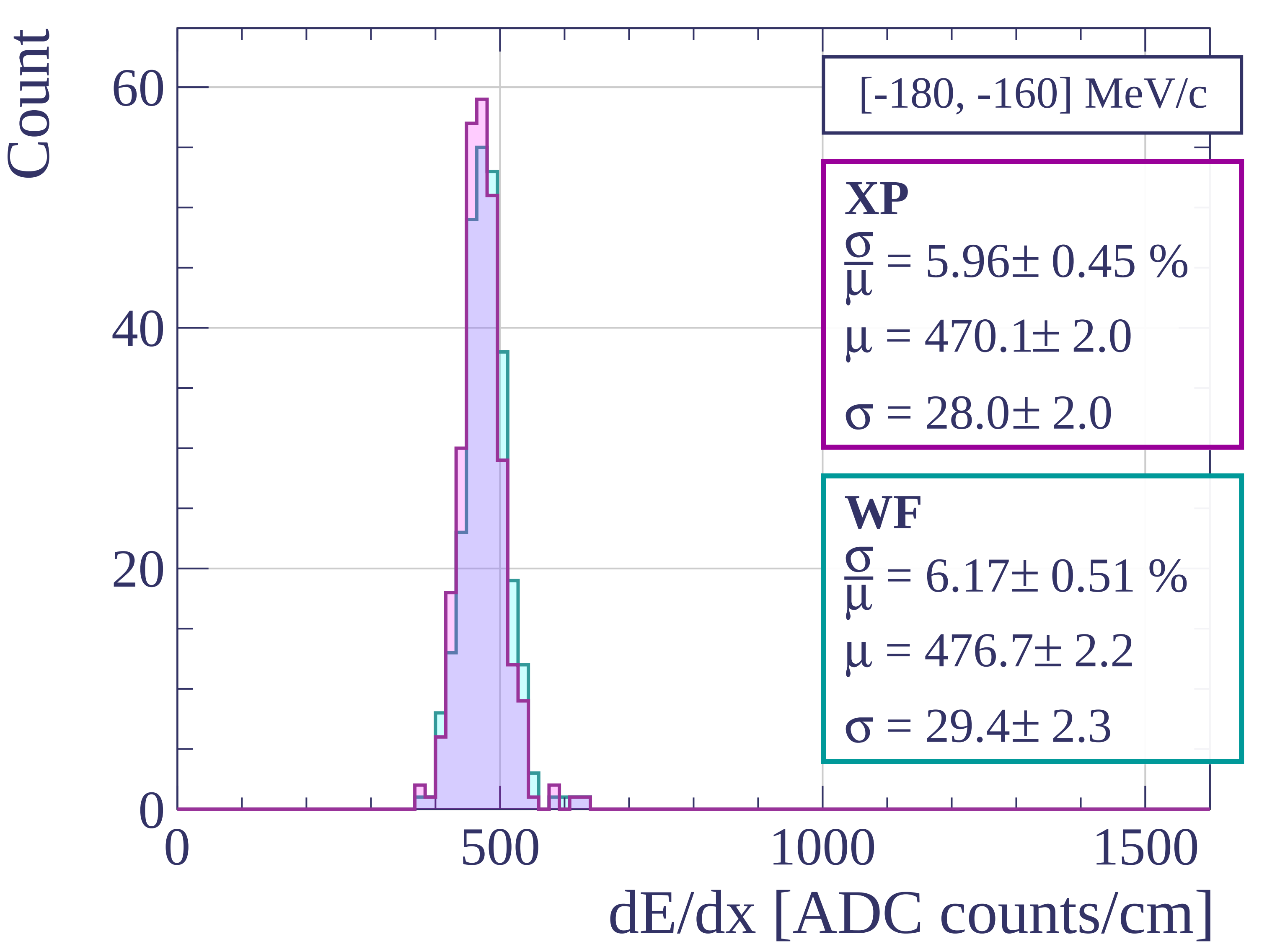

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-240, -220] MeV/c
XP
σ/μ = 6.03± 0.48 %
μ = 426.5± 1.6
σ = 25.7± 1.9
WF
σ/μ = 5.88± 0.40 %
μ = 431.5± 1.5
σ = 25.4± 1.6

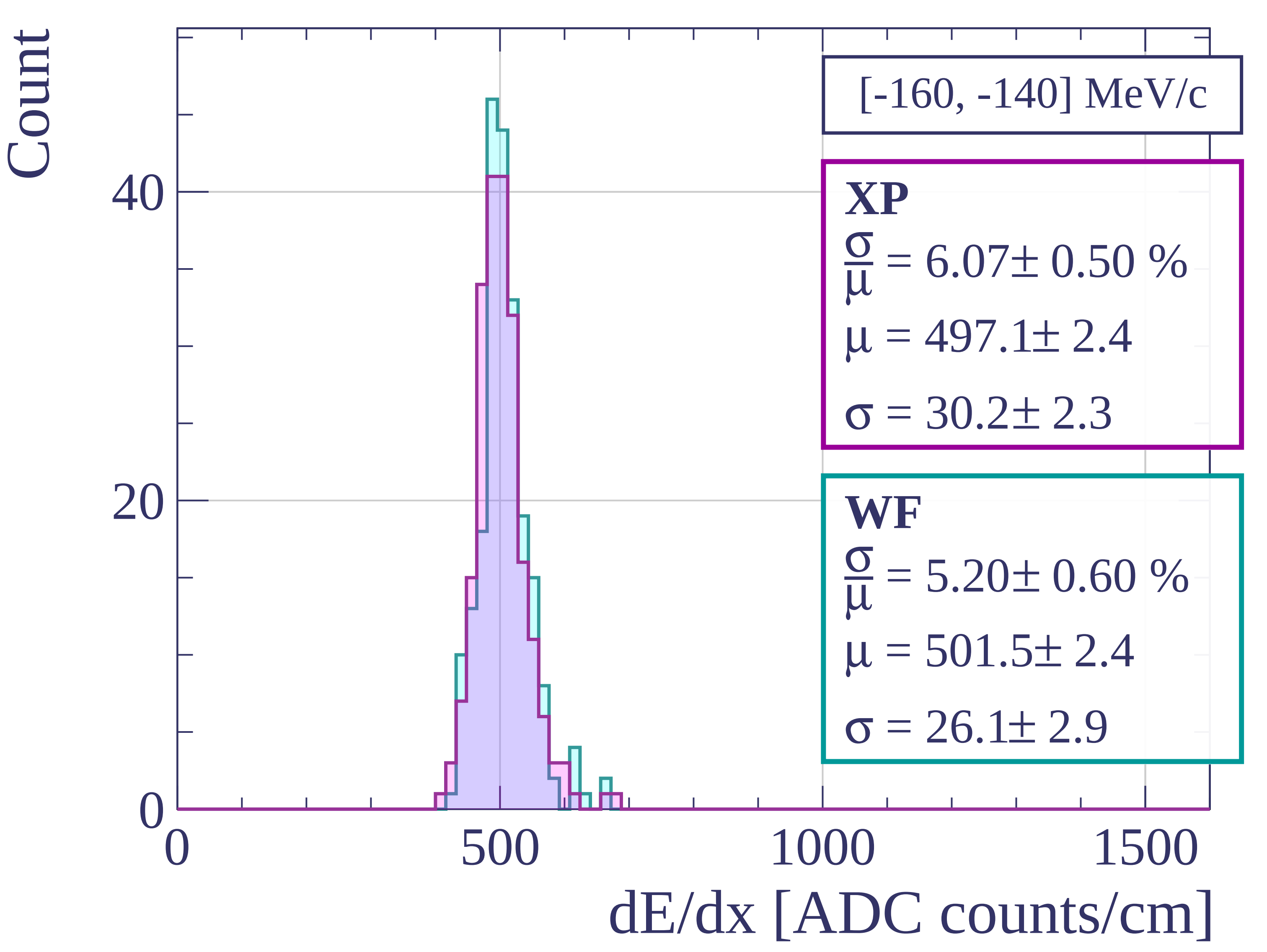


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-220, -200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.07± 0.35 %
μ = 439.6± 1.5
σ = 26.7± 1.4
WF
$\frac{\sigma}{\mu}$ = 6.40± 0.42 %
μ = 445.4± 1.7
σ = 28.5± 1.7

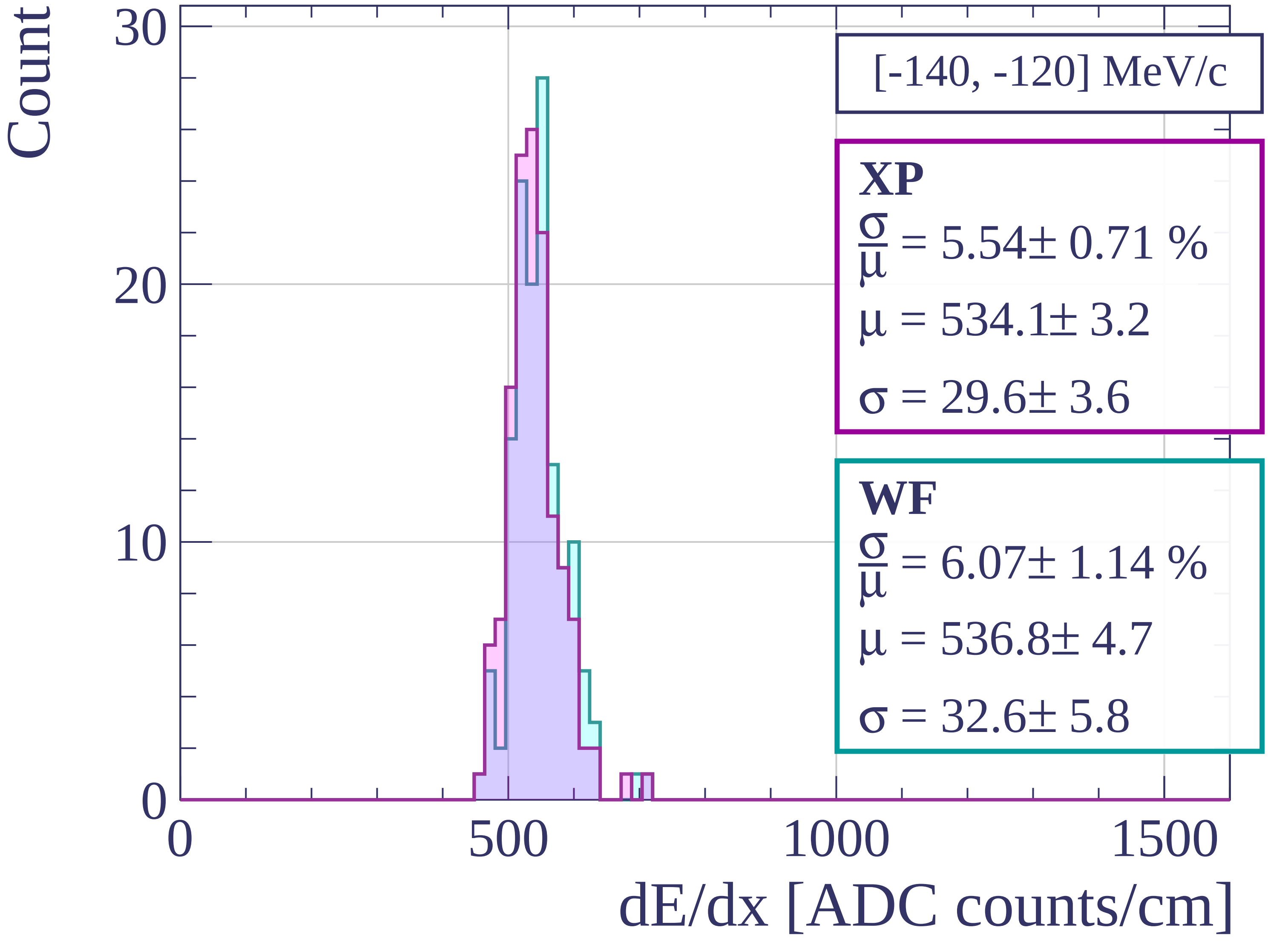


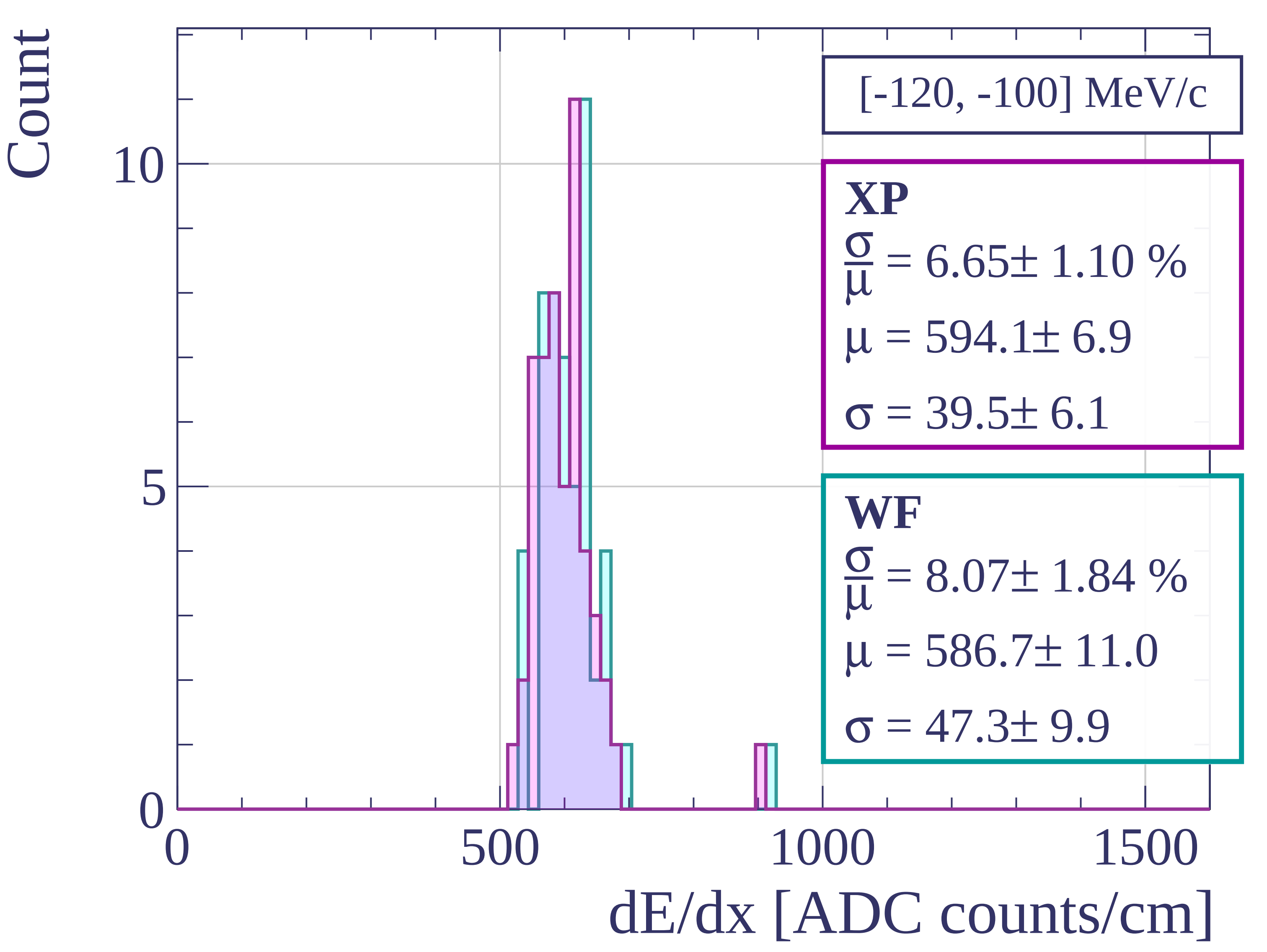
Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
[-200, -180] MeV/c
XP
σ/μ = 5.69± 0.42 %
μ = 449.2± 1.7
σ = 25.6± 1.8
WF
σ/μ = 6.15± 0.42 %
μ = 455.6± 1.8
σ = 28.0± 1.8


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-180, -160] MeV/c
XP
σ/μ = 5.96± 0.45 %
μ = 470.1± 2.0
σ = 28.0± 2.0
WF
σ/μ = 6.17± 0.51 %
μ = 476.7± 2.2
σ = 29.4± 2.3

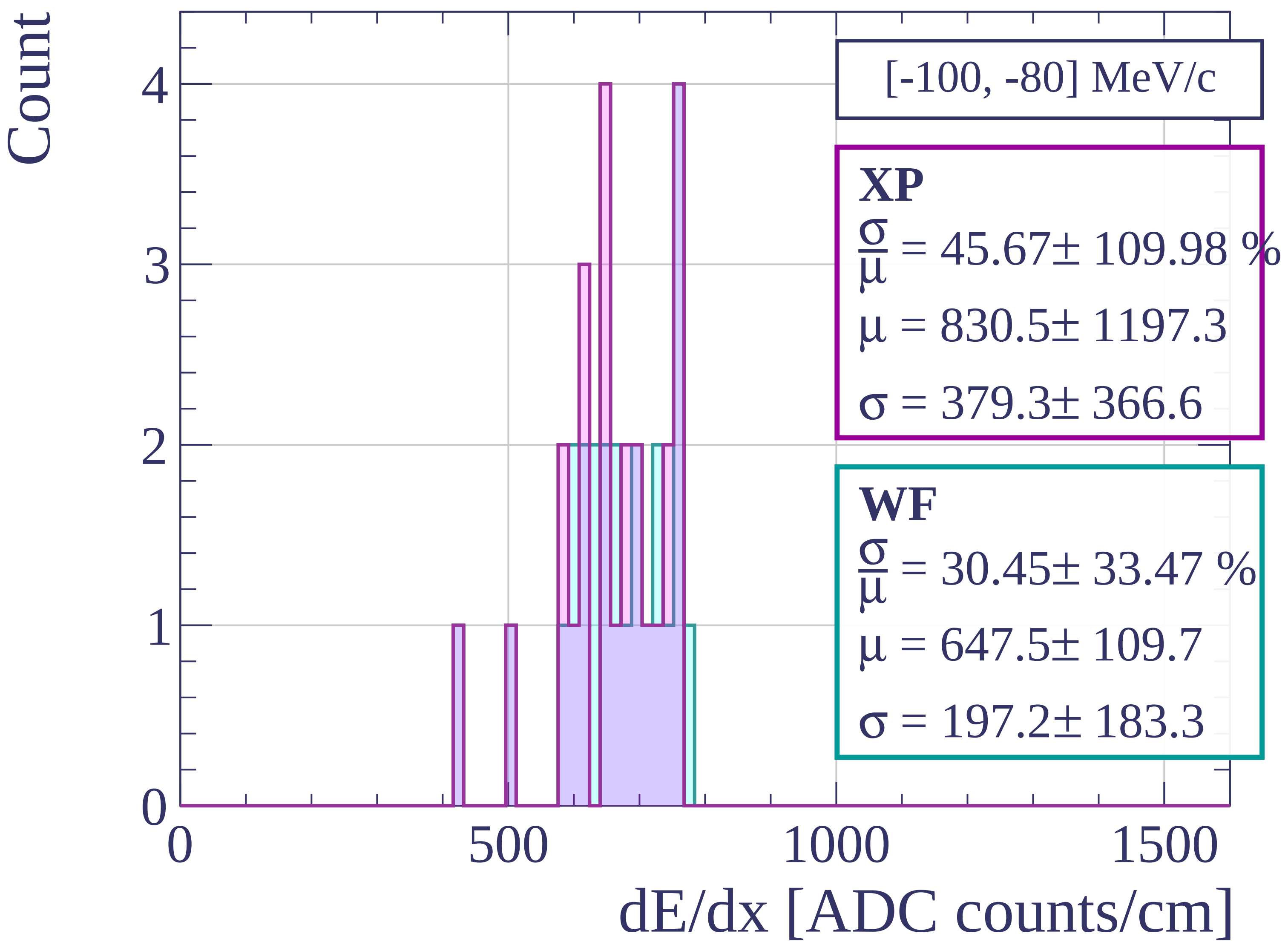


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-160, -140] MeV/c
XP
σ/μ = 6.07± 0.50 %
μ = 497.1± 2.4
σ = 30.2± 2.3
WF
σ/μ = 5.20± 0.60 %
μ = 501.5± 2.4
σ = 26.1± 2.9

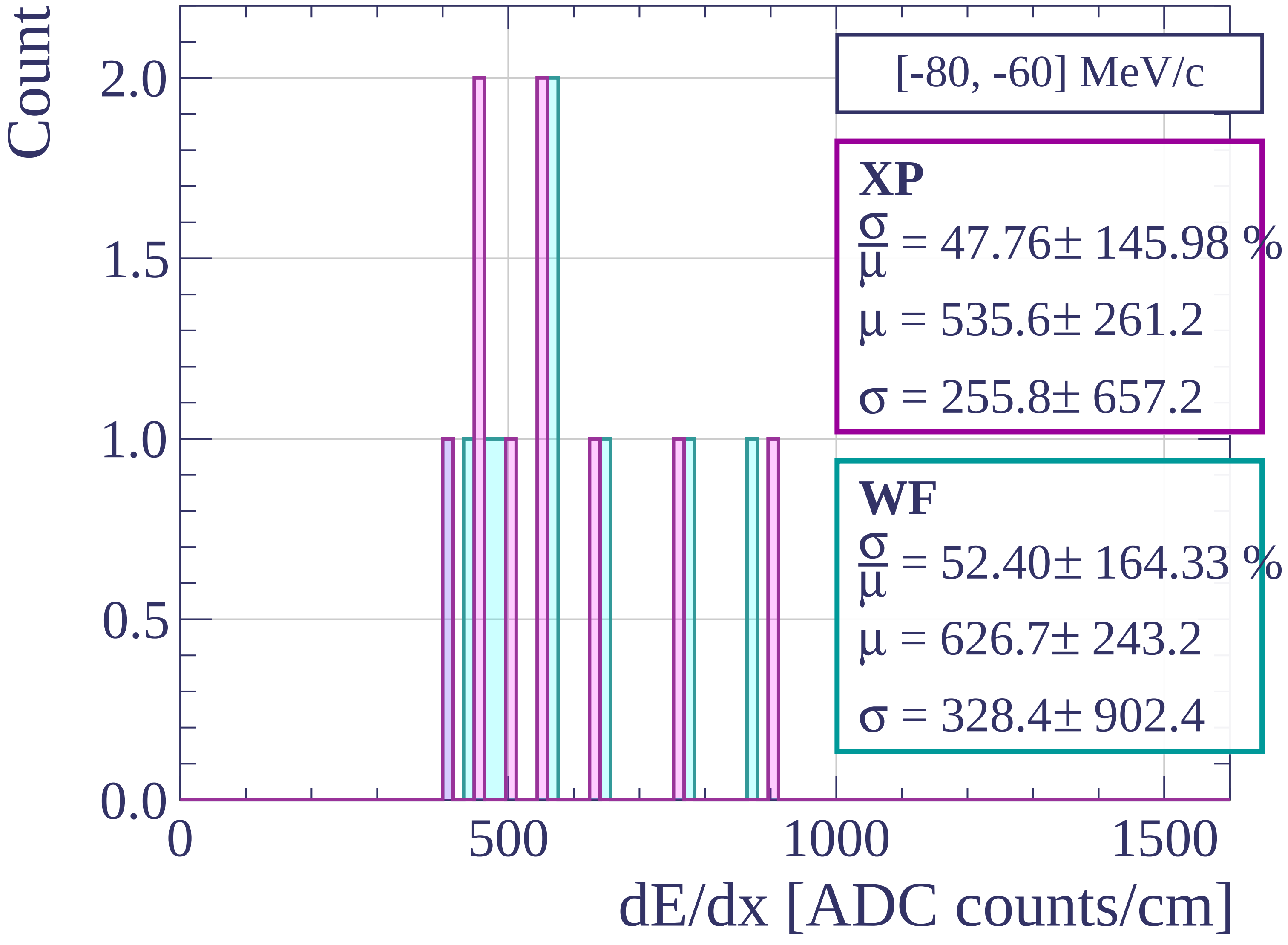

Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-140, -120] MeV/c
XP
σ/μ = 5.54± 0.71 %
μ = 534.1± 3.2
σ = 29.6± 3.6
WF
σ/μ = 6.07± 1.14 %
μ = 536.8± 4.7
σ = 32.6± 5.8



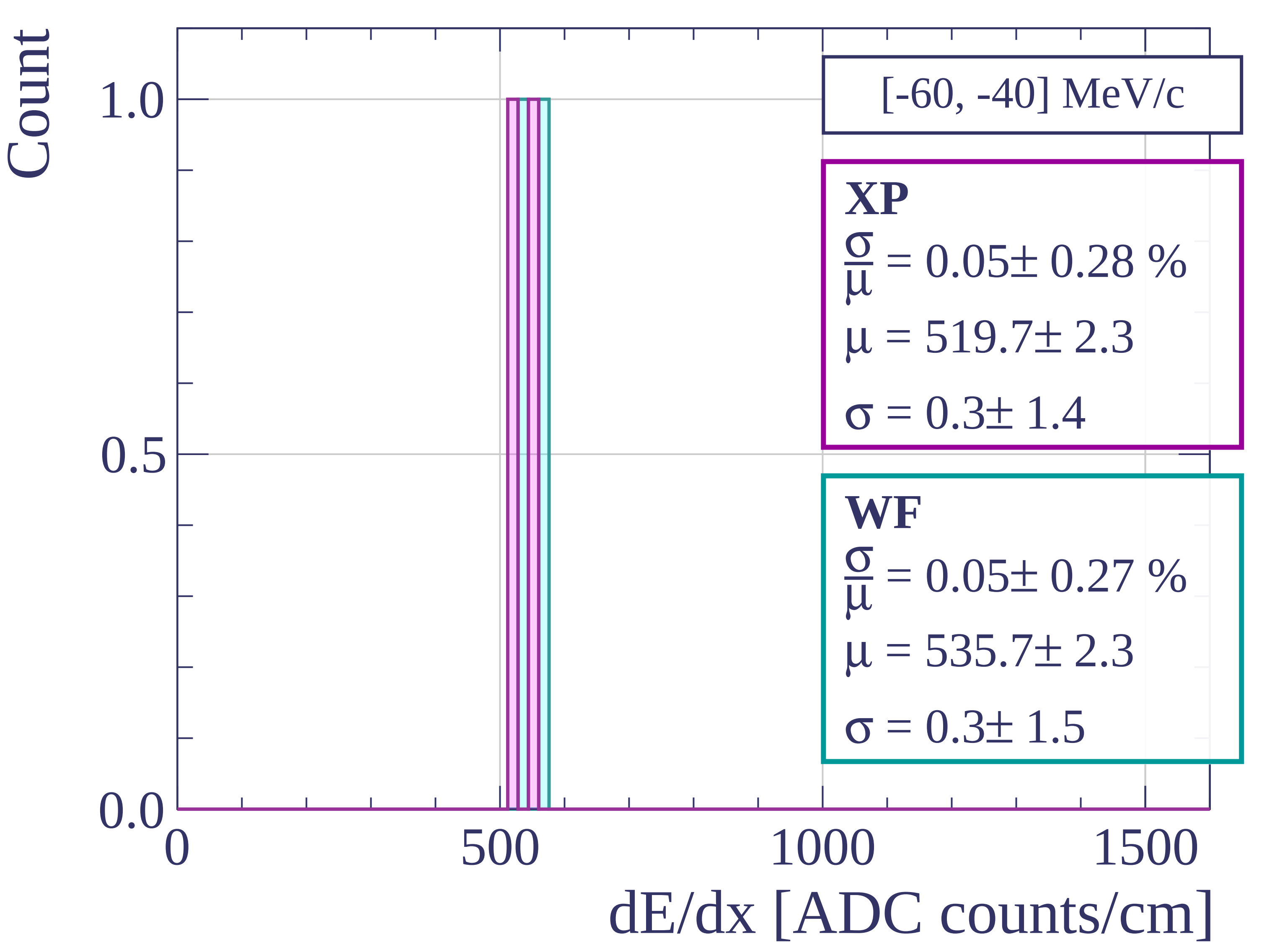
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-120, -100] MeV/c
XP
σ/μ = 6.65± 1.10 %
μ = 594.1± 6.9
σ = 39.5± 6.1
WF
σ/μ = 8.07± 1.84 %
μ = 586.7± 11.0
σ = 47.3± 9.9

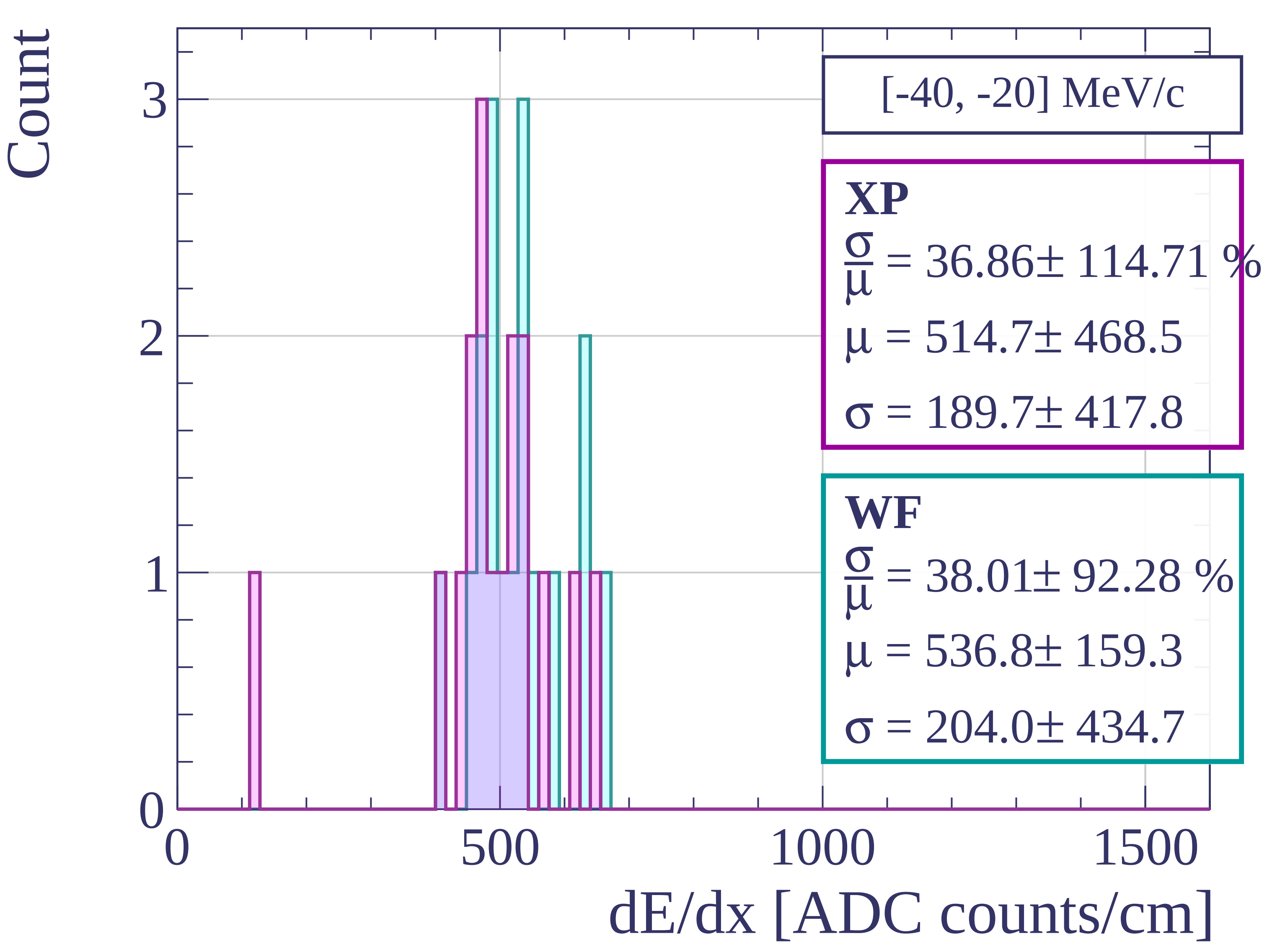


Count
[-100, -80] MeV/c
XP
σ/μ = 45.67± 109.98 %
μ = 830.5± 1197.3
σ = 379.3± 366.6
WF
σ/μ = 30.45± 33.47 %
μ = 647.5± 109.7
σ = 197.2± 183.3
0
1
2
3
4
0
500
1000
1500
dE/dx [ADC counts/cm]

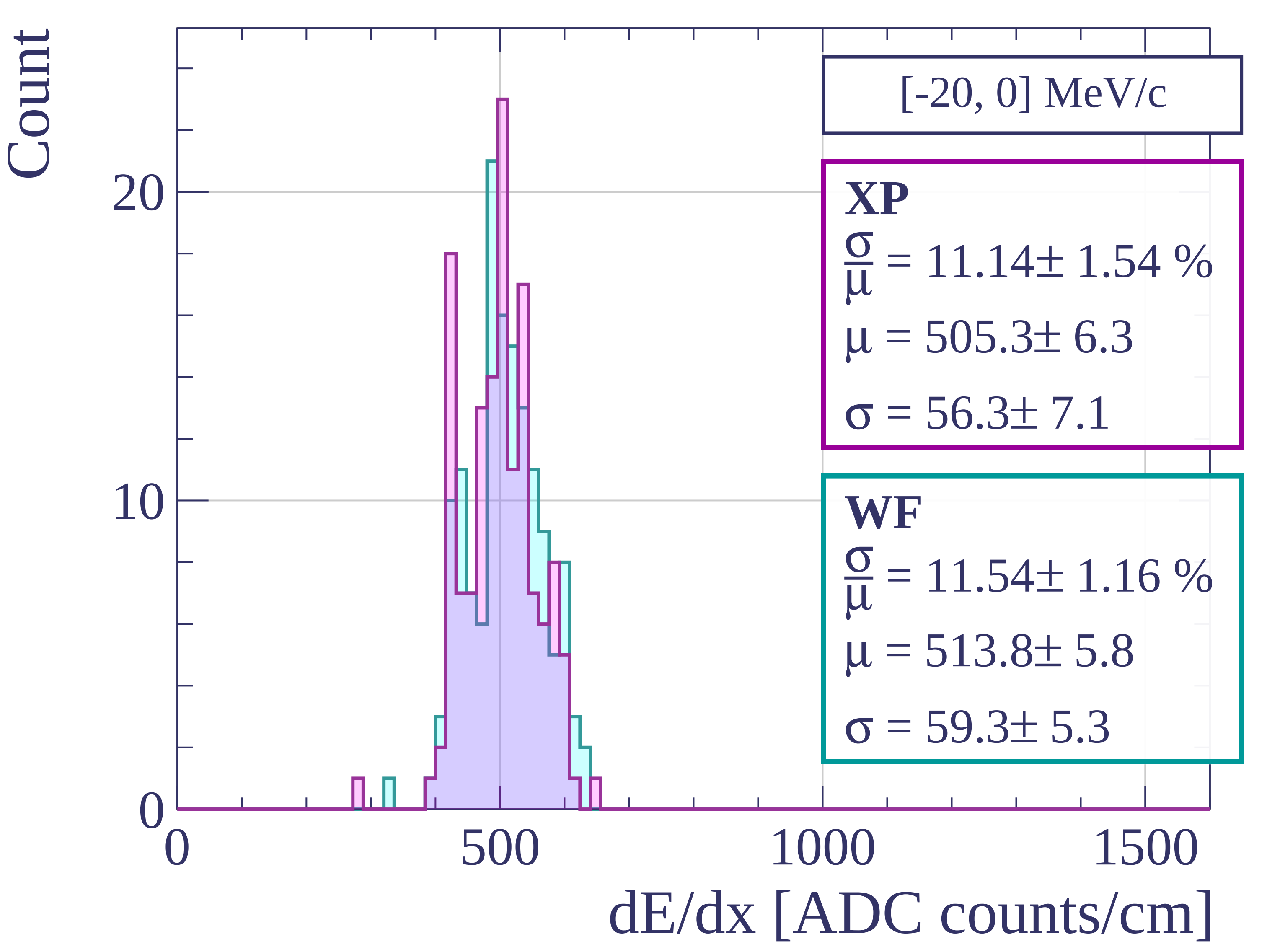


Count
dE/dx [ADC counts/cm]
[-80, -60] MeV/c
XP
$\frac{\sigma}{\mu}$ = 47.76± 145.98 %
μ = 535.6± 261.2
σ = 255.8± 657.2
WF
$\frac{\sigma}{\mu}$ = 52.40± 164.33 %
μ = 626.7± 243.2
σ = 328.4± 902.4

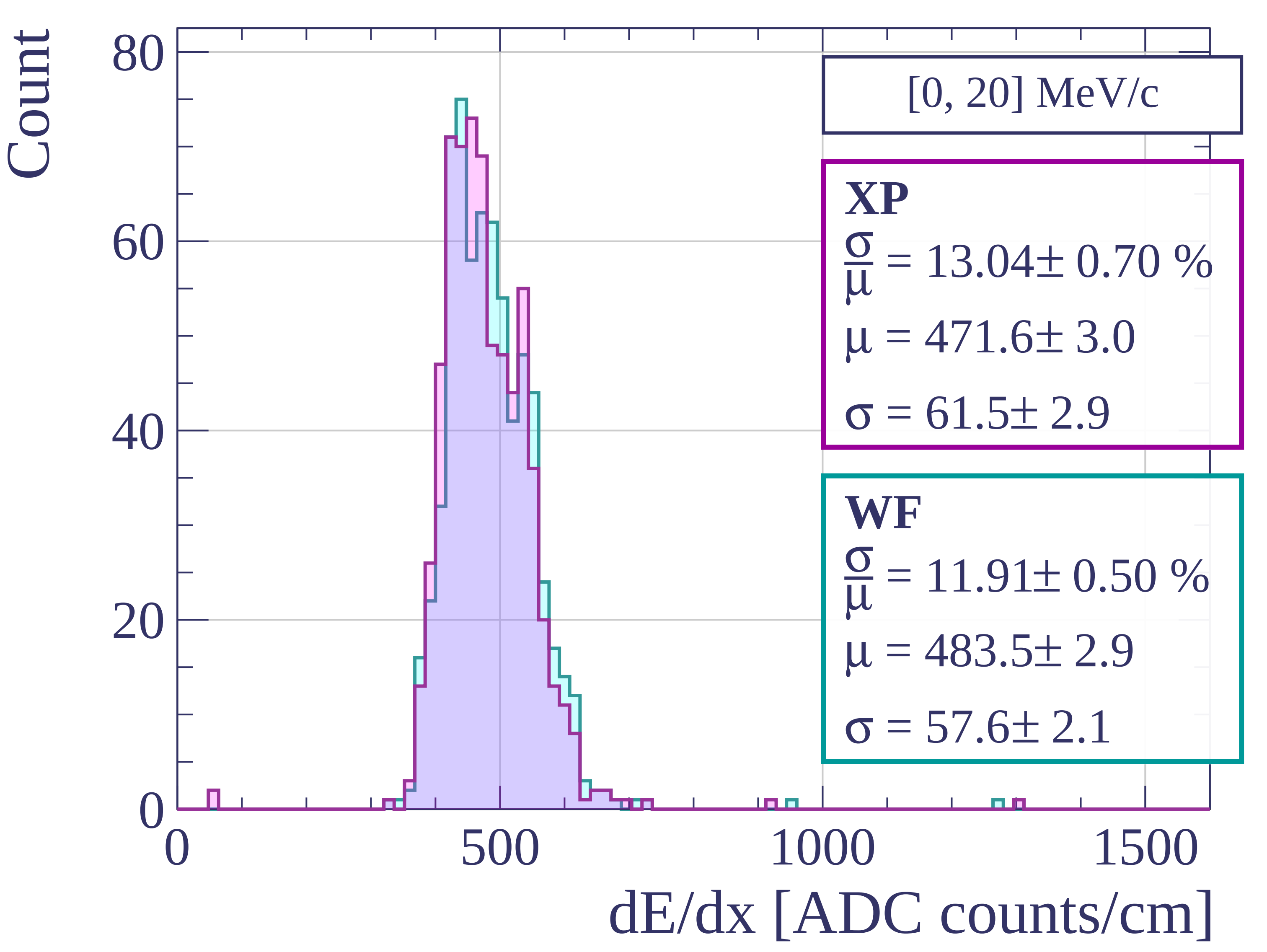

Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-60, -40] MeV/c
XP
σ/μ = 0.05± 0.28 %
μ = 519.7± 2.3
σ = 0.3± 1.4
WF
σ/μ = 0.05± 0.27 %
μ = 535.7± 2.3
σ = 0.3± 1.5



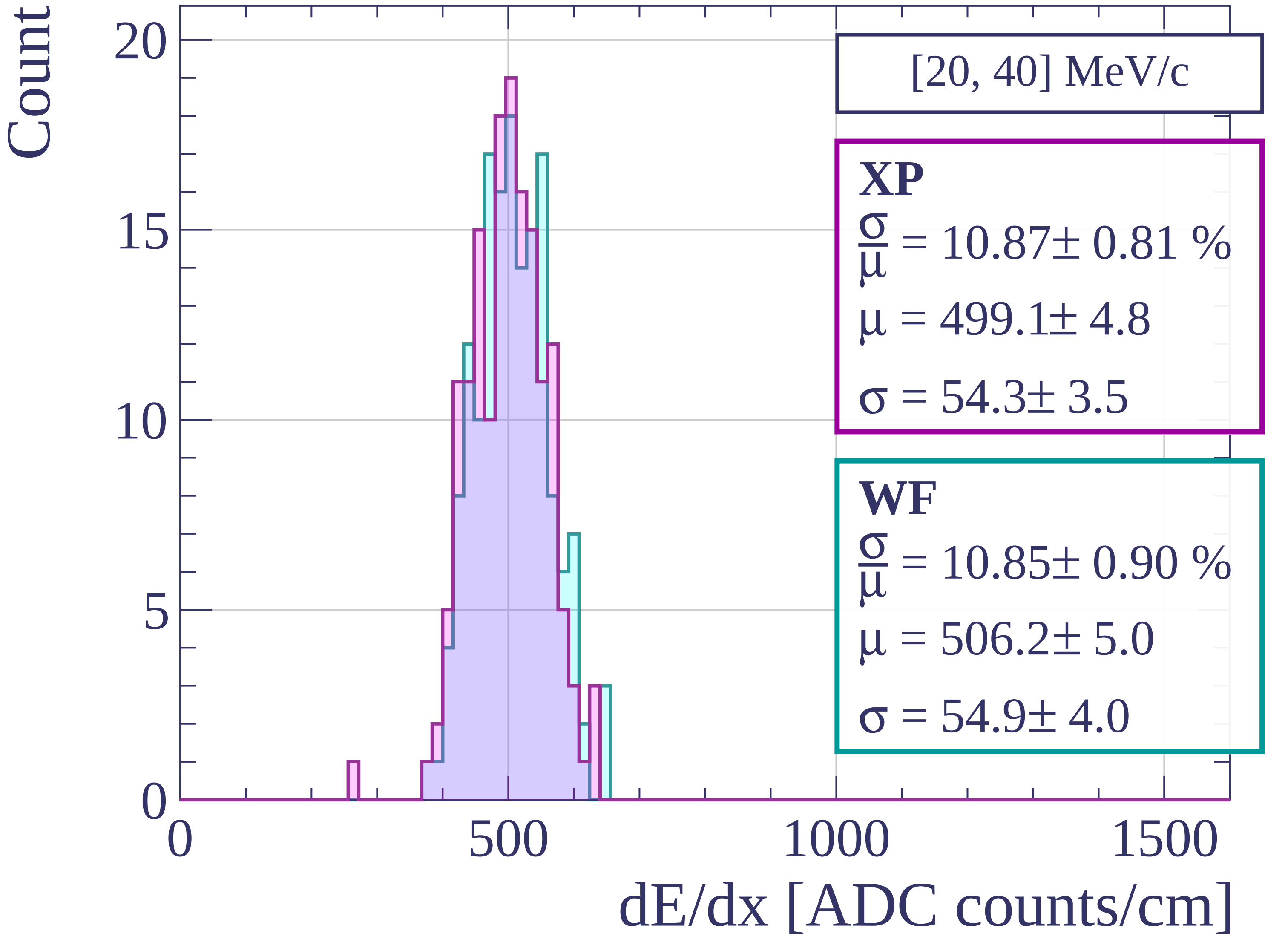
Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-40, -20] MeV/c
XP
σ/μ = 36.86± 114.71 %
μ = 514.7± 468.5
σ = 189.7± 417.8
WF
σ/μ = 38.01± 92.28 %
μ = 536.8± 159.3
σ = 204.0± 434.7



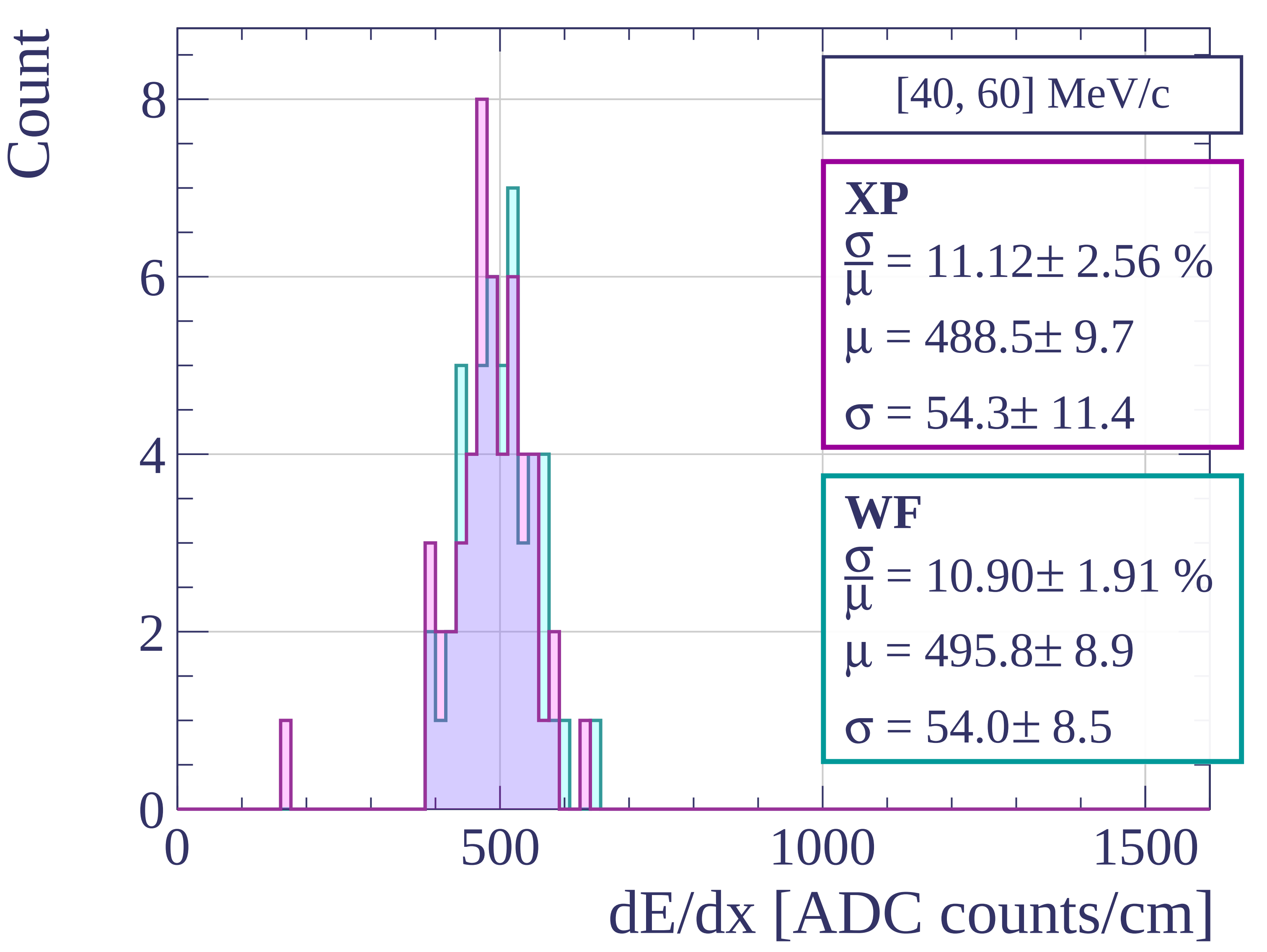
Count
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-20, 0] MeV/c
XP
σ/μ = 11.14± 1.54 %
μ = 505.3± 6.3
σ = 56.3± 7.1
WF
σ/μ = 11.54± 1.16 %
μ = 513.8± 5.8
σ = 59.3± 5.3



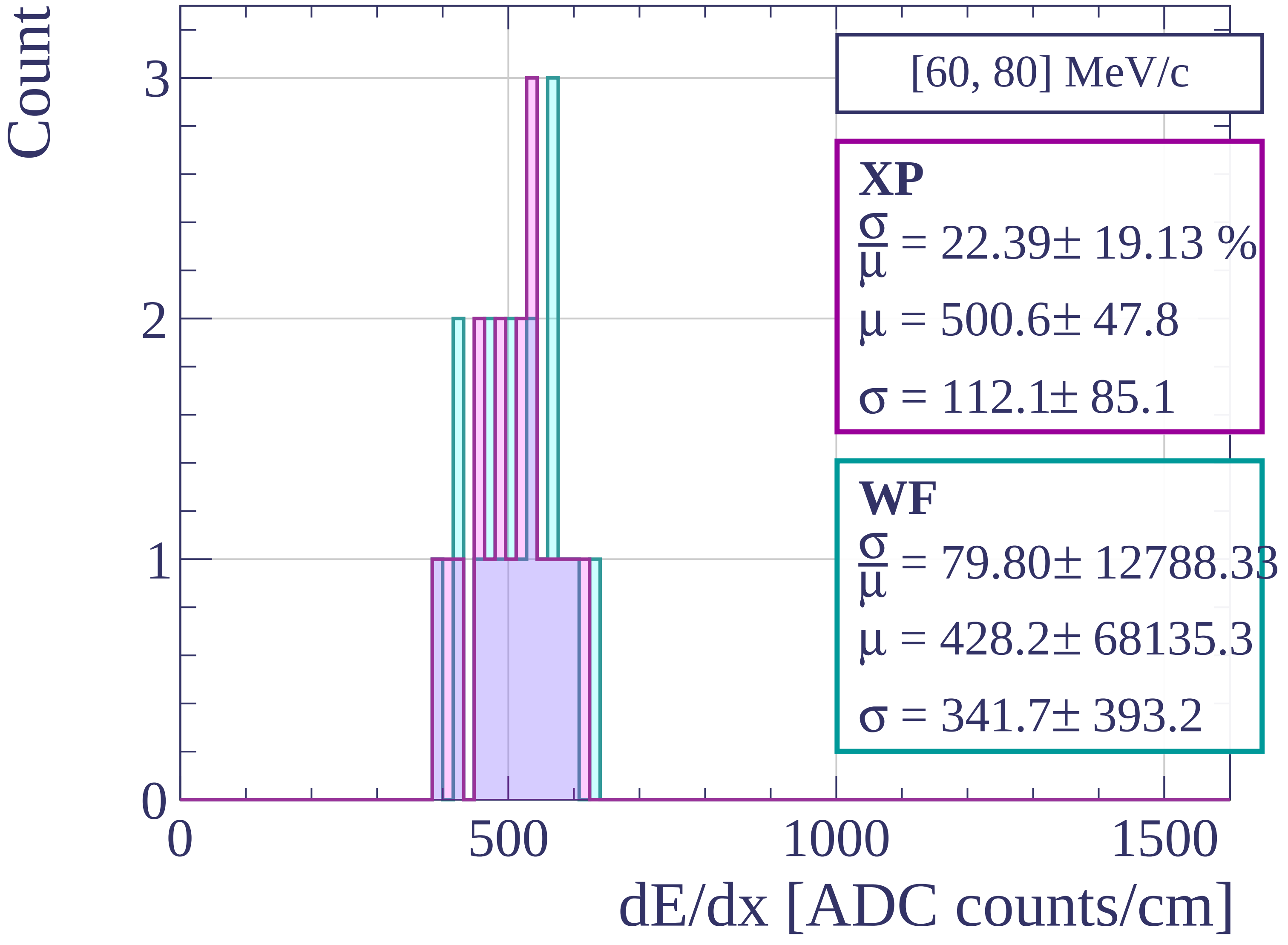
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 20] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.04± 0.70 %
μ = 471.6± 3.0
σ = 61.5± 2.9
WF
$\frac{\sigma}{\mu}$ = 11.91± 0.50 %
μ = 483.5± 2.9
σ = 57.6± 2.1

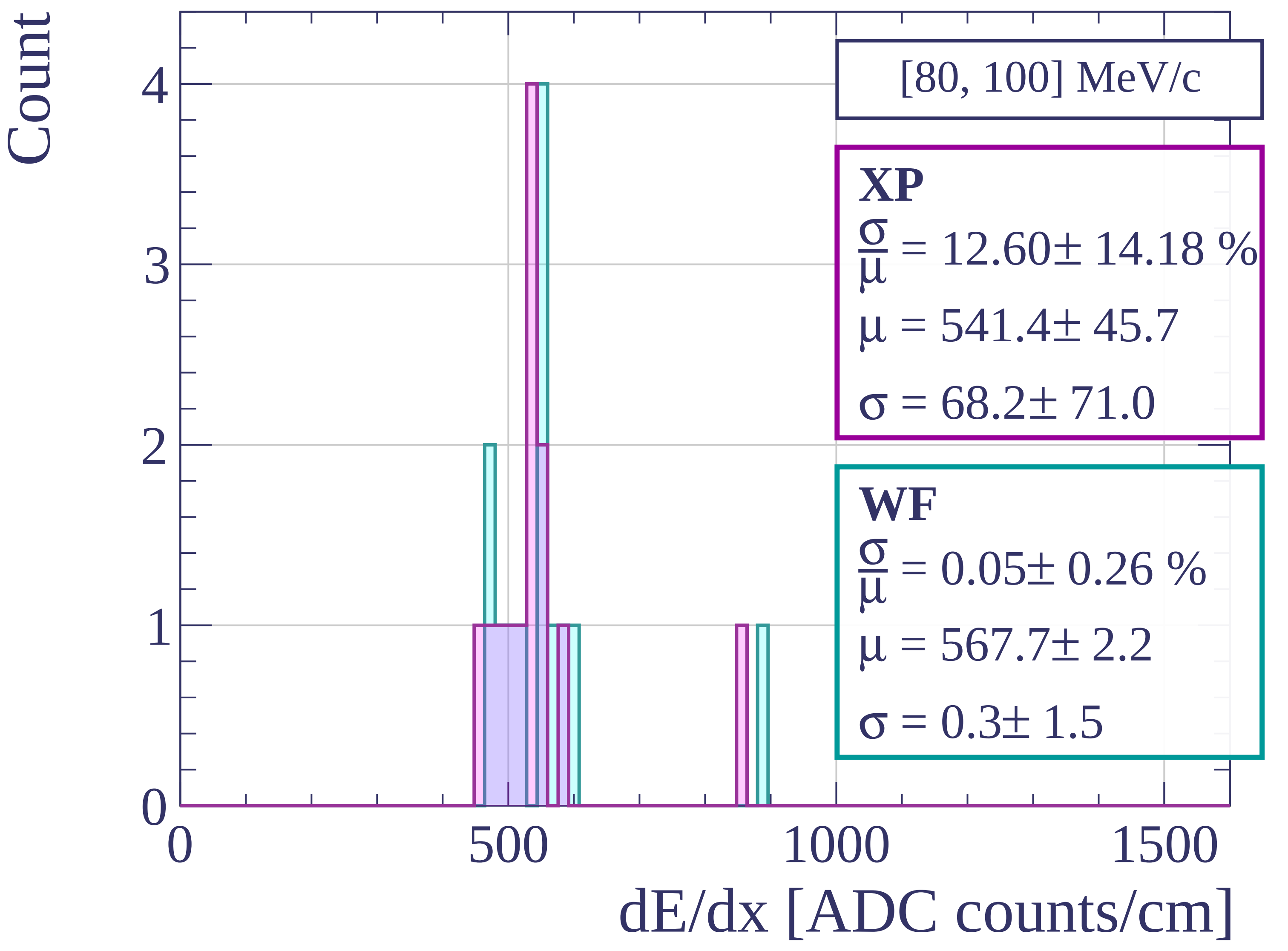


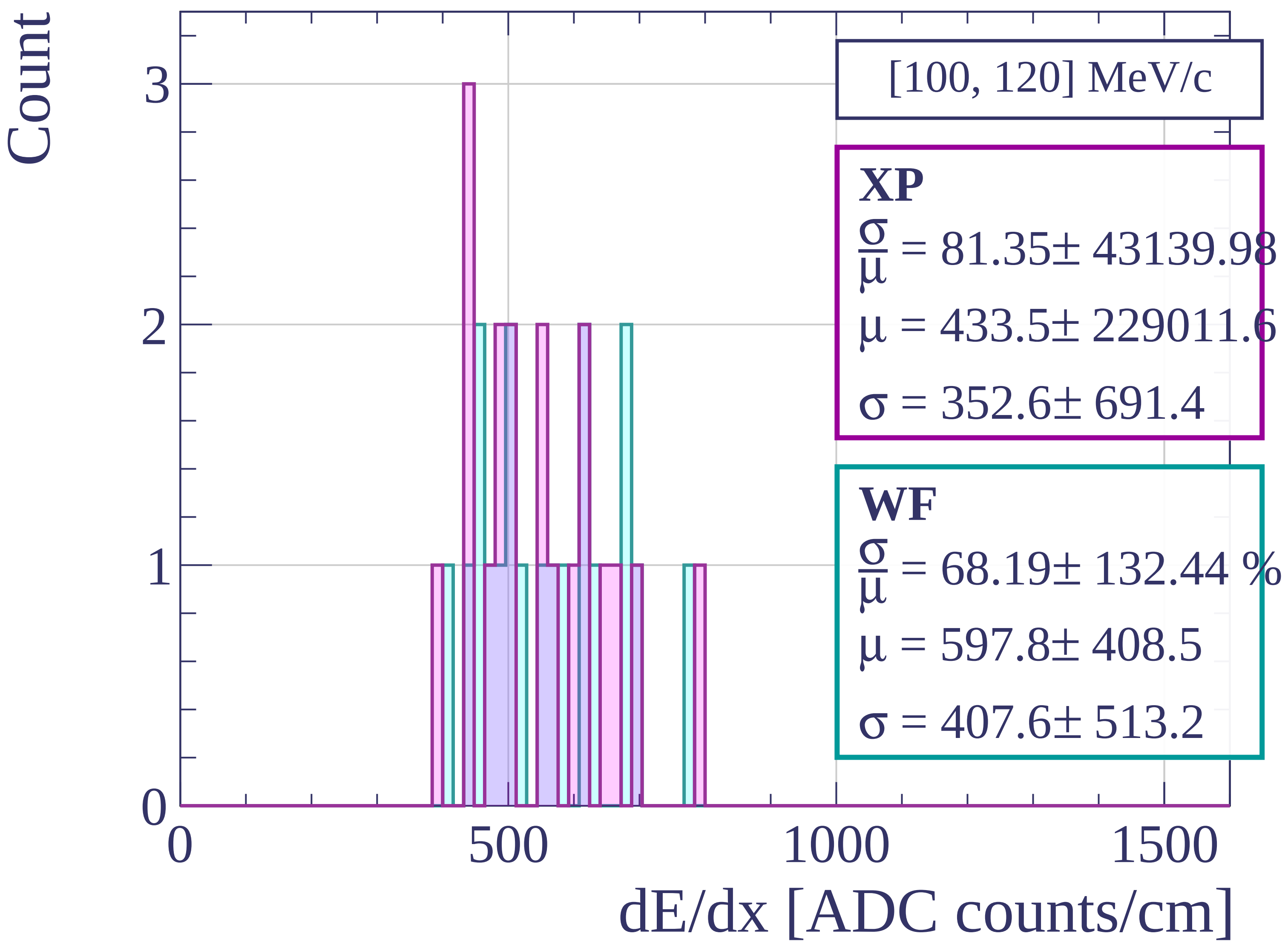
Count
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[20, 40] MeV/c
XP
σ/μ = 10.87± 0.81 %
μ = 499.1± 4.8
σ = 54.3± 3.5
WF
σ/μ = 10.85± 0.90 %
μ = 506.2± 5.0
σ = 54.9± 4.0



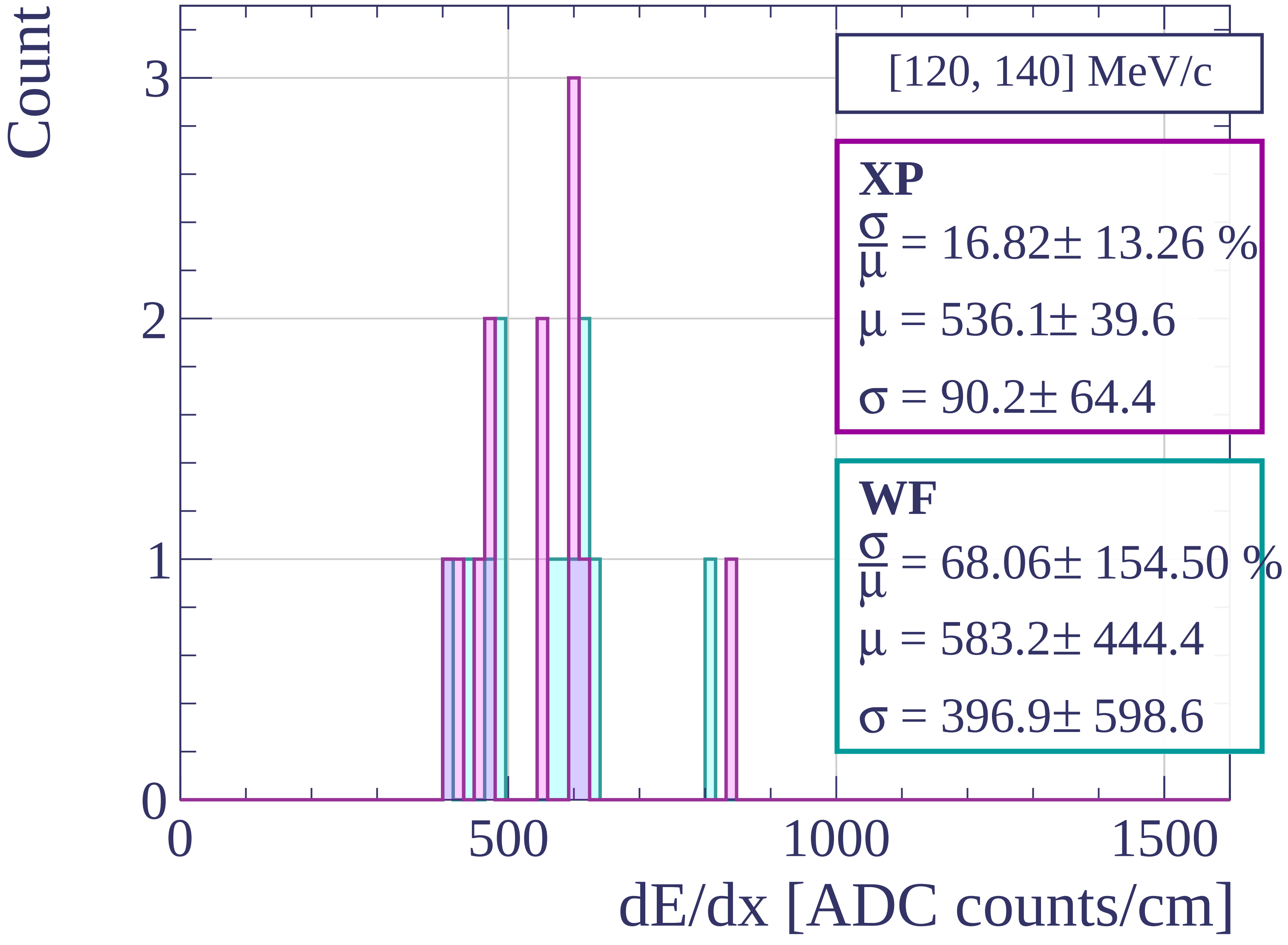
Count
dE/dx [ADC counts/cm]
[40, 60] MeV/c
XP
σ/μ = 11.12± 2.56 %
μ = 488.5± 9.7
σ = 54.3± 11.4
WF
σ/μ = 10.90± 1.91 %
μ = 495.8± 8.9
σ = 54.0± 8.5
0
2
4
6
8
0
500
1000
1500

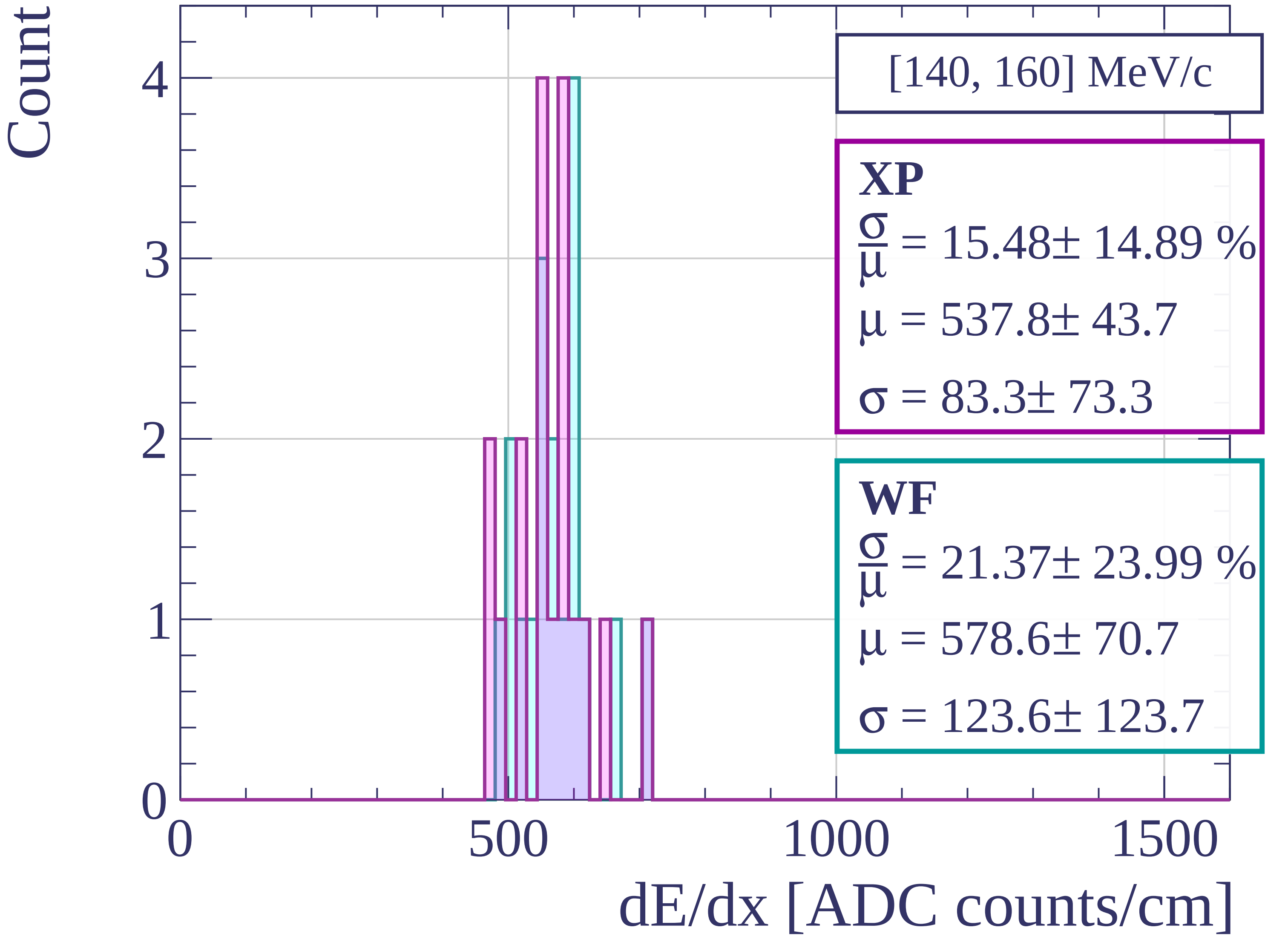

Count
0
1
2
3
0
500
1000
1500
dE/dx [ADC counts/cm]
[60, 80] MeV/c
XP
σ/μ = 22.39± 19.13 %
μ = 500.6± 47.8
σ = 112.1± 85.1
WF
σ/μ = 79.80± 12788.33
μ = 428.2± 68135.3
σ = 341.7± 393.2


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
1
2
3
4
[80, 100] MeV/c
XP
σ/μ = 12.60± 14.18 %
μ = 541.4± 45.7
σ = 68.2± 71.0
WF
σ/μ = 0.05± 0.26 %
μ = 567.7± 2.2
σ = 0.3± 1.5

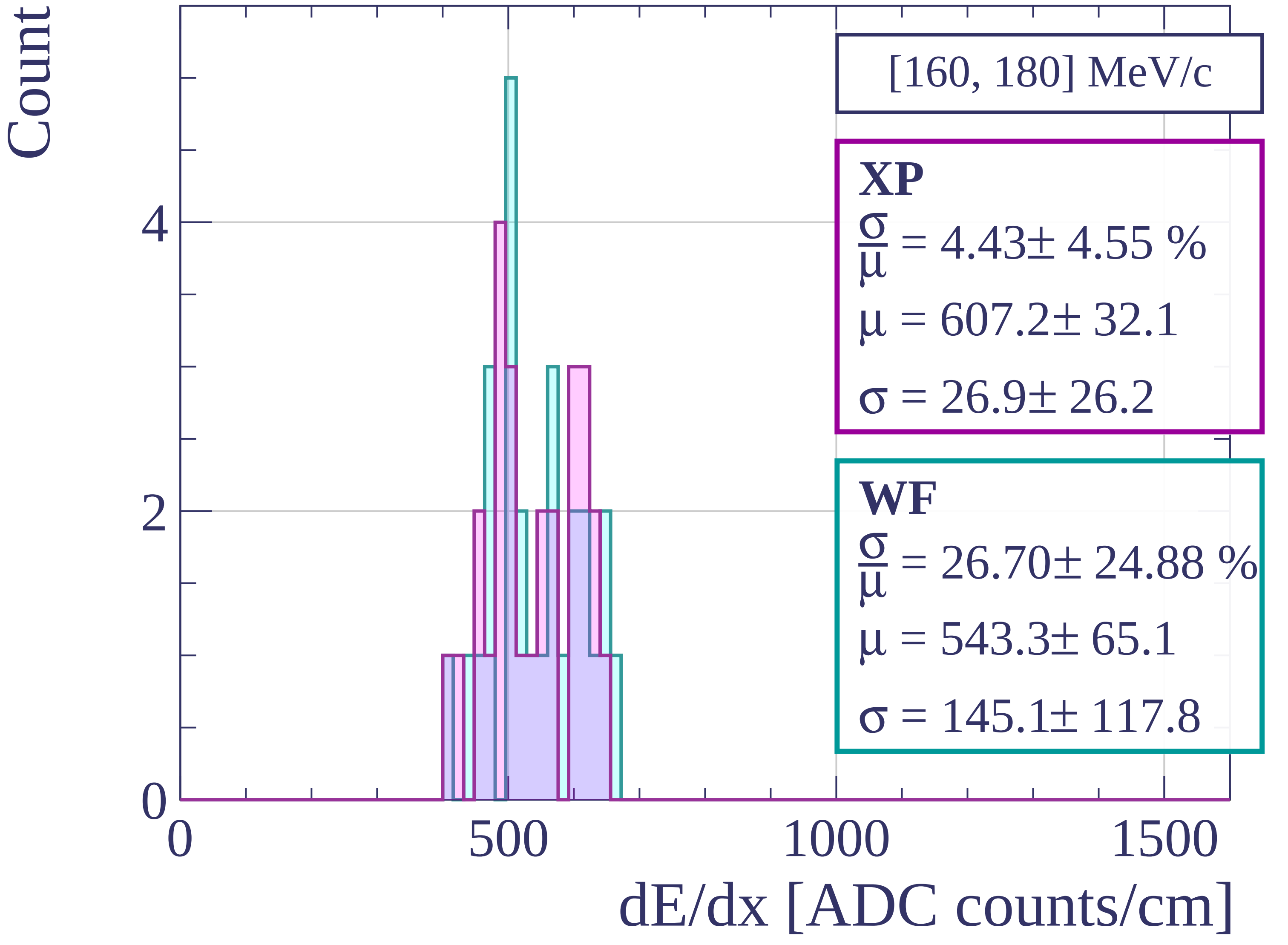


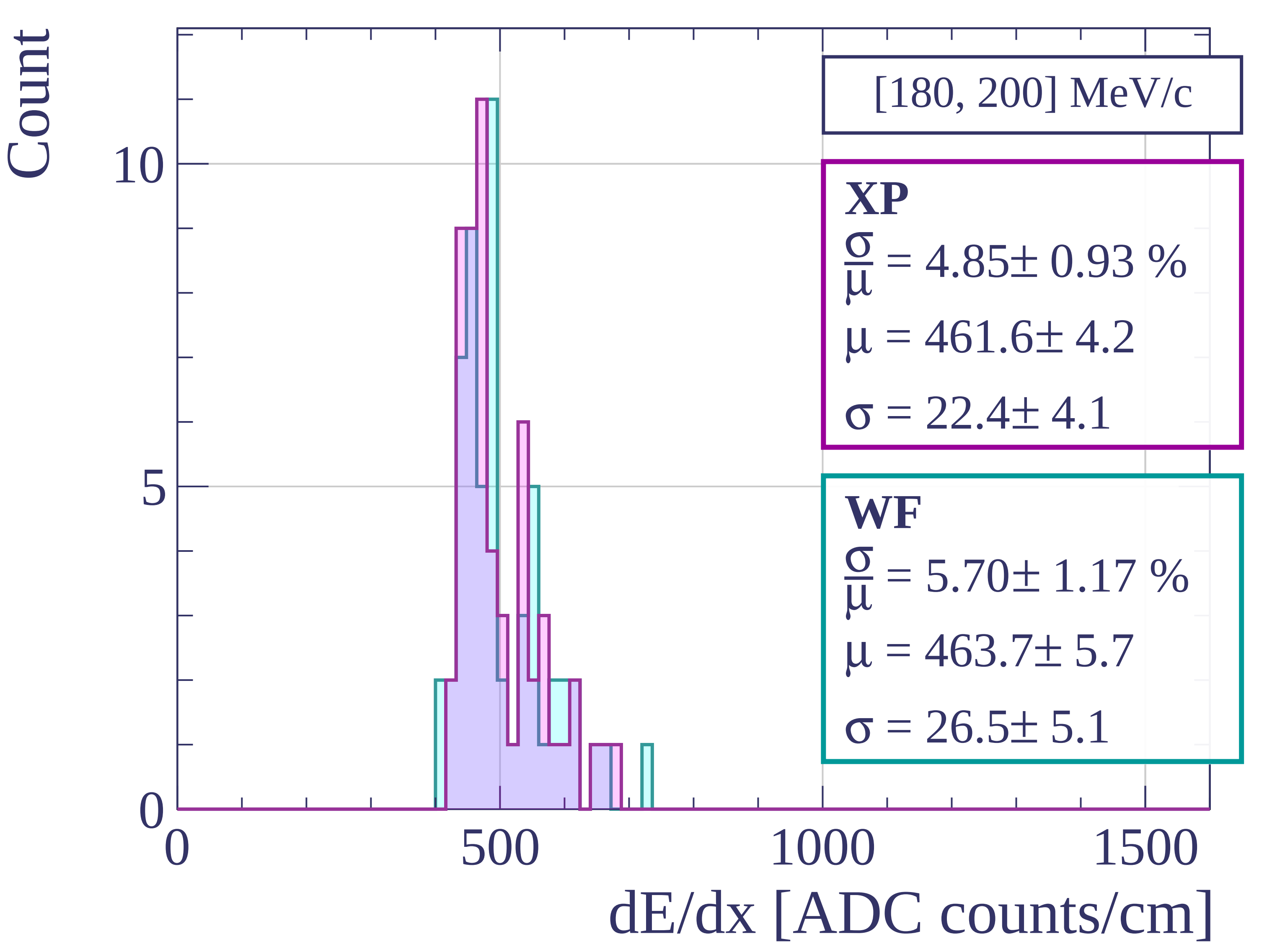
Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[100, 120] MeV/c
XP
σ/μ = 81.35± 43139.98
μ = 433.5± 229011.6
σ = 352.6± 691.4
WF
σ/μ = 68.19± 132.44 %
μ = 597.8± 408.5
σ = 407.6± 513.2


Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[120, 140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 16.82± 13.26 %
μ = 536.1± 39.6
σ = 90.2± 64.4
WF
$\frac{\sigma}{\mu}$ = 68.06± 154.50 %
μ = 583.2± 444.4
σ = 396.9± 598.6

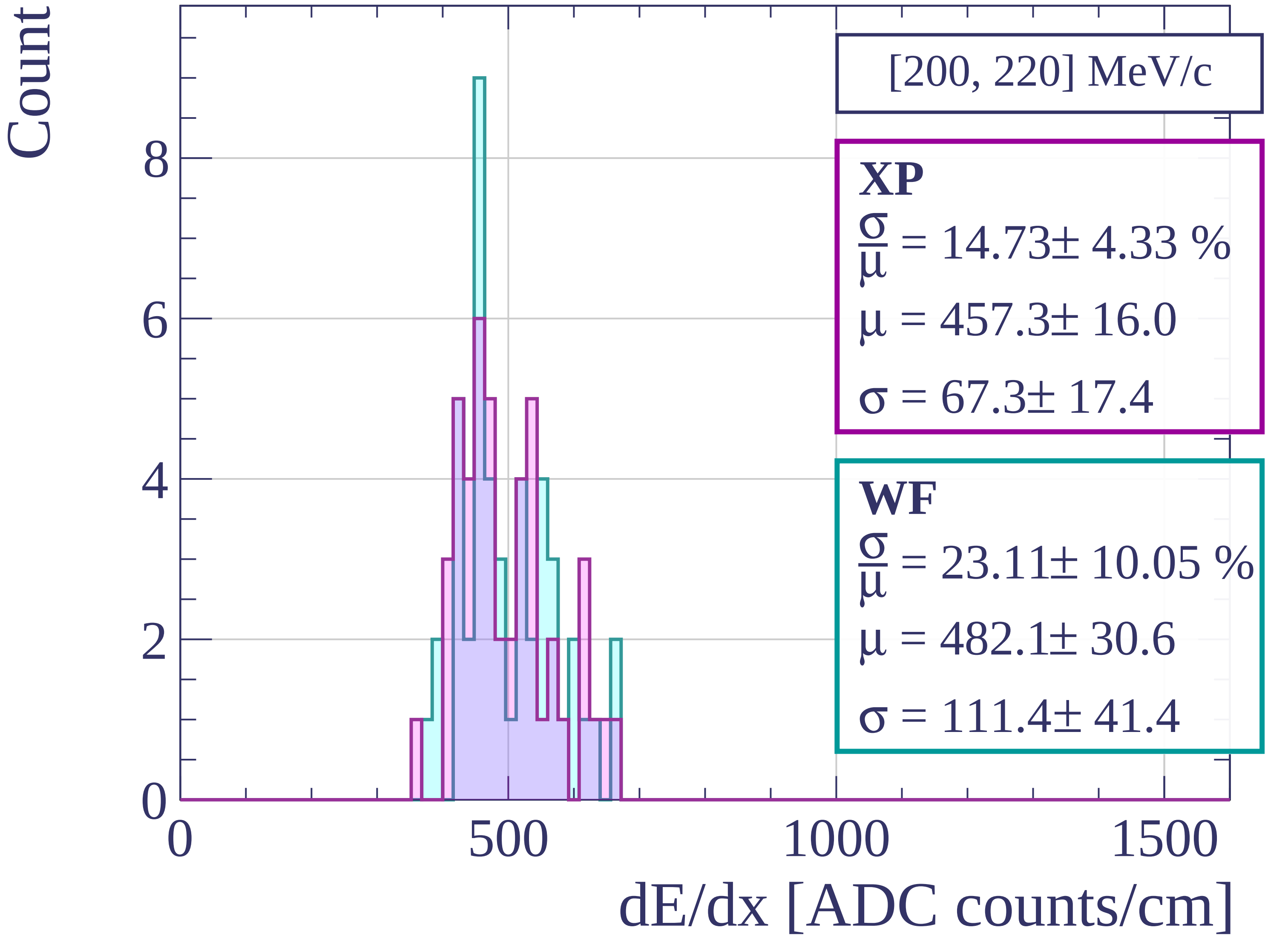


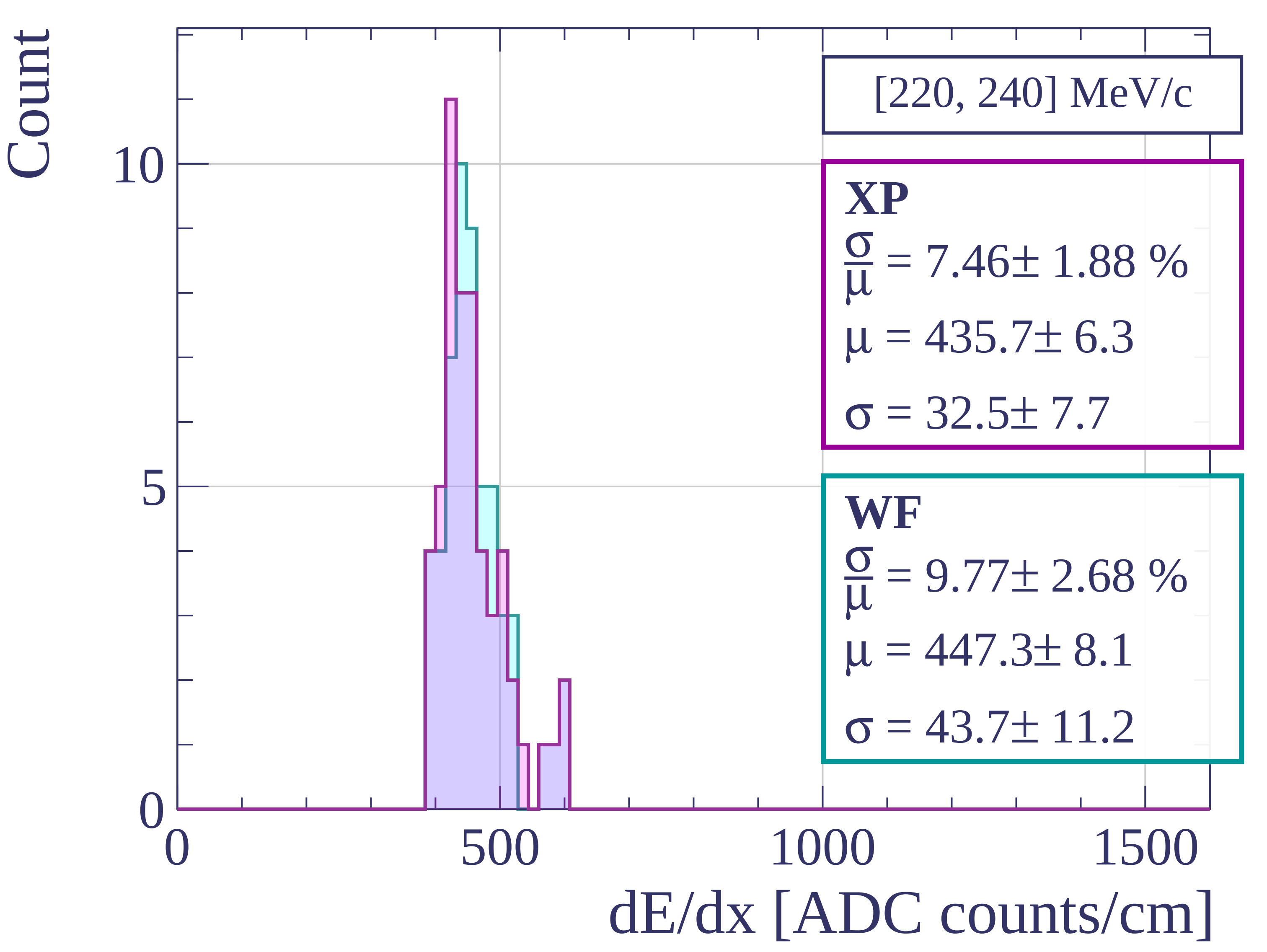
Count
4
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[140, 160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 15.48± 14.89 %
μ = 537.8± 43.7
σ = 83.3± 73.3
WF
$\frac{\sigma}{\mu}$ = 21.37± 23.99 %
μ = 578.6± 70.7
σ = 123.6± 123.7


Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 180] MeV/c
XP
σ/μ = 4.43± 4.55 %
μ = 607.2± 32.1
σ = 26.9± 26.2
WF
σ/μ = 26.70± 24.88 %
μ = 543.3± 65.1
σ = 145.1± 117.8

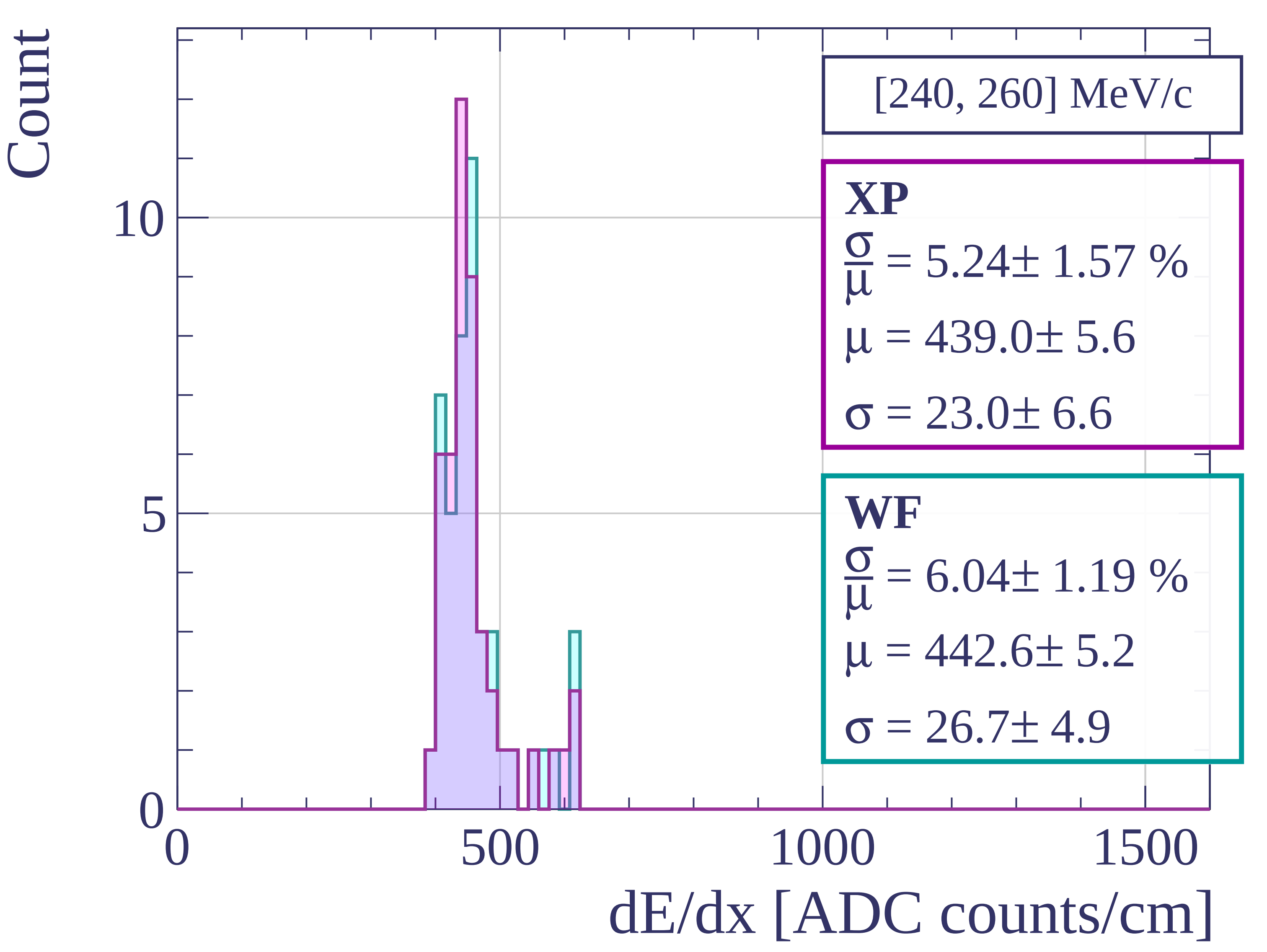


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[180, 200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.85± 0.93 %
μ = 461.6± 4.2
σ = 22.4± 4.1
WF
$\frac{\sigma}{\mu}$ = 5.70± 1.17 %
μ = 463.7± 5.7
σ = 26.5± 5.1

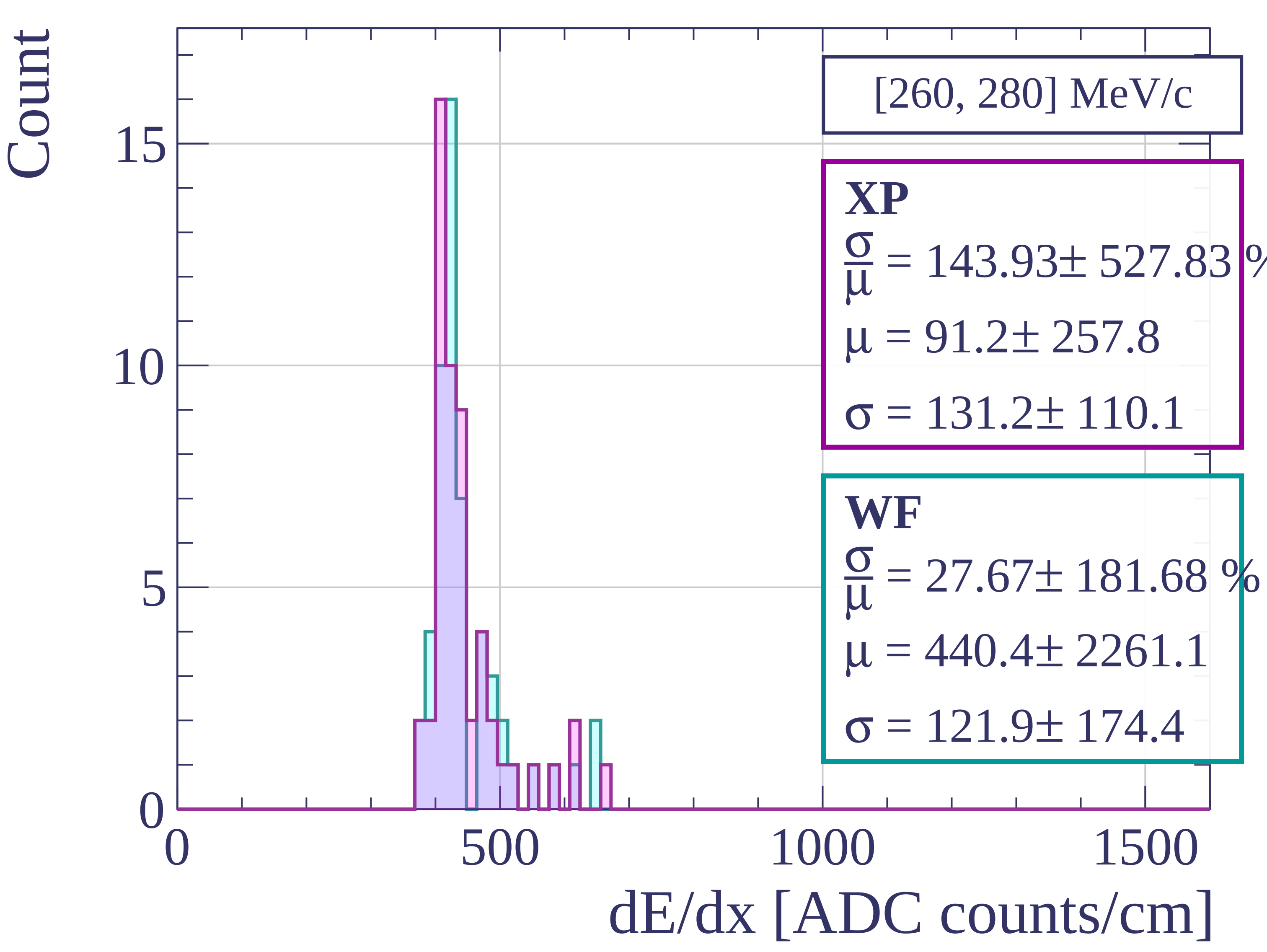

Count
dE/dx [ADC counts/cm]
[200, 220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.73± 4.33 %
μ = 457.3± 16.0
σ = 67.3± 17.4
WF
$\frac{\sigma}{\mu}$ = 23.11± 10.05 %
μ = 482.1± 30.6
σ = 111.4± 41.4
0
2
4
6
8
0
500
1000
1500



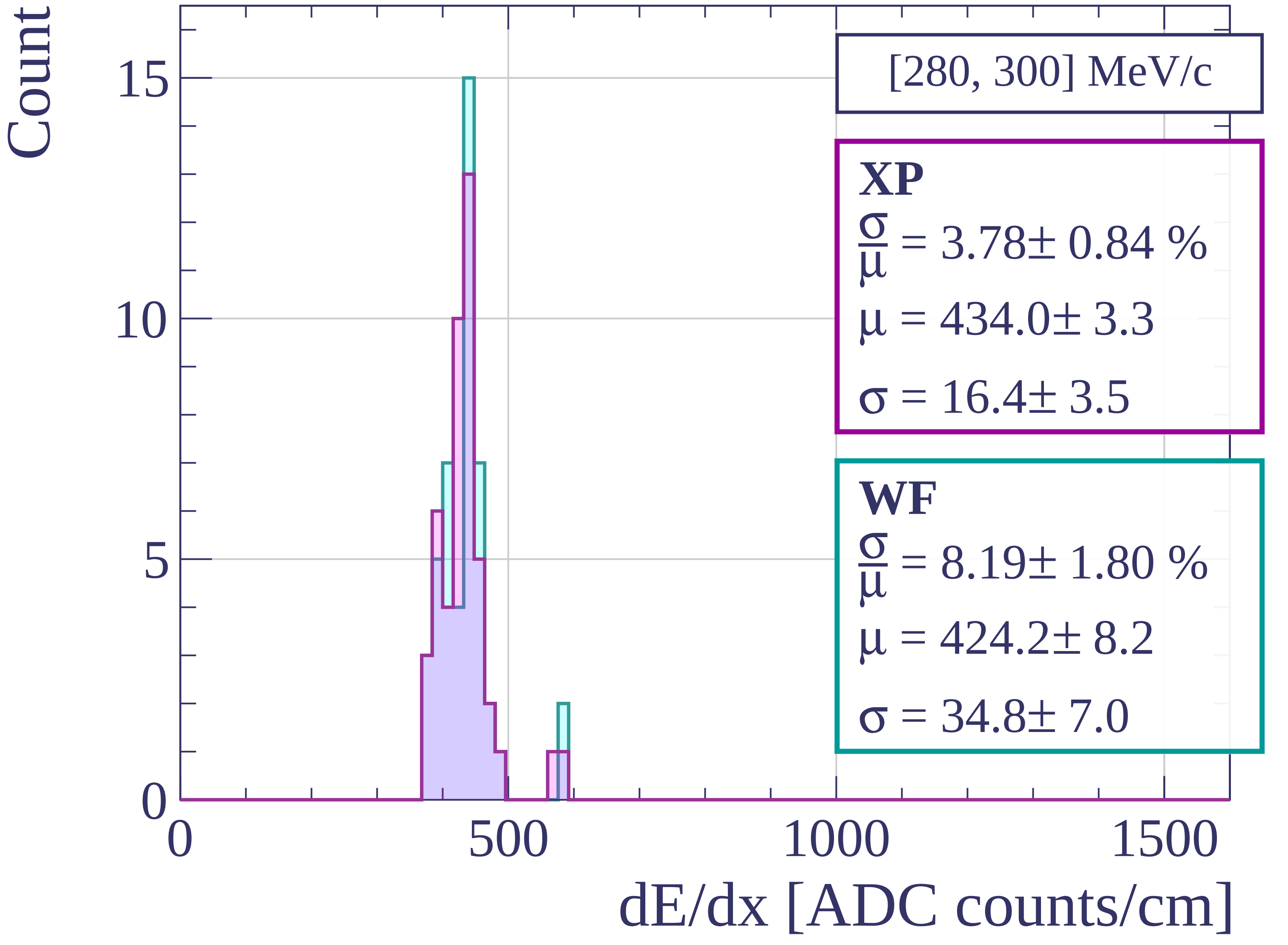
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[220, 240] MeV/c
XP
σ/μ = 7.46± 1.88 %
μ = 435.7± 6.3
σ = 32.5± 7.7
WF
σ/μ = 9.77± 2.68 %
μ = 447.3± 8.1
σ = 43.7± 11.2

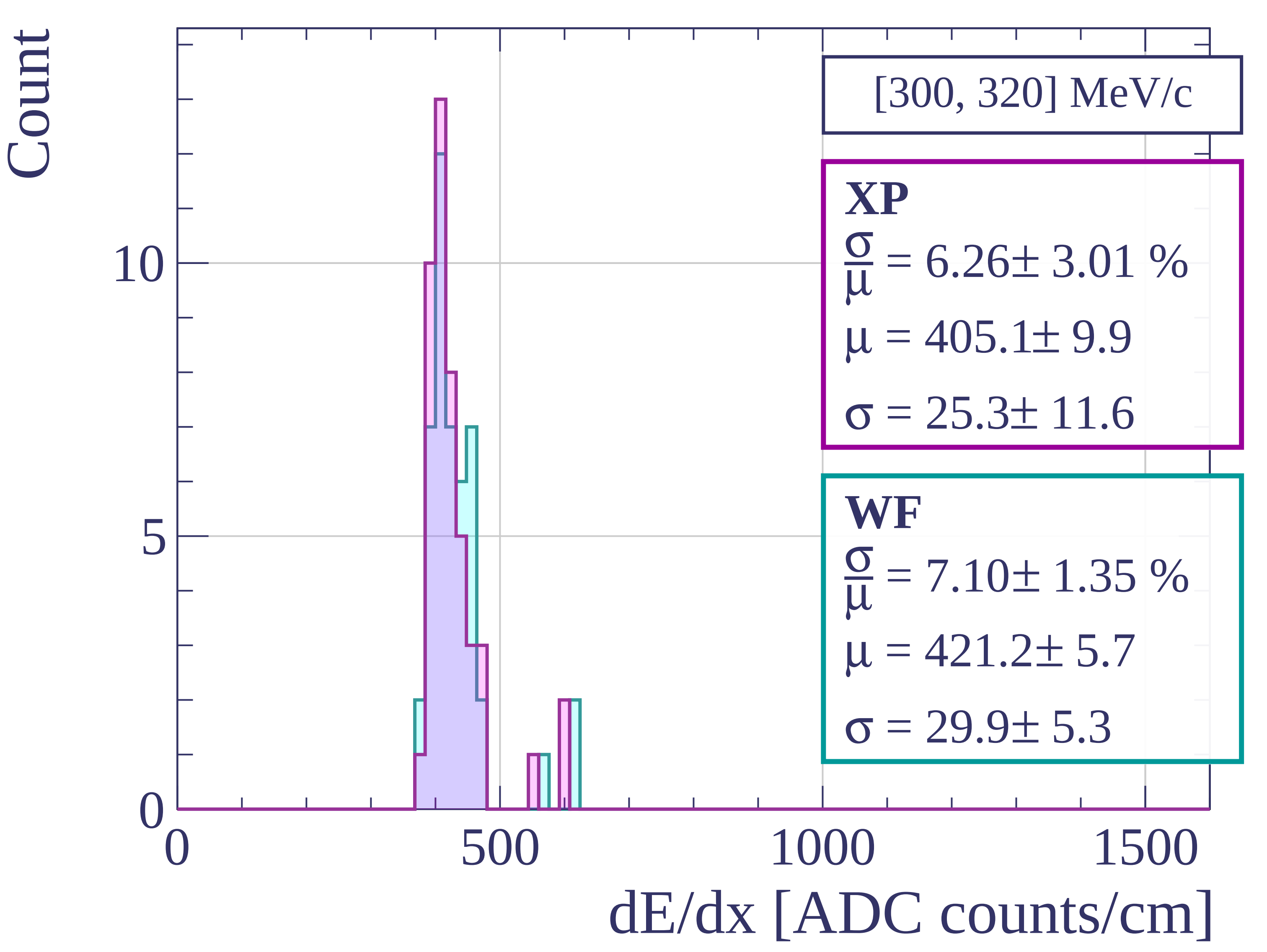


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.24± 1.57 %
μ = 439.0± 5.6
σ = 23.0± 6.6
WF
$\frac{\sigma}{\mu}$ = 6.04± 1.19 %
μ = 442.6± 5.2
σ = 26.7± 4.9

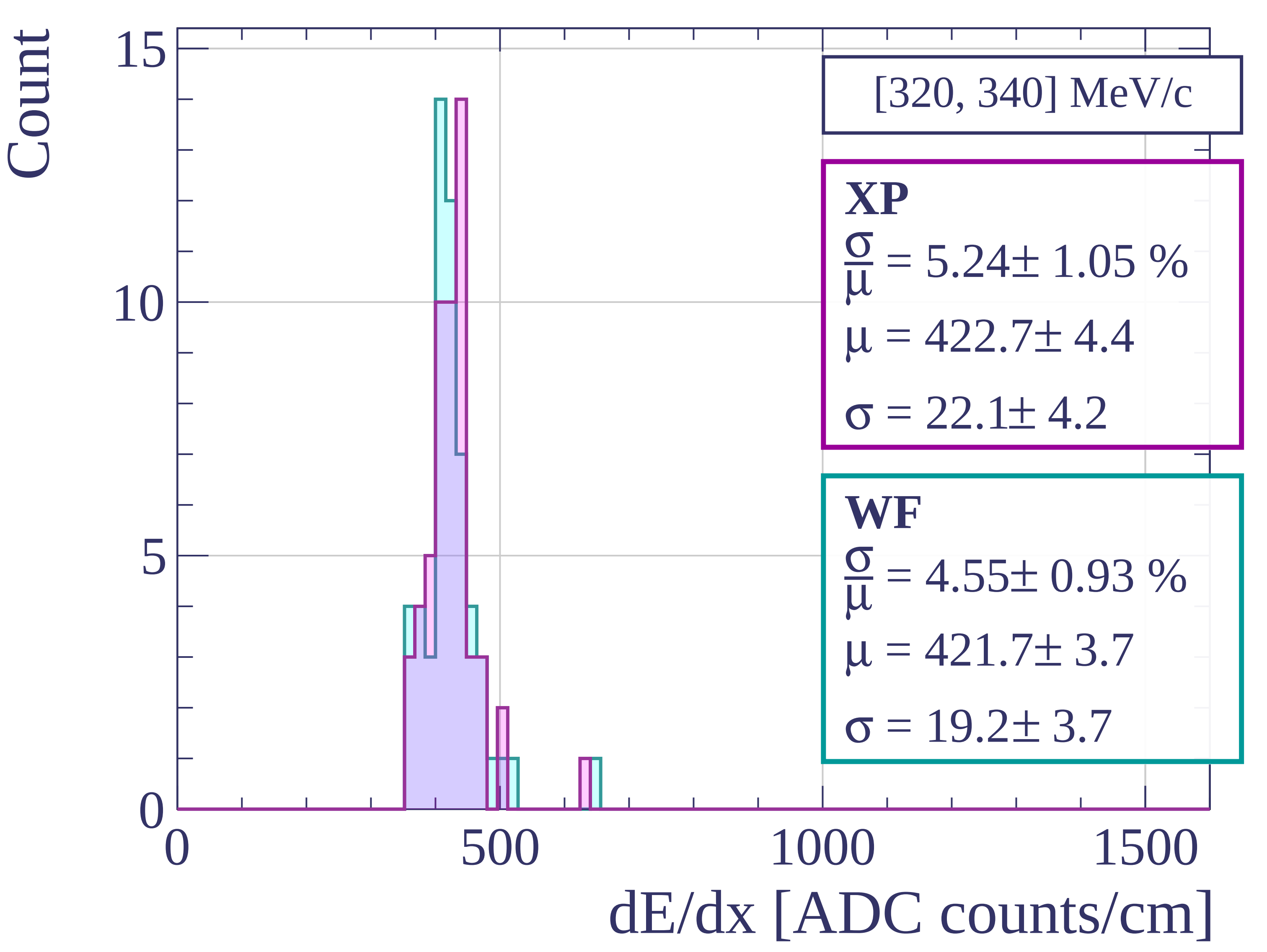


Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
[260, 280] MeV/c
XP
σ/μ = 143.93± 527.83 %
μ = 91.2± 257.8
σ = 131.2± 110.1
WF
σ/μ = 27.67± 181.68 %
μ = 440.4± 2261.1
σ = 121.9± 174.4

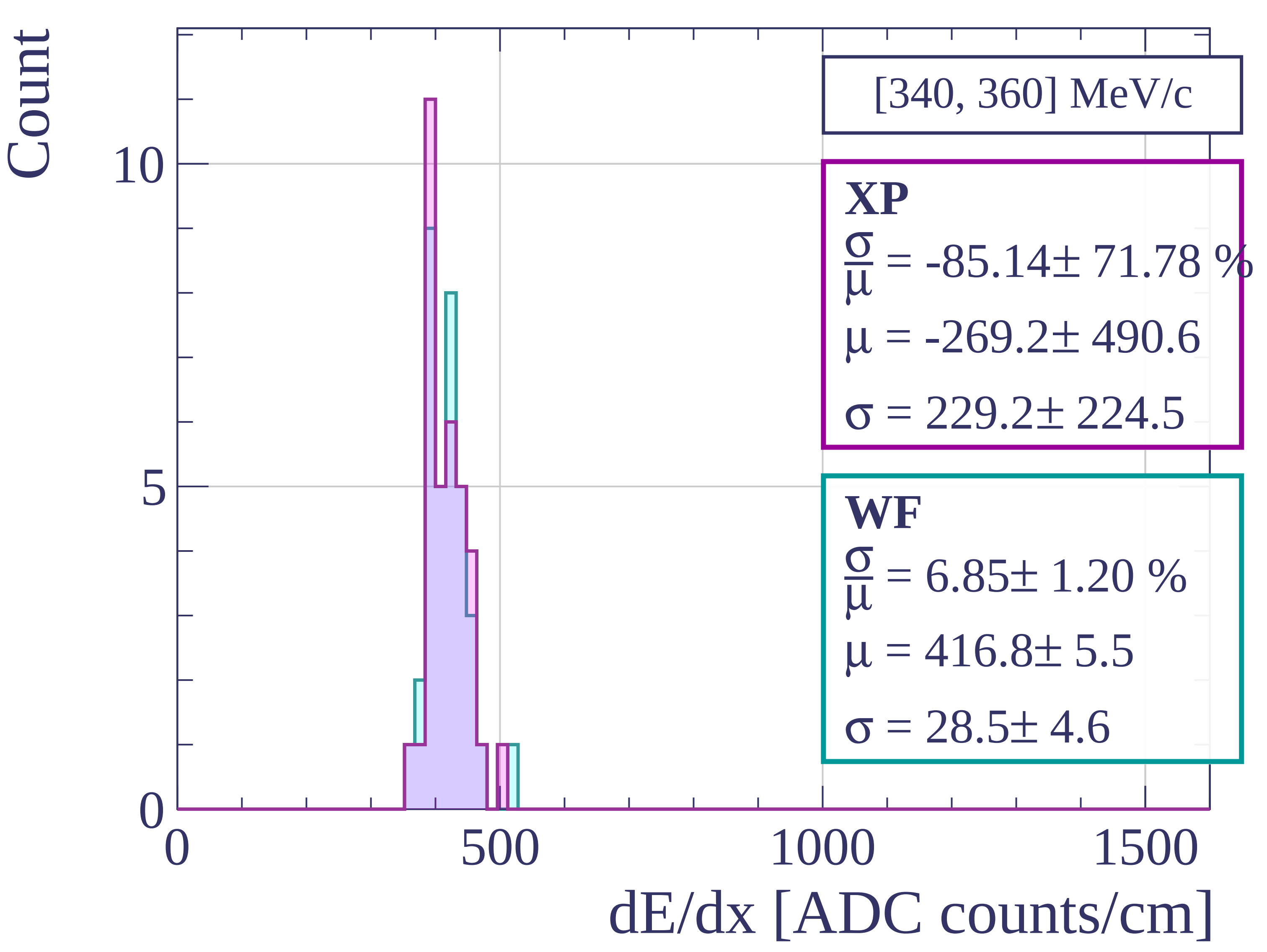

Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[280, 300] MeV/c
XP
σ/μ = 3.78± 0.84 %
μ = 434.0± 3.3
σ = 16.4± 3.5
WF
σ/μ = 8.19± 1.80 %
μ = 424.2± 8.2
σ = 34.8± 7.0

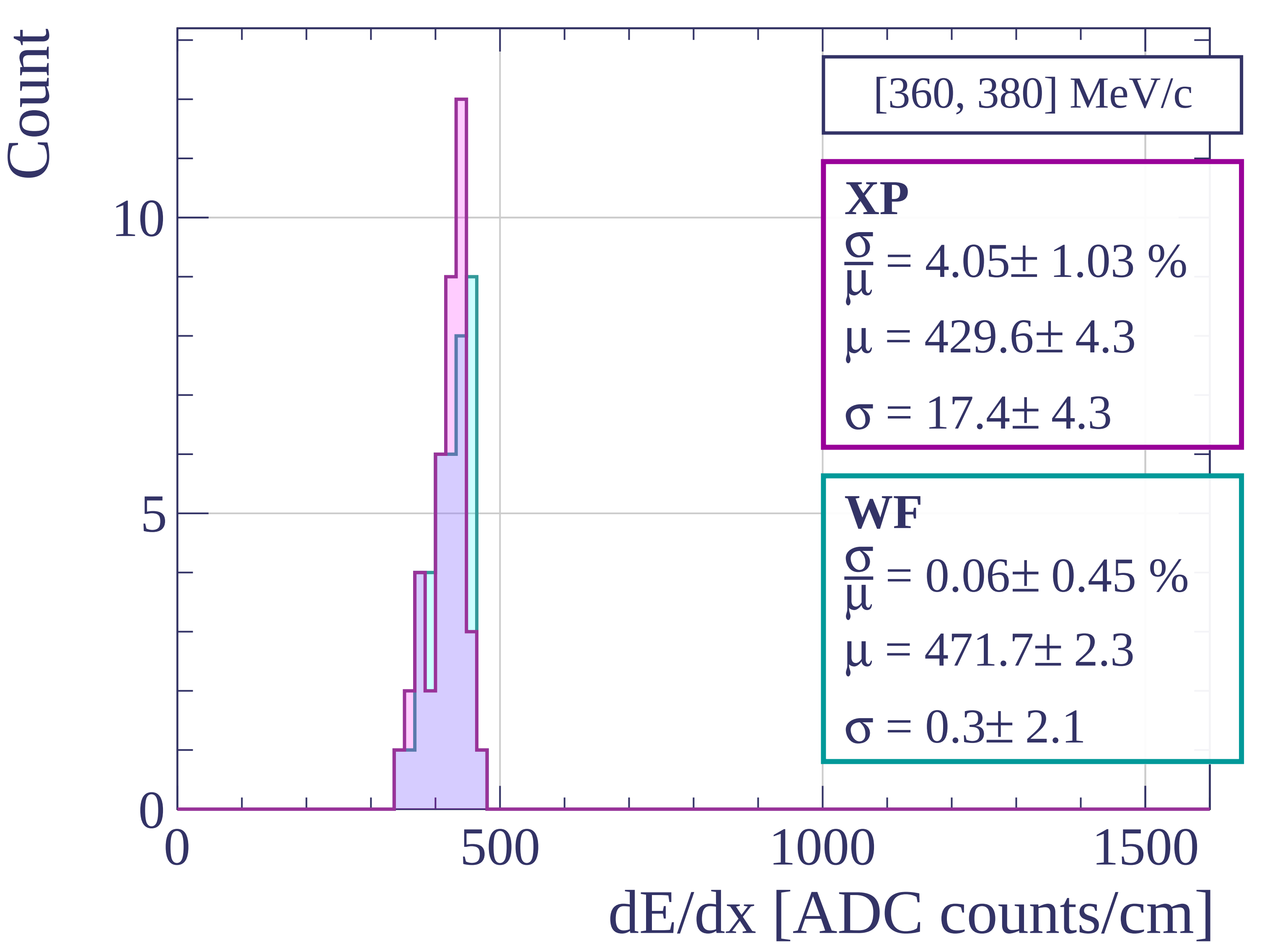


Count
0
5
10
0
500
1000
1500
dE/dx [ADC counts/cm]
[300, 320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.26± 3.01 %
μ = 405.1± 9.9
σ = 25.3± 11.6
WF
$\frac{\sigma}{\mu}$ = 7.10± 1.35 %
μ = 421.2± 5.7
σ = 29.9± 5.3

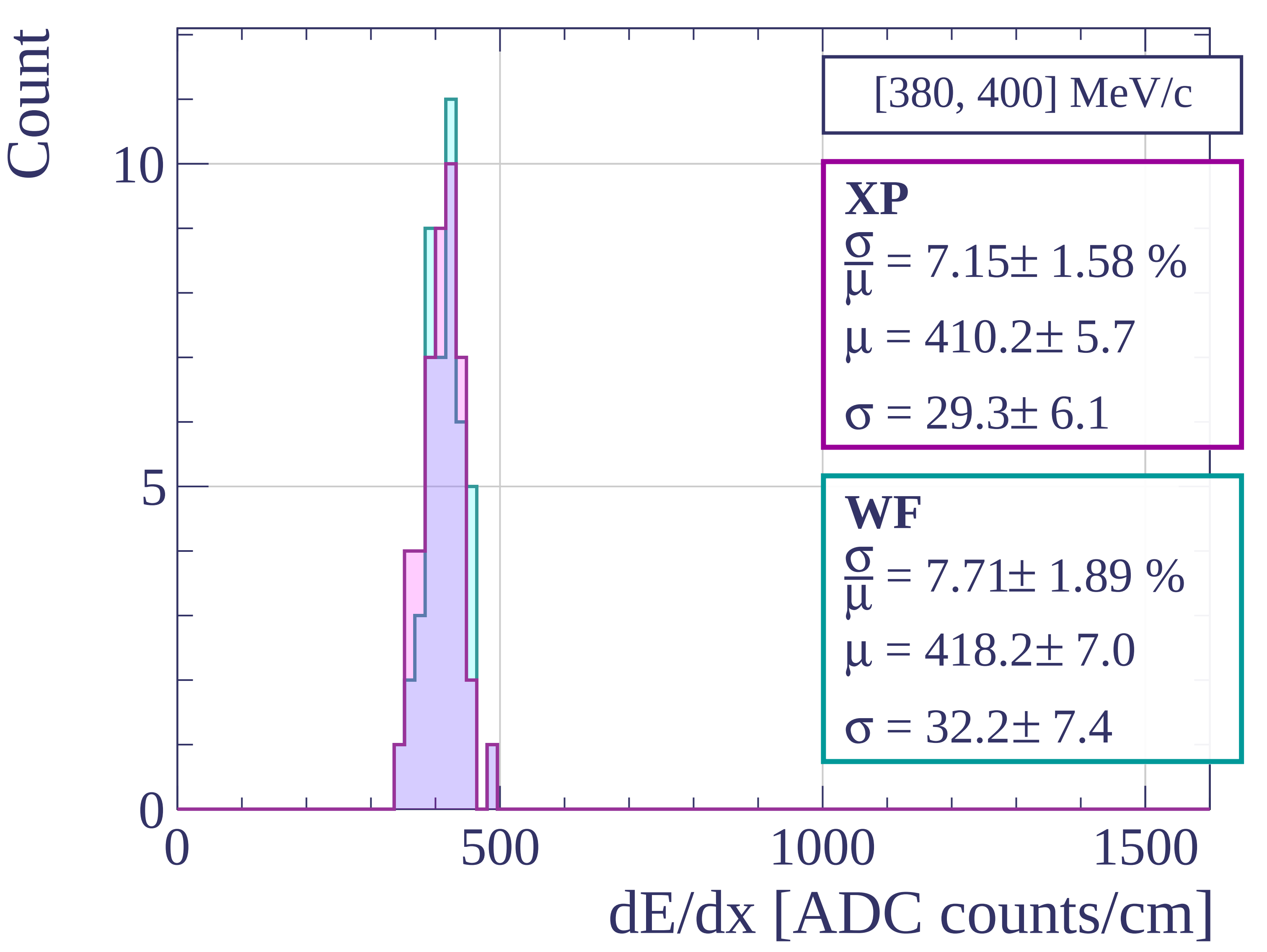


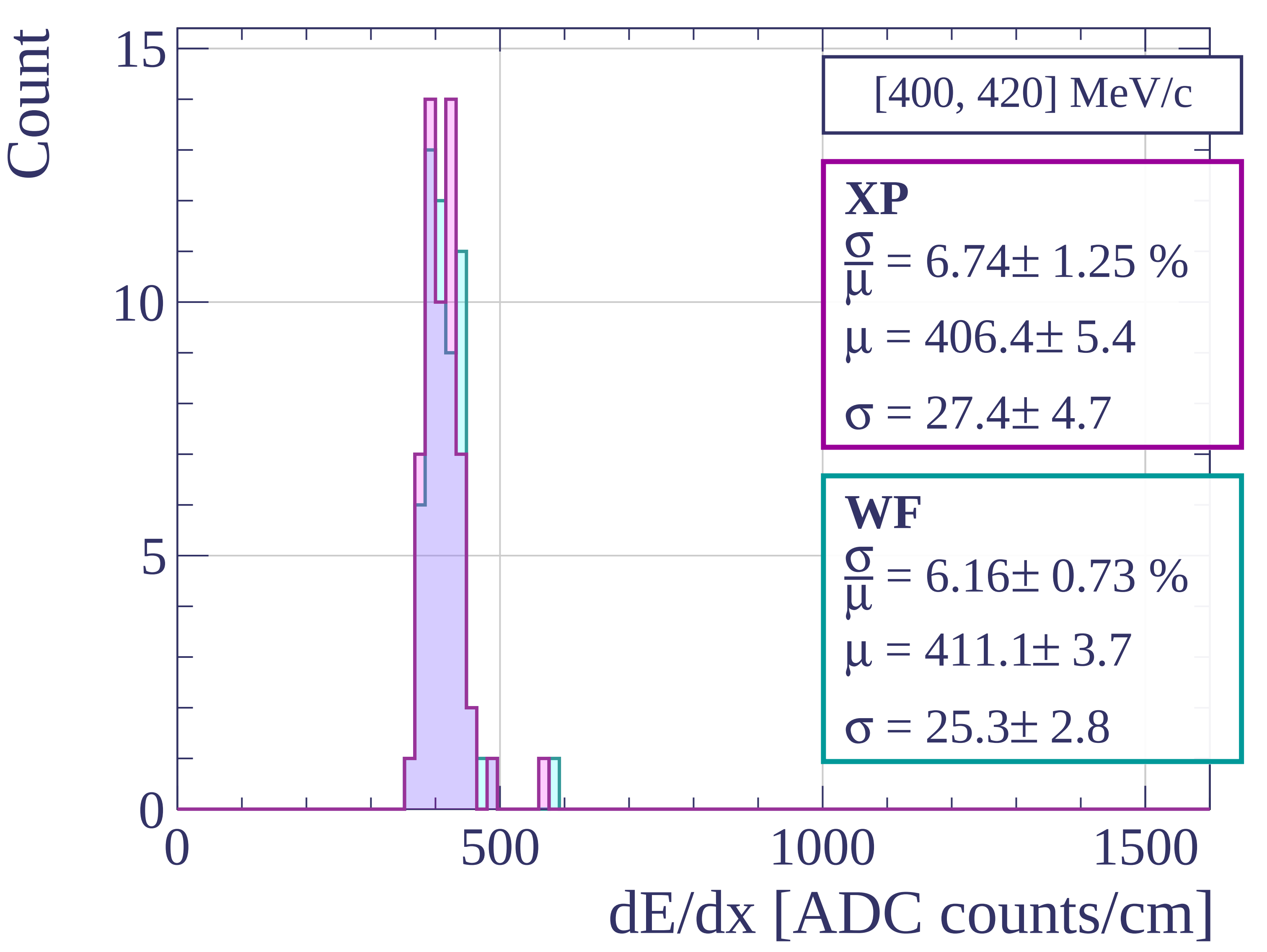
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[320, 340] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.24± 1.05 %
μ = 422.7± 4.4
σ = 22.1± 4.2
WF
$\frac{\sigma}{\mu}$ = 4.55± 0.93 %
μ = 421.7± 3.7
σ = 19.2± 3.7


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[340, 360] MeV/c
XP
$\frac{\sigma}{\mu}$ = -85.14± 71.78 %
μ = -269.2± 490.6
σ = 229.2± 224.5
WF
$\frac{\sigma}{\mu}$ = 6.85± 1.20 %
μ = 416.8± 5.5
σ = 28.5± 4.6

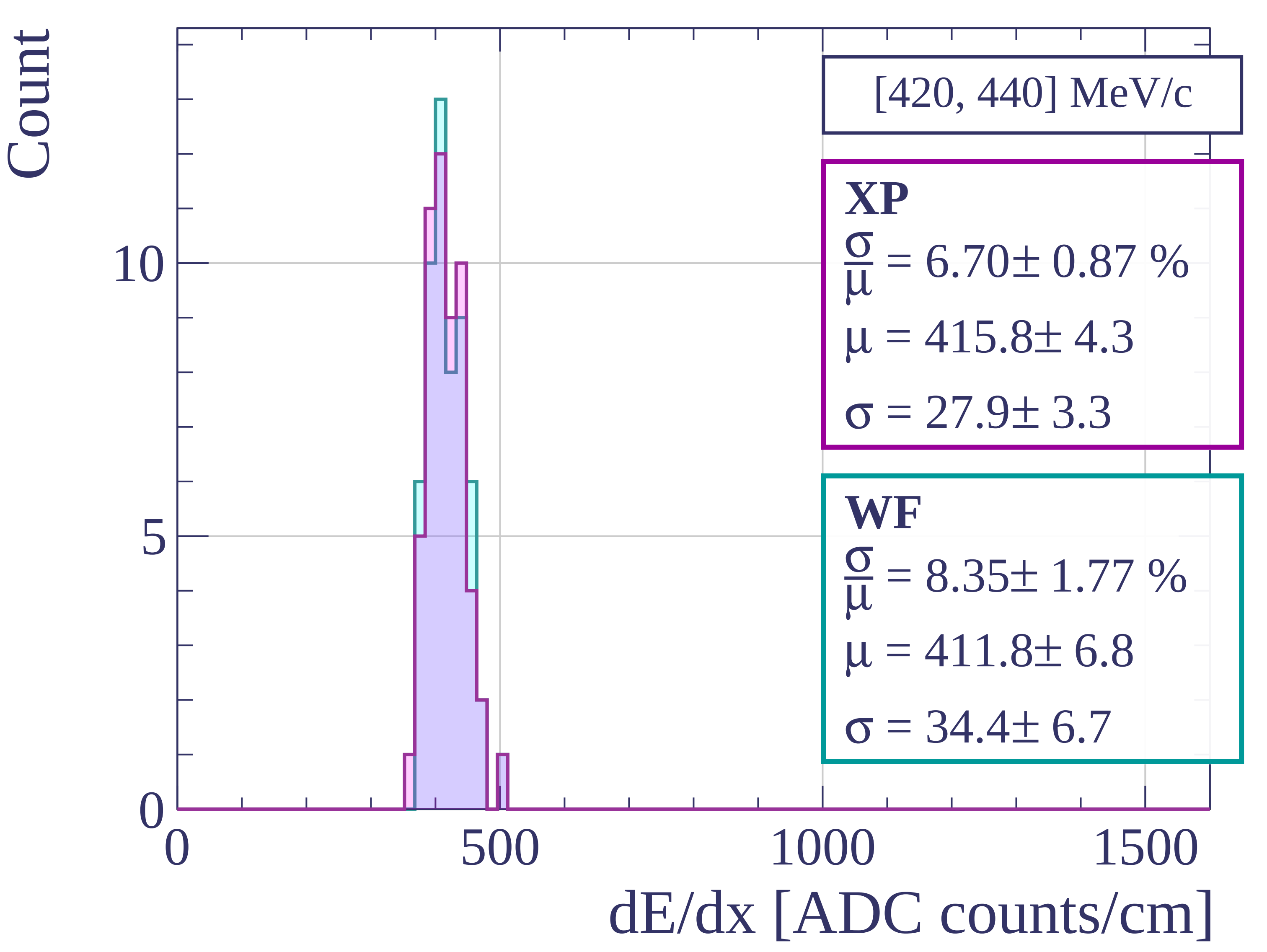


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[360, 380] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.05± 1.03 %
μ = 429.6± 4.3
σ = 17.4± 4.3
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.45 %
μ = 471.7± 2.3
σ = 0.3± 2.1

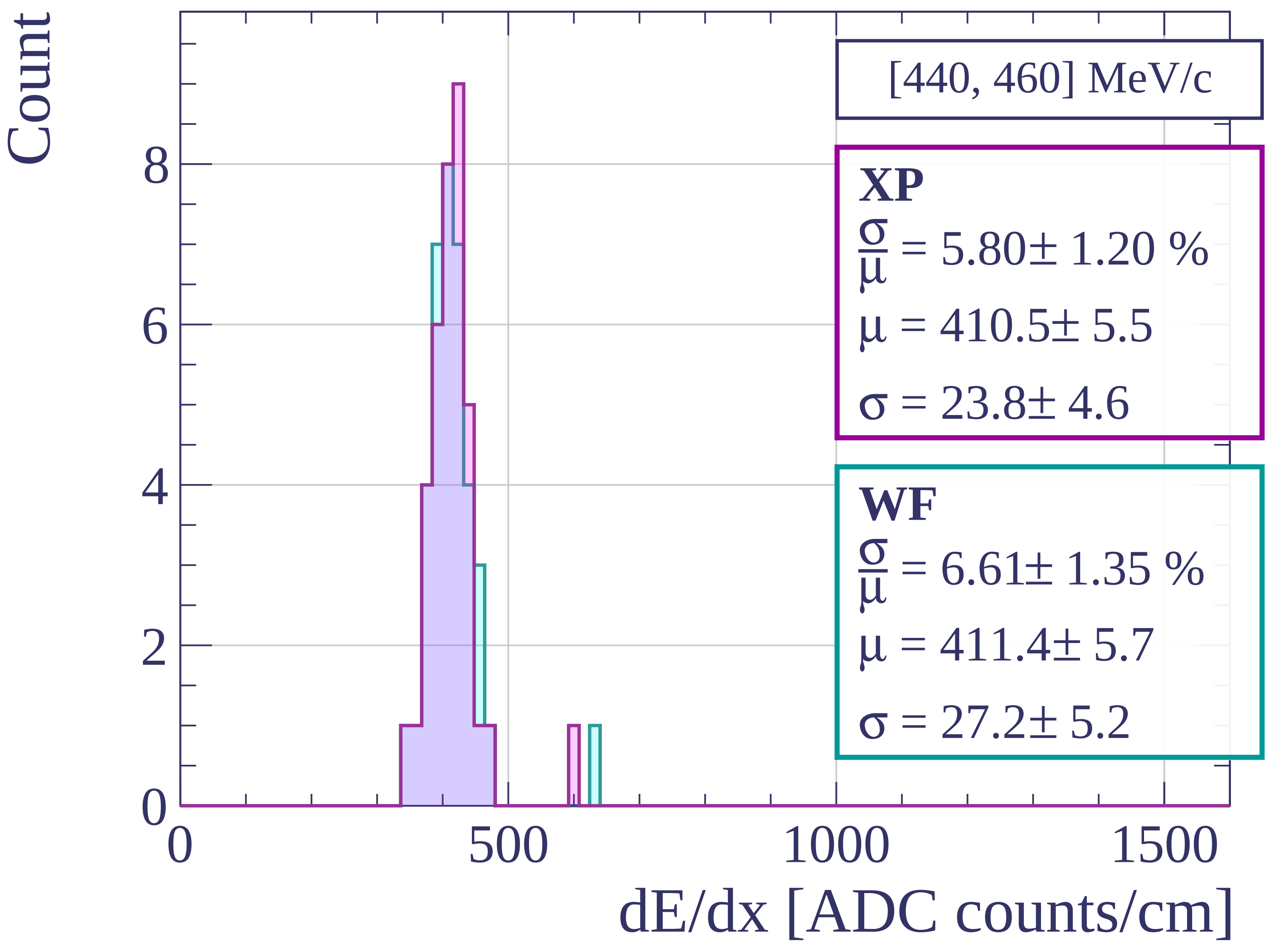

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[380, 400] MeV/c
XP
σ/μ = 7.15± 1.58 %
μ = 410.2± 5.7
σ = 29.3± 6.1
WF
σ/μ = 7.71± 1.89 %
μ = 418.2± 7.0
σ = 32.2± 7.4



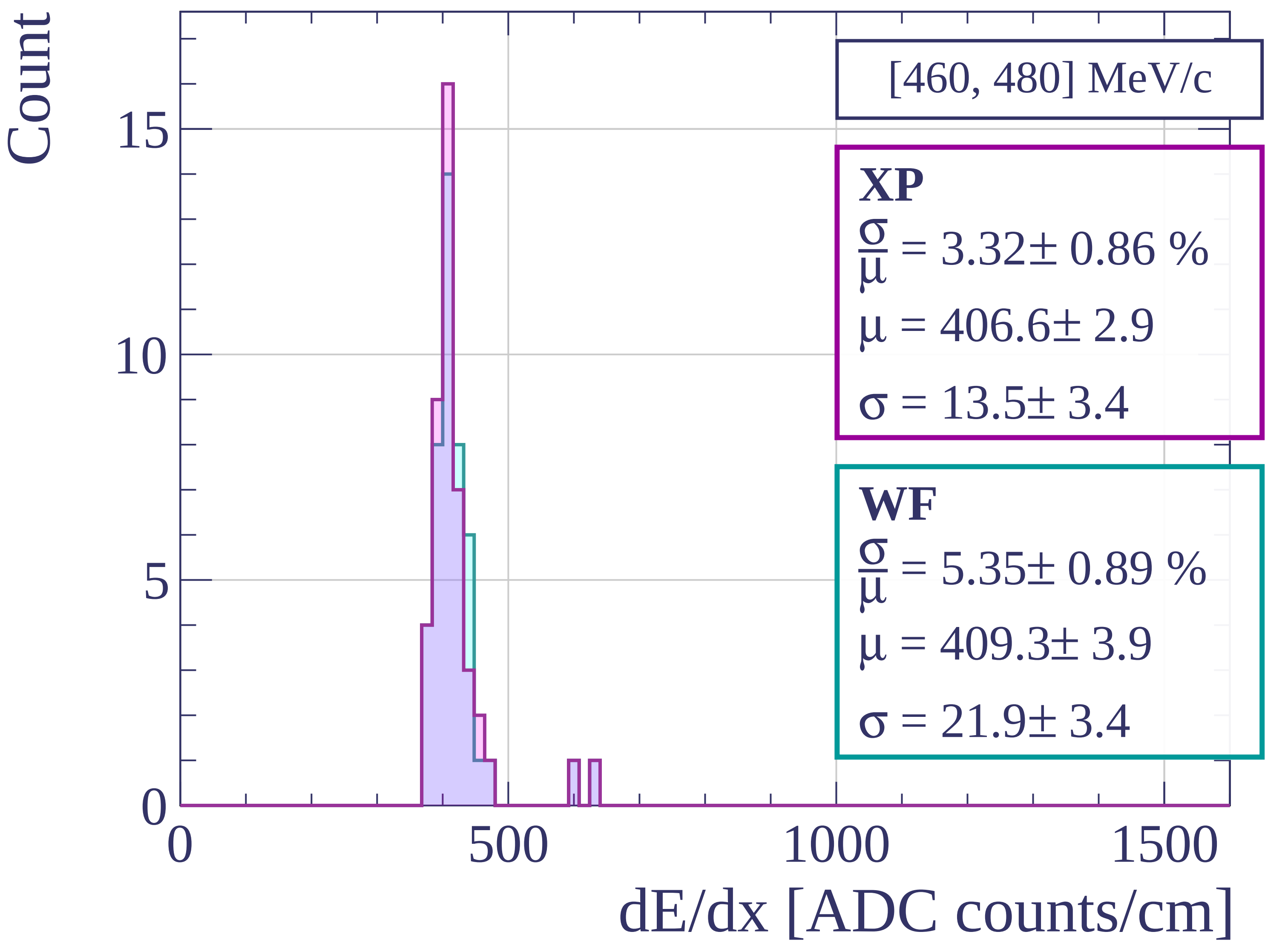
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 420] MeV/c
XP
σ/μ = 6.74± 1.25 %
μ = 406.4± 5.4
σ = 27.4± 4.7
WF
σ/μ = 6.16± 0.73 %
μ = 411.1± 3.7
σ = 25.3± 2.8



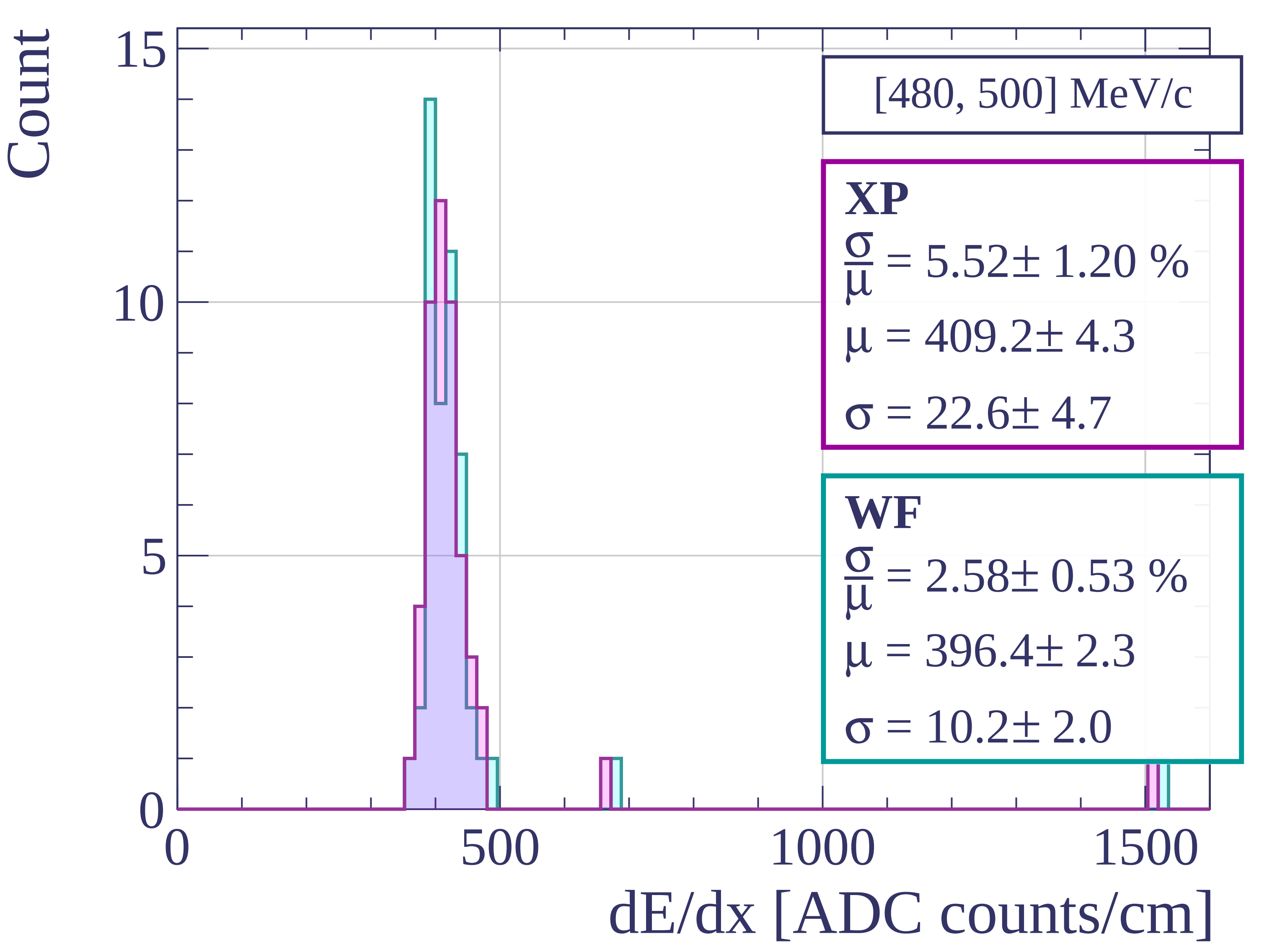
Count
[420, 440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.70± 0.87 %
μ = 415.8± 4.3
σ = 27.9± 3.3
WF
$\frac{\sigma}{\mu}$ = 8.35± 1.77 %
μ = 411.8± 6.8
σ = 34.4± 6.7
0
5
10
0
500
1000
1500
dE/dx [ADC counts/cm]



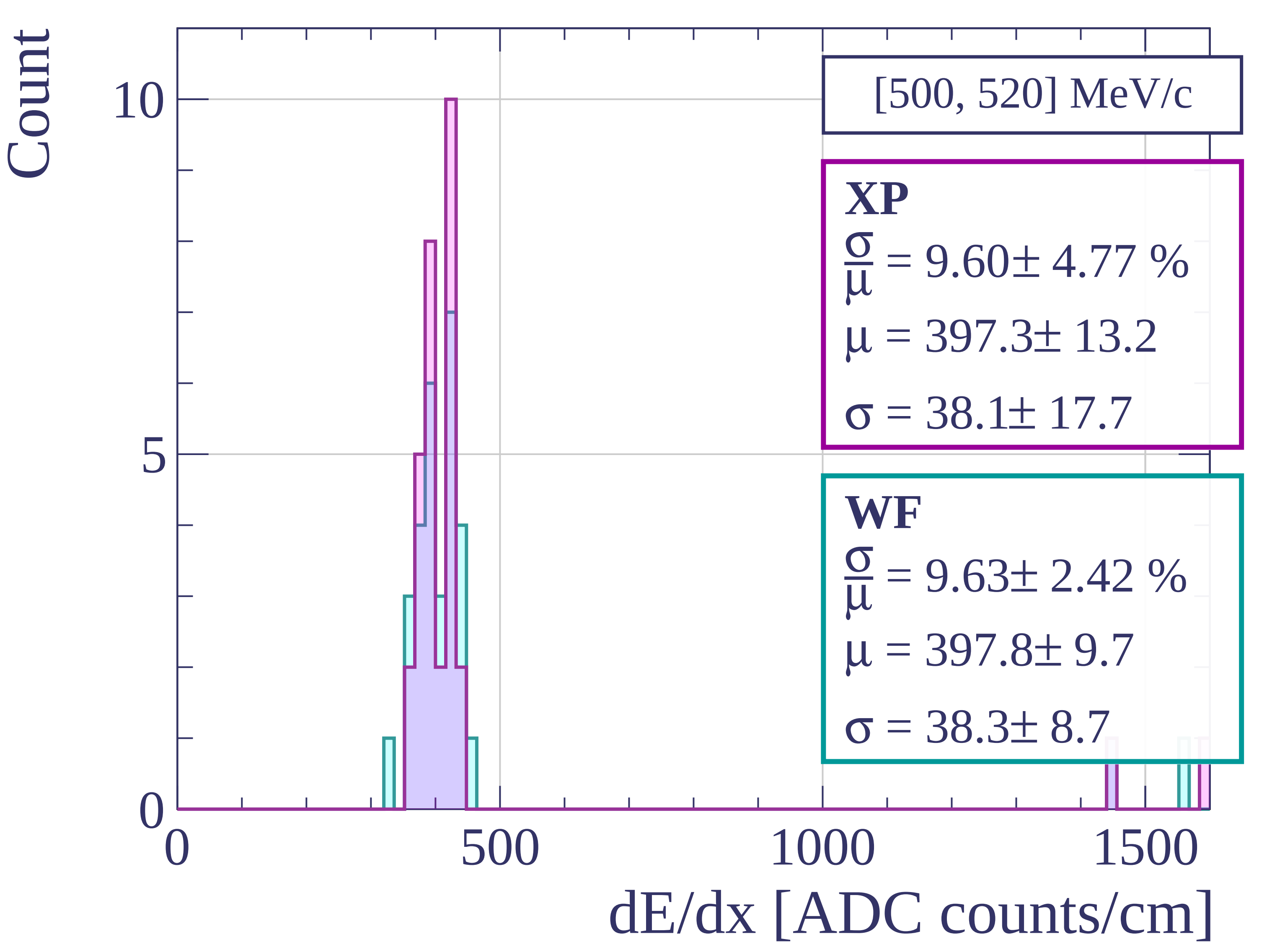
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[440, 460] MeV/c
XP
σ/μ = 5.80± 1.20 %
μ = 410.5± 5.5
σ = 23.8± 4.6
WF
σ/μ = 6.61± 1.35 %
μ = 411.4± 5.7
σ = 27.2± 5.2



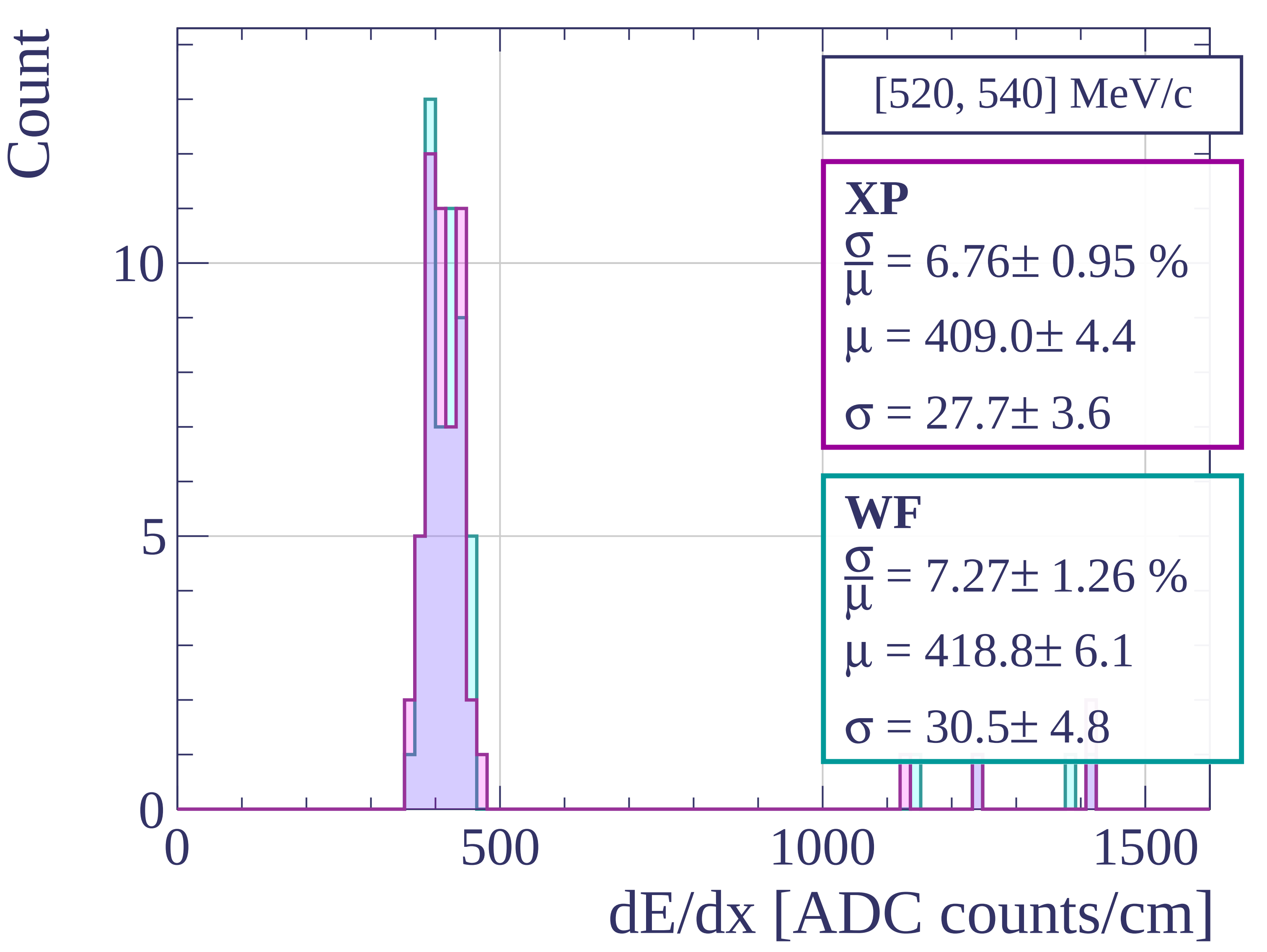
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[460, 480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 3.32± 0.86 %
μ = 406.6± 2.9
σ = 13.5± 3.4
WF
$\frac{\sigma}{\mu}$ = 5.35± 0.89 %
μ = 409.3± 3.9
σ = 21.9± 3.4



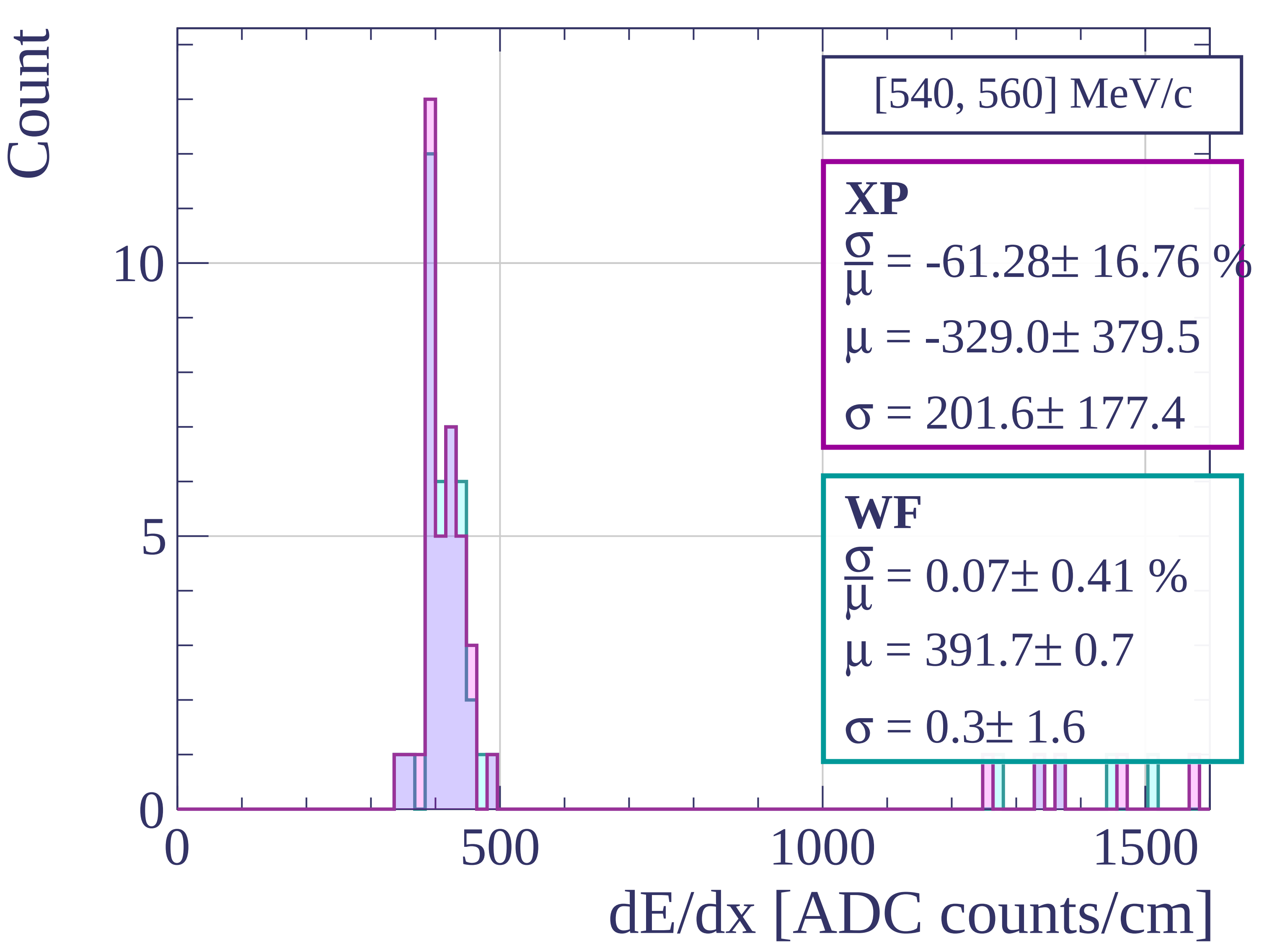
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 500] MeV/c
XP
σ/μ = 5.52± 1.20 %
μ = 409.2± 4.3
σ = 22.6± 4.7
WF
σ/μ = 2.58± 0.53 %
μ = 396.4± 2.3
σ = 10.2± 2.0

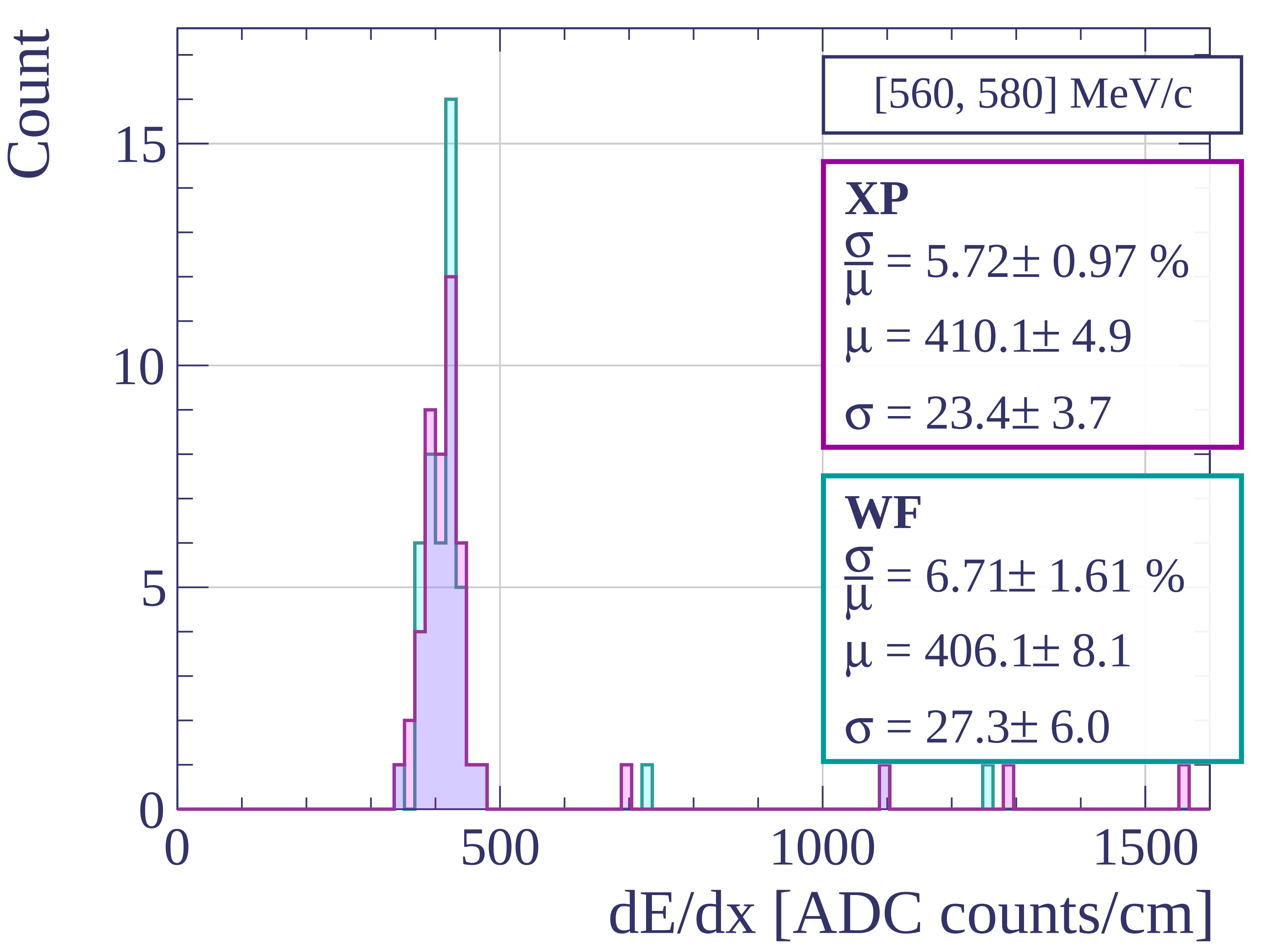


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[500, 520] MeV/c
XP
σ/μ = 9.60± 4.77 %
μ = 397.3± 13.2
σ = 38.1± 17.7
WF
σ/μ = 9.63± 2.42 %
μ = 397.8± 9.7
σ = 38.3± 8.7

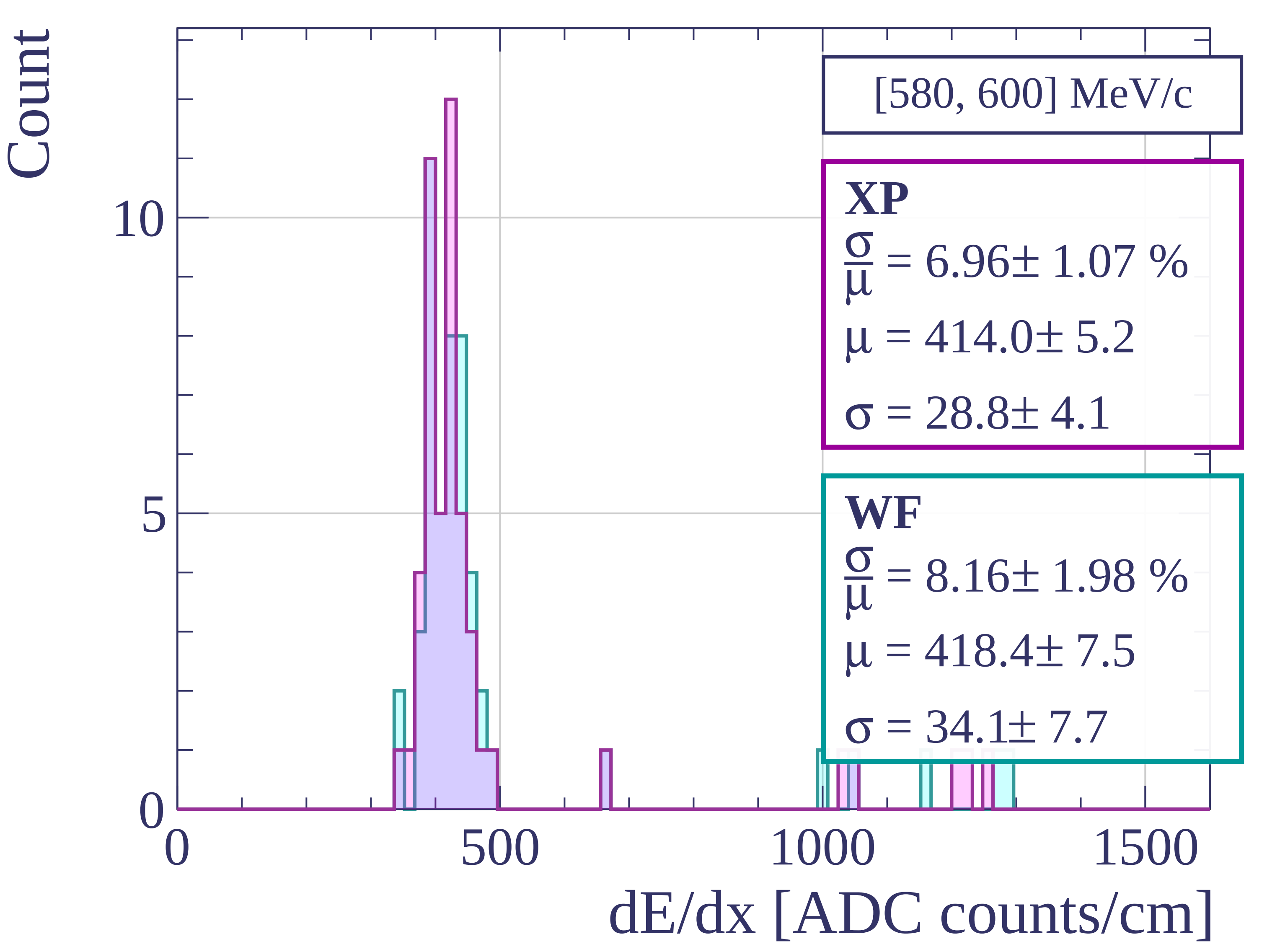


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[520, 540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.76± 0.95 %
μ = 409.0± 4.4
σ = 27.7± 3.6
WF
$\frac{\sigma}{\mu}$ = 7.27± 1.26 %
μ = 418.8± 6.1
σ = 30.5± 4.8

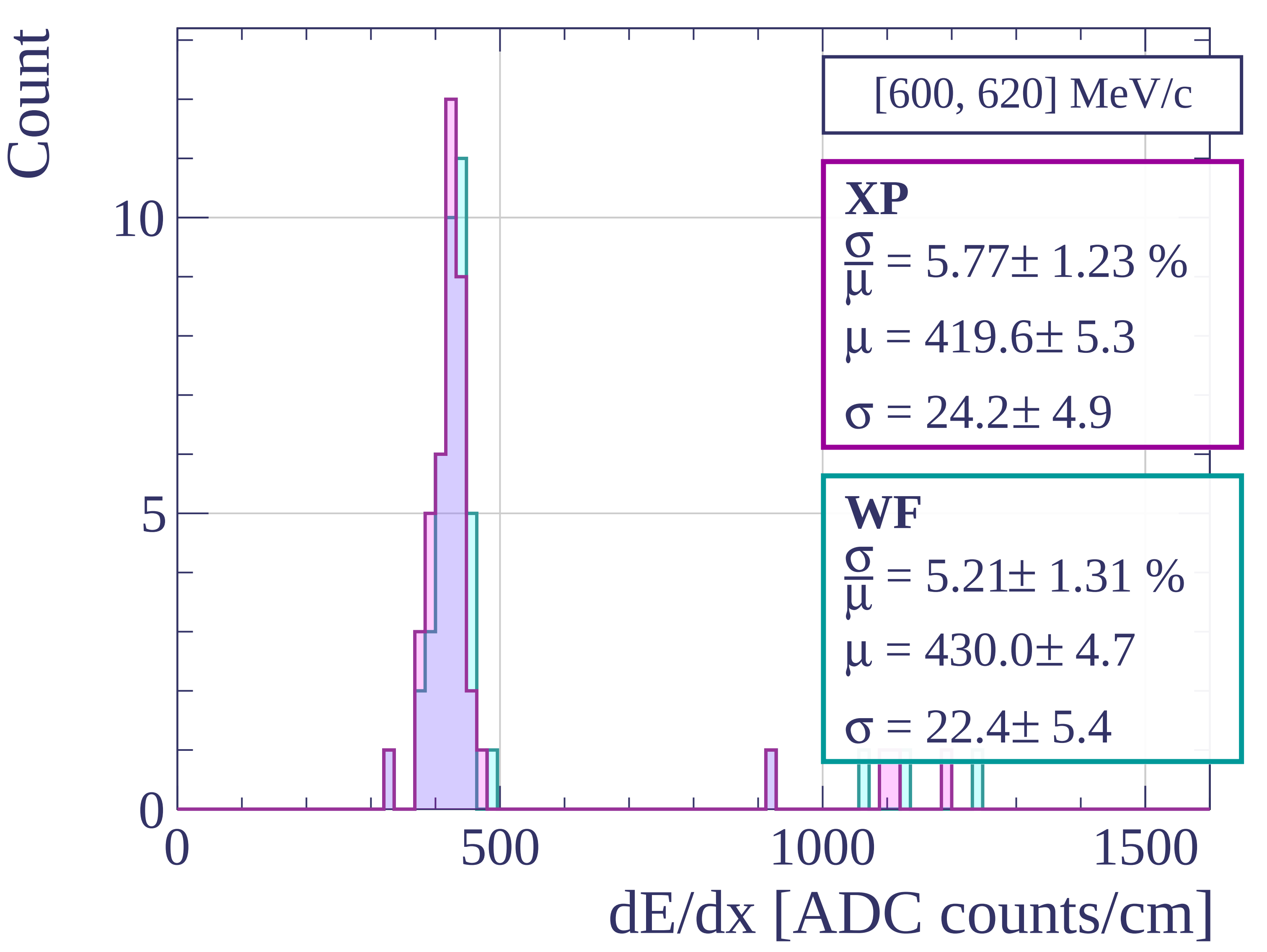

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[540, 560] MeV/c
XP
σ/μ = -61.28± 16.76 %
μ = -329.0± 379.5
σ = 201.6± 177.4
WF
σ/μ = 0.07± 0.41 %
μ = 391.7± 0.7
σ = 0.3± 1.6

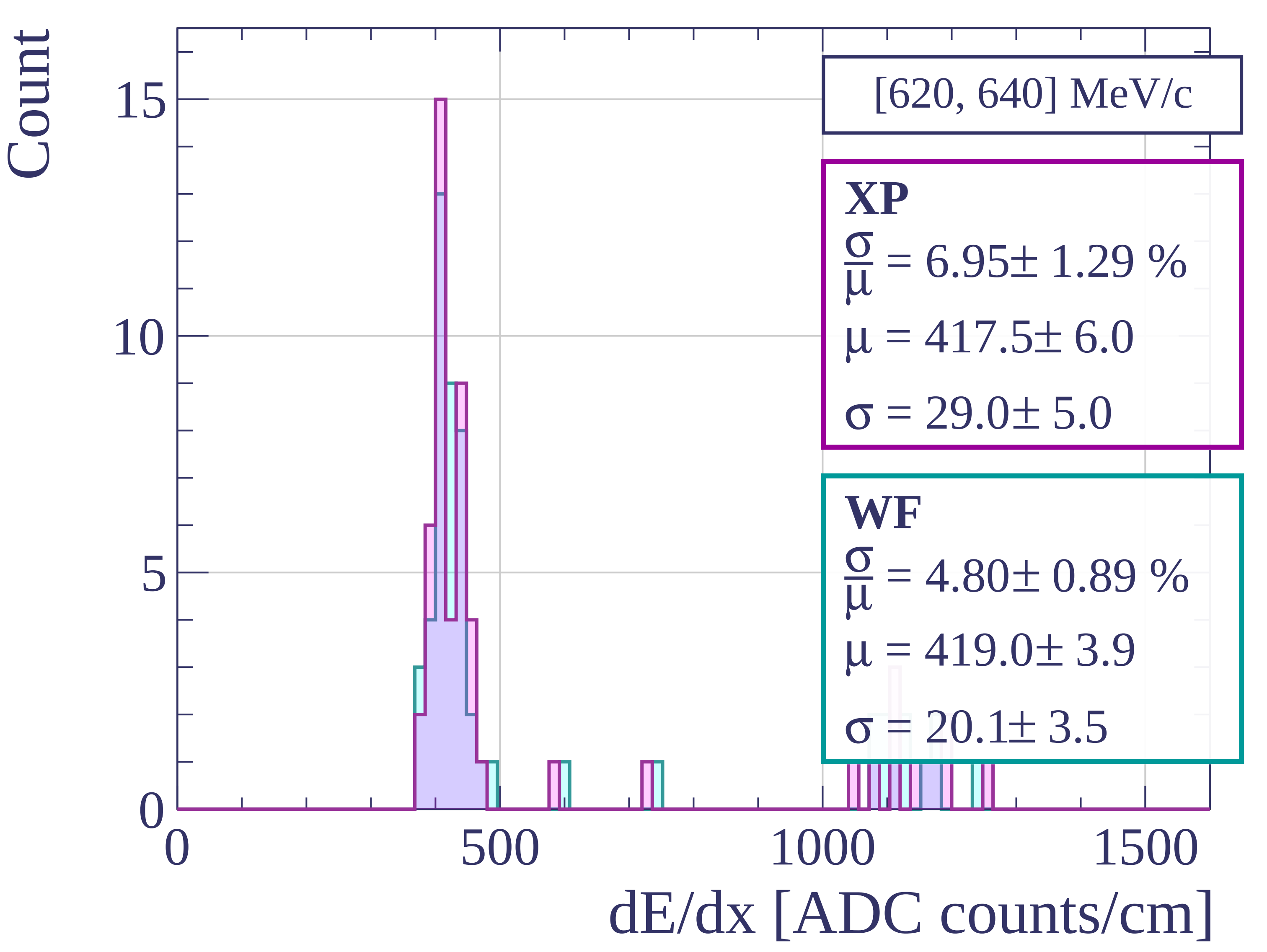


Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
[560, 580] MeV/c
XP
σ/μ = 5.72± 0.97 %
μ = 410.1± 4.9
σ = 23.4± 3.7
WF
σ/μ = 6.71± 1.61 %
μ = 406.1± 8.1
σ = 27.3± 6.0

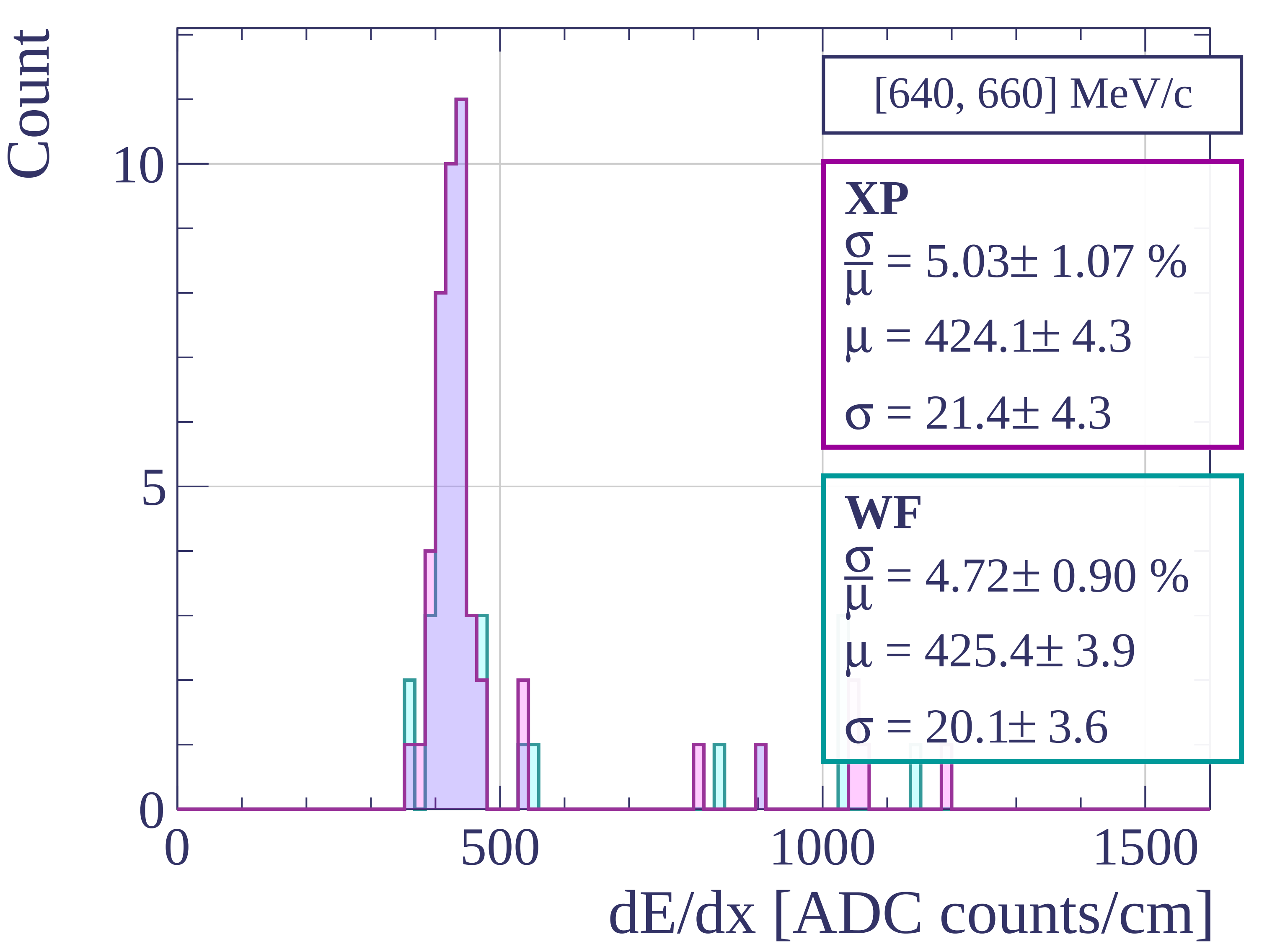


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[580, 600] MeV/c
XP
σ/μ = 6.96± 1.07 %
μ = 414.0± 5.2
σ = 28.8± 4.1
WF
σ/μ = 8.16± 1.98 %
μ = 418.4± 7.5
σ = 34.1± 7.7

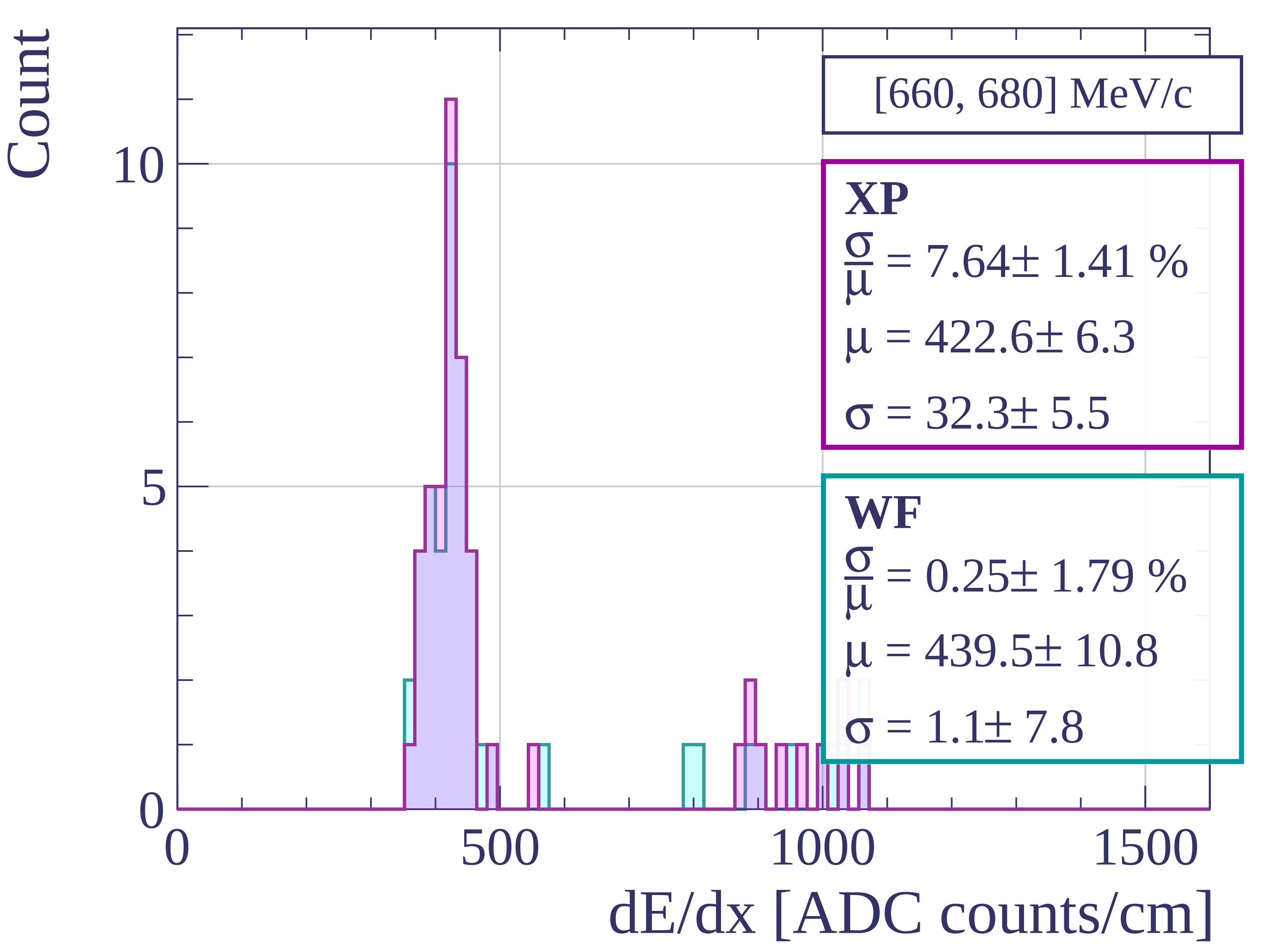

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[600, 620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.77± 1.23 %
μ = 419.6± 5.3
σ = 24.2± 4.9
WF
$\frac{\sigma}{\mu}$ = 5.21± 1.31 %
μ = 430.0± 4.7
σ = 22.4± 5.4



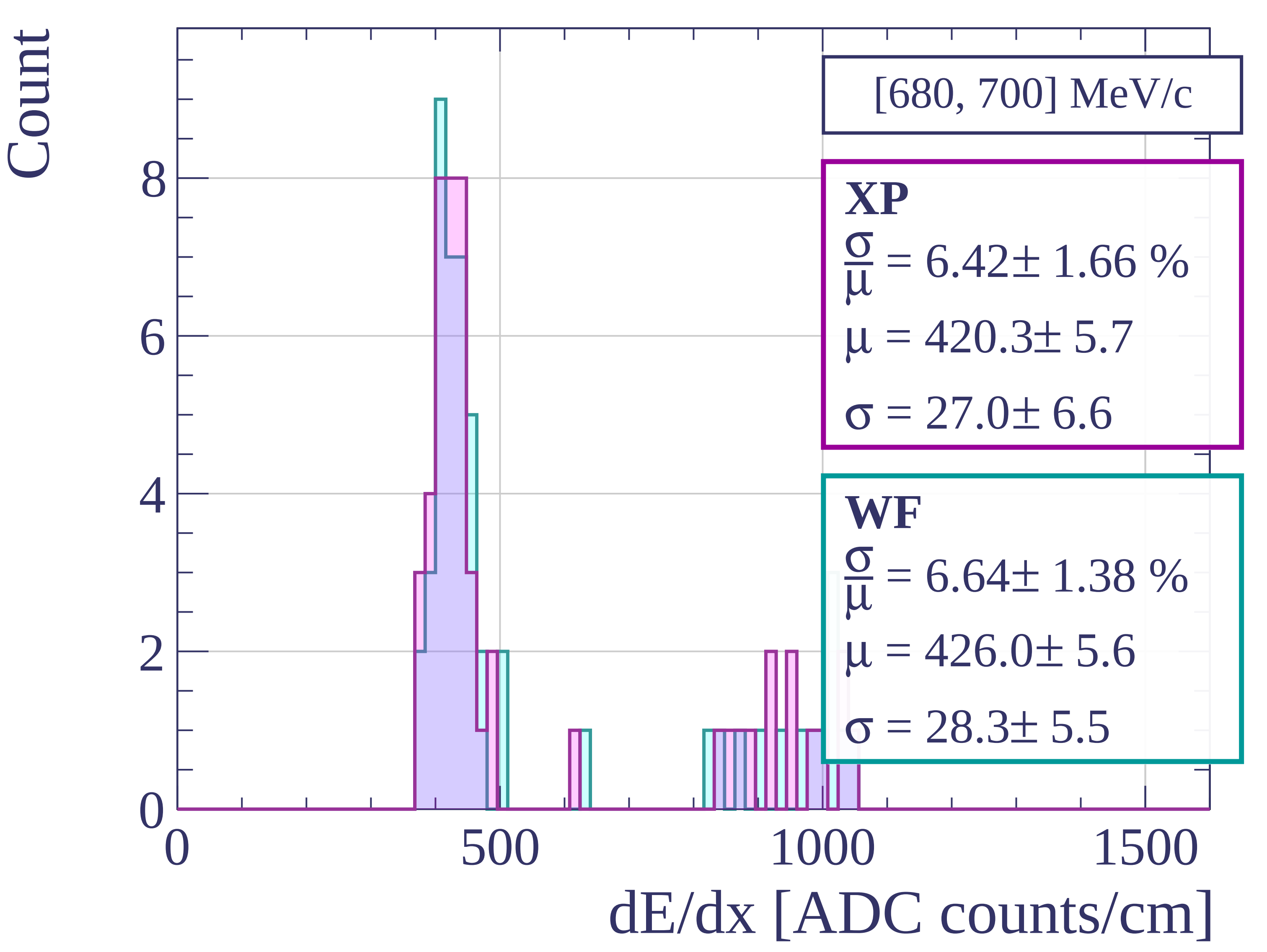
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[620, 640] MeV/c
XP
σ/μ = 6.95± 1.29 %
μ = 417.5± 6.0
σ = 29.0± 5.0
WF
σ/μ = 4.80± 0.89 %
μ = 419.0± 3.9
σ = 20.1± 3.5



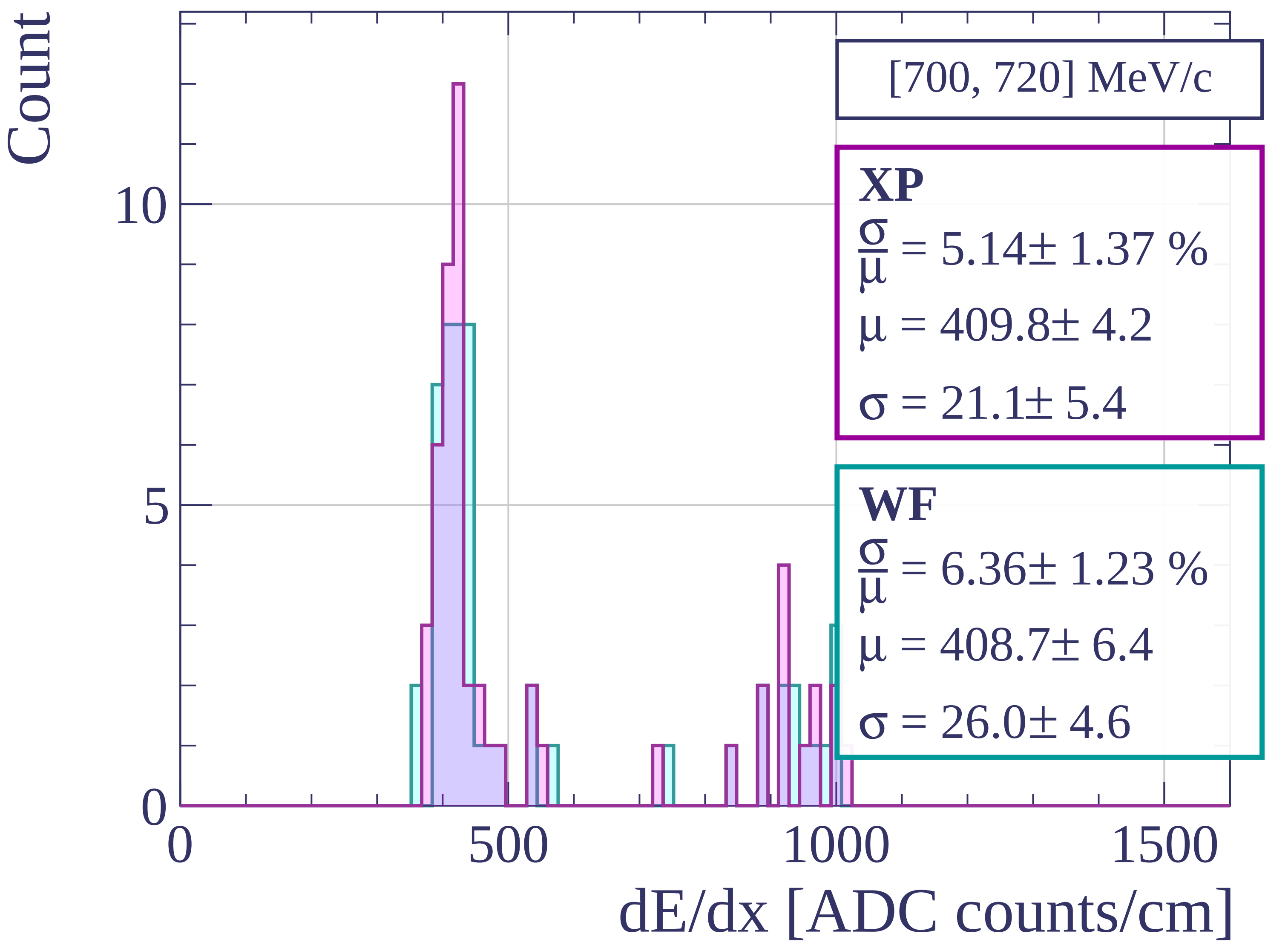
Count
dE/dx [ADC counts/cm]
[640, 660] MeV/c
XP
σ/μ = 5.03± 1.07 %
μ = 424.1± 4.3
σ = 21.4± 4.3
WF
σ/μ = 4.72± 0.90 %
μ = 425.4± 3.9
σ = 20.1± 3.6
0
500
1000
1500
5
10



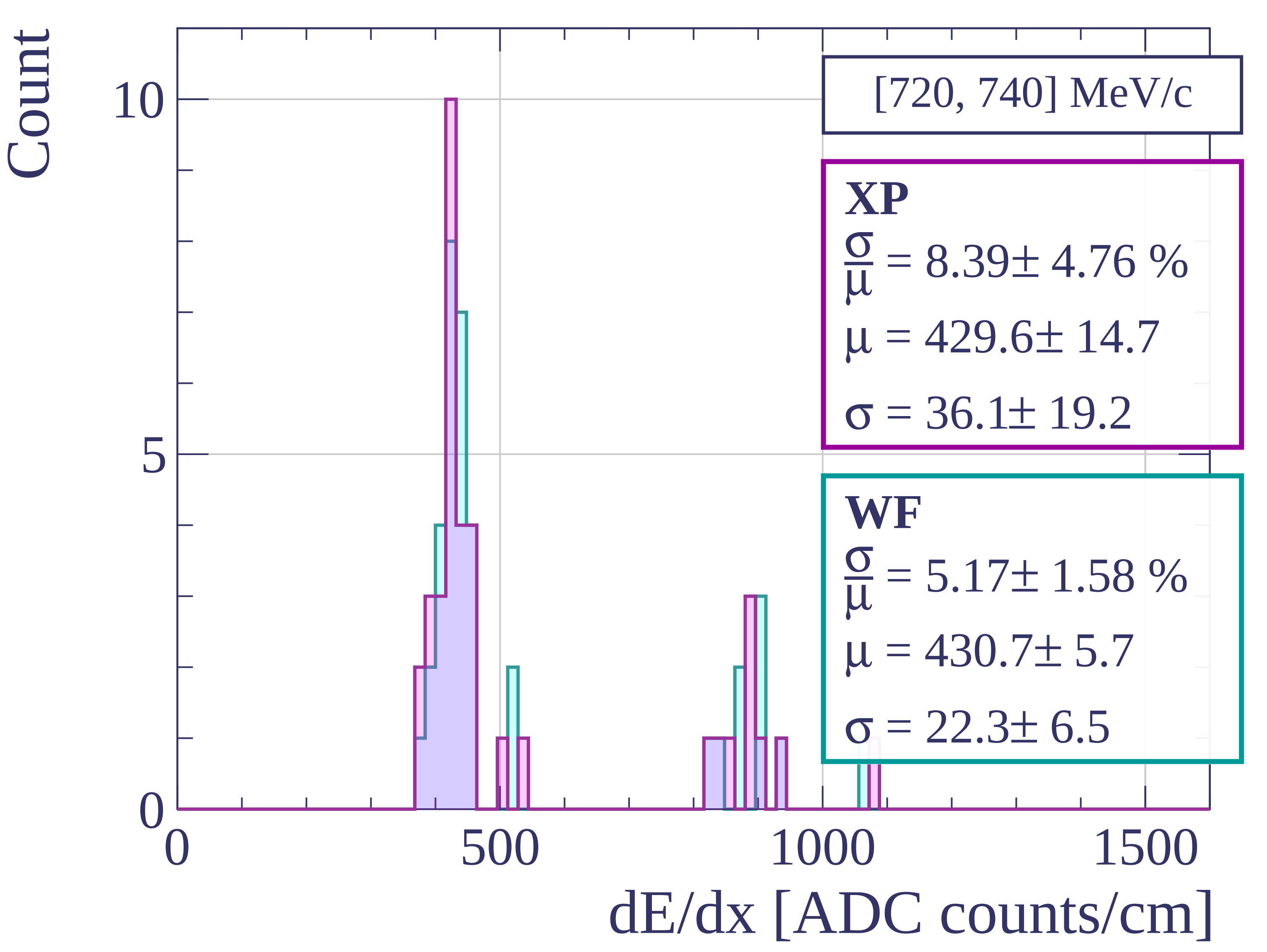
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[660, 680] MeV/c
XP
σ/μ = 7.64± 1.41 %
μ = 422.6± 6.3
σ = 32.3± 5.5
WF
σ/μ = 0.25± 1.79 %
μ = 439.5± 10.8
σ = 1.1± 7.8



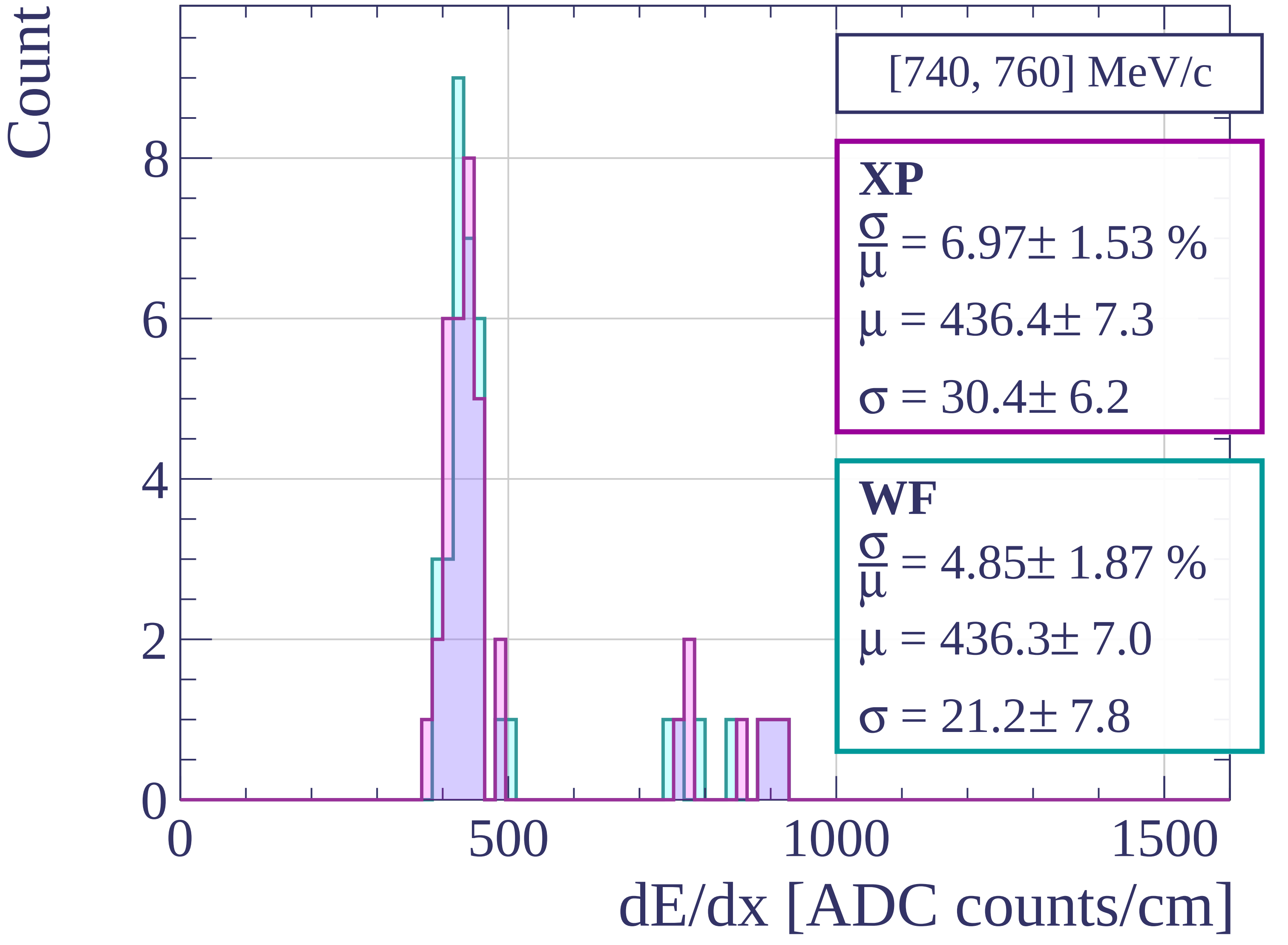
Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
[680, 700] MeV/c
XP
σ/μ = 6.42± 1.66 %
μ = 420.3± 5.7
σ = 27.0± 6.6
WF
σ/μ = 6.64± 1.38 %
μ = 426.0± 5.6
σ = 28.3± 5.5



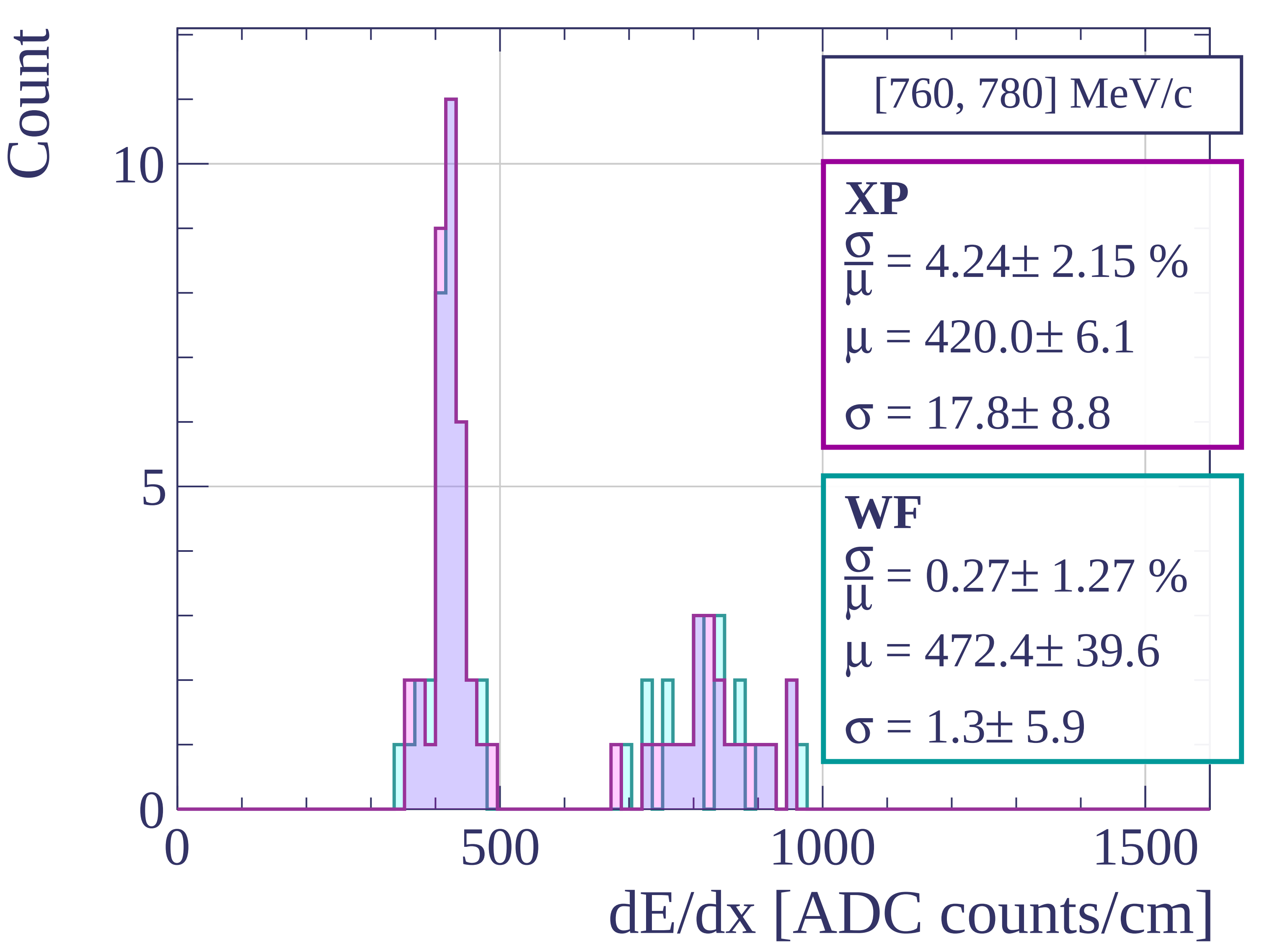
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[700, 720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.14± 1.37 %
μ = 409.8± 4.2
σ = 21.1± 5.4
WF
$\frac{\sigma}{\mu}$ = 6.36± 1.23 %
μ = 408.7± 6.4
σ = 26.0± 4.6



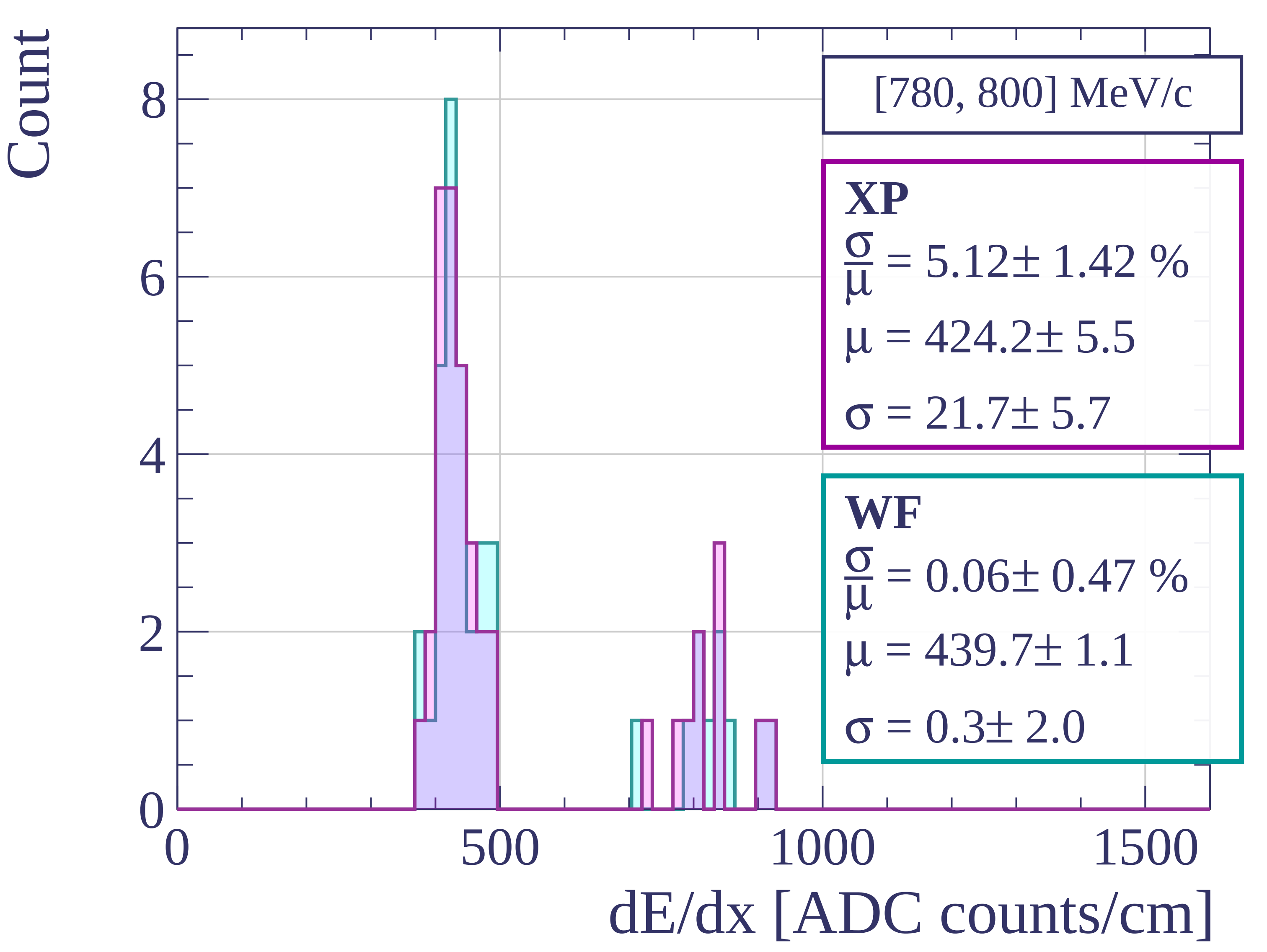
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[720, 740] MeV/c
XP
σ/μ = 8.39± 4.76 %
μ = 429.6± 14.7
σ = 36.1± 19.2
WF
σ/μ = 5.17± 1.58 %
μ = 430.7± 5.7
σ = 22.3± 6.5



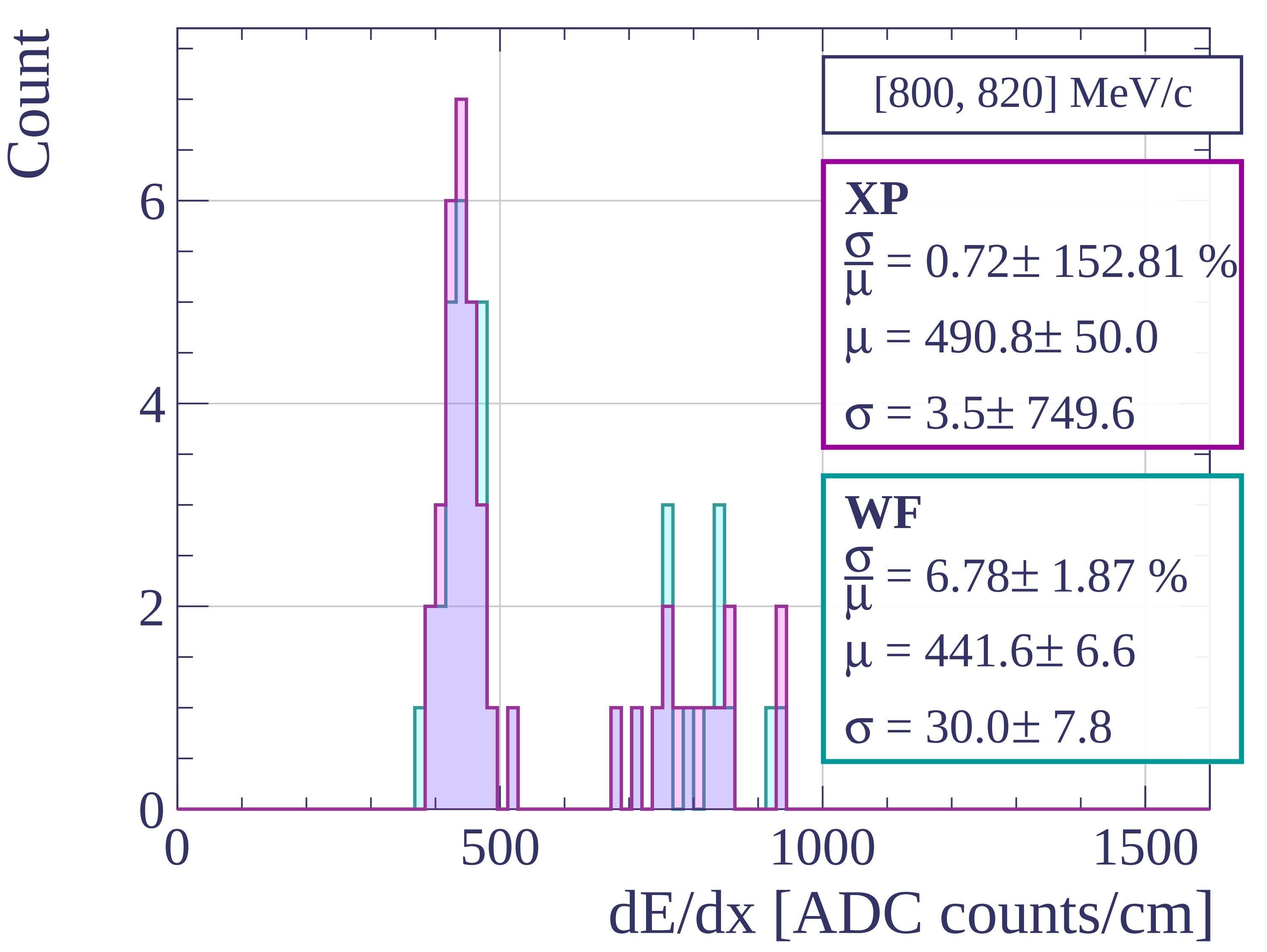
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[740, 760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.97± 1.53 %
μ = 436.4± 7.3
σ = 30.4± 6.2
WF
$\frac{\sigma}{\mu}$ = 4.85± 1.87 %
μ = 436.3± 7.0
σ = 21.2± 7.8



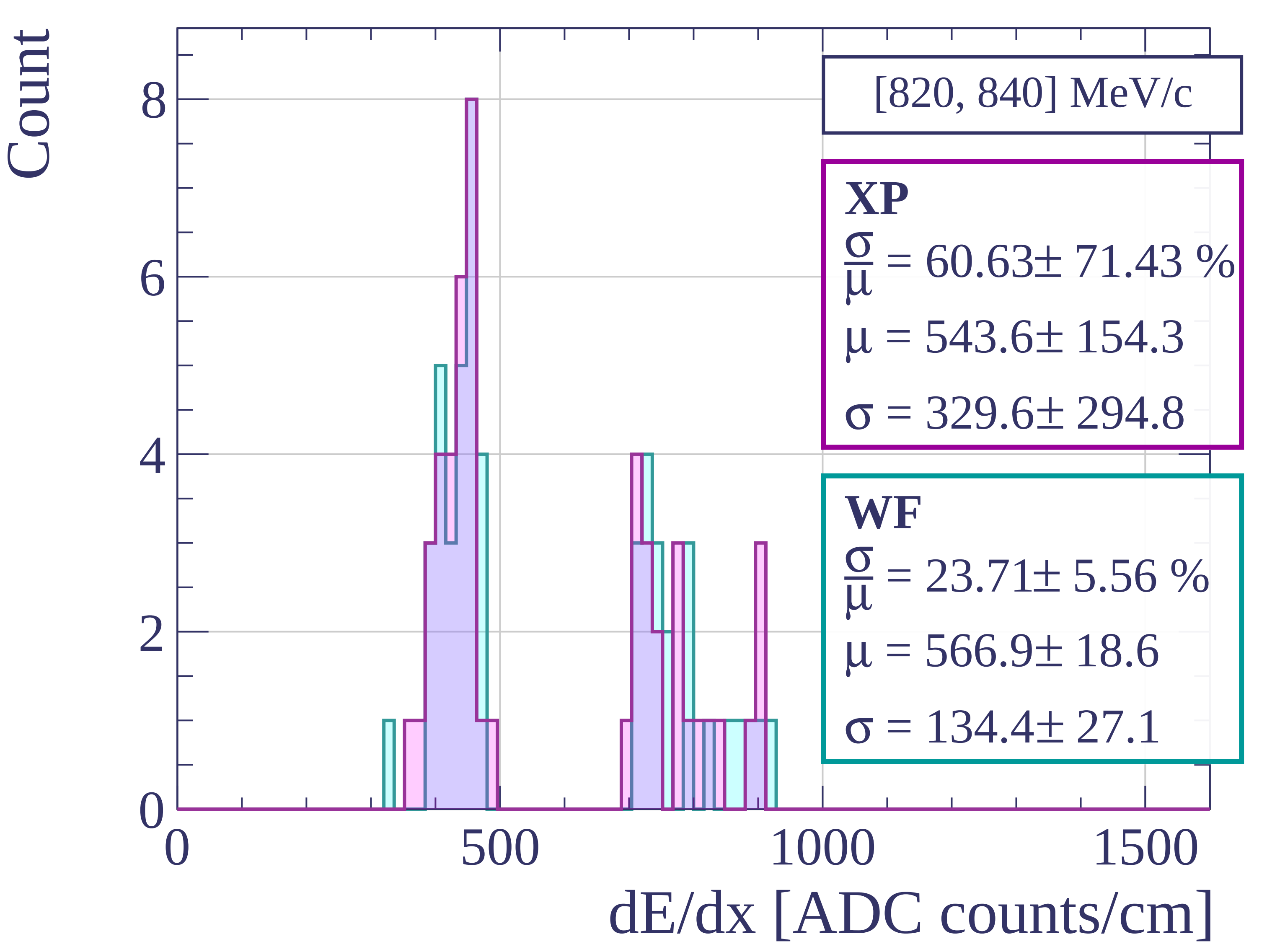
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[760, 780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.24± 2.15 %
μ = 420.0± 6.1
σ = 17.8± 8.8
WF
$\frac{\sigma}{\mu}$ = 0.27± 1.27 %
μ = 472.4± 39.6
σ = 1.3± 5.9



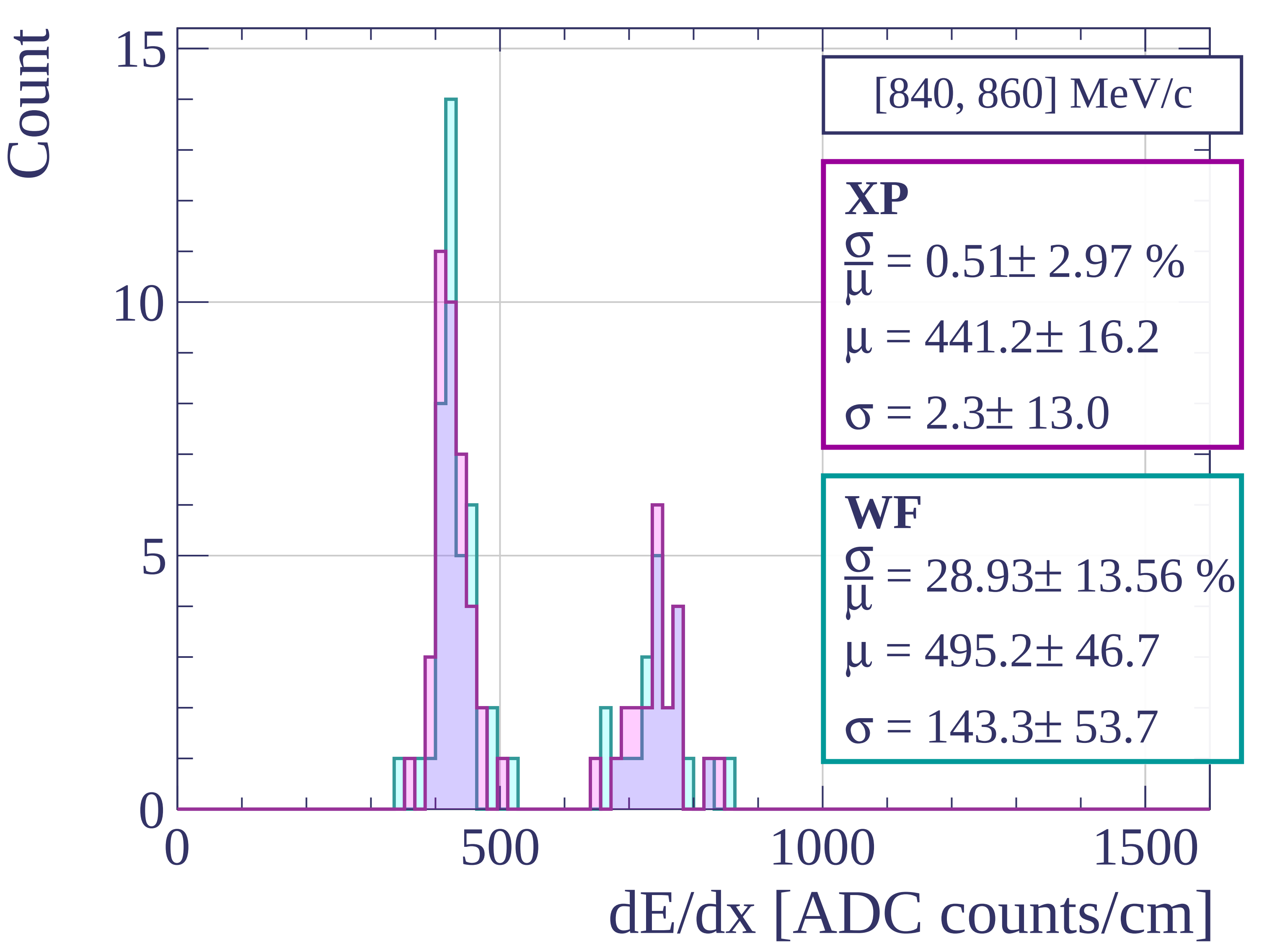
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[780, 800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.12± 1.42 %
μ = 424.2± 5.5
σ = 21.7± 5.7
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.47 %
μ = 439.7± 1.1
σ = 0.3± 2.0

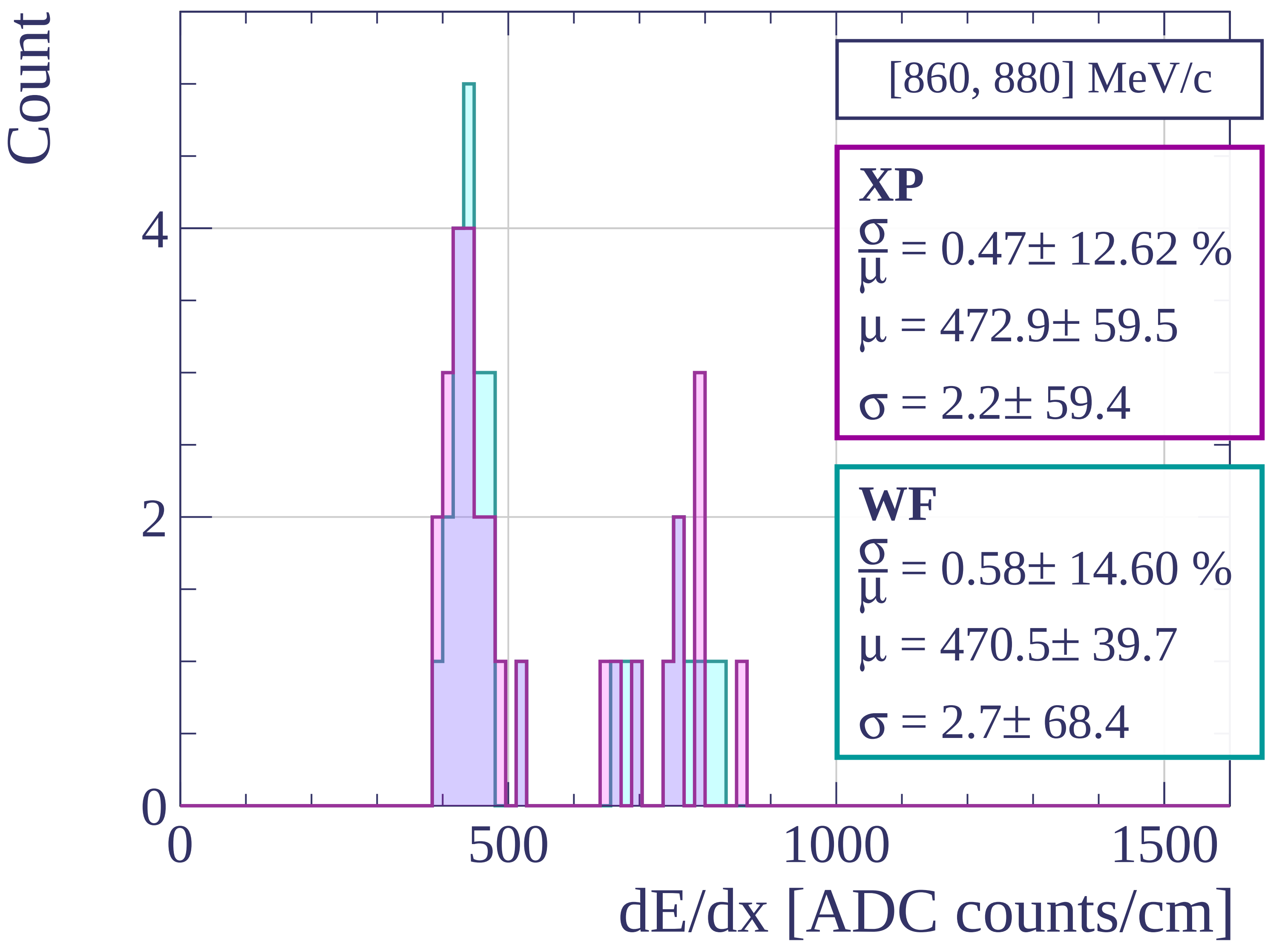


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
[800, 820] MeV/c
XP
σ/μ = 0.72± 152.81 %
μ = 490.8± 50.0
σ = 3.5± 749.6
WF
σ/μ = 6.78± 1.87 %
μ = 441.6± 6.6
σ = 30.0± 7.8

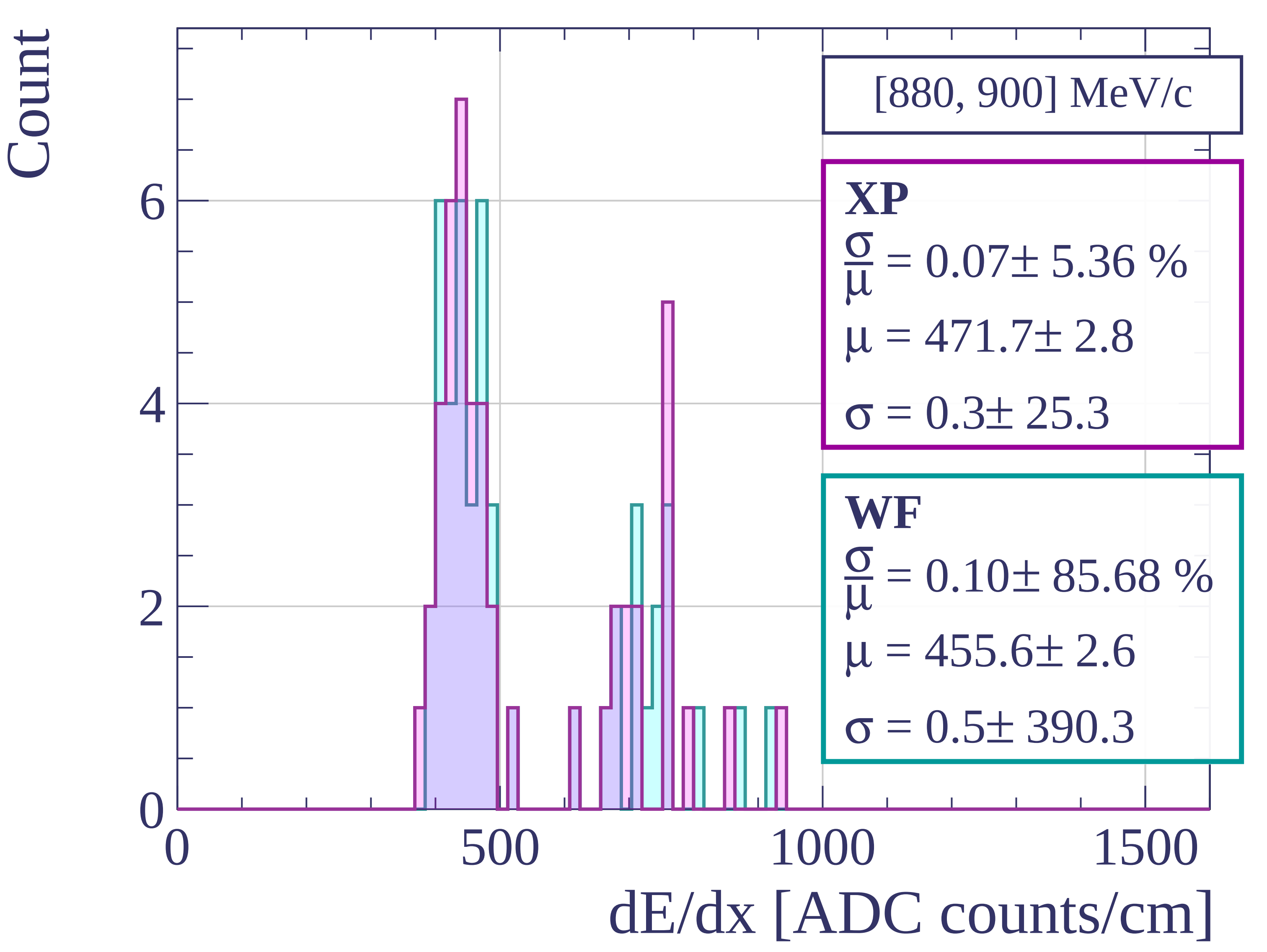


Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[820, 840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 60.63± 71.43 %
μ = 543.6± 154.3
σ = 329.6± 294.8
WF
$\frac{\sigma}{\mu}$ = 23.71± 5.56 %
μ = 566.9± 18.6
σ = 134.4± 27.1

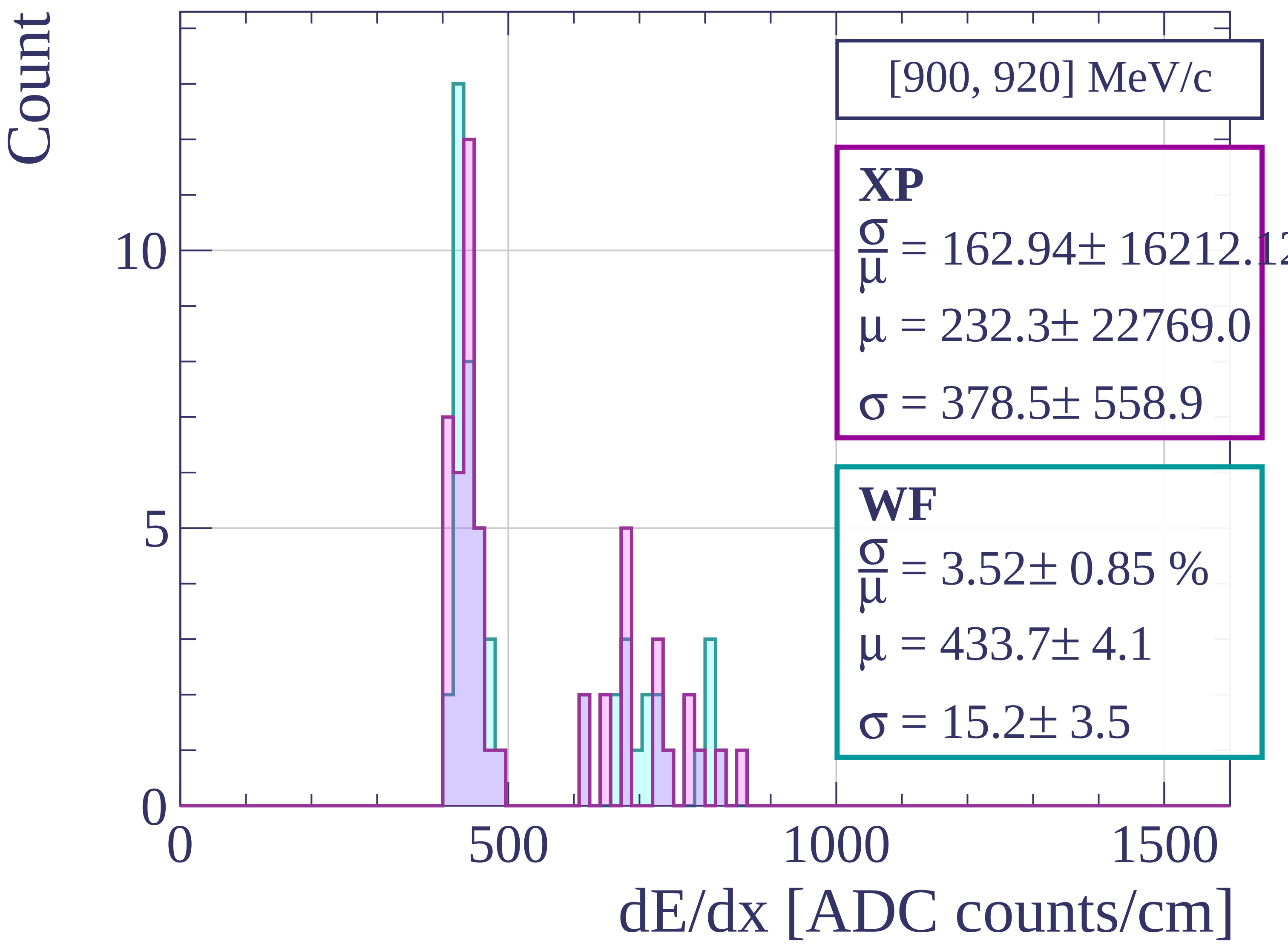

Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[840, 860] MeV/c
XP
σ/μ = 0.51± 2.97 %
μ = 441.2± 16.2
σ = 2.3± 13.0
WF
σ/μ = 28.93± 13.56 %
μ = 495.2± 46.7
σ = 143.3± 53.7



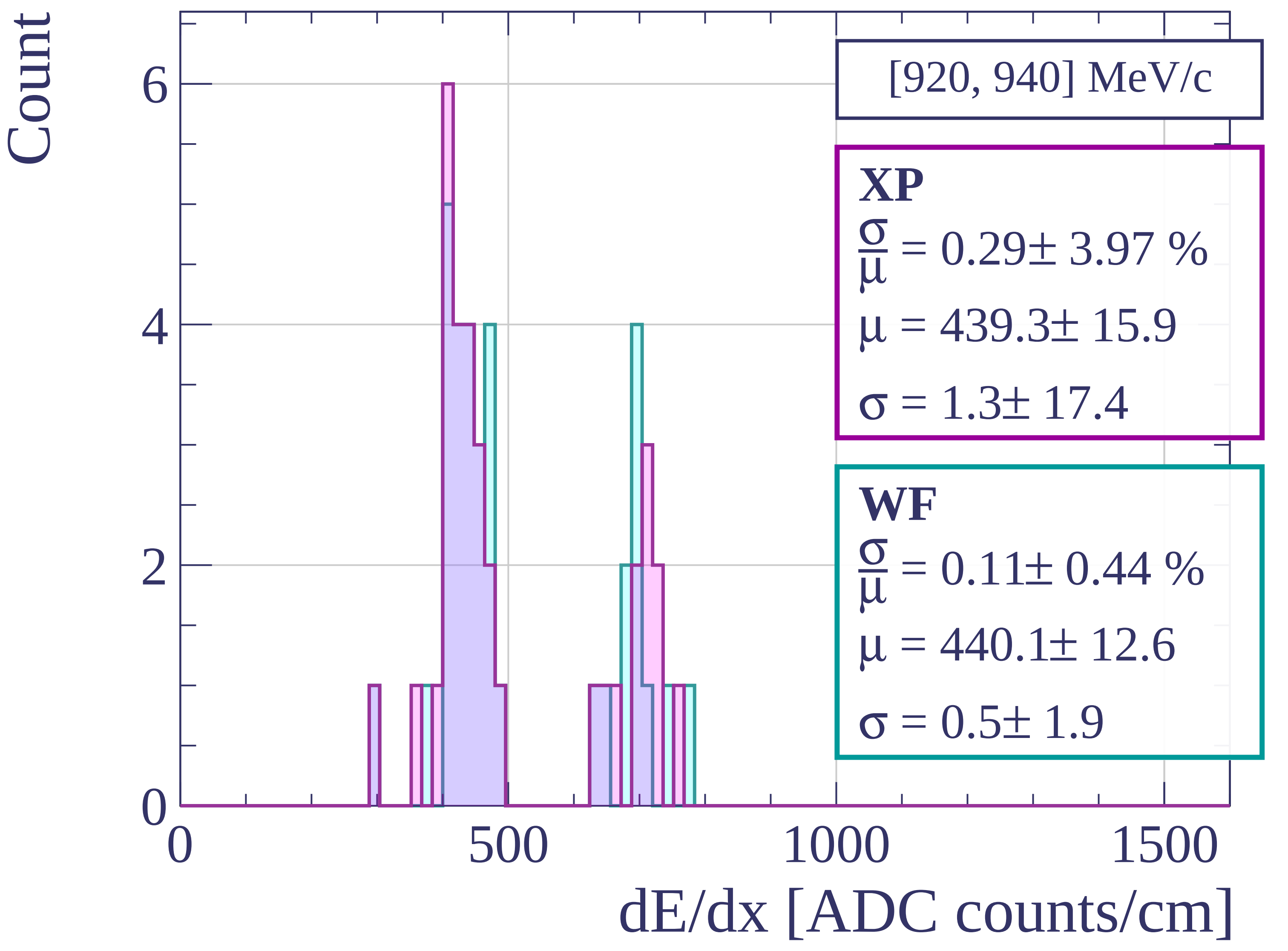
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[860, 880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 0.47± 12.62 %
μ = 472.9± 59.5
σ = 2.2± 59.4
WF
$\frac{\sigma}{\mu}$ = 0.58± 14.60 %
μ = 470.5± 39.7
σ = 2.7± 68.4



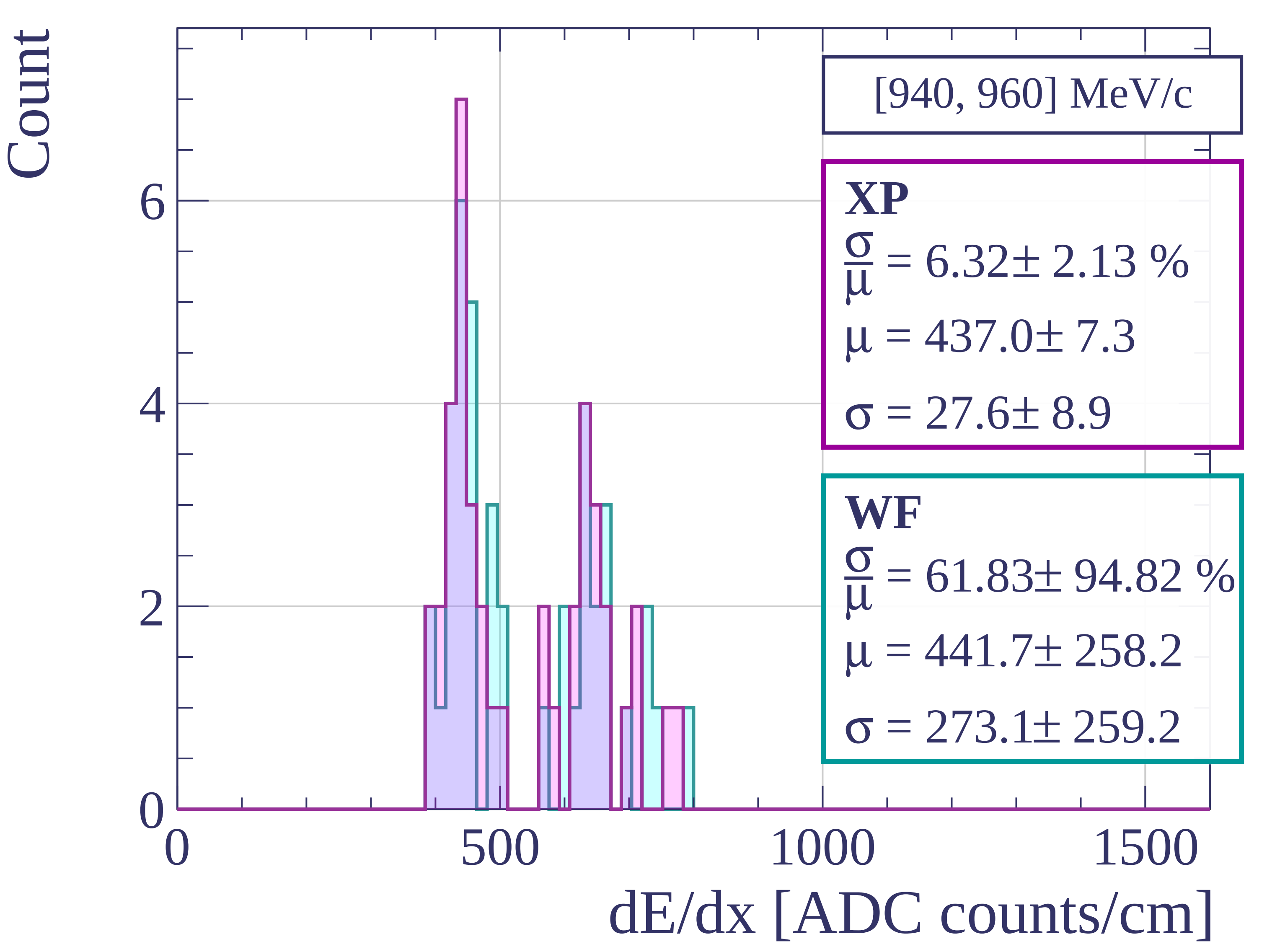
Count
[880, 900] MeV/c
XP
σ/μ = 0.07± 5.36 %
μ = 471.7± 2.8
σ = 0.3± 25.3
WF
σ/μ = 0.10± 85.68 %
μ = 455.6± 2.6
σ = 0.5± 390.3
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]



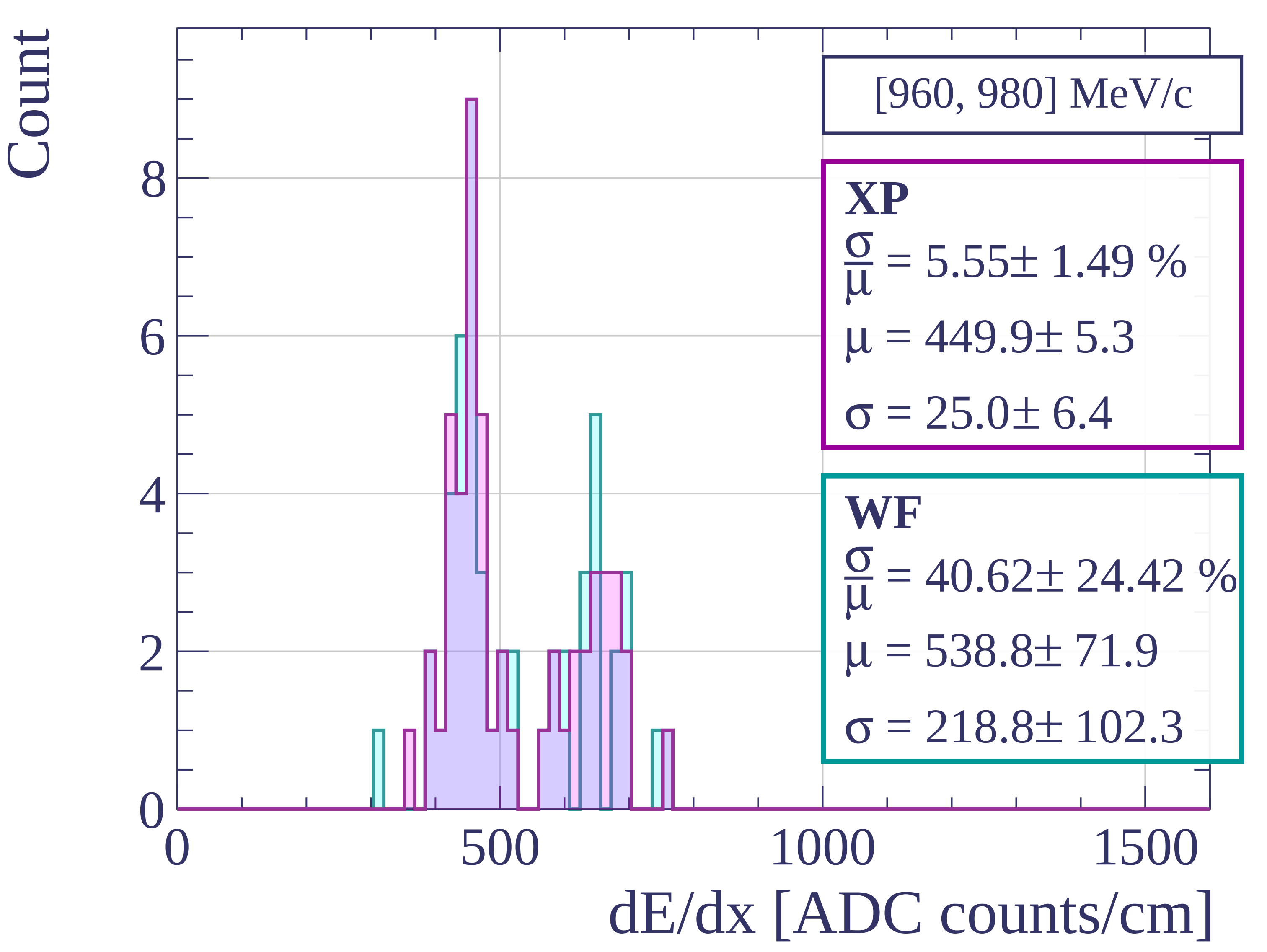
Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
[900, 920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 162.94± 16212.1
μ = 232.3± 22769.0
σ = 378.5± 558.9
WF
$\frac{\sigma}{\mu}$ = 3.52± 0.85 %
μ = 433.7± 4.1
σ = 15.2± 3.5



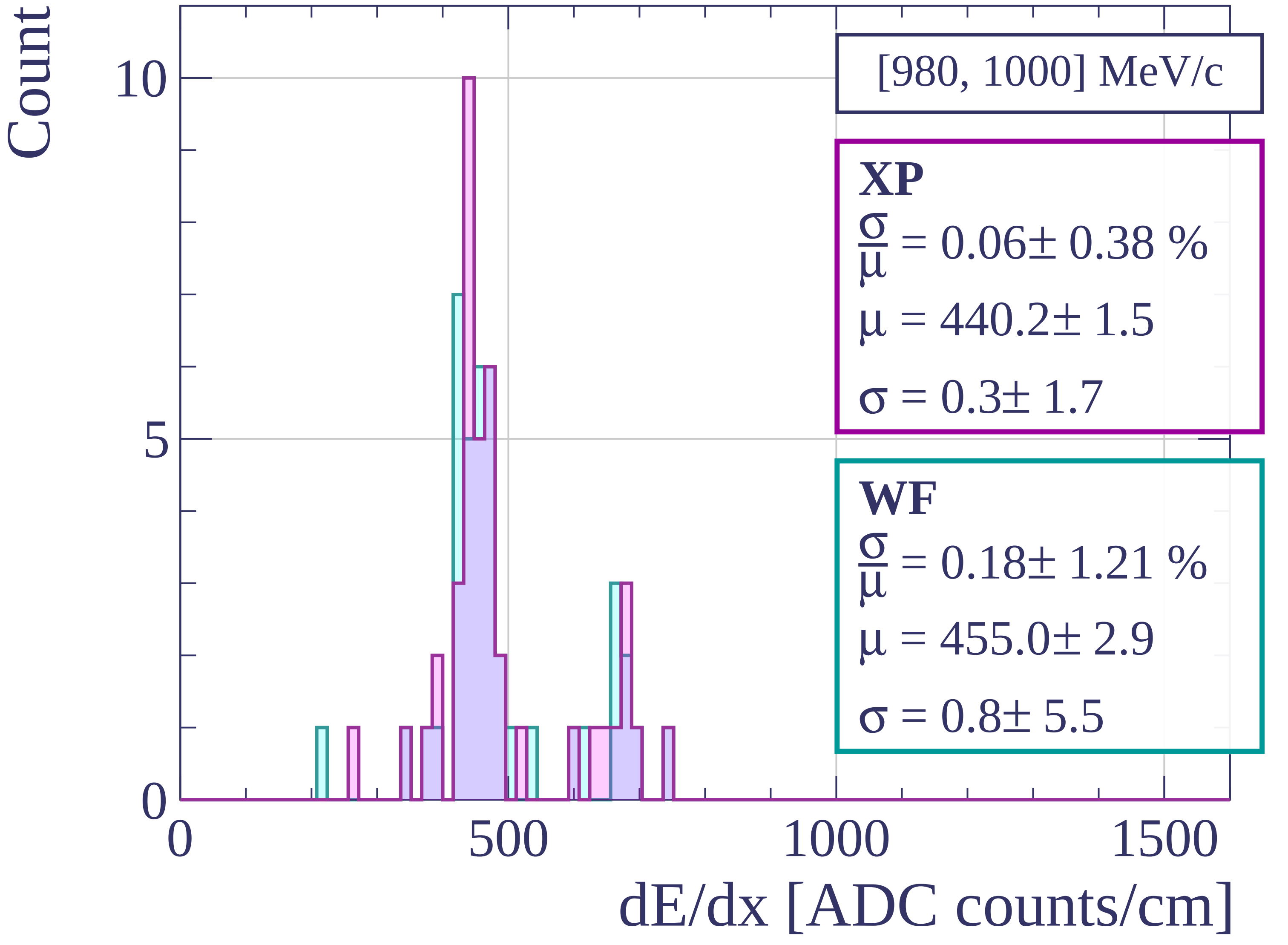
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[920, 940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 0.29± 3.97 %
μ = 439.3± 15.9
σ = 1.3± 17.4
WF
$\frac{\sigma}{\mu}$ = 0.11± 0.44 %
μ = 440.1± 12.6
σ = 0.5± 1.9



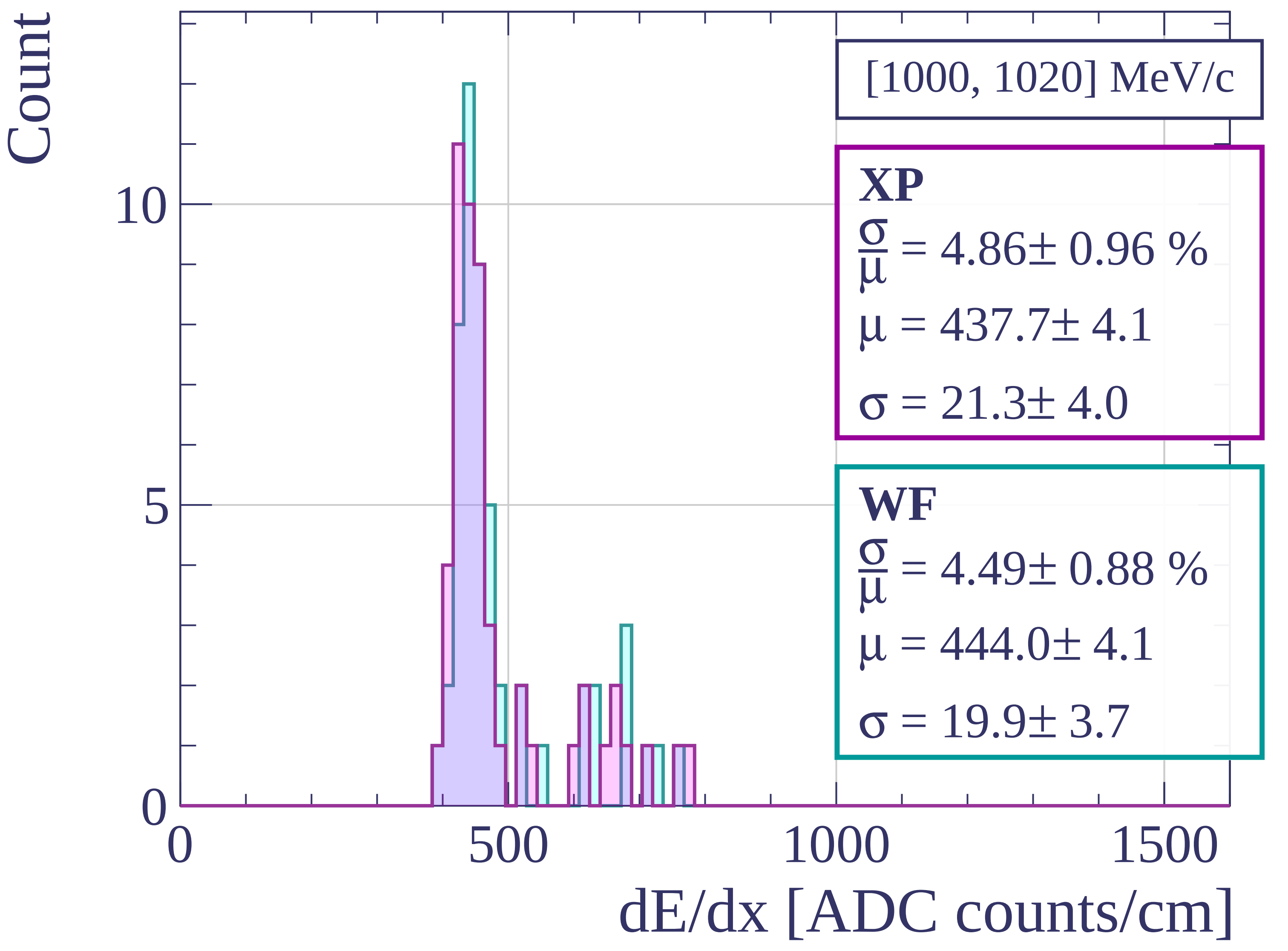
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[940, 960] MeV/c
XP
σ/μ = 6.32± 2.13 %
μ = 437.0± 7.3
σ = 27.6± 8.9
WF
σ/μ = 61.83± 94.82 %
μ = 441.7± 258.2
σ = 273.1± 259.2

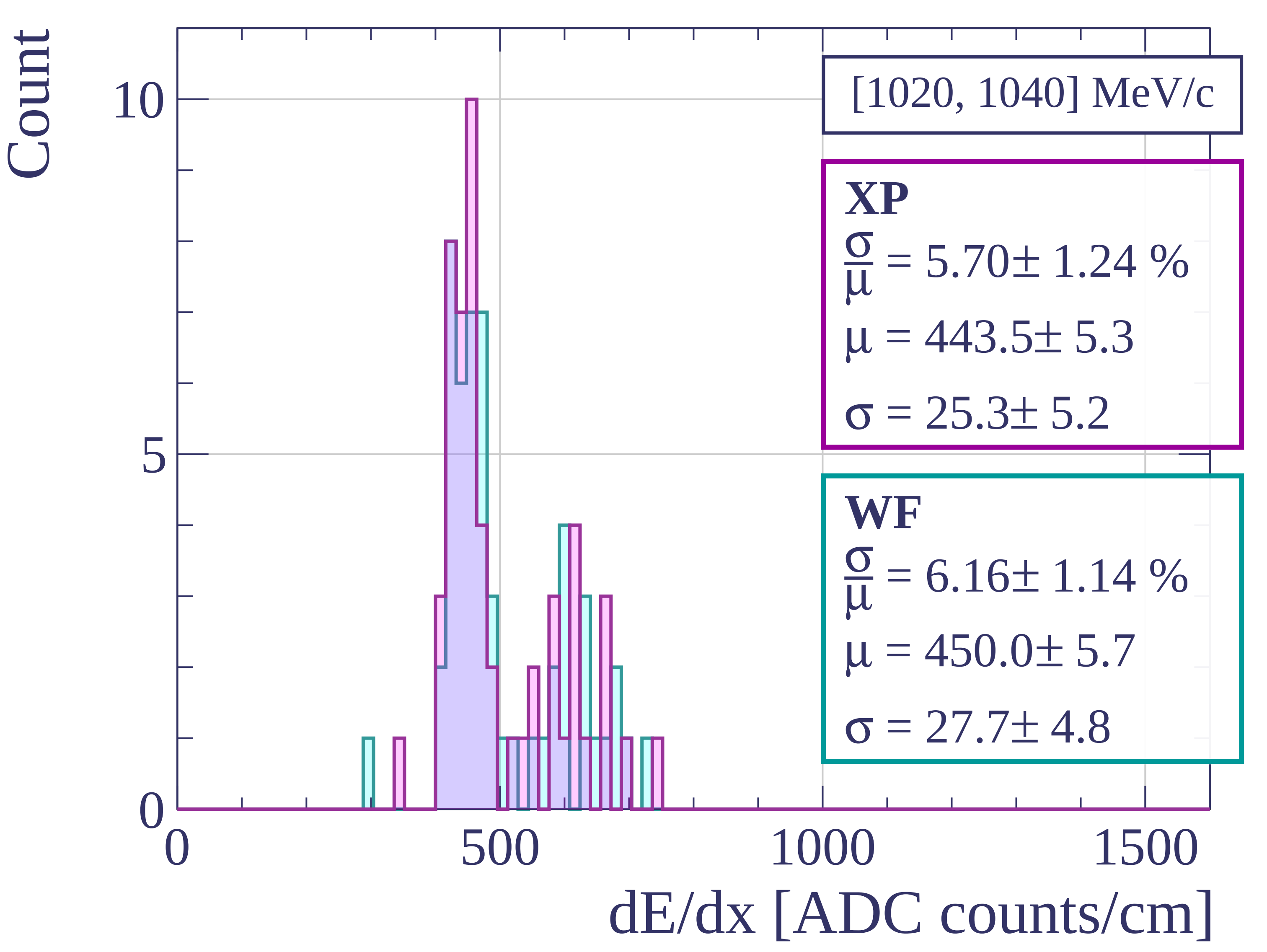


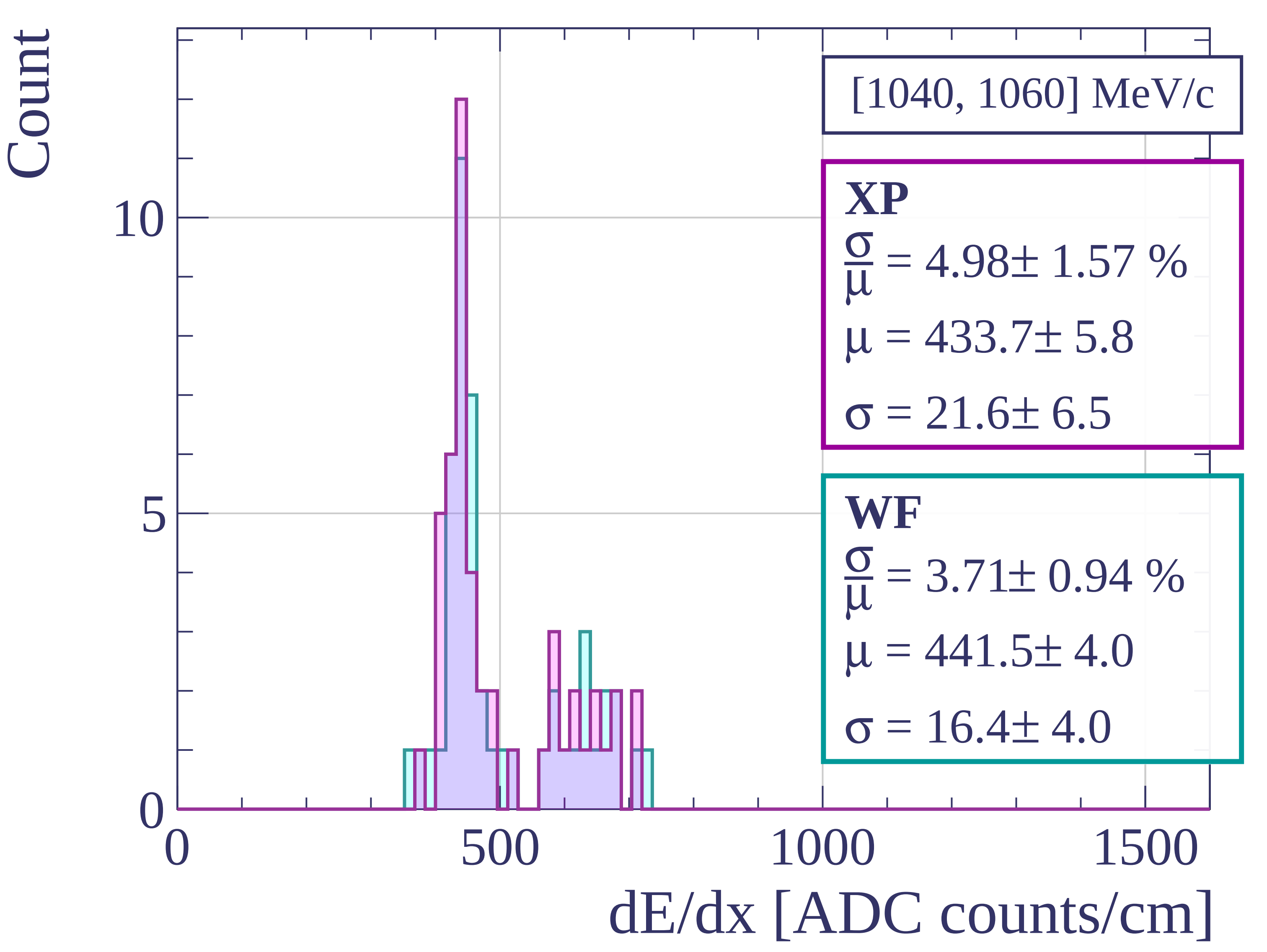
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[960, 980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.55± 1.49 %
μ = 449.9± 5.3
σ = 25.0± 6.4
WF
$\frac{\sigma}{\mu}$ = 40.62± 24.42 %
μ = 538.8± 71.9
σ = 218.8± 102.3

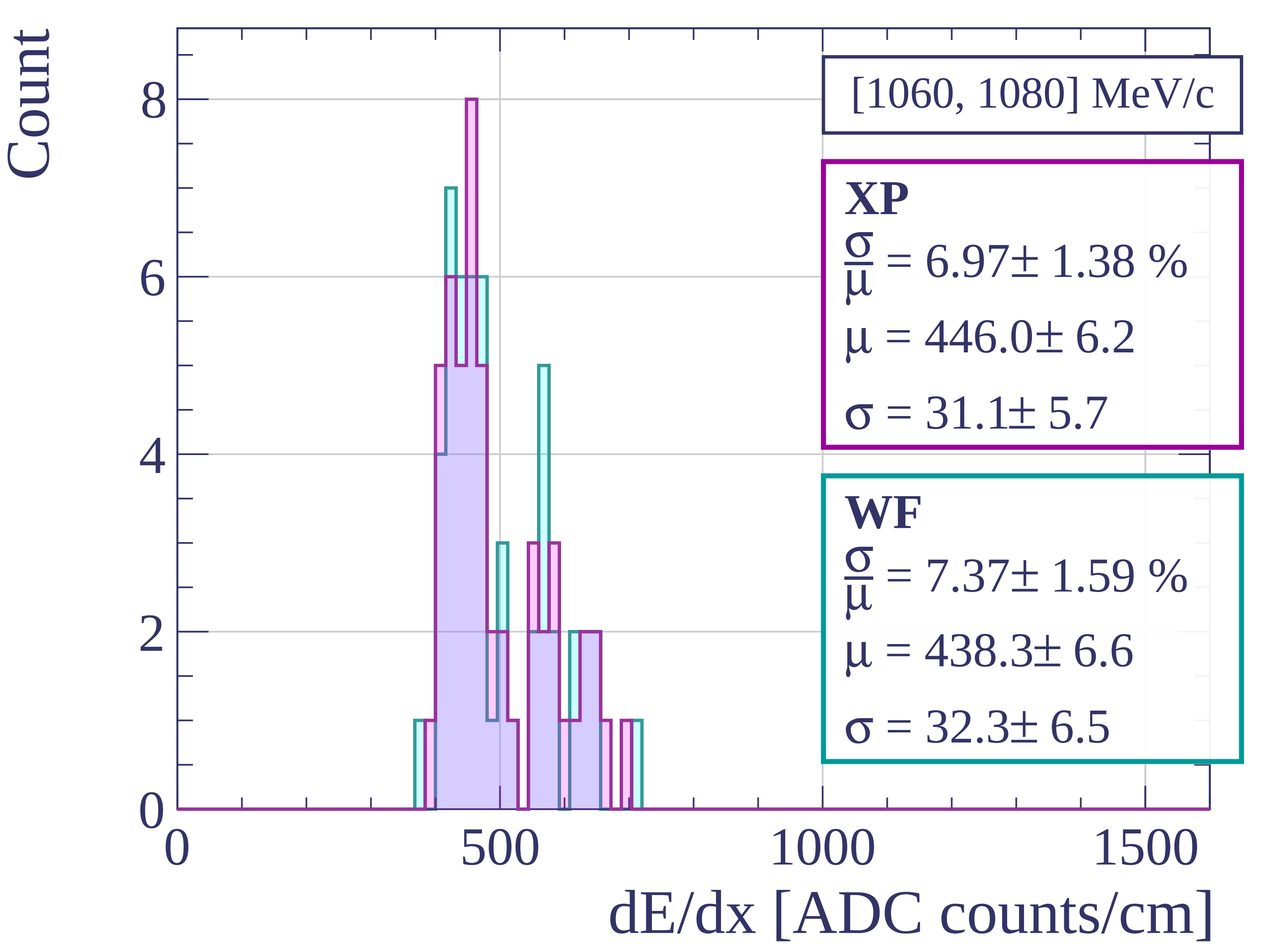


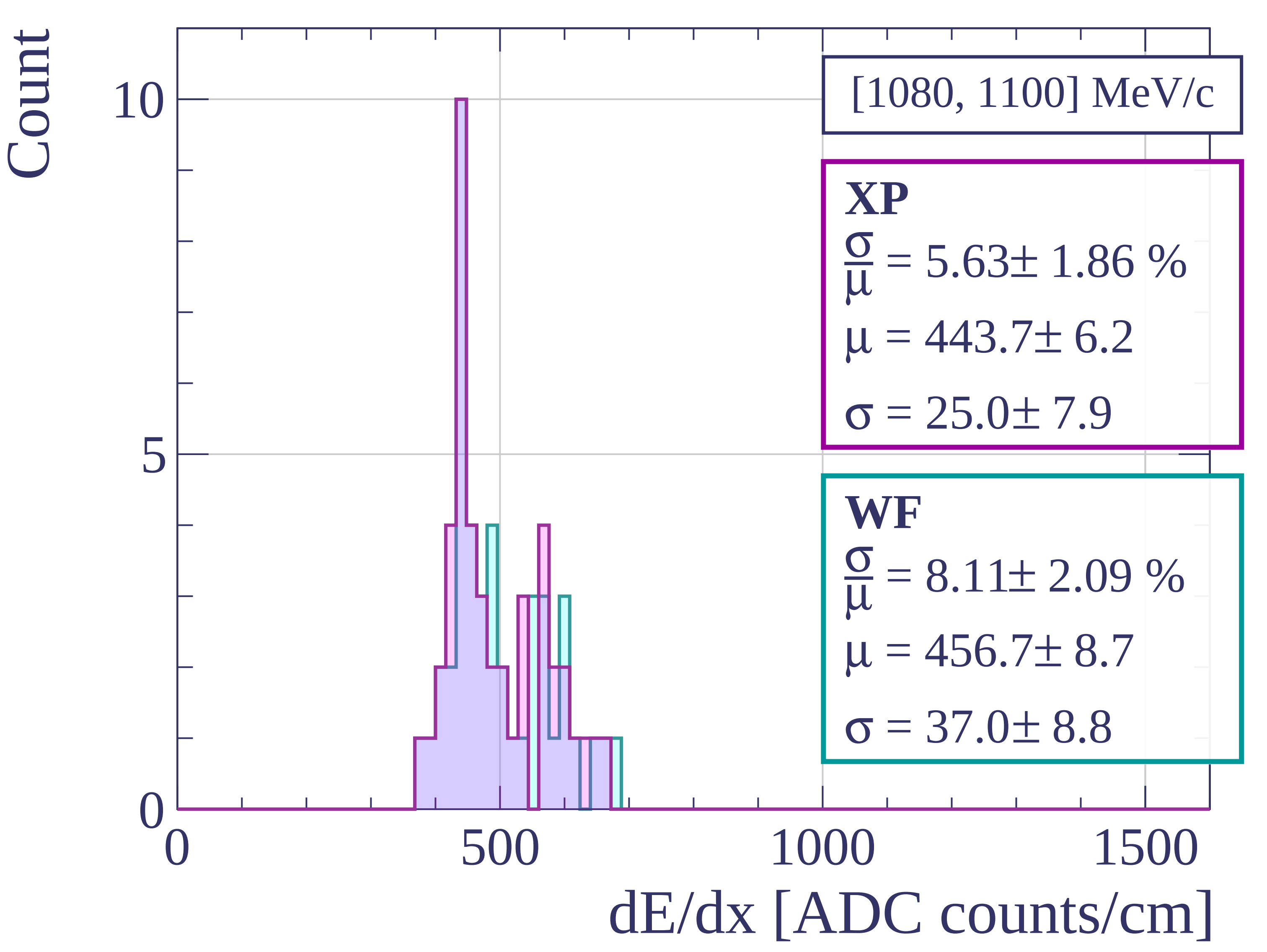
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[980, 1000] MeV/c
XP
σ/μ = 0.06± 0.38 %
μ = 440.2± 1.5
σ = 0.3± 1.7
WF
σ/μ = 0.18± 1.21 %
μ = 455.0± 2.9
σ = 0.8± 5.5


Count
0
5
10
0
500
1000
1500
dE/dx [ADC counts/cm]
[1000, 1020] MeV/c
XP
σ/μ = 4.86± 0.96 %
μ = 437.7± 4.1
σ = 21.3± 4.0
WF
σ/μ = 4.49± 0.88 %
μ = 444.0± 4.1
σ = 19.9± 3.7

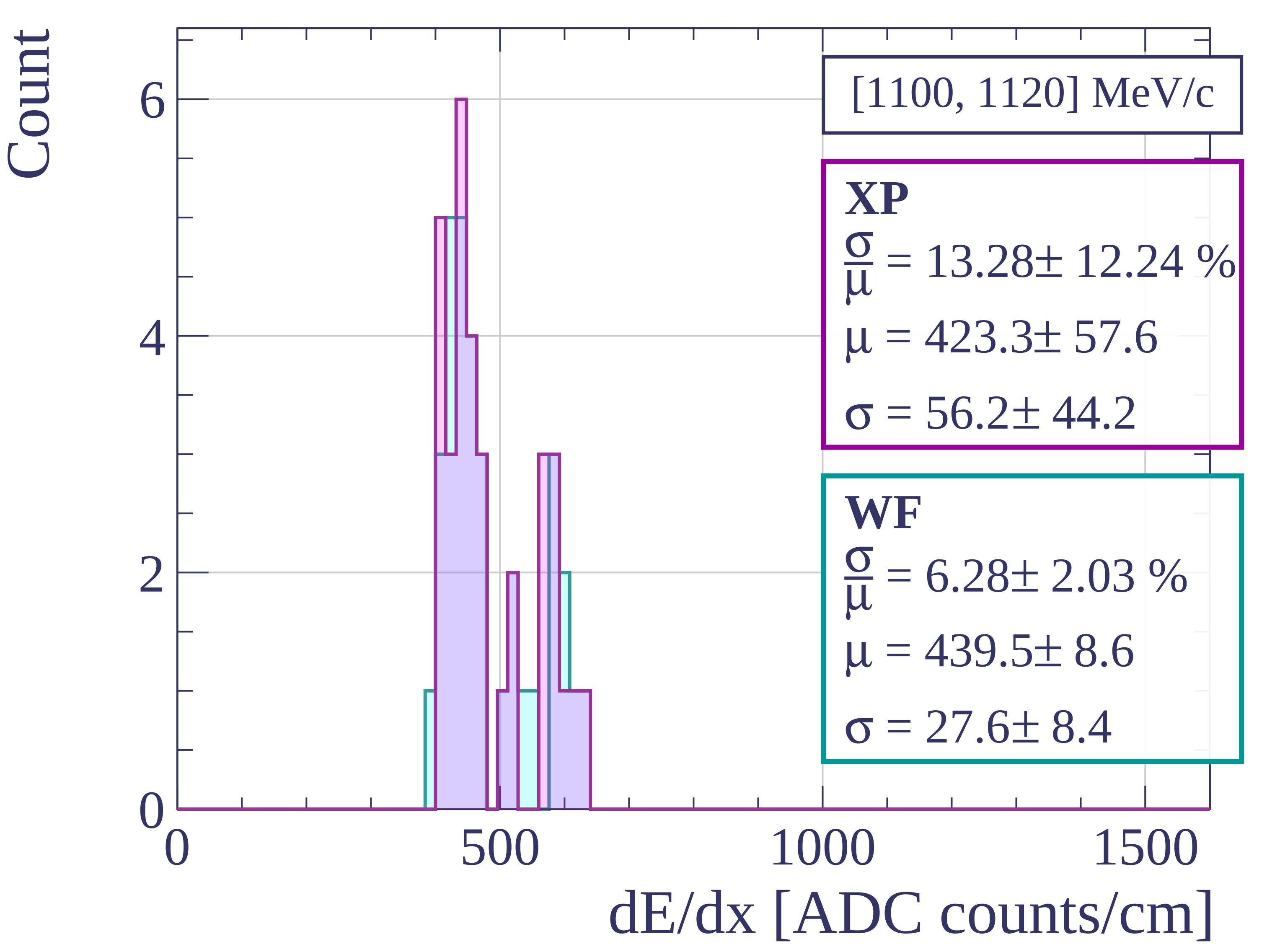

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1020, 1040] MeV/c
XP
σ/μ = 5.70± 1.24 %
μ = 443.5± 5.3
σ = 25.3± 5.2
WF
σ/μ = 6.16± 1.14 %
μ = 450.0± 5.7
σ = 27.7± 4.8


Count
dE/dx [ADC counts/cm]
[1040, 1060] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.98± 1.57 %
μ = 433.7± 5.8
σ = 21.6± 6.5
WF
$\frac{\sigma}{\mu}$ = 3.71± 0.94 %
μ = 441.5± 4.0
σ = 16.4± 4.0

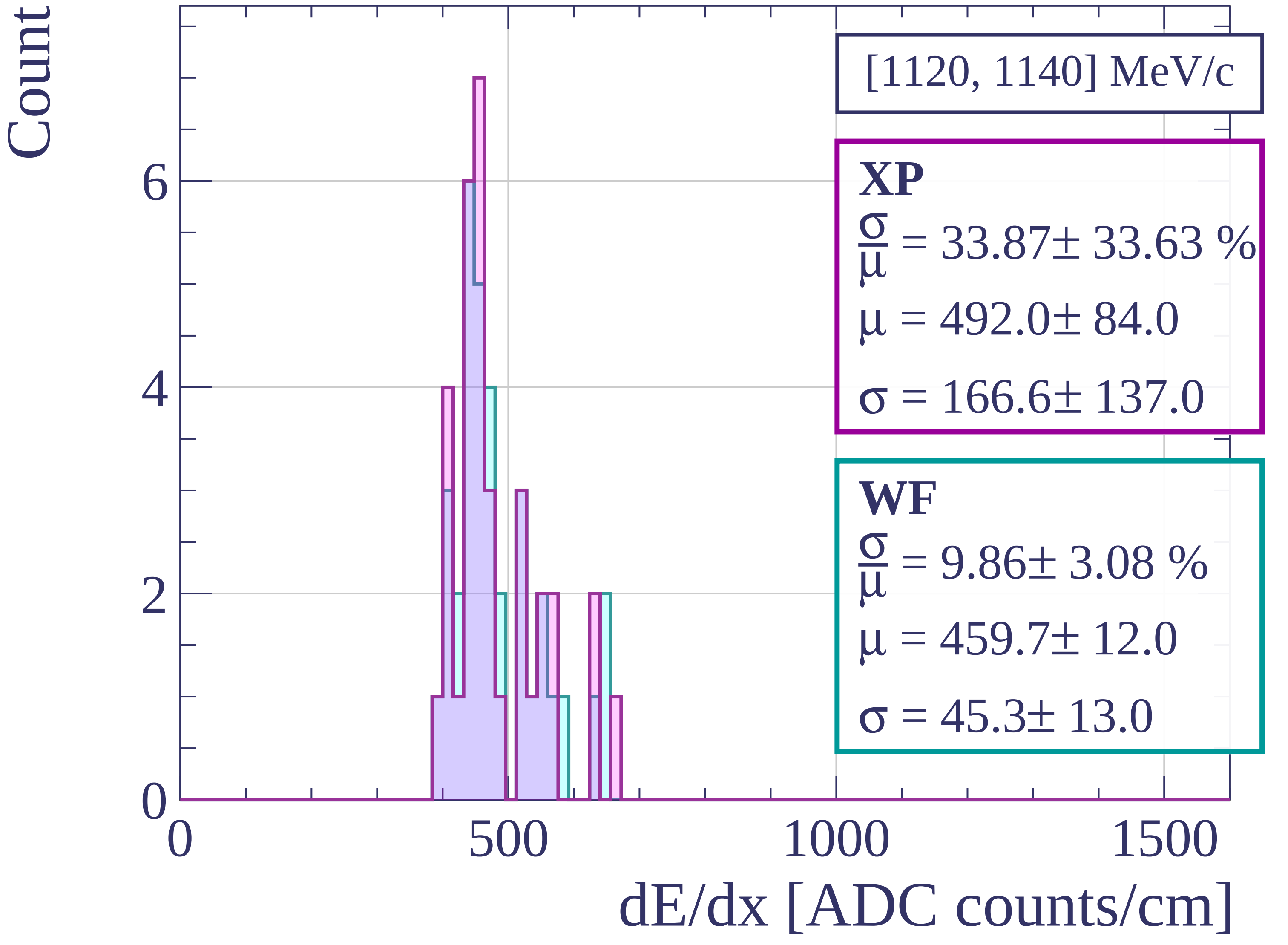

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1060, 1080] MeV/c
XP
σ/μ = 6.97± 1.38 %
μ = 446.0± 6.2
σ = 31.1± 5.7
WF
σ/μ = 7.37± 1.59 %
μ = 438.3± 6.6
σ = 32.3± 6.5

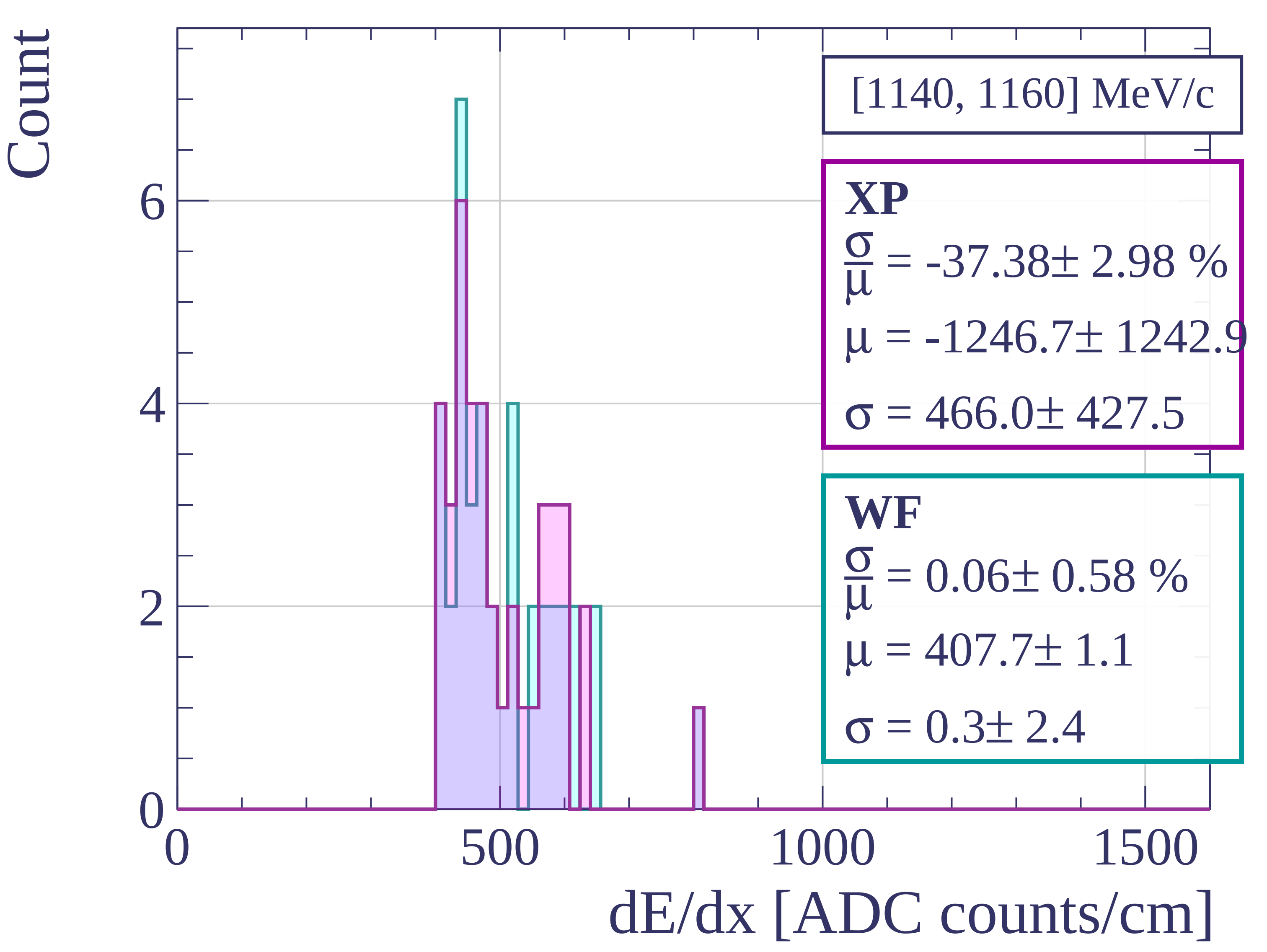


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1080, 1100] MeV/c
XP
σ/μ = 5.63± 1.86 %
μ = 443.7± 6.2
σ = 25.0± 7.9
WF
σ/μ = 8.11± 2.09 %
μ = 456.7± 8.7
σ = 37.0± 8.8

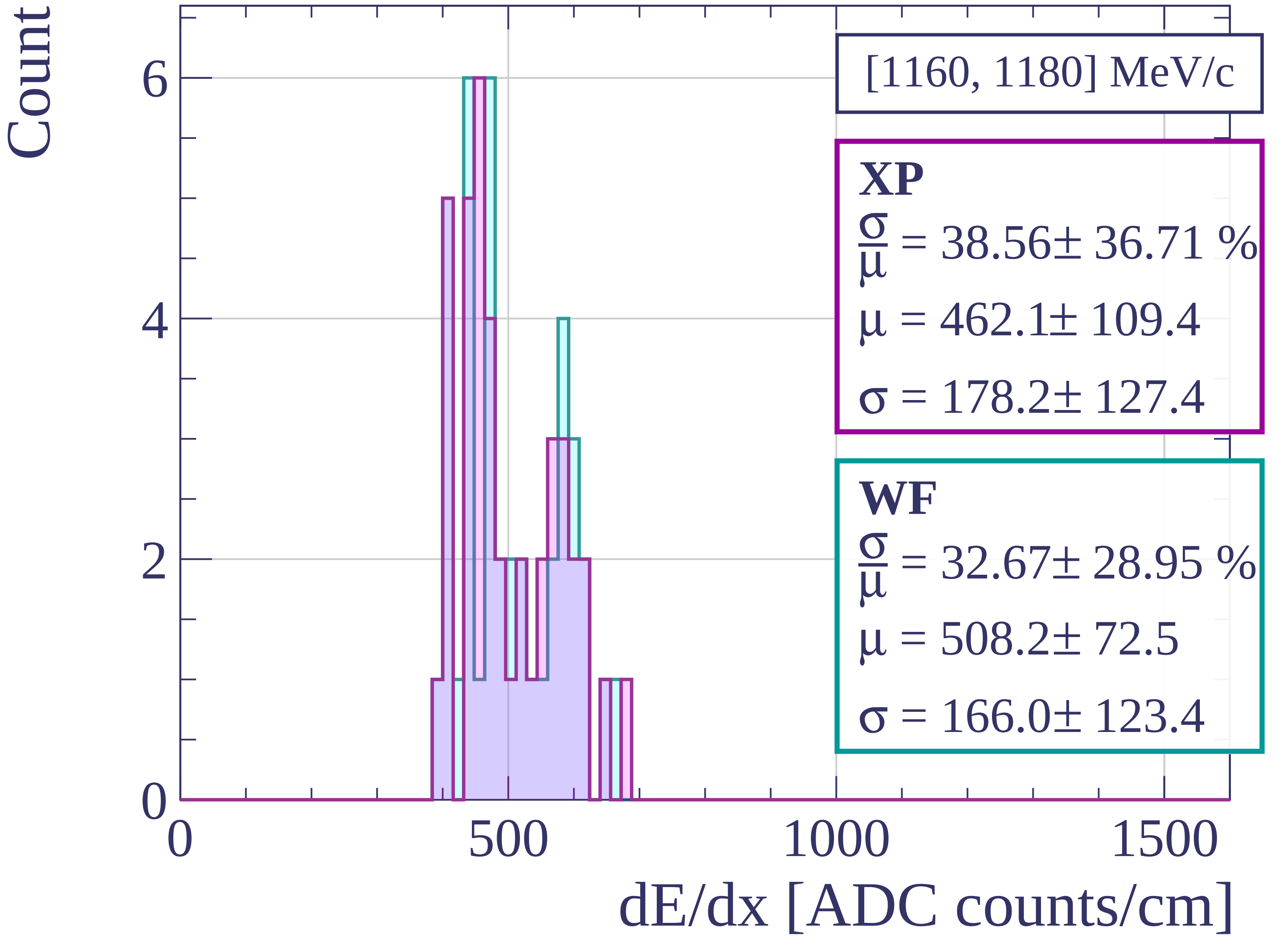


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[1100, 1120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.28± 12.24 %
μ = 423.3± 57.6
σ = 56.2± 44.2
WF
$\frac{\sigma}{\mu}$ = 6.28± 2.03 %
μ = 439.5± 8.6
σ = 27.6± 8.4

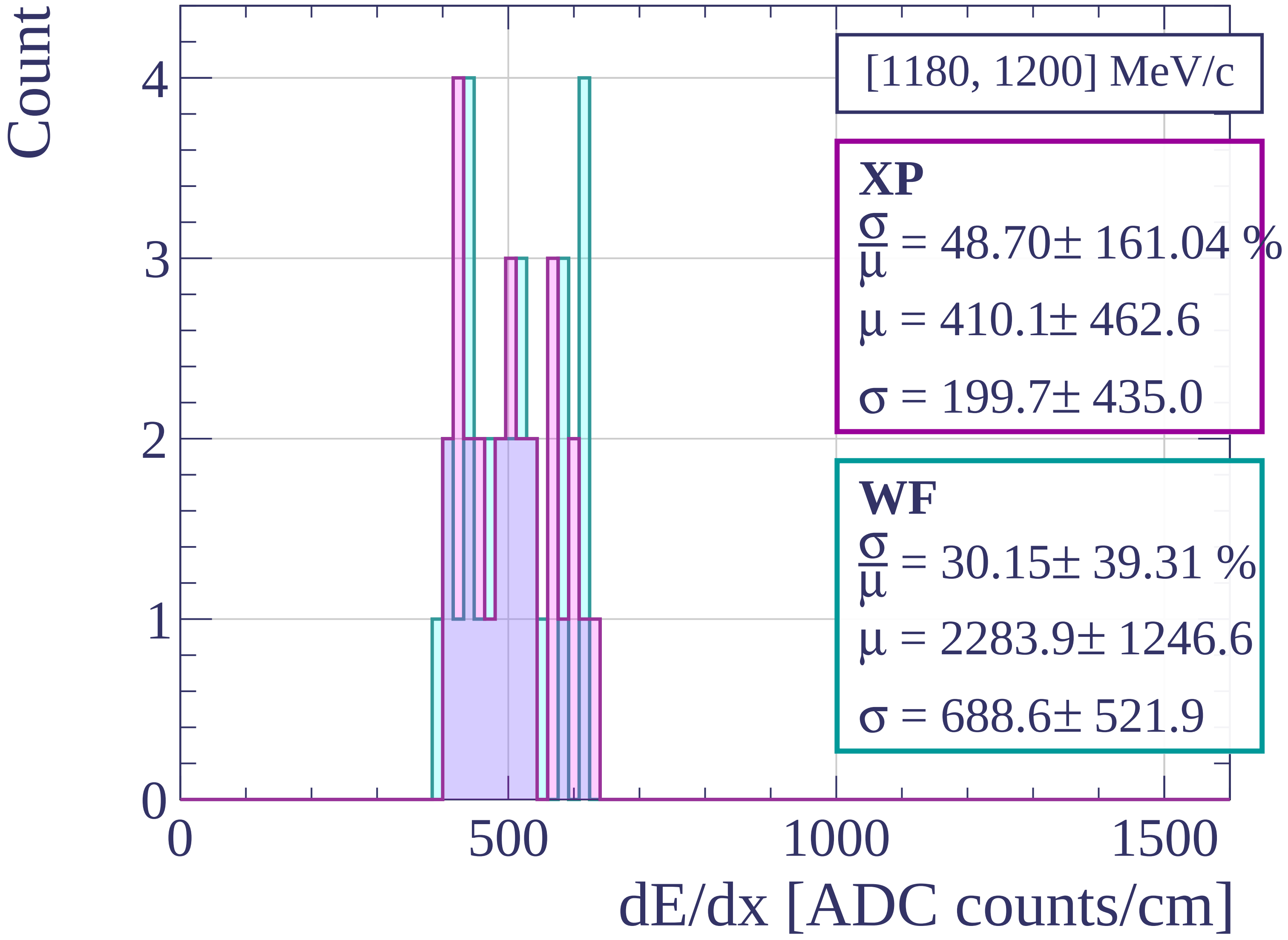

Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1120, 1140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 33.87± 33.63 %
μ = 492.0± 84.0
σ = 166.6± 137.0
WF
$\frac{\sigma}{\mu}$ = 9.86± 3.08 %
μ = 459.7± 12.0
σ = 45.3± 13.0



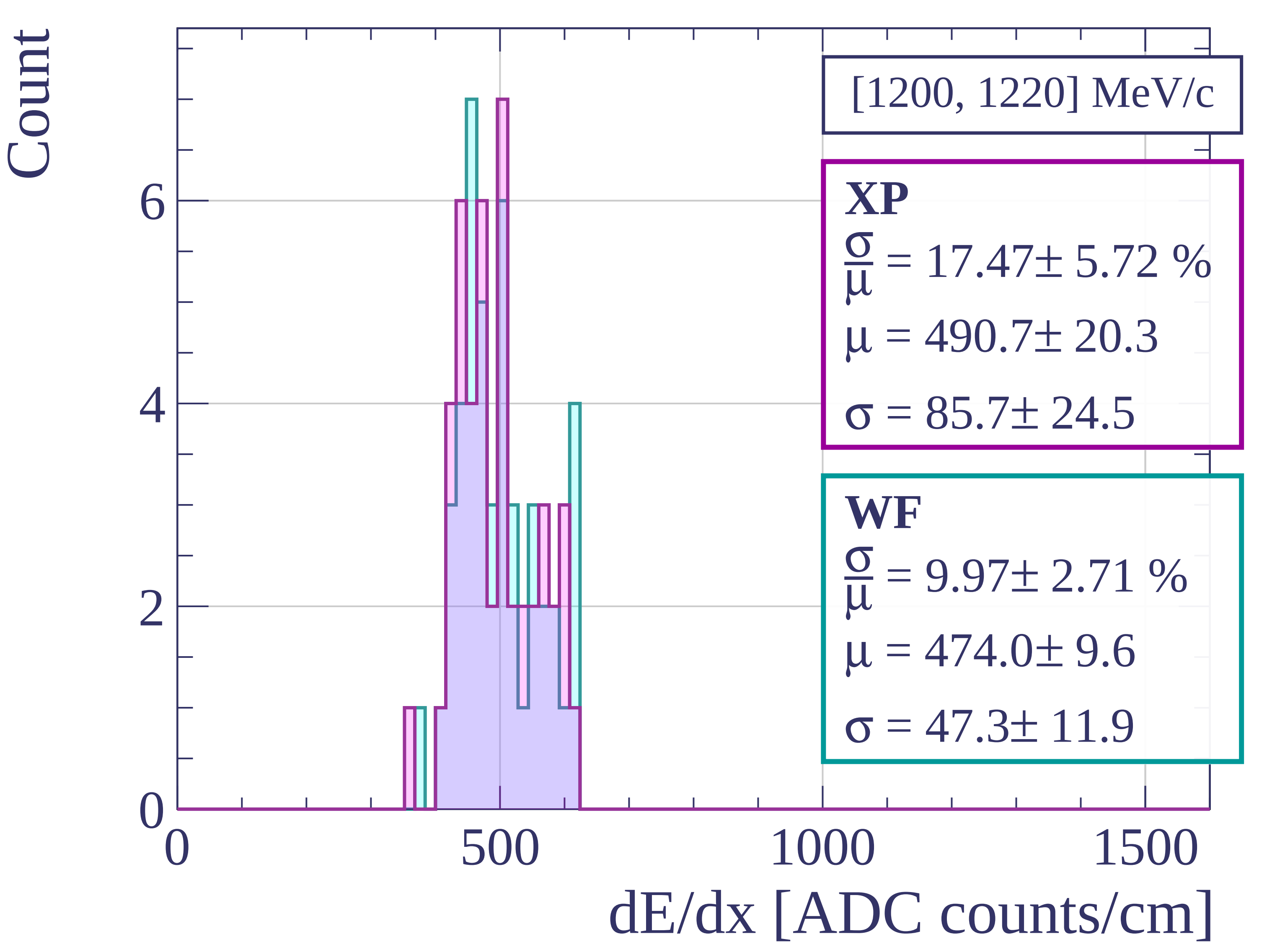
Count
dE/dx [ADC counts/cm]
[1140, 1160] MeV/c
XP
σ/μ = -37.38± 2.98 %
μ = -1246.7± 1242.9
σ = 466.0± 427.5
WF
σ/μ = 0.06± 0.58 %
μ = 407.7± 1.1
σ = 0.3± 2.4
0
2
4
6
0
500
1000
1500



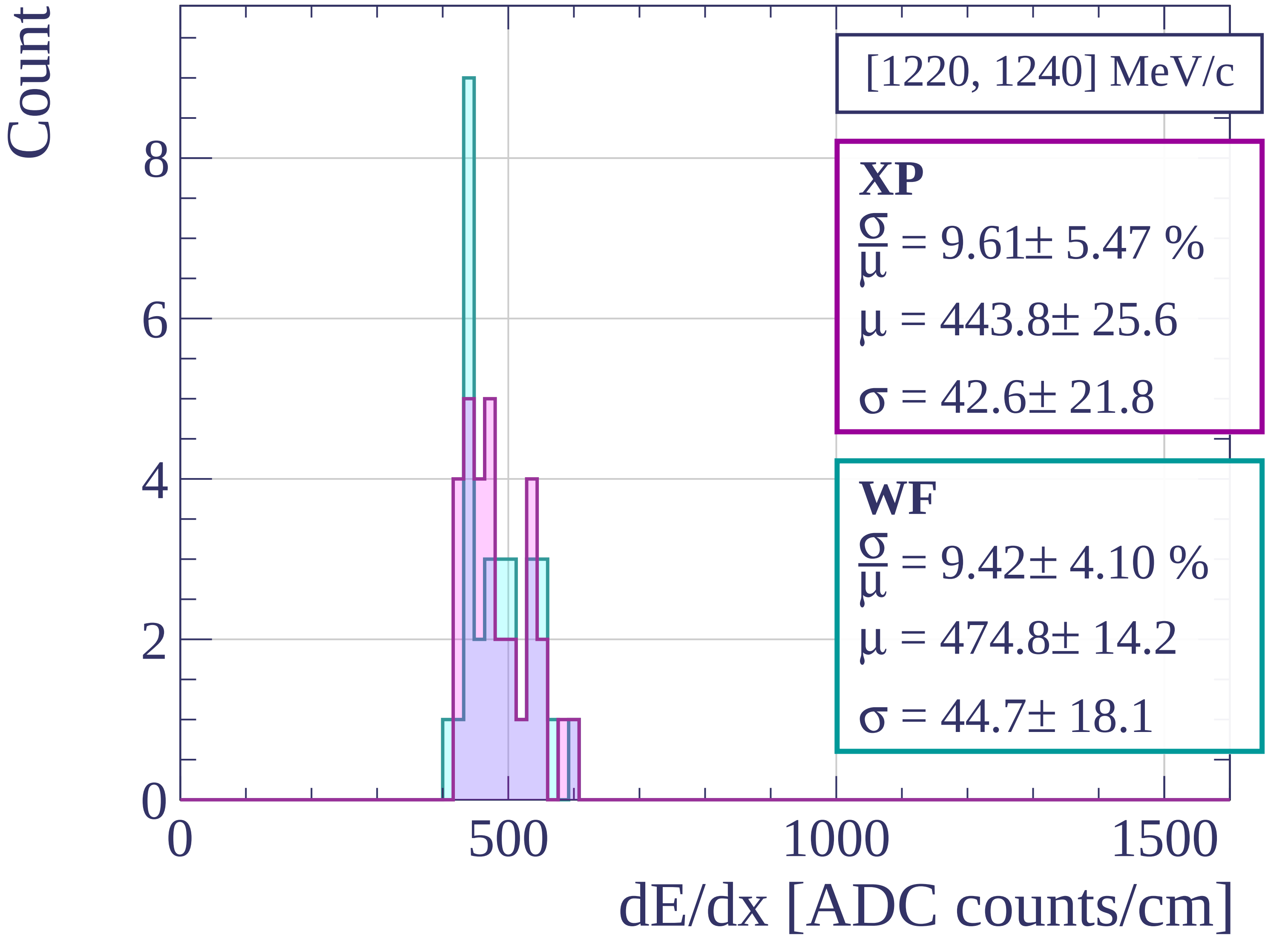
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[1160, 1180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 38.56± 36.71 %
μ = 462.1± 109.4
σ = 178.2± 127.4
WF
$\frac{\sigma}{\mu}$ = 32.67± 28.95 %
μ = 508.2± 72.5
σ = 166.0± 123.4



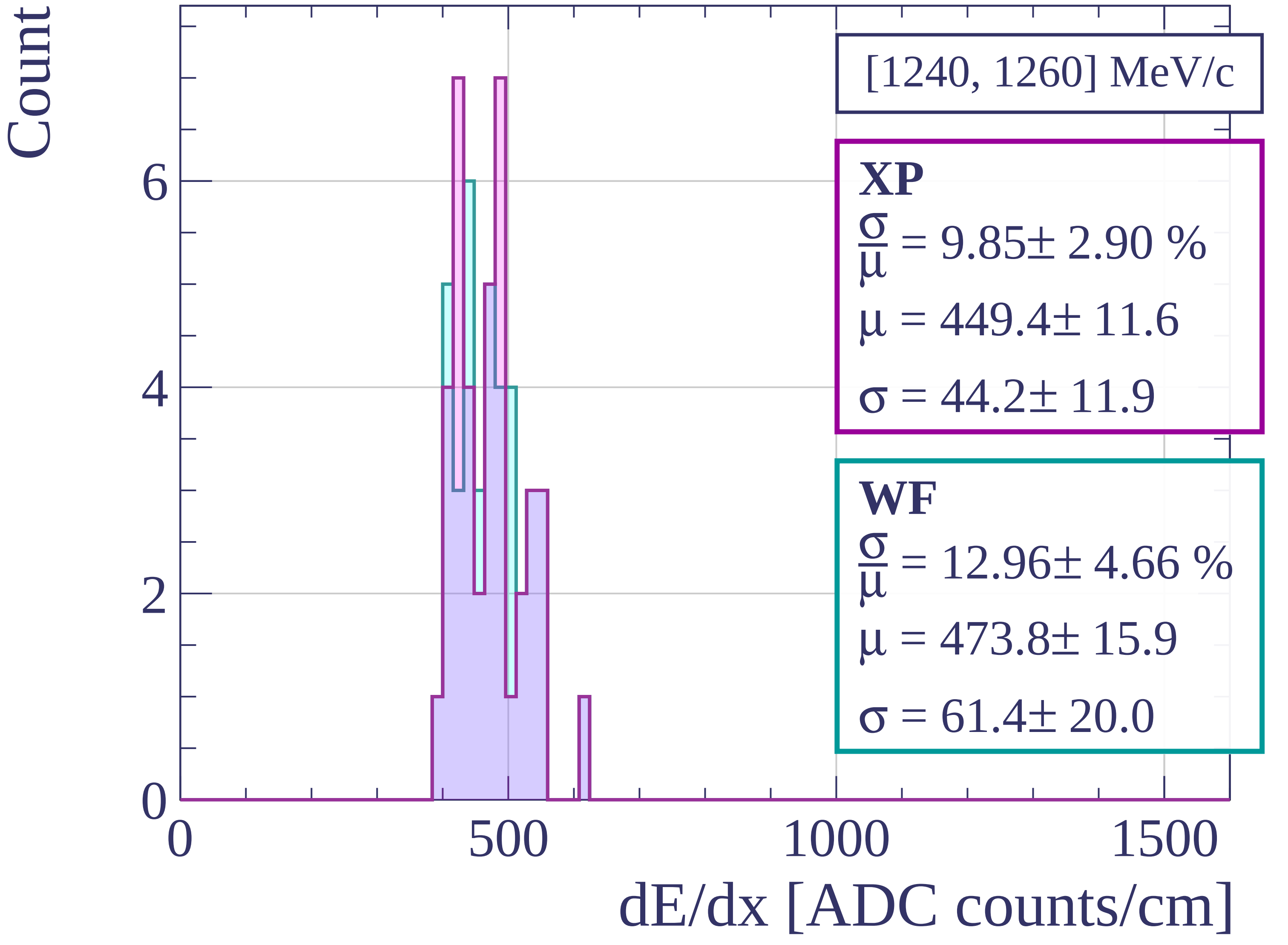
Count
dE/dx [ADC counts/cm]
[1180, 1200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 48.70± 161.04 %
μ = 410.1± 462.6
σ = 199.7± 435.0
WF
$\frac{\sigma}{\mu}$ = 30.15± 39.31 %
μ = 2283.9± 1246.6
σ = 688.6± 521.9
0
500
1000
1500
1
2
3
4



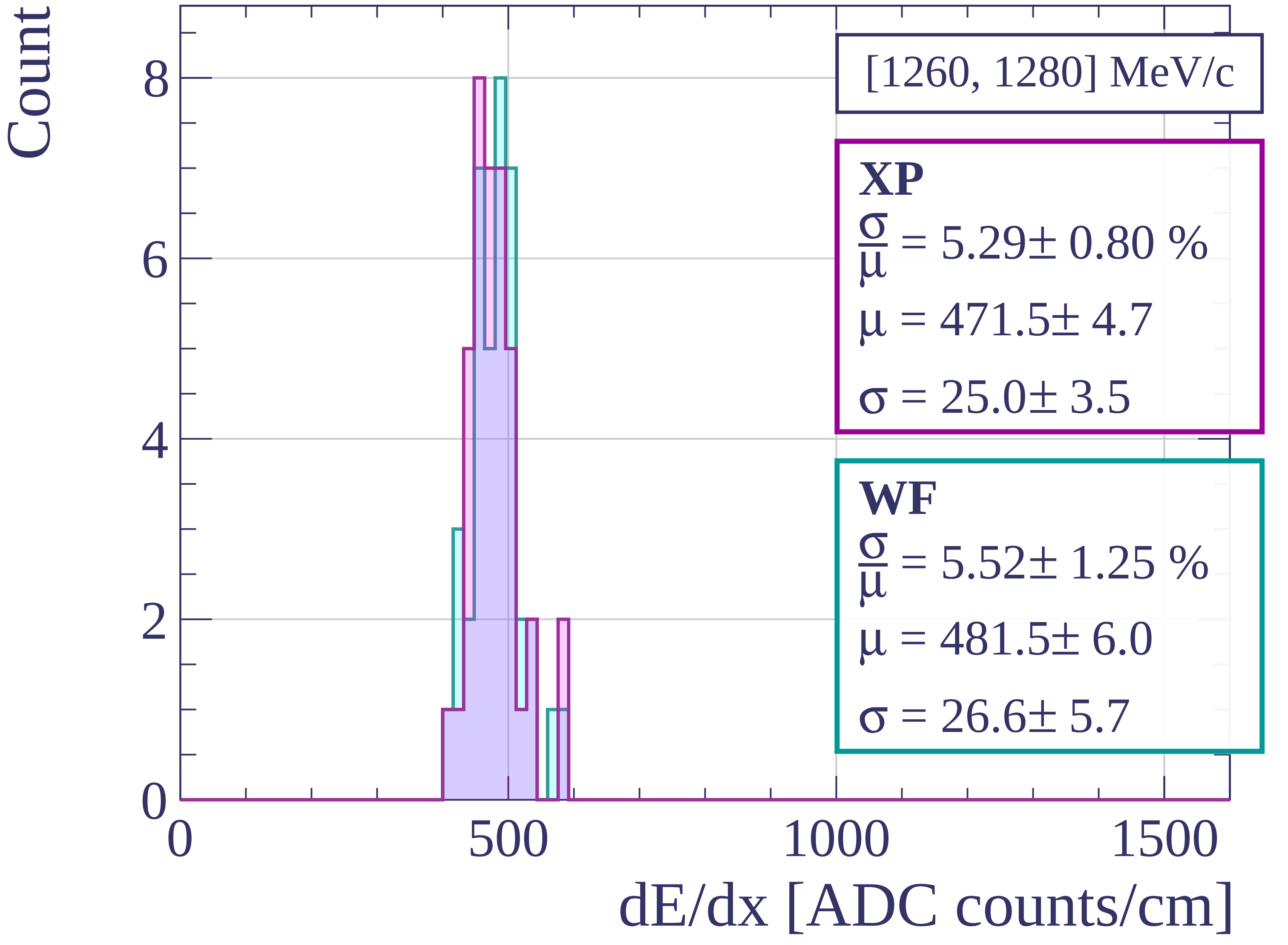
Count
[1200, 1220] MeV/c
XP
σ/μ = 17.47± 5.72 %
μ = 490.7± 20.3
σ = 85.7± 24.5
WF
σ/μ = 9.97± 2.71 %
μ = 474.0± 9.6
σ = 47.3± 11.9
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]



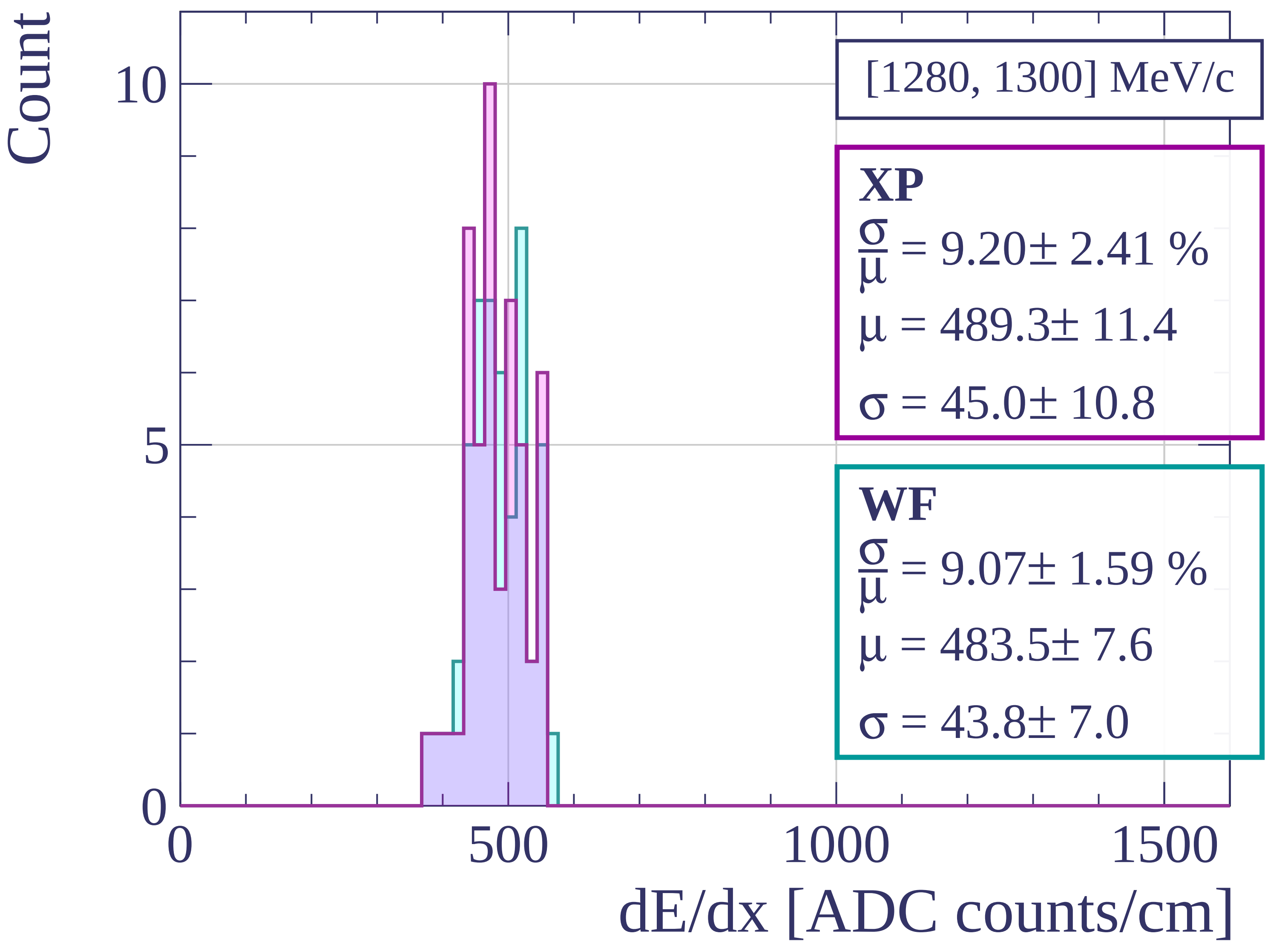
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[1220, 1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.61± 5.47 %
μ = 443.8± 25.6
σ = 42.6± 21.8
WF
$\frac{\sigma}{\mu}$ = 9.42± 4.10 %
μ = 474.8± 14.2
σ = 44.7± 18.1



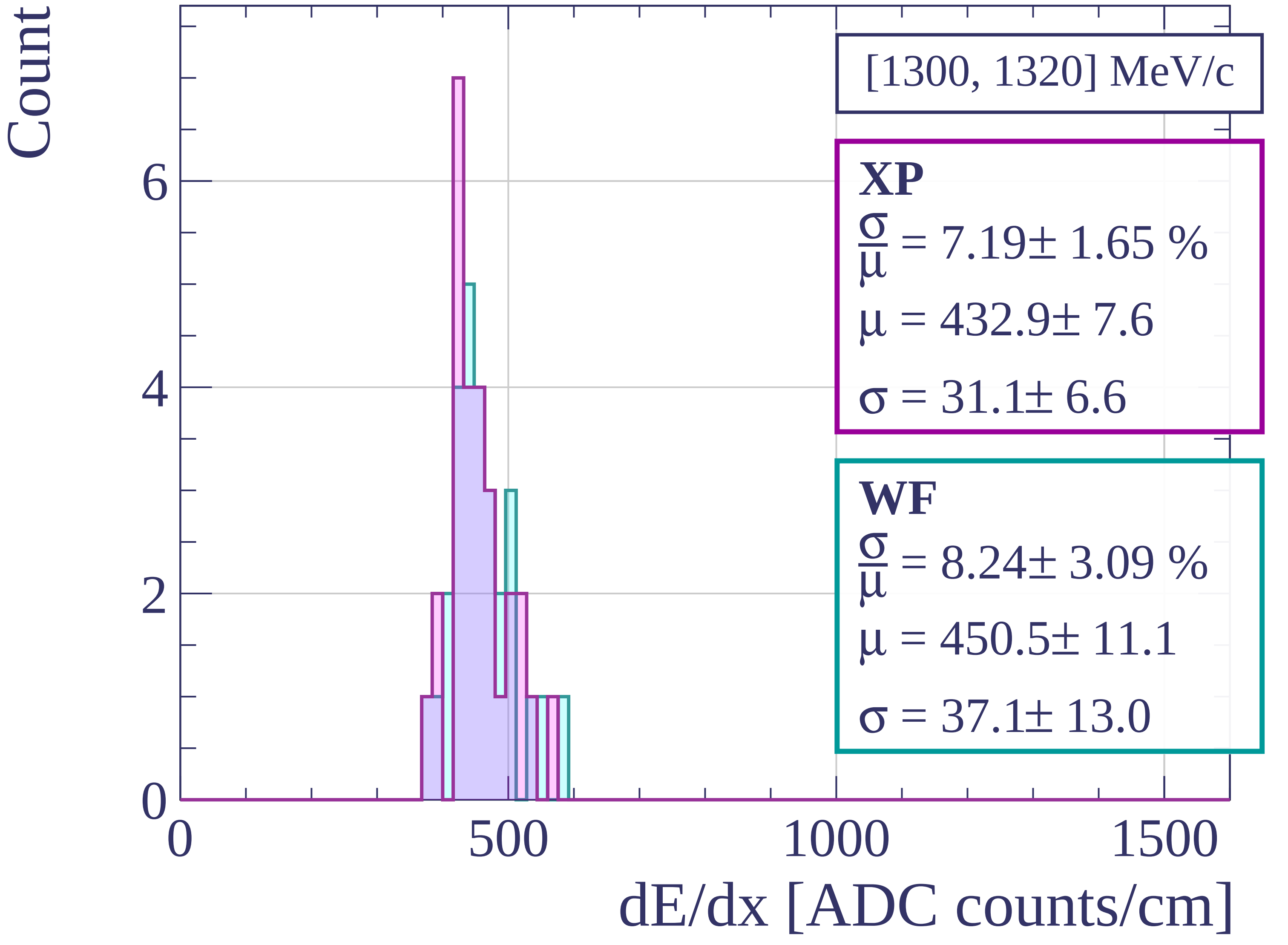
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1240, 1260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.85± 2.90 %
μ = 449.4± 11.6
σ = 44.2± 11.9
WF
$\frac{\sigma}{\mu}$ = 12.96± 4.66 %
μ = 473.8± 15.9
σ = 61.4± 20.0



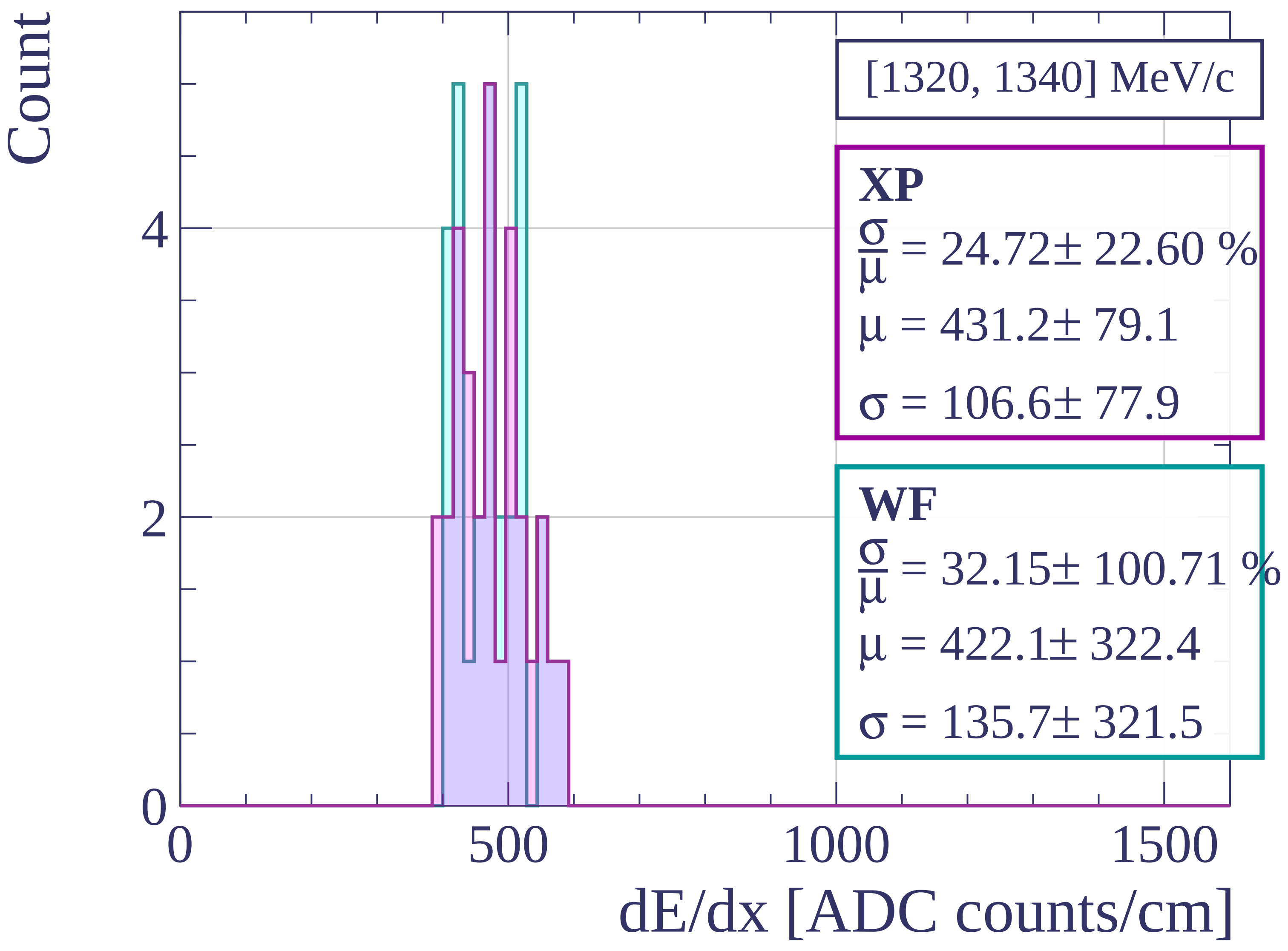
Count
dE/dx [ADC counts/cm]
[1260, 1280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.29± 0.80 %
μ = 471.5± 4.7
σ = 25.0± 3.5
WF
$\frac{\sigma}{\mu}$ = 5.52± 1.25 %
μ = 481.5± 6.0
σ = 26.6± 5.7

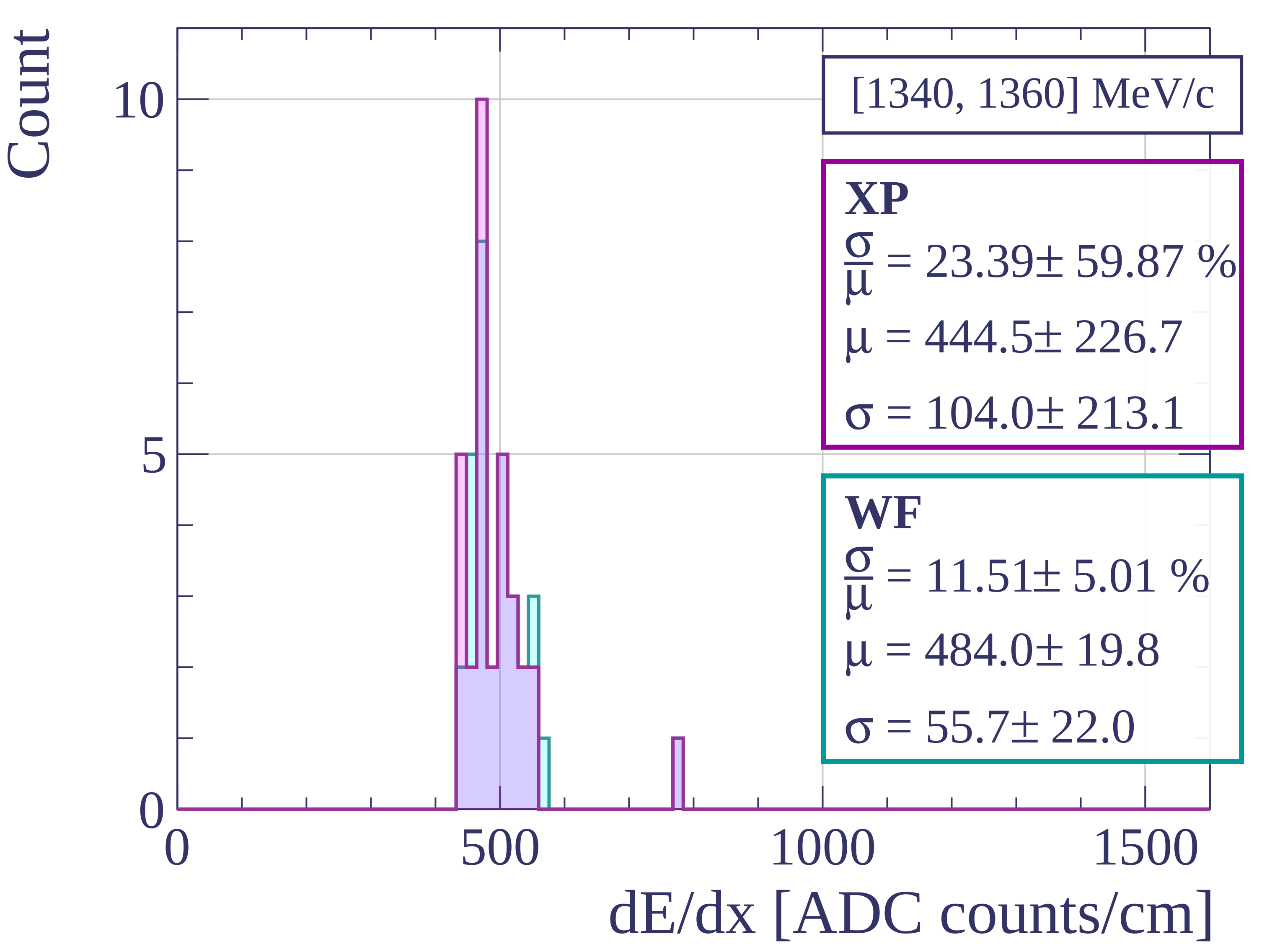


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1280, 1300] MeV/c
XP
σ/μ = 9.20± 2.41 %
μ = 489.3± 11.4
σ = 45.0± 10.8
WF
σ/μ = 9.07± 1.59 %
μ = 483.5± 7.6
σ = 43.8± 7.0

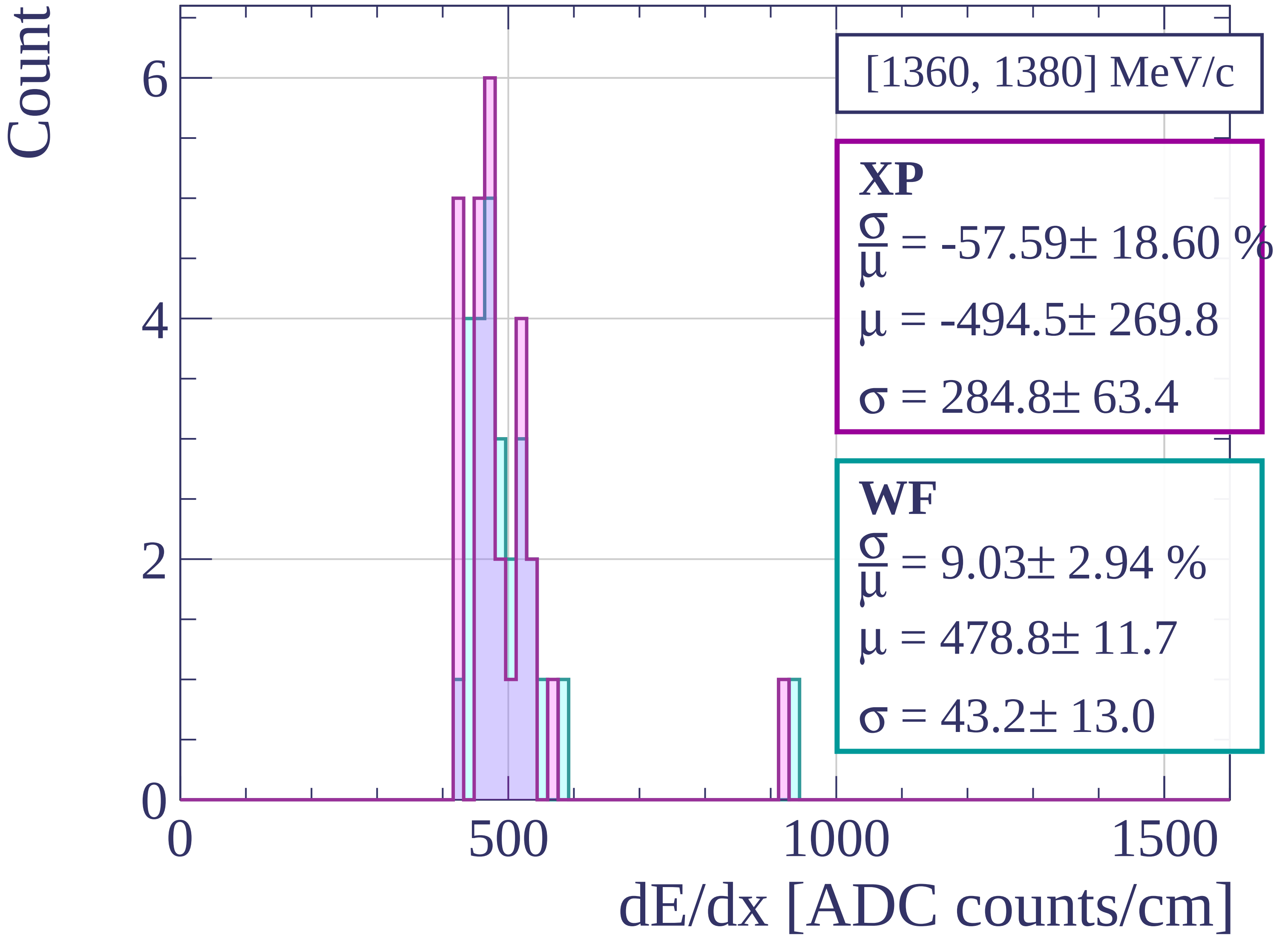


Count
[1300, 1320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.19± 1.65 %
μ = 432.9± 7.6
σ = 31.1± 6.6
WF
$\frac{\sigma}{\mu}$ = 8.24± 3.09 %
μ = 450.5± 11.1
σ = 37.1± 13.0
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]

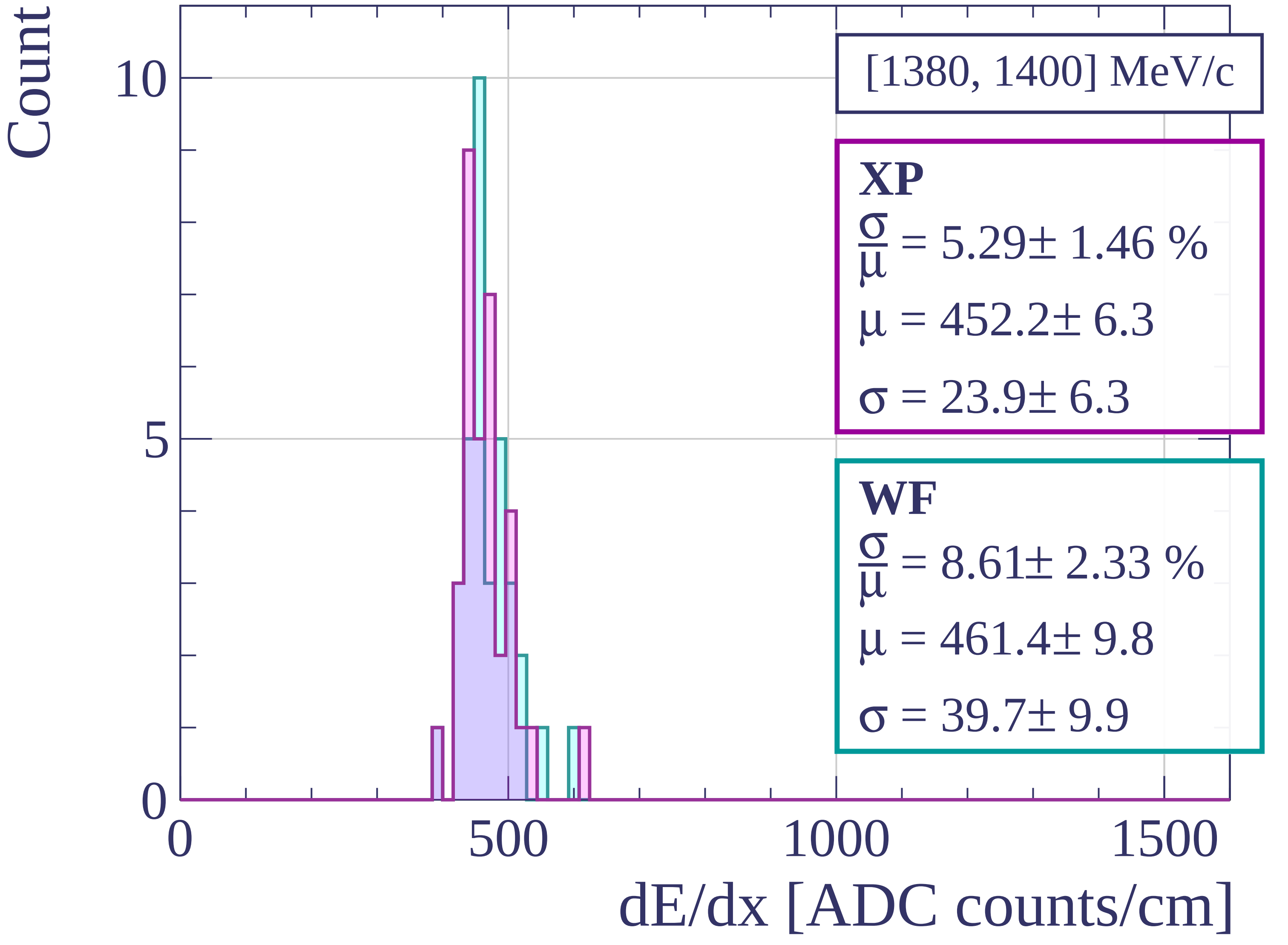

Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[1320, 1340] MeV/c
XP
σ/μ = 24.72± 22.60 %
μ = 431.2± 79.1
σ = 106.6± 77.9
WF
σ/μ = 32.15± 100.71 %
μ = 422.1± 322.4
σ = 135.7± 321.5



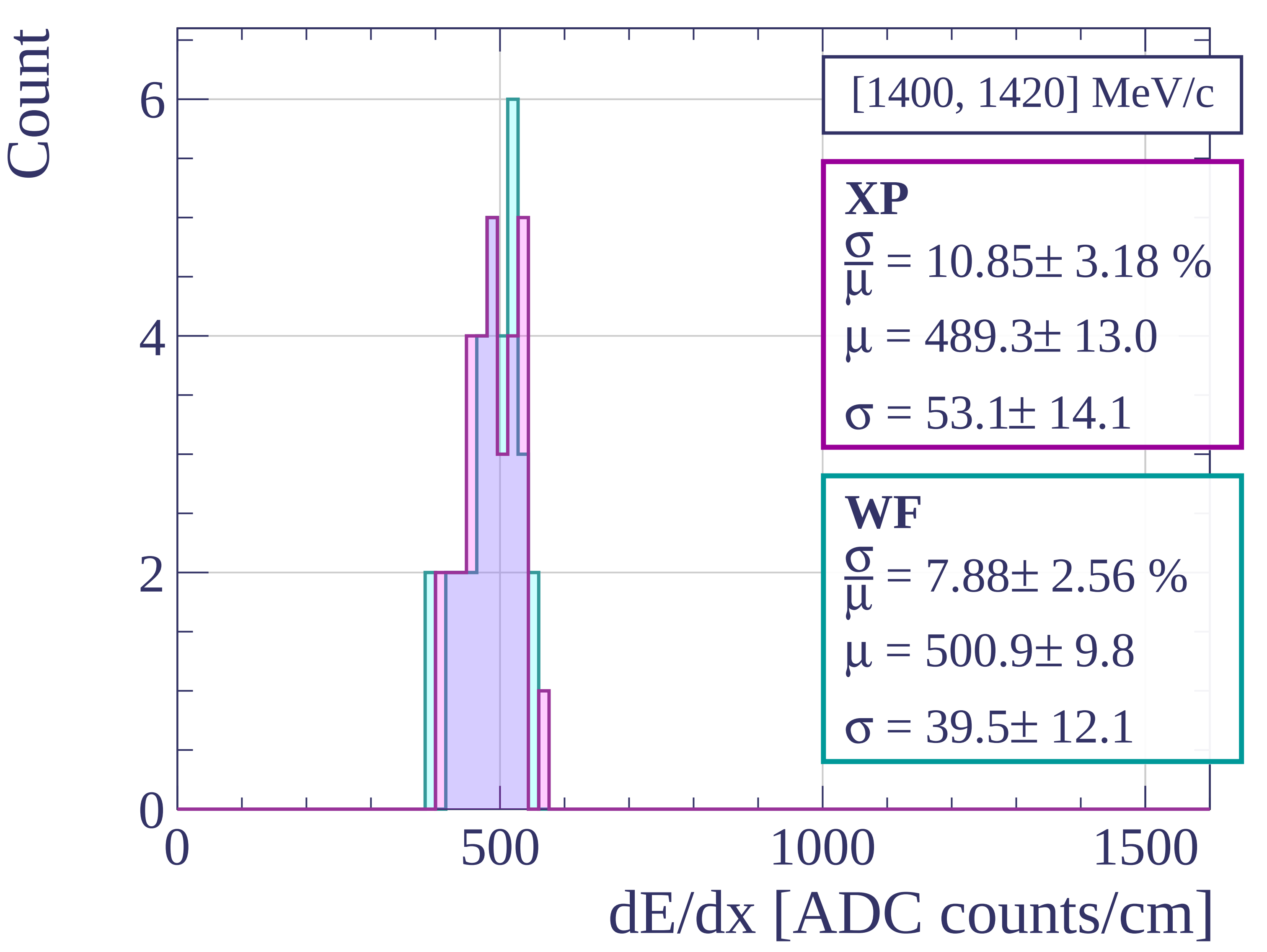
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1340, 1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 23.39± 59.87 %
μ = 444.5± 226.7
σ = 104.0± 213.1
WF
$\frac{\sigma}{\mu}$ = 11.51± 5.01 %
μ = 484.0± 19.8
σ = 55.7± 22.0

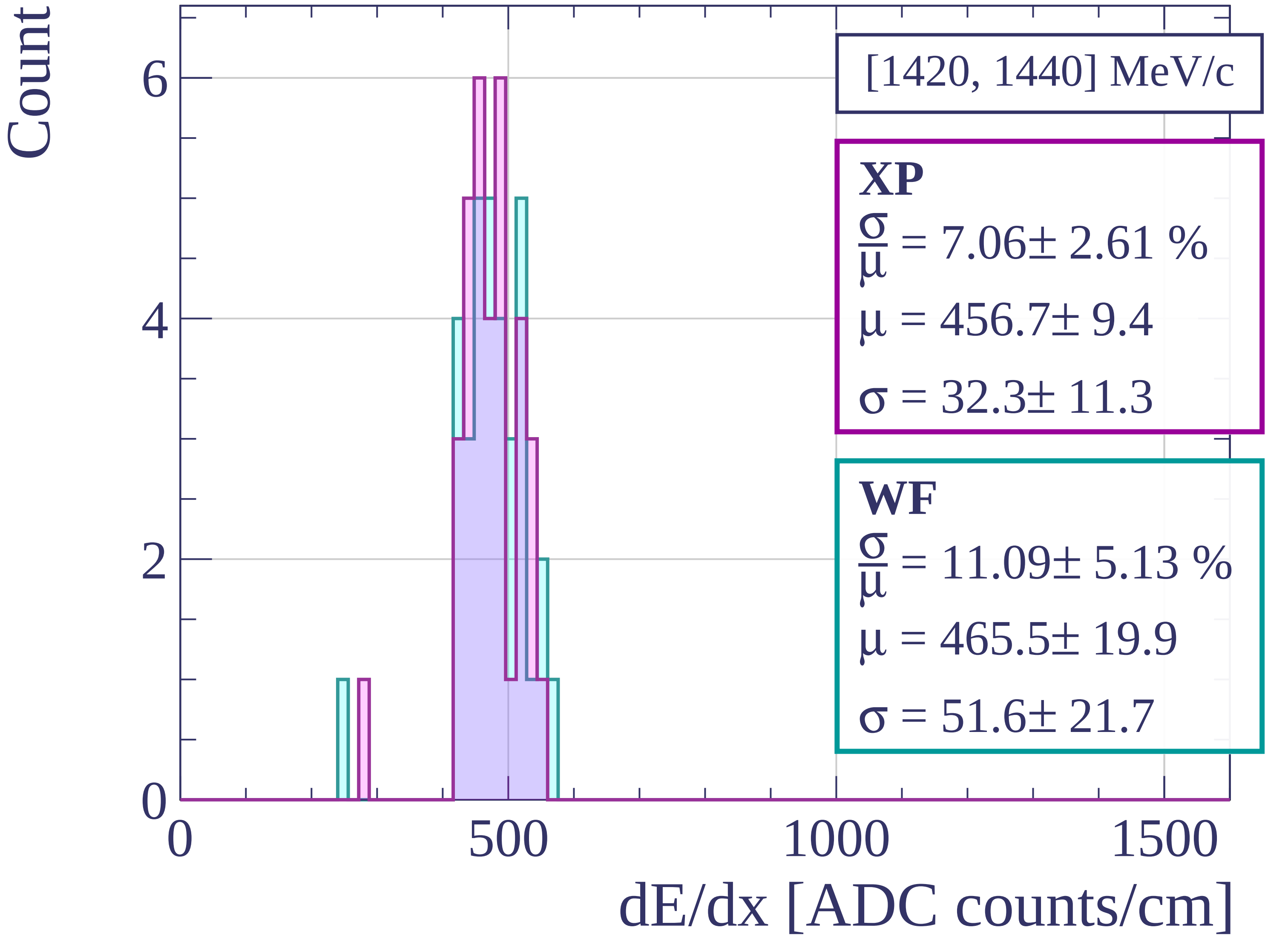


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[1360, 1380] MeV/c
XP
σ/μ = -57.59± 18.60 %
μ = -494.5± 269.8
σ = 284.8± 63.4
WF
σ/μ = 9.03± 2.94 %
μ = 478.8± 11.7
σ = 43.2± 13.0

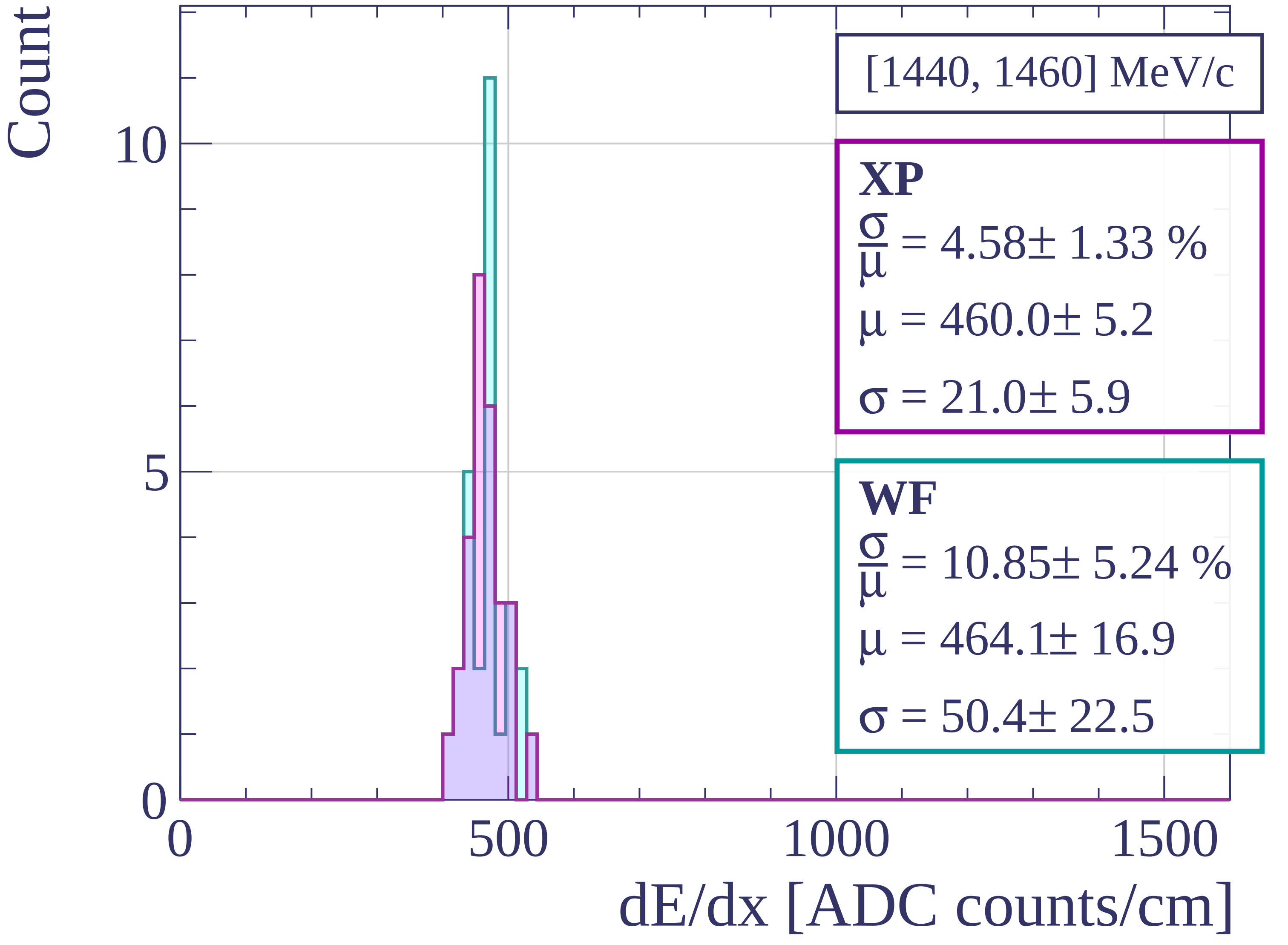


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1380, 1400] MeV/c
XP
σ/μ = 5.29± 1.46 %
μ = 452.2± 6.3
σ = 23.9± 6.3
WF
σ/μ = 8.61± 2.33 %
μ = 461.4± 9.8
σ = 39.7± 9.9

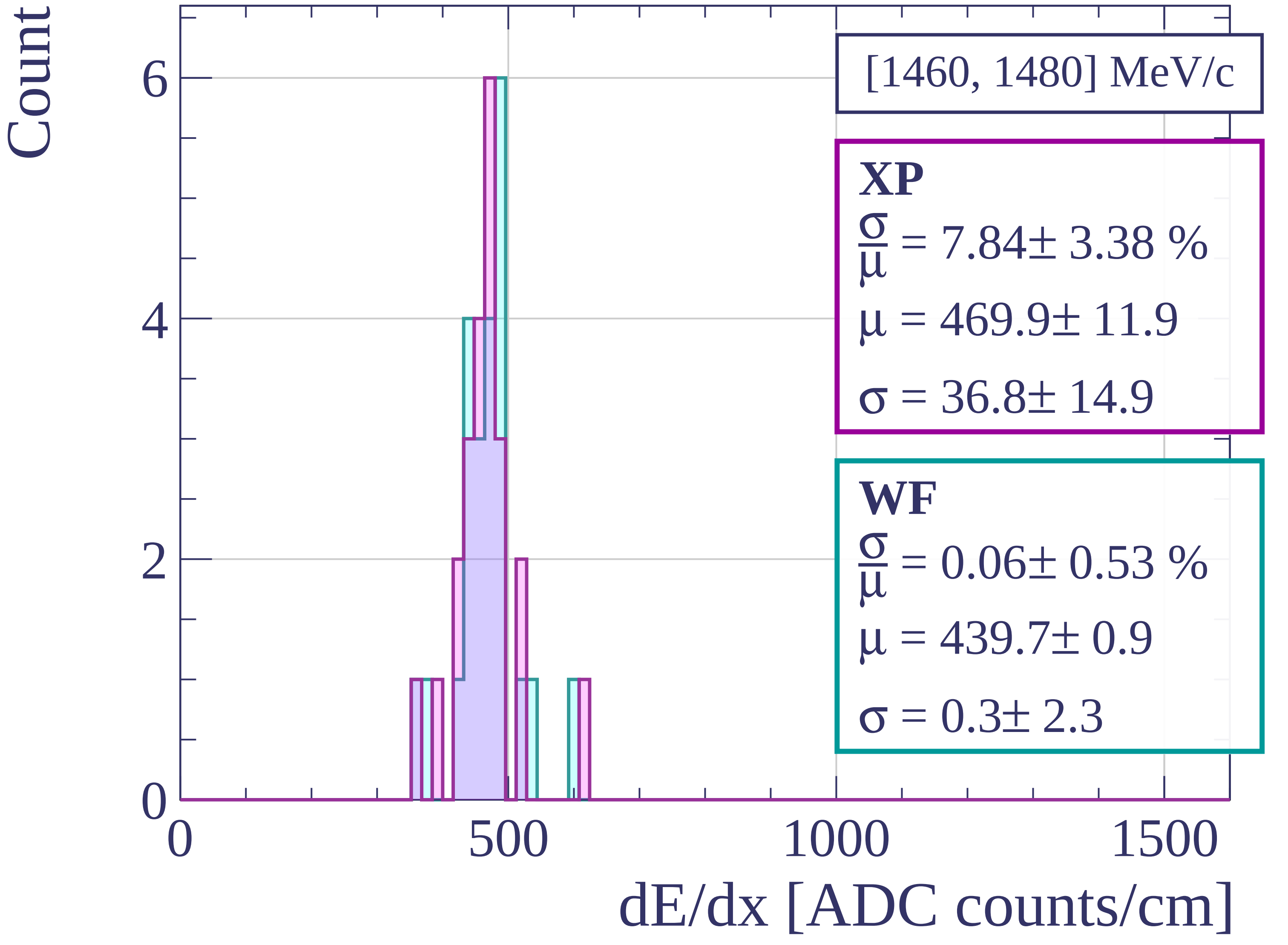

Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1400, 1420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.85± 3.18 %
μ = 489.3± 13.0
σ = 53.1± 14.1
WF
$\frac{\sigma}{\mu}$ = 7.88± 2.56 %
μ = 500.9± 9.8
σ = 39.5± 12.1



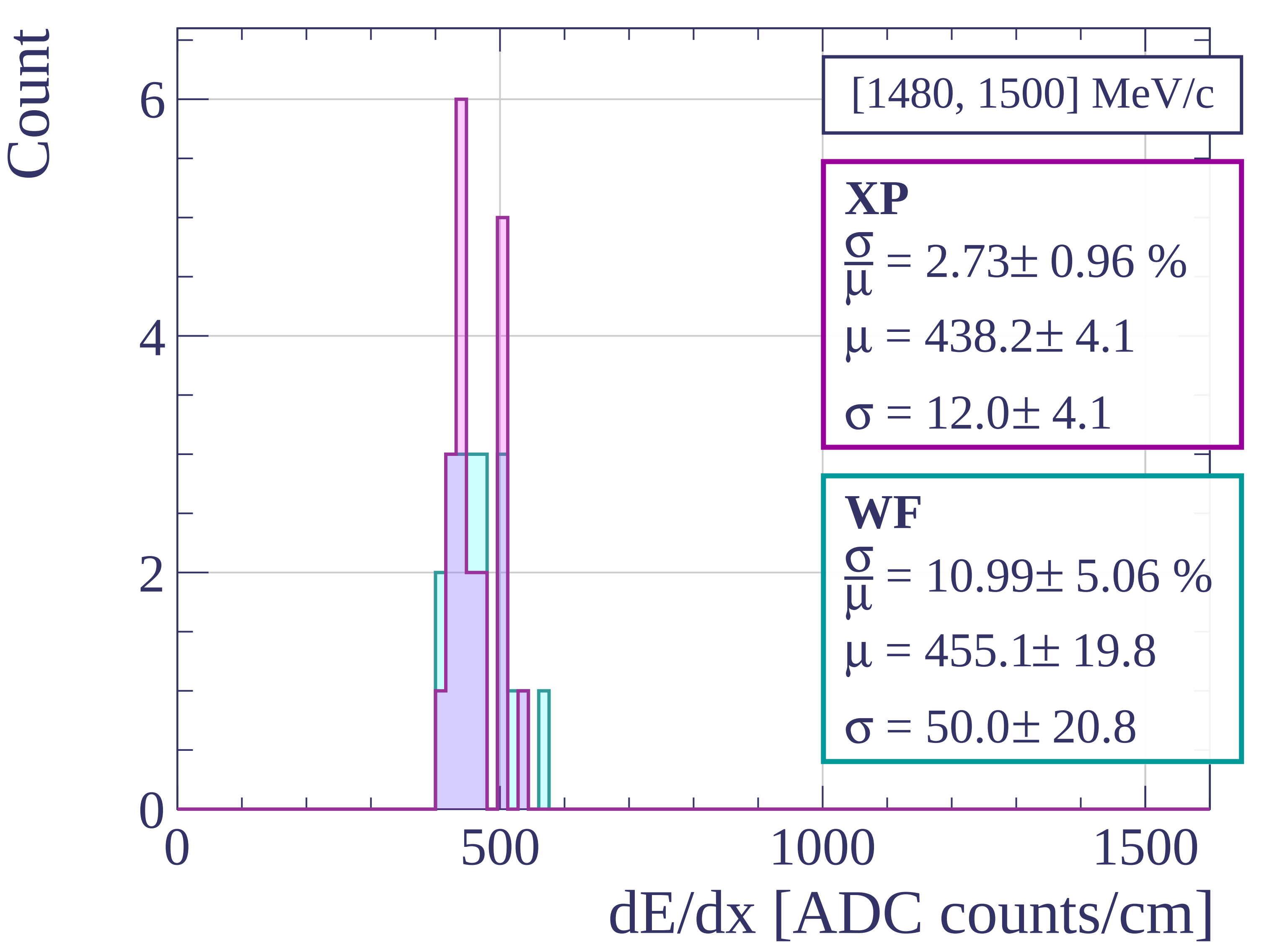
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[1420, 1440] MeV/c
XP
σ/μ = 7.06± 2.61 %
μ = 456.7± 9.4
σ = 32.3± 11.3
WF
σ/μ = 11.09± 5.13 %
μ = 465.5± 19.9
σ = 51.6± 21.7



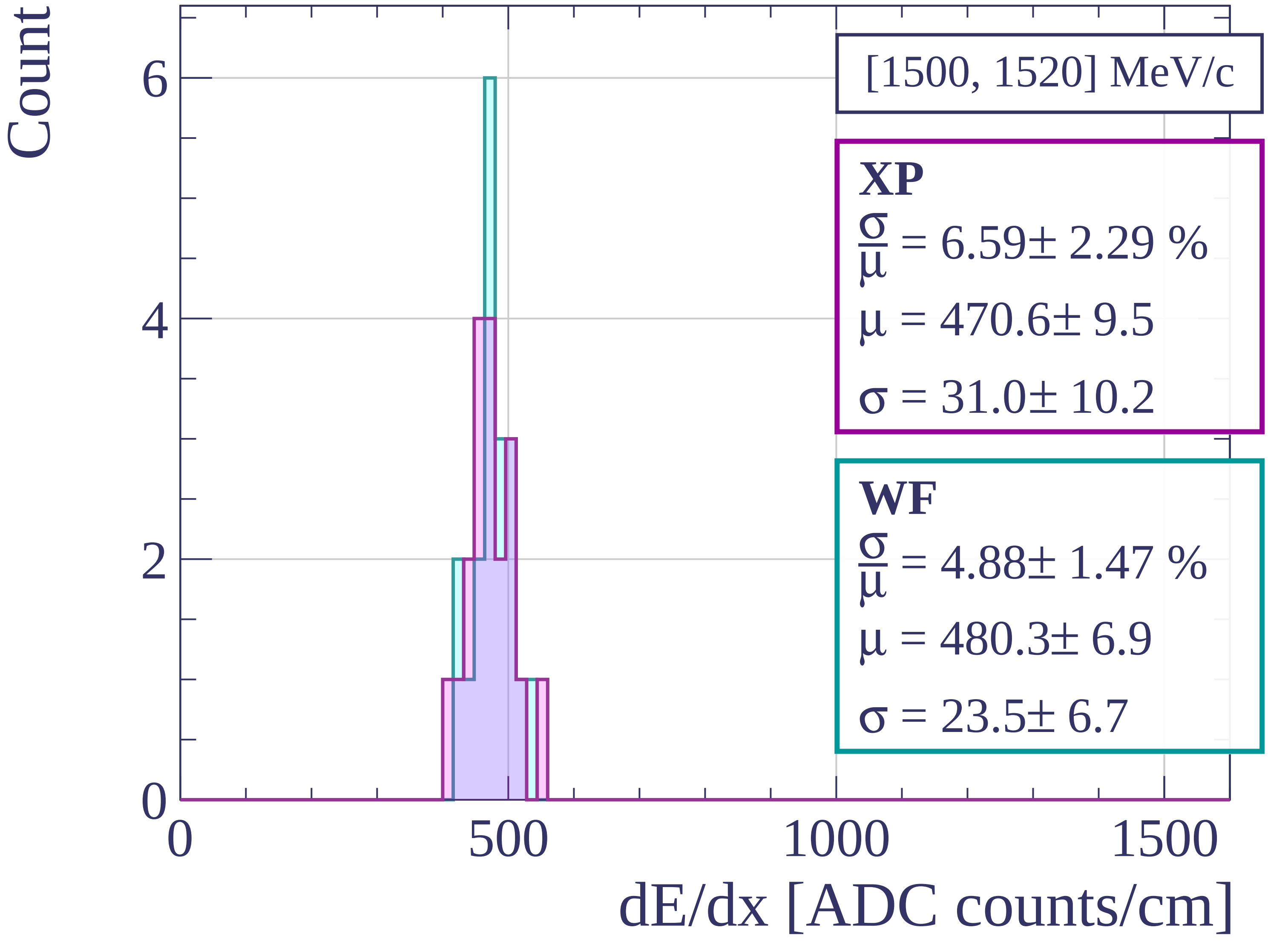
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1440, 1460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.58± 1.33 %
μ = 460.0± 5.2
σ = 21.0± 5.9
WF
$\frac{\sigma}{\mu}$ = 10.85± 5.24 %
μ = 464.1± 16.9
σ = 50.4± 22.5



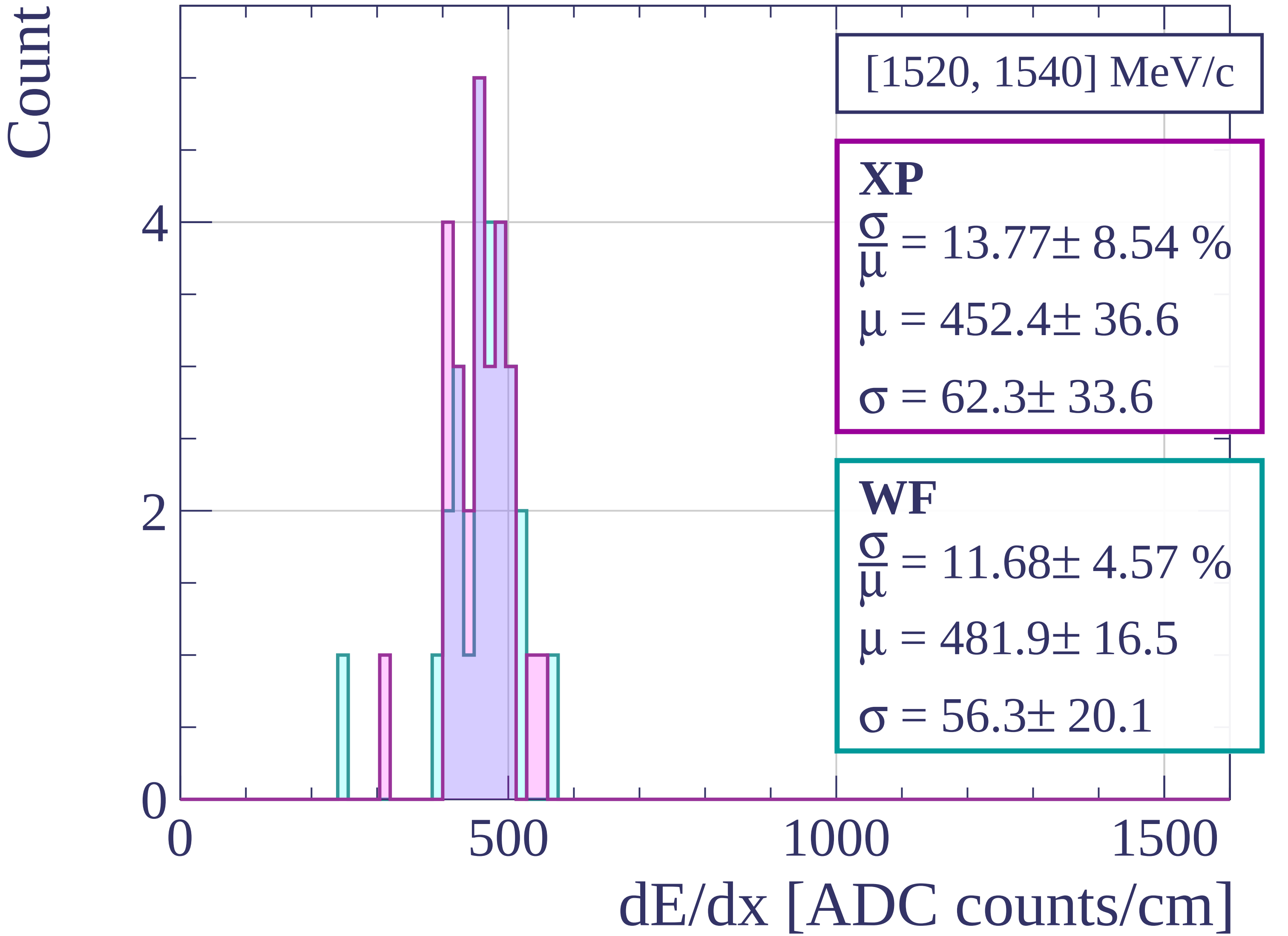
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1460, 1480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.84± 3.38 %
μ = 469.9± 11.9
σ = 36.8± 14.9
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.53 %
μ = 439.7± 0.9
σ = 0.3± 2.3



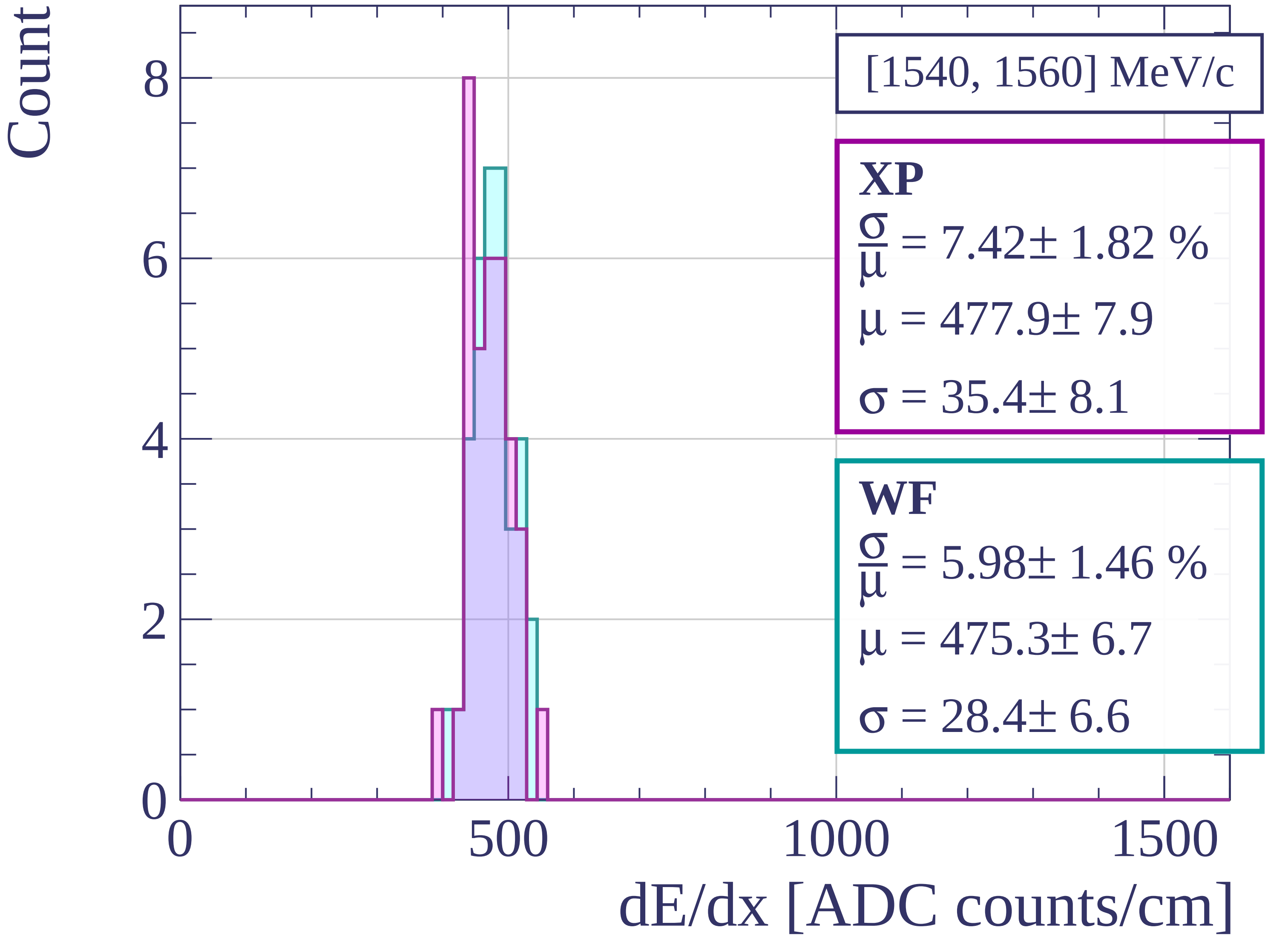
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1480, 1500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 2.73± 0.96 %
μ = 438.2± 4.1
σ = 12.0± 4.1
WF
$\frac{\sigma}{\mu}$ = 10.99± 5.06 %
μ = 455.1± 19.8
σ = 50.0± 20.8

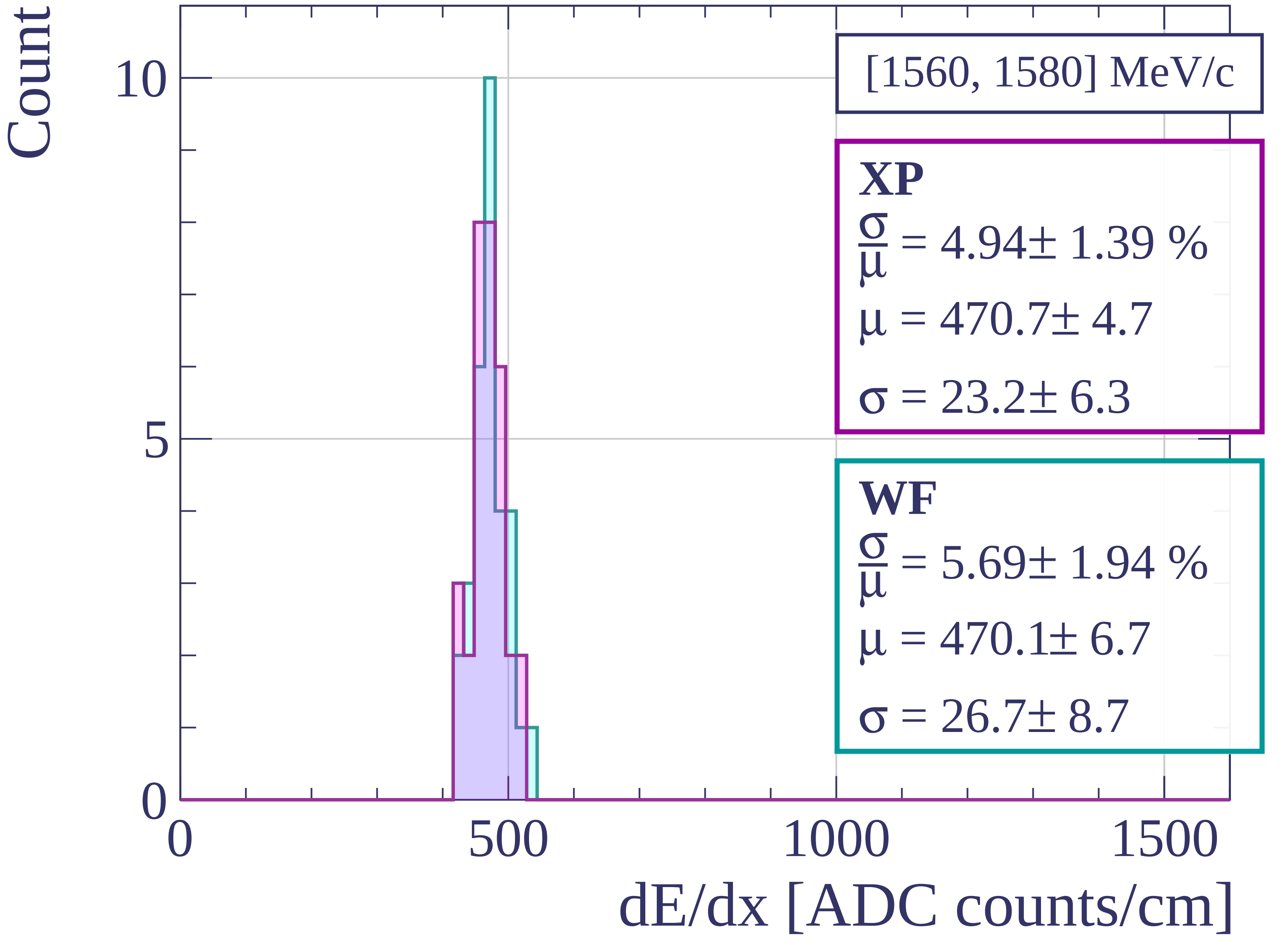


Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[1500, 1520] MeV/c
XP
σ/μ = 6.59± 2.29 %
μ = 470.6± 9.5
σ = 31.0± 10.2
WF
σ/μ = 4.88± 1.47 %
μ = 480.3± 6.9
σ = 23.5± 6.7

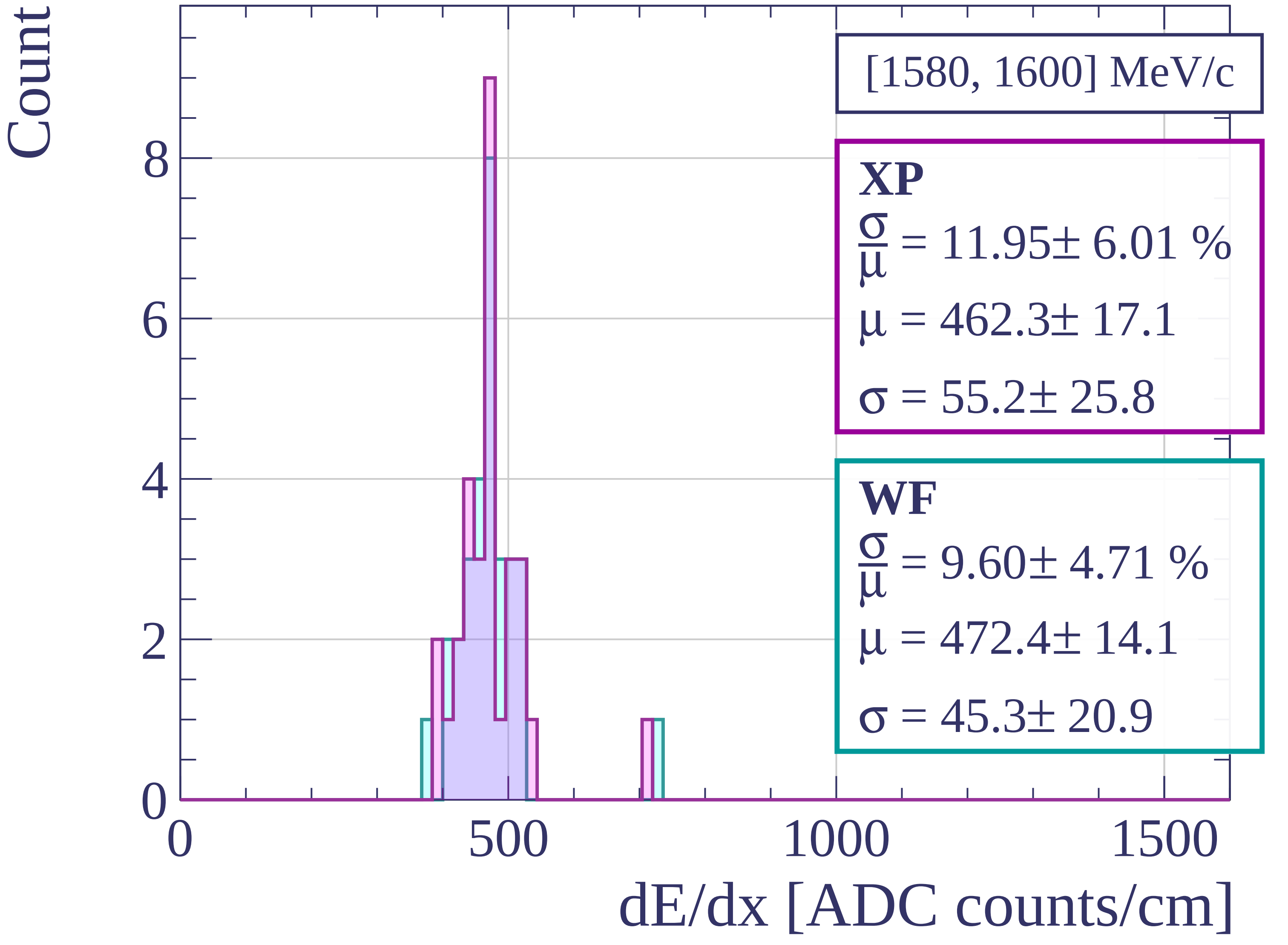


Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1520, 1540] MeV/c
XP
σ/μ = 13.77± 8.54 %
μ = 452.4± 36.6
σ = 62.3± 33.6
WF
σ/μ = 11.68± 4.57 %
μ = 481.9± 16.5
σ = 56.3± 20.1

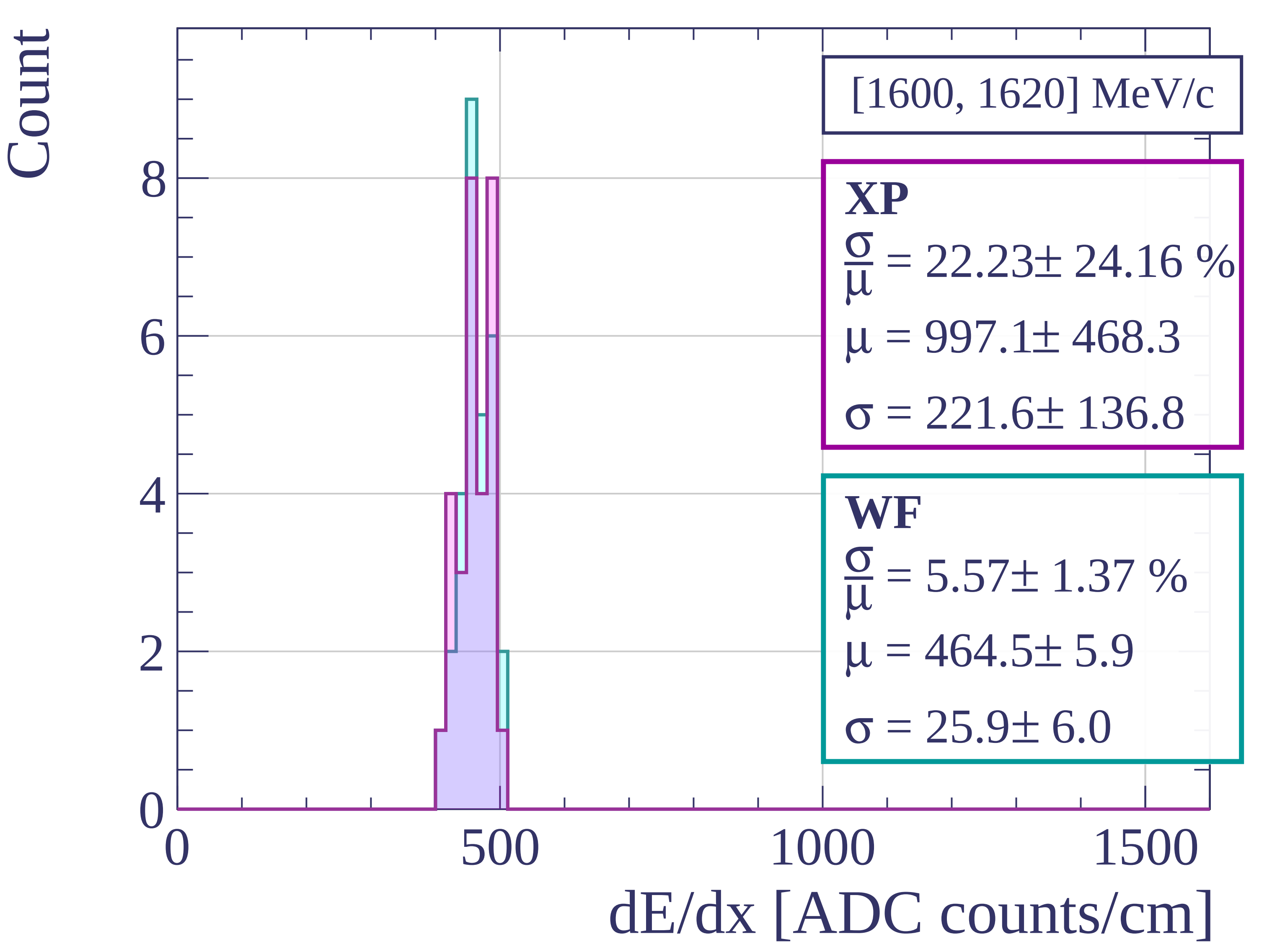

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1540, 1560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.42± 1.82 %
μ = 477.9± 7.9
σ = 35.4± 8.1
WF
$\frac{\sigma}{\mu}$ = 5.98± 1.46 %
μ = 475.3± 6.7
σ = 28.4± 6.6



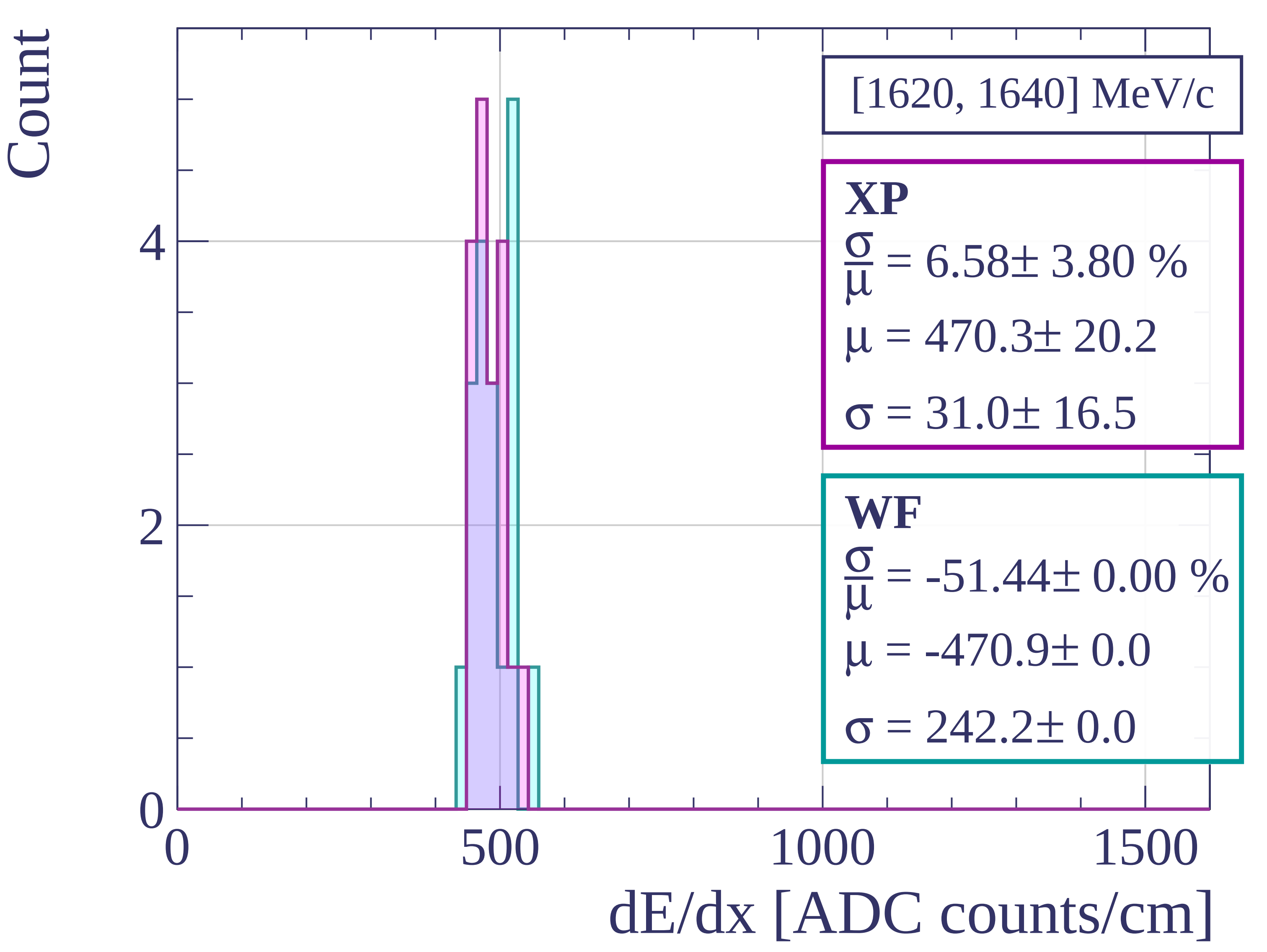
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1560, 1580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.94± 1.39 %
μ = 470.7± 4.7
σ = 23.2± 6.3
WF
$\frac{\sigma}{\mu}$ = 5.69± 1.94 %
μ = 470.1± 6.7
σ = 26.7± 8.7

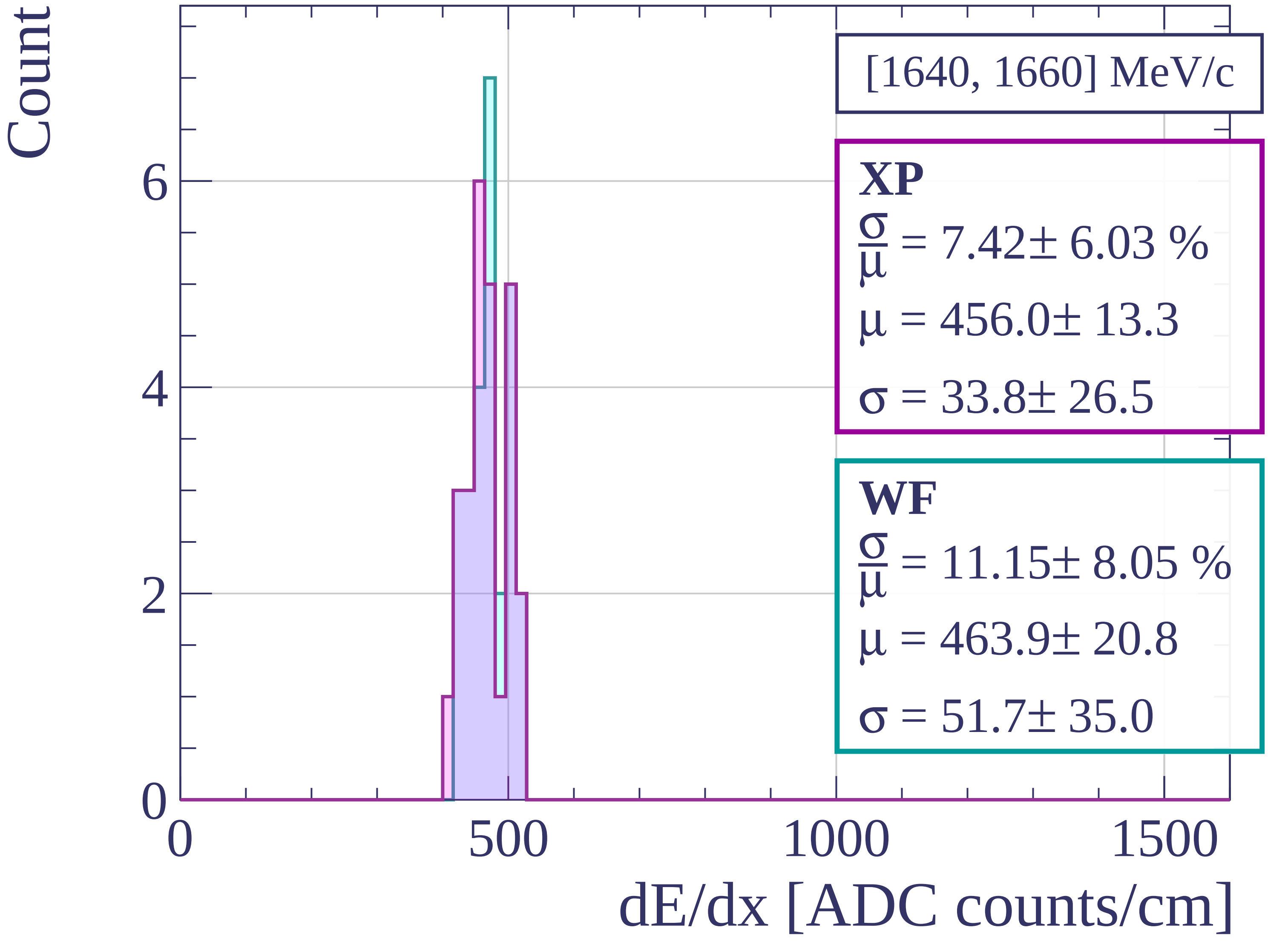


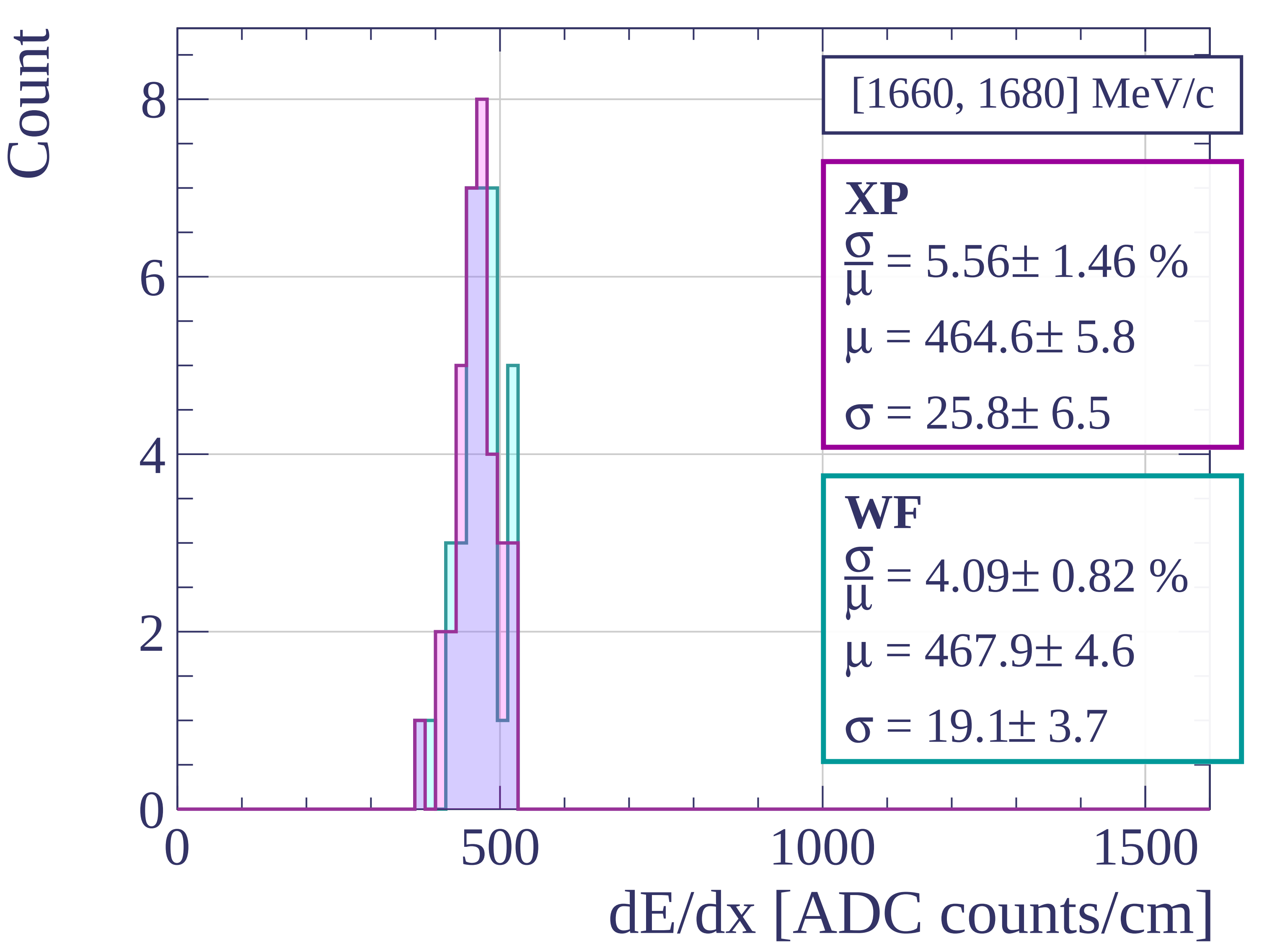
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[1580, 1600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.95± 6.01 %
μ = 462.3± 17.1
σ = 55.2± 25.8
WF
$\frac{\sigma}{\mu}$ = 9.60± 4.71 %
μ = 472.4± 14.1
σ = 45.3± 20.9



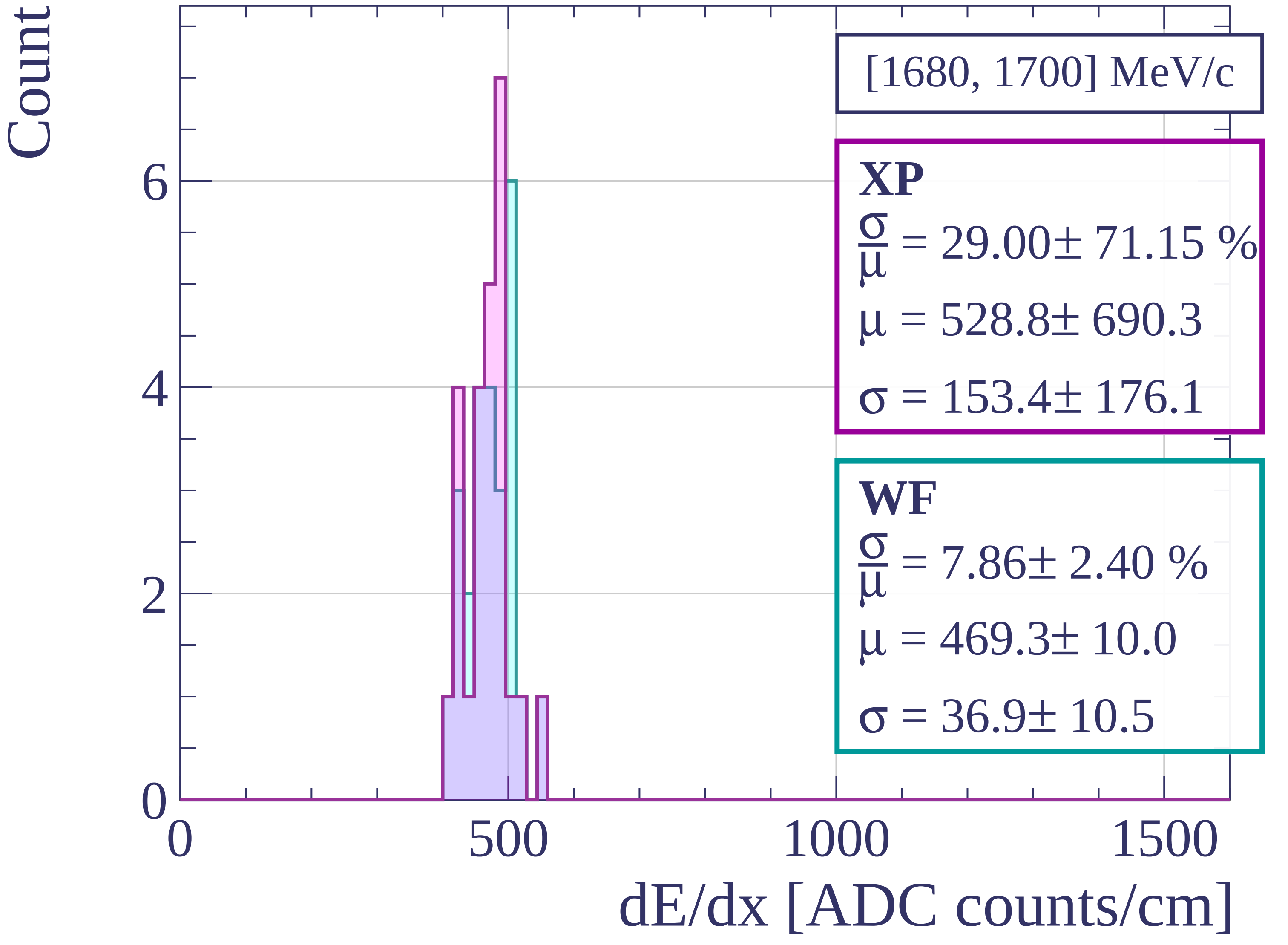
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1620] MeV/c
XP
σ/μ = 22.23± 24.16 %
μ = 997.1± 468.3
σ = 221.6± 136.8
WF
σ/μ = 5.57± 1.37 %
μ = 464.5± 5.9
σ = 25.9± 6.0


[1620, 1640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.58± 3.80 %
μ = 470.3± 20.2
σ = 31.0± 16.5
WF
$\frac{\sigma}{\mu}$ = -51.44± 0.00 %
μ = -470.9± 0.0
σ = 242.2± 0.0
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500

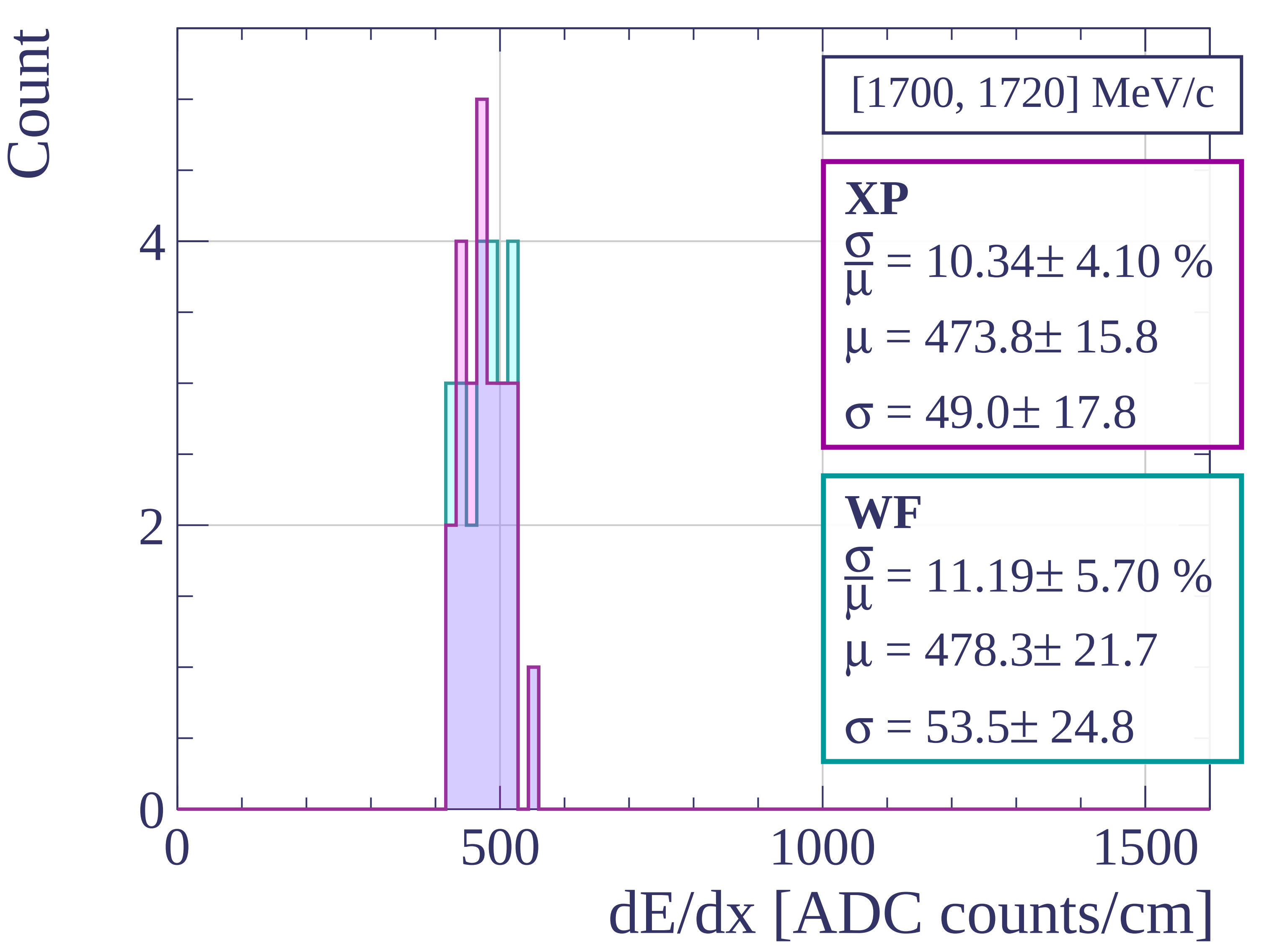

Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1640, 1660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.42± 6.03 %
μ = 456.0± 13.3
σ = 33.8± 26.5
WF
$\frac{\sigma}{\mu}$ = 11.15± 8.05 %
μ = 463.9± 20.8
σ = 51.7± 35.0



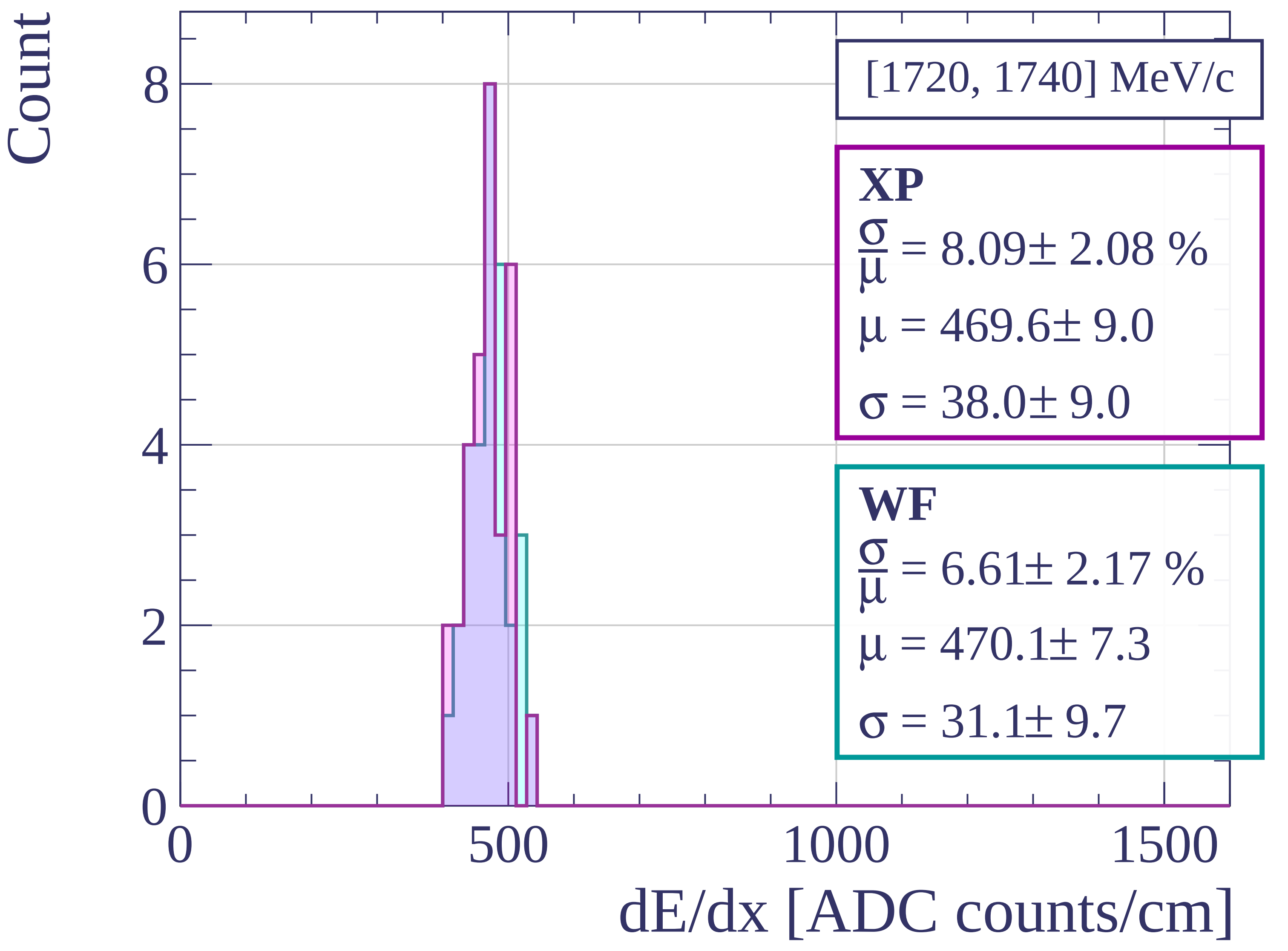
Count
dE/dx [ADC counts/cm]
[1660, 1680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.56± 1.46 %
μ = 464.6± 5.8
σ = 25.8± 6.5
WF
$\frac{\sigma}{\mu}$ = 4.09± 0.82 %
μ = 467.9± 4.6
σ = 19.1± 3.7
0
2
4
6
8
0
500
1000
1500



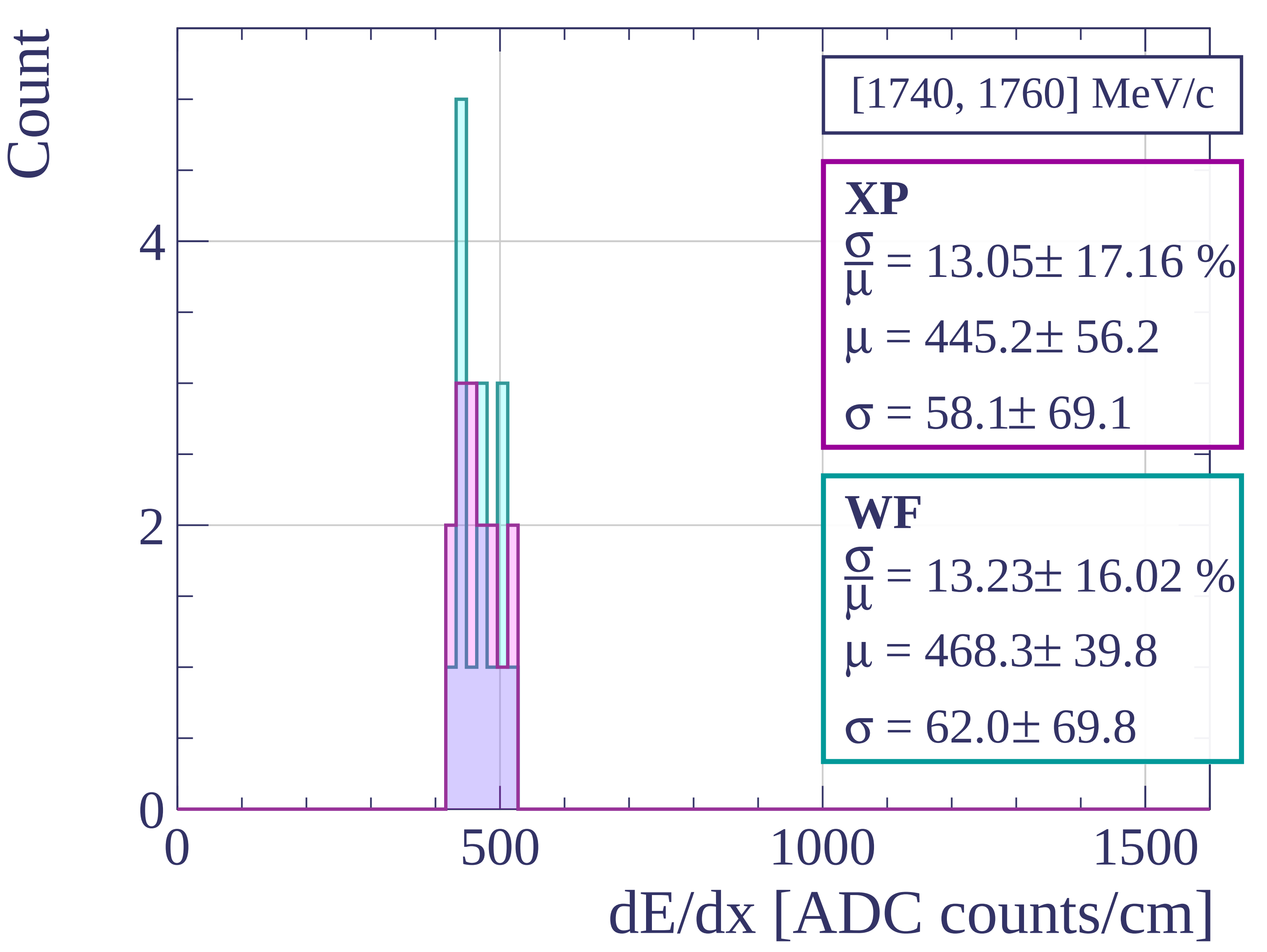
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[1680, 1700] MeV/c
XP
σ/μ = 29.00± 71.15 %
μ = 528.8± 690.3
σ = 153.4± 176.1
WF
σ/μ = 7.86± 2.40 %
μ = 469.3± 10.0
σ = 36.9± 10.5



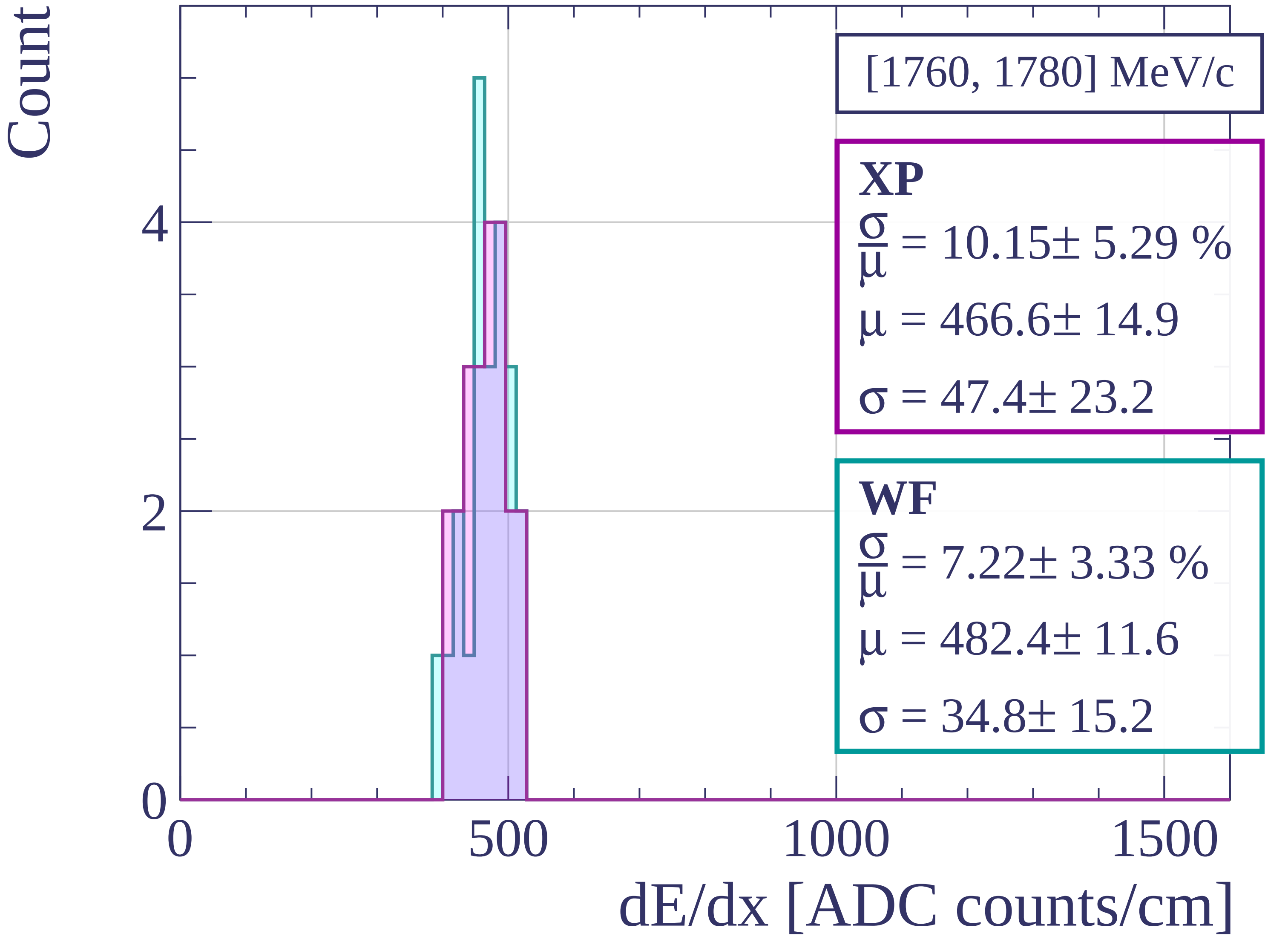
Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[1700, 1720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.34± 4.10 %
μ = 473.8± 15.8
σ = 49.0± 17.8
WF
$\frac{\sigma}{\mu}$ = 11.19± 5.70 %
μ = 478.3± 21.7
σ = 53.5± 24.8



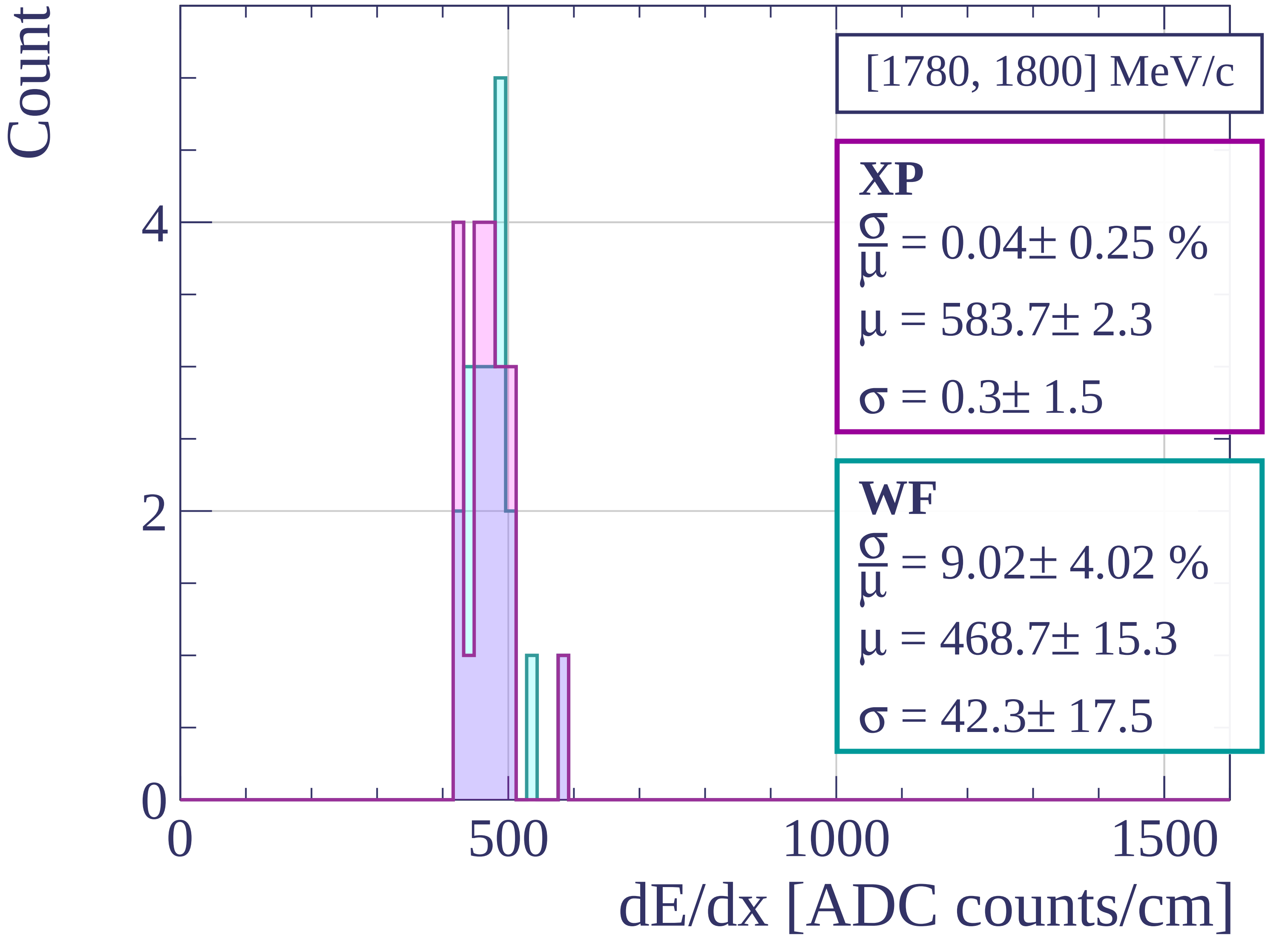
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[1720, 1740] MeV/c
XP
σ/μ = 8.09± 2.08 %
μ = 469.6± 9.0
σ = 38.0± 9.0
WF
σ/μ = 6.61± 2.17 %
μ = 470.1± 7.3
σ = 31.1± 9.7



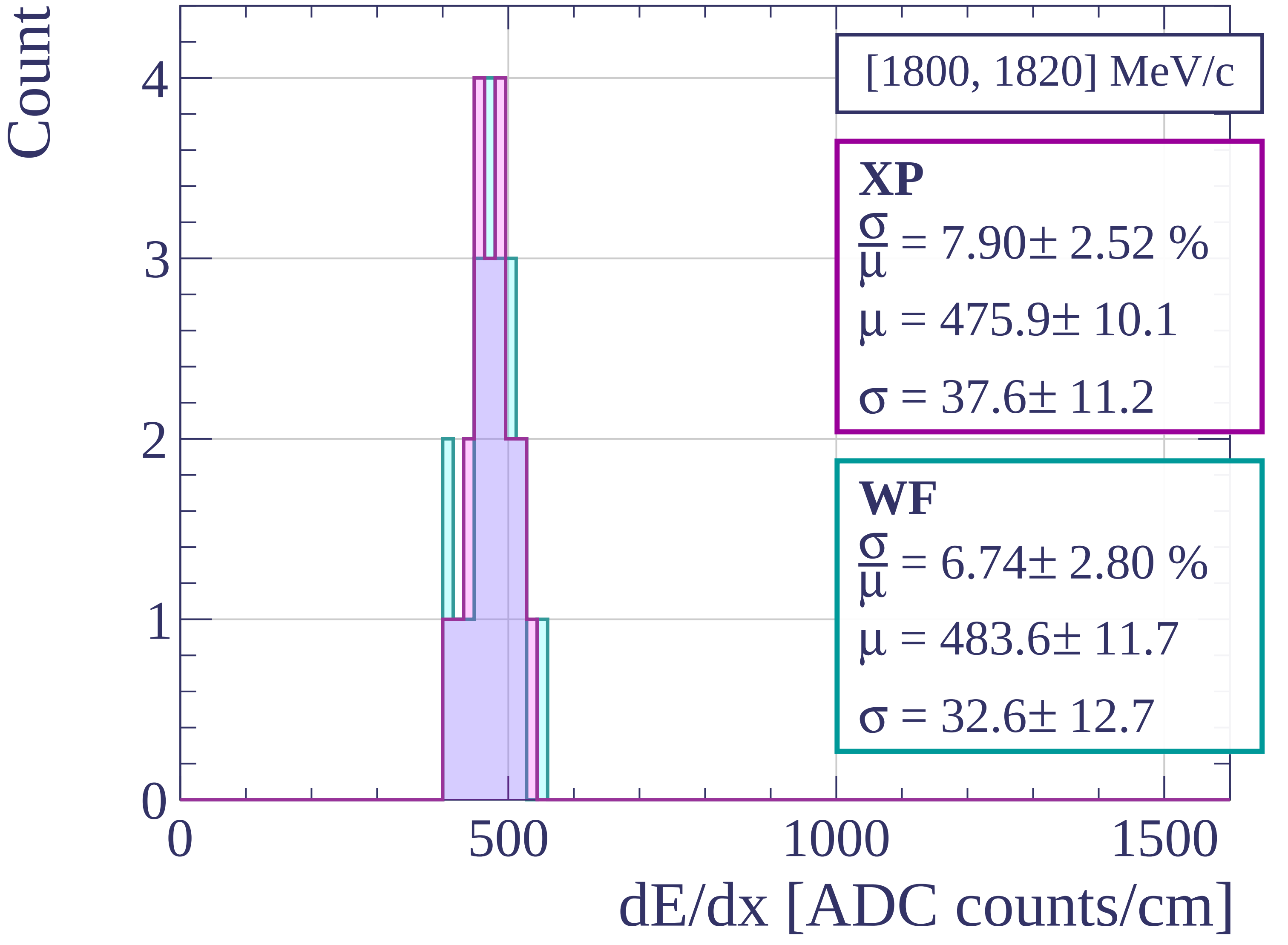
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1740, 1760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.05± 17.16 %
μ = 445.2± 56.2
σ = 58.1± 69.1
WF
$\frac{\sigma}{\mu}$ = 13.23± 16.02 %
μ = 468.3± 39.8
σ = 62.0± 69.8



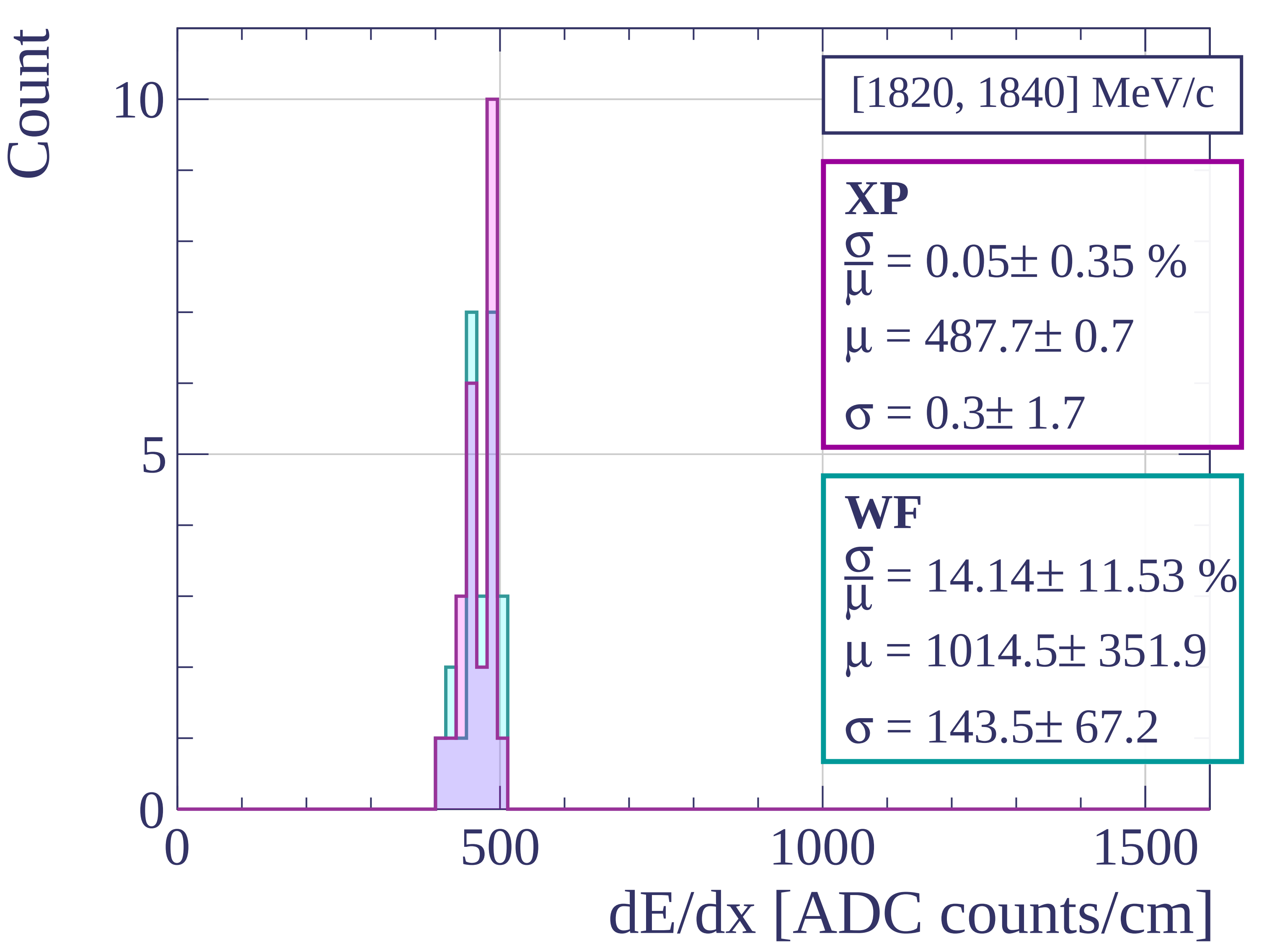
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[1760, 1780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.15± 5.29 %
μ = 466.6± 14.9
σ = 47.4± 23.2
WF
$\frac{\sigma}{\mu}$ = 7.22± 3.33 %
μ = 482.4± 11.6
σ = 34.8± 15.2



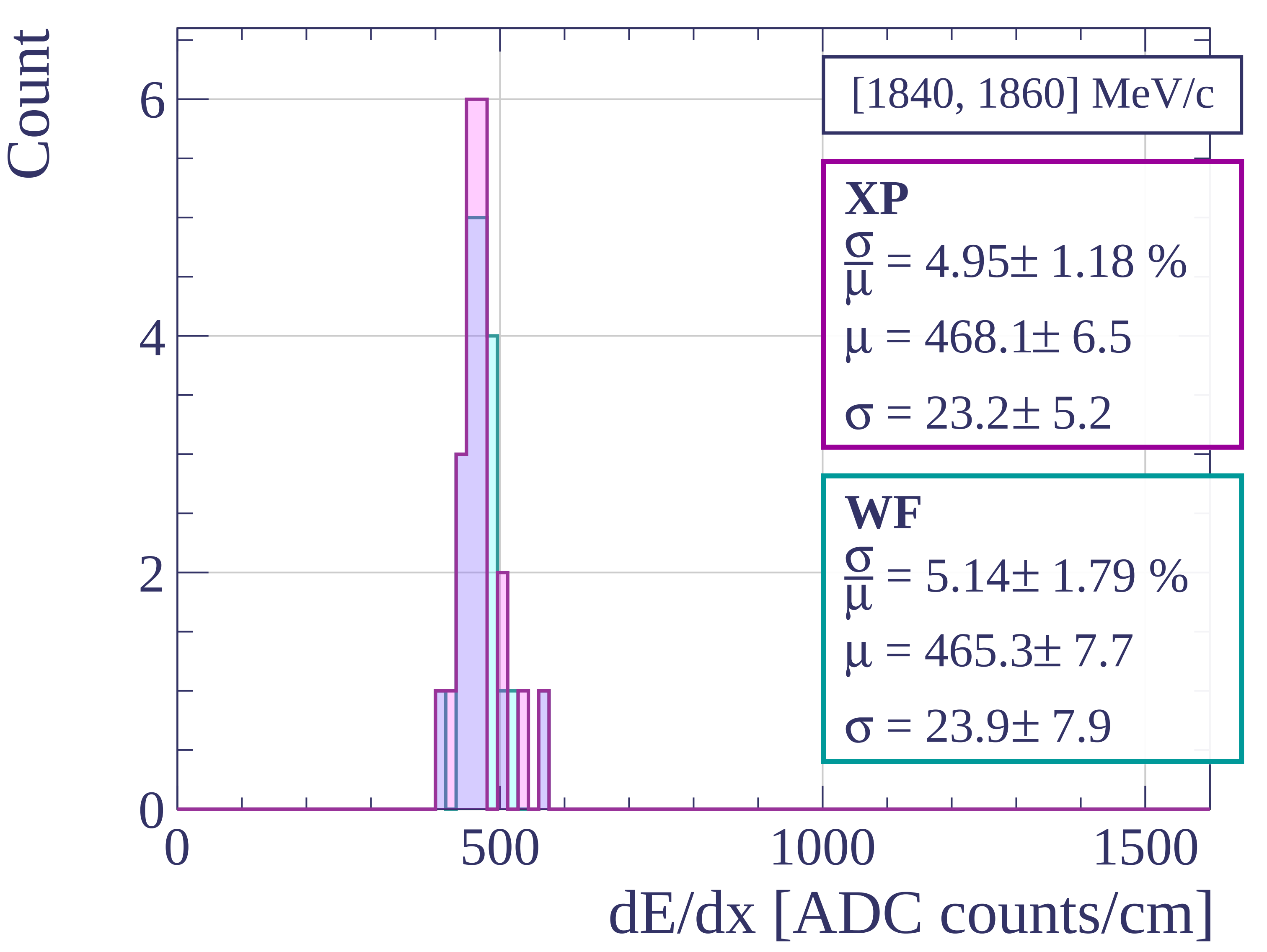
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1780, 1800] MeV/c
XP
σ/μ = 0.04± 0.25 %
μ = 583.7± 2.3
σ = 0.3± 1.5
WF
σ/μ = 9.02± 4.02 %
μ = 468.7± 15.3
σ = 42.3± 17.5



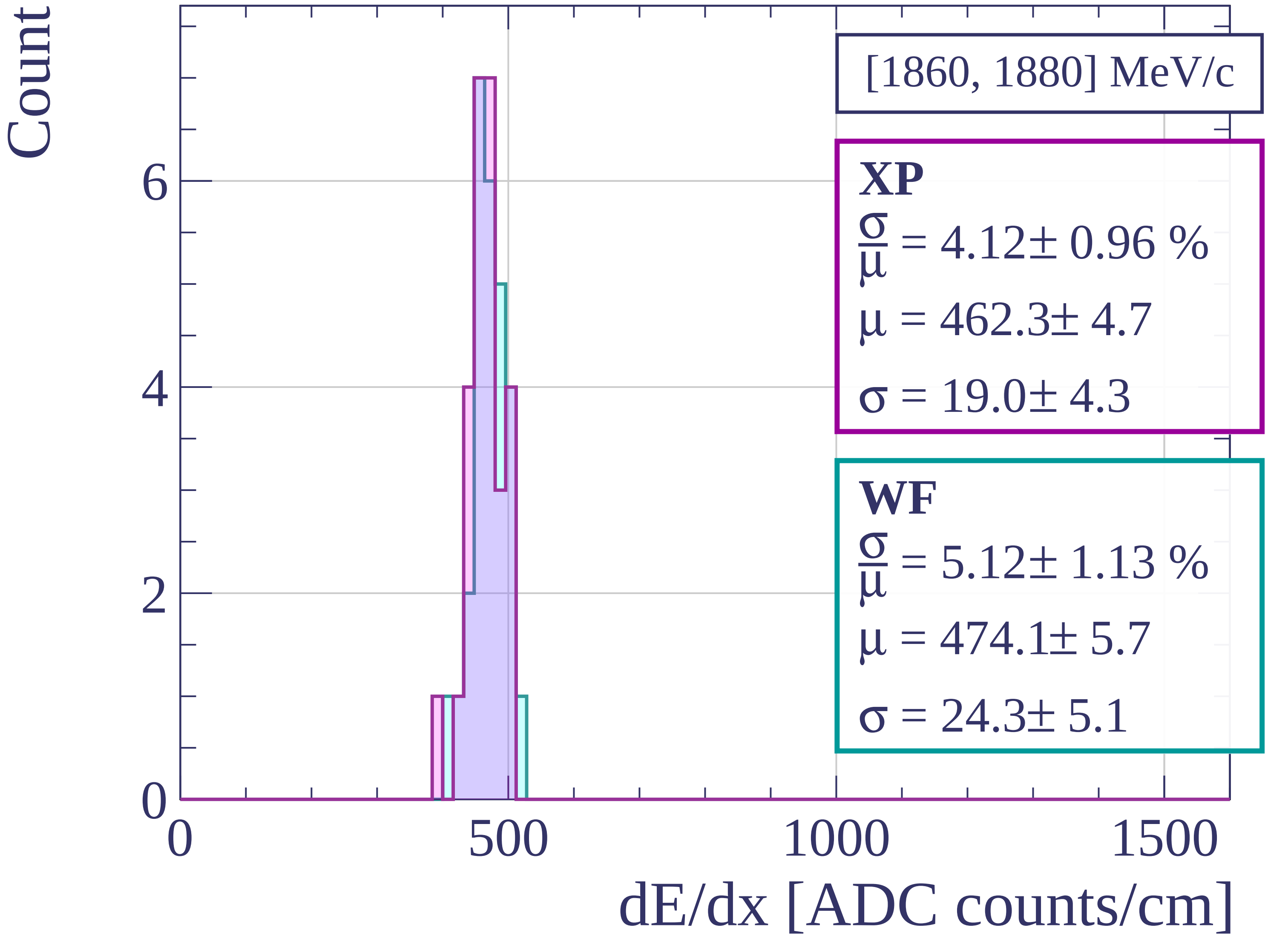
Count
dE/dx [ADC counts/cm]
[1800, 1820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.90± 2.52 %
μ = 475.9± 10.1
σ = 37.6± 11.2
WF
$\frac{\sigma}{\mu}$ = 6.74± 2.80 %
μ = 483.6± 11.7
σ = 32.6± 12.7
0
1
2
3
4
0
500
1000
1500

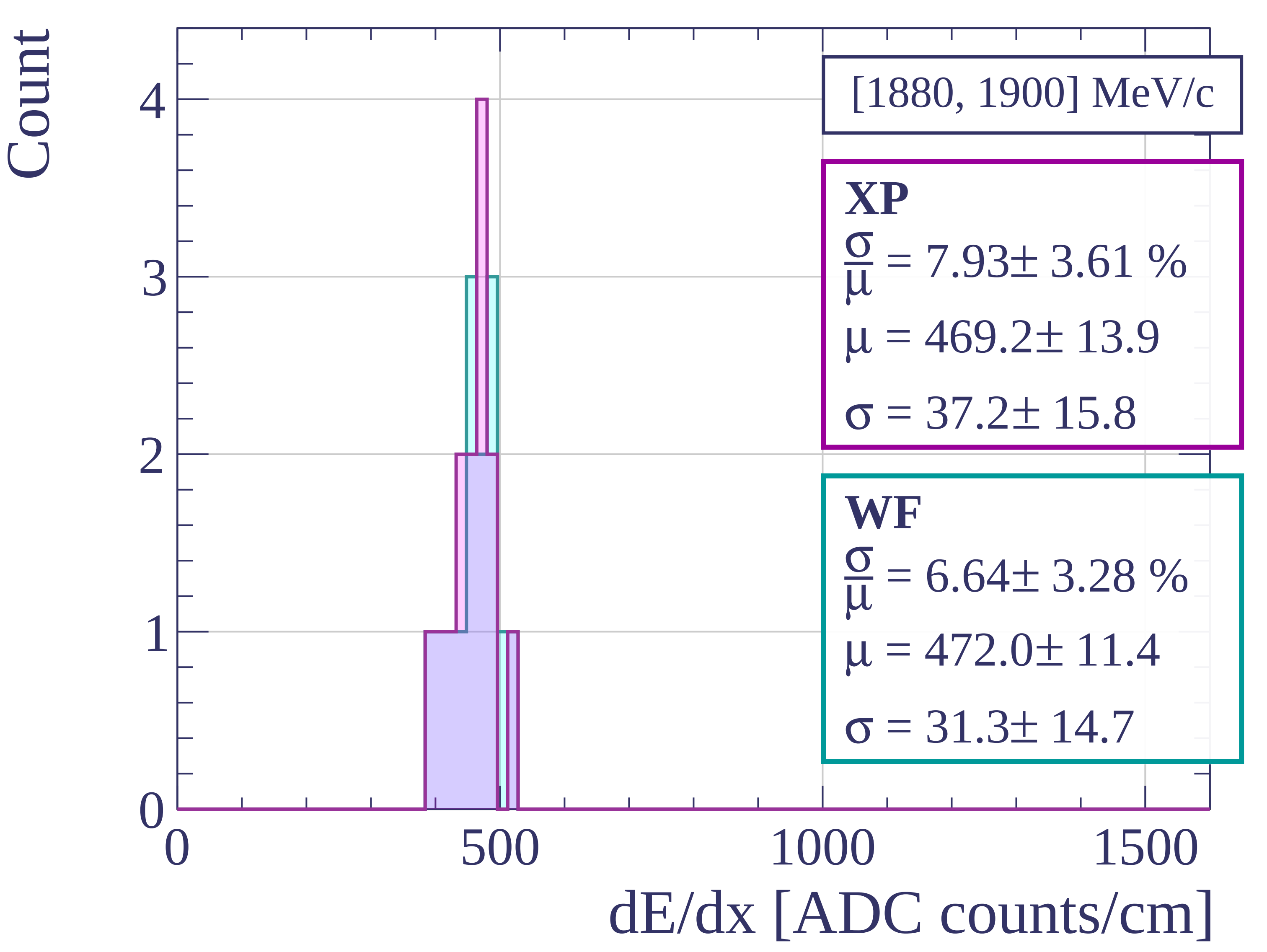


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1820, 1840] MeV/c
XP
σ/μ = 0.05± 0.35 %
μ = 487.7± 0.7
σ = 0.3± 1.7
WF
σ/μ = 14.14± 11.53 %
μ = 1014.5± 351.9
σ = 143.5± 67.2

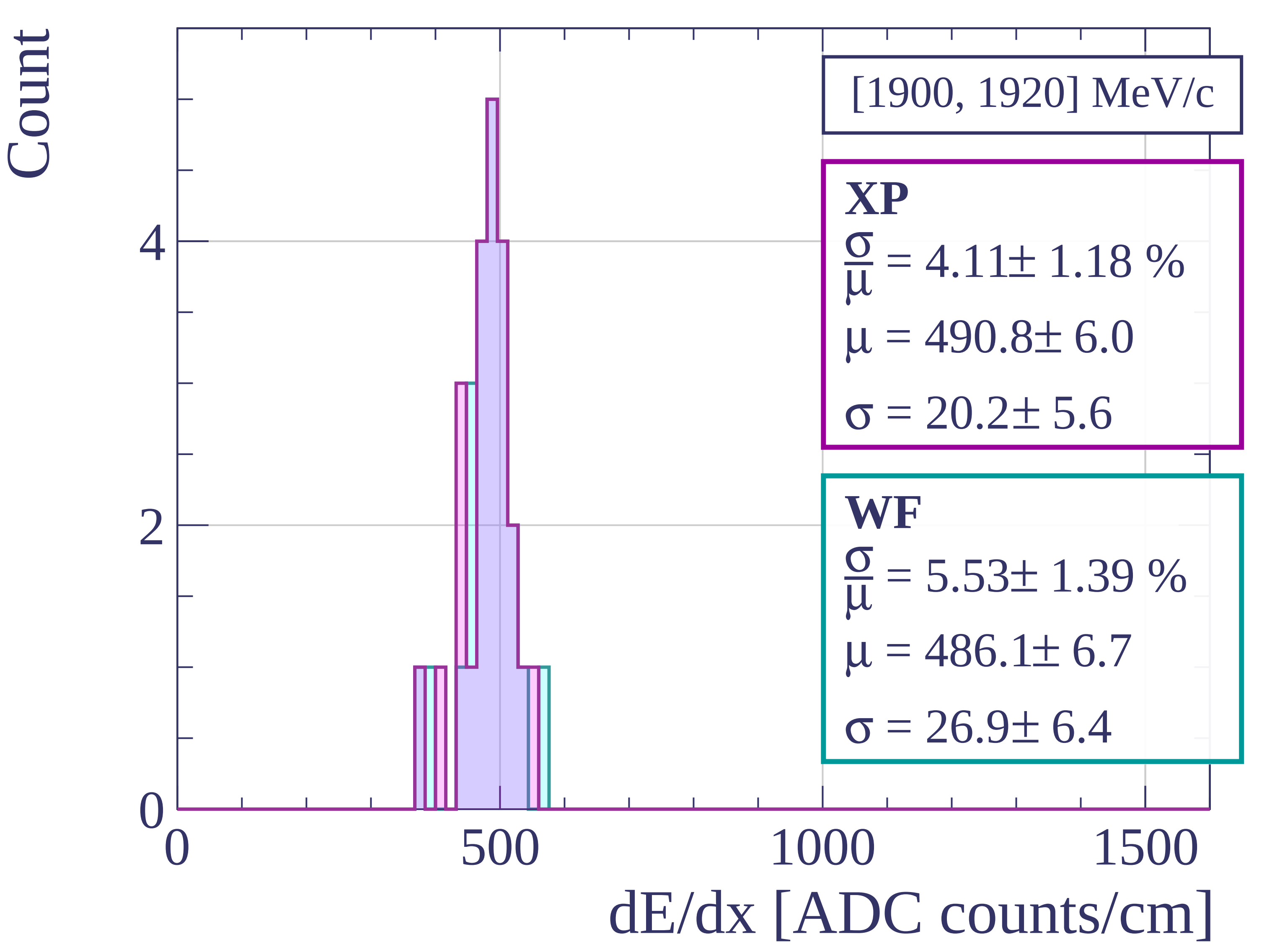


Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[1840, 1860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.95± 1.18 %
μ = 468.1± 6.5
σ = 23.2± 5.2
WF
$\frac{\sigma}{\mu}$ = 5.14± 1.79 %
μ = 465.3± 7.7
σ = 23.9± 7.9

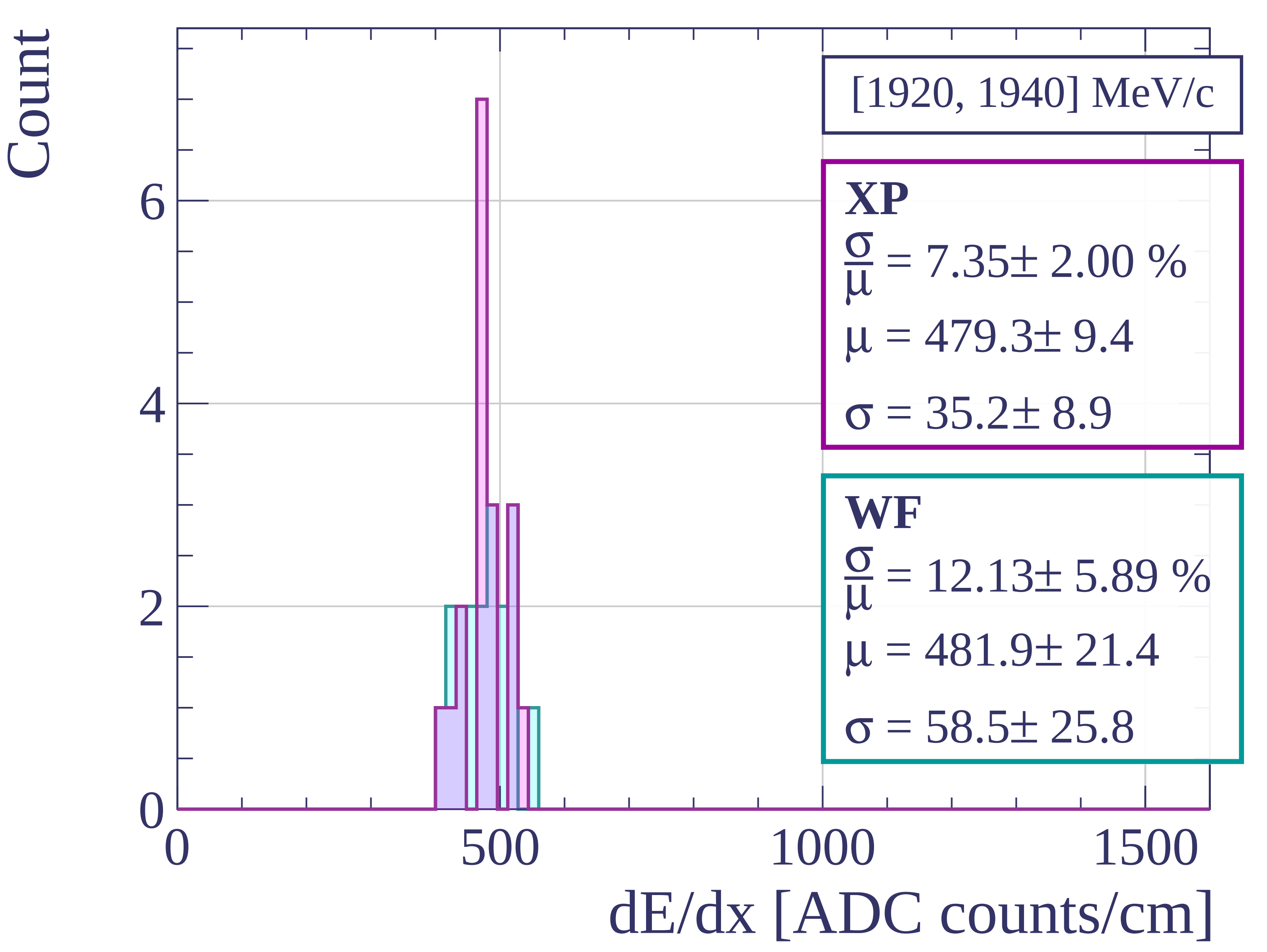

Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1860, 1880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.12± 0.96 %
μ = 462.3± 4.7
σ = 19.0± 4.3
WF
$\frac{\sigma}{\mu}$ = 5.12± 1.13 %
μ = 474.1± 5.7
σ = 24.3± 5.1

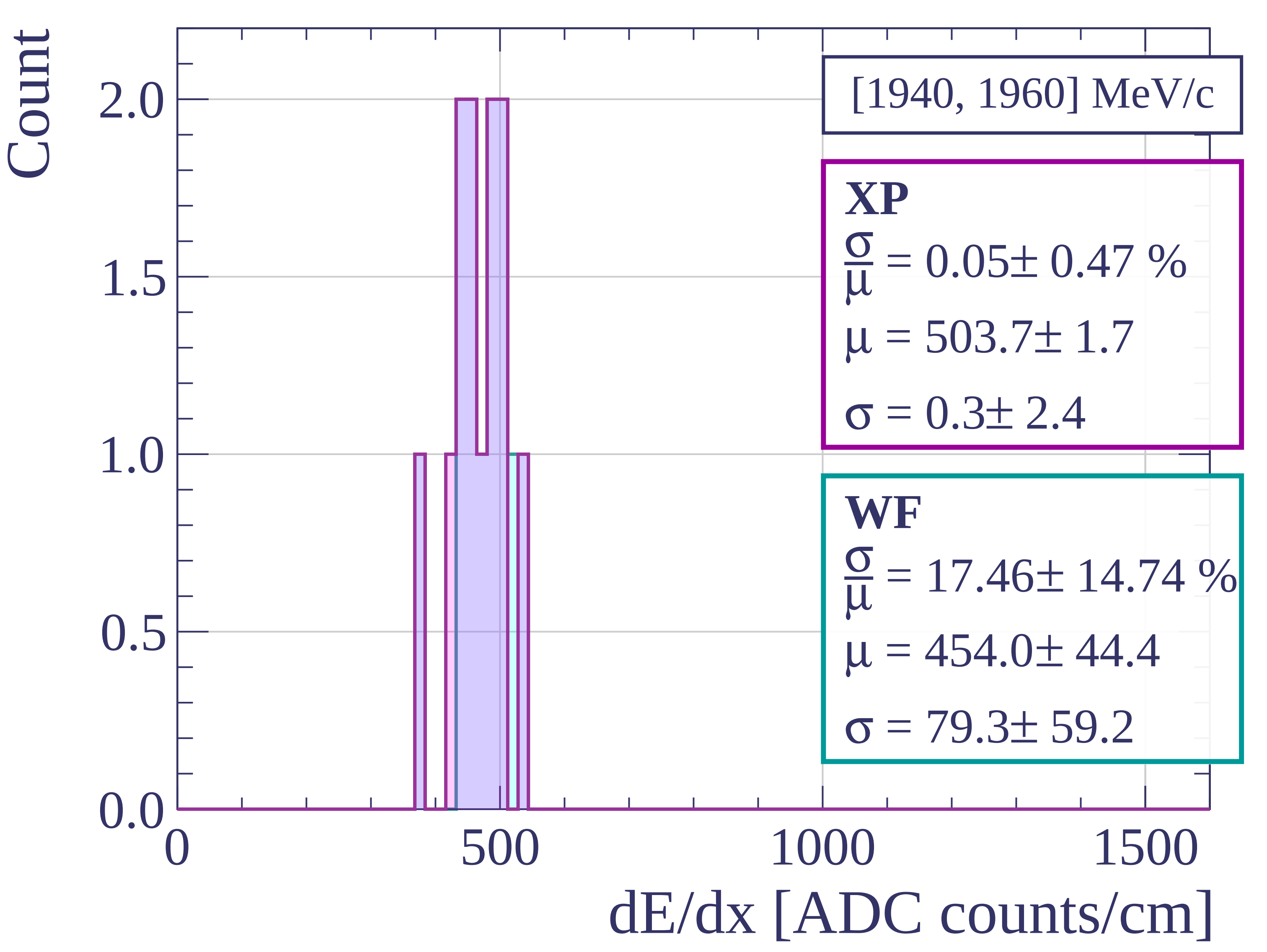


Count
4
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1880, 1900] MeV/c
XP
σ/μ = 7.93± 3.61 %
μ = 469.2± 13.9
σ = 37.2± 15.8
WF
σ/μ = 6.64± 3.28 %
μ = 472.0± 11.4
σ = 31.3± 14.7

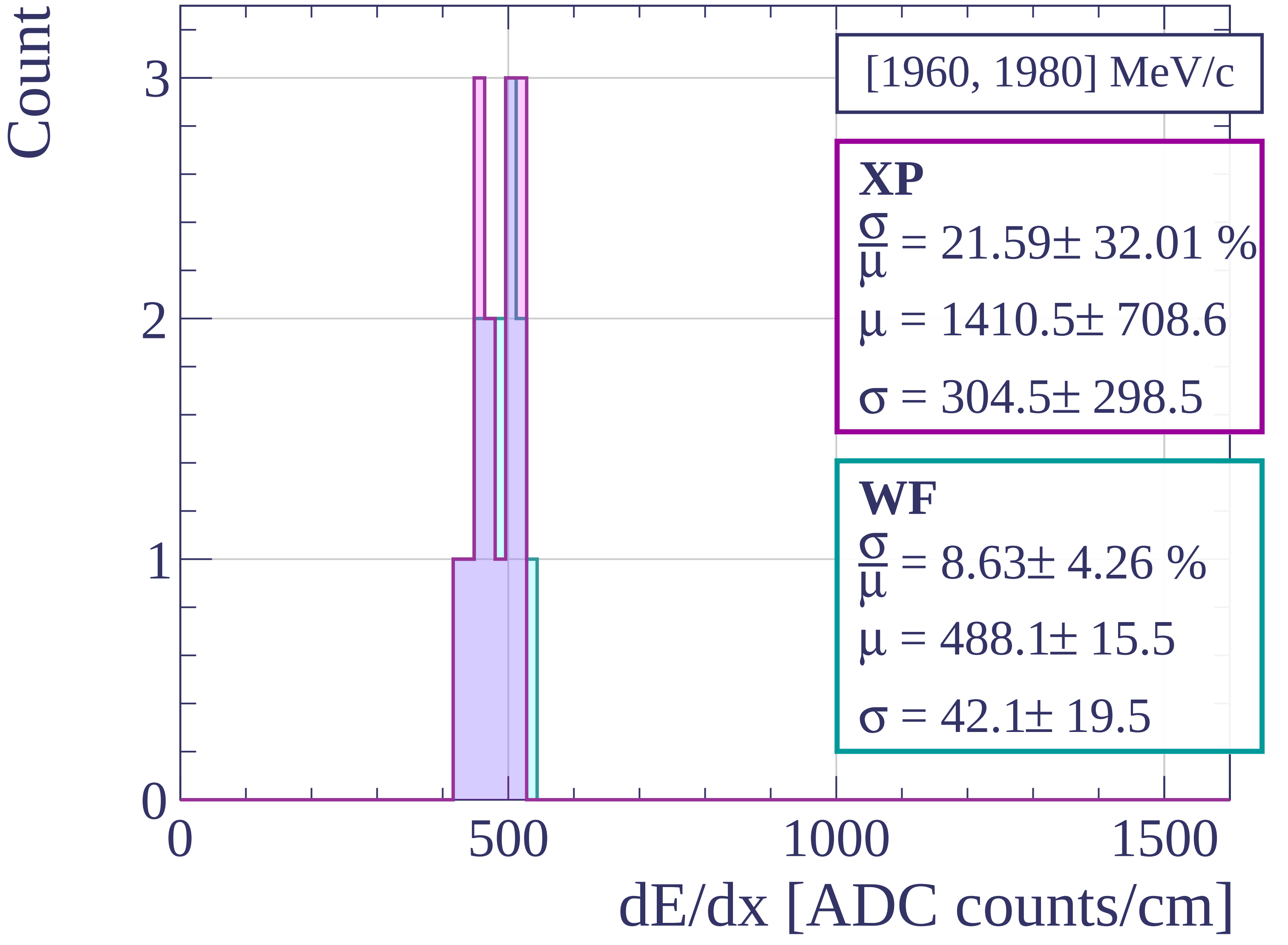


Count
[1900, 1920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.11± 1.18 %
μ = 490.8± 6.0
σ = 20.2± 5.6
WF
$\frac{\sigma}{\mu}$ = 5.53± 1.39 %
μ = 486.1± 6.7
σ = 26.9± 6.4
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]

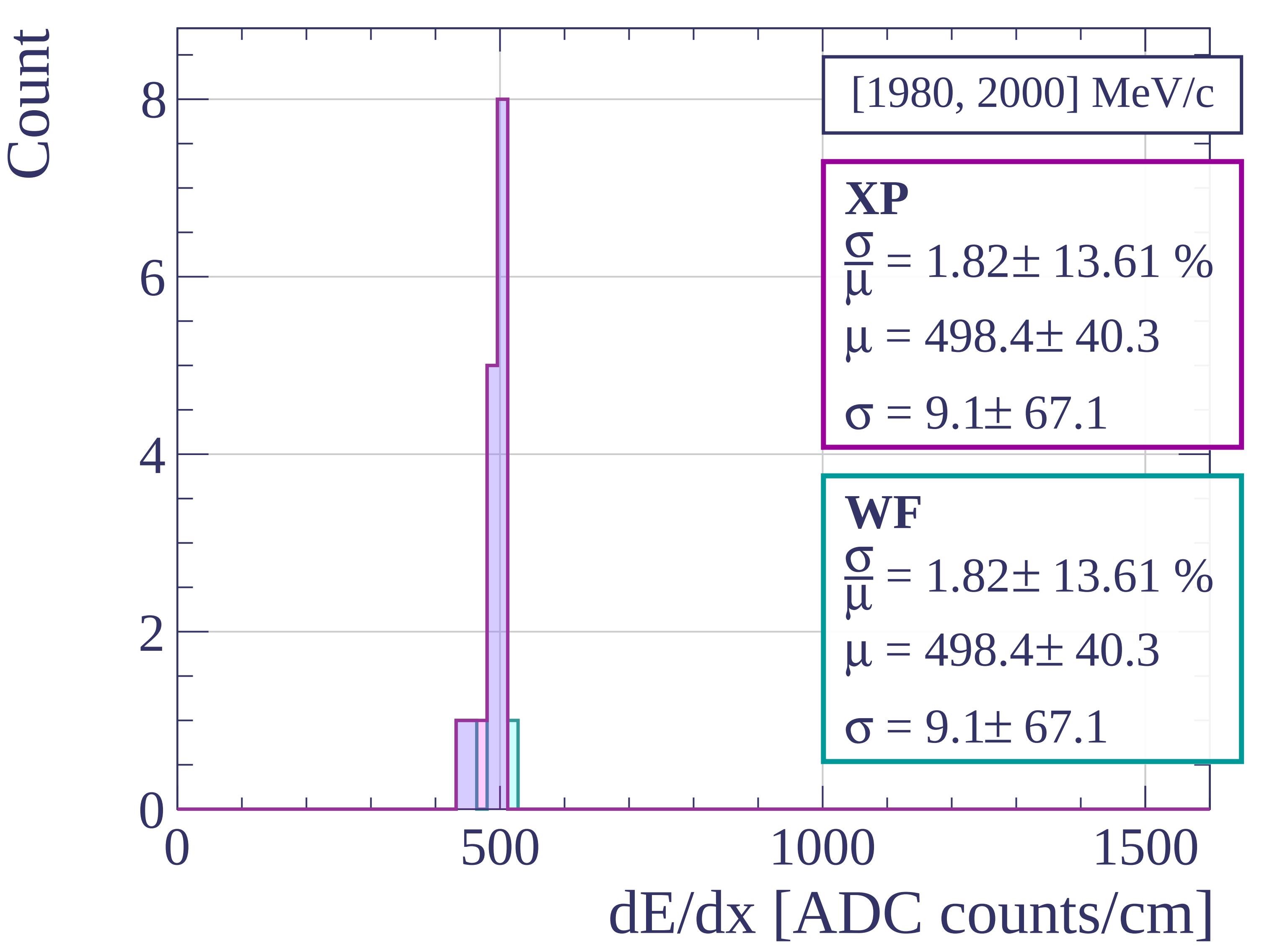

Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1920, 1940] MeV/c
XP
σ/μ = 7.35± 2.00 %
μ = 479.3± 9.4
σ = 35.2± 8.9
WF
σ/μ = 12.13± 5.89 %
μ = 481.9± 21.4
σ = 58.5± 25.8



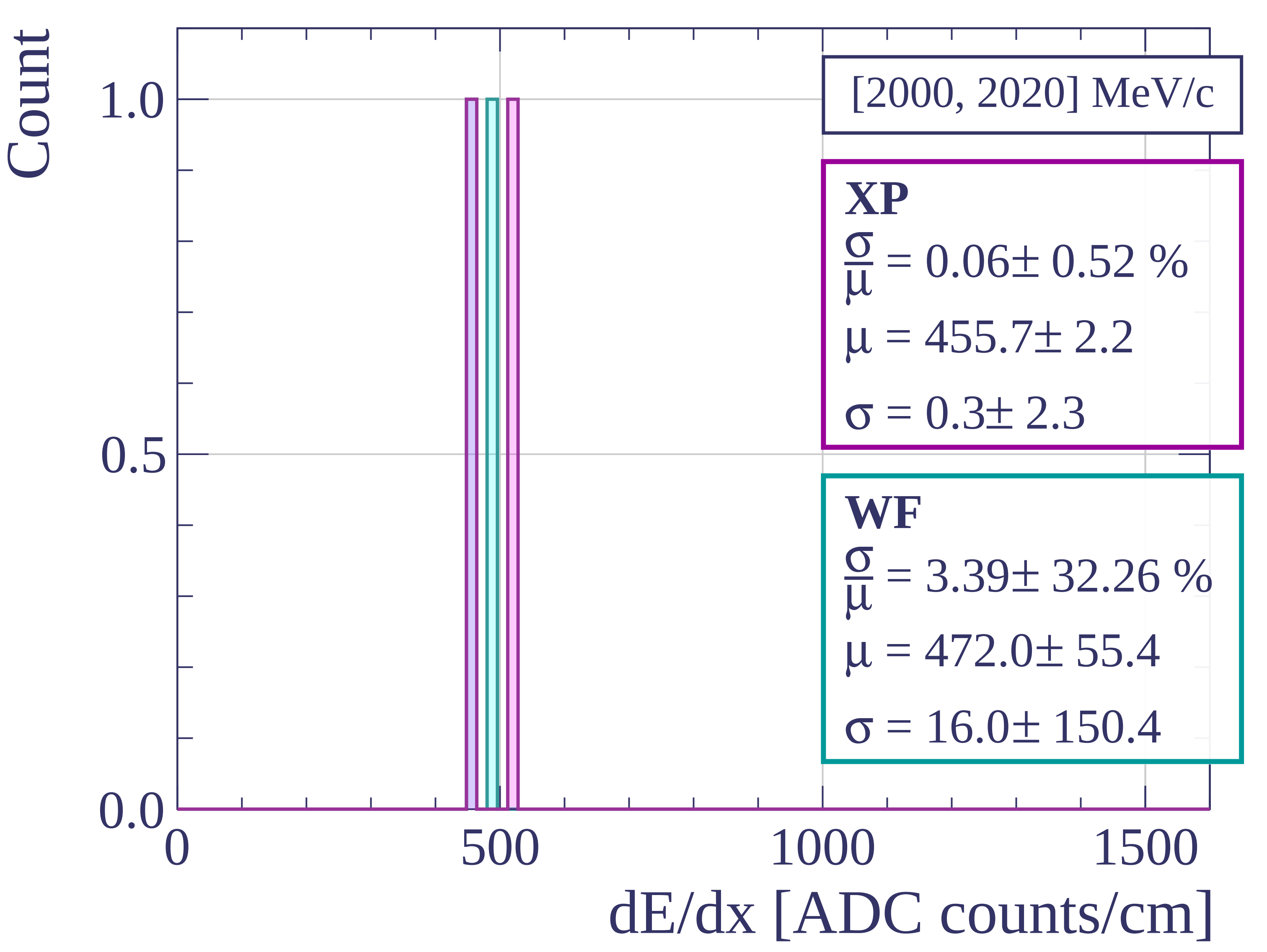
Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1940, 1960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 0.05± 0.47 %
μ = 503.7± 1.7
σ = 0.3± 2.4
WF
$\frac{\sigma}{\mu}$ = 17.46± 14.74 %
μ = 454.0± 44.4
σ = 79.3± 59.2



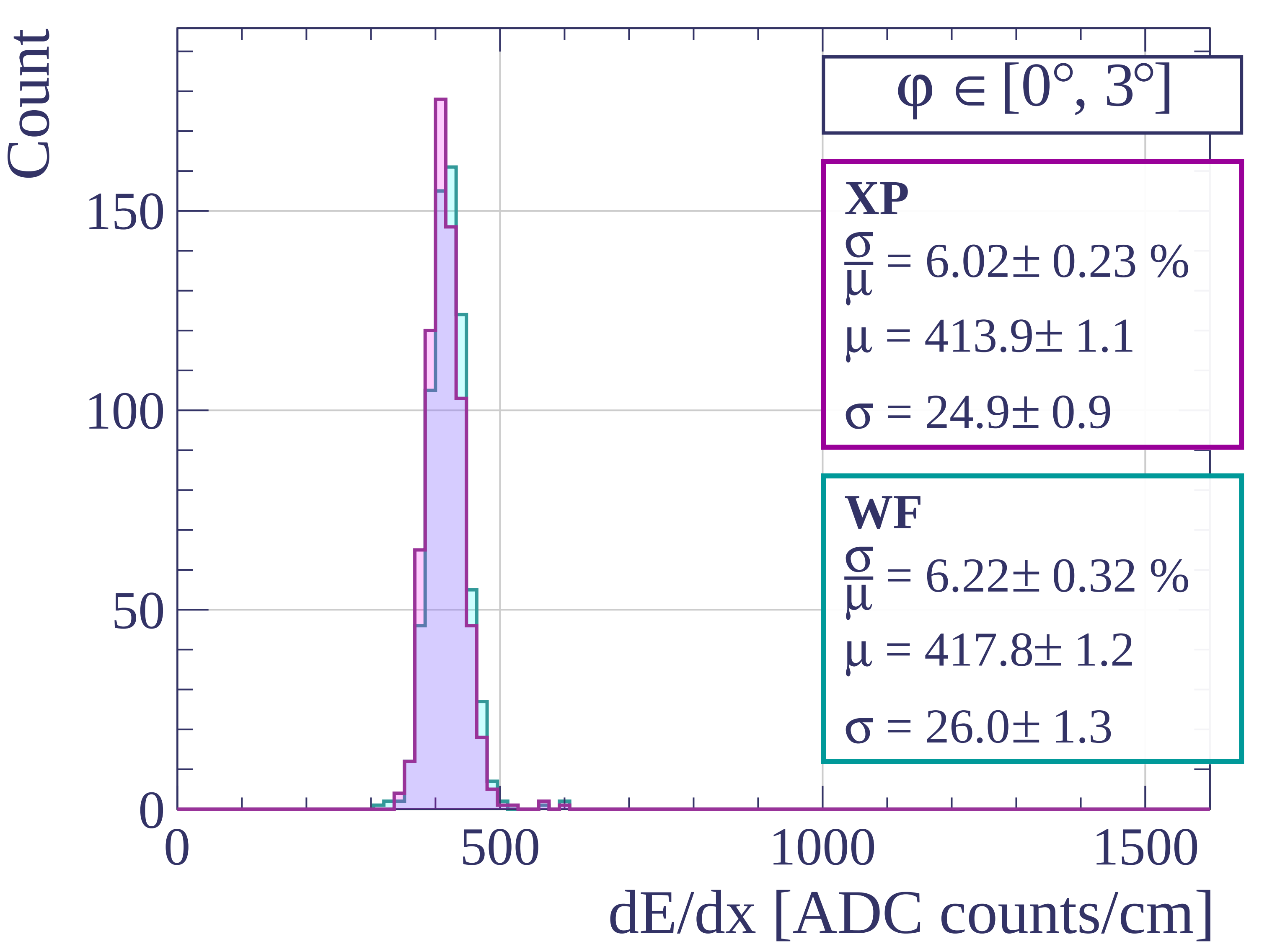
Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1960, 1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 21.59± 32.01 %
μ = 1410.5± 708.6
σ = 304.5± 298.5
WF
$\frac{\sigma}{\mu}$ = 8.63± 4.26 %
μ = 488.1± 15.5
σ = 42.1± 19.5



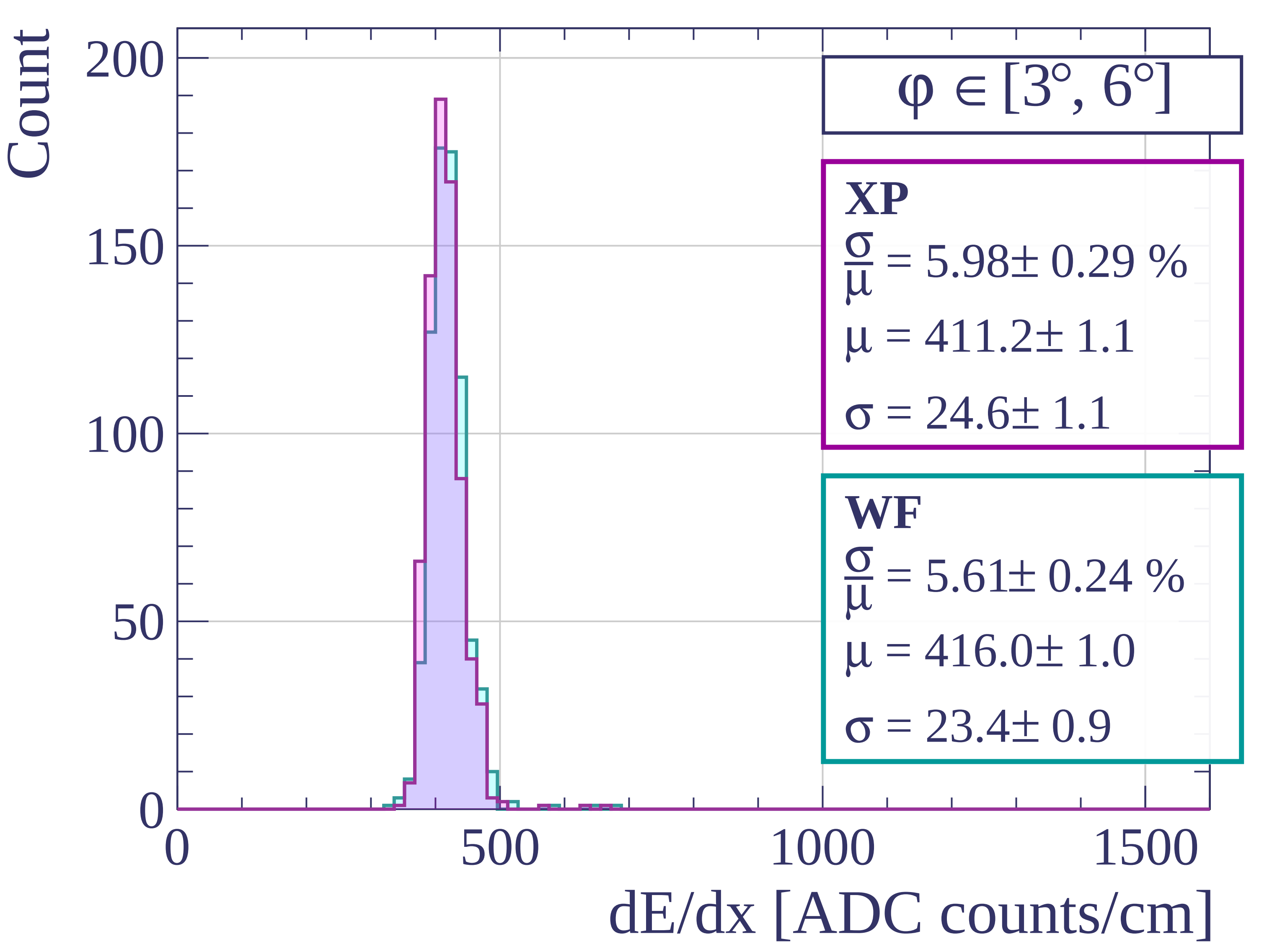
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[1980, 2000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 1.82± 13.61 %
μ = 498.4± 40.3
σ = 9.1± 67.1
WF
$\frac{\sigma}{\mu}$ = 1.82± 13.61 %
μ = 498.4± 40.3
σ = 9.1± 67.1



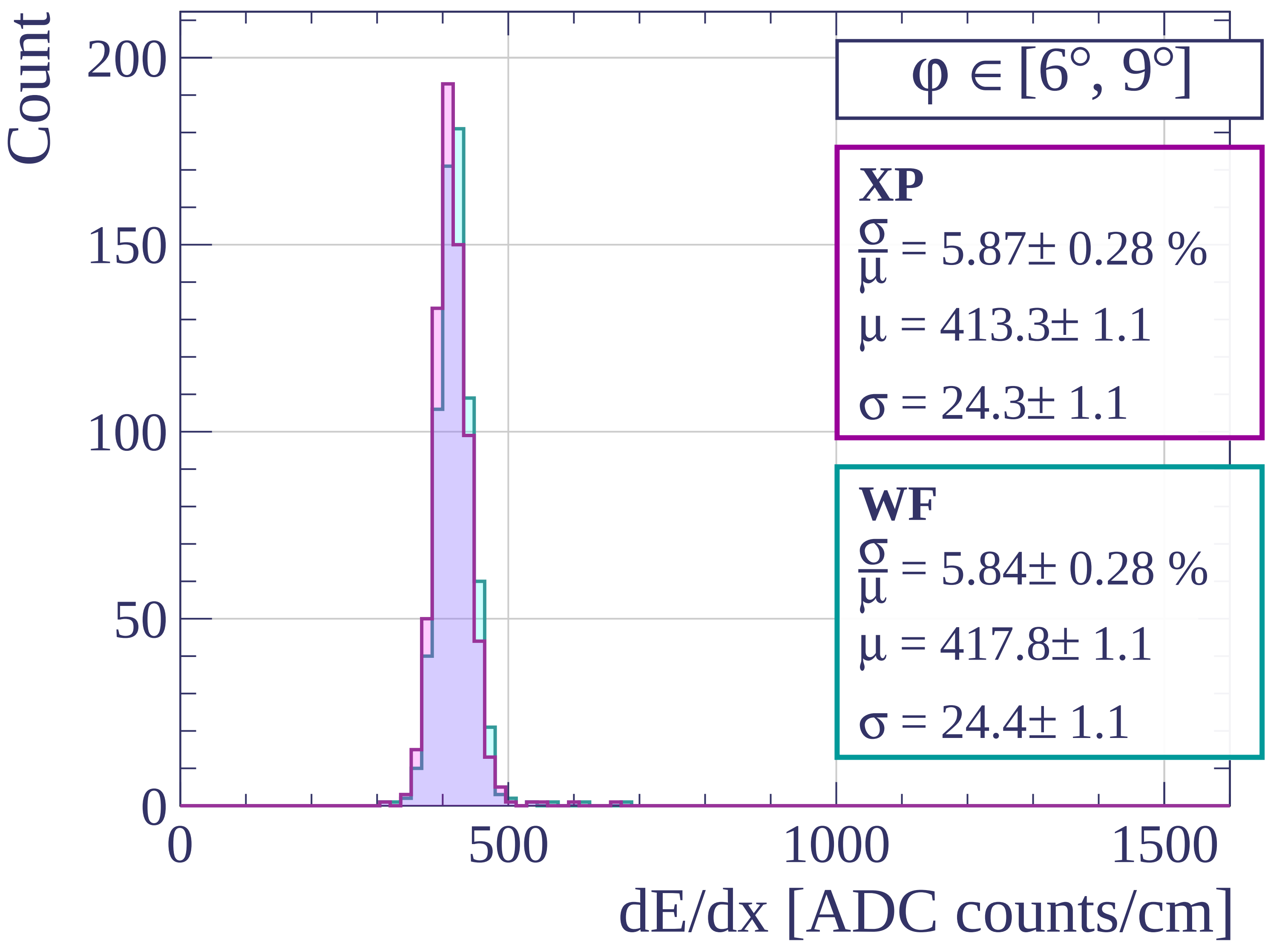
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
[2000, 2020] MeV/c
XP
$\frac{\sigma}{\mu}$ = 0.06± 0.52 %
μ = 455.7± 2.2
σ = 0.3± 2.3
WF
$\frac{\sigma}{\mu}$ = 3.39± 32.26 %
μ = 472.0± 55.4
σ = 16.0± 150.4



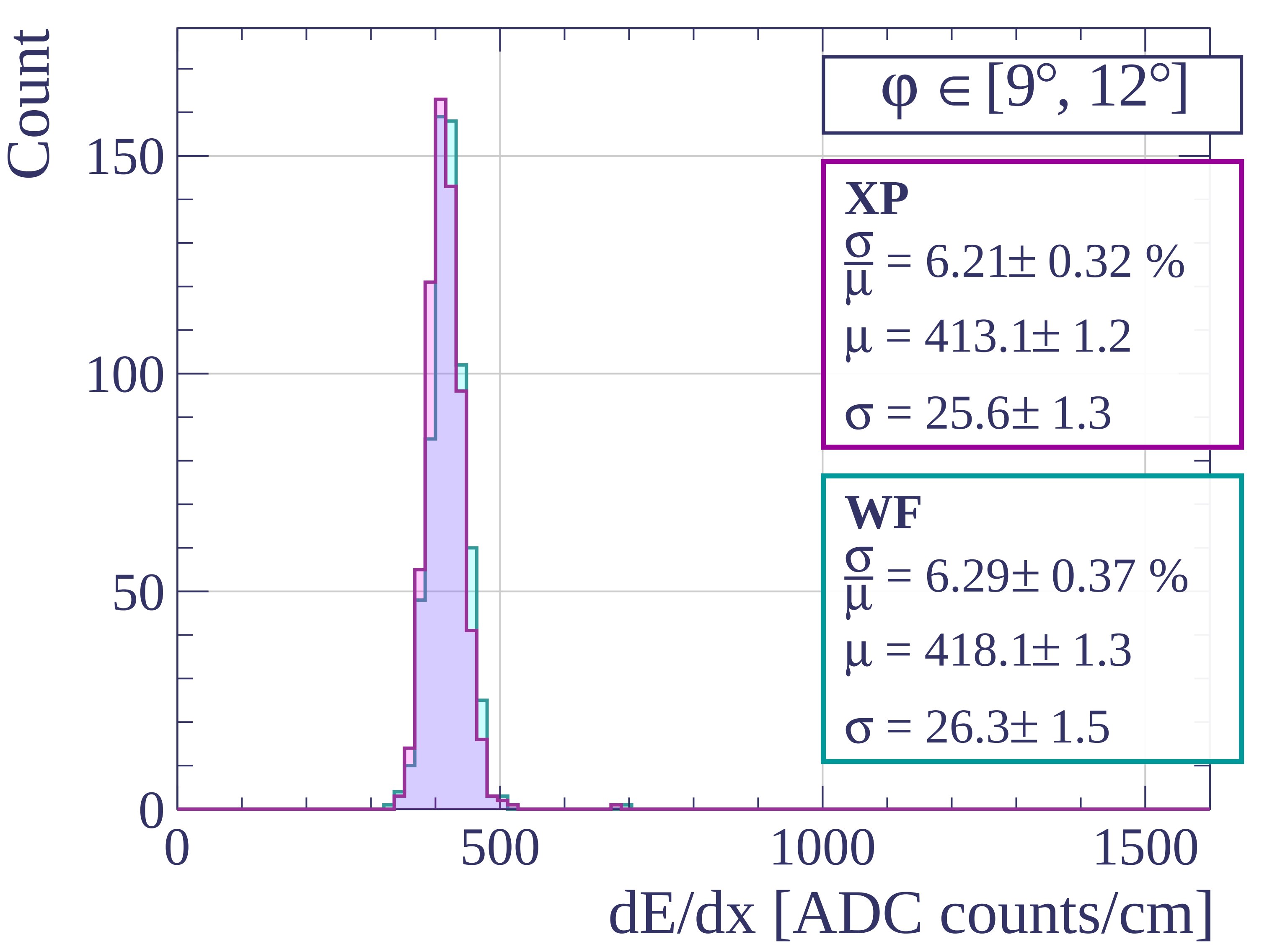
Count
0
50
100
150
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [0°, 3°]
XP
σ/μ = 6.02± 0.23 %
μ = 413.9± 1.1
σ = 24.9± 0.9
WF
σ/μ = 6.22± 0.32 %
μ = 417.8± 1.2
σ = 26.0± 1.3



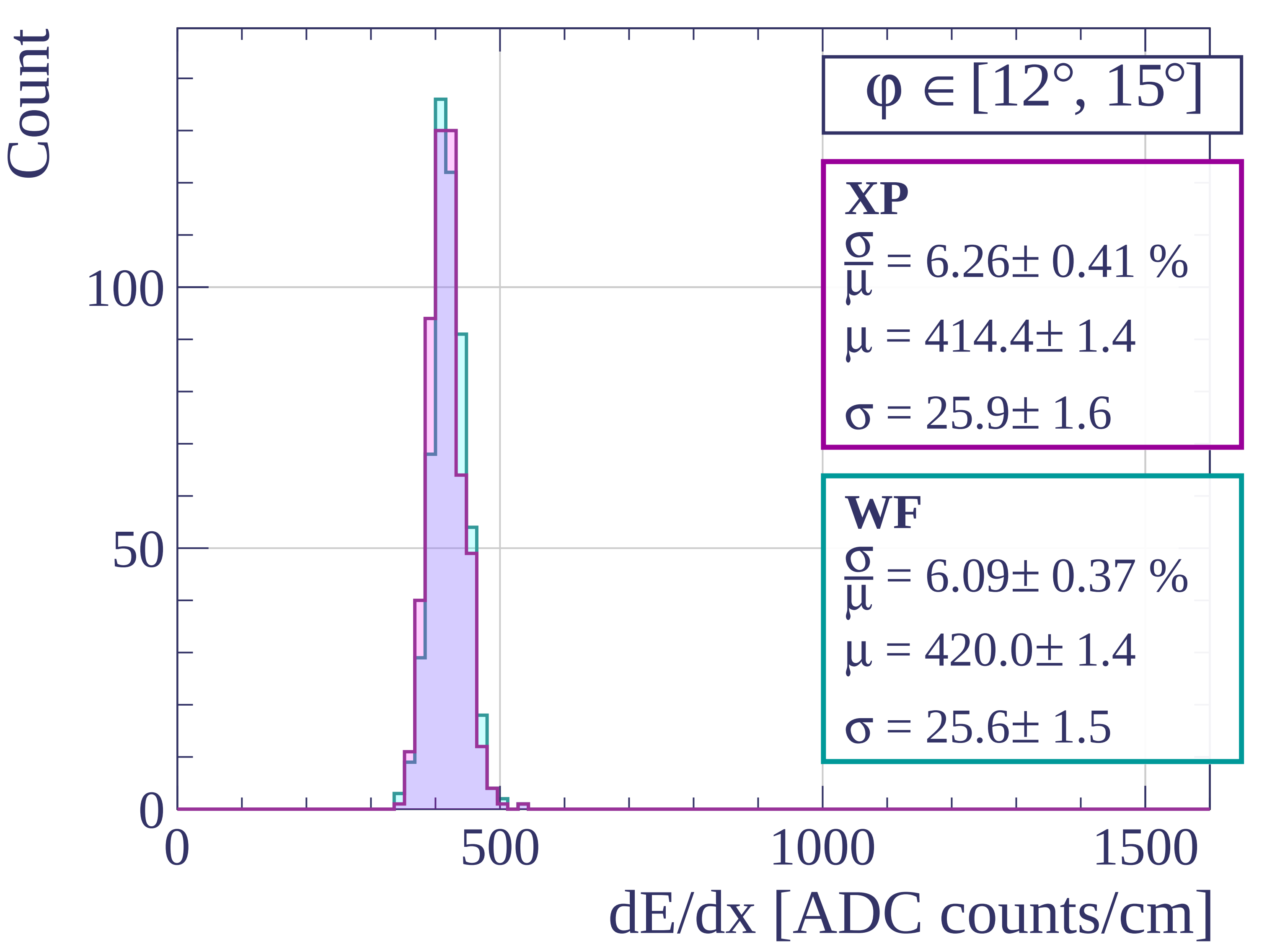
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [3°, 6°]
XP
σ/μ = 5.98± 0.29 %
μ = 411.2± 1.1
σ = 24.6± 1.1
WF
σ/μ = 5.61± 0.24 %
μ = 416.0± 1.0
σ = 23.4± 0.9

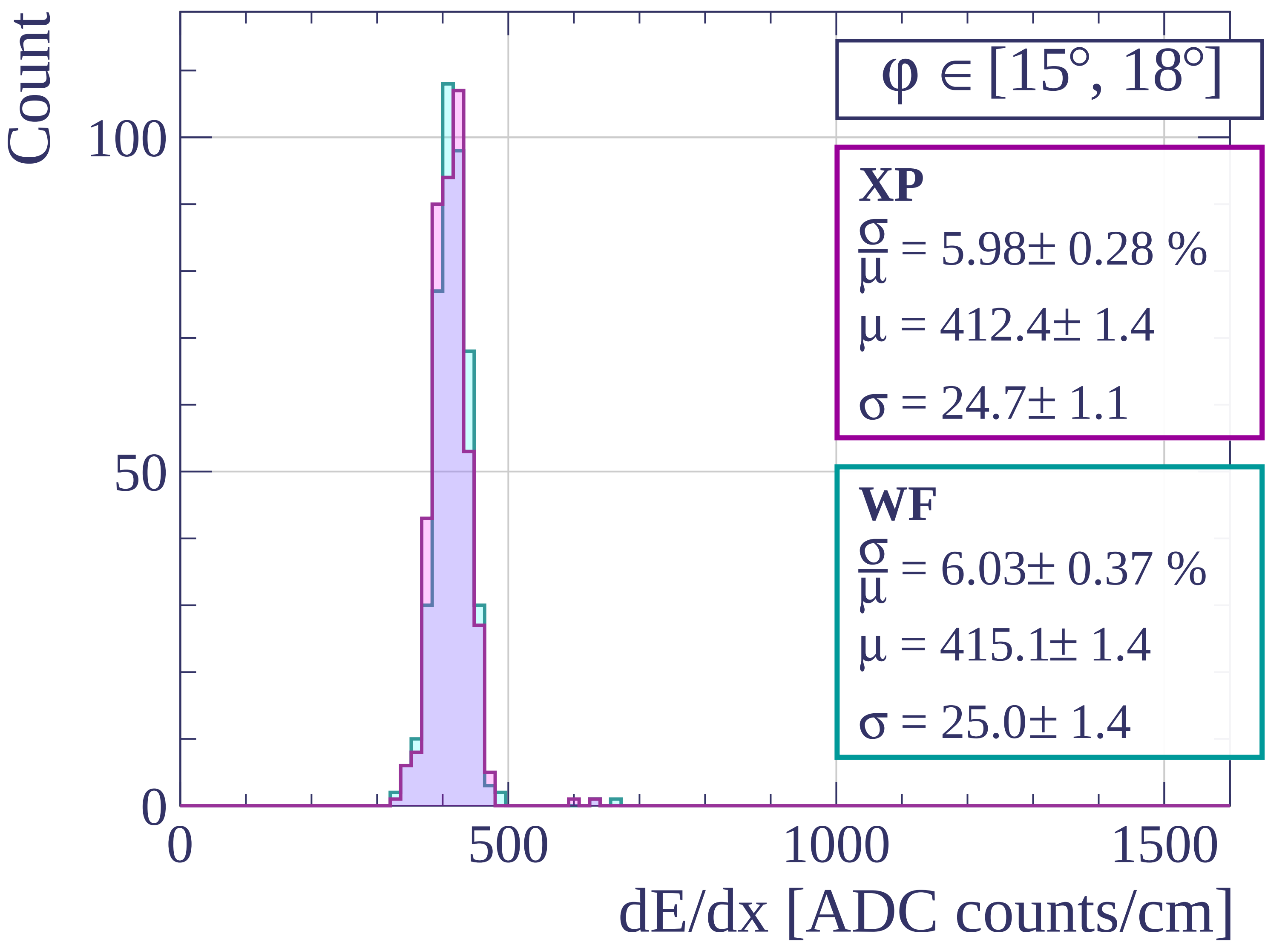


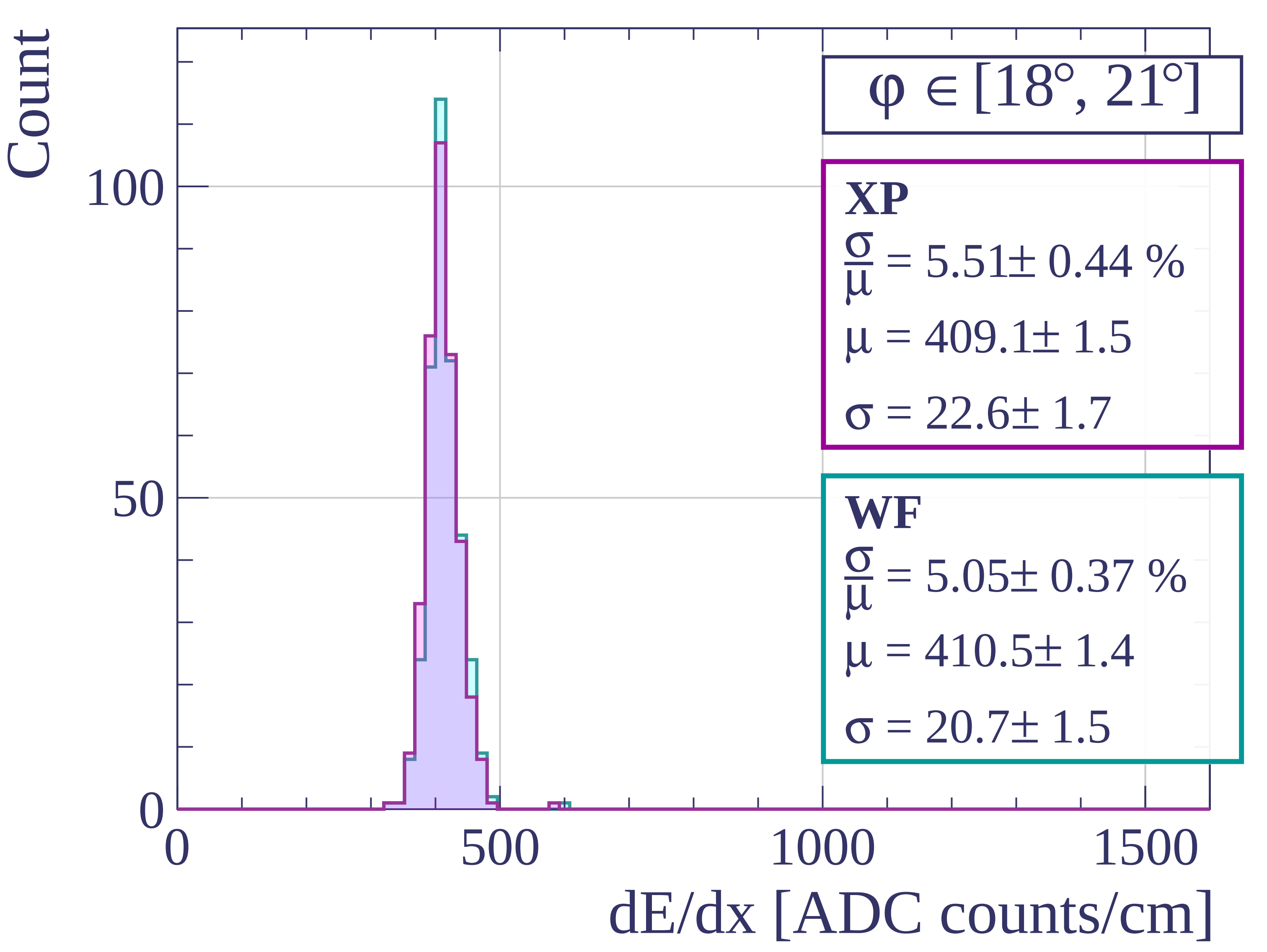
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [6°, 9°]
XP
σ/μ = 5.87± 0.28 %
μ = 413.3± 1.1
σ = 24.3± 1.1
WF
σ/μ = 5.84± 0.28 %
μ = 417.8± 1.1
σ = 24.4± 1.1



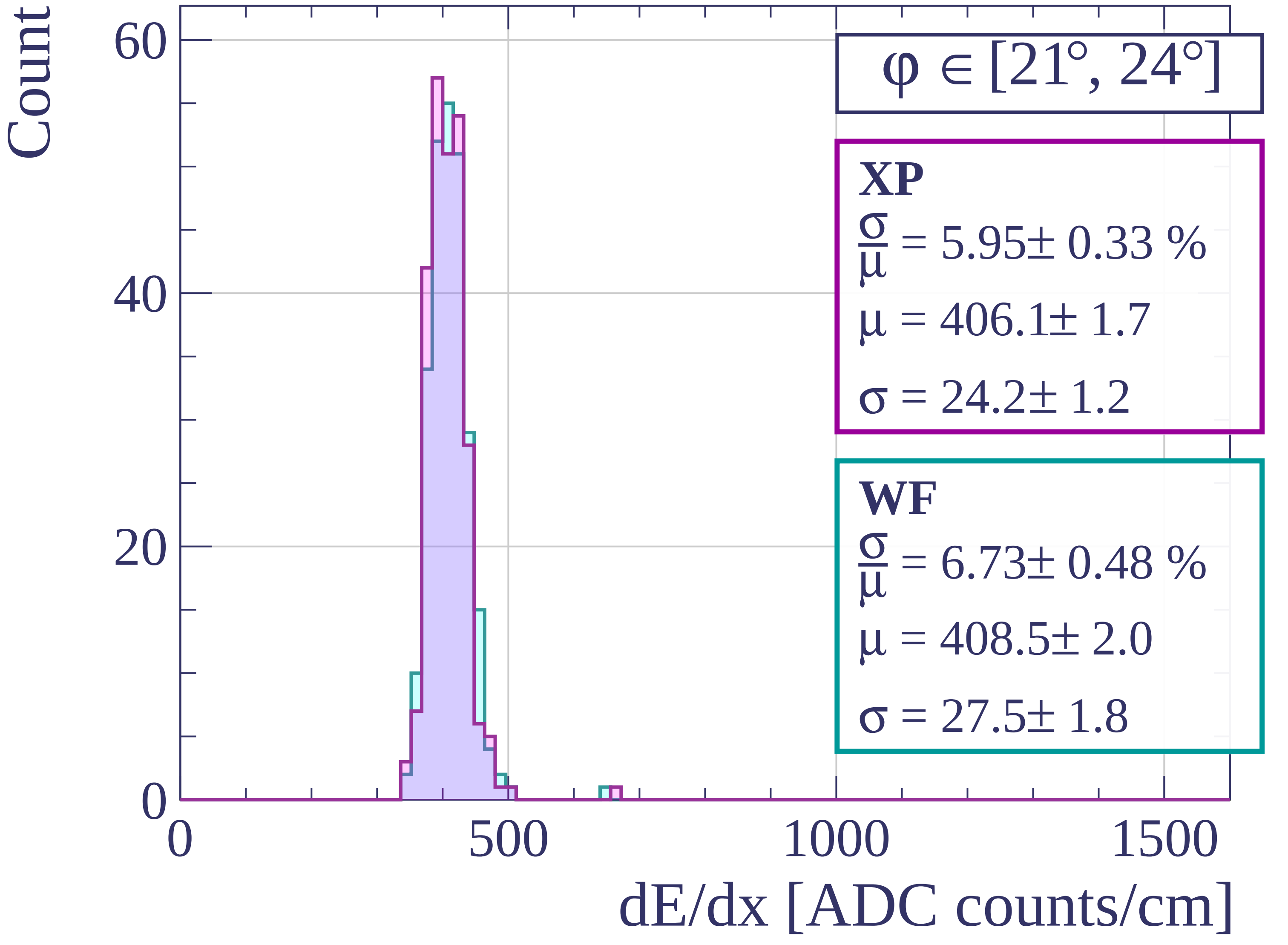
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [9°, 12°]
XP
σ/μ = 6.21± 0.32 %
μ = 413.1± 1.2
σ = 25.6± 1.3
WF
σ/μ = 6.29± 0.37 %
μ = 418.1± 1.3
σ = 26.3± 1.5

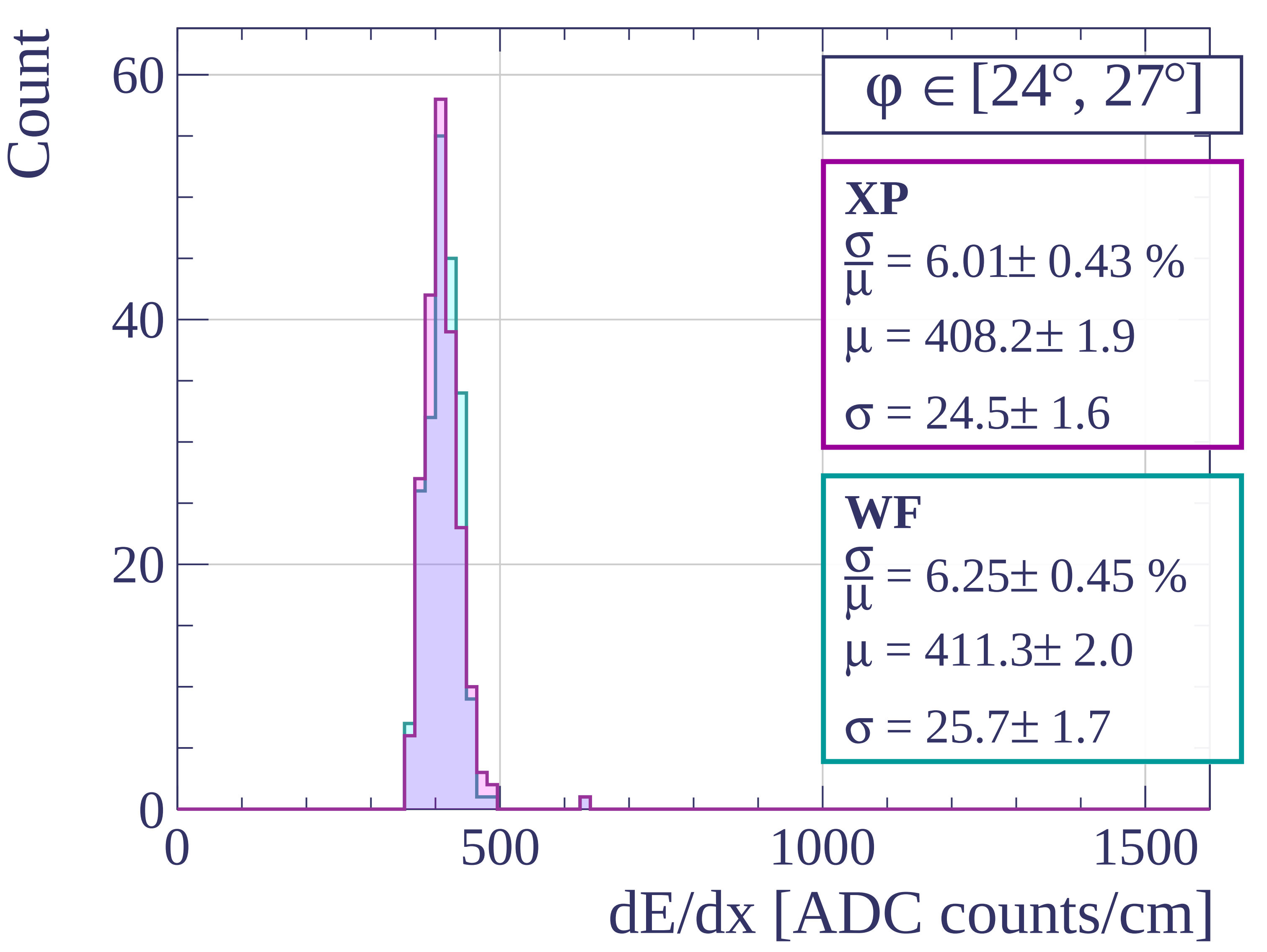

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [12°, 15°]
XP
σ/μ = 6.26± 0.41 %
μ = 414.4± 1.4
σ = 25.9± 1.6
WF
σ/μ = 6.09± 0.37 %
μ = 420.0± 1.4
σ = 25.6± 1.5


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [15°, 18°]
XP
σ/μ = 5.98± 0.28 %
μ = 412.4± 1.4
σ = 24.7± 1.1
WF
σ/μ = 6.03± 0.37 %
μ = 415.1± 1.4
σ = 25.0± 1.4

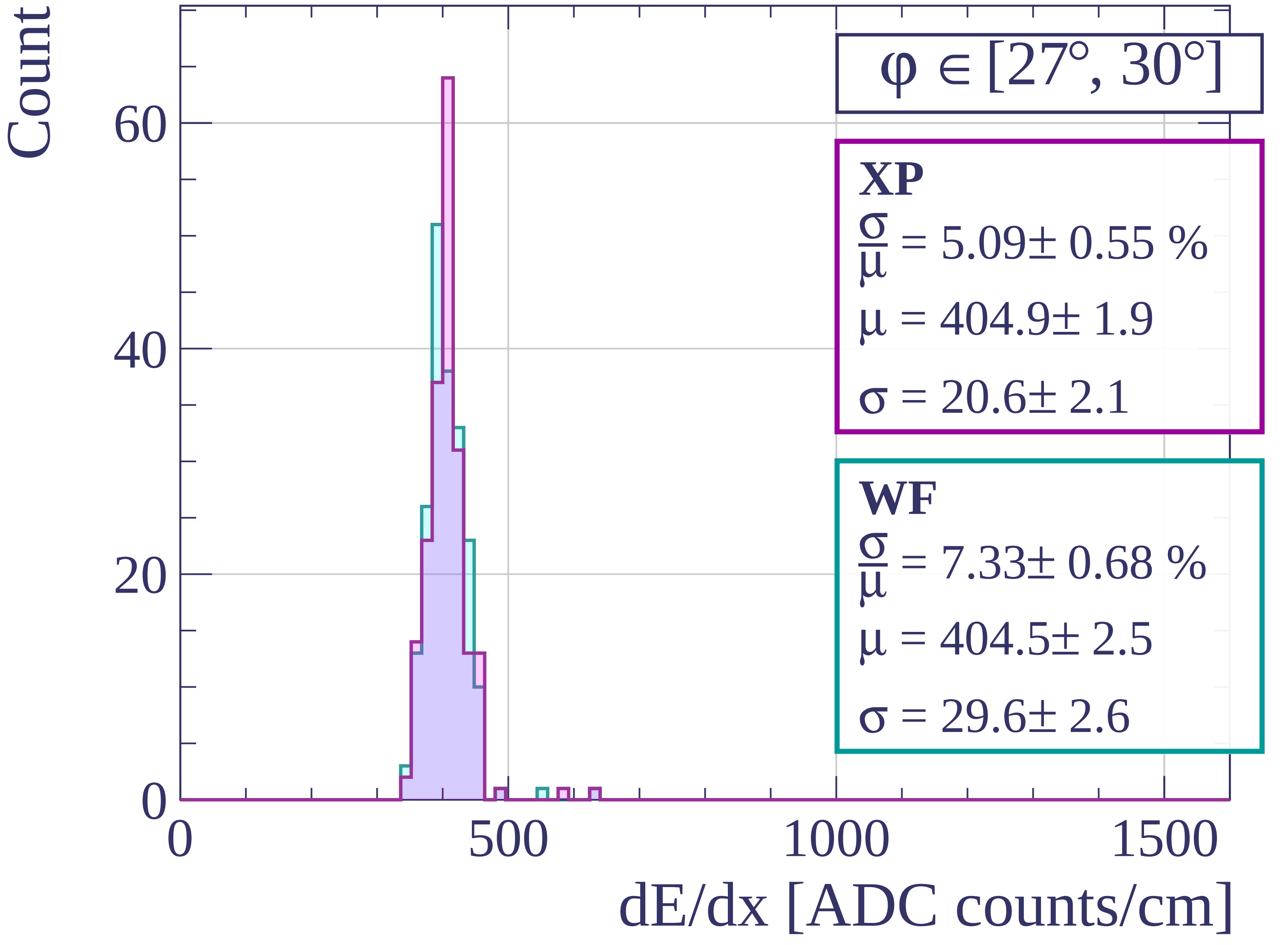


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [18°, 21°]
XP
σ/μ = 5.51± 0.44 %
μ = 409.1± 1.5
σ = 22.6± 1.7
WF
σ/μ = 5.05± 0.37 %
μ = 410.5± 1.4
σ = 20.7± 1.5

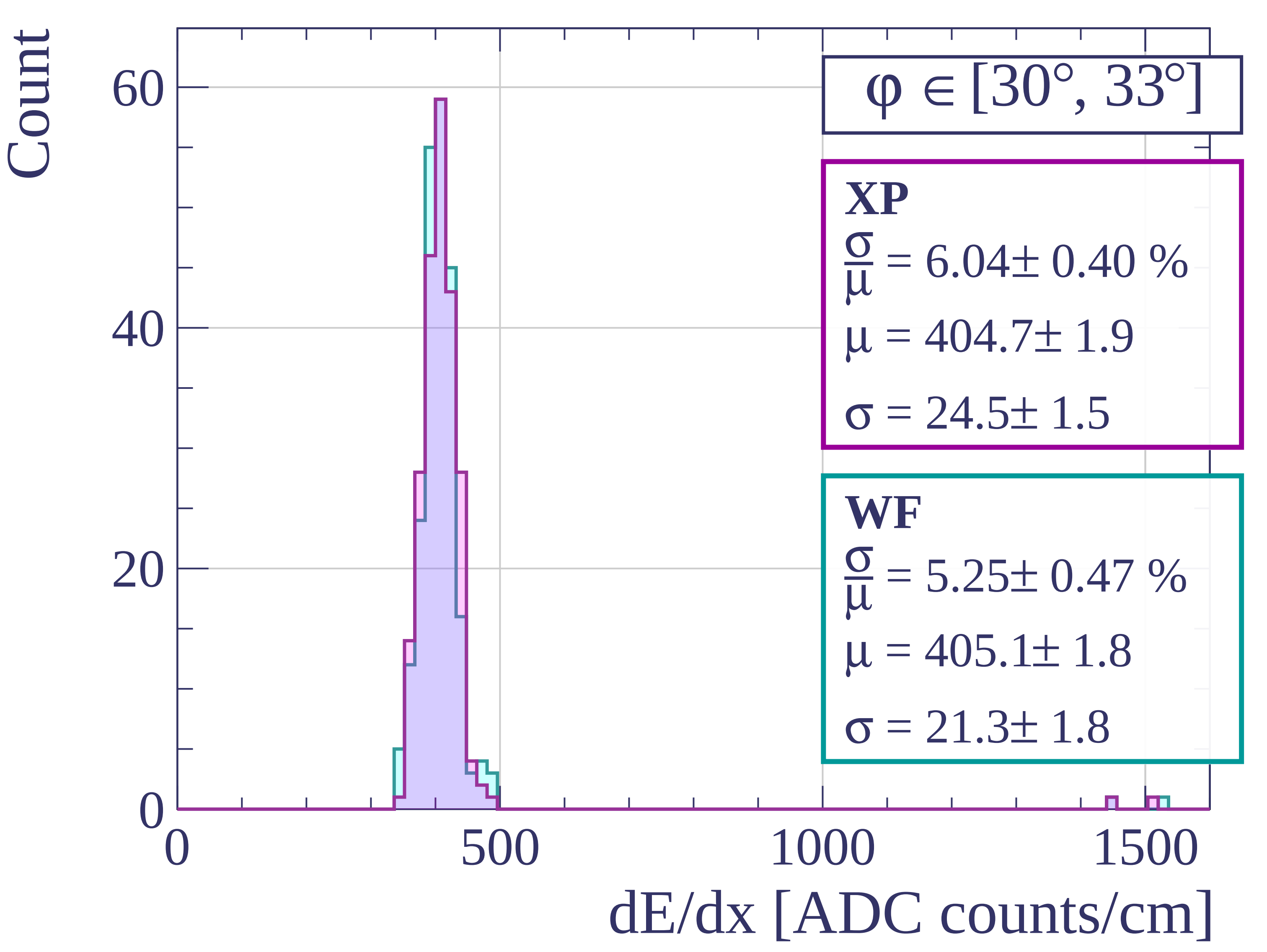

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [21°, 24°]
XP
σ/μ = 5.95± 0.33 %
μ = 406.1± 1.7
σ = 24.2± 1.2
WF
σ/μ = 6.73± 0.48 %
μ = 408.5± 2.0
σ = 27.5± 1.8



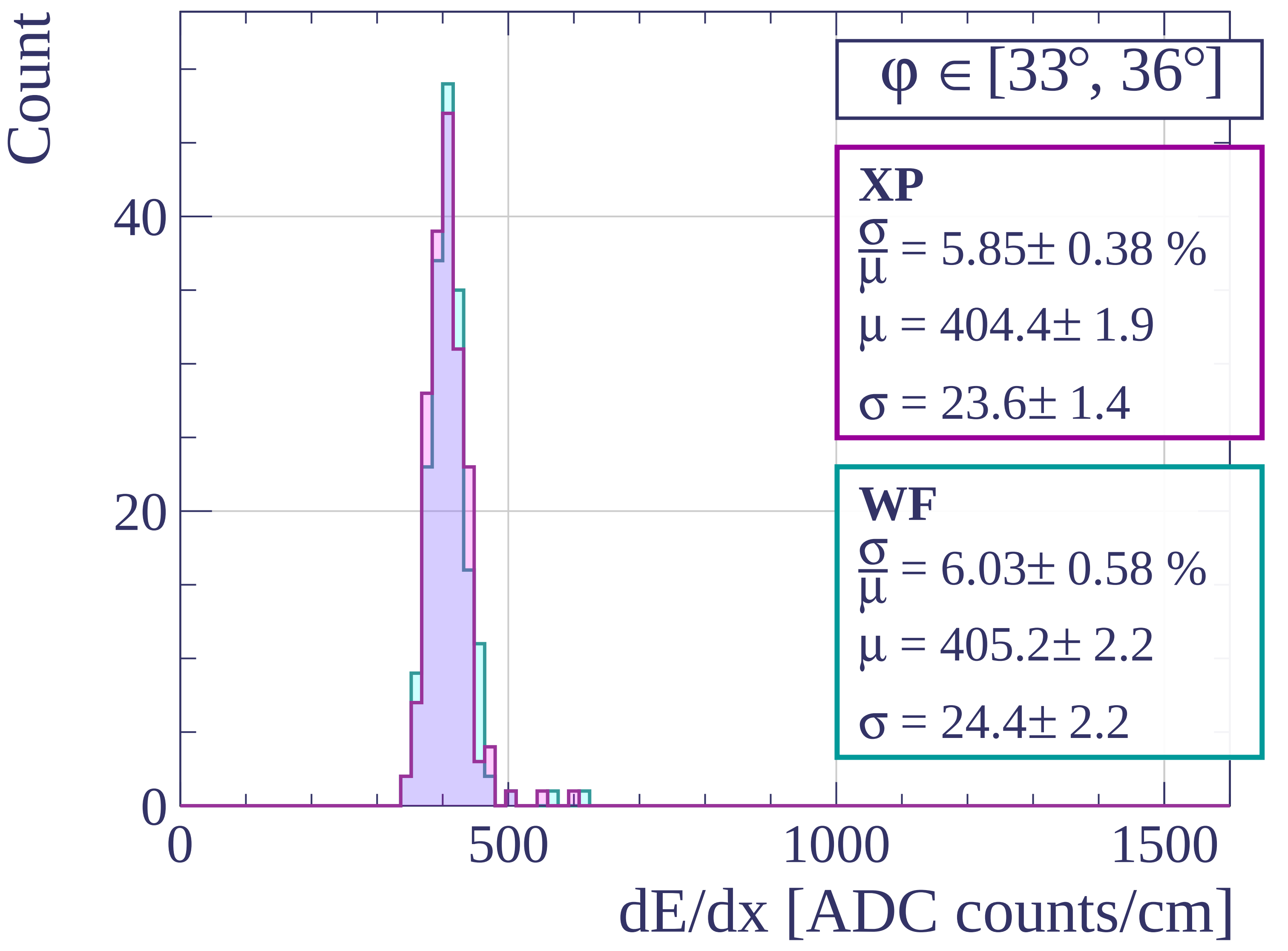
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [24°, 27°]
XP
σ/μ = 6.01± 0.43 %
μ = 408.2± 1.9
σ = 24.5± 1.6
WF
σ/μ = 6.25± 0.45 %
μ = 411.3± 2.0
σ = 25.7± 1.7



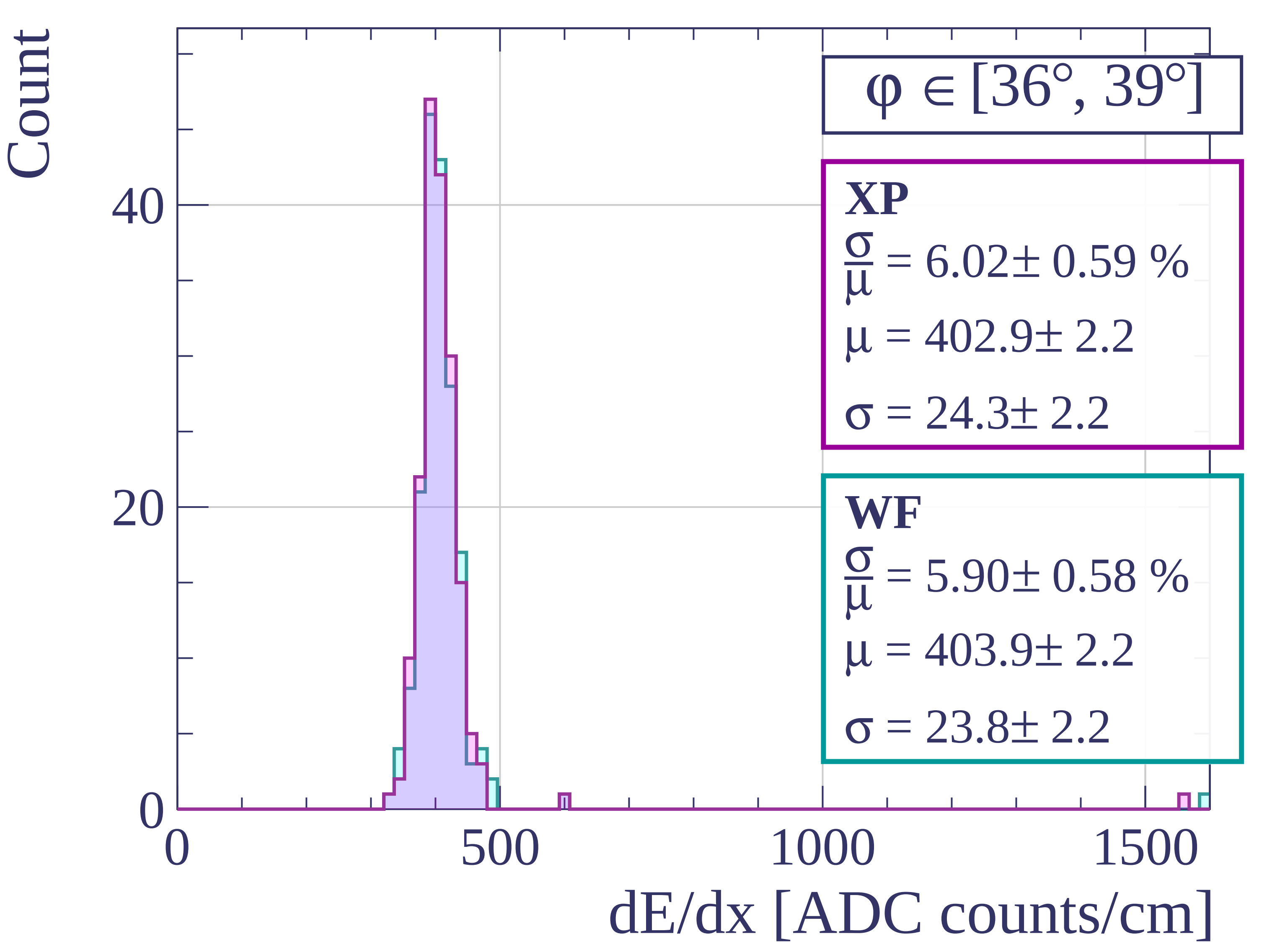
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [27°, 30°]
XP
σ/μ = 5.09± 0.55 %
μ = 404.9± 1.9
σ = 20.6± 2.1
WF
σ/μ = 7.33± 0.68 %
μ = 404.5± 2.5
σ = 29.6± 2.6



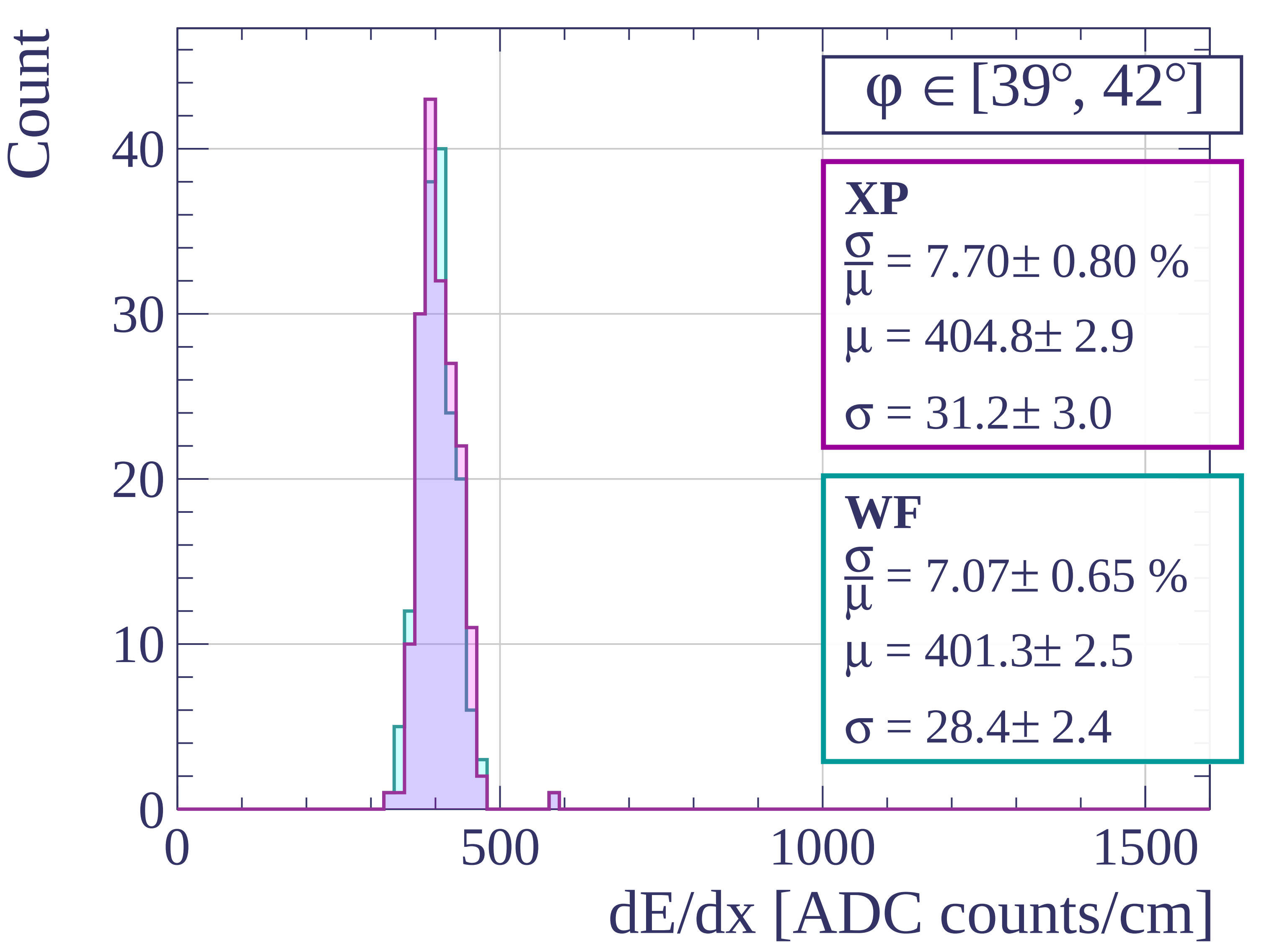
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [30°, 33°]
XP
σ/μ = 6.04± 0.40 %
μ = 404.7± 1.9
σ = 24.5± 1.5
WF
σ/μ = 5.25± 0.47 %
μ = 405.1± 1.8
σ = 21.3± 1.8



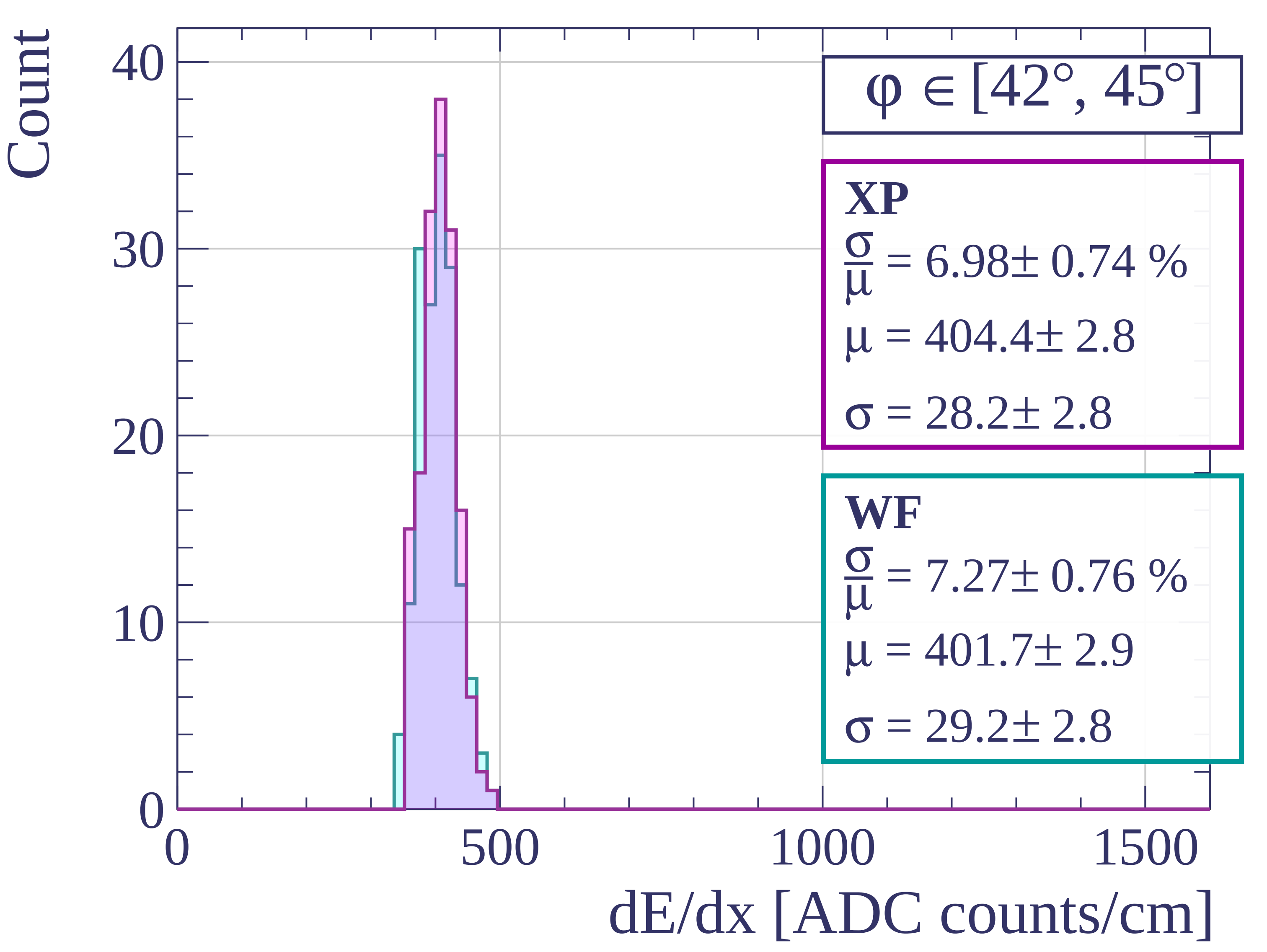
Count
dE/dx [ADC counts/cm]
φ ∈ [33°, 36°]
XP
σ/μ = 5.85± 0.38 %
μ = 404.4± 1.9
σ = 23.6± 1.4
WF
σ/μ = 6.03± 0.58 %
μ = 405.2± 2.2
σ = 24.4± 2.2
0
20
40
0
500
1000
1500



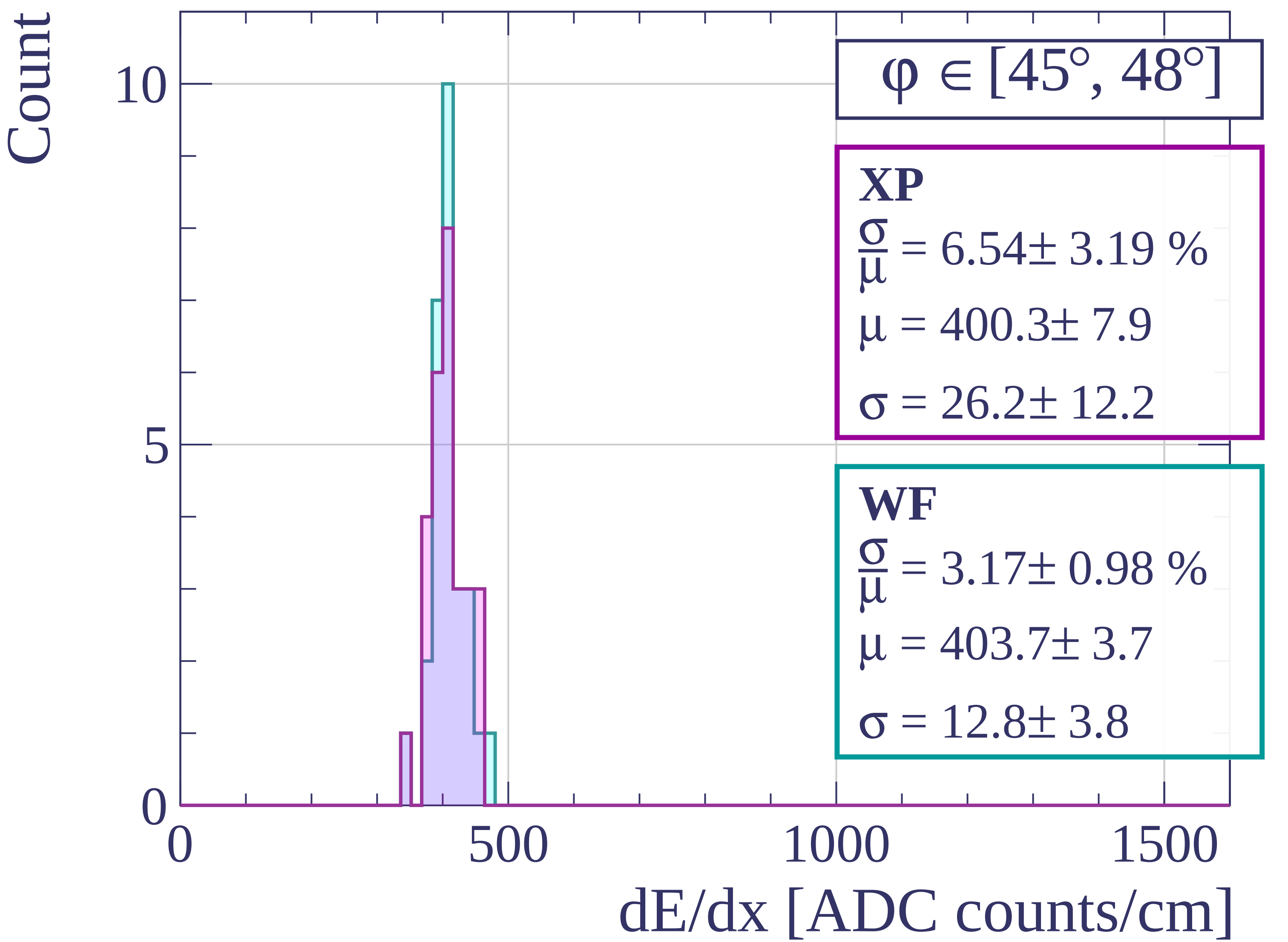
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [36°, 39°]
XP
σ/μ = 6.02± 0.59 %
μ = 402.9± 2.2
σ = 24.3± 2.2
WF
σ/μ = 5.90± 0.58 %
μ = 403.9± 2.2
σ = 23.8± 2.2



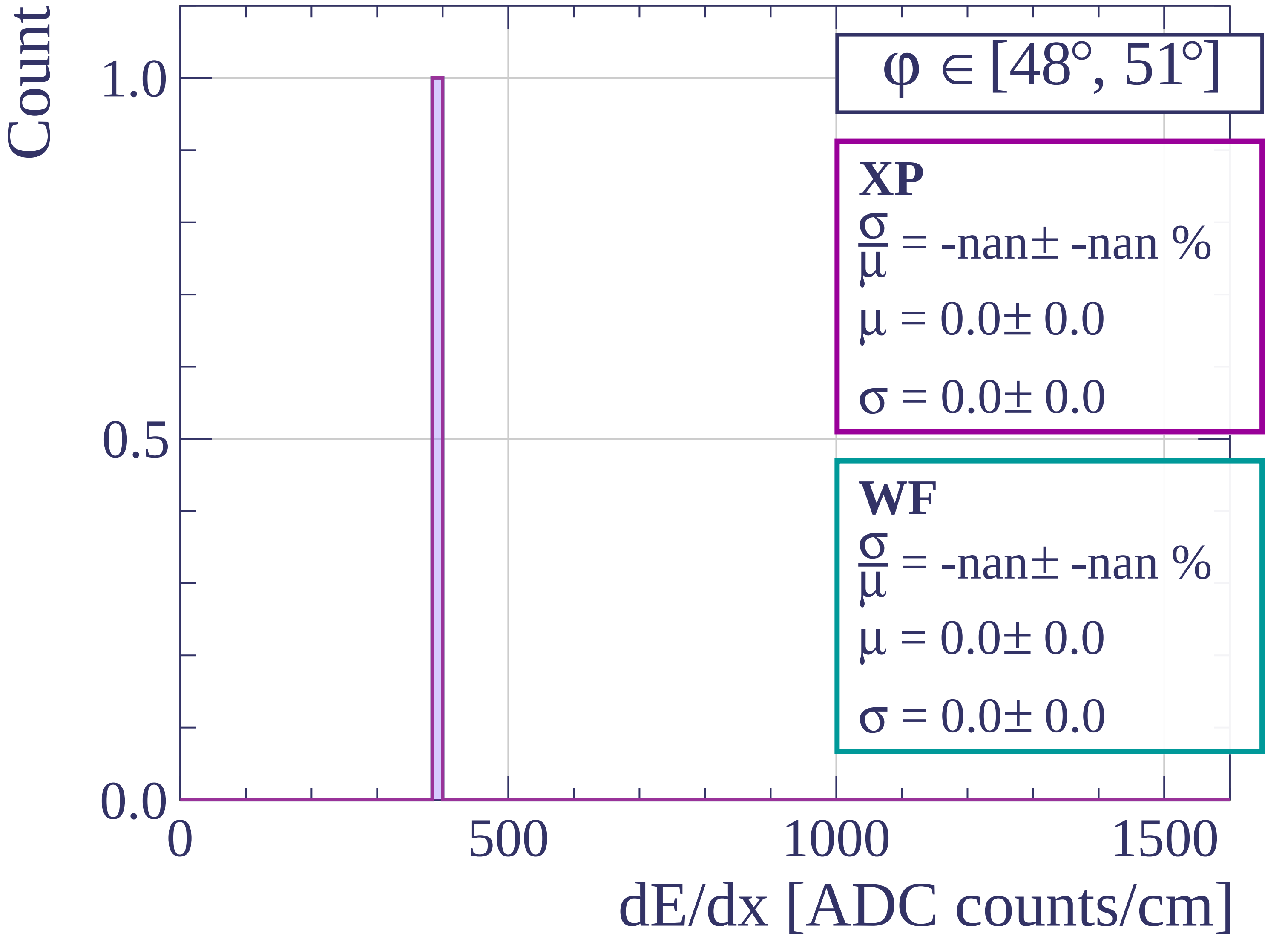
Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [39°, 42°]
XP
σ/μ = 7.70± 0.80 %
μ = 404.8± 2.9
σ = 31.2± 3.0
WF
σ/μ = 7.07± 0.65 %
μ = 401.3± 2.5
σ = 28.4± 2.4



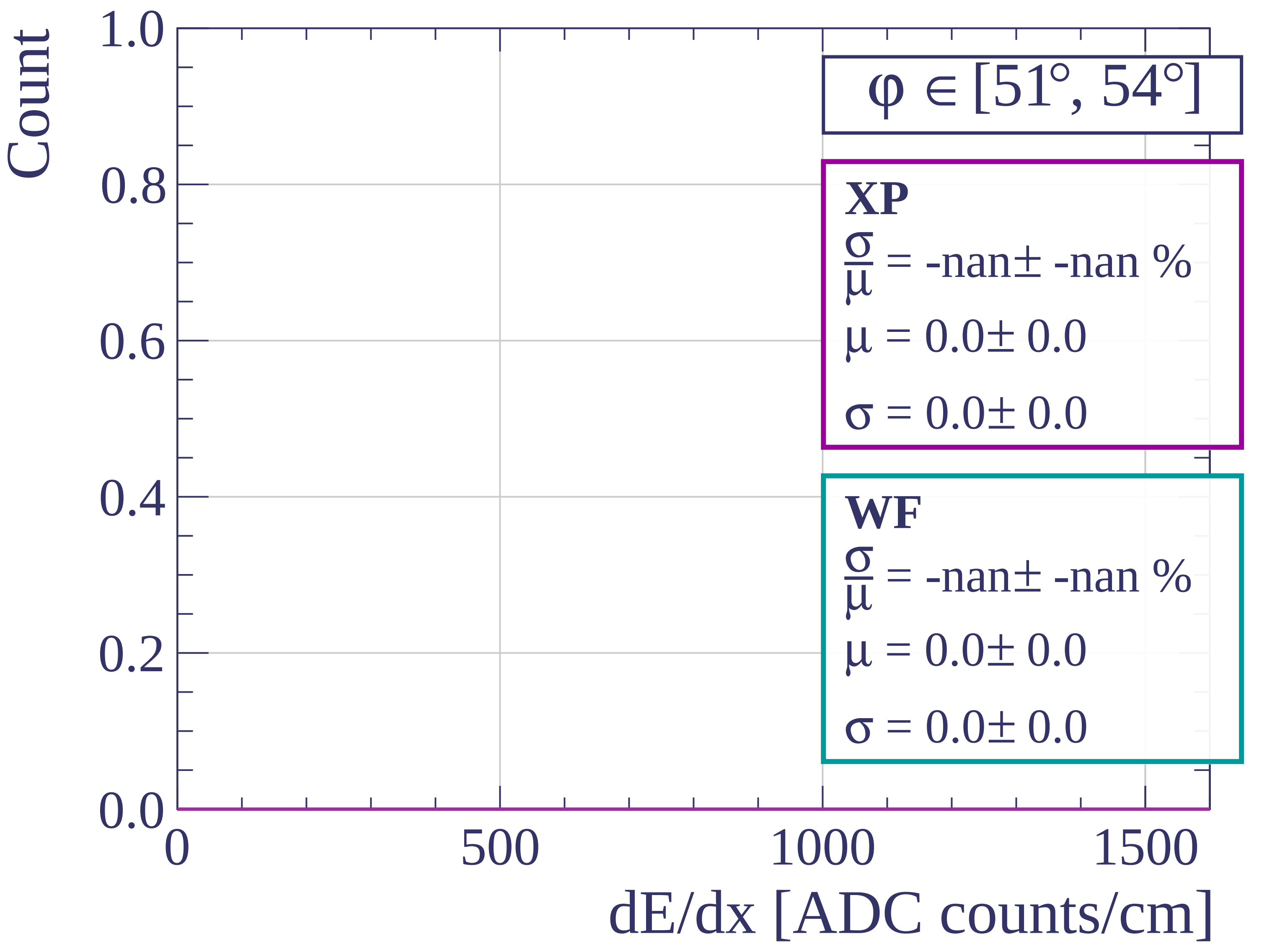
Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [42°, 45°]
XP
σ/μ = 6.98± 0.74 %
μ = 404.4± 2.8
σ = 28.2± 2.8
WF
σ/μ = 7.27± 0.76 %
μ = 401.7± 2.9
σ = 29.2± 2.8



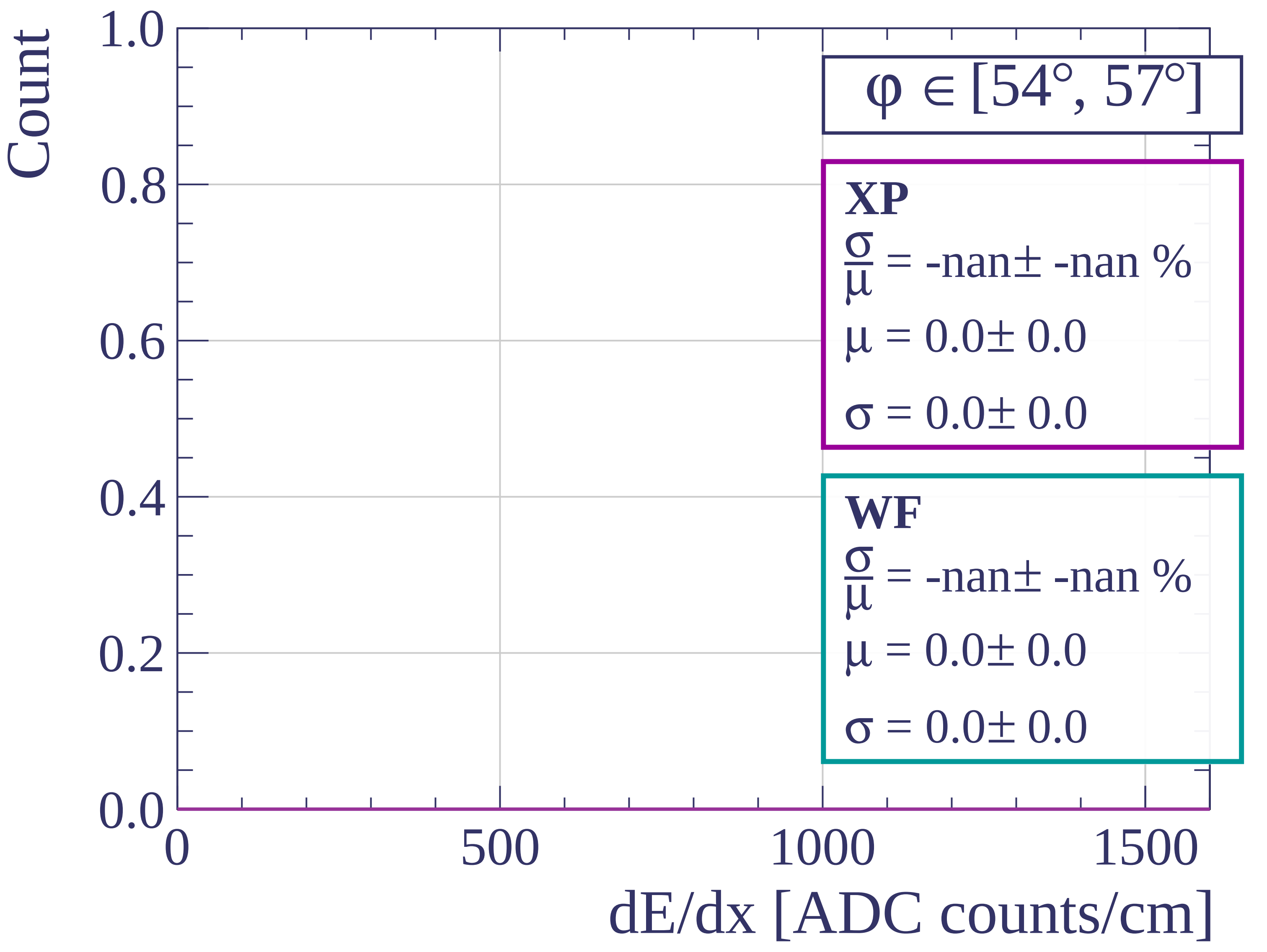
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [45°, 48°]
XP
σ/μ = 6.54± 3.19 %
μ = 400.3± 7.9
σ = 26.2± 12.2
WF
σ/μ = 3.17± 0.98 %
μ = 403.7± 3.7
σ = 12.8± 3.8



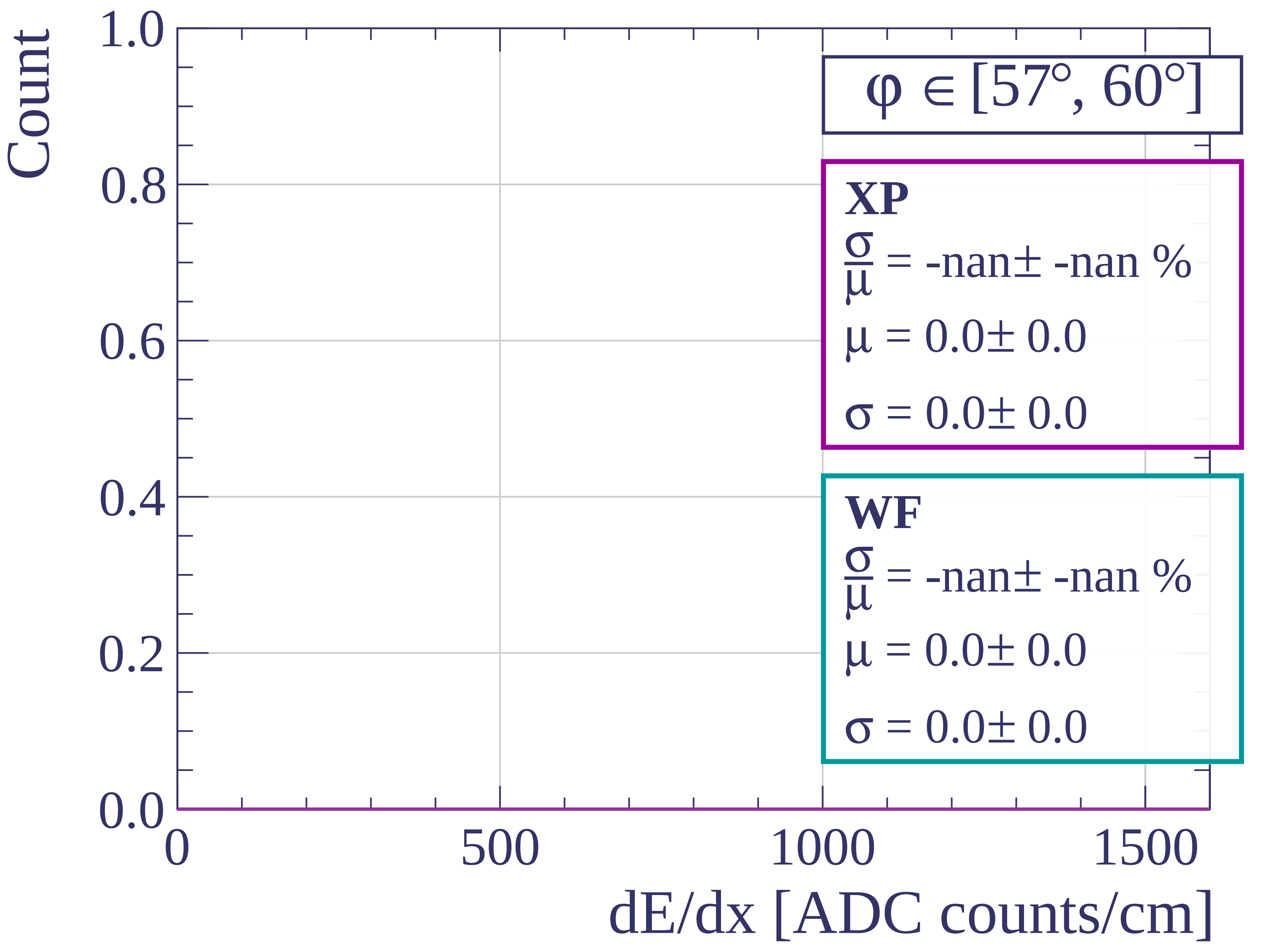
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [48°, 51°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



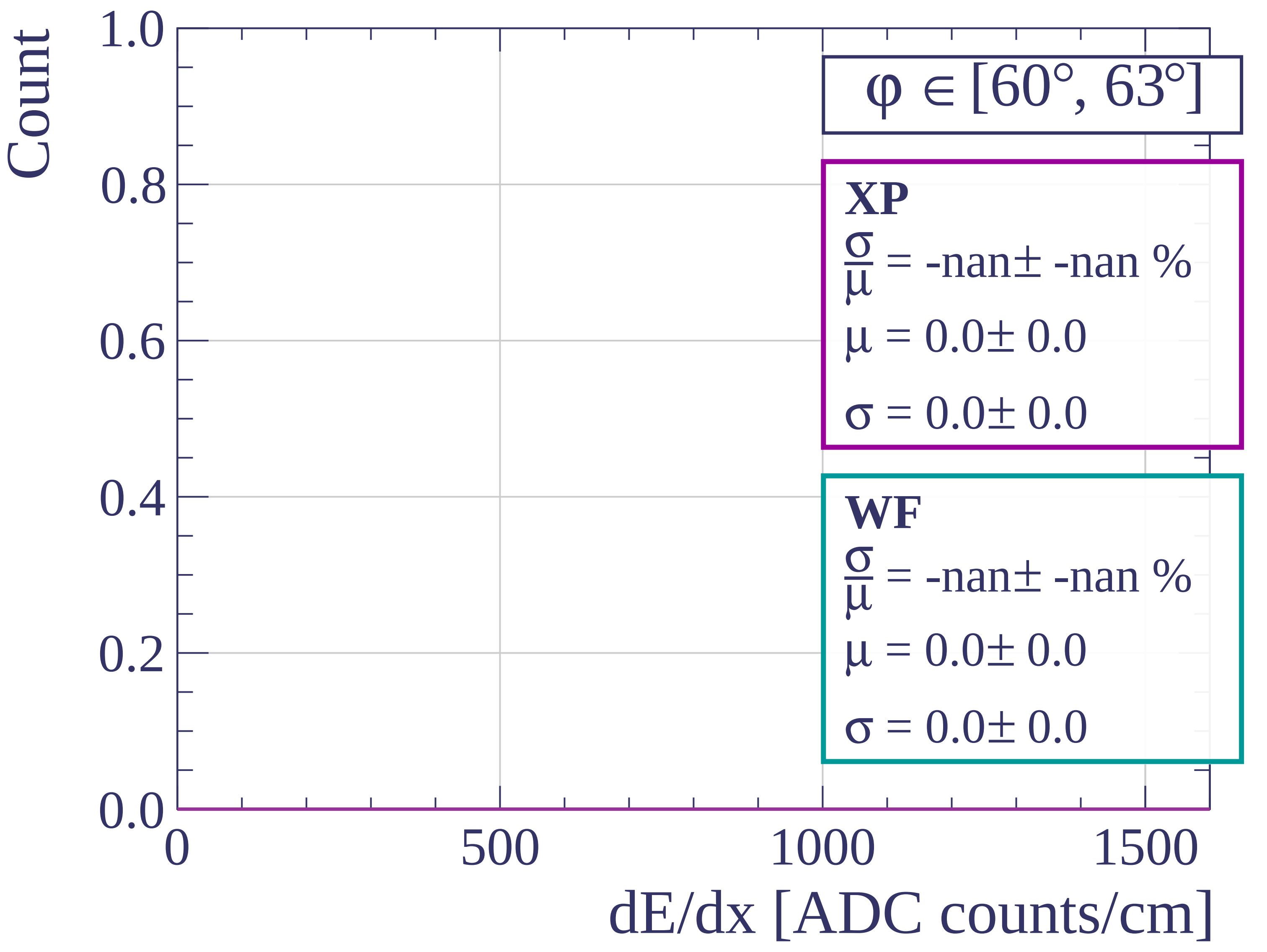
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [51°, 54°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



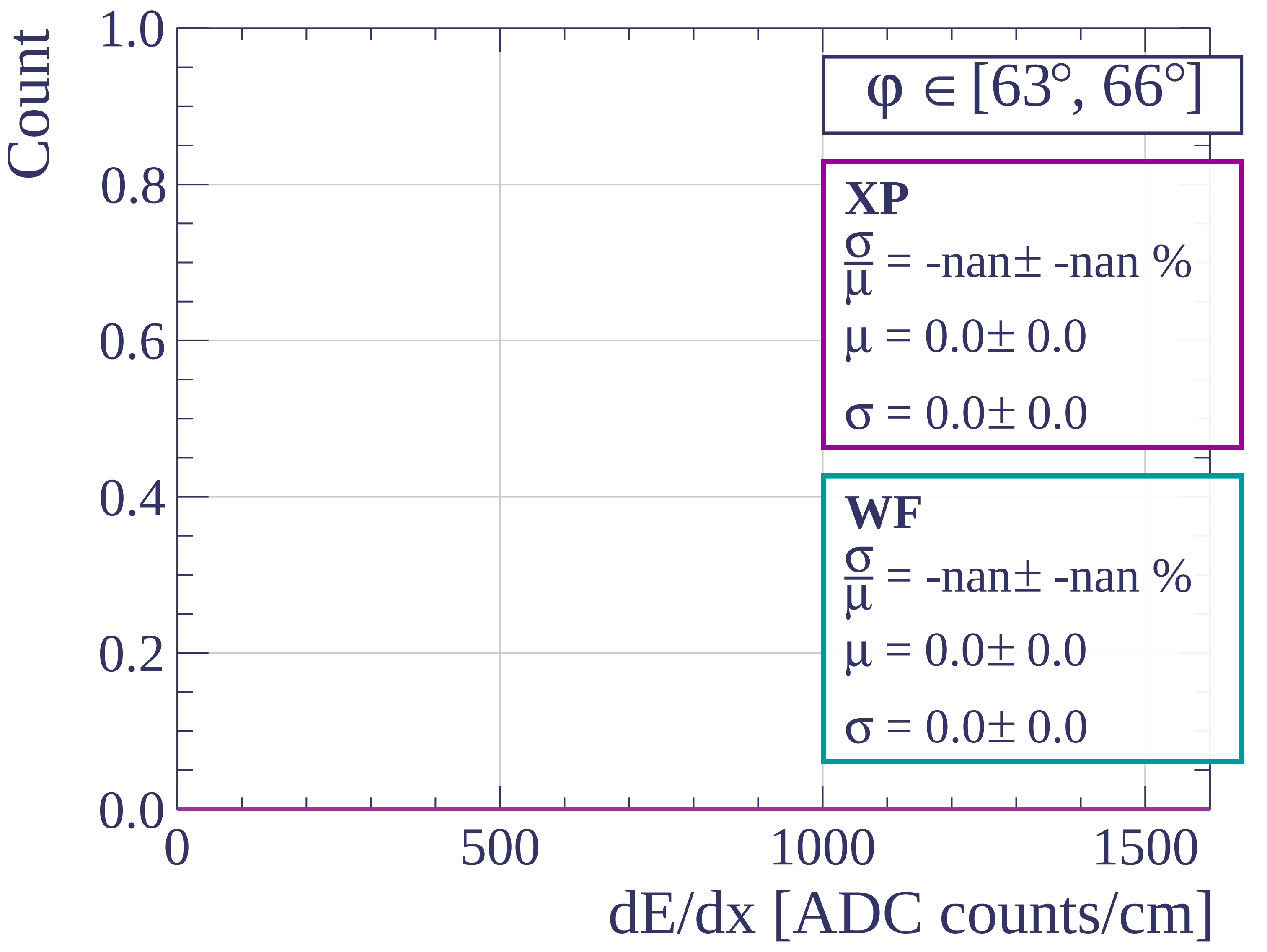
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [54°, 57°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



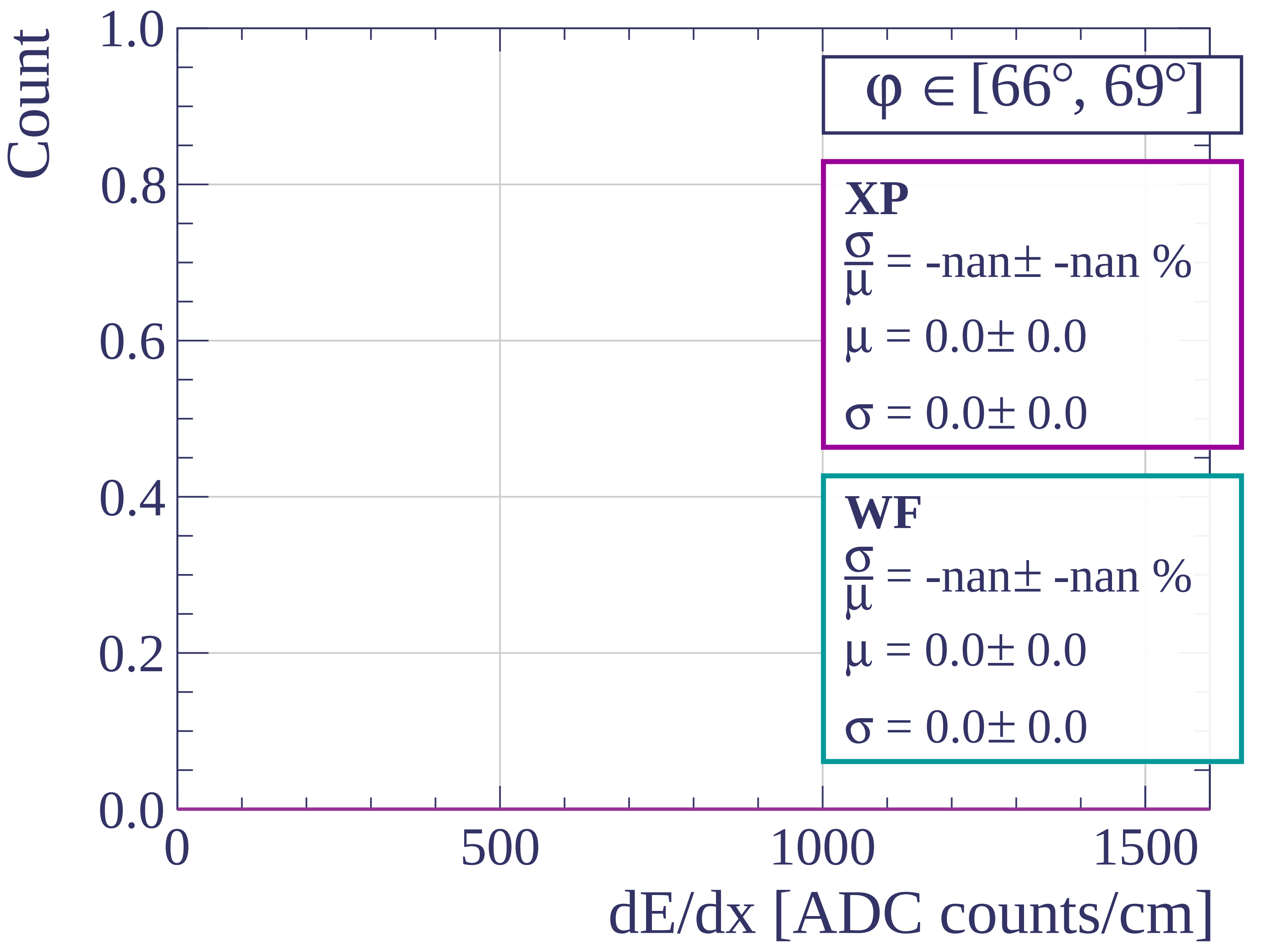
φ ∈ [57°, 60°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0.0
0.2
0.4
0.6
0.8
1.0

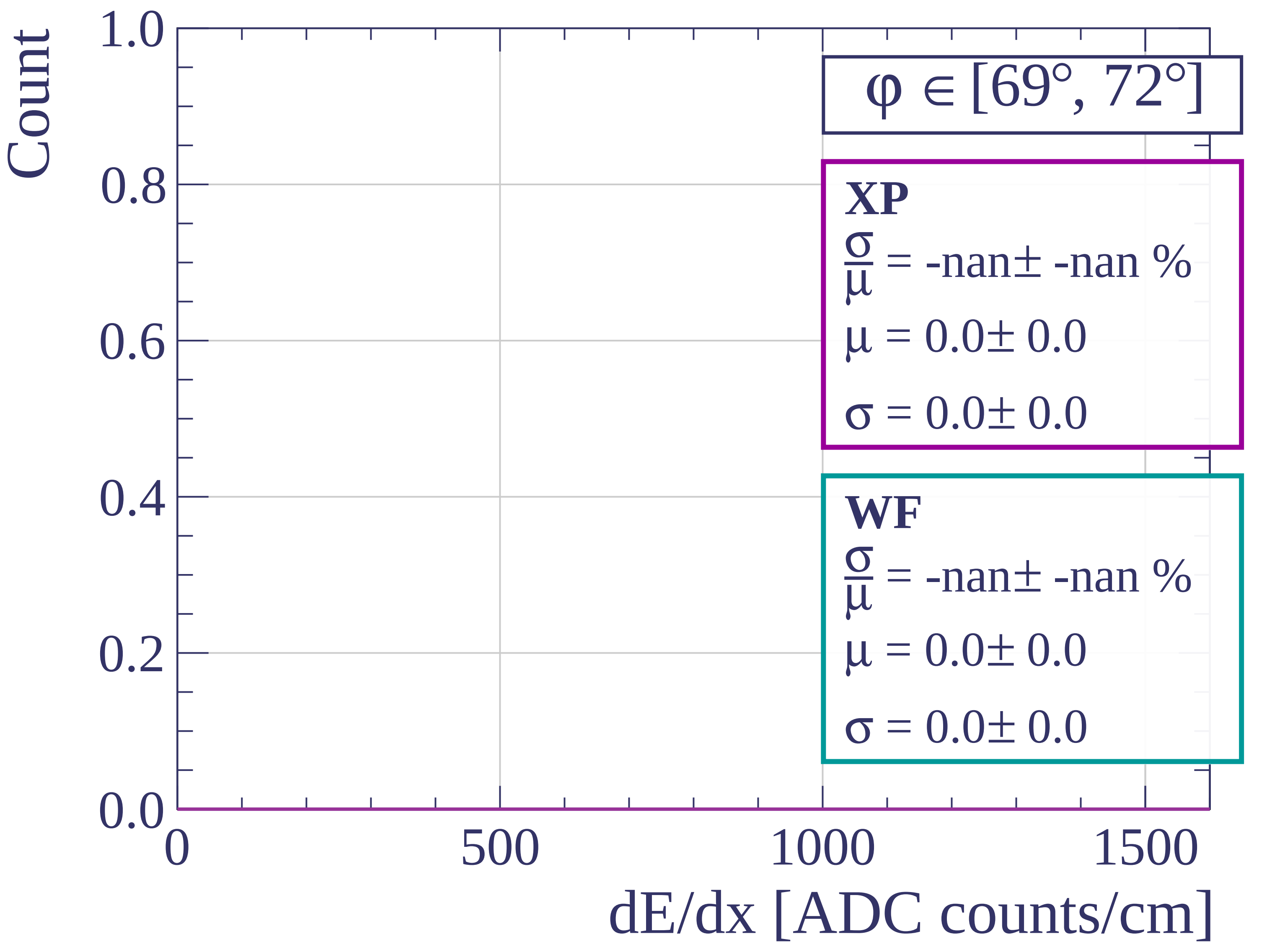


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [60°, 63°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

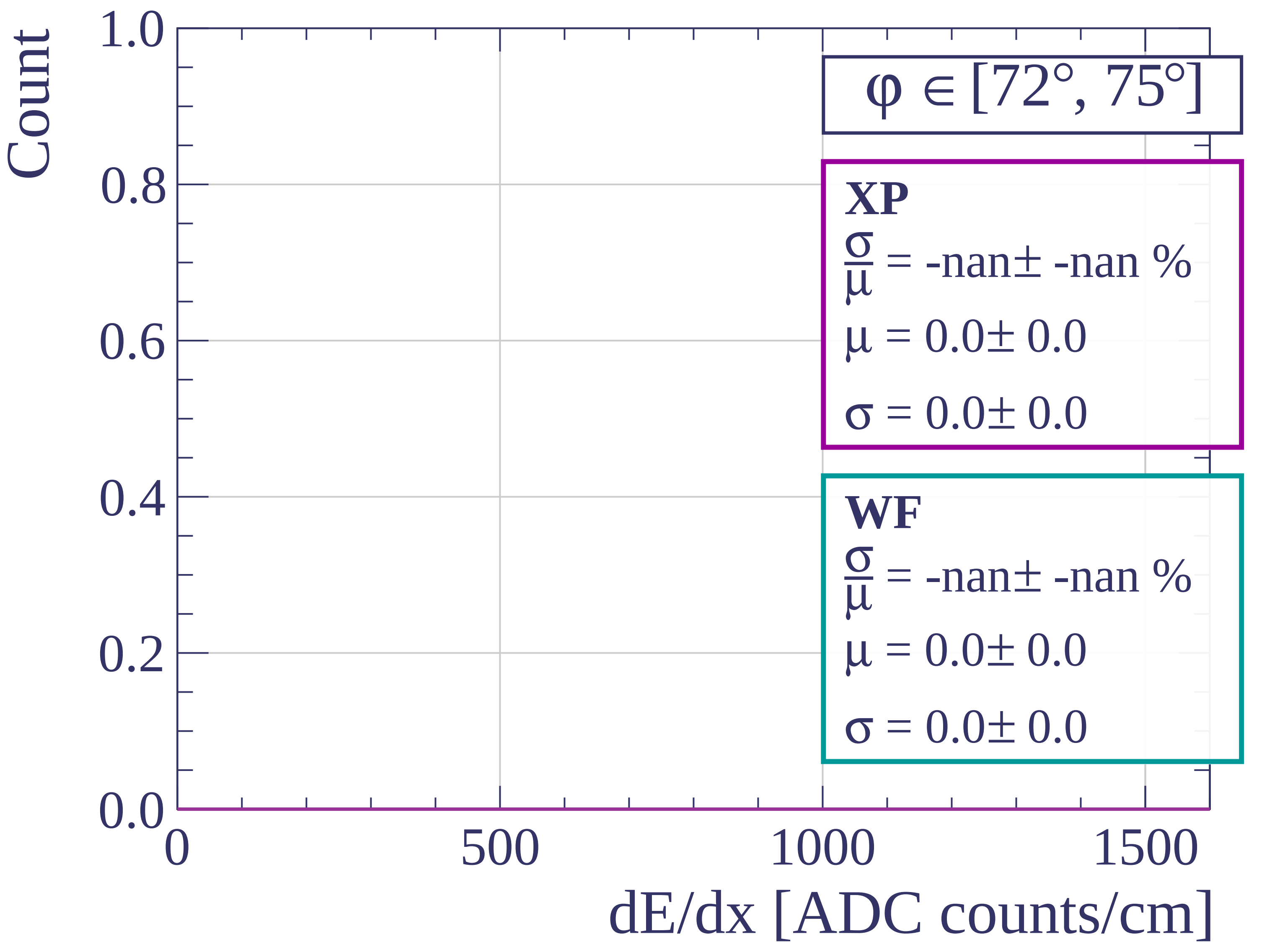


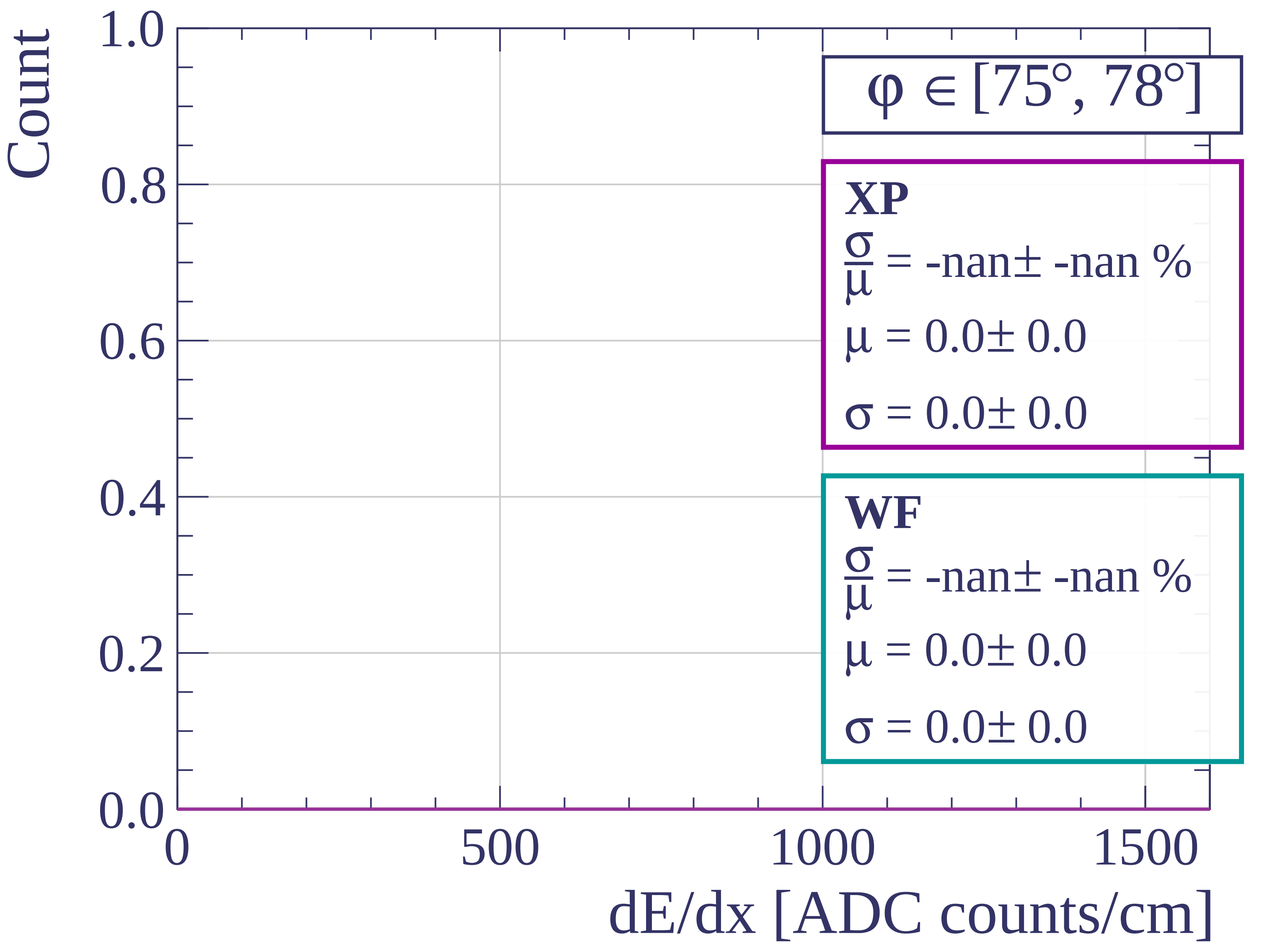
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [63°, 66°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [66°, 69°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

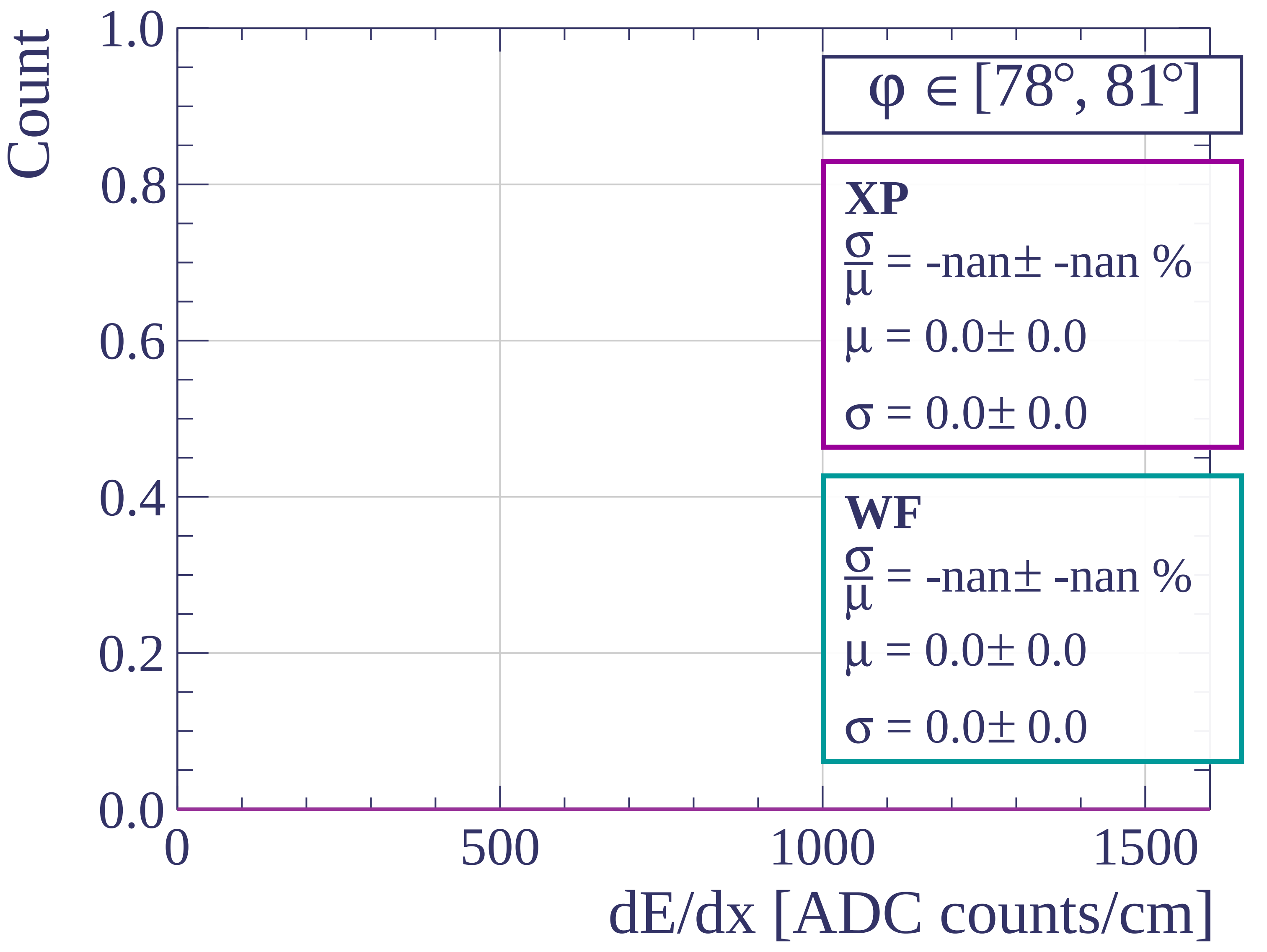


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [69°, 72°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

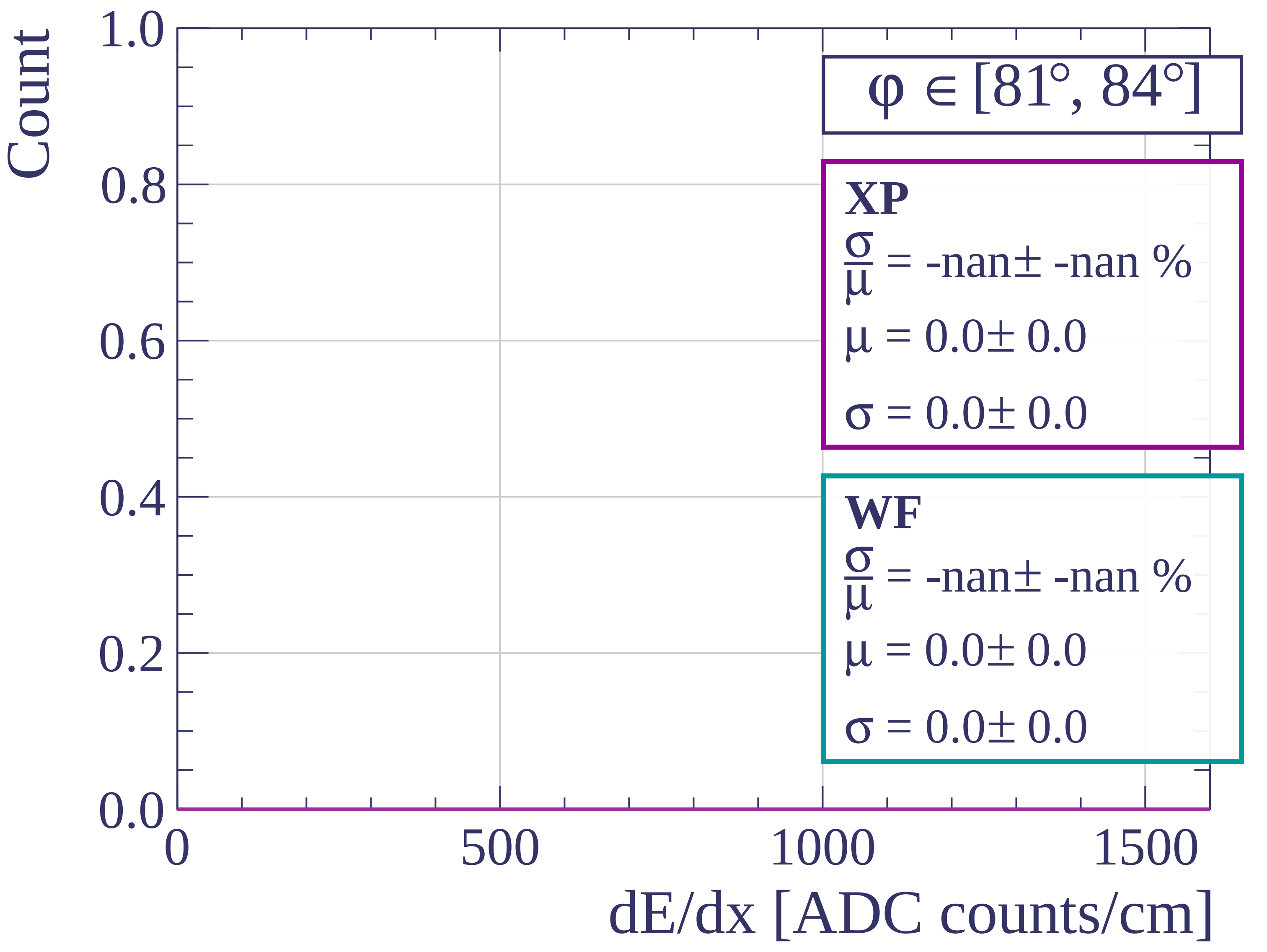

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [72°, 75°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

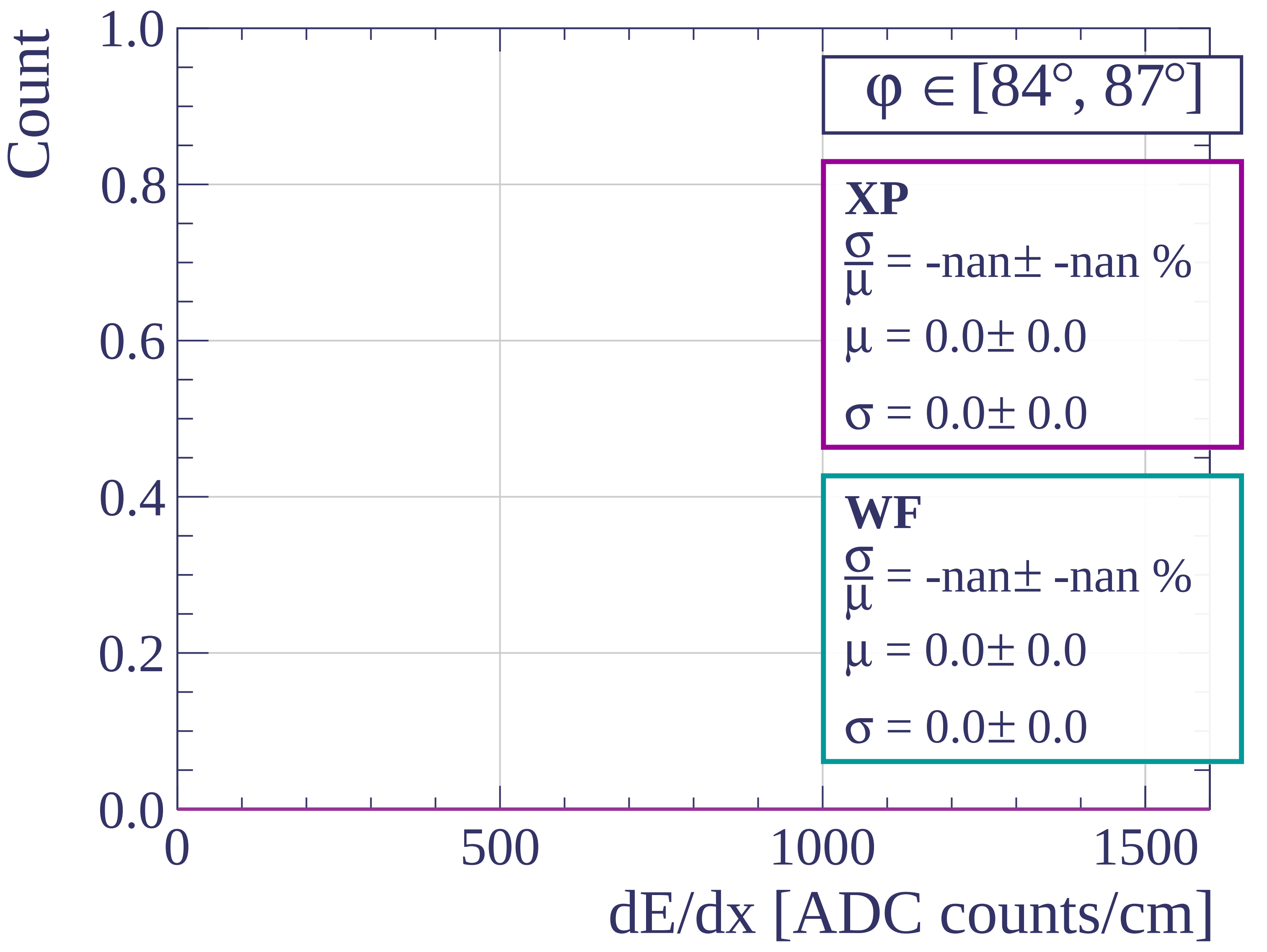


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [75°, 78°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

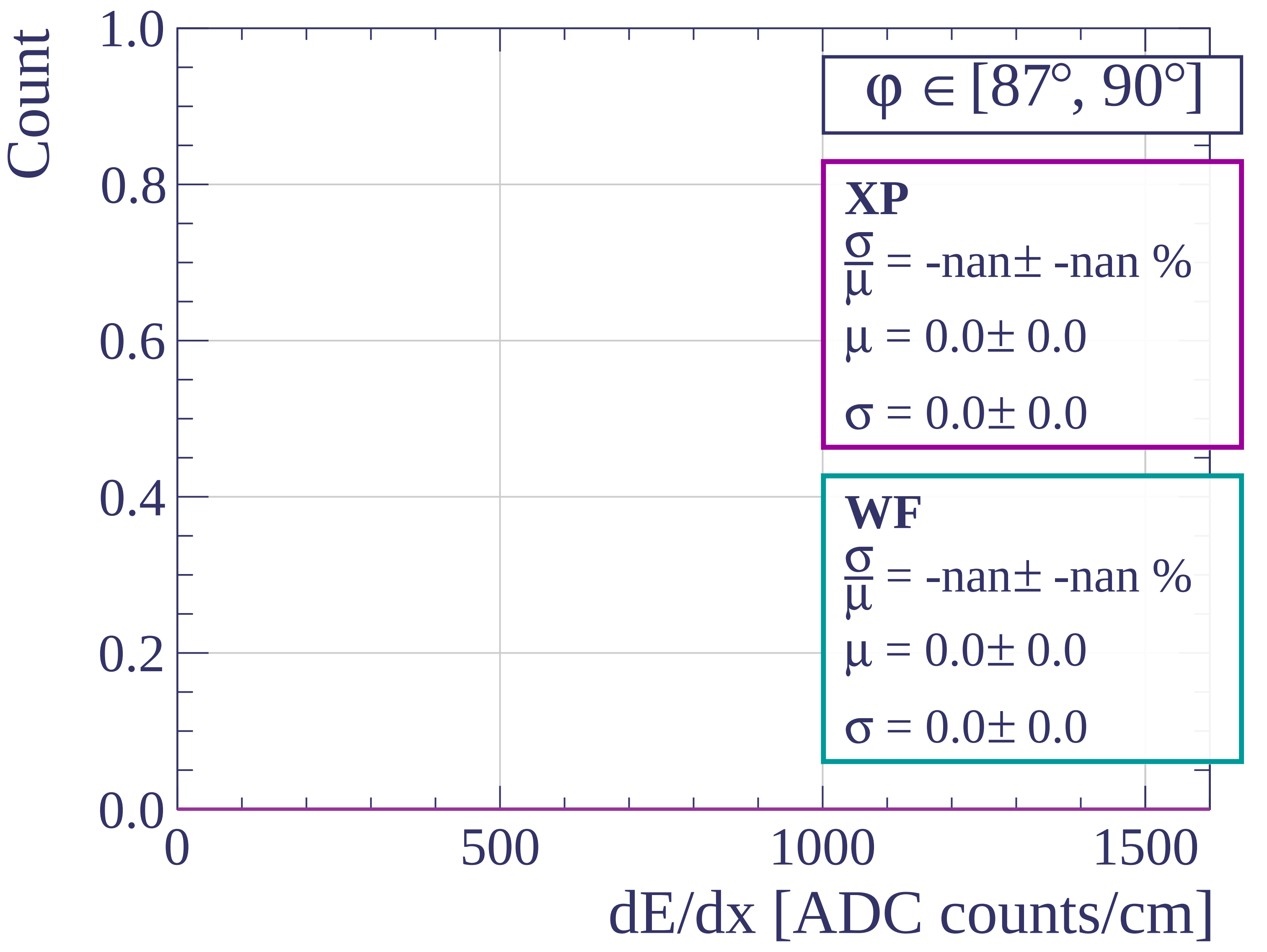


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [78°, 81°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

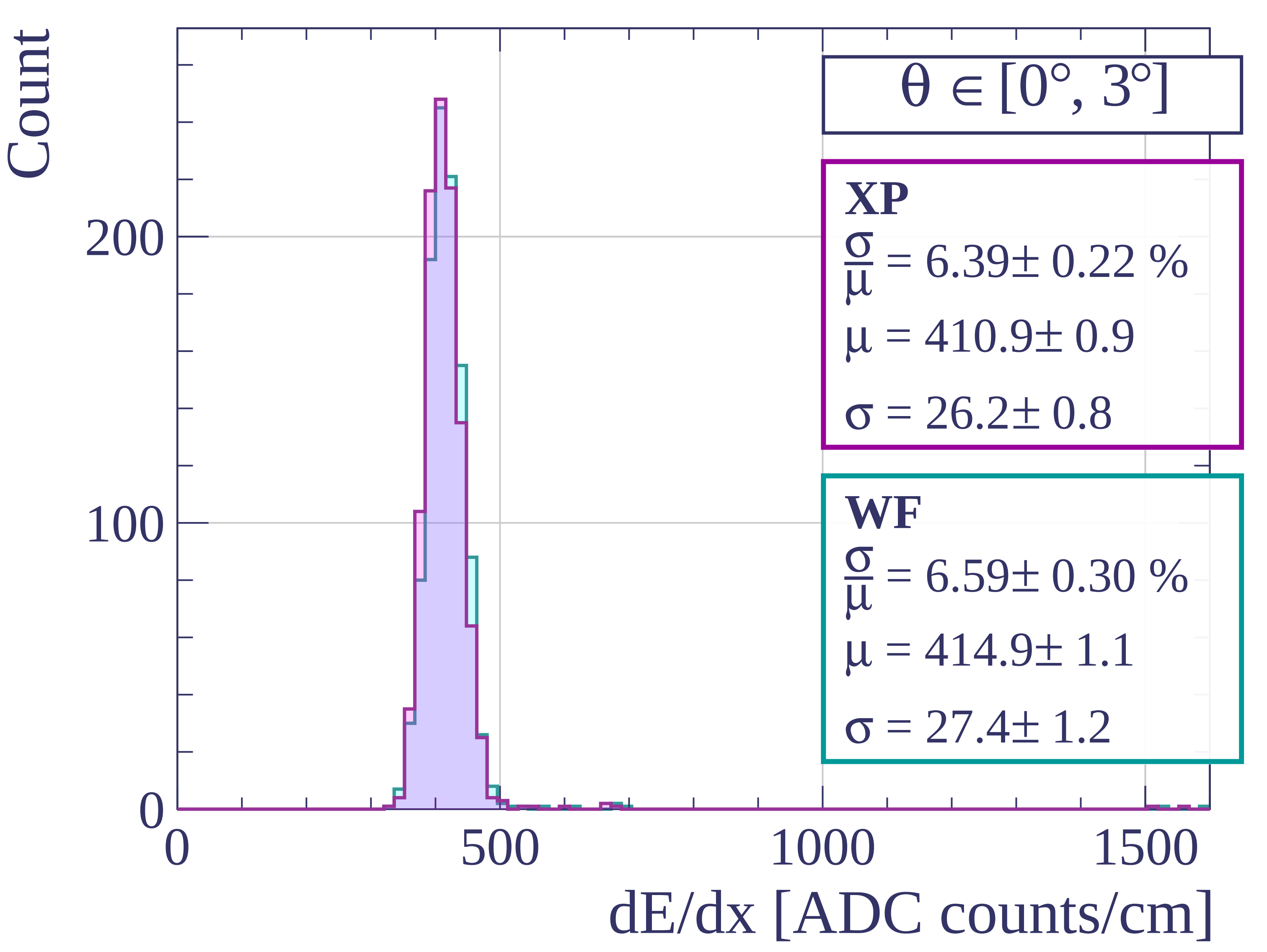

φ ∈ [81°, 84°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
Count
dE/dx [ADC counts/cm]
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500



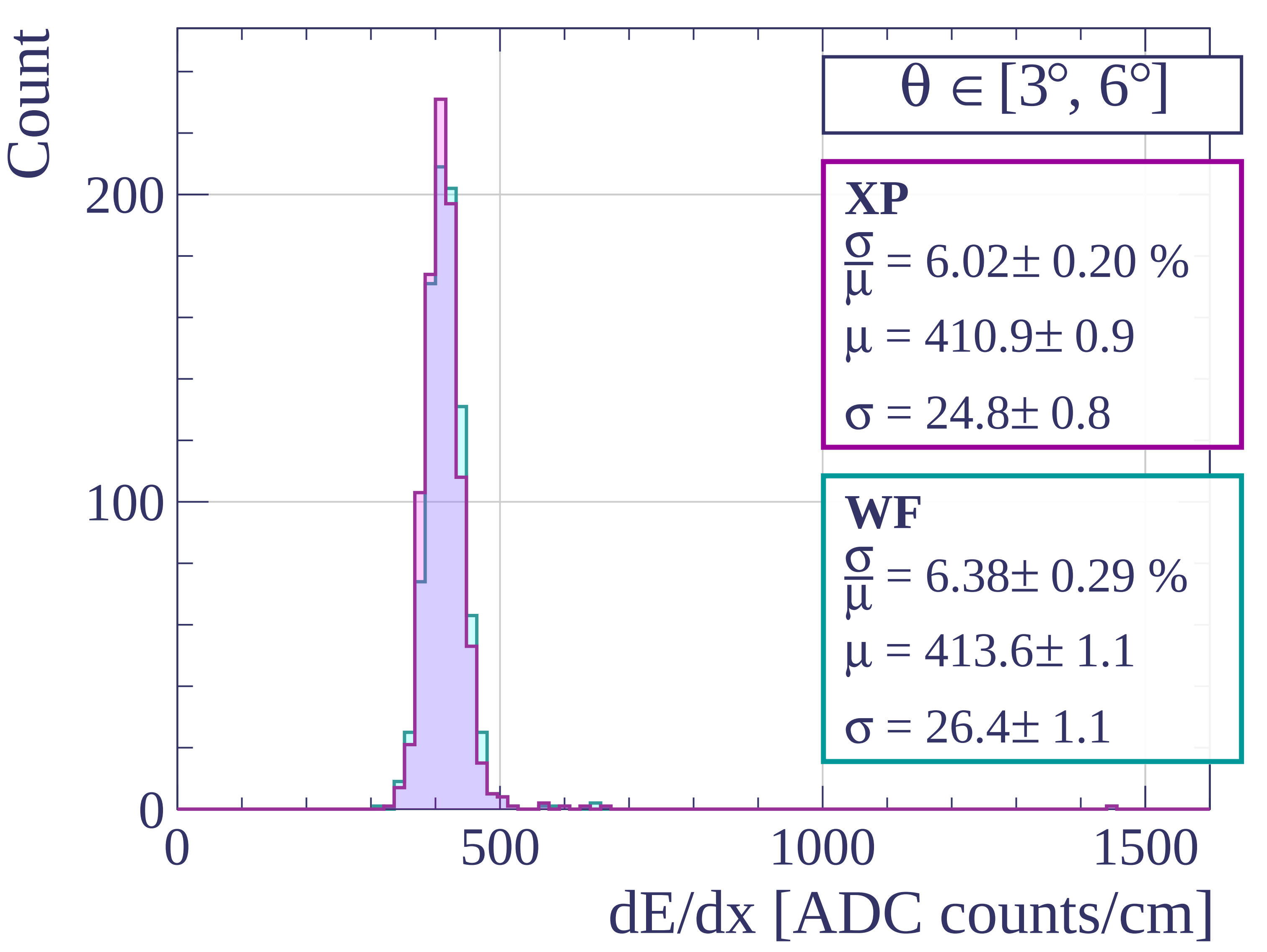
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [84°, 87°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



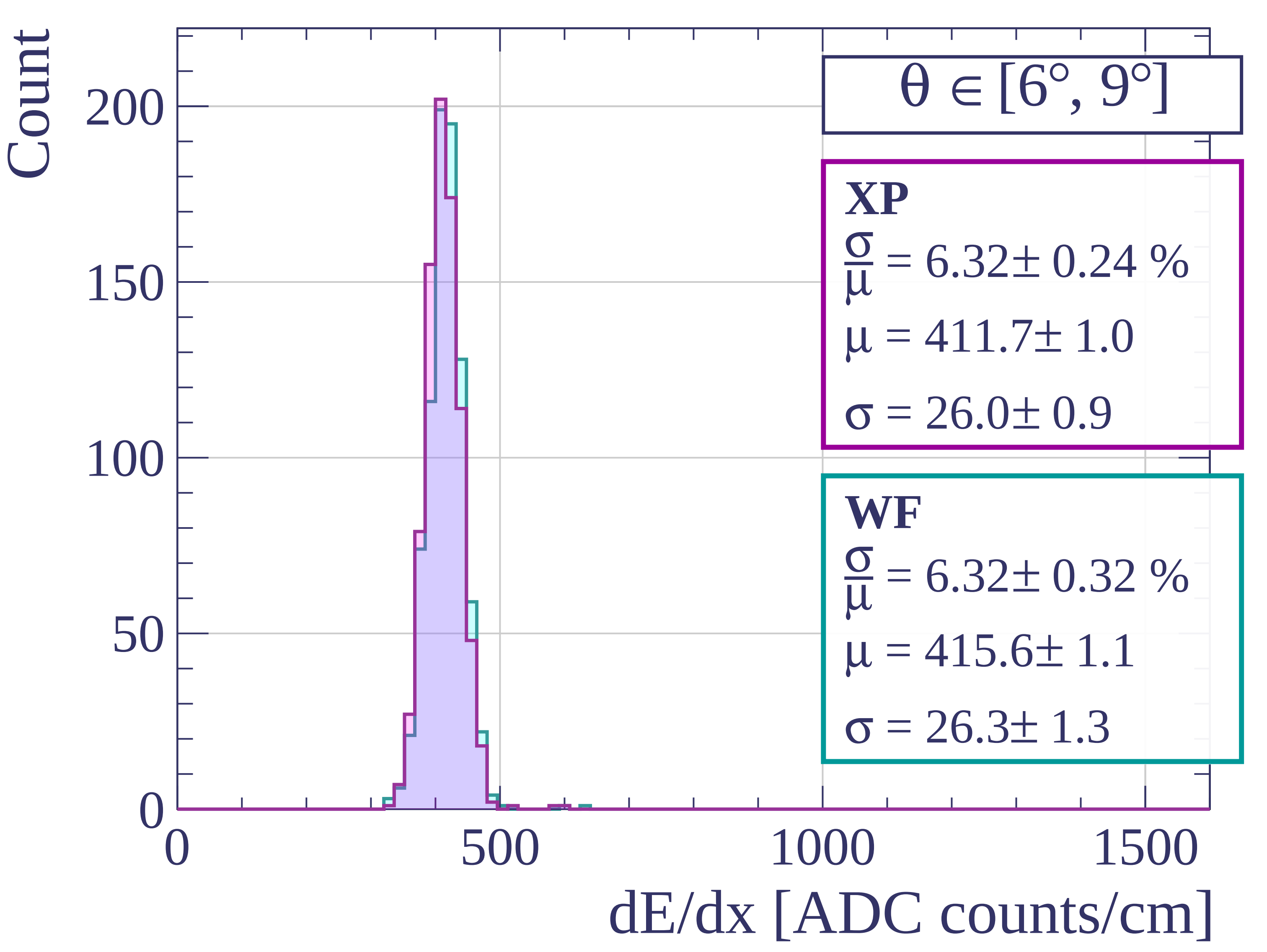
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [87°, 90°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

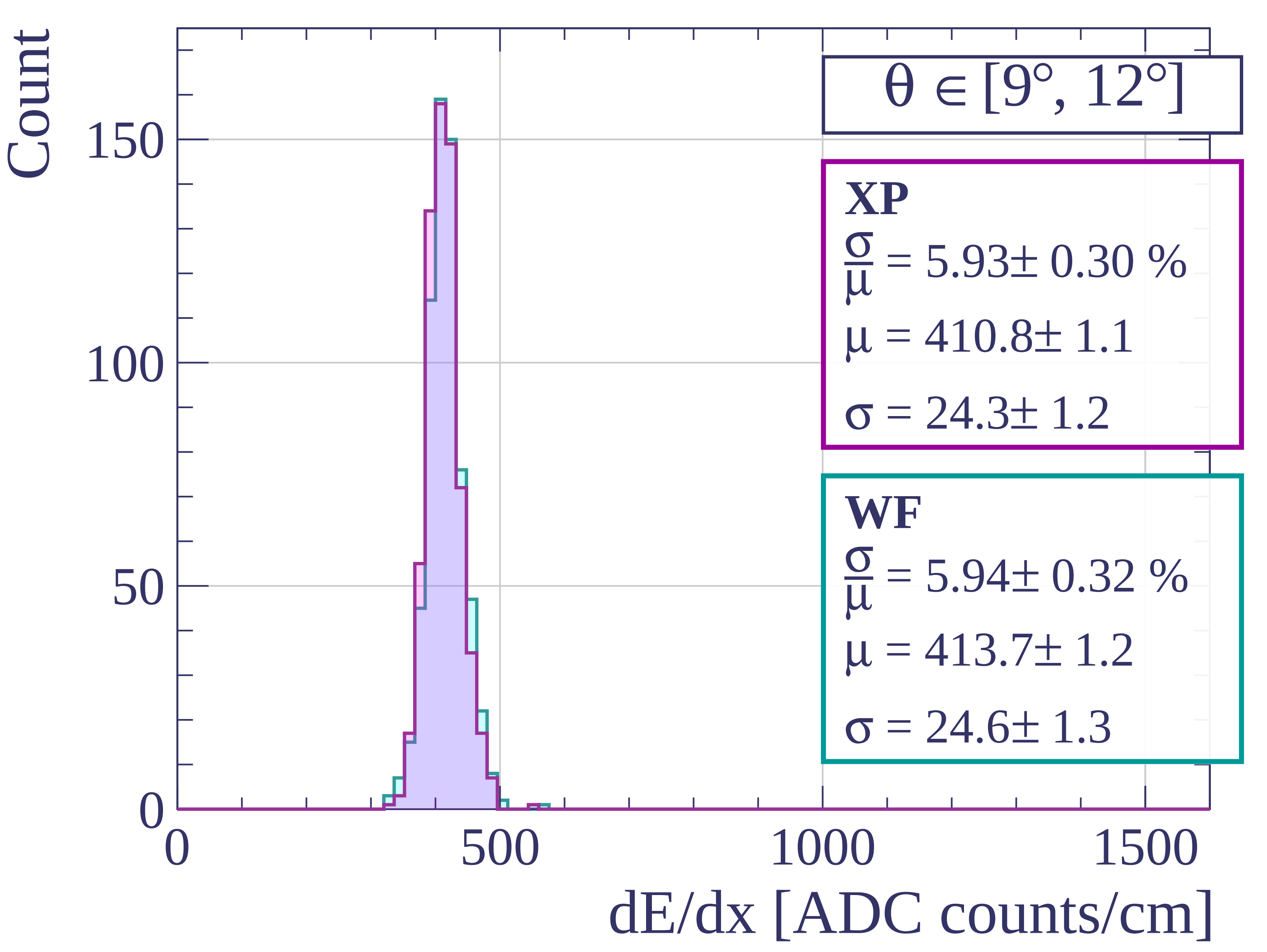


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
$\theta \in [0°, 3°]$
XP
$\frac{\sigma}{\mu} = 6.39 \pm 0.22$ %
$\mu = 410.9 \pm 0.9$
$\sigma = 26.2 \pm 0.8$
WF
$\frac{\sigma}{\mu} = 6.59 \pm 0.30$ %
$\mu = 414.9 \pm 1.1$
$\sigma = 27.4 \pm 1.2$

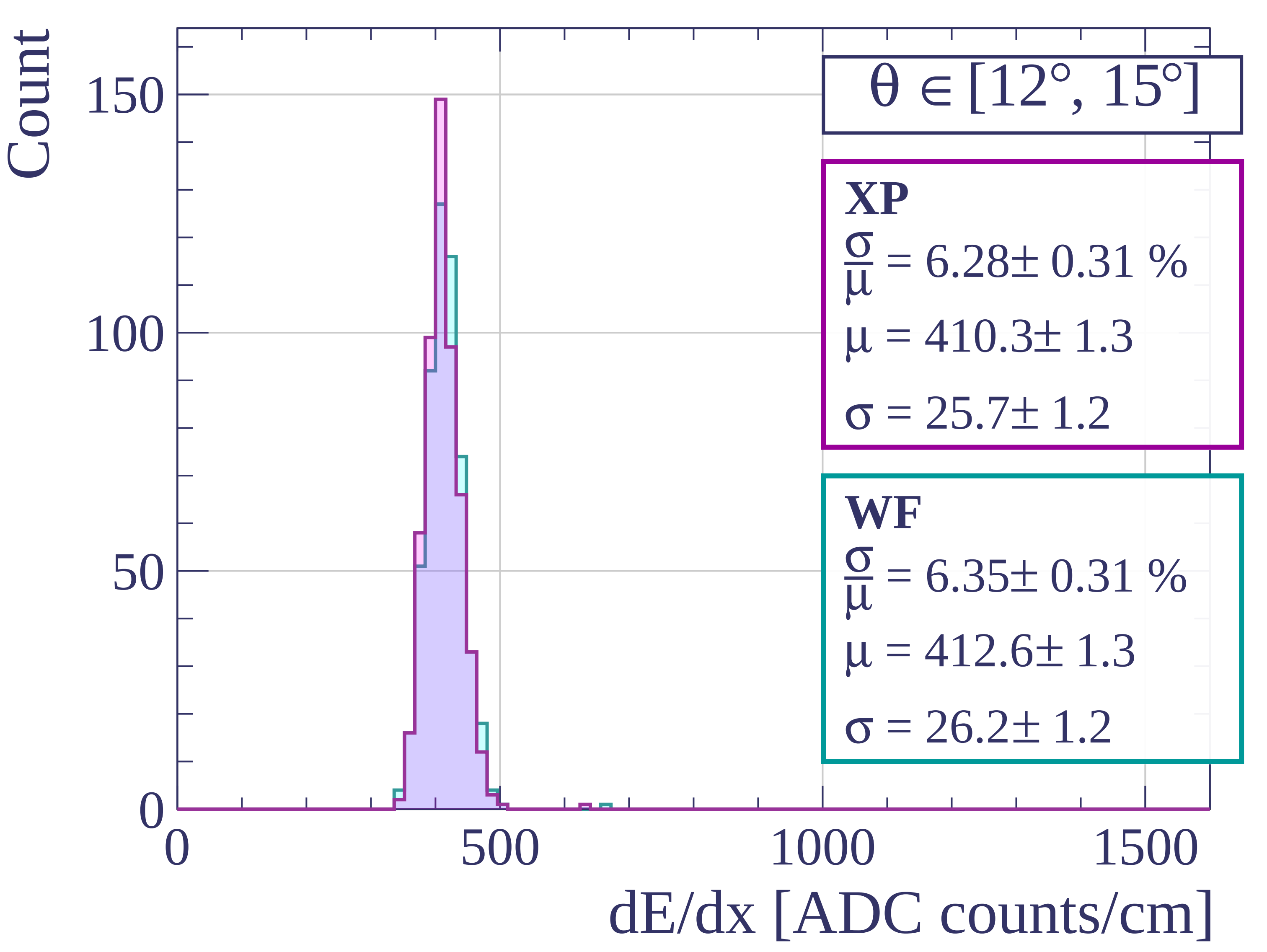


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [3°, 6°]
XP
σ/μ = 6.02± 0.20 %
μ = 410.9± 0.9
σ = 24.8± 0.8
WF
σ/μ = 6.38± 0.29 %
μ = 413.6± 1.1
σ = 26.4± 1.1

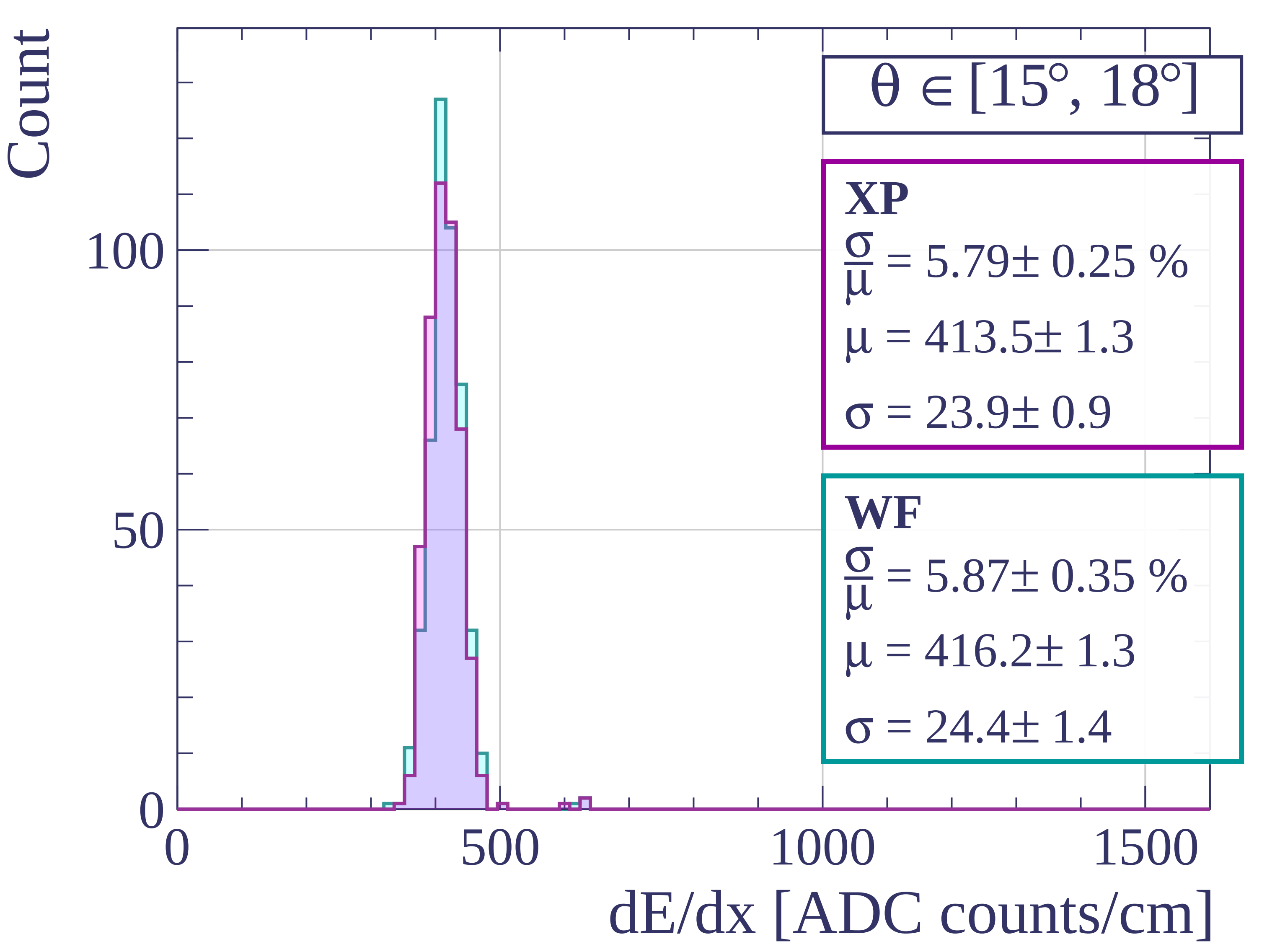

Count
0
50
100
150
200
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [6°, 9°]
XP
σ/μ = 6.32± 0.24 %
μ = 411.7± 1.0
σ = 26.0± 0.9
WF
σ/μ = 6.32± 0.32 %
μ = 415.6± 1.1
σ = 26.3± 1.3



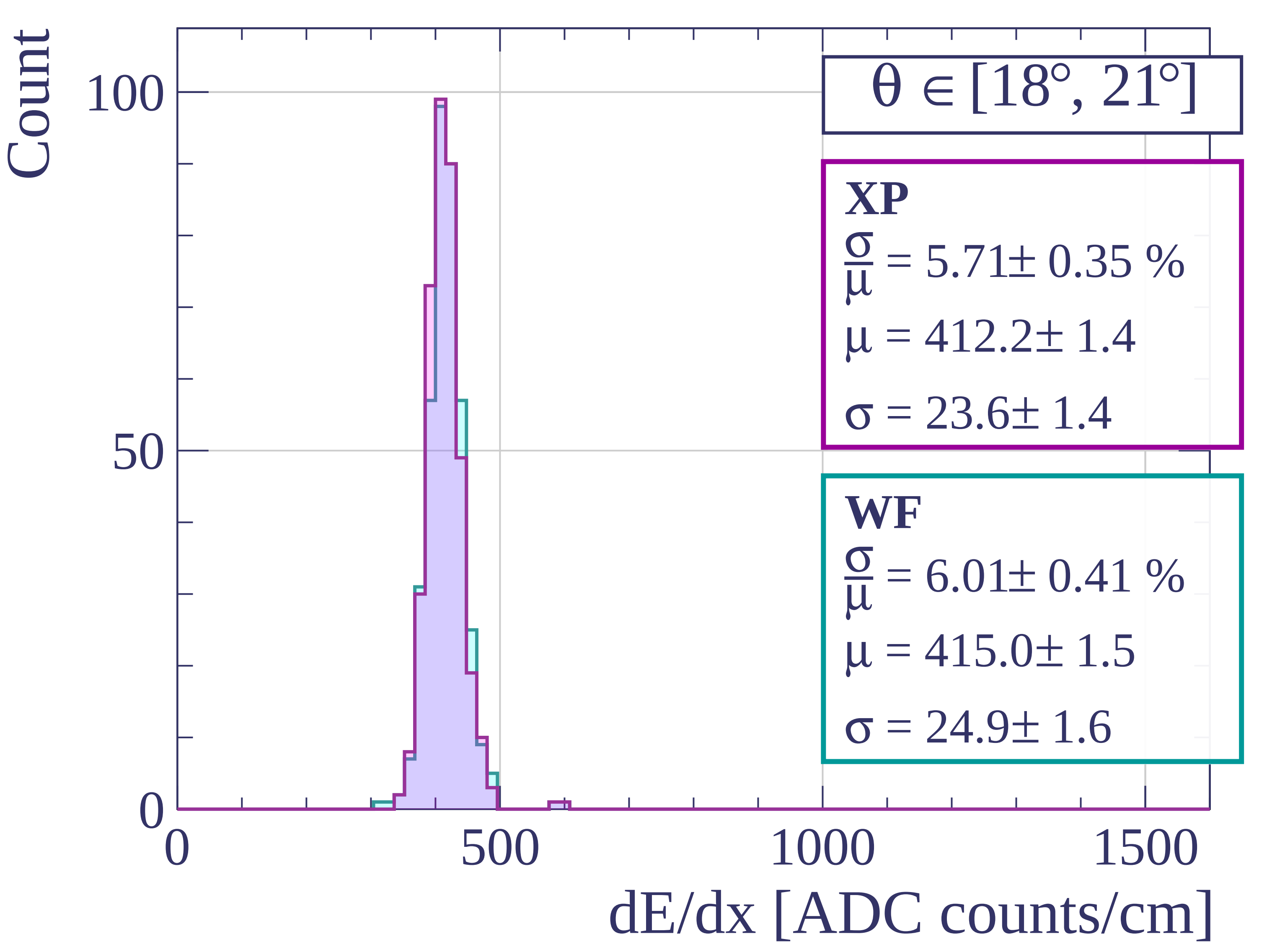
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [9°, 12°]
XP
σ/μ = 5.93± 0.30 %
μ = 410.8± 1.1
σ = 24.3± 1.2
WF
σ/μ = 5.94± 0.32 %
μ = 413.7± 1.2
σ = 24.6± 1.3



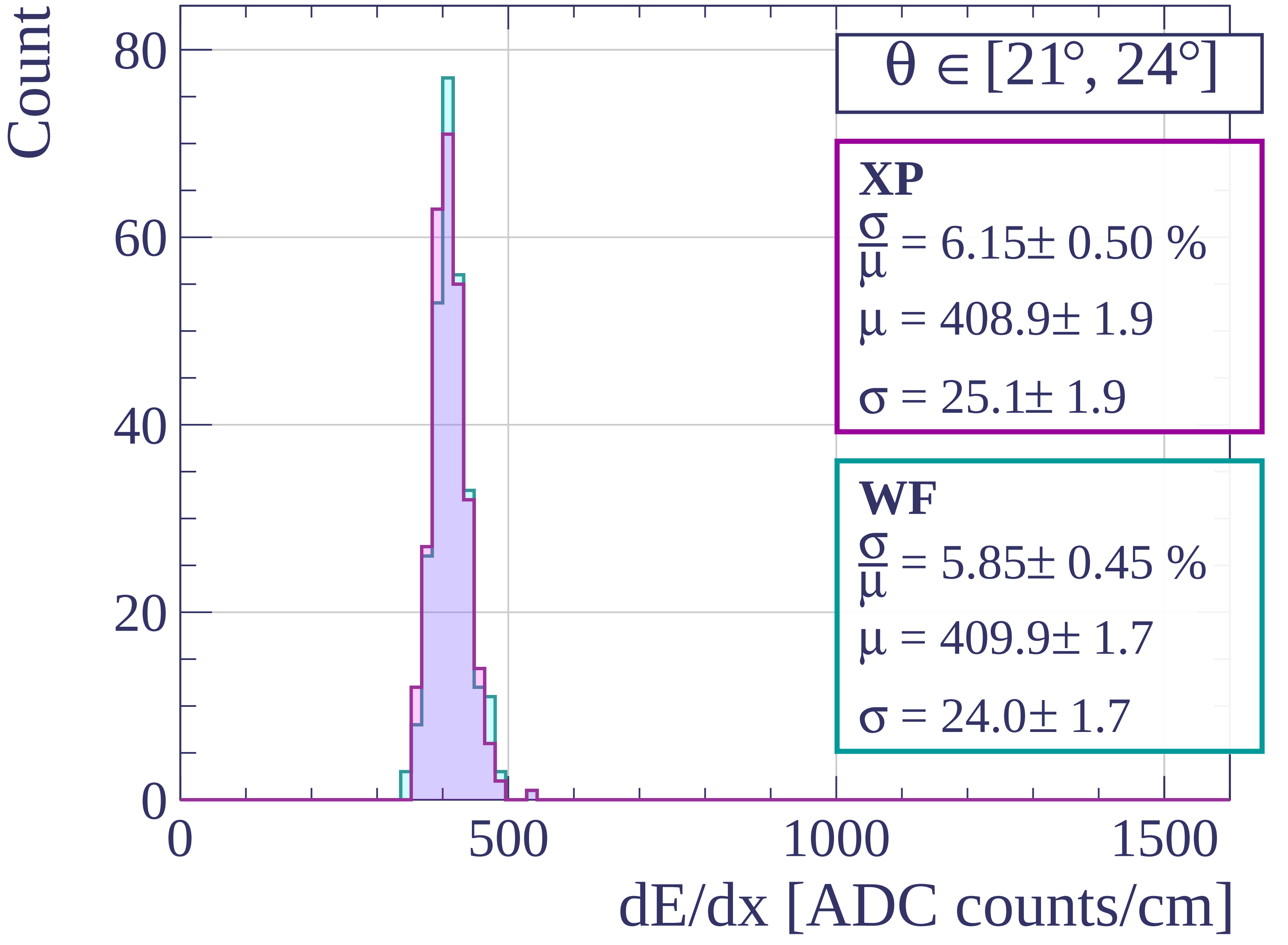
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [12°, 15°]
XP
σ/μ = 6.28± 0.31 %
μ = 410.3± 1.3
σ = 25.7± 1.2
WF
σ/μ = 6.35± 0.31 %
μ = 412.6± 1.3
σ = 26.2± 1.2



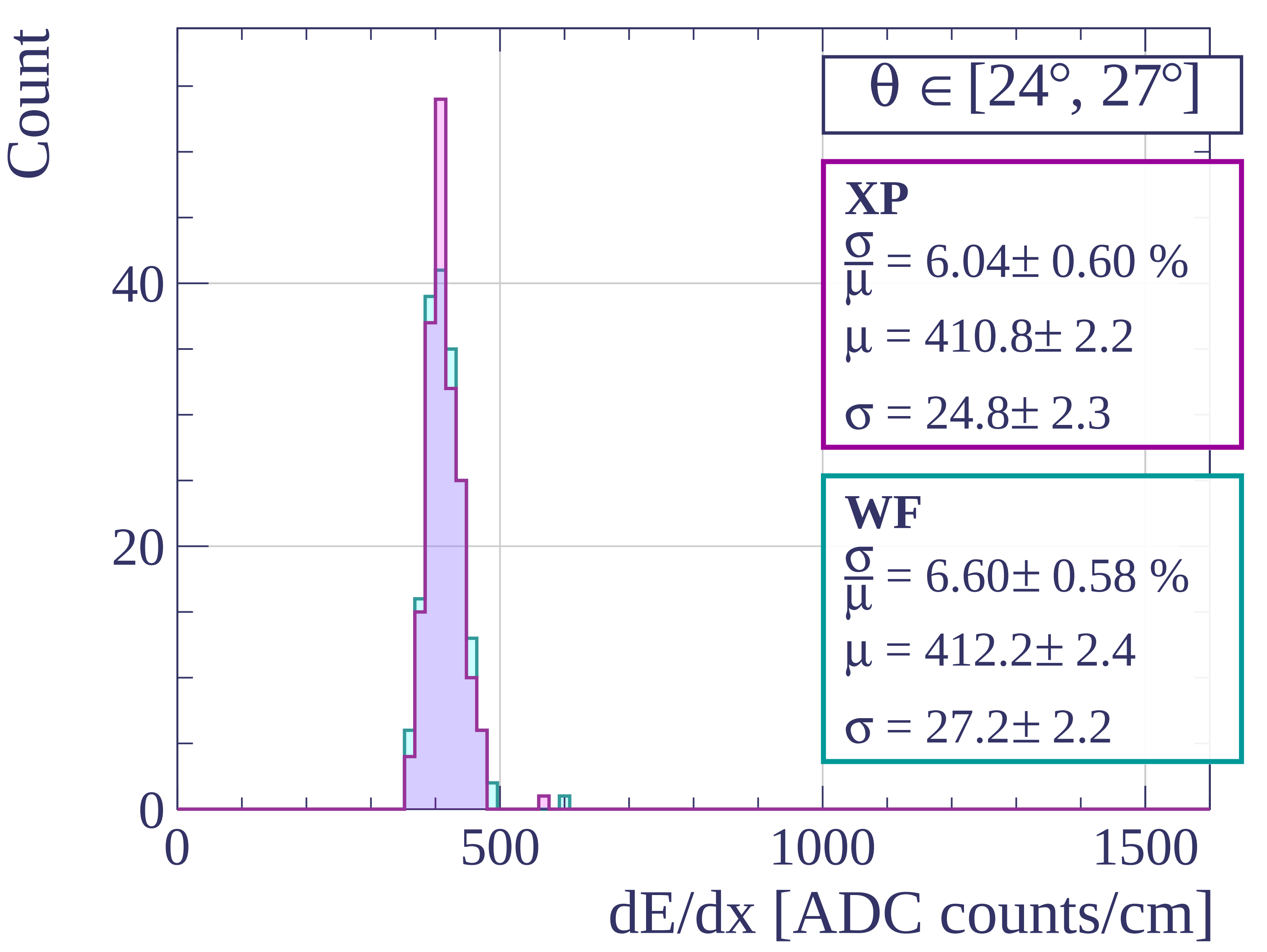
Count
dE/dx [ADC counts/cm]
θ ∈ [15°, 18°]
XP
σ/μ = 5.79± 0.25 %
μ = 413.5± 1.3
σ = 23.9± 0.9
WF
σ/μ = 5.87± 0.35 %
μ = 416.2± 1.3
σ = 24.4± 1.4

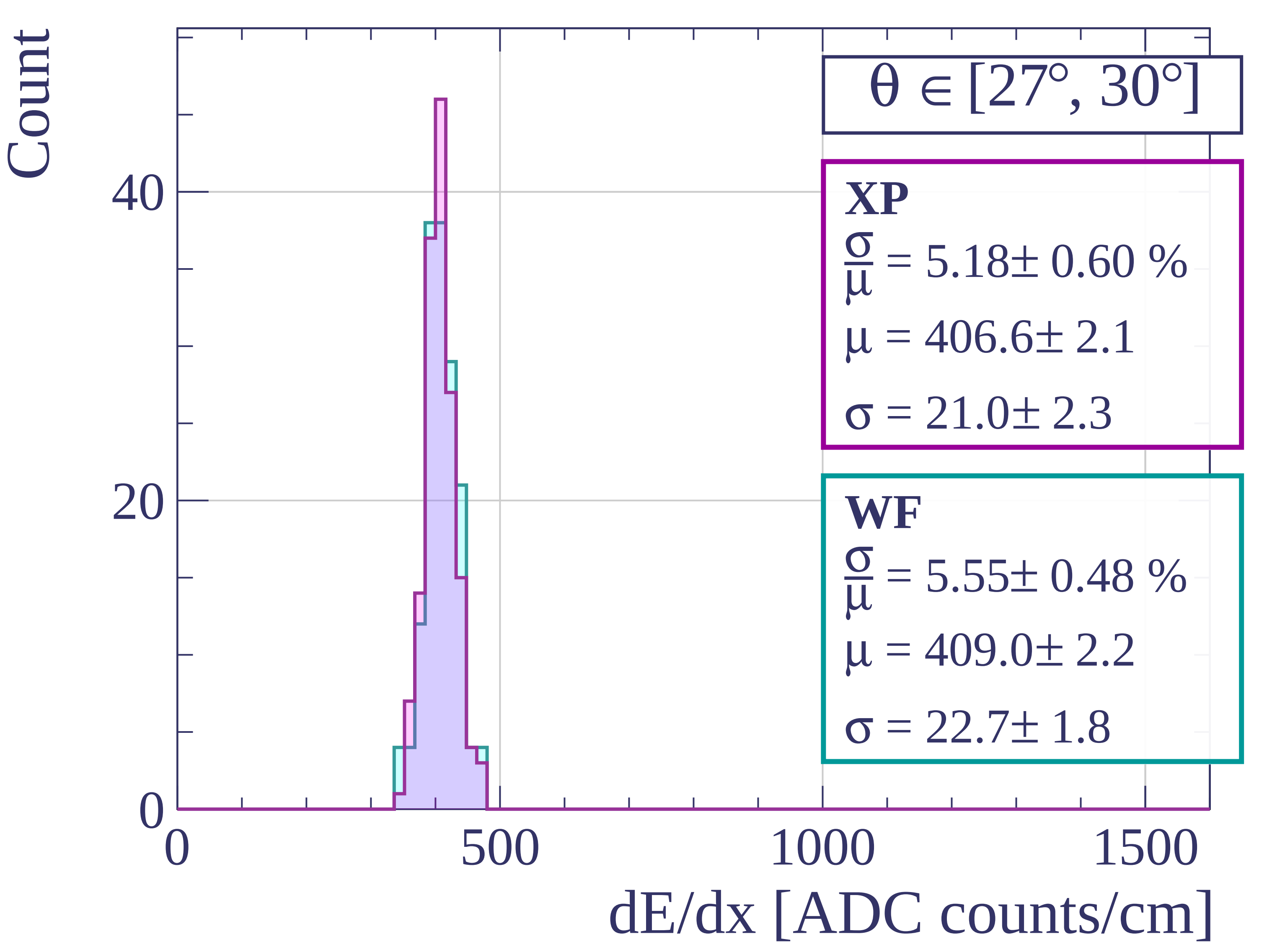


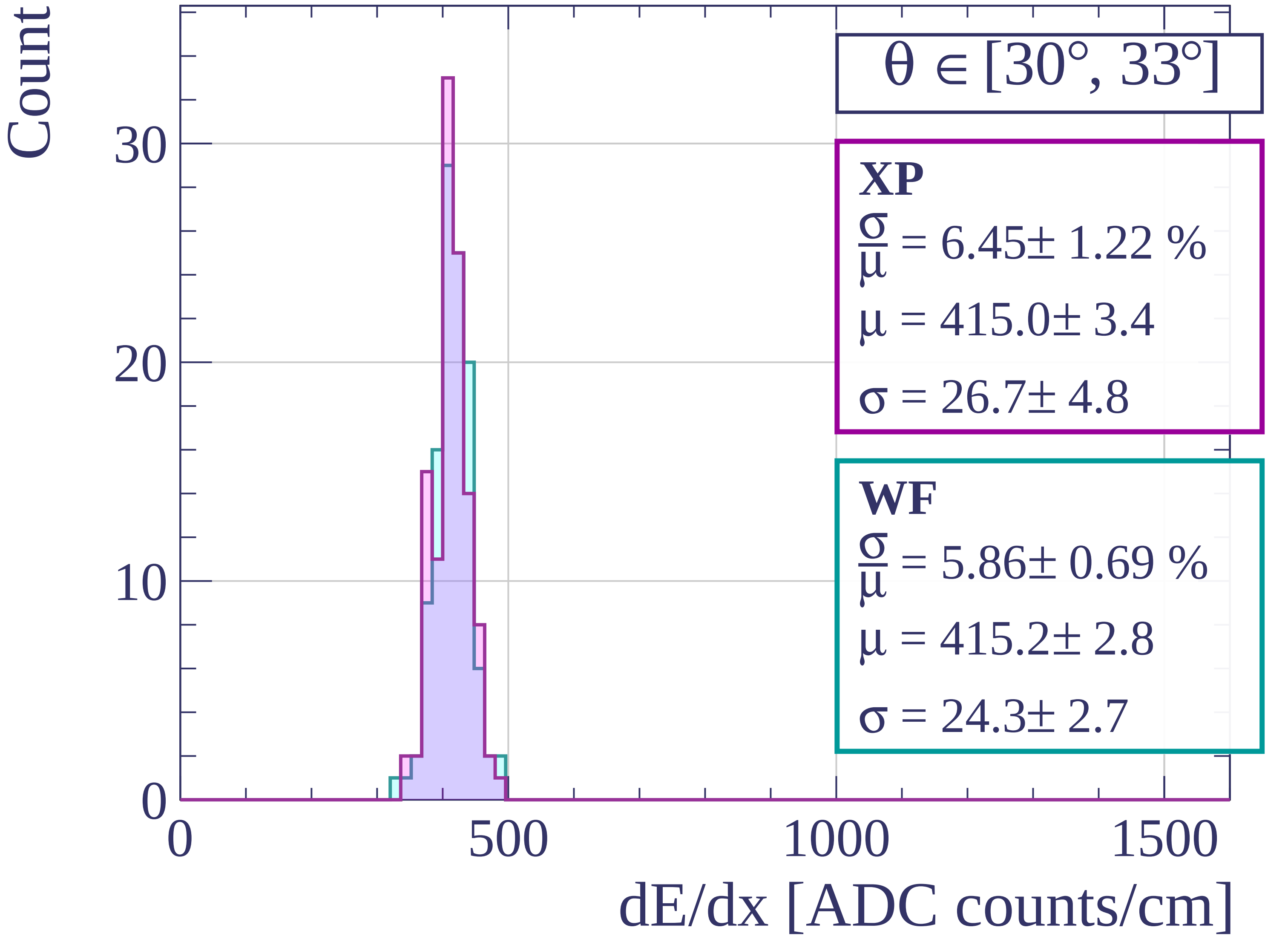
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [18°, 21°]
XP
σ/μ = 5.71± 0.35 %
μ = 412.2± 1.4
σ = 23.6± 1.4
WF
σ/μ = 6.01± 0.41 %
μ = 415.0± 1.5
σ = 24.9± 1.6



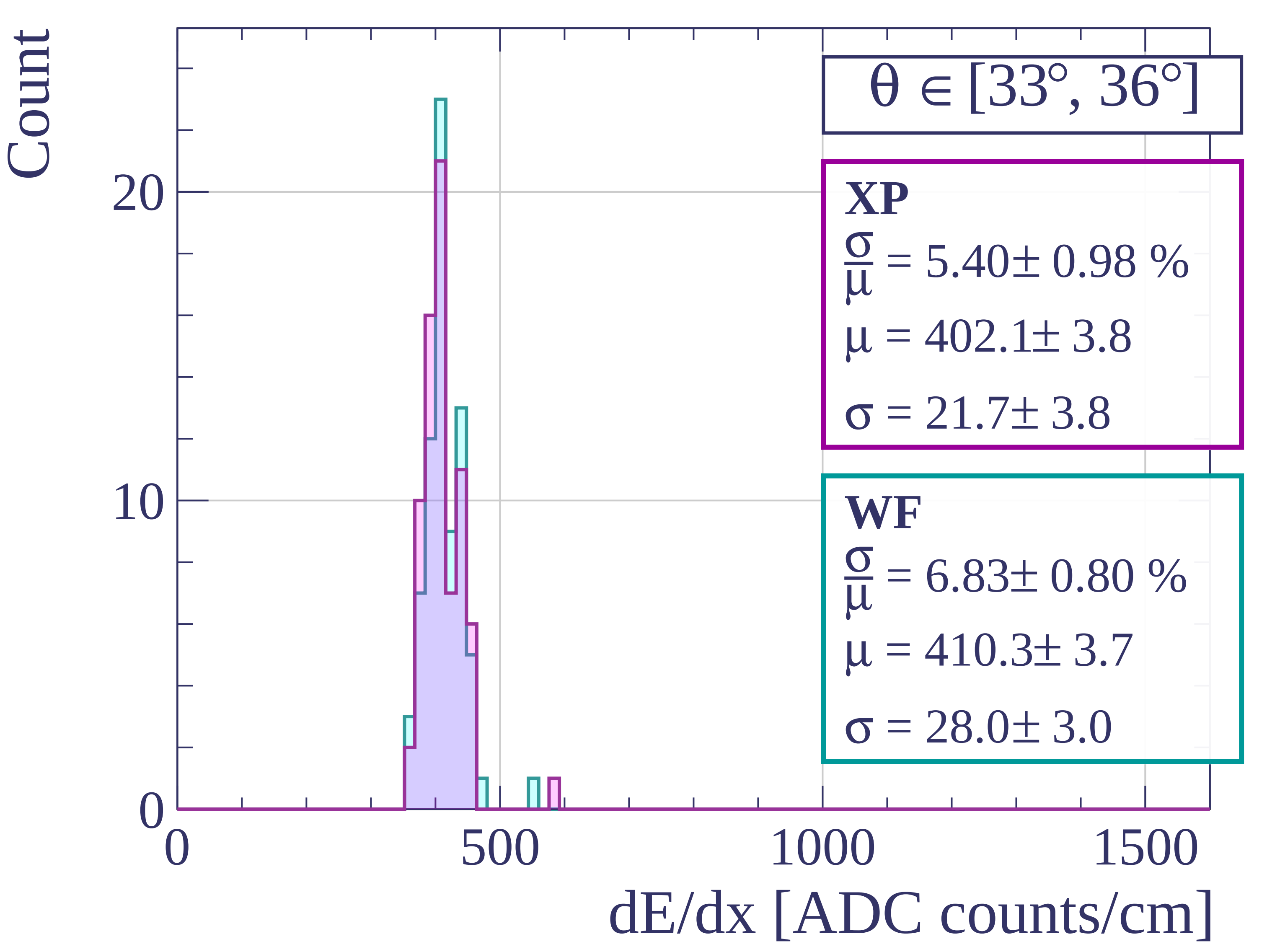
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [21°, 24°]
XP
σ/μ = 6.15± 0.50 %
μ = 408.9± 1.9
σ = 25.1± 1.9
WF
σ/μ = 5.85± 0.45 %
μ = 409.9± 1.7
σ = 24.0± 1.7


Count
dE/dx [ADC counts/cm]
0
20
40
0
500
1000
1500
θ ∈ [24°, 27°]
XP
σ/μ = 6.04± 0.60 %
μ = 410.8± 2.2
σ = 24.8± 2.3
WF
σ/μ = 6.60± 0.58 %
μ = 412.2± 2.4
σ = 27.2± 2.2

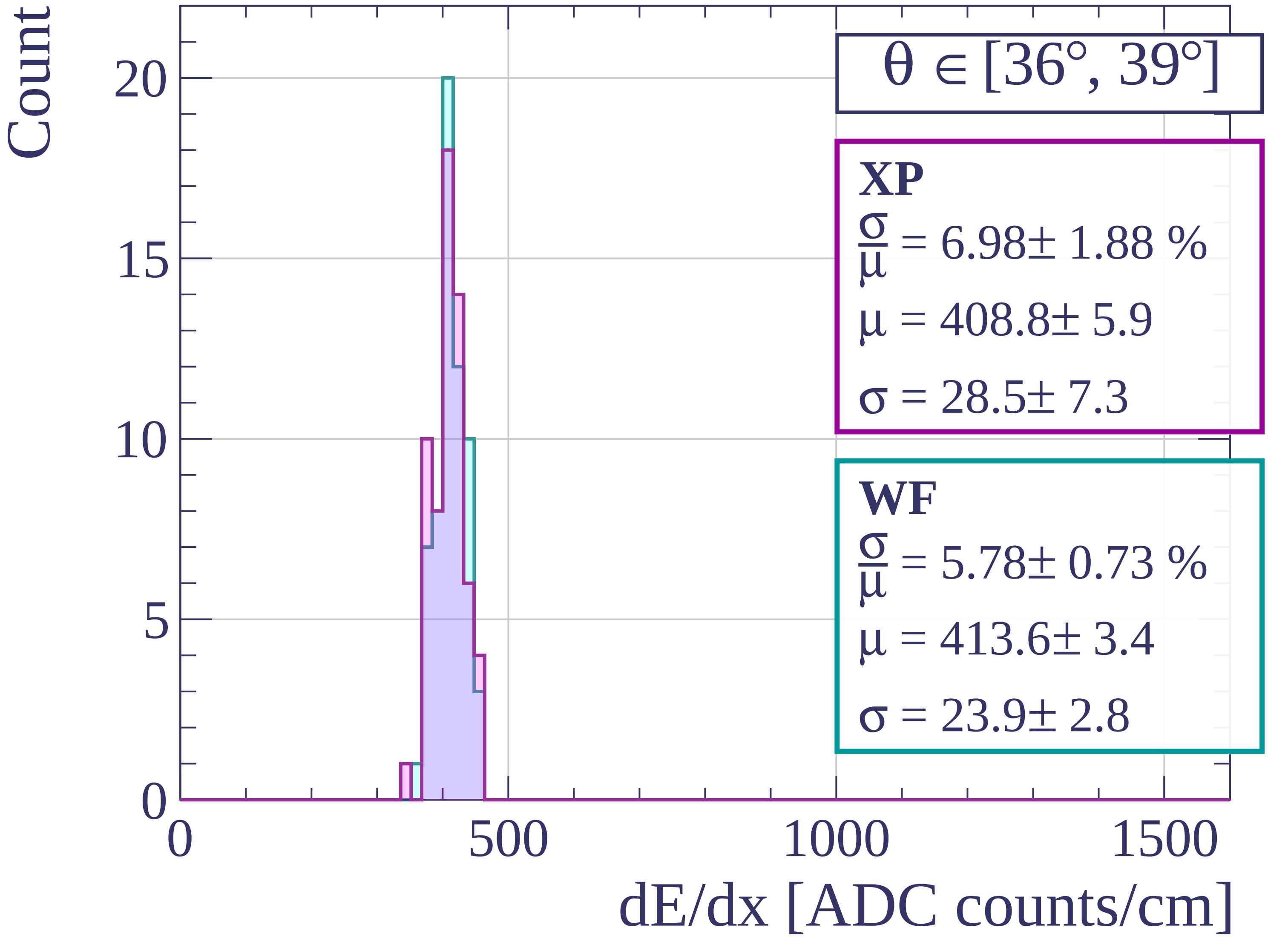

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [27°, 30°]
XP
σ/μ = 5.18± 0.60 %
μ = 406.6± 2.1
σ = 21.0± 2.3
WF
σ/μ = 5.55± 0.48 %
μ = 409.0± 2.2
σ = 22.7± 1.8



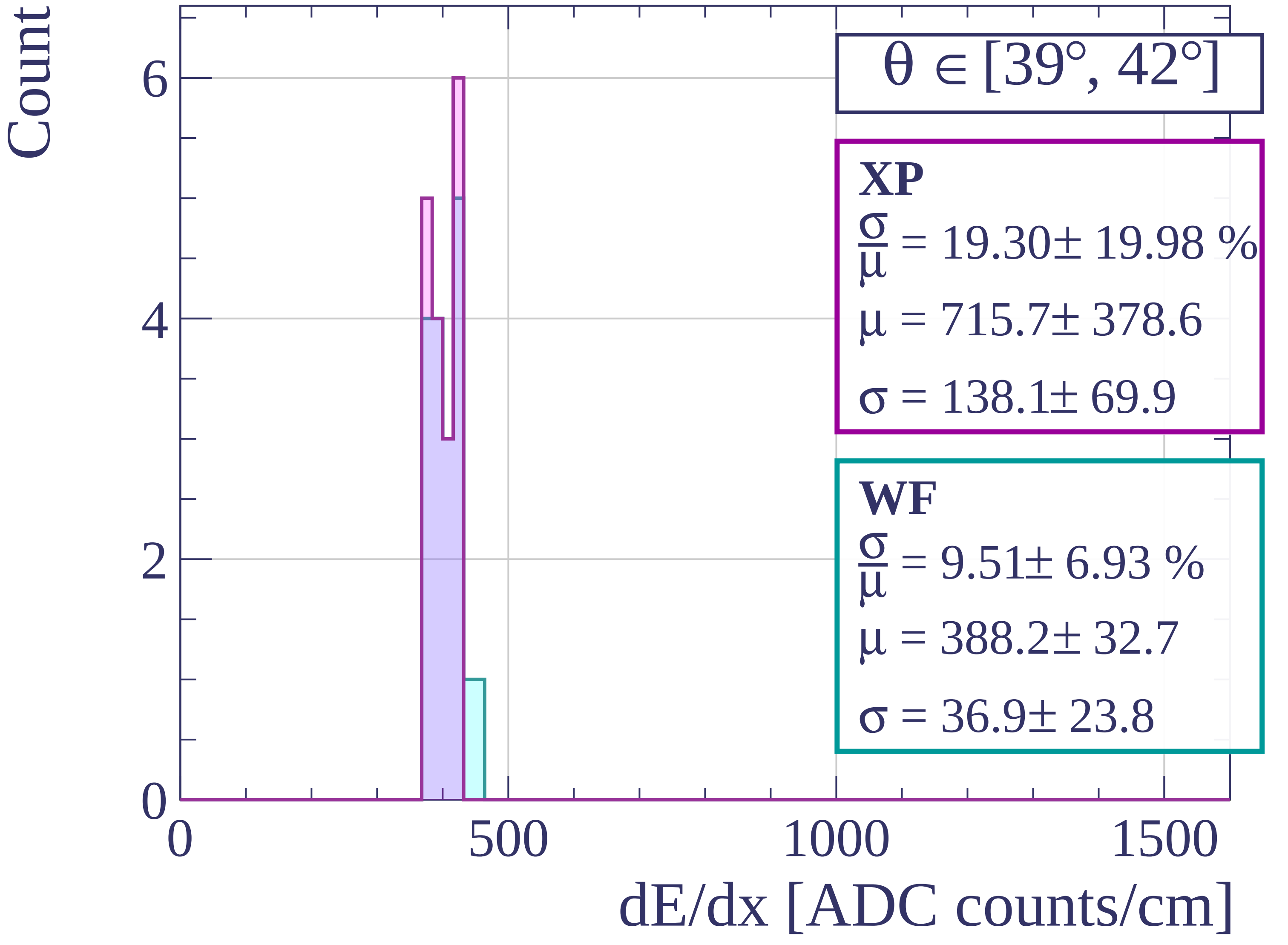
Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [30°, 33°]
XP
σ/μ = 6.45± 1.22 %
μ = 415.0± 3.4
σ = 26.7± 4.8
WF
σ/μ = 5.86± 0.69 %
μ = 415.2± 2.8
σ = 24.3± 2.7



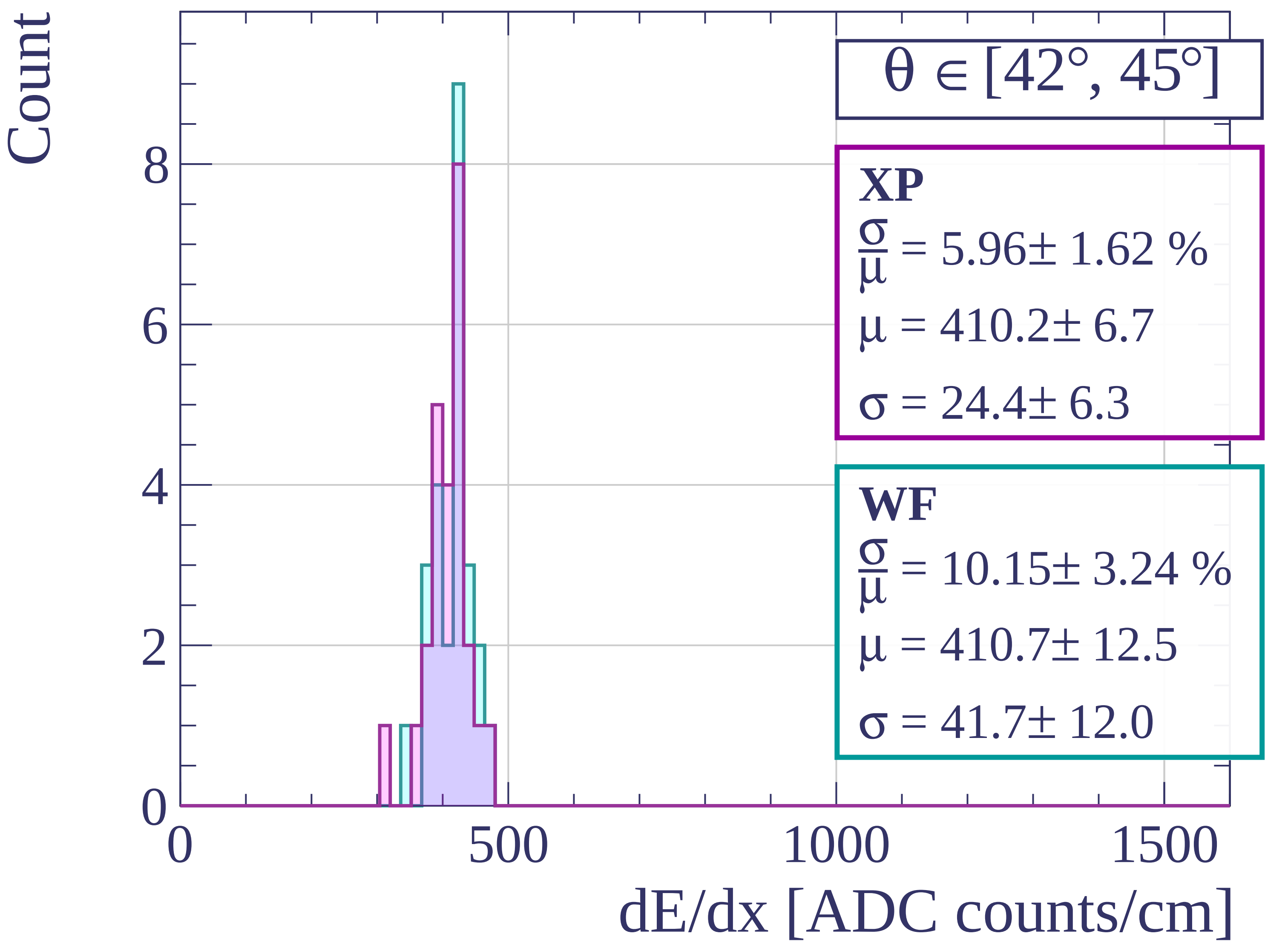
Count
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [33°, 36°]
XP
σ/μ = 5.40± 0.98 %
μ = 402.1± 3.8
σ = 21.7± 3.8
WF
σ/μ = 6.83± 0.80 %
μ = 410.3± 3.7
σ = 28.0± 3.0



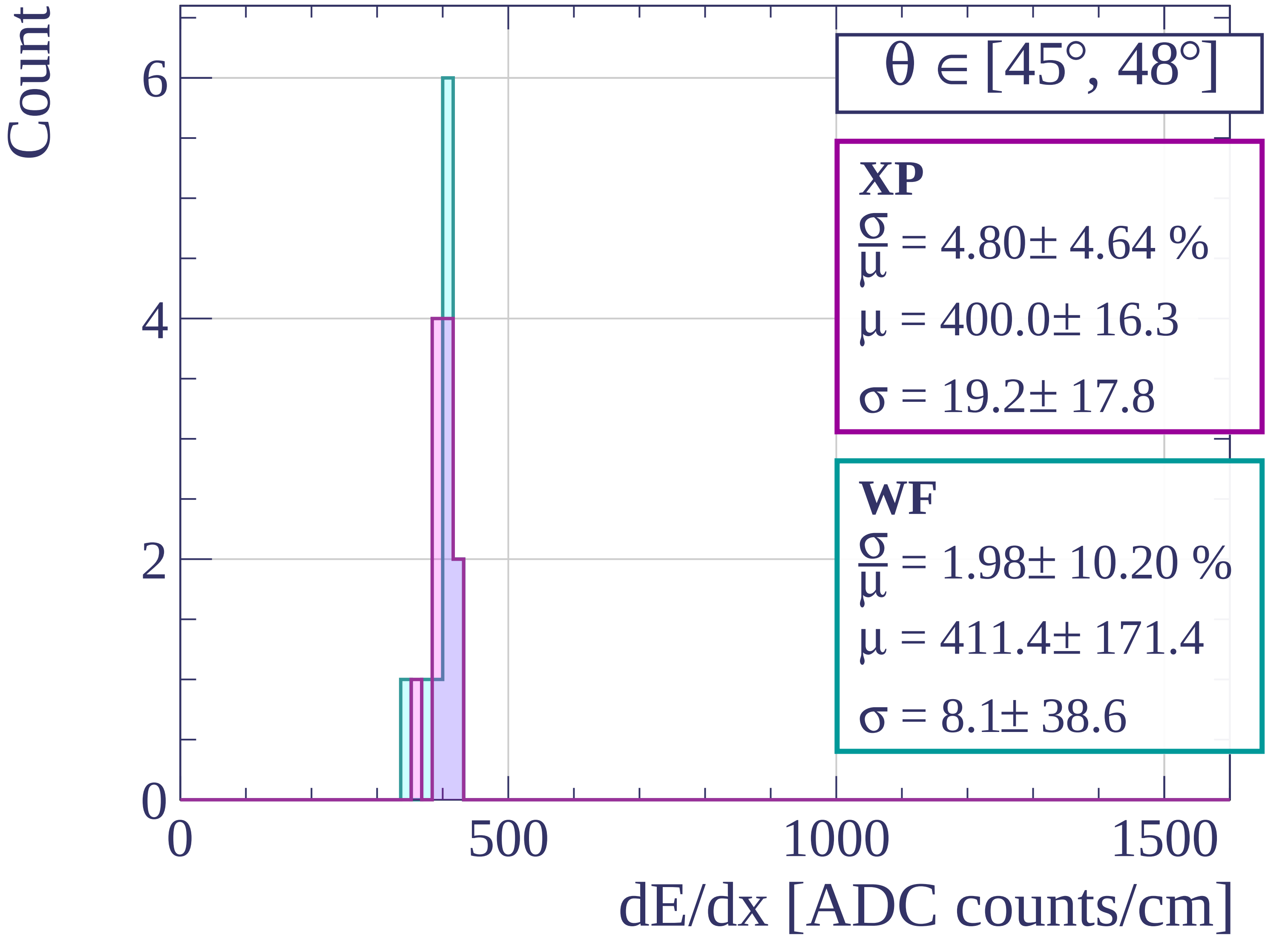
Count
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
$\theta \in [36°, 39°]$
XP
$\frac{\sigma}{\mu} = 6.98 \pm 1.88$ %
$\mu = 408.8 \pm 5.9$
$\sigma = 28.5 \pm 7.3$
WF
$\frac{\sigma}{\mu} = 5.78 \pm 0.73$ %
$\mu = 413.6 \pm 3.4$
$\sigma = 23.9 \pm 2.8$

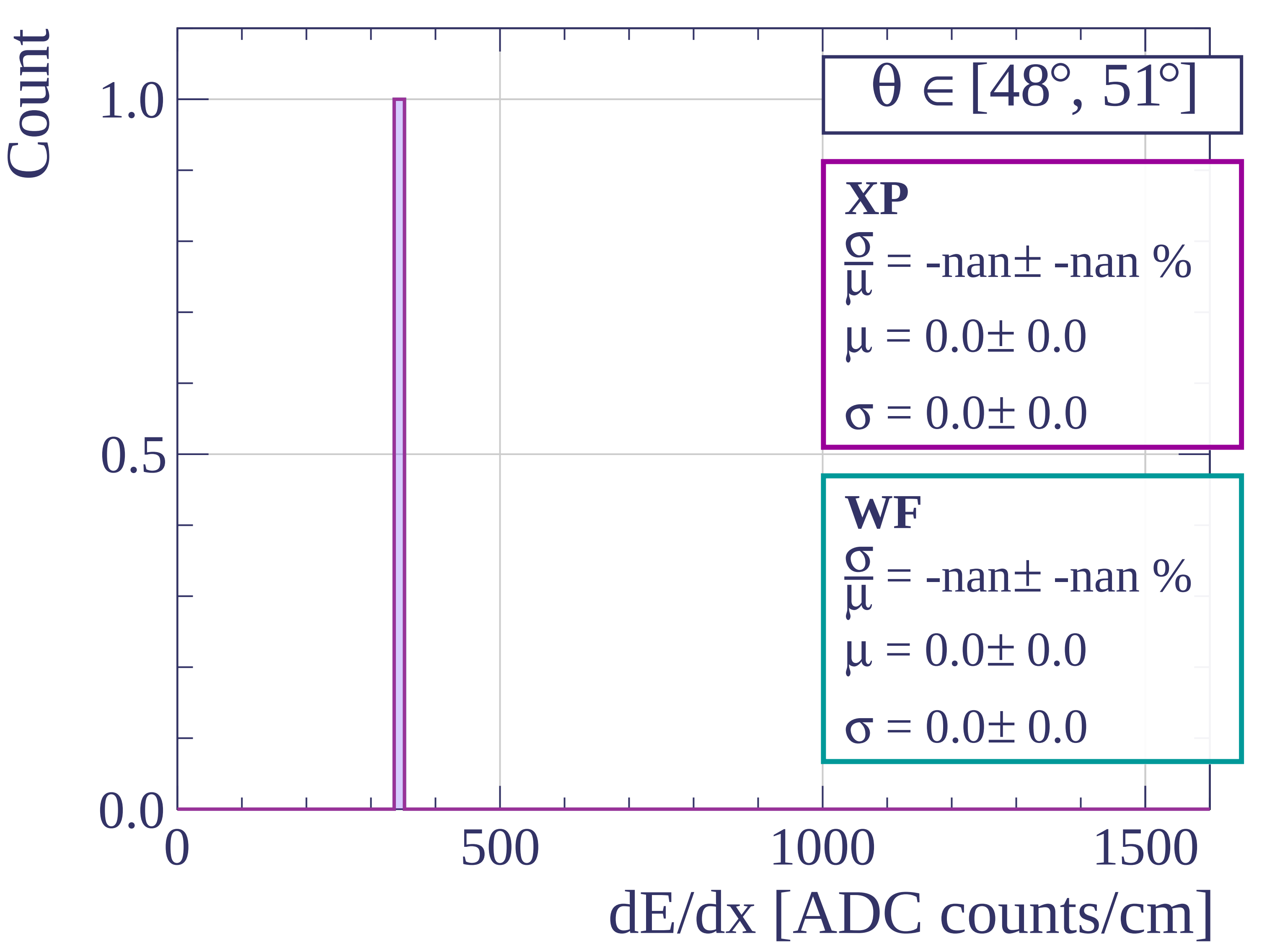


Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
θ ∈ [39°, 42°]
XP
σ/μ = 19.30± 19.98 %
μ = 715.7± 378.6
σ = 138.1± 69.9
WF
σ/μ = 9.51± 6.93 %
μ = 388.2± 32.7
σ = 36.9± 23.8

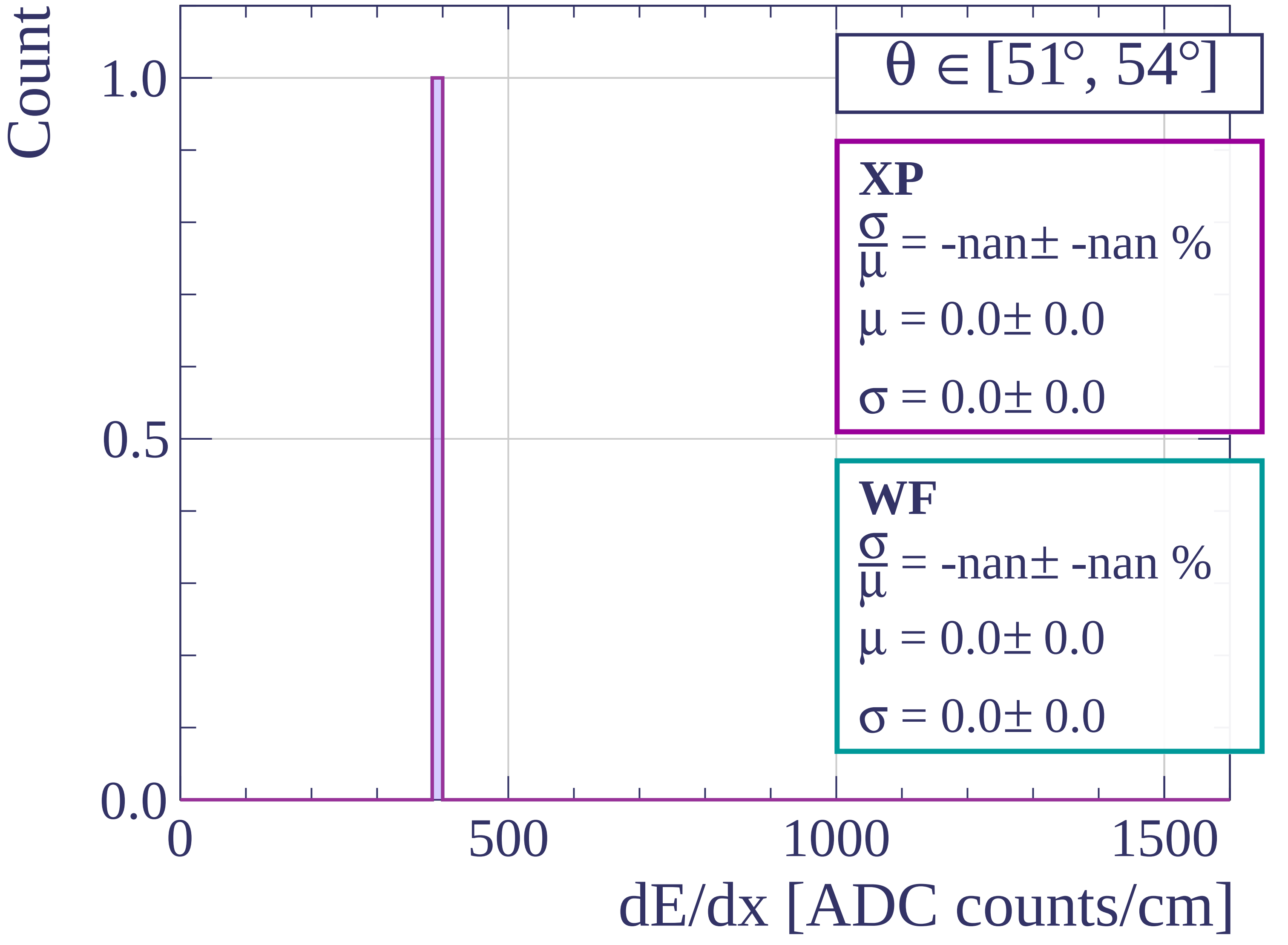


Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [42°, 45°]
XP
σ/μ = 5.96± 1.62 %
μ = 410.2± 6.7
σ = 24.4± 6.3
WF
σ/μ = 10.15± 3.24 %
μ = 410.7± 12.5
σ = 41.7± 12.0

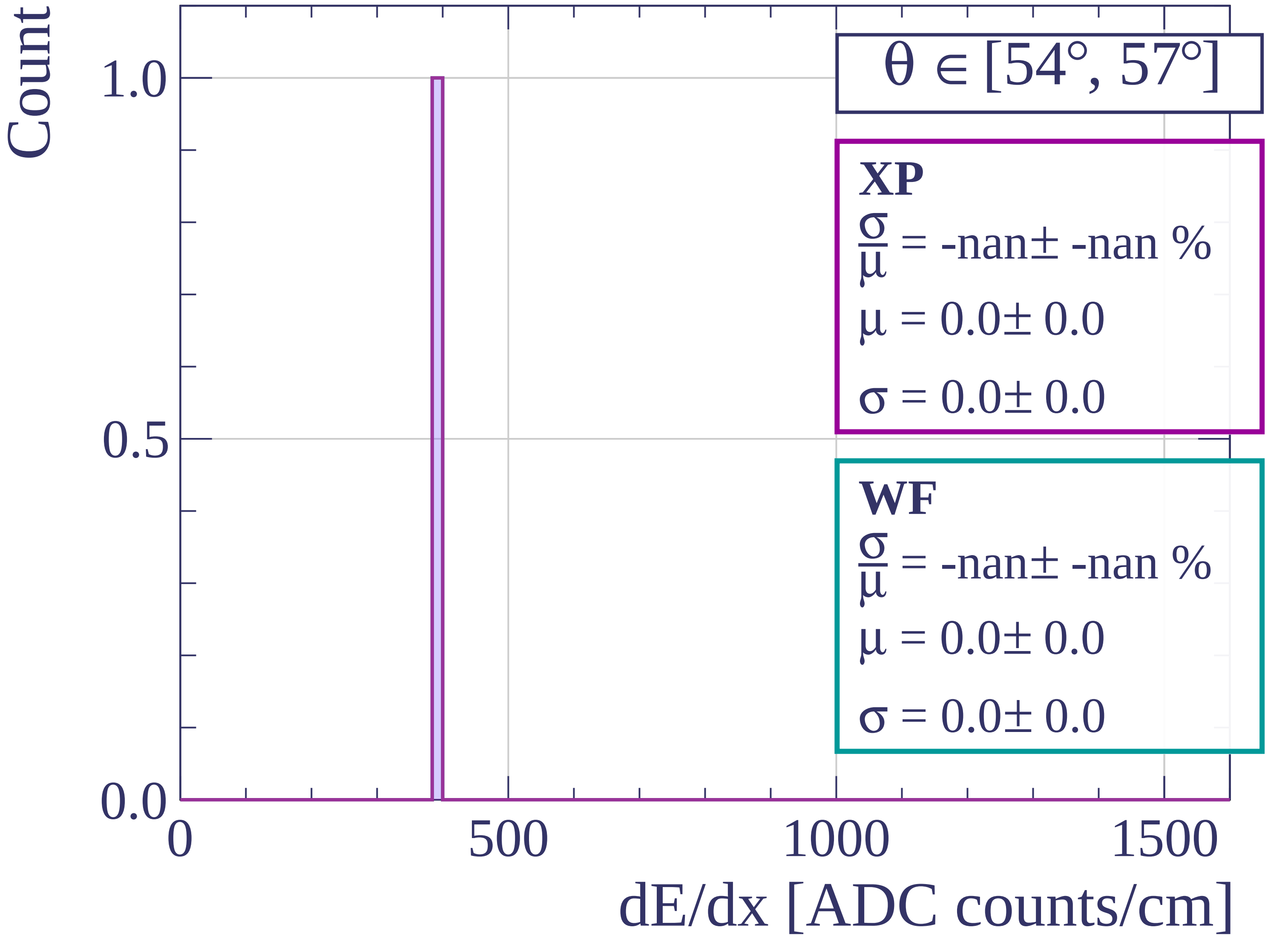

Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
θ ∈ [45°, 48°]
XP
σ/μ = 4.80± 4.64 %
μ = 400.0± 16.3
σ = 19.2± 17.8
WF
σ/μ = 1.98± 10.20 %
μ = 411.4± 171.4
σ = 8.1± 38.6



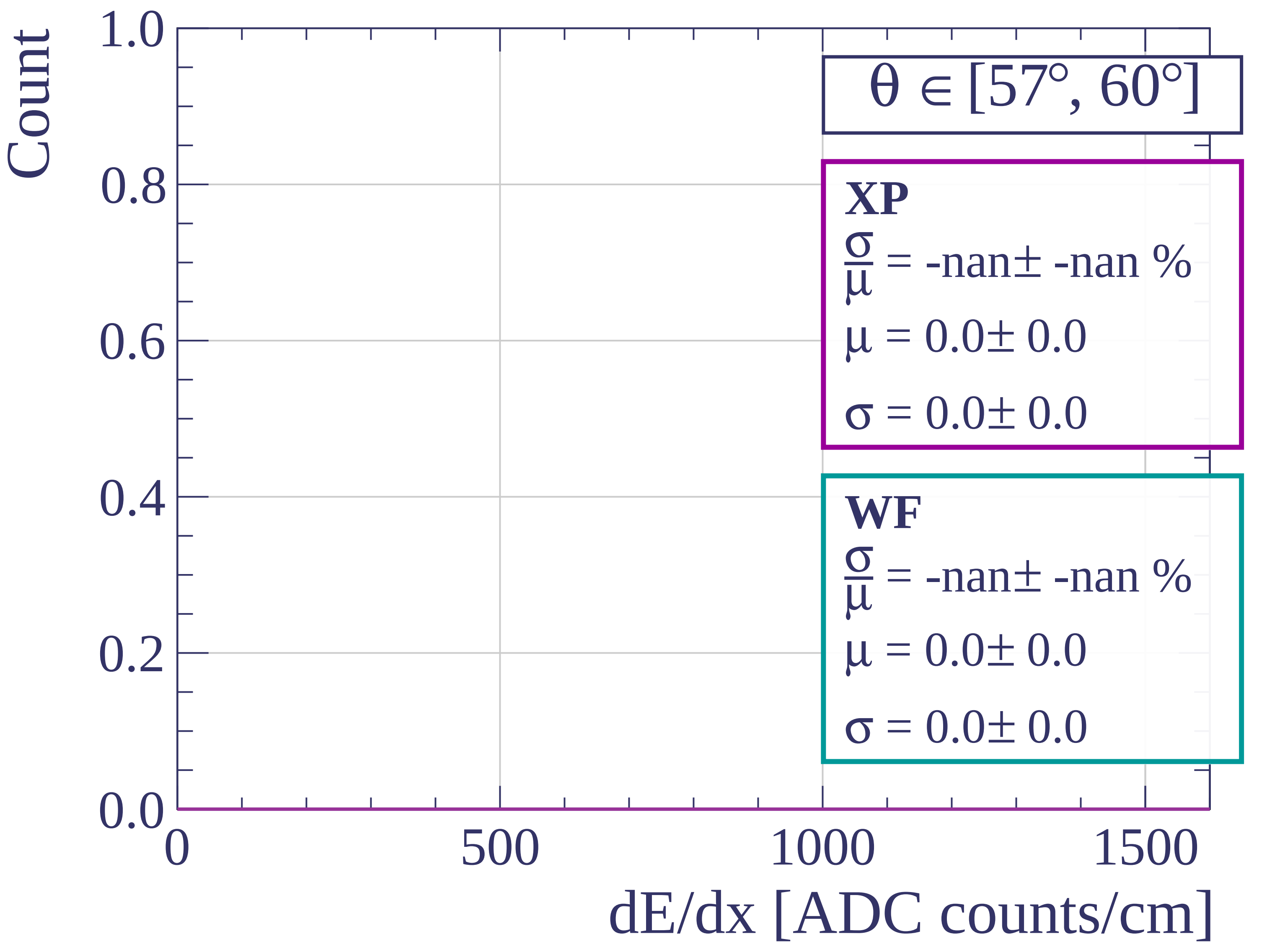
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [48°, 51°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



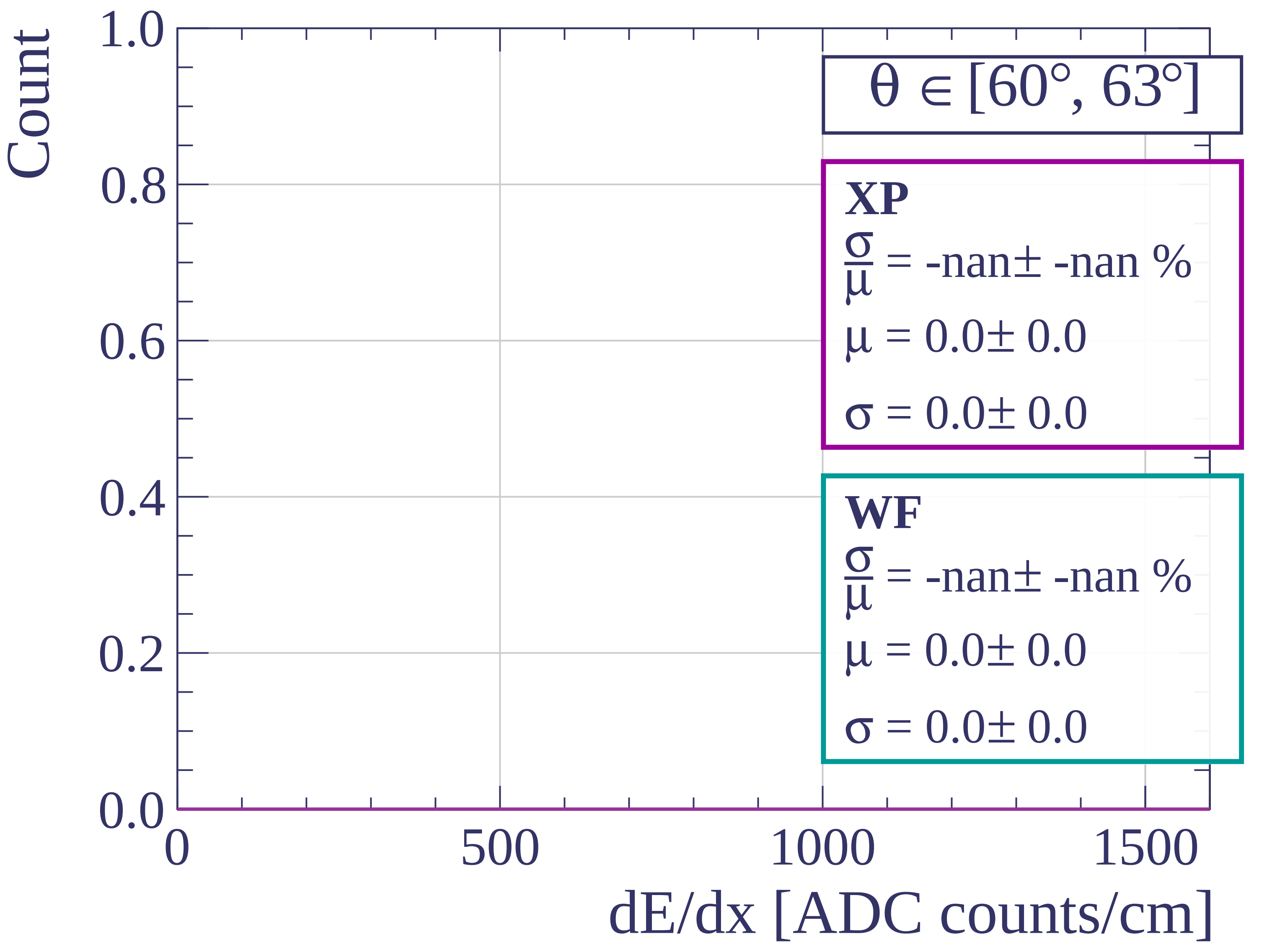
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [51°, 54°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



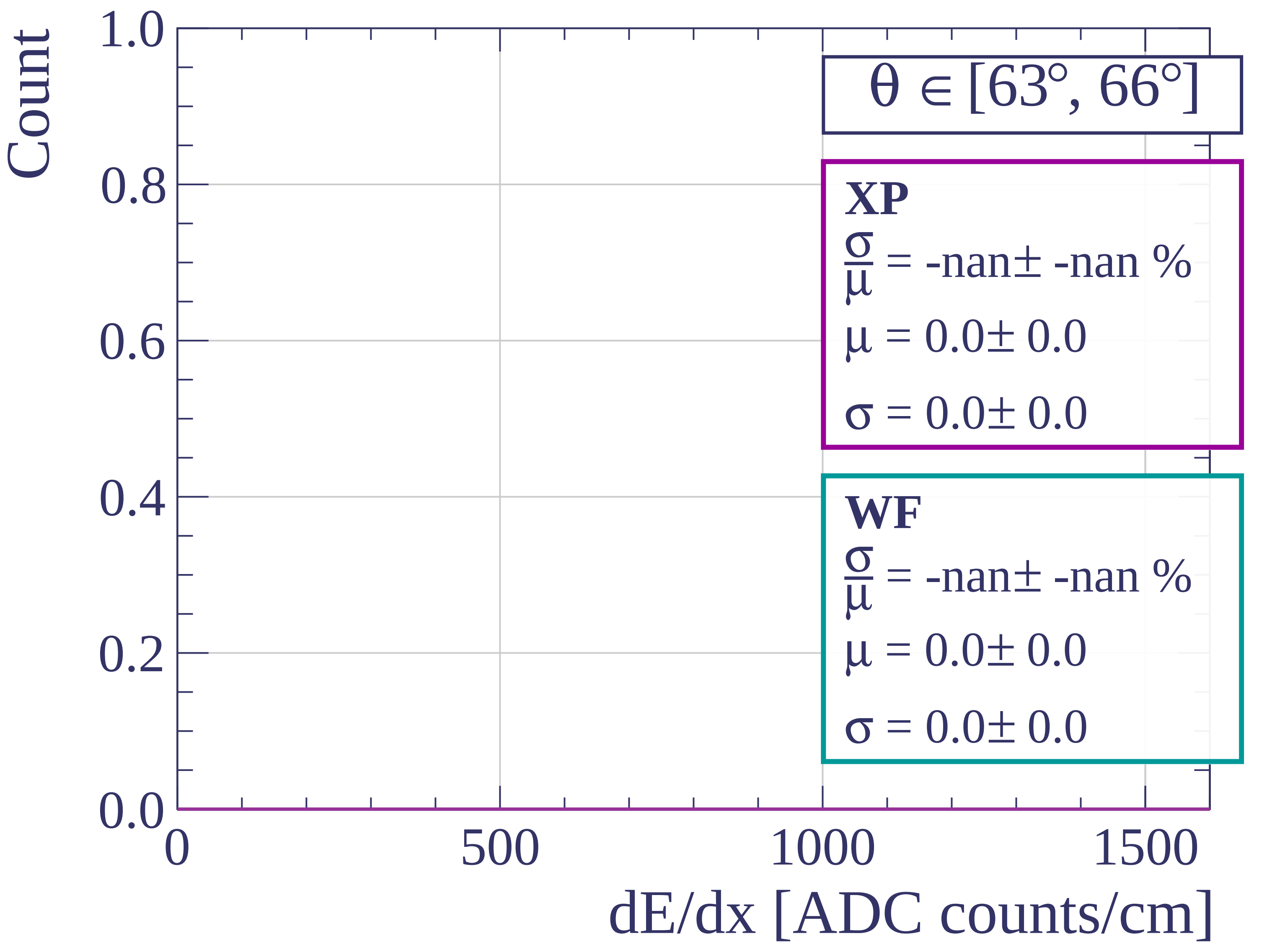
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [54°, 57°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



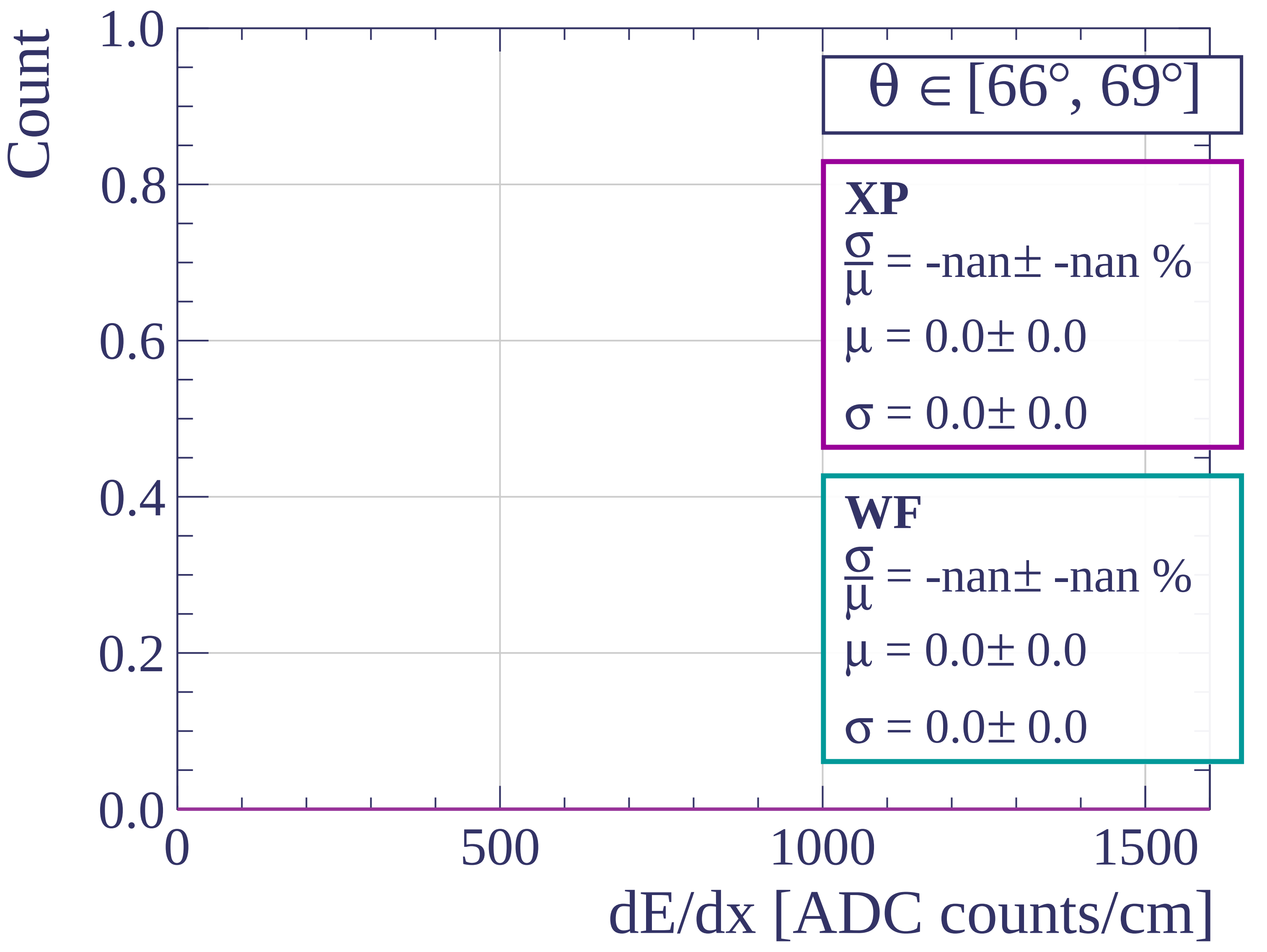
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [57°, 60°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



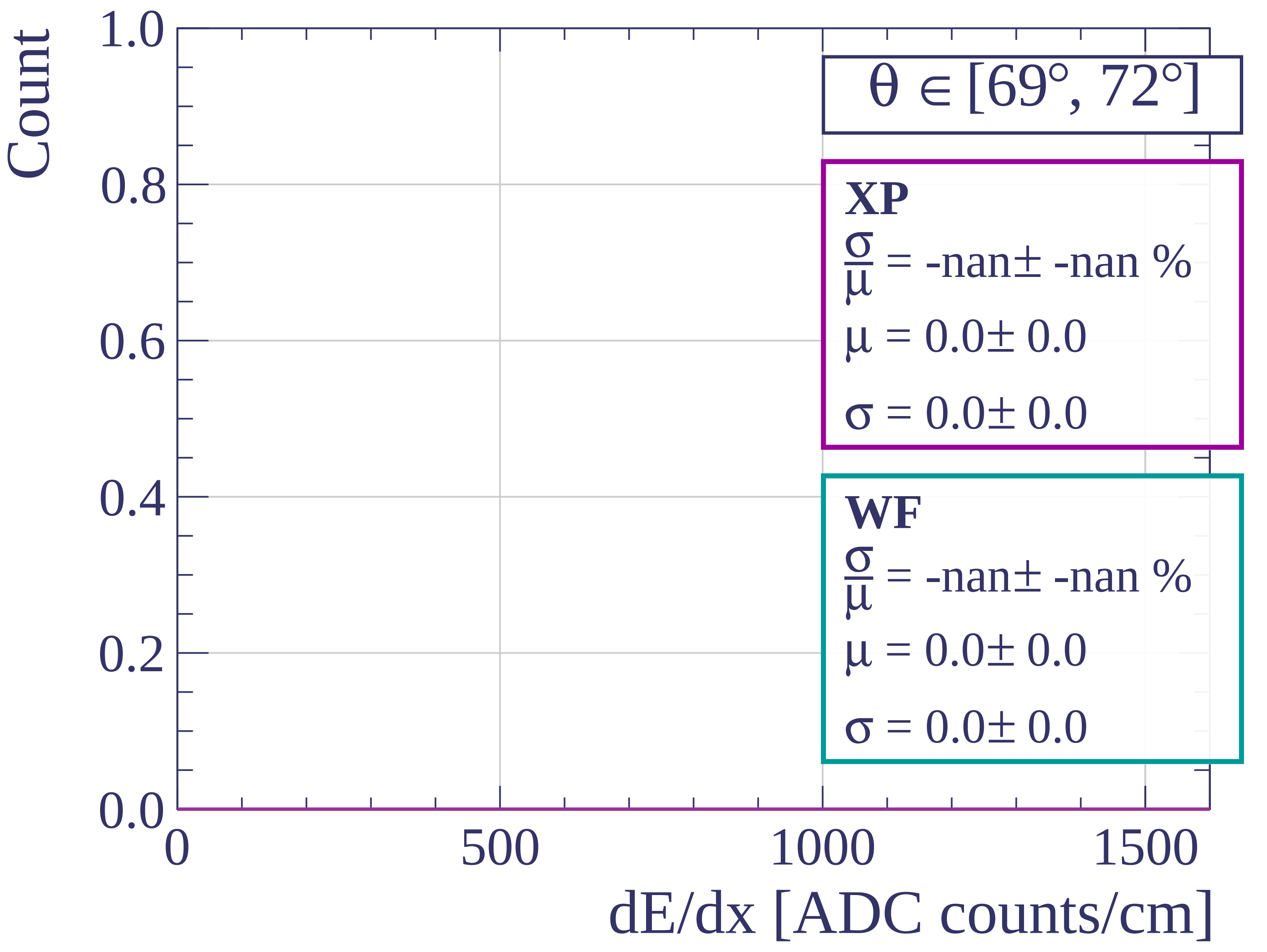
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [60°, 63°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



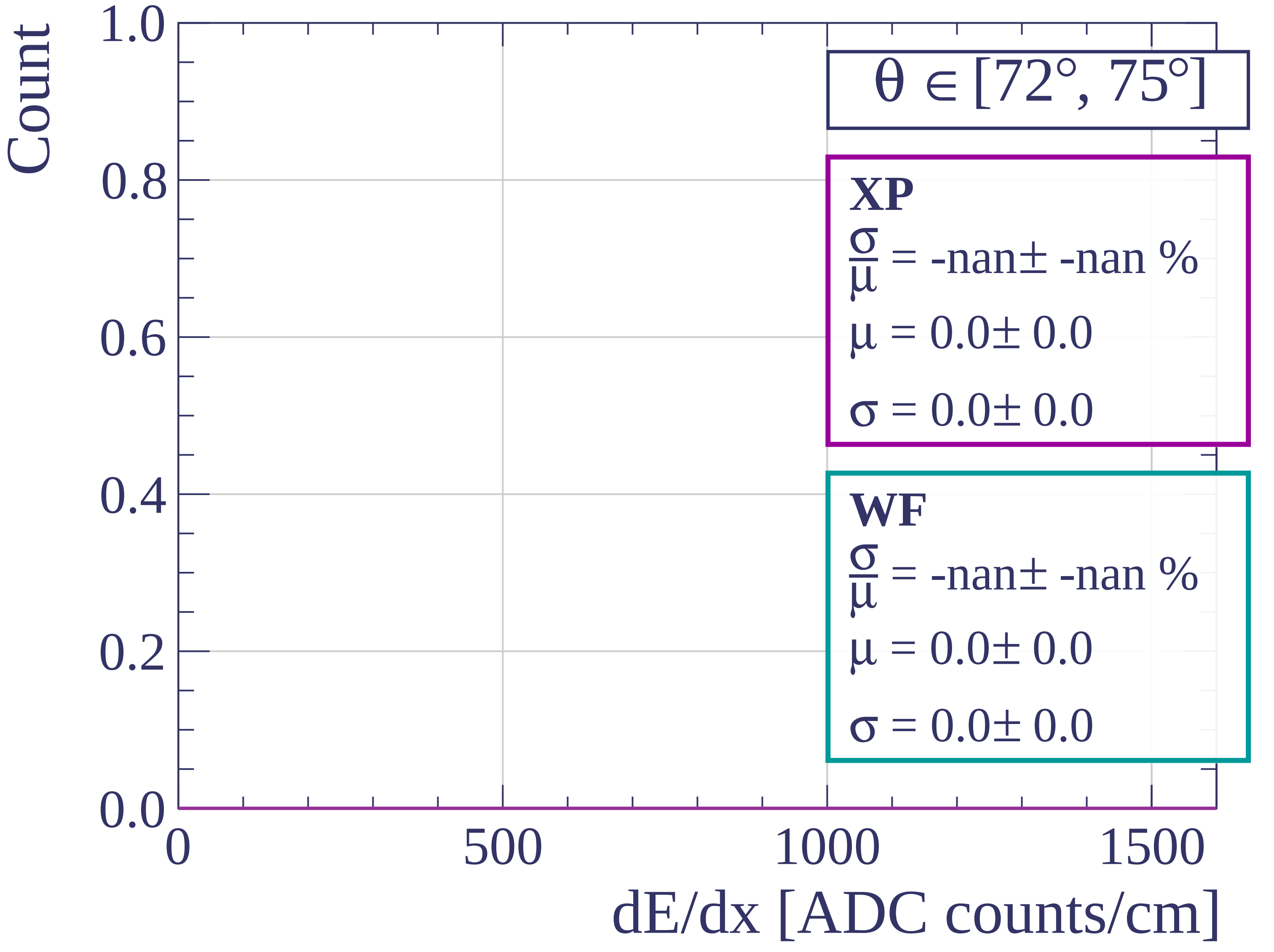
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [63°, 66°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



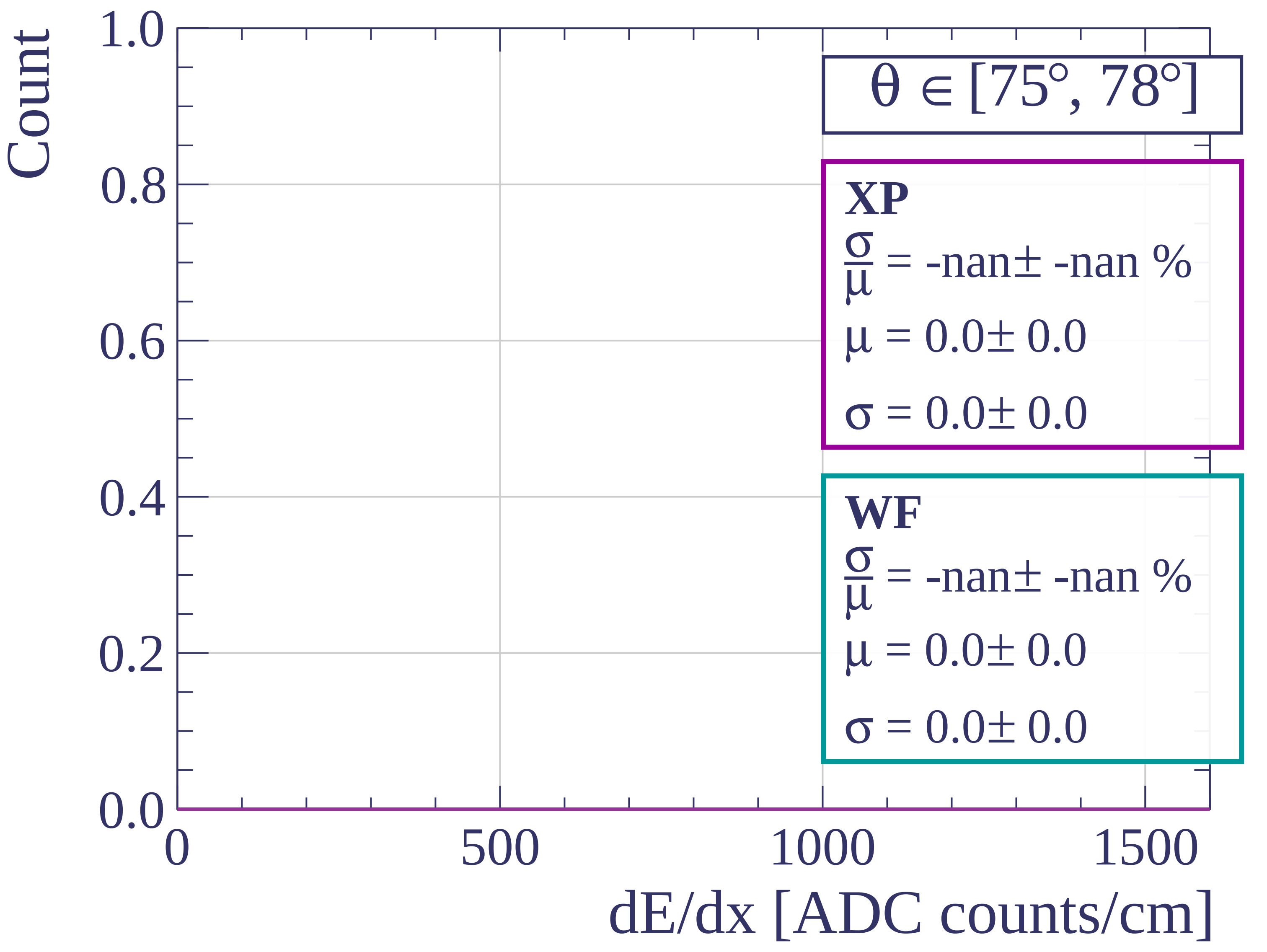
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [66°, 69°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



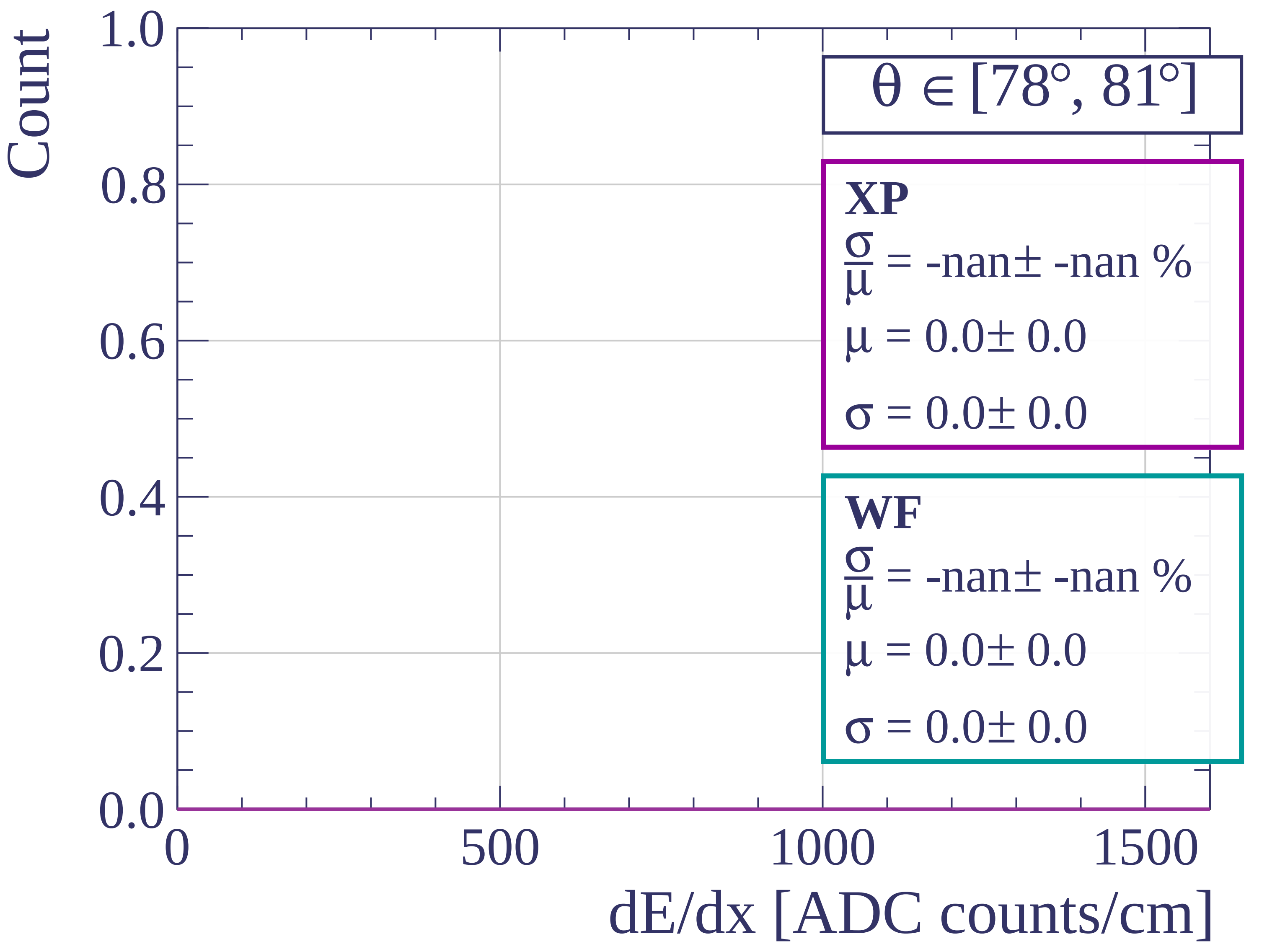
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [69°, 72°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



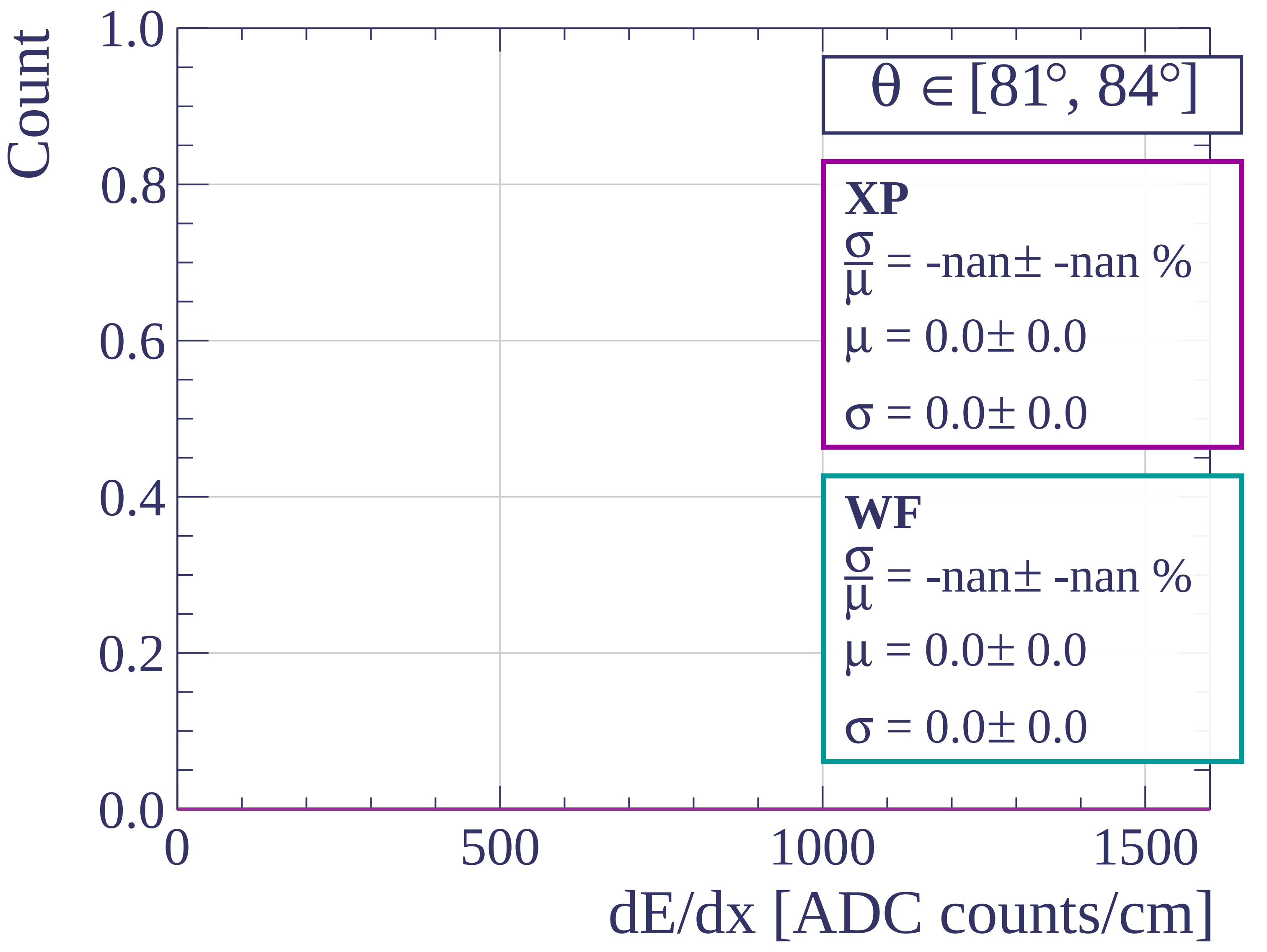
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [72°, 75°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



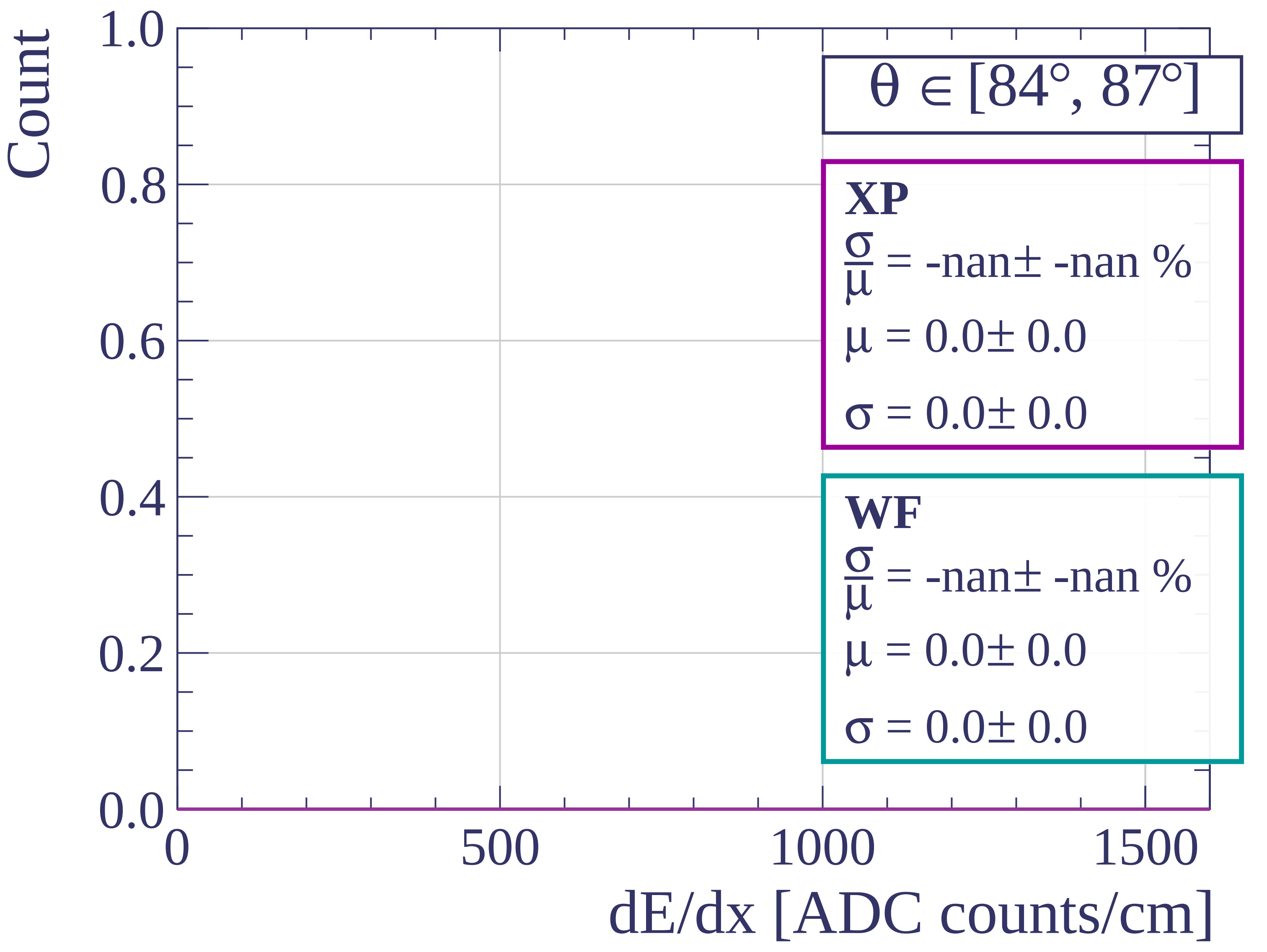
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [75°, 78°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



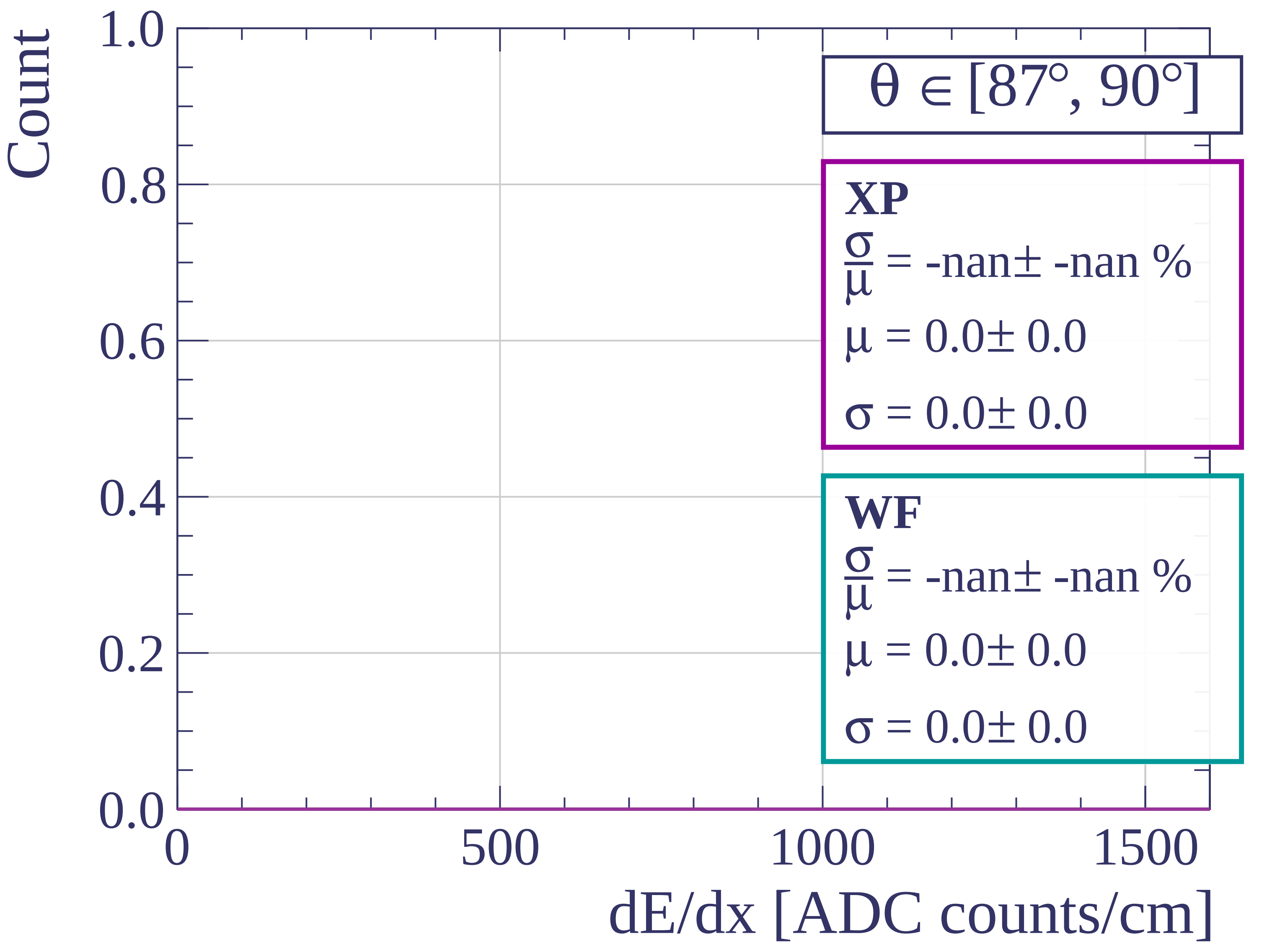
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [78°, 81°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



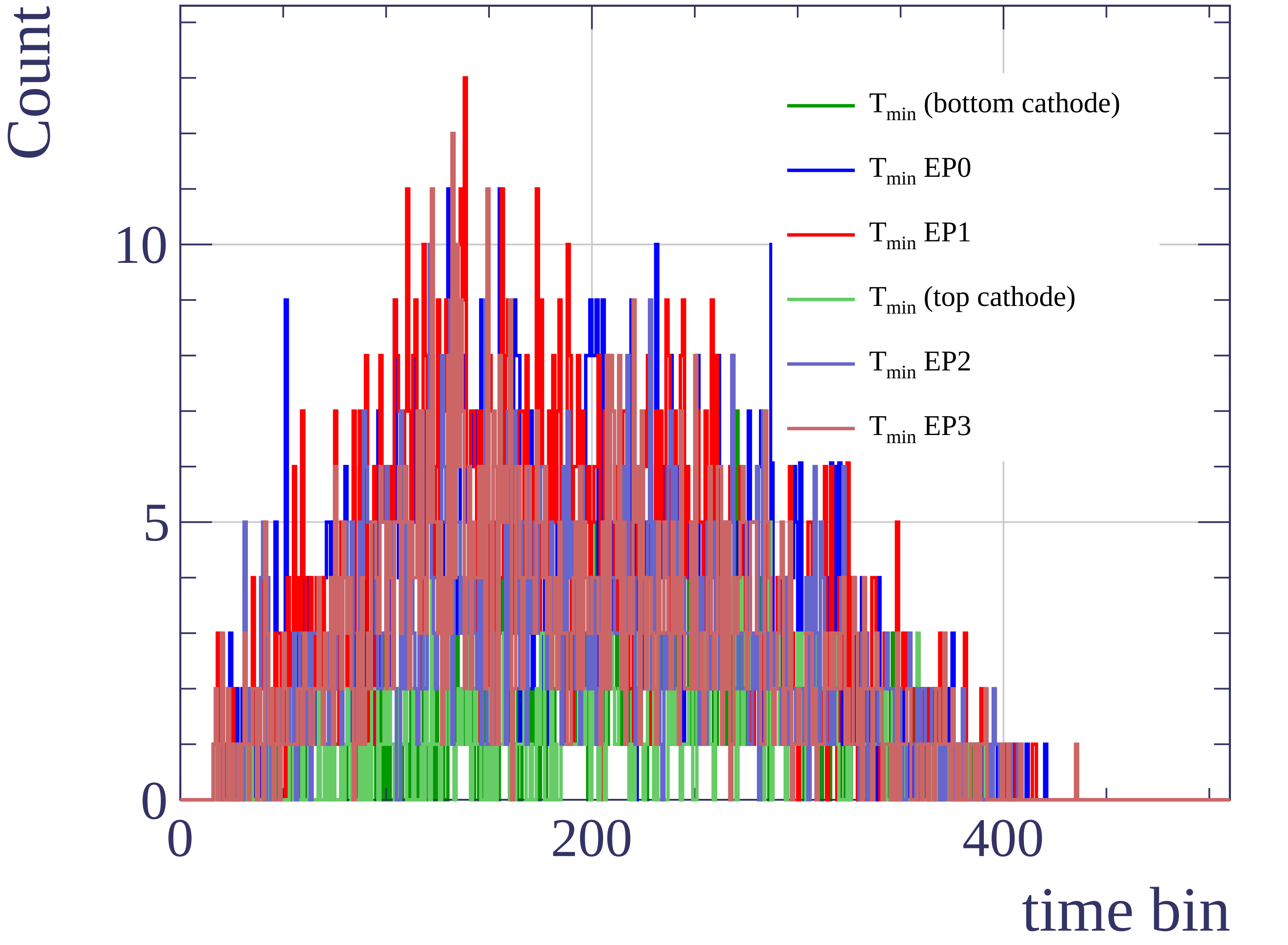
Count
dE/dx [ADC counts/cm]
θ ∈ [81°, 84°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

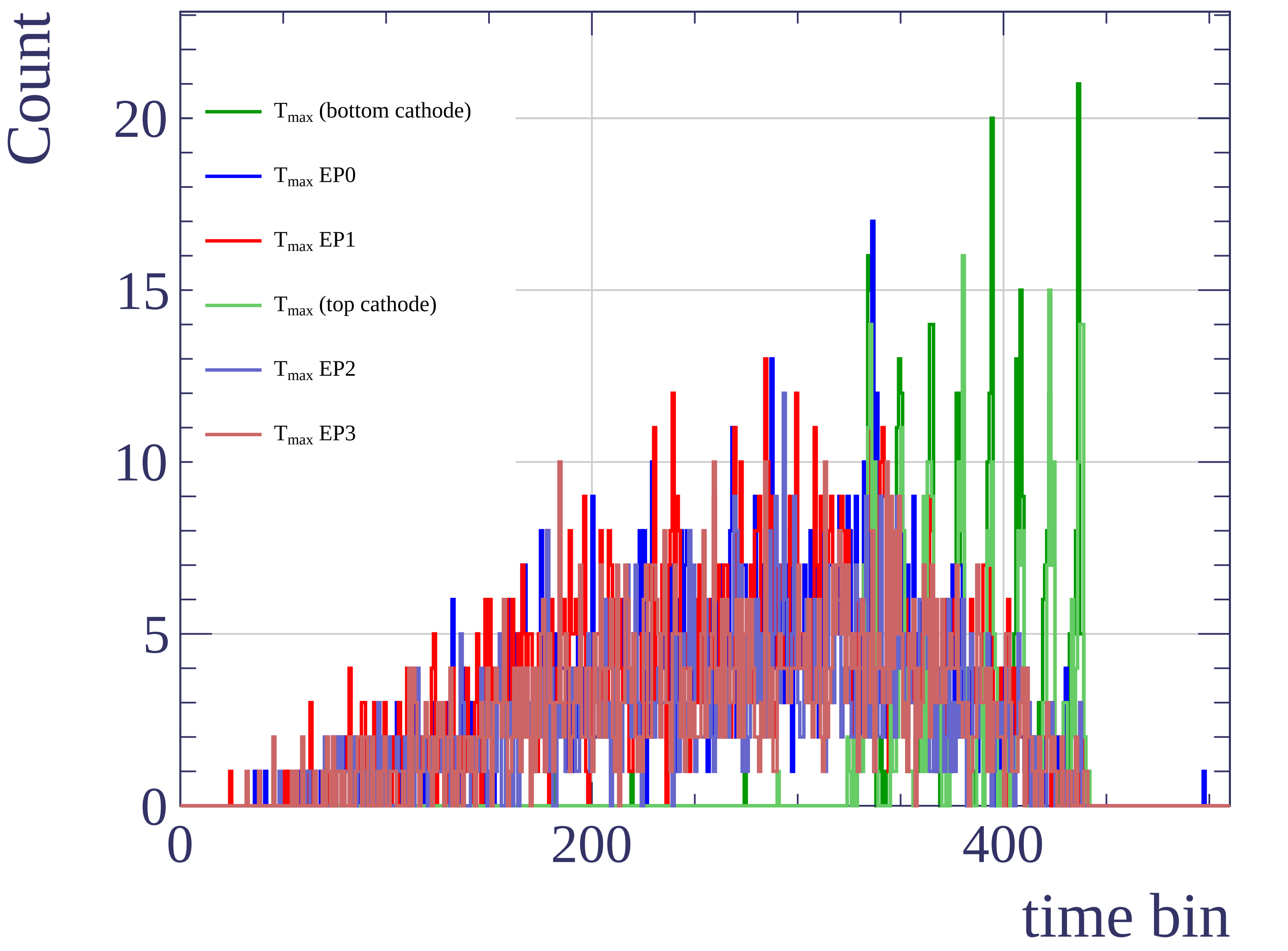


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [84°, 87°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [87°, 90°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
time bin
0
200
400
0
5
10
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
time bin
0
200
400
0
5
10
15
20
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3